*Research report after round 8. The run is still in progress: this is the report as it stands after this round, with charts drawn from the run's own numbers. The results and the writing are not final.*

# Where a Prompt Ban Sends a Code Model, and What Else Moves Its Algorithm Choice

This report is the internal record of a study of algorithmic diversity in an open-weight instruction-tuned model. The question that started the run is how "creativity" shows up inside a language model and whether it can be influenced: if a model is asked to implement a path search algorithm, does it always write the same one, and which knobs (sampling temperature, random context seeds, instructions, prohibitions, or edits to the activations) change the algorithm rather than only its wording. The behavioural puzzle comes from the inspiration repository by Drejc Pesjak that the request names, in which a closed model wrote Dijkstra's algorithm for a weighted shortest path task in 125 of 125 programs across five prompting strategies, and cumulative bans moved it to SPFA and Bellman-Ford rather than to a spread of alternatives. The same collapse appears in controlled measurements across 15 model configurations, where a no cue prompt gives Dijkstra 94.1% of the time [1], and in broader benchmarks of algorithmic diversity among correct programs [2, 3].

The hypothesis entering the run is mechanistic. When an instruction-tuned model is told not to use its default solution, the prohibition is expected to do two things: suppress the banned algorithm and promote the alternative the model associates with it, so the output collapses again on one predictable substitute instead of spreading over the repertoire. This mirrors the suppression plus promotion account of negation on factual questions [4]. Competing accounts say the substitute is chosen by baseline typicality, by proximity on an activation space solution manifold, by a few early forking tokens during generation, by a single list versus implement commitment direction, or that the whole collapse was installed by post-training. The report records, iteration by iteration, which of these the evidence supported and which it did not. All work so far uses Qwen3-8B [5] with thinking mode off unless stated, on the request's two path search tasks. *(Note added in iteration 2: iteration 2 adds Llama-3.1-8B-Instruct [22] and a bench of 13 further coding tasks; Sections 2.7 and 2.8.) (Note added in iteration 6: iteration 6 adds Qwen2.5-7B-Instruct, read with natural language autoencoders, and gemma-3-4b-it, read with Gemma Scope 2 sparse autoencoders; Sections 6.5 and 6.6.) (Note added in iteration 8: iteration 8 adds Qwen3-8B-Base at the fork, Llama-3.1-8B-Instruct as a held-out family for the temperature law, and Qwen3-14B, Qwen3-32B, gemma-3-27b-it and Qwen2.5-Coder-32B-Instruct in a screen under a ban of five names; Sections 8.7 and 8.8.)* The hypothesis reached its tested form after seven rounds of drafting and review before any experiment ran; the reviewer's overall score across the rounds was 5, 6, 6, 7, 7, 7 and 7, and the final round scored soundness 3, presentation 3, contribution 3 and fidelity 4.

# Iteration 1: a wide screen of six accounts on the two path search tasks

## 1.1 Why this iteration ran what it ran

The first iteration was planned as a screen rather than a test. The hypothesis carried four alternates that disagree with it about what decides the algorithm written after a ban, and none of them had been measured. The strategy therefore ran every account cheaply and in parallel on the same events, with one shared protocol, one per-sample output schema and a selection rule fixed before

any artifact ran, and it left inference to a later confirmation on reserved tasks. The selection rule had two columns. The PREDICT column scores each account's forecast of the distribution after a ban, committed by sha256 hash before sampling, by its log score on the realised compliant samples; an account survives if its mean log score beats the next best account by at least 0.10 nats on at least 3 events where the accounts' top-1 nominations disagree. The BUY column scores each knob by the correctness conditioned Hill number it reaches while keeping at least 80% of the baseline pass rate; a knob survives if it beats the next best knob by at least 0.5 effective algorithms. *(Correction, iteration 2: as written in the strategy, the PREDICT rule scores only informative events, those with the banned mode present at 10% or more of the latent baseline, compliance of at least 50% and at least two survivors, and the BUY rule names a winner only if it leads on the pooled tasks and in direction on each task. This paragraph omitted both conditions, and Section 1.11 applied the rules without them. Sections 2.3.6 and 2.3.7 recompute both.)* A third check, the locus diagnostic, asks where the algorithm decision is made: it would move every geometric analysis from the read position (the last prompt token, Section 1.2) to generated-token states if a decoder at a forking token beat the read position decoder by 20 points over a shuffled label control. The strategy also stated in advance that nothing in this iteration would be inferential.

Seven artifacts ran: one dataset and six experiments, each owning one account (Table 1). We call each experiment by the account it screens: the association, geometry, forking token, decommitment and instruction tuning experiments, and the decoding experiment, which covers sampling temperature, random seeds and creativity instructions. The strategy asked the dataset artifact to build a reserved bench of 12 further tasks with hidden tests, labellers and forecasts frozen before any ban, so that the account that survived the screen could be confirmed out of sample in the next iteration.

**Table 1.** The seven artifacts of iteration 1, the account each was asked to screen, and the knob or predictor it contributed.

| Artifact | Account screened | Predictor (PREDICT column) | Knob (BUY column) |
|---|---|---|---|
| Dataset: source data for an algorithm-choice bench | none (reserved bench for confirmation) | none | none |
| Association experiment | substitution by association vs typicality ladder | Borda association map; latent baseline frequency | single and cumulative prompt bans |
| Geometry experiment | proximity on the read position manifold; shift decomposition | geodesic kernel; read position decoder | affine directional ablation of the modal algorithm |
| Forking token experiment | algorithm fixed at forking tokens during generation | none | temperature raised only around forking tokens |
| Decommitment experiment | one list versus implement direction controls collapse | decommitted survivor distribution | activation addition of the decommitment direction |
| Instruction tuning experiment | post-training installed the collapse | base-model survivor distribution | subtraction of the instruct minus base direction |
| Decoding experiment | temperature and random seeds change wording, not ideas | none | temperature and top-k, seeds, creativity instructions, generate then select |

## 1.2 Shared setup and the conventions used in this report

**Tasks.** Both tasks come from the request. The grid task asks for `solve(grid, start, goal)` returning any valid 4-neighbour path on a grid of free and blocked cells, or an empty list when the goal is unreachable. Its catalogue has six algorithms, that is, breadth-first search (BFS), depth-first search (DFS), bidirectional BFS, iterative-deepening DFS (IDDFS), A* or greedy best-first search with a Manhattan heuristic (A* in tables), and unit weight Dijkstra. The shortest path task asks for the shortest distance from s to t in a directed graph with nonnegative integer weights and cycles, or -1. Its catalogue has five algorithms: Dijkstra, Bellman-Ford or SPFA (counted as one algorithm), bidirectional Dijkstra, Floyd-Warshall and Dial's bucket queue. Every experiment wrote its own hidden tests (between 39 and 48 cases per task) with a time limit of ten times the slowest reference implementation and a 1 s minimum, and every catalogue reference had to pass before any model sample was scored.

**Model and decoding.** Qwen3-8B in bf16 with the chat template and thinking off, sampled at temperature 1.0 with top-p 1.0 and no top-k unless a table says otherwise. Engines differed between artifacts because the containers had no C compiler: vLLM 0.11.0 [6] in eager mode in five experiments (association, forking token, decommitment, instruction tuning and decoding), and HF transformers `generate` in the geometry experiment and in the hooked steering arms. GPUs were an RTX 5090 (association and decoding experiments), an NVIDIA GPU with 23 GB of memory (geometry and instruction tuning experiments) or an RTX A4500 (forking token and decommitment experiments).

**Prompts.** Every prompt is a condition segment, then a fixed task specification of at least 50 tokens, then the assistant header. The read position is the last prompt token, so no token specific to a condition sits at or near it. The conditions shared by all experiments are the baseline (no condition segment or a neutral one), a ban (`Do not use X.`), the mention twin and the listing control. The mention twin (`X is one well-known approach to this task; implement any correct algorithm.`) puts the same name at the same place without forbidding it, because naming a forbidden item can itself prime it [7]. We call the condition that names every catalogue algorithm the listing control. Each condition segment has 10 paraphrases, and N = 100 means 10 paraphrases with 10 samples each. Each experiment wrote its own paraphrases.

**Labels and correctness.** Each program is labelled with one catalogue algorithm or "other" by a runtime signature tracer (the program runs on crafted inputs with instrumented heaps, queues, stacks and grid reads) and by AST rules; disagreements go to a blind LLM judge on OpenRouter. Recursion versus iteration never counts as a separate algorithm. A program is correct when it passes the hidden tests within the limit. Intervals on proportions are Wilson score intervals [8] unless stated.

**Quantities.** We use the following names throughout. The *banned mode share* is the fraction of all coded samples labelled as the banned algorithm. A *compliant sample* is a correct program not labelled as a banned algorithm. The *realised mode* of a ban is the most common label among compliant samples if it leads the next most common label by at least 15 points; otherwise we call the event one with no clear mode. The *correctness conditioned Hill number* (Hill_cc) is the exponential of the Shannon entropy of the label distribution over correct programs, the effective number of algorithms [9]; a value of 1 means one algorithm and 2 means two equally common ones. The *log score* of a predictor on a ban event is the mean natural logarithm of the probability it assigned to the labels of the realised compliant samples, after mixing in 5% uniform mass over the surviving algorithms and a 2% floor for "other". It is a strictly proper scoring rule [10]; higher is

better, and a uniform predictor over four survivors scores -1.406. Artifacts reported a quantity they called compliance in two different ways (one minus the banned mode share, or the fraction of compliant samples), so this report states the banned mode share and the number of compliant samples instead, recomputed from each artifact's per-sample rows.

**Labeller validation.** Each experiment validated its own labeller against a gold set adjudicated by two or three LLM judges, with Cohen's kappa [11] between the runtime and AST labels as the main agreement statistic. Table 2 collects the results. Four labellers passed their gates and two did not. Both failures were driven by runtime versus AST disagreement on small gold sets. The combined label (runtime, AST and judge) still agreed with the adjudicated label on 97% of gold programs in the association experiment, and on 93% (grid) and 70% (shortest path) in the decoding experiment, 92% on its correct gold programs.

**Table 2.** Labeller validation per experiment, as each artifact reported it (gold set sizes per task).

| Experiment | Gold programs (Grid / Shortest path) | Kappa runtime vs AST | Runtime vs adjudicated | Combined label vs adjudicated | References correct | Gate |
|---|---|---|---|---|---|---|
| Association | 50 / 42 (evaluation split; 92 pooled) | 0.753 pooled | 0.843 pooled | 0.966 pooled | 22/22 | failed (targets 0.8 and 0.9) |
| Geometry | 28 / 26 | 0.902 / 0.938 | 0.964 / 0.923 | not reported | 100% | passed |
| Forking token | 24 / 23 | 1.0 / 0.847 | 1.0 / 0.913 | 1.0 / 1.0 | 9/9, 8/8 (AST; Dial has no runtime signature) | passed |
| Decommitment | 10 baseline + 6 steered + references + 2 outside the catalogue per task | 1.00 / 0.82 | 1.00 / 0.92 | 1.00 / 1.00 | 12/12 | passed after two revision rounds on Shortest path |
| Instruction tuning | 25 references and mutants; 40 audit programs | 0.77 over all parseable programs | runtime never wrong on references | kappa between judges 0.93 | AST 100% | passed (reference gate) |
| Decoding | 30 / 30 | 0.337 / 0.111 on gold; 0.881 / 0.851 over all programs | not reported | 0.933 / 0.70 (0.917 / 0.923 on correct gold programs) | 9/9, 7/7 | failed (strict criteria) |

## 1.3 The dataset artifact: source data, not yet a bench

The dataset artifact was asked for a validated 12-task bench and delivered the raw material for one. It ran 98 Hugging Face Hub queries (574 unique datasets), screened 25 candidates plus the inspiration repository, and kept ten sources at pinned revisions, 61,469 examples in total (Table 3). For the 20 bench problems that LeetCode covers it parsed every I/O pair and reran the reference solutions, which passed 20 of 20. The LRU cache and streaming median problems are absent from LeetCodeDataset, so their tests must be generated.

**Table 3.** Sources kept by the dataset artifact (counts recomputed from `full_data_out.json`).

| Source | Examples | Role in the planned bench |
|---|---|---|
| LeetCodeDataset [12] | 2,869 | problem texts, references and parsed I/O pairs for 20 bench problems |
| Mercury [13] | 440 | human solutions with runtimes for 20 bench problems, including LRU cache and median |
| inspiration repository (Drejc Pesjak) | 225 | labels of the repository's shortest path programs (Dijkstra 132, Bellman-Ford/SPFA 34, bidirectional 30, other 21, Dial 5, A* 2, Floyd-Warshall 1); code not redistributed (no licence) |
| TheAlgorithms/Python | 1,354 | MIT implementations, 55 labelled by hand by task and algorithm |
| CLRS-Text [14] | 2,910 | execution traces of 15 catalogue algorithms for tracer tests |
| HumanEval+ [15] | 164 | harness validation |
| ENAMEL [16] | 164 | efficient references with input generators for time limits |
| Concreteness norms [17, 18] | 39,954 | 718 neutral nouns for neutral twins and context seeds |
| KodCode [19] | 2,767 | tested solutions, 253 naming a catalogue algorithm |
| XLCoST [20] | 10,622 | GeeksforGeeks programs, 209 naming a catalogue algorithm |

The frozen task specifications, hidden tests filtered against the specification, labellers, gold sets and forecasts frozen before any ban were not built. The reserved bench on which the strategy meant to confirm the surviving account next iteration therefore does not exist yet, and the confirmation step has to build it first. No LLM calls were made and the spend was zero.

## 1.4 The baseline collapse, measured six times

Every experiment drew its own baseline, so the collapse was measured six times with different paraphrases, test suites and engines (Table 4; association, geometry, forking token, decommitment, instruction tuning and decoding experiments). Of 1,200 baseline programs on the grid task, 1,199 are BFS. Of 1,200 on the shortest path task, 1,174 are Dijkstra (97.8%), above the 94.1% no cue prior [1]. The few non-modal programs came from three experiments, and the decommitment experiment's baseline was the least collapsed. A further 200 samples per task at temperature 1.3 in the association experiment were also 100% BFS and 100% Dijkstra.

**Table 4.** Baseline label counts per experiment (all coded samples; pass = share passing the hidden tests).

| Experiment (engine) | Grid N | Grid labels | Grid pass | Shortest path N | Shortest path labels | Shortest path pass |
|---|---|---|---|---|---|---|
| Association (vLLM) | 300 | BFS 300 | 0.933 | 300 | Dijkstra 300 | 1.00 |
| Geometry (HF) | 200 | BFS 200 | 0.99 | 200 | Dijkstra 200 | 1.00 |
| Forking token (vLLM) | 200 | BFS 200 | 0.91 | 200 | Dijkstra 200 | 0.995 |
| Decommitment (vLLM) | 200 | BFS 200 | 1.00 | 200 | Dijkstra 181, Bellman-Ford/SPFA 14, Dial 4, other 1 | 0.885 |
| Instruction tuning (vLLM) | 200 | BFS 200 | 1.00 | 200 | Dijkstra 193, Bellman-Ford/SPFA 7 | 1.00 |
| Decoding (vLLM) | 100 | BFS 99, unit weight Dijkstra 1 | 0.97 | 100 | Dijkstra 100 | 0.99 |

The collapse makes one of the planned predictors uninformative before any ban runs. The typicality ladder predicts that a ban redistributes mass in proportion to each survivor's baseline frequency. Every survivor's baseline frequency was zero in the association, geometry and forking token experiments and at most 7% elsewhere (Bellman-Ford/SPFA, 14 of 200 in the decommitment experiment), so the frequency forecast is uniform wherever the baseline is fully collapsed and its

top-1 nomination is a tie. The preregistered presence strata are empty for the same reason: no catalogue algorithm other than the default reached the 10% presence cut, so every non-modal single ban is a ban of an absent algorithm.

## 1.5 Association experiment: does the alternative the model names predict where a ban lands?

**Why.** This experiment screened the main hypothesis against its behavioural rival. The main hypothesis predicts the algorithm written after a ban from the model's association between algorithms; the typicality ladder predicts it from baseline frequencies restricted to the survivors. Both predictors were elicited and frozen before any ban prompt was generated, and every prediction file was chained by sha256 hash before the samples it predicts. The commitment audit found the chain intact and every event committed in advance (commit at 15:18:30 UTC, first ban generation at 15:20:05 UTC).

**Design.** The latent baseline was N = 300 per task at temperature 1.0, plus 200 samples at temperature 1.3 and 50 without the neutral condition segment. The association map was elicited on separate queries with thinking off: for each of the 11 catalogue algorithms X, the model returned 20 sampled complete rankings of the other algorithms by the criterion `how standard an alternative to X for this task`, aggregated by Borda count (each ranking gives an alternative points according to its rank) with one extra point per algorithm. We call the normalised result the *association map* A(X); its winner is the association nomination. The *clarity* of X is the share of the 20 rankings whose alternative ranked first is the Borda winner, and Kendall's W [21] measures agreement across the 20 rankings. The model also gave 20 free-form answers to the question `what is the standard alternative to X?`, and Llama-3.1-8B-Instruct [22] repeated the whole elicitation as an external map. The ban arms were all 11 single bans at N = 100, the mention twin of each at N = 50, the listing control at N = 100 and a greedy succession that bans the realised mode of each step cumulatively, in the manner of denial prompting [23]. Generation used vLLM on one RTX 5090 (about 6 GPU minutes); OpenRouter spend was USD 0.27 for the judge, gold set adjudication and the external elicitation.

**The association maps.** Table 5 gives the map for every catalogue algorithm. The two maps that matter are those of the defaults. For BFS, the model's rankings spread the Borda mass almost evenly over unit weight Dijkstra, bidirectional BFS and A*, with DFS ranked last; clarity was 0.20, the lowest of all 11 algorithms. Yet 19 of its 20 free-form answers named DFS. For Dijkstra, all 20 rankings put Bellman-Ford/SPFA first (clarity 1.0). The map from Llama agreed with the Borda winner of Qwen for 5 of 6 grid algorithms and 4 of 5 shortest path algorithms, and its map for BFS tied DFS and IDDFS on top.

**Table 5.** Association maps from the association experiment. Borda shares of the top alternative, clarity (share of 20 rankings whose first choice is the Borda winner), Kendall's W across the 20 rankings, the modal free-form answer (out of 20), and the external (Llama-3.1-8B) Borda winner and clarity.

| Task | X | Qwen Borda winner (share) | Clarity | Kendall's W | Qwen free-form (count/20) | Llama Borda winner | Llama clarity |
|---|---|---|---|---|---|---|---|
| Grid | BFS | unit weight Dijkstra (0.293); bidir BFS 0.283, A* 0.263, IDDFS 0.107, DFS 0.054 | 0.20 | 0.52 | DFS (19) | DFS = IDDFS (0.251) | 0.368 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Grid | DFS | BFS (0.395) | 1.0 | 0.577 | BFS (20) | BFS | 0.684 |
| Grid | bidirectional BFS | BFS (0.38) | 0.9 | 0.454 | A* (11) | BFS | 0.70 |
| Grid | IDDFS | BFS (0.395) | 1.0 | 0.513 | BFS (18) | BFS | 0.60 |
| Grid | A* | BFS (0.385) | 0.9 | 0.70 | BFS (20) | BFS | 0.60 |
| Grid | unit weight Dijkstra | BFS (0.395) | 1.0 | 0.589 | BFS (18) | BFS | 0.75 |
| Shortest path | Dijkstra | Bellman-Ford/SPFA (0.492); Floyd-Warshall 0.226, bidir Dijkstra 0.153, Dial 0.129 | 1.0 | 0.639 | Bellman-Ford/ SPFA 8, other 8, self 4 | Bellman-Ford/SPFA | 0.65 |
| Shortest path | Bellman-Ford/ SPFA | Dijkstra (0.435) | 0.65 | 0.573 | self 15, Dijkstra 5 | Dijkstra | 0.65 |
| Shortest path | bidirectional Dijkstra | Dijkstra (0.492) | 1.0 | 0.981 | Dijkstra (20) | Bellman-Ford/SPFA | 0.40 |
| Shortest path | Floyd-Warshall | Dijkstra (0.444) | 0.7 | 0.916 | Dijkstra (20) | Dijkstra | 0.55 |
| Shortest path | Dial | Dijkstra (0.492) | 1.0 | 0.841 | Dijkstra (20) | Dijkstra | 0.65 |

The Borda winners of the two models matched at a rate of 0.833 on the grid task and 0.8 on the shortest path task, with a mean Kendall tau (rank agreement between the two models' rankings) of 0.362 and 0.765. A post hoc map from gpt-4.1-mini, collected after the freeze and so not a contest predictor, showed the same split for BFS: its Borda winner was bidirectional BFS while its free-form answer was DFS. The artifact's reading is that the ranking question elicits the closest shortest path variant of BFS rather than the alternative a programmer reaches for.

**What the bans did.** Table 6 gives every condition of the experiment. The two modal bans were largely obeyed in this experiment: 10 of 100 grid programs and 9 of 100 shortest path programs still used the banned algorithm. Every compliant program landed on one substitute, DFS for the grid task (38 of 38) and Bellman-Ford/SPFA for the shortest path task (86 of 86). The low pass rate of the grid ban (0.46) comes from the DFS programs: 52 of 90 failed the hidden tests (43 wrong answers, 7 timeouts, 2 runtime errors), and forced DFS implementations passed only 3 of 10 times.

**Table 6.** Every condition in the association experiment (labels over all coded samples; Hill_cc over correct programs).

| Task | Condition | N | Labels | Pass rate | Hill_cc |
|---|---|---|---|---|---|
| Grid | baseline | 300 | BFS 300 | 0.933 | 1.0 |
| Grid | baseline, T = 1.3 | 200 | BFS 200 | 0.895 | 1.0 |
| Grid | baseline, no condition segment | 50 | BFS 50 | 0.94 | 1.0 |
| Grid | ban BFS (modal) | 100 | DFS 90, BFS 10 | 0.46 | 1.587 |
| Grid | ban DFS | 100 | BFS 96, DFS 4 | 0.88 | 1.115 |
| Grid | ban bidirectional BFS | 100 | BFS 99, other 1 | 0.92 | 1.0 |
| Grid | ban IDDFS | 100 | BFS 88, DFS 12 | 0.80 | 1.220 |
| Grid | ban A* | 100 | BFS 100 | 0.72 | 1.0 |
| Grid | ban unit weight Dijkstra | 100 | BFS 92, DFS 8 | 0.71 | 1.137 |
| Grid | listing control | 100 | BFS 97, DFS 3 | 0.92 | 1.0 |
| Grid | mention BFS | 50 | BFS 50 | 1.00 | 1.0 |

| | | | | | |
|---|---|---|---|---|---|
| Grid | mention DFS | 50 | DFS 50 | 0.30 | 1.0 |
| Grid | mention bidirectional BFS | 50 | BFS 43, bidirectional BFS 5, other 2 | 0.72 | 1.405 |
| Grid | mention IDDFS | 50 | IDDFS 36, DFS 14 | 0.18 | 1.417 |
| Grid | mention A* | 50 | A* 28, BFS 12, unit weight Dijkstra 8, DFS 2 | 0.82 | 2.756 |
| Grid | mention unit weight Dijkstra | 50 | BFS 50 | 0.90 | 1.0 |
| Grid | succession step 2 (ban BFS and DFS) | 100 | DFS 79, BFS 15, unit weight Dijkstra 4, A* 1, IDDFS 1 | 0.53 | 2.278 |
| Shortest path | baseline | 300 | Dijkstra 300 | 1.00 | 1.0 |
| Shortest path | baseline, T = 1.3 | 200 | Dijkstra 200 | 1.00 | 1.0 |
| Shortest path | baseline, no condition segment | 50 | Dijkstra 50 | 1.00 | 1.0 |
| Shortest path | ban Dijkstra (modal) | 100 | Bellman-Ford/SPFA 90, Dijkstra 9, other 1 | 0.95 | 1.368 |
| Shortest path | ban Bellman-Ford/SPFA | 100 | Dijkstra 94, Bellman-Ford/SPFA 5, other 1 | 0.93 | 1.061 |
| Shortest path | ban bidirectional Dijkstra | 100 | Dijkstra 97, Bellman-Ford/SPFA 3 | 1.00 | 1.144 |
| Shortest path | ban Dial | 100 | Dijkstra 95, Bellman-Ford/SPFA 5 | 0.97 | 1.106 |
| Shortest path | ban Floyd-Warshall | 100 | Dijkstra 77, Bellman-Ford/SPFA 23 | 0.71 | 1.191 |
| Shortest path | listing control | 100 | Dijkstra 97, Bellman-Ford/SPFA 3 | 1.00 | 1.144 |
| Shortest path | mention Dijkstra | 50 | Dijkstra 50 | 1.00 | 1.0 |
| Shortest path | mention Bellman-Ford/SPFA | 50 | Bellman-Ford/SPFA 50 | 0.84 | 1.0 |
| Shortest path | mention bidirectional Dijkstra | 50 | Dijkstra 50 | 0.30 | 1.0 |
| Shortest path | mention Dial | 50 | Dijkstra 50 | 1.00 | 1.0 |
| Shortest path | mention Floyd-Warshall | 50 | Floyd-Warshall 43, Bellman-Ford/SPFA 7 | 0.88 | 1.115 |
| Shortest path | succession step 2 (ban Dijkstra and Bellman-Ford/SPFA) | 100 | Bellman-Ford/SPFA 65, Dijkstra 35 | 0.85 | 1.960 |

*(Added in iteration 2: the caption says every condition, but the experiment also ran eleven forced implementation conditions of 10 samples each, which this table omits; they are given in Table 50, Section 2.3.10. Forced IDDFS passed 0 of 10, forced Dial 0 of 10, and forced bidirectional Dijkstra was written as plain Dijkstra 10 of 10 times.)*

**The forecast contest.** Only the two modal bans were informative events, since every other ban removed an algorithm the model never produced. Table 7 scores every predictor on them. The association map missed the grid event, where it nominated unit weight Dijkstra and ranked DFS last, and hit the shortest path event. Against the uniform frequency forecast it lost by 1.188 nats per sample on the grid event and won by 0.652 on the shortest path event, a mean of -0.268 (association wins 1 of 2). The two predictors that hit both events were not in the preregistered

contest: the model's own free-form answer and what we define as the listing control distribution, the label distribution the model produces when every catalogue name is listed. Their log scores were close to zero.

**Table 7.** Forecast contest on the two informative events of the association experiment (raw log score in nats per compliant sample; top-1 nomination in brackets; chance top-1 is 1/5 and 1/4).

| Predictor | Grid ban BFS (38 compliant, all DFS) | Shortest path ban Dijkstra (86 compliant, all Bellman-Ford/SPFA) | Mean log score |
|---|---|---|---|
| Association map, Borda (preregistered) | -2.817 (unit weight Dijkstra, miss) | -0.755 (Bellman-Ford/SPFA, hit) | -1.786 |
| Association map, first rank counts | -2.474 (bidirectional BFS, miss) | -0.190 (hit) | -1.332 |
| External map, Llama-3.1-8B Borda | -1.412 (DFS/IDDFS tie) | -0.928 (hit) | -1.170 |
| Latent frequency (typicality ladder) | -1.630 (tie over all survivors) | -1.406 (tie) | -1.518 |
| Free-form answer (standard alternative question) | -0.112 (DFS, hit) | -0.058 (hit) | -0.085 |
| Listing control distribution | -0.061 (DFS, hit) | -0.058 (hit) | -0.060 |

The family test, which the hypothesis treats as its direct test of substitution against subtraction, was degenerate here. It fits a substitution family (a point mass on the association winner with a uniform leak) and a subtraction family (the baseline restricted to the survivors) on half of an event's compliant samples and scores both on the other half. Because the baseline gave zero mass to every survivor, the subtraction family was uniform: the grid event was a tie at -1.63 and the shortest path event went to substitution (-0.058 against -1.406). With one decisive event the family test says nothing about the hypothesis.

**Mention twins are recommendations.** The mention twin was meant as a placebo that carries the name without the prohibition. It is not one. Mentioning an absent algorithm moved 100 points of mass to DFS and to Bellman-Ford/SPFA, 86 points to Floyd-Warshall, 72 to IDDFS and 56 to A*, and 10 to bidirectional BFS; it moved nothing for unit weight Dijkstra on grids, Dial and bidirectional Dijkstra. Several mentioned algorithms were then implemented badly (mention DFS passed 0.30, mention IDDFS 0.18, mention bidirectional Dijkstra 0.30).

**The sham ban control.** Bans of absent algorithms were planned as a negative control that should match the mention twin. Because the mention twin recommends, the comparison inverts (Table 8): a sham ban kept the default at 77-100%, while its mention twin moved the model to the mentioned algorithm whenever the model could write it. None of the nine sham events was equivalent to its twin under the paraphrase null, whose 95th percentile was 0.003 bits. Against the baseline, sham bans moved some mass toward a related algorithm: banning Floyd-Warshall gave 23% Bellman-Ford/SPFA, banning IDDFS 12% DFS.

**Table 8.** Sham bans (bans of absent algorithms) against their mention twins. JS is the Jensen-Shannon divergence, that is, a symmetric and bounded divergence in bits between the two label distributions; the default's share is BFS on the grid task and Dijkstra on the shortest path task.

| Event | JS (ban vs mention) | Default share under ban | Default share under mention | Shift (points) | Compliant samples (realised mode) |
|---|---|---|---|---|---|
| Grid ban DFS | 0.801 | 0.96 | 0.0 | +96 | 86 (BFS) |

| | | | | | |
|---|---|---|---|---|---|
| Grid ban bidirectional BFS | 0.053 | 0.99 | 0.86 | +13 | 92 (BFS) |
| Grid ban IDDFS | 0.741 | 0.88 | 0.0 | +88 | 80 (BFS 76, DFS 4) |
| Grid ban A* | 0.506 | 1.0 | 0.24 | +76 | 72 (BFS) |
| Grid ban unit weight Dijkstra | 0.027 | 0.92 | 1.0 | -8 | 71 (BFS 69, DFS 2) |
| Shortest path ban Bellman-Ford/SPFA | 0.780 | 0.94 | 0.0 | +94 | 92 (Dijkstra) |
| Shortest path ban bidirectional Dijkstra | 0.007 | 0.97 | 1.0 | -3 | 100 (Dijkstra 97, Bellman-Ford/SPFA 3) |
| Shortest path ban Floyd-Warshall | 0.745 | 0.77 | 0.0 | +77 | 71 (Dijkstra 68, Bellman-Ford/SPFA 3) |
| Shortest path ban Dial | 0.015 | 0.95 | 1.0 | -5 | 97 (Dijkstra 95, Bellman-Ford/SPFA 2) |

**The succession stalled at step 2.** Banning the first substitute as well (BFS and DFS on the grid task, Dijkstra and Bellman-Ford/SPFA on the shortest path task) failed outright. 94 of 100 and 100 of 100 programs used a banned algorithm, leaving 5 and 0 compliant samples, and the preregistered stop rule (fewer than 20 compliant) ended both successions. The model reimplemented the banned mechanism under another description (Table 9). On the grid task 41 of 94 recurrent programs never named the banned algorithm and 20 called their DFS `backtracking`; on the shortest path task 58 of 100 never named it, and 19 described SPFA as `BFS with pruning`.

**Table 9.** Recurrence of banned algorithms and how often recurrent programs avoid naming them (regular expression counts over comments and identifiers).

| Event | Recurrent programs | Naming the banned algorithm | Not naming it | Most common descriptors |
|---|---|---|---|---|
| Grid ban BFS | 10 | 1 | 9 | `queue` 10 |
| Grid ban DFS (sham) | 4 | 2 | 2 | `backtracking` 2 |
| Grid succession step 2 (BFS, DFS) | 94 | 53 | 41 | `backtracking` 20, `queue` 16, `BFS` 7 |
| Shortest path ban Dijkstra | 9 | 8 | 1 | `priority queue` 9, `queue` 8 |
| Shortest path ban Bellman-Ford/SPFA (sham) | 5 | 1 | 4 | `queue` 3 |
| Shortest path succession step 2 (Dijkstra, Bellman-Ford/SPFA) | 100 | 42 | 58 | `BFS` 98, `queue` 91, `priority queue` 48, `pruning` 19 |

**The ladder limit test.** The hypothesis predicts that a ban fails more often when the banned algorithm's alternative is unclear. With four present algorithm events the relation cannot be estimated (Table 10). The grid modal ban had the lowest clarity and was obeyed; the two step-2 events had higher clarity and failed. The pooled Spearman rank correlation between recurrence and clarity was -0.2 over 4 events.

**Table 10.** Recurrence of the banned algorithm against clarity of its alternative.

| Event | Recurrence | Clarity | Clarity corrected for chance | Kendall's W |
|---|---|---|---|---|
| Grid ban BFS | 0.10 | 0.20 | 0.0 | 0.52 |
| Grid succession step 2 | 0.94 | 0.45 | 1.0 | 0.577 |
| Shortest path ban Dijkstra | 0.09 | 1.0 | 1.0 | 0.639 |

| | | | | |
|---|---|---|---|---|
| Shortest path succession step 2 | 1.00 | 0.65 | 0.533 | 0.573 |

**What this experiment settled.** On the one event where the model's ranked association map and its free-form answer disagreed (the grid task), the free-form answer was right and the map was wrong; on the other event both were right. The ranking instrument, which the design chose because it gives full support where a free-form answer gives a point mass, does not measure the same association the model acts on. The typicality predictor built from baseline frequencies is uninformative under a full collapse, while the listing control distribution, a typicality measure taken under a prompt that names every algorithm, predicted both events. Bans naming several algorithms fail by disguise, so compliance has to be checked at the level of mechanism, not name. The labeller gate failed its runtime accuracy targets (Table 2), and every runtime and AST disagreement was resolved by the judge. Both task suites passed their acceptance checks: all 22 reference implementations passed, and every planted mutant (an off by one distance, a hop count in place of a weighted distance, a path through a wall, a diagonal step) was rejected. Resending 6 requests (64 samples) with the same request seeds reproduced 39 of 64 texts exactly, so results reproduce in distribution rather than token for token. An independent script rederived all 36 headline numbers from the raw files.

## 1.6 Geometry experiment: the read position manifold, the ban shift, and activation removal

**Why.** The hypothesis places the algorithm decision at the read position and predicts that the ban shift there, the change in the read position state that the ban causes, has a larger component toward the associated alternative than away from the banned algorithm. This experiment built the solution manifold at the read position, tested whether the read position encodes the algorithm at all, nominated the geodesically nearest survivor as a third predictor (the geodesic kernel, a forecast that weights each survivor by the exponential of minus its geodesic distance from the banned algorithm's centroid, scaled by the median distance between centroids), decomposed the ban shift, and tried to remove the modal algorithm with an activation edit instead of a prompt. It ran HF `generate` on one NVIDIA GPU with 23 GB of memory, since vLLM could not start, for about 2.2 GPU hours; OpenRouter spend was about USD 0.09.

**Design.** Forced implementation prompts (`Implement X for this task`) with 30 paraphrases per catalogue algorithm, plus a name masked set that describes X without naming it, gave a point cloud of residual stream activations at the read position for each algorithm for layers 8 to 28 of 36. A logistic linear probe from the read position state to the algorithm label was trained with one paraphrase held out at a time and compared with 20 shuffled label refits, the control-task logic of probe selectivity [24]. Two gate statistics were computed per layer: transfer accuracy from named to name masked prompts, and the Jensen-Shannon divergence, that is, a symmetric and bounded divergence, between the probe's decoded distribution on the baseline prompts and the realised label distribution of the baseline. The manifold used principal component analysis down to 95% of the variance, a k-nearest-neighbour graph with $k = 10$, geodesic distances along the graph as in Isomap [25], two dimensional multidimensional scaling and the TwoNN intrinsic-dimension estimator [26], following the manifold-steering instrument [27].

**Manipulation check.** The model can write most catalogue algorithms when told to, but not all of them correctly (Table 11). Forced IDDFS was implemented as IDDFS every time on the grid task and never passed the tests; forced bidirectional Dijkstra and forced Dial were implemented as asked in only 33% and 37% of samples. These two shortest path algorithms were flagged, and their centroids rest on fewer valid points.

**Table 11.** Forced implementation prompts in the geometry experiment (30 samples each): share labelled as the requested algorithm, and pass rate, for named and name masked prompts.

| Task | Requested algorithm | Named: share as asked / pass | Name masked: share as asked / pass |
|---|---|---|---|
| Grid | BFS | 1.0 / 1.0 | 1.0 / 0.933 |
| Grid | DFS | 1.0 / 0.30 | 0.933 / 0.167 |
| Grid | bidirectional BFS | 0.667 / 0.233 | 0.967 / 0.067 |
| Grid | IDDFS | 1.0 / 0.0 | 0.767 / 0.0 |
| Grid | A* | 0.967 / 0.967 | 1.0 / 0.933 |
| Grid | unit weight Dijkstra | 1.0 / 0.90 | 1.0 / 0.833 |
| Shortest path | Dijkstra | 1.0 / 1.0 | 1.0 / 0.967 |
| Shortest path | Bellman-Ford/SPFA | 1.0 / 0.767 | 0.767 / 0.667 |
| Shortest path | bidirectional Dijkstra | 0.333 / 0.50 | 0.20 / 0.433 |
| Shortest path | Floyd-Warshall | 1.0 / 1.0 | 0.733 / 0.667 |
| Shortest path | Dial | 0.367 / 0.30 | 0.567 / 0.70 |

**Does the read position encode the algorithm?** Named probes reached 0.98-1.0 accuracy at every layer, which says only that the algorithm's name is in the context. The informative statistic is transfer to name masked prompts (Figure 1). It rose with depth and peaked at layer 27 on the grid task (0.672 against a shuffled label 95th percentile of 0.218) and at layer 24 on the shortest path task (0.54 against 0.307). The decoder criterion split the tasks. On the shortest path task the probe decoded the baseline state as 98% Dijkstra (JS 0.011 against a null 5th percentile of 0.030). On the grid task it decoded the baseline as DFS 0.60, unit weight Dijkstra 0.27 and BFS 0.12, while the model writes BFS every time, so the grid read position gate failed and the grid geometry below carries that flag.

*[Figure fig_probe_layers: generated in the report phase, not in this draft]*

**The manifold.** The point clouds are low-dimensional: TwoNN intrinsic dimension 4.16 on the grid task and 5.24 on the shortest path task, against 11.92 and 11.58 for nulls with shuffled columns (5th-95th percentiles 10.78-13.46 and 10.64-12.71). MDS stress-1, the residual misfit of the two dimensional embedding, was 0.22 on both tasks and geodesic and Euclidean distances had Spearman correlation 0.87 and 0.90. Figure 2 draws the two maps with every condition's read position point, and Table 12 gives the geodesic distances from the default's centroid. On the grid task, BFS's two nearest centroids are DFS (41.19) and unit weight Dijkstra (58.39); A*, IDDFS and bidirectional BFS lie 71-95 away. On the shortest path task Dijkstra's nearest centroid is Floyd-Warshall (25.49), with Bellman-Ford/SPFA and bidirectional Dijkstra tied at about 42.

*[Figure fig_manifold: generated in the report phase, not in this draft]*

**Table 12.** Geodesic distances between centroids at the read position (grid task, layer 27; shortest path task, layer 24), from the default algorithm's centroid.

| Task | From | To: distance |
|---|---|---|
| Grid | BFS | DFS 41.19; unit weight Dijkstra 58.39; A* 71.06; IDDFS 81.66; bidirectional BFS 94.95 |
| Shortest path | Dijkstra | Floyd-Warshall 25.49; Bellman-Ford/SPFA 42.18; bidirectional Dijkstra 42.46; Dial 54.96 |

The ban points do not sit near the centroids of the algorithms the model then wrote. Under the grid ban the read position point moved far from both BFS and DFS (from (-20.26, -58.22) for the baseline to (-24.67, 10.46)), while the mention point of BFS stayed close to the BFS centroid; the ban points of all six grid algorithms clustered together in one region of the map regardless of which algorithm was banned. This matches the decomposition below: most of the ban shift is not along

the algorithm directions.

**The bans in this experiment.** Here the modal bans mostly failed (Table 13). On the grid task 62 of 100 programs were still BFS and the 18 compliant programs split into DFS 10 and unit weight Dijkstra 8, so there was no clear realised mode. On the shortest path task 72 of 100 were still Dijkstra and all 22 compliant programs were Bellman-Ford/SPFA. Compliance depended on the paraphrase: on the grid task four of the ten ban wordings gave 10 of 10 BFS while two gave 0 of 10 (Section 1.11). Some programs acknowledged the ban in a comment and used the algorithm anyway (`# Dijkstra's algorithm (but since the user said not to use Dijkstra's algorithm, this is a workaround)`). Three step-2 bans were run as exploratory events after the preregistered succession stopped. On the grid task the displaced mass landed only on BFS's two geodesic neighbours, DFS and unit weight Dijkstra, and banning one of them moved the compliant programs to the other; it never reached A*, IDDFS or bidirectional BFS. In the words of the request, mass moved to the edge of the banned region rather than to the opposite side. On the shortest path task the mass went to the second nearest centroid, Bellman-Ford/SPFA, and the nearest, Floyd-Warshall, received none. *(Correction, iteration 2: the claim that Bellman-Ford/SPFA is second nearest does not survive the sensitivity matrix of this experiment, which puts bidirectional Dijkstra (40.91) closer to Dijkstra than Bellman-Ford/SPFA (42.18) once the two flagged centroids are recomputed from correctly labelled forced samples. Under paraphrase level resampling Floyd-Warshall is nearest in every subsample, and Bellman-Ford/SPFA is second at layers 23 to 27 and third at layers 21 and 22; Section 2.3.2.)*

**Table 13.** Ban events in the geometry experiment (all coded samples; compliant = correct and not banned).

| Event | N | Labels | Compliant (labels) | Realised mode | Status |
|---|---|---|---|---|---|
| Grid ban BFS | 100 | BFS 62, DFS 29, unit weight Dijkstra 9 | 18 (DFS 10, unit weight Dijkstra 8) | no clear mode | primary |
| Grid ban BFS and DFS | 50 | BFS 29, unit weight Dijkstra 19, DFS 2 | 18 (unit weight Dijkstra 18) | unit weight Dijkstra | exploratory |
| Grid ban BFS and unit weight Dijkstra | 50 | BFS 31, DFS 17, unit weight Dijkstra 2 | 5 (DFS 5) | DFS | exploratory |
| Shortest path ban Dijkstra | 100 | Dijkstra 72, Bellman-Ford/SPFA 27, other 1 | 22 (Bellman-Ford/ SPFA 22) | Bellman-Ford/SPFA | primary |
| Shortest path ban Dijkstra and Bellman-Ford/SPFA | 50 | Dijkstra 37, Bellman-Ford/SPFA 12, other 1 | 0 | no clear mode | exploratory |

The ban raised the correctness conditioned Hill number from 1.00 to 2.034 on the grid task at a pass-rate cost of 0.99 to 0.76, and from 1.00 to 1.723 on the shortest path task at 1.00 to 0.94. Most of this diversity is the banned algorithm mixed with its substitute, since banned programs that pass the tests count toward Hill_cc.

**The forecast contest.** Forecasts for 45 planned events were committed by hash (sha256 186f7182...) at 16:36:57 UTC, before any ban or removal sample. Table 14 scores them. This experiment's own association elicitation produced different maps from the association experiment's: it nominated A* for BFS (clarity 0.80, Kendall's W 0.886) and Dial for Dijkstra (clarity 0.579). Both nominations missed and the association map was the worst predictor, worse than uniform. The geodesic kernel was the best on the two primary events (mean -1.433 against -1.563 for uniform), by a margin of 0.13 nats per sample, and no predictor scored a top-1 hit on a primary event.

**Table 14.** Forecast contest in the geometry experiment (raw mean log score; top-1 nomination; the

samples scored are all labels of the event other than the banned algorithm).

| Event (samples scored) | Association map | Latent frequency | Geodesic kernel | Read position decoder |
|---|---|---|---|---|
| Grid ban BFS (38) | -3.543 (A*) | -1.630 (tie) | -1.337 (DFS) | -1.287 (DFS) |
| Shortest path ban Dijkstra (28) | -1.647 (Dial) | -1.496 (tie) | -1.530 (Floyd-Warshall) | -1.587 (bidirectional Dijkstra) |
| Grid ban BFS and DFS, exploratory (19) | -0.998 (unit weight Dijkstra, hit) | -1.406 (tie) | -1.073 (unit weight Dijkstra, hit) | -4.124 (IDDFS) |
| Grid ban BFS and unit weight Dijkstra, exploratory (17) | -3.966 (A*) | -1.406 (tie) | -1.280 (A*) | -1.372 (IDDFS) |
| Shortest path ban Dijkstra and Bellman-Ford/SPFA, exploratory (1) | -3.912 | -3.912 | -3.912 | -3.912 |
| Mean, 2 primary events | -2.595 | -1.563 | -1.433 | -1.437 |
| Mean, all 5 events; top-1 hits | -2.813; 1/5 | -1.970; ties | -1.826; 1/5 | -2.456; 0/5 |

**The shift decomposition.** The ban shift is the read position state under the ban minus the state under its reference, computed against the mention twin and against the listing control. It was decomposed in a preregistered order into a component away from the banned algorithm (u_ren), toward the association winner (u_sub), toward the most frequent survivor (u_typ) and toward the geodesically nearest other survivor (u_rel), each orthogonalised against the previous ones. The preregistered share test needed the u_sub share to exceed the u_ren share and the u_ren share to exceed a split half paraphrase null under both references. Table 15 gives the shares at the gate layer. The four named components together explained 1-39% of the squared norm; the rest was residual. The u_ren share exceeded its null in only 7 of 130 cells across the layer band (Dial under the mention reference at layers 23-27, and bidirectional Dijkstra twice), never under both references, so every algorithm was inconclusive. The cosine of each shift with the banned name's input embedding and unembedding row was at most 0.034 in absolute value. A synthetic test with a planted 36/64 split recovered it, so the decomposition code works; the shifts simply do not lie along the algorithm directions.

**Table 15.** Squared norm shares of the ban shift at the gate layer (layer 27 for the grid task, layer 24 for the shortest path task); u_typ is undefined when the most frequent survivor is also the association winner or no survivor has baseline mass.

| Task | Banned X | Reference | u_ren | u_sub | u_typ | u_rel | Residual | u_ren null p95 |
|---|---|---|---|---|---|---|---|---|
| Grid | BFS | mention | 0.067 | 0.054 | 0.012 | 0.003 | 0.863 | 0.248 |
| Grid | BFS | listing control | 0.058 | 0.061 | 0.034 | 0.004 | 0.844 | 0.248 |
| Grid | DFS | mention | 0.042 | 0.038 | n/a | 0.000 | 0.920 | 0.170 |
| Grid | DFS | listing control | 0.052 | 0.017 | n/a | 0.002 | 0.929 | 0.170 |
| Grid | bidirectional BFS | mention | 0.078 | 0.020 | n/a | 0.000 | 0.901 | 0.263 |
| Grid | bidirectional BFS | listing control | 0.001 | 0.099 | n/a | 0.003 | 0.897 | 0.263 |
| Grid | IDDFS | mention | 0.090 | 0.005 | n/a | 0.029 | 0.876 | 0.274 |
| Grid | IDDFS | listing control | 0.003 | 0.034 | n/a | 0.054 | 0.909 | 0.274 |
| Grid | A* | mention | 0.086 | 0.005 | n/a | 0.014 | 0.895 | 0.134 |
| Grid | A* | listing control | 0.003 | 0.004 | n/a | 0.003 | 0.990 | 0.134 |
| Grid | unit weight Dijkstra | mention | 0.017 | 0.041 | 0.014 | 0.000 | 0.927 | 0.287 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Grid | unit weight Dijkstra | listing control | 0.130 | 0.076 | 0.000 | 0.011 | 0.783 | 0.287 |
| Shortest path | Dijkstra | mention | 0.004 | 0.002 | 0.041 | 0.013 | 0.940 | 0.311 |
| Shortest path | Dijkstra | listing control | 0.047 | 0.003 | 0.050 | 0.000 | 0.900 | 0.311 |
| Shortest path | Bellman-Ford/SPFA | mention | 0.000 | 0.069 | n/a | 0.021 | 0.909 | 0.298 |
| Shortest path | Bellman-Ford/SPFA | listing control | 0.104 | 0.094 | n/a | 0.005 | 0.797 | 0.298 |
| Shortest path | bidirectional Dijkstra | mention | 0.107 | 0.079 | n/a | 0.001 | 0.812 | 0.126 |
| Shortest path | bidirectional Dijkstra | listing control | 0.003 | 0.027 | n/a | 0.000 | 0.970 | 0.126 |
| Shortest path | Floyd-Warshall | mention | 0.000 | 0.019 | n/a | 0.004 | 0.976 | 0.212 |
| Shortest path | Floyd-Warshall | listing control | 0.075 | 0.005 | n/a | 0.001 | 0.919 | 0.212 |
| Shortest path | Dial | mention | 0.329 | 0.056 | n/a | 0.001 | 0.614 | 0.267 |
| Shortest path | Dial | listing control | 0.014 | 0.015 | n/a | 0.001 | 0.970 | 0.267 |

**Activation removal of the modal algorithm.** The removal used affine directional ablation [28, 29], anchored at the mean of the survivors: the component of each activation along a difference-in-means direction d_X for the modal algorithm was moved to the survivors' mean, scaled by a coefficient alpha. Neither direction check reached its 0.9 target: the held-out AUC of d_X was 0.89 and 0.79 at the read position and at best 0.66 and 0.76 at generated positions. The fallback declared in advance therefore ran. It edited the prompt suffix including the read position, over a 5-layer band on the grid task and a 7-layer band on the shortest path task, at alpha 0.5, 1 and 2, with two random directions of identical edit norm under common random numbers (the same sampling seeds for edited and unedited runs). It moved 0 of 50 samples on either task at every alpha (Table 16). Between 78% and 90% of the edited outputs were byte-identical to their alpha = 0 twins, as were 80-96% of the random direction outputs. The hook tests passed on both tasks: at alpha = 0 the logits were identical to the unhooked model, the remaining projection on d_X after the edit was below 1e-6 of its value before, and random and d_X edits had exactly equal norms. A post hoc full variant that also edited every generated position still left 25 of 25 programs on BFS and on Dijkstra at alpha 2; at alpha 4 the d_X edit destroyed generation (25 of 25 without code on both tasks) and the random direction broke most outputs as well. The verdict, as declared, was untestable, because the removal was unsuccessful at the tested coefficients. The comparison between prompt bans and activation removal, which the hypothesis treats as a headline, could not be made.

**Table 16.** Affine directional ablation of the modal algorithm (prompt suffix variant declared in advance unless marked full; HF engine; pass rates over all samples).

| Task | Arm | Alpha | N | Labels | Pass | Moved off the modal algorithm | Identical text to alpha = 0 |
|---|---|---|---|---|---|---|---|
| Grid | hooked control | 0 | 50 | BFS 50 | 1.0 | 0/50 | 1.0 |
| Grid | d_X | 0.5 / 1.0 / 2.0 | 50 each | BFS 50 each | 1.0 each | 0/50 each | 0.80 / 0.86 / 0.78 |
| Grid | random 1 | 0.5 / 1.0 / 2.0 | 25 each | BFS 25 each | 1.0 each | 0/25 each | 0.80 / 0.80 / 0.84 |
| Grid | random 2 | 0.5 / 1.0 | 25 each | BFS 25 each | 1.0 each | 0/25 each | 0.80 / 0.84 |
| Grid | d_X, full (post hoc) | 2.0 | 25 | BFS 25 | 0.76 | 0/25 | 0.0 |
| Grid | d_X, full (post hoc) | 4.0 | 25 | no code 25 | 0.0 | 0/25 | 0.0 |
| Grid | random 1, full (post hoc) | 4.0 | 25 | BFS 18, other 6, no code 1 | 0.0 | 6/25 | 0.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Shortest path | hooked control | 0 | 50 | Dijkstra 50 | 1.0 | 0/50 | 1.0 |
| Shortest path | d_X | 0.5 / 1.0 / 2.0 | 50 each | Dijkstra 50 each | 1.0 each | 0/50 each | 0.90 / 0.86 / 0.82 |
| Shortest path | random 1 | 0.5 / 1.0 / 2.0 | 25 each | Dijkstra 25 each | 1.0 each | 0/25 each | 0.88 / 0.96 / 0.92 |
| Shortest path | random 2 | 0.5 / 1.0 | 25 each | Dijkstra 25 each | 1.0 each | 0/25 each | 0.96 / 0.92 |
| Shortest path | d_X, full (post hoc) | 2.0 | 25 | Dijkstra 25 | 1.0 | 0/25 | 0.04 |
| Shortest path | d_X, full (post hoc) | 4.0 | 25 | no code 25 | 0.0 | 0/25 | 0.0 |
| Shortest path | random 1, full (post hoc) | 4.0 | 25 | no code 20, Dijkstra 3, other 2 | 0.0 | 2/25 | 0.0 |

**What this experiment settled.** The read position holds a low-dimensional map of the catalogue, and on the grid task the mass displaced by the ban went to the geodesic neighbours of the default; the geodesic kernel was the least bad forecaster on the primary events. *(Correction, iteration 2: those neighbours, DFS and unit weight Dijkstra, are also second choices of the model, so the observation is consistent with geodesic proximity and with typicality and does not separate them; both events called primary here fail the forecast contest's compliance gate; Sections 2.3.2 and 2.3.7.)* The read position does not, however, decode the grid task's actual default, the ban shift is almost entirely orthogonal to the algorithm directions, and an activation edit of the modal algorithm did nothing until it broke the model. The labeller passed its gate on both tasks. An independent recomputation reproduced 99 headline numbers with no mismatch.

## 1.7 Forking token experiment: is the algorithm decided during generation?

**Why.** If the read position does not fix the algorithm, the decision may be made at a few high entropy tokens during generation, as forking token analyses find for other outcomes [30, 31]. The knob this account implies is a sampler that raises temperature only around those tokens, in the spirit of adaptive temperature sampling for code [32]. This experiment screened that account on its own pool of about 10,000 labelled generations, with vLLM 0.11.0 in eager mode, its own tests (40 per task, 17 reference solutions) and a judge used only on 6 disputed programs (USD 0.024).

**Design.** A resampling screen took 10 base paths per task for the baseline and for the modal ban and drew 8 continuations at every fourth generated token over the first 92 tokens. Baseline paths were screened only at tokens 4, 8, 16, 32 and 64, a fallback declared in advance because the baseline was already fixed at the read position. Change points in the series of continuation outcome distributions were detected against a permutation threshold (a PELT detector [33] run alongside found none on any path), localised to single tokens, and confirmed with 30 continuations at each candidate and at two control positions without a change point. A prompt swap cut a baseline path just before or just after its commitment token (the first token of evidence that identifies the algorithm, such as the `deque` import) and continued it under the ban prompt. The knob arm compared global temperature 0.3-1.3 with temperature 1.3 applied only in a window around detected forking tokens, which we call the forking window knob, against a random window of the same size and an anchored window at the start of the code block. A planted change sensitivity test found that the detector caught a total variation change of 0.2, 0.3 and 0.5 in 82.5%, 95% and 99% of trials, with a 6% false-positive rate at no change.

**Table 17.** Sample pools in the forking token experiment.

| Task | Pool | N | Labels | Pass | Hill_cc |
|---|---|---|---|---|---|
| Grid | baseline, T = 1.0 | 200 | BFS 200 | 0.91 | 1.0 |
| Grid | ban BFS, T = 1.0 | 100 | DFS 78, BFS 21, unit weight Dijkstra 1 | 0.56 | 1.945 |
| Grid | baseline, T = 0.3 / 0.7 / 1.3 | 100 each | BFS 100 each | 0.91 / 0.82 / 0.96 | 1.0 |
| Shortest path | baseline, T = 1.0 | 200 | Dijkstra 200 | 0.995 | 1.0 |
| Shortest path | ban Dijkstra, T = 1.0 | 100 | Bellman-Ford/SPFA 80, Dijkstra 20 | 1.00 | 1.649 |
| Shortest path | baseline, T = 0.3 / 0.7 / 1.3 | 100 each | Dijkstra 100 each | 1.00 each | 1.0 |

**Where the outcome is fixed.** The sample pools behind the screen (Table 17) repeat the collapse: every baseline pool at every temperature was 100% BFS or 100% Dijkstra, and the ban pools were the only ones with a second algorithm. Under the baseline prompt every base path was fixed at the read position on both tasks: all continuations from token 0 onward produced the path's own algorithm (Table 18). Under the ban, outcomes were open for longer. On the grid task 3 of 10 ban paths were fixed from the read position and 5 of 10 by token 16; on the shortest path task none was fixed at the read position and only 3 of 10 by token 64. Change points were confirmed only on the shortest path ban paths: 6 forking tokens on 5 of the 10 paths, while the 3 grid candidates failed confirmation and the 10 control positions produced no false positive.

**Table 18.** Resampling screen. We call a base path decided at the read position when its continuations from token 0 agree with its final label at least 80% of the time, and fixed by token k when at least 7 of 8 continuations agree with it at every screened position from k on; the columns give the share of base paths of each kind.

| Condition | Base paths | Decided at the read position | Fixed by token 0 / 8 / 16 / 32 / 64 | Paths not fixed | Forking token candidates | Confirmed forking tokens | Final labels |
|---|---|---|---|---|---|---|---|
| Grid baseline | 10 | 1.0 | 1.0 / 1.0 / 1.0 / 1.0 / 1.0 | 0 | 0 | 0 | BFS 10 |
| Grid ban BFS | 10 | 0.8 | 0.3 / 0.3 / 0.5 / 0.5 / 0.5 | 7 | 3 | 0 | DFS 9, BFS 1 |
| Shortest path baseline | 10 | 1.0 | 1.0 / 1.0 / 1.0 / 1.0 / 1.0 | 0 | 0 | 0 | Dijkstra 10 |
| Shortest path ban Dijkstra | 10 | 0.4 | 0.0 / 0.1 / 0.1 / 0.1 / 0.3 | 10 | 10 | 6 | Bellman-Ford/ SPFA 6, Dijkstra 4 |

The confirmed forking tokens are not the punctuation tokens that studies of forking paths report on other tasks (Table 19). Two are the `import` token at position 3, where the model decides which module to import, two are algorithm names written in comments (the tokens `Bell` and `D` at position 68, each with a leading space), one is `BFS` inside the description of an SPFA queue, and one is a newline at position 7. Across confirmed forking tokens the outcome entropy fell by a mean of -0.131 bits, that is, it rose slightly, because several forking tokens moved a path from a fixed

outcome to a mixed one.

**Table 19.** Confirmed forking tokens (all on shortest path ban paths). TVD is the total variation distance between continuation outcome distributions on either side.

| Base path | Position | Token | Token type | TVD | Permutation p | Outcome entropy before / after (bits) |
|---|---|---|---|---|---|---|
| Shortest path ban 0 | 3 | `import` | import | 0.80 | 0.0005 | 0.567 / 0.353 |
| Shortest path ban 1 | 3 | `import` | import | 0.767 | 0.0005 | 0.211 / 0.784 |
| Shortest path ban 1 | 77 | `BFS` (leading space) | identifier | 0.333 | 0.013 | 1.0 / 0.65 |
| Shortest path ban 3 | 68 | `Bell` (leading space) | identifier | 0.333 | 0.002 | 0.948 / 0.211 |
| Shortest path ban 4 | 68 | `D` (leading space) | identifier | 0.533 | 0.0005 | 0.0 / 1.368 |
| Shortest path ban 7 | 7 | newline | whitespace | 0.90 | 0.0005 | 0.211 / 0.353 |

**Does the commitment token lock the algorithm in?** No. The prompt swap predicted that a ban would work better on a prefix cut before the commitment token than after it. The ban effect was the same on both sides (Table 20): on the grid task 75% of continuations avoided BFS before the cut and 73.5% after it, a difference of 0.015 with a 95% interval of -0.02 to 0.055. Writing `from collections import deque` did not commit the model to BFS. On the shortest path task the ban was weak on both sides of the cut (25.5% and 27% avoided Dijkstra).

**Table 20.** Prompt swap at the commitment token (10 base paths per task; ban effect = share of continuations avoiding the banned algorithm; 95% bootstrap intervals).

| Task | Ban effect, cut before commitment | Ban effect, cut after commitment | Before minus after | Banned share after the cut (lock in) |
|---|---|---|---|---|
| Grid | 0.75 [0.545, 0.91] | 0.735 [0.54, 0.895] | 0.015 [-0.02, 0.055] | 0.265 [0.095, 0.45] |
| Shortest path | 0.255 [0.155, 0.365] | 0.27 [0.155, 0.40] | -0.015 [-0.095, 0.05] | 0.73 [0.61, 0.84] |
| Pooled | 0.5025 [0.347, 0.648] | 0.5025 [0.347, 0.642] | 0.0 [-0.045, 0.04] | 0.498 [0.343, 0.638] |

**The forking window temperature knob.** Raising temperature only around detected forking tokens changed nothing about the algorithm (Table 21): every arm on both tasks was 100% BFS or 100% Dijkstra, and the difference in Hill_cc between the forking token window and a random window was 0 with a [0, 0] interval. What temperature did change was the number of distinct phrasings. On the grid task global temperature took the number of phrasing clusters (defined in Section 1.10) at a Jaccard threshold of 0.8 from 11 at T = 0.3 to 63 at T = 1.3, while the forking token window gave 10, about the same as T = 0.3.

**Table 21.** Temperature arms of the forking token experiment (N = 100 each; all programs BFS on the grid task and Dijkstra on the shortest path task). Phrasing clusters at Jaccard thresholds 0.7 / 0.8 / 0.9 over identifier normalised AST shingles; phrasing Hill number at 0.8.

| Task | Arm | Pass | Hill_cc | Phrasing clusters (0.7 / 0.8 / 0.9) | Phrasing Hill (0.8) | Mean length (tokens) |
|---|---|---|---|---|---|---|
| Grid | global T = 0.3 | 0.91 | 1.0 | 8 / 11 / 18 | 6.28 | 335.15 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Grid | global T = 0.7 | 0.82 | 1.0 | 17 / 25 / 35 | 14.36 | 345.11 |
| Grid | global T = 1.0 | 0.94 | 1.0 | 24 / 41 / 60 | 24.58 | 331.6 |
| Grid | global T = 1.3 | 0.96 | 1.0 | 39 / 63 / 80 | 44.72 | 342.62 |
| Grid | forking token window | 0.83 | 1.0 | 8 / 10 / 23 | 6.06 | 346.67 |
| Grid | random window | 0.85 | 1.0 | 6 / 11 / 20 | 5.63 | 349.55 |
| Grid | anchored window | 0.85 | 1.0 | 9 / 13 / 25 | 5.85 | 337.87 |
| Shortest path | global T = 0.3 | 1.00 | 1.0 | 1 / 1 / 1 | 1.0 | 191.88 |
| Shortest path | global T = 0.7 | 1.00 | 1.0 | 1 / 2 / 2 | 1.10 | 192.28 |
| Shortest path | global T = 1.0 | 0.99 | 1.0 | 2 / 4 / 5 | 1.47 | 193.62 |
| Shortest path | global T = 1.3 | 1.00 | 1.0 | 3 / 5 / 7 | 1.84 | 195.08 |
| Shortest path | forking token window | 1.00 | 1.0 | 1 / 2 / 2 | 1.06 | 192.01 |
| Shortest path | random window | 0.99 | 1.0 | 1 / 2 / 2 | 1.10 | 192.5 |
| Shortest path | anchored window | 1.00 | 1.0 | 1 / 1 / 2 | 1.0 | 192.3 |

**Dead end: the locus probes never ran to completion.** The decisive measurement for this account was a probe from residual-stream states at the read position, at generated tokens 8, 16, 32 and 64, at the forking token and at the commitment token, to the final label. The states were extracted (300 samples per task at 8 positions and 21 layers, with read position cosine to a separate pass of at least 0.9997), but the probe stage wrote an empty `probes.json` and no probe result exists. The strategy's locus diagnostic is therefore undetermined. *(Correction, iteration 2: the cause is now recorded. `probes.json` (28 bytes, an empty summary) was written at 18:29:22, while the state log records the state phase starting at 18:31:55 and the grid and shortest path states saved at 18:34:34 and 18:34:50: the probe stage ran before its inputs existed. The states (`states/T{1,2}_part_001-005.npz`, about 790 MB, kept on the run volume) were intact, and the audit ran the probe on them in iteration 2; Section 2.3.1.)* The preregistered verdict rule returned INCONCLUSIVE: no confirmed forking token on the grid task, confirmed forking tokens on half of the shortest path ban paths, and no effect of the forking window knob.

**What this experiment settled.** Without a ban the algorithm is fixed before the first generated token on both tasks, so there is no forking token for temperature to act on. Under a ban the outcome stays open for dozens of tokens and a few forking tokens exist on the shortest path task, including the choice of import; the first token that identifies the algorithm does not lock the algorithm in. A temperature knob aimed at forking tokens adds neither ideas nor phrasings.

## 1.8 Decommitment experiment: does one list versus implement direction control the collapse?

**Why.** An alternate account holds that the bottleneck is commitment, not the choice of a particular algorithm: a model asked to list ten approaches knows many, and a model asked to implement one collapses. If a single direction separates the two framings, adding it while the model implements should spread its output over the repertoire instead of moving it to one substitute. This experiment built that direction, steered with it under random direction and prompting controls [34], and tested

whether a ban moves the read position along it.

**Design.** The decommitment direction is a difference-in-means direction at the read position: the mean state over 50 paraphrases of `list ten different approaches to this task`, which we call the list framing, minus the mean over 50 paraphrases of `implement one correct algorithm`, the implement framing, per layer. Held-out separation AUC was 1.00 at every layer from 9 to 28, and layer 18 was chosen by the held-out separation statistic d' (8.55). At layer 18 the direction's norm was 0.21 (grid) and 0.19 (shortest path) times the state norm, its split half cosine was 0.99, and the two tasks' directions had cosine 0.98. Its cosine was 0.53-0.55 with a breadth direction that contrasts listing approaches with naming one, -0.08 to -0.04 with a variable name placebo direction, and 0.59-0.62 with the shift produced by the enumeration prompt. It was added as c times the state norm times the unit direction at the read position and at every generated token, with c in {0.25, 0.5, 1, 1.5} plus a preregistered bisection. The correctness preserving coefficient c* was the largest c with at least 80% of the c = 0 pass rate: 0.1 on the grid task and 0.25 on the shortest path task. Controls were two random directions, the placebo and breadth directions at c*, the other task's direction, the prompt we call the enumeration prompt (`list 10 approaches, then implement one`), the modal ban and its mention twin, and a two-step succession. Labels came from hidden tests (43 and 39 cases), the tracer and AST rules, with a judge on disagreements; the labeller gate passed (Table 2). An engine check compared the vLLM baseline with the hooked HF arm at c = 0: identical on the grid task, total variation 0.03 on the shortest path task (Fisher p 0.40).

**Steering moved the shortest path task and never the grid task.** On the grid task every arm except the bans wrote BFS (Table 22). At c = 0.25 the pass rate fell to 0.57 and every passing program was still BFS; at c = 0.5 it fell to 0.03, and from c = 1 all output was broken (pass 0, 28-85% of samples truncated). On the shortest path task the c* = 0.25 arm moved correct programs from Dijkstra to Bellman-Ford/SPFA (Bellman-Ford/SPFA 54, Dijkstra 23, other 1, against Dijkstra 88 and Bellman-Ford/SPFA 7 at c = 0), a rarefied Hill-number gain of +0.65 (paraphrase-cluster 90% interval 0.31 to 0.97) at a 17-point pass-rate cost. A random direction at the same strength did almost the same (+0.59), and the difference between the decommitment direction and that random direction was +0.07 (-0.26 to 0.39). The gain therefore is not specific to the direction, although it exceeded the second random direction (+0.70, 0.38 to 1.01) and the placebo (+0.47, 0.01 to 0.84). Per-sample flips against c = 0 under common random numbers were 0.66 for the decommitment direction, 0.42 and 0.18 for the two random directions and 0.22 for the placebo. On the shortest path task the off-mode rate peaked at c = 0.25 (0.55) and fell again at c = 0.5 (0.12) as programs broke. A Hill number of 2 was reached at c of about 0.40 for a 34.5-point pass cost; a Hill number of 3 was not reached in the tested range.

**Table 22.** BUY table of the decommitment experiment. Delta Hill is the change in correctness conditioned Hill number, rarefied to a common sample depth, against the HF c = 0 arm for steering and against the vLLM baseline for prompts, with a 90% paraphrase-cluster interval; efficiency is Delta Hill per pass-rate point lost.

| Task | Knob | Pass | Pass points lost | Labels of correct programs | Delta Hill [90% interval] | Efficiency |
|---|---|---|---|---|---|---|
| Grid | vLLM baseline (N = 200) | 1.00 | n/a | BFS 200 | n/a | n/a |
| Grid | HF c = 0 | 1.00 | n/a | BFS 100 | n/a | n/a |
| Grid | decommitment c = 0.1 (c*) | 0.98 | 2 | BFS 98 | 0.0 [0.0, 0.0] | 0.0 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Grid | decommitment c = 0.25 | 0.57 | 43 | BFS 57 | 0.0 | 0.0 |
| Grid | decommitment c = 0.5 | 0.03 | 97 | BFS 3 | 0.0 | 0.0 |
| Grid | decommitment c = 1 / 1.5 | 0.0 / 0.0 | 100 | none | n/a | n/a |
| Grid | random 1 / random 2 / placebo / breadth at c* | 0.98 / 1.00 / 1.00 / 0.99 | 2 / 0 / 0 / 1 | all BFS | 0.0 | 0.0 |
| Grid | other task's direction at c* (N = 50) | 0.98 | 2 | all BFS | 0.0 | 0.0 |
| Grid | enumeration prompt | 1.00 | 0 | BFS 100 | 0.0 | n/a |
| Grid | mention BFS | 1.00 | 0 | BFS 100 | 0.0 | n/a |
| Grid | ban BFS | 0.64 | 36 | BFS 46, DFS 16, unit weight Dijkstra 2 | +0.99 [0.55, 1.34] | 0.028 |
| Grid | succession (ban BFS and DFS) | 0.88 | 12 | BFS 65, unit weight Dijkstra 13, DFS 7, A* 3 | +1.27 [0.64, 1.80] | 0.106 |
| Shortest path | vLLM baseline (N = 200) | 0.89 | n/a | Dijkstra 169, Bellman-Ford/SPFA 8 | n/a | n/a |
| Shortest path | HF c = 0 | 0.95 | n/a | Dijkstra 88, Bellman-Ford/SPFA 7 | n/a | n/a |
| Shortest path | decommitment c = 0.25 (c*) | 0.78 | 17 | Bellman-Ford/SPFA 54, Dijkstra 23, other 1 | +0.65 [0.31, 0.97] | 0.038 |
| Shortest path | decommitment c = 0.375 | 0.67 | 28 | Bellman-Ford/SPFA 47, Dijkstra 19 | +0.65 [0.29, 0.94] | 0.023 |
| Shortest path | decommitment c = 0.5 | 0.37 | 58 | Dijkstra 25, Bellman-Ford/SPFA 10, other 2 | +0.87 [0.09, 1.38] | 0.015 |
| Shortest path | decommitment c = 1 / 1.5 | 0.0 / 0.0 | 95 | none | n/a | n/a |
| Shortest path | random 1 at c* | 0.79 | 16 | Dijkstra 53, Bellman-Ford/SPFA 26 | +0.59 [0.21, 0.85] | 0.037 |
| Shortest path | random 2 at c* | 0.87 | 8 | Dijkstra 82, Bellman-Ford/SPFA 5 | -0.05 [-0.31, 0.20] | -0.006 |
| Shortest path | placebo at c* | 0.66 | 29 | Dijkstra 60, Dial 3, Bellman-Ford/SPFA 2 | +0.19 [-0.30, 0.70] | 0.006 |
| Shortest path | breadth direction at c* | 0.60 | 35 | Bellman-Ford/SPFA 31, Dijkstra 28, Dial 1 | +0.85 [0.46, 1.17] | 0.024 |
| Shortest path | other task's direction at c* (N = 50) | 0.72 | 23 | Bellman-Ford/SPFA 28, Dijkstra 8 | +0.40 [-0.01, 0.69] | 0.018 |
| Shortest path | enumeration prompt | 0.82 | 6.5 | Dijkstra 80, Bellman-Ford/SPFA 2 | -0.08 [-0.27, 0.13] | -0.013 |
| Shortest path | mention Dijkstra | 0.94 | -5.5 | Dijkstra 92, Bellman-Ford/SPFA 2 | -0.09 [-0.29, 0.12] | n/a |
| Shortest path | ban Dijkstra | 0.87 | 1.5 | Dijkstra 49, Bellman-Ford/SPFA 38 | +0.79 [0.60, 0.90] | 0.524 |
| Shortest path | succession (ban Dijkstra and Bellman-Ford/SPFA) | 0.68 | 20.5 | Bellman-Ford/SPFA 34, Dijkstra 33, other 1 | +0.93 [0.66, 1.17] | 0.045 |

The prompting baselines decide the comparison. The enumeration prompt, which states the decommitment framing in words, produced no diversity on either task. The modal ban bought +0.79 effective algorithms on the shortest path task for 1.5 pass points, an efficiency 14 times that of steering, but most banned samples still used the banned algorithm: 46 of 64 correct grid programs were BFS and 49 of 87 correct shortest path programs were Dijkstra. The ban's Hill gain is therefore mostly a mixture of ban failures and Bellman-Ford/SPFA, not two algorithms chosen freely.

**Does a ban decommit?** The commitment test projects the ban minus mention shift at layer 18 onto the axis from the implement framing (0) to the list framing (1). The ban moved the read position toward the list framing by 0.039 on the grid task and 0.053 on the shortest path task (Table 23), outside the preregistered paraphrase split null band (plus or minus 0.024 and 0.025), so the preregistered label is that the ban decommits. An audit judged that band too narrow; exact permutation tests gave $p = 0.021$ and 0.007 (label permutation) and 0.025 and 0.016 (paired sign flip), with 8 of 10 paraphrase pairs positive on each task. The shift is small: the enumeration prompt, the positive control, moved the state by 0.544 and 0.582 of the same distance, and the listing control sat at 0.816 and 0.806. The share of the ban shift along the direction ranked at the 78th and 87th percentiles of a null built from the top 50 principal components.

**Table 23.** Position along the implement (0) to list (1) axis at layer 18 for each prompt condition.

| Condition | Grid | Shortest path |
|---|---|---|
| baseline | -0.054 | -0.075 |
| enumeration prompt | 0.483 | 0.495 |
| listing control | 0.816 | 0.806 |
| framing that names one approach | 0.653 | 0.650 |
| ban of the modal algorithm / its mention twin | 0.040 / 0.001 | 0.035 / -0.018 |
| ban DFS / mention DFS (Grid); ban Bellman-Ford/SPFA / mention (Shortest path) | 0.023 / -0.001 | 0.047 / -0.018 |
| ban bidirectional BFS / mention (Grid); ban bidirectional Dijkstra / mention (Shortest path) | 0.007 / -0.004 | 0.017 / 0.001 |
| ban IDDFS / mention (Grid); ban Floyd-Warshall / mention (Shortest path) | 0.036 / 0.012 | 0.030 / -0.013 |
| ban A* / mention (Grid); ban Dial / mention (Shortest path) | 0.073 / 0.004 | 0.052 / -0.002 |
| ban unit weight Dijkstra / mention (Grid) | 0.051 / 0.010 | n/a |
| Ban minus mention for the modal algorithm (null band) | 0.039 (±0.024) | 0.053 (±0.025) |
| Enumeration minus baseline, positive control (null band) | 0.544 (±0.047) | 0.582 (±0.048) |

**The forecast record.** Forecasts were committed at 16:26:17 UTC, before the ban samples. On the shortest path modal ban all 38 compliant programs were Bellman-Ford/SPFA; the decommitted survivor distribution scored -0.058 and hit, the vLLM frequency forecast -0.219 (hit), the HF frequency forecast -0.238 (hit) and the uniform forecast -1.406. On the grid modal ban (18 compliant: DFS 16, unit weight Dijkstra 2) every forecast was degenerate, because no correct program other than BFS existed before the ban, and all tied at -1.630. The succession events tied as well (16 compliant grid programs, unit weight Dijkstra 13 and A* 3; 1 compliant shortest path program).

**What this experiment settled.** The decommitment direction exists as a clean difference-in-means direction, shared across tasks, but adding it is a generic perturbation: it moves the shortest path task toward Bellman-Ford/SPFA as a random direction does and costs a sixth of the pass rate, and it

never moves the grid task. Its prompt counterpart, the enumeration prompt, does nothing. A ban moves the read position measurably but slightly toward the list framing. The preregistered verdict was PARTIAL (the diversity gain held on one task, the specificity test failed). vLLM same seed determinism failed on the grid task, so vLLM arms are reproducible in distribution only, and every common random number analysis used HF arms.

## 1.9 Instruction tuning experiment: did post-training install the collapse?

**Why.** If instruction tuning installed the collapse, the base model should spread over several algorithms at matched prompts, and subtracting the instruct minus base activation difference should restore that spread inside the instruct model. Base models are reported to be more diverse than aligned ones [35], and chat formatting alone narrows output diversity [36]; this experiment separated the two causes on the two tasks.

**Design.** A two by two design crossed weights (Qwen3-8B-Base, Qwen3-8B) with format (code completion without a chat template, which we call plain completion, or the instruct chat template), all at temperature 1.0 with top-p 1.0, thinking off. Each program ran on 47 (grid) or 46 (shortest path) hidden tests and was labelled by the tracer and AST rules with Gemini 2.5 Flash Lite as judge on disagreements; a second judge agreed with the first on 95% of 40 audit programs (kappa 0.93). The runtime and AST labels agreed with kappa 0.77 over all parseable programs, but much lower on base-model output (0.20-0.50), where 217 of 292 grid programs in plain completion needed the judge. The sharpening direction is the paired difference between instruct and base read position states over the same chat-templated prompts, per layer from 8 to 28. It was subtracted at c in {0.5, 1.0, 1.5} over the whole band, at the read position only, and at layer 18 only, against two norm-matched random directions. Hardware was one NVIDIA GPU with 23 GB of memory (2.1 GPU hours); judge spend was USD 0.21.

**The collapse is sharpened by post-training, not invented by it.** Table 24 gives every arm. The instruct model in chat format is collapsed on both tasks. The base model under the same chat tokens writes mostly the same algorithm but less exclusively: BFS in 49 of 57 correct grid programs and Dijkstra in 63 of 90 correct shortest path programs, with Bellman-Ford/SPFA second (24). In plain completion the base model rarely writes a correct program (pass 0.05 and 0.08), and its correct shortest path programs are mostly Bellman-Ford/SPFA (16 of 25).

**Table 24.** Arms of the instruction tuning experiment (vLLM, T = 1.0). Hill_cc with 95% cluster bootstrap interval; Hill_all over all parseable labelled programs.

| Arm | Task | N | Parse rate | Pass [95% Wilson] | Correct | Labels of correct programs | Hill_cc [95%] | Hill_all |
|---|---|---|---|---|---|---|---|---|
| base, plain completion | Grid | 300 | 0.97 | 0.05 [0.03, 0.08] | 15 | BFS 12, DFS 3 | 1.65 [1.00, 2.00] | 3.34 |
| base, plain completion | Shortest path | 300 | 0.97 | 0.08 [0.06, 0.12] | 25 | Bellman-Ford/SPFA 16, Dijkstra 8, Floyd-Warshall 1 | 2.18 [1.00, 2.73] | 3.24 |
| base, chat template | Grid | 300 | 0.58 | 0.19 [0.15, 0.24] | 57 | BFS 49, DFS 7, A* 1 | 1.58 [1.10, 2.06] | 3.13 |
| base, chat template | Shortest path | 300 | 0.66 | 0.30 [0.25, 0.35] | 90 | Dijkstra 63, Bellman-Ford/SPFA 24, Floyd-Warshall 1, other 2 | 2.09 [1.63, 2.54] | 2.67 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| instruct, plain completion | Grid | 100 | 0.93 | 0.73 [0.64, 0.81] | 73 | BFS 73 | 1.00 [1.00, 1.00] | 1.41 |
| instruct, plain completion | Shortest path | 100 | 0.89 | 0.72 [0.63, 0.80] | 72 | Dijkstra 57, Floyd-Warshall 8, Bellman-Ford/SPFA 7 | 1.93 [1.41, 2.37] | 2.60 |
| instruct, chat template | Grid | 200 | 1.00 | 1.00 [0.98, 1.00] | 200 | BFS 200 | 1.00 [1.00, 1.00] | 1.00 |
| instruct, chat template | Shortest path | 200 | 1.00 | 1.00 [0.98, 1.00] | 200 | Dijkstra 193, Bellman-Ford/SPFA 7 | 1.16 [1.03, 1.31] | 1.16 |
| base, plain completion, ban | Grid | 100 | 0.95 | 0.01 [0.00, 0.05] | 1 | BFS 1 | 1.00 | 3.50 |
| base, plain completion, ban | Shortest path | 100 | 0.98 | 0.08 [0.04, 0.15] | 8 | Dijkstra 4, Bellman-Ford/ SPFA 4 | 2.00 | 3.43 |
| instruct, chat template, ban | Grid | 100 | 1.00 | 0.70 [0.60, 0.78] | 70 | BFS 57, DFS 12, unit weight Dijkstra 1 | 1.70 [1.22, 2.25] | 2.10 |
| instruct, chat template, ban | Shortest path | 100 | 1.00 | 1.00 [0.96, 1.00] | 100 | Dijkstra 54, Bellman-Ford/ SPFA 46 | 1.99 [1.65, 2.00] | 1.99 |

The labels of base-model output lean heavily on the judge, which matters for the base arms' Hill numbers (Table 24b): runtime and AST rules agreed with kappa 0.20-0.50 on base-model programs, and the judge labelled 217 of 292 grid programs in plain completion. The final labels agreed with the judge on 97-100% of programs in every arm, and a 15-program audit per arm found them correct in every arm but one (0.93 on the base chat shortest path arm).

**Table 24b.** Labeller agreement per generation arm in the instruction tuning experiment.

| Arm | Task | Parseable programs | Kappa runtime vs AST | Final label vs judge | Runtime + AST label vs judge | Labelled by the judge |
|---|---|---|---|---|---|---|
| base, plain completion | Grid | 292 | 0.45 | 0.98 | 0.91 | 217 |
| base, plain completion | Shortest path | 291 | 0.20 | 0.99 | 0.92 | 266 |
| base, chat template | Grid | 175 | 0.40 | 0.99 | 0.99 | 94 |
| base, chat template | Shortest path | 199 | 0.50 | 0.97 | 0.94 | 94 |
| base, plain completion, ban | Grid | 95 | 0.43 | 0.98 | 0.93 | 68 |
| base, plain completion, ban | Shortest path | 98 | 0.34 | 0.99 | 0.93 | 83 |
| instruct, chat template | Grid | 200 | 1.00 | 1.00 | 1.00 | 0 |
| instruct, chat template | Shortest path | 200 | 1.00 | 1.00 | 1.00 | 7 |
| instruct, plain completion | Grid | 93 | 0.71 | 1.00 | 1.00 | 17 |

| | | | | | | |
|---|---|---|---|---|---|---|
| instruct, plain completion | Shortest path | 89 | 0.89 | 1.00 | 1.00 | 23 |
| instruct, chat template, ban | Grid | 100 | 0.64 | 1.00 | 1.00 | 20 |
| instruct, chat template, ban | Shortest path | 100 | 0.98 | 1.00 | 1.00 | 46 |

Table 25 gives the preregistered contrasts. The weights effect (instruct chat minus base chat) is negative on both tasks: -0.58 effective algorithms on the grid task and -0.93 on the shortest path task, with intervals excluding zero. The format effect inside the base model is near zero on both tasks, while inside the instruct model the chat format cuts the shortest path Hill number by 0.76. The preregistered verdict was INSTALLED_BY_POST_TRAINING on the shortest path task and MIXED on the grid task, where the gap in modal share of 14.0 points fell just short of the 15-point bar; under Hill_all both tasks read INSTALLED_BY_POST_TRAINING. At the originally planned N, before the base arms were extended to 300, both tasks read MIXED. The chat format also switched the base model's shortest path mode from Bellman-Ford/SPFA to Dijkstra.

**Table 25.** Collapse screen: differences in Hill numbers with paraphrase cluster bootstrap 95% intervals.

| Task | Weights effect: instruct chat minus base chat | Format effect in base: chat minus plain completion | Format effect in instruct | Interaction | Verdict (Hill_cc) | Verdict (Hill_all) |
|---|---|---|---|---|---|---|
| Grid (Hill_cc) | -0.58 [-1.01, -0.15] | -0.07 [-0.76, 0.88] | 0.00 [0.00, 0.00] | 0.07 [-0.83, 0.74] | MIXED | INSTALLED_BY_POST_TRAINING |
| Shortest path (Hill_cc) | -0.93 [-1.38, -0.46] | -0.09 [-0.79, 1.12] | -0.76 [-1.26, -0.23] | -0.67 [-1.73, 0.10] | INSTALLED_BY_POST_TRAINING | INSTALLED_BY_POST_TRAINING |
| Grid (Hill_all) | -2.13 [-2.66, -1.43] | -0.21 [-1.05, 0.55] | -0.41 [-0.66, -0.15] | -0.19 [-0.94, 0.63] | | |
| Shortest path (Hill_all) | -1.50 [-1.91, -1.06] | -0.57 [-1.04, -0.10] | -1.44 [-2.03, -0.71] | -0.87 [-1.60, 0.04] | | |

**The sharpening direction is a format direction.** The instruct minus base difference at the read position was stable (split half cosine 0.998-0.999 across the band) and largely shared between tasks (cosine between the two tasks' differences 0.84-0.93 across layers 8-28), with a relative norm of about 1.0-1.1 of the state at layer 18. The logit lens (reading an intermediate state through the model's unembedding matrix) at layer 28 showed what subtracting it promotes: `def` and `class` on the grid prompts, and `Conversation`, `Claude`, `Prompt` and `chat` on the shortest path prompts. Subtracting it over the band destroyed generation at every coefficient (0 of 100 parseable programs, Table 26); the downstream layers amplified the edit, so the gap to the base state was closed at layer 8 but overshot at layers 18 and 28 (gap closed -0.38 and -0.40 at $c = 1.5$). Read position only subtraction turned the model into thinking style prose (73% and 95% of outputs opened like a thinking trace), and the code it still wrote named BFS and Dijkstra. Subtraction at layer 18 alone kept 94-98% of the pass rate at $c = 0.25$ with no algorithm change. No coefficient of any variant restored diversity.

**Table 26.** Steering arms of the instruction tuning experiment (HF, coupled seeds; deltas against the hooked c = 0 control).

| Task | Arm | N | Parse | Pass | Hill_cc | Delta Hill_cc | Delta pass (points) | Flip rate |
|---|---|---|---|---|---|---|---|---|
| Grid | control c = 0 | 100 | 1.00 | 1.00 | 1.00 | n/a | n/a | n/a |
| Grid | random 1 | 100 | 0.96 | 0.41 | 1.00 | 0.00 | -59.0 | 0.04 |
| Grid | random 1, read position only, c = 1.0 | 100 | 1.00 | 1.00 | 1.00 | 0.00 | 0.0 | 0.00 |
| Grid | random 2 | 100 | 0.98 | 0.68 | 1.00 | 0.00 | -32.0 | 0.01 |
| Grid | sharpening, band, c = 0.5 / 1.0 / 1.5 | 100 each | 0.00 | 0.00 | n/a | n/a | -100.0 | n/a |
| Grid | sharpening, read position only, c = 1.0 | 100 | 0.28 | 0.14 | 1.00 | 0.00 | -86.0 | 0.07 |
| Grid | same, no 512-token cutoff | 20 | 0.55 | 0.15 | 1.00 | 0.00 | -85.0 | 0.00 |
| Grid | sharpening, layer 18 only, c = 0.25 / 0.5 / 1.0 | 50 each | 1.00 / 1.00 / 0.00 | 0.94 / 0.52 / 0.00 | 1.00 / 1.00 / n/a | 0.00 / 0.00 / n/a | -6.0 / -48.0 / -100.0 | 0.00 / 0.00 / n/a |
| Shortest path | control c = 0 | 100 | 1.00 | 1.00 | 1.00 | n/a | n/a | n/a |
| Shortest path | random 1 | 100 | 1.00 | 0.00 | n/a | n/a | -100.0 | 0.00 |
| Shortest path | random 1, read position only, c = 1.0 | 100 | 1.00 | 1.00 | 1.43 | +0.43 [0.12, 0.74] | 0.0 | 0.10 |
| Shortest path | random 2 | 100 | 1.00 | 0.33 | 1.00 | 0.00 | -67.0 | 0.36 |
| Shortest path | sharpening, band, c = 0.5 / 1.0 / 1.5 | 100 each | 0.00 | 0.00 | n/a | n/a | -100.0 | n/a |
| Shortest path | sharpening, read position only, c = 1.0 | 100 | 0.05 | 0.05 | 1.00 | 0.00 | -95.0 | 0.00 |
| Shortest path | same, no 512-token cutoff | 20 | 0.40 | 0.20 | 1.75 | +0.75 [0.00, 1.00] | -80.0 | 0.50 |
| Shortest path | sharpening, layer 18 only, c = 0.25 / 0.5 / 1.0 | 50 each | 1.00 / 1.00 / 0.00 | 0.98 / 0.98 / 0.00 | 1.00 / 1.10 / n/a | 0.00 / +0.10 / n/a | -2.0 / -2.0 / -100.0 | 0.00 / 0.02 / n/a |

*(Correction, iteration 2: Table 26 reports Hill numbers and flip rates but not what the correct programs were, which hides a complete algorithm switch. On the shortest path task band random direction 2 turned all 33 of its correct programs into Bellman-Ford/SPFA, against Dijkstra 100 of 100 in the control, and band random direction 1 left no correct program; the read position only random edit gave Dijkstra 90, Bellman-Ford/SPFA 9 and other 1; layer-18 sharpening at c = 0.5 gave Dijkstra 48 and Bellman-Ford/SPFA 1; read position sharpening without the cutoff gave Dijkstra 3 and Bellman-Ford/SPFA 1. On the grid task every correct program in every arm was BFS (41 under band random direction 1, 68 under band random direction 2, 100 under the read position random edit). Source: audit* `tables/perturbation_table.csv`*; Section 2.3.5.)*

Table 26b shows why the band subtraction broke the model. The edit closed the gap between instruct and base states completely at layer 8 at c = 1.0, but by layers 18 and 28 the downstream

computation had erased most of the correction, and at c = 1.5 it overshot past the base state.

**Table 26b.** Manipulation check at the read position: gap closed is 1 minus the ratio of two distances, from the mean steered state to the mean base state and from the mean instruct state to the mean base state, at layers 8, 18 and 28.

| Arm | Grid: layers 8 / 18 / 28 | Shortest path: layers 8 / 18 / 28 |
|---|---|---|
| sharpening, band, c = 0.5 | 0.50 / 0.34 / 0.38 | 0.50 / 0.37 / 0.38 |
| sharpening, band, c = 1.0 | 1.00 / 0.05 / 0.01 | 1.00 / 0.08 / 0.04 |
| sharpening, band, c = 1.5 | 0.50 / -0.38 / -0.40 | 0.50 / -0.32 / -0.37 |
| random 1 | -0.12 / 0.06 / -0.07 | -0.12 / 0.05 / -0.08 |
| random 2 | -0.12 / 0.07 / -0.05 | -0.12 / 0.06 / -0.05 |
| sharpening, layer 18 only, c = 0.25 / 0.5 / 1.0 | 0.00 / 0.25 / 0.04; 0.00 / 0.50 / 0.11; 0.00 / 1.00 / 0.36 | 0.00 / 0.25 / 0.06; 0.00 / 0.50 / 0.13; 0.00 / 1.00 / 0.34 |
| sharpening, read position only, c = 1.0 | 1.00 / 0.05 / 0.01 | 1.00 / 0.08 / 0.04 |
| random 1, read position only, c = 1.0 | -0.42 / 0.03 / -0.33 | -0.42 / 0.03 / -0.31 |

The ban prompt did better than any steering arm: on the shortest path task it raised Hill_cc from 1.16 to 1.99 with no loss of pass rate, the best price of any knob in the run (the artifact's efficiency score was 0.830), and on the grid task to 1.70 for 30 pass points. The base model, by contrast, did not obey bans: the banned share of the base model in plain completion moved from 0.46 to 0.36 on the grid task and from 0.25 to 0.27 on the shortest path task, with 0 and 4 compliant programs.

**The forecast record.** The base model's own survivor distribution, committed before the instruct ban samples, nominated DFS and Bellman-Ford/SPFA and hit both (log scores -0.470 and -0.220 against uniform -1.630 and -1.406). The base chat distribution did as well (-0.412 and -0.058), and on the shortest path task so did the instruct chat distribution (-0.058), because Bellman-Ford/SPFA was the second algorithm in every reference distribution.

**What this experiment settled.** Post-training concentrated an ordering the base model already had (BFS first, Dijkstra first in chat format, Bellman-Ford/SPFA second on the shortest path task); it did not install a new attractor. *(Correction, iteration 2: the preregistered verdicts stay as recorded above, INSTALLED_BY_POST_TRAINING on the shortest path task and MIXED on the grid task, and MIXED on both at the planned sample size; the base ordering matches the instruct ordering only under the chat template, and in plain completion the base model's shortest path mode is Bellman-Ford/SPFA, first rather than second.)* The direction that separates instruct from base states is about answer format, and subtracting it breaks the model rather than freeing it. This alternate passed its screen under the preregistered rule, but the pass rests on the shortest path task and on extended sample sizes, and the knob it implies does not work.

## 1.10 Decoding experiment: temperature, top-k, random seeds and creativity instructions

**Why.** The request asks three things directly: whether a tight temperature with top-k = 10 yields one idea in many phrasings or many ideas, whether loosening temperature multiplies ideas or phrasings, and whether a random "seed" in the prompt (an entropy string or a list of unrelated words) pushes the model into a different region of its solution space. The account screened here says that temperature and random seeds change wording and leave the algorithm alone, and that only context with meaning moves it, in line with the finding that temperature correlates only

weakly with novelty [37]. This experiment also produced the request's literal first figure, the BFS fraction by temperature.

**Design.** The experiment drew 5,220 generations on one RTX 5090 with vLLM in eager mode, N = 100 per cell (10 paraphrases of 10 samples). Besides the no cue baseline, a temperature grid crossed T in {0.3, 0.7, 1.0, 1.3} with top-k = 10 and top-p = 0.95, once with thinking off and once with thinking on (thinking capped at 2,048 tokens). Three seed conditions added an entropy seed (`seed=d32qxk4nt43`), a word seed (`seed=(horse chair roof love)`) or a semantic seed, a paragraph that implies without naming a search from both ends on the grid task and a relaxation over all edges on the shortest path task. Four instructions followed, which we call the creativity instruction (`be creative`), the new algorithm instruction (`make a new algorithm`), the novelty instruction (`make a fully novel algorithm never seen before`) and, as in Section 1.8, the enumeration prompt (`list 10 approaches, then implement one`). In the generate then select arm the model listed ten approaches and an external random draw picked the one it had to implement. For the read position analyses, forced implementations of every catalogue algorithm (20 framings each) defined the mode subspace, and three further conditions were encoded without generation: a fresh entropy seed and a fresh word seed (a different random string or word list for each paraphrase) and a neutral sentence of matched length. OpenRouter spend was USD 0.11. The labeller failed its strict gate on both tasks (Table 2), so every number here is descriptive.

**The BFS fraction does not depend on temperature or sampler.** Table 27 answers the request's BFS question. With thinking off, 797 of 800 grid programs across the eight temperature and sampler cells were BFS and every cell was at least 0.99 BFS; with thinking on, 800 of 800 were. On the shortest path task 797 of 800 were Dijkstra with thinking off and 800 of 800 with thinking on. A chart of these shares would be a flat line at 1.0, so the table stands alone.

**Table 27.** Share of programs that are BFS (grid task) or Dijkstra (shortest path task) across the temperature grid (N = 100 per cell; all other labels listed; pass = share passing the hidden tests).

| Task | Thinking | Sampler | T = 0.3 | T = 0.7 | T = 1.0 | T = 1.3 |
|---|---|---|---|---|---|---|
| Grid | off | top-k = 10 | BFS 100 (pass 1.00) | BFS 99, DFS 1 (0.98) | BFS 100 (0.98) | BFS 99, DFS 1 (0.95) |
| Grid | off | top-p = 0.95 | BFS 100 (1.00) | BFS 100 (1.00) | BFS 100 (1.00) | BFS 99, DFS 1 (0.97) |
| Grid | on | top-k = 10 | BFS 100 (1.00) | BFS 100 (1.00) | BFS 100 (0.98) | BFS 100 (1.00) |
| Grid | on | top-p = 0.95 | BFS 100 (1.00) | BFS 100 (1.00) | BFS 100 (0.99) | BFS 100 (1.00) |
| Shortest path | off | top-k = 10 | Dijkstra 100 (1.00) | Dijkstra 100 (0.99) | Dijkstra 99, Bellman-Ford/SPFA 1 (0.98) | Dijkstra 99, Bellman-Ford/SPFA 1 (1.00) |
| Shortest path | off | top-p = 0.95 | Dijkstra 100 (1.00) | Dijkstra 100 (1.00) | Dijkstra 100 (0.97) | Dijkstra 99, Bellman-Ford/SPFA 1 (0.99) |
| Shortest path | on | top-k = 10 | Dijkstra 100 (1.00) | Dijkstra 100 (1.00) | Dijkstra 100 (1.00) | Dijkstra 100 (1.00) |
| Shortest path | on | top-p = 0.95 | Dijkstra 100 (1.00) | Dijkstra 100 (1.00) | Dijkstra 100 (1.00) | Dijkstra 100 (1.00) |

The no cue baseline at T = 1.0 without top-k or top-p gave BFS 99 and unit weight Dijkstra 1 (pass 0.97) and Dijkstra 100 (pass 0.99).

**Ideas and phrasings.** We count ideas as distinct runtime labelled algorithms among correct programs. A *phrasing cluster* is a connected component, within one algorithm, of programs whose

identifier normalised AST 4-gram shingle sets have Jaccard similarity at or above a threshold theta. Identifiers are renamed in order of appearance, and comments, docstrings and constants are dropped. Theta 0.8 is primary, 0.7 and 0.9 are sensitivity values, and complete linkage is a robustness check. The *phrasings per idea* is the rarefied number of phrasing clusters divided by the rarefied number of ideas at a depth of 25 programs. Table 28 gives the full ideas by phrasings table, which answers the request's question 1a-c. At the tightest setting (T = 0.3, top-k = 10, thinking off) the model wrote one idea in 4 phrasing clusters on the grid task and 1 on the shortest path task. Loosening temperature multiplied phrasings and left ideas at one or two: on the grid task the clusters at theta 0.8 rose from 4 to 36 (top-k) and from 3 to 24 (top-p) between T = 0.3 and T = 1.3. Output with thinking on was more varied in wording at every temperature (26 clusters at T = 0.3 with top-k) but never in algorithm. The shortest path programs are shorter and more uniform (one to six clusters with thinking off). Figure 3 plots the clusters at theta 0.8 against temperature.

**Table 28.** Ideas by phrasings (correct programs only). Phrasing clusters at theta 0.7 / 0.8 / 0.9 with single linkage, at 0.8 with complete linkage, and phrasings per idea at theta 0.8.

| Task | Thinking | Sampler | T | Correct | Ideas | Clusters 0.7 / 0.8 / 0.9 | Complete linkage 0.8 | Phrasings per idea |
|---|---|---|---|---|---|---|---|---|
| Grid | off | none (baseline) | 1.0 | 97 | 2 | 15 / 21 / 32 | 29 | 7.48 |
| Grid | off | top-k = 10 | 0.3 | 100 | 1 | 4 / 4 / 6 | 7 | 3.21 |
| Grid | off | top-k = 10 | 0.7 | 98 | 1 | 11 / 13 / 17 | 18 | 6.20 |
| Grid | off | top-k = 10 | 1.0 | 98 | 1 | 19 / 22 / 36 | 32 | 9.57 |
| Grid | off | top-k = 10 | 1.3 | 95 | 2 | 31 / 36 / 54 | 50 | 10.96 |
| Grid | off | top-p = 0.95 | 0.3 | 100 | 1 | 3 / 3 / 5 | 5 | 2.91 |
| Grid | off | top-p = 0.95 | 0.7 | 100 | 1 | 7 / 8 / 10 | 12 | 4.73 |
| Grid | off | top-p = 0.95 | 1.0 | 100 | 1 | 11 / 15 / 28 | 26 | 7.44 |
| Grid | off | top-p = 0.95 | 1.3 | 97 | 1 | 21 / 24 / 41 | 36 | 9.58 |
| Grid | on | top-k = 10 | 0.3 | 100 | 1 | 18 / 26 / 41 | 36 | 11.69 |
| Grid | on | top-k = 10 | 0.7 | 100 | 1 | 24 / 36 / 57 | 48 | 14.36 |
| Grid | on | top-k = 10 | 1.0 | 98 | 1 | 36 / 49 / 67 | 60 | 16.43 |
| Grid | on | top-k = 10 | 1.3 | 100 | 1 | 39 / 55 / 73 | 68 | 17.60 |
| Grid | on | top-p = 0.95 | 0.3 | 100 | 1 | 25 / 31 / 46 | 41 | 12.32 |
| Grid | on | top-p = 0.95 | 0.7 | 100 | 1 | 29 / 41 / 57 | 51 | 15.21 |
| Grid | on | top-p = 0.95 | 1.0 | 99 | 1 | 26 / 35 / 56 | 47 | 13.51 |
| Grid | on | top-p = 0.95 | 1.3 | 100 | 1 | 41 / 53 / 78 | 66 | 17.65 |
| Shortest path | off | none (baseline) | 1.0 | 99 | 1 | 3 / 3 / 17 | 10 | 2.24 |
| Shortest path | off | top-k = 10 | 0.3 | 100 | 1 | 1 / 1 / 3 | 2 | 1.00 |
| Shortest path | off | top-k = 10 | 0.7 | 99 | 1 | 2 / 2 / 7 | 5 | 1.59 |
| Shortest path | off | top-k = 10 | 1.0 | 98 | 2 | 3 / 3 / 14 | 10 | 1.62 |
| Shortest path | off | top-k = 10 | 1.3 | 100 | 2 | 5 / 6 / 15 | 11 | 2.26 |
| Shortest path | off | top-p = 0.95 | 0.3 | 100 | 1 | 2 / 2 / 3 | 2 | 1.25 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Shortest path | off | top-p = 0.95 | 0.7 100 | 1 | 3 / 3 / 7 | 4 | 1.69 |
| Shortest path | off | top-p = 0.95 | 1.0 97 | 1 | 2 / 2 / 9 | 5 | 1.70 |
| Shortest path | off | top-p = 0.95 | 1.3 99 | 1 | 3 / 3 / 9 | 6 | 1.84 |
| Shortest path | on | top-k = 10 | 0.3 100 | 1 | 2 / 3 / 18 | 14 | 2.25 |
| Shortest path | on | top-k = 10 | 0.7 100 | 1 | 3 / 4 / 17 | 11 | 2.69 |
| Shortest path | on | top-k = 10 | 1.0 100 | 1 | 4 / 7 / 24 | 19 | 3.25 |
| Shortest path | on | top-k = 10 | 1.3 100 | 1 | 10 / 12 / 32 | 26 | 5.46 |
| Shortest path | on | top-p = 0.95 | 0.3 100 | 1 | 3 / 3 / 13 | 9 | 2.25 |
| Shortest path | on | top-p = 0.95 | 0.7 100 | 1 | 2 / 4 / 16 | 12 | 2.50 |
| Shortest path | on | top-p = 0.95 | 1.0 100 | 1 | 2 / 3 / 16 | 13 | 2.25 |
| Shortest path | on | top-p = 0.95 | 1.3 100 | 1 | 3 / 4 / 20 | 15 | 3.02 |

*[Figure fig_phrasings: generated in the report phase, not in this draft]*

Table 29 gives the bootstrap slopes of log phrasings per idea in temperature. The slope was positive with a 95% interval above zero in six of eight task-sampler-thinking combinations at theta 0.8; the two exceptions were the shortest path task with top-p, thinking off (0.365, interval -0.077 to 0.759) and thinking on (0.232, interval -0.0 to 0.484). The slope of log ideas was zero everywhere except the thinking off top-k cells (0.203 on the grid task, 0.266 on the shortest path task), where single non-modal programs appeared at the highest temperatures. The preregistered criterion (positive slope at every theta and sampler, with the paired interval above zero) held on the grid task and failed on the shortest path task.

**Table 29.** Slope of log phrasings per idea against temperature (per unit T; 1,000 bootstrap draws; single linkage) at theta 0.7, 0.8 and 0.9, with 95% intervals, and the slope of log ideas (identical at every theta).

| Task | Sampler | Thinking | Theta 0.7 | Theta 0.8 | Theta 0.9 | Slope, ideas |
|---|---|---|---|---|---|---|
| Grid | top-k = 10 | off | 1.138 [0.808, 1.409] | 1.265 [0.902, 1.547] | 1.440 [0.990, 1.727] | 0.203 [0.0, 0.41] |
| Grid | top-p = 0.95 | off | 1.083 [0.835, 1.253] | 1.223 [0.975, 1.368] | 1.435 [1.182, 1.620] | 0.0 |
| Grid | top-k = 10 | on | 0.563 [0.282, 0.784] | 0.417 [0.213, 0.567] | 0.233 [0.072, 0.321] | 0.0 |
| Grid | top-p = 0.95 | on | 0.289 [0.113, 0.602] | 0.293 [0.134, 0.486] | 0.247 [0.123, 0.351] | 0.0 |
| Shortest path | top-k = 10 | off | 0.693 [0.442, 1.001] | 0.758 [0.453, 1.044] | 1.295 [0.916, 1.849] | 0.266 [0.0, 0.486] |
| Shortest path | top-p = 0.95 | off | 0.365 [-0.102, 0.757] | 0.365 [-0.077, 0.759] | 1.101 [0.780, 1.475] | 0.0 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Shortest path | top-k = 10 | on | 0.752 [0.449, 0.956] | 0.846 [0.520, 1.075] | 0.408 [0.213, 0.567] | 0.0 |
| Shortest path | top-p = 0.95 | on | -0.011 [-0.245, 0.222] | 0.232 [-0.0, 0.484] | 0.205 [0.017, 0.355] | 0.0 |

**Thinking traces do not explore.** Every thinking on trace hit the 2,048-token cap, so no trace completed. In the 800 grid traces the first approach named was BFS in all but one, and DFS was mentioned as a second option in 67-90% of traces depending on the cell; the implemented algorithm equalled the first approach named in 99-100% of traces. In the shortest path traces the first approach was Dijkstra in 98-100% and Bellman-Ford/SPFA was mentioned in 1-5% (Table 30). The plan-statement lexicon used to read the traces had precision 0.62 and recall 0.482 against a judge on 30 traces, so these counts are approximate.

**Table 30.** Approaches named in thinking traces (N = 100 per cell; all traces truncated at 2,048 tokens).

| Task | Sampler | T | First approach named | Also considered | Mean approaches considered | First equals implemented |
|---|---|---|---|---|---|---|
| Grid | top-k = 10 | 0.3 | BFS 100 | DFS 67 | 1.67 | 1.00 |
| Grid | top-k = 10 | 0.7 | BFS 100 | DFS 78 | 1.78 | 1.00 |
| Grid | top-k = 10 | 1.0 | BFS 100 | DFS 84, A* 1 | 1.85 | 1.00 |
| Grid | top-k = 10 | 1.3 | BFS 99, DFS 1 | DFS 90 | 1.90 | 0.99 |
| Grid | top-p = 0.95 | 0.3 | BFS 100 | DFS 70 | 1.70 | 1.00 |
| Grid | top-p = 0.95 | 0.7 | BFS 100 | DFS 77 | 1.77 | 1.00 |
| Grid | top-p = 0.95 | 1.0 | BFS 100 | DFS 84 | 1.84 | 1.00 |
| Grid | top-p = 0.95 | 1.3 | BFS 100 | DFS 85, A* 1 | 1.86 | 1.00 |
| Shortest path | top-k = 10 | 0.3 | Dijkstra 100 | Bellman-Ford/SPFA 2 | 1.02 | 1.00 |
| Shortest path | top-k = 10 | 0.7 | Dijkstra 100 | Bellman-Ford/SPFA 2 | 1.02 | 1.00 |
| Shortest path | top-k = 10 | 1.0 | Dijkstra 100 | Bellman-Ford/SPFA 3 | 1.03 | 1.00 |
| Shortest path | top-k = 10 | 1.3 | Dijkstra 98, Bellman-Ford/ SPFA 2 | Bellman-Ford/SPFA 4 | 1.04 | 0.98 |
| Shortest path | top-p = 0.95 | 0.3 | Dijkstra 100 | Bellman-Ford/SPFA 1 | 1.01 | 1.00 |
| Shortest path | top-p = 0.95 | 0.7 | Dijkstra 100 | Bellman-Ford/SPFA 3 | 1.03 | 1.00 |
| Shortest path | top-p = 0.95 | 1.0 | Dijkstra 100 | Bellman-Ford/SPFA 4, Floyd-Warshall 1 | 1.05 | 1.00 |
| Shortest path | top-p = 0.95 | 1.3 | Dijkstra 100 | Bellman-Ford/SPFA 3 | 1.03 | 1.00 |

**Seeds and instructions.** Random seeds did nothing to the output (Table 31): the entropy and word seeds shifted the modal share by about 1 point on the grid task and 0 on the shortest path task, with 90% intervals inside the preregistered 15-point equivalence bound. The semantic seed moved the shortest path task by 66.3 points toward Bellman-Ford/SPFA among correct programs (61 of 92

correct programs, while the Dijkstra share fell by 67.39 points; permutation $p < 0.001$; corrected in iteration 2 from 67 points, which is the fall in the Dijkstra share) and the grid task by 2.2 points toward bidirectional BFS (2 of 93 correct programs; permutation $p = 0.078$ for the shift toward the implied algorithm). Among the instructions, the creativity instruction did nothing and the new algorithm instruction moved 1-4 points. The novelty instruction moved 3.7 points on the grid task and 14.6 points on the shortest path task, toward Bellman-Ford/SPFA. The enumeration prompt moved 10.7 points on the shortest path task at a 24-point pass-rate cost.

**Table 31.** Seeds and instructions (N = 100 each; labels of correct programs; shift of the default's share against the baseline, with 90% interval; cluster permutation p).

| Task | Condition | Pass | Labels of correct programs | Default share shift (points) [90%] | p |
|---|---|---|---|---|---|
| Grid | baseline (reference) | 0.97 | BFS 96, unit weight Dijkstra 1 | n/a | n/a |
| Grid | entropy seed | 1.00 | BFS 100 | +1.03 [0.0, 4.08] | 0.086 |
| Grid | word seed | 1.00 | BFS 100 | +1.03 [0.0, 4.04] | 0.096 |
| Grid | semantic seed | 0.93 | BFS 91, bidirectional BFS 2 | -1.12 [-5.47, 3.03] | 0.402 |
| Grid | creativity instruction | 1.00 | BFS 100 | +1.03 [0.0, 4.04] | 0.085 |
| Grid | new algorithm instruction | 0.94 | BFS 92, DFS 2 | -1.10 [-5.26, 3.0] | 0.329 |
| Grid | novelty instruction | 0.85 | BFS 81, unit weight Dijkstra 2, DFS 1, A* 1 | -3.68 [-9.19, 0.98] | 0.031 |
| Grid | enumeration prompt | 0.99 | BFS 99 | +1.03 [0.0, 4.0] | 0.246 |
| Shortest path | baseline (reference) | 0.99 | Dijkstra 99 | n/a | n/a |
| Shortest path | entropy seed | 1.00 | Dijkstra 100 | 0.0 [0.0, 0.0] | 1.0 |
| Shortest path | word seed | 1.00 | Dijkstra 100 | 0.0 [0.0, 0.0] | 1.0 |
| Shortest path | semantic seed | 0.92 | Bellman-Ford/SPFA 61, Dijkstra 30, Floyd-Warshall 1 | -67.39 [-87.18, -46.39] | <0.001 |
| Shortest path | creativity instruction | 0.99 | Dijkstra 99 | 0.0 [0.0, 0.0] | 1.0 |
| Shortest path | new algorithm instruction | 0.94 | Dijkstra 90, Bellman-Ford/SPFA 4 | -4.26 [-9.47, 0.0] | 0.012 |
| Shortest path | novelty instruction | 0.96 | Dijkstra 82, Bellman-Ford/SPFA 14 | -14.58 [-25.26, -5.44] | 0.001 |
| Shortest path | enumeration prompt | 0.75 | Dijkstra 67, Bellman-Ford/SPFA 8 | -10.67 [-20.38, -3.57] | 0.001 |

The generate then select arm was the only one to reach several algorithms. On the grid task its correct programs were BFS 26, A* 14, unit weight Dijkstra 9 and DFS 6 (Hill_cc 3.457) at a pass rate of 0.55; on the shortest path task Bellman-Ford/SPFA 31, Dijkstra 29 and Floyd-Warshall 15 (Hill_cc 2.87) at 0.75. The rarefied number of ideas at depth 25 was 3.977 and 2.999, against 1.258 and 1.0 for the baseline. Every one of the 100 stage-1 lists named all six grid algorithms, so the model knows the repertoire it does not use.

**Seeds at the read position.** The subspace test asks whether a condition moves the read position state along the mode subspace (the span of the forced implementation centroids, dimension 5 on

the grid task and 4 on the shortest path task) more than switching between baseline paraphrases does. The statistic is the ratio of the median displacement under the condition to the median paraphrase displacement (Table 32, Figure 4). The entropy seed stayed within the paraphrase noise (0.734 and 1.165). *(Correction, iteration 2: on the shortest path task its 90% interval reaches 1.596, above the 1.5 equivalence bound, so equivalence with paraphrase noise was not shown there.)* The word seed moved the grid state 1.555 times the noise (90% interval 1.309 to 1.85), above the preregistered 1.5 bound, although a fresh word seed with a different word list per paraphrase moved it only 1.055 times and a neutral sentence of the same length 0.812 times. The semantic seed moved the state 3.4 and 3.7 times the noise, and on the grid task its displacement pointed toward the implied centroid (cosine 0.493) while the output did not follow. *(Added in iteration 2: the other half of the same result was omitted. On the shortest path task the semantic seed's displacement did not point toward the Bellman-Ford/SPFA centroid (cosine -0.087), yet the output moved 66.3 points toward it, and the fixed word seed moved the grid probe's decoded distribution 0.88 toward A* with no behavioural change. Displacement along the mode subspace predicted behavioural movement in neither direction; Section 2.3.8.)* The instructions that changed behaviour also moved the state: the novelty instruction 1.809 and 3.063, the enumeration prompt 2.757 and 2.959. A probe from the read position to the label distribution did not beat its shuffled label null on the shortest path task (mean JS 0.122 against a 5th percentile of 0.121), so the mode subspace there rests on forced prompts that the natural output never visits.

**Table 32.** Subspace test: median displacement along the mode subspace relative to the paraphrase null, with 90% interval (grid task at layer 27, subspace dimension 5; shortest path task at layer 22, dimension 4). Conditions marked * were encoded without generation.

| Condition | Grid ratio [90%] | Shortest path ratio [90%] |
|---|---|---|
| entropy seed | 0.734 [0.579, 0.875] | 1.165 [1.0, 1.596] |
| word seed | 1.555 [1.309, 1.85] | 1.012 [0.865, 1.401] |
| fresh entropy seed* | 0.664 [0.53, 0.825] | 0.968 [0.796, 1.365] |
| fresh word seed* | 1.055 [0.842, 1.314] | 0.927 [0.756, 1.252] |
| neutral sentence* | 0.812 [0.561, 1.073] | 1.288 [1.002, 1.932] |
| semantic seed | 3.417 [2.876, 4.211] | 3.70 [3.005, 4.951] |
| creativity instruction | 1.041 [0.846, 1.436] | 1.684 [1.304, 2.35] |
| new algorithm instruction | 1.302 [0.979, 1.637] | 1.017 [0.79, 1.963] |
| novelty instruction | 1.809 [1.509, 2.173] | 3.063 [2.523, 4.158] |
| enumeration prompt | 2.757 [2.251, 3.435] | 2.959 [2.382, 4.016] |

*[Figure fig_subspace: generated in the report phase, not in this draft]*

The same ratios four layers before and after the analysis layer keep the ordering (Table 32b). The semantic seed and the enumeration prompt stay at two to four times the noise and the random seeds stay near 1. The exception is the shortest path task at layer 18, where the neutral sentence (1.812) and the entropy seed (1.484) also exceed 1.4.

**Table 32b.** Subspace test ratios at layers four before and four after the analysis layer (90% intervals).

| Condition | Grid layer 23 | Grid layer 31 | Shortest path layer 18 | Shortest path layer 26 |
|---|---|---|---|---|
| entropy seed | 0.846 [0.771, 1.023] | 0.777 [0.648, 0.962] | 1.484 [1.153, 1.953] | 0.953 [0.707, 1.411] |

| | | | | |
|---|---|---|---|---|
| word seed | 1.921 [1.665, 2.170] | 1.504 [1.311, 1.897] | 1.331 [1.031, 1.786] | 1.072 [0.793, 1.508] |
| fresh entropy seed* | 0.883 [0.708, 1.079] | 0.737 [0.634, 0.942] | 1.278 [1.053, 1.687] | 0.791 [0.604, 1.187] |
| fresh word seed* | 1.294 [1.082, 1.632] | 1.067 [0.918, 1.434] | 1.238 [0.903, 1.581] | 0.871 [0.655, 1.343] |
| neutral sentence* | 0.777 [0.607, 1.044] | 0.879 [0.577, 1.203] | 1.812 [1.238, 2.674] | 0.901 [0.667, 1.348] |
| semantic seed | 3.806 [3.314, 4.437] | 3.237 [2.809, 4.270] | 3.427 [2.801, 4.364] | 4.009 [3.166, 5.873] |
| creativity instruction | 1.525 [1.301, 1.822] | 1.048 [0.894, 1.607] | 2.128 [1.672, 2.813] | 1.093 [0.843, 1.721] |
| new algorithm instruction | 1.611 [1.259, 2.062] | 1.220 [0.900, 1.687] | 1.770 [1.349, 2.274] | 0.848 [0.672, 1.698] |
| novelty instruction | 2.717 [2.234, 3.159] | 1.838 [1.590, 2.363] | 3.410 [2.647, 4.329] | 2.291 [1.675, 3.270] |
| enumeration prompt | 3.242 [2.827, 3.858] | 2.614 [2.303, 3.308] | 4.257 [3.537, 5.363] | 2.005 [1.541, 3.101] |

**Where temperature acts in the trajectory.** Over the first 64 generated tokens, the variance of the residual-stream states rose with temperature almost entirely outside the mode subspace (Table 33). On the grid task the complement variance rose from 36,267 to 46,308 between T = 0.3 and T = 1.3 while the share in the mode subspace estimated from generated tokens stayed at 0.061-0.062. The preregistered prediction was a nonpositive slope of the share in excess of a null matched to the data; the slope was slightly positive (0.006 and 0.015 per unit T, intervals above zero), so the prediction failed, although the excess itself stayed negative at every temperature on both tasks. In the mode subspace estimated at the read position the share was 0.004 (grid) and 0.003 (shortest path) at every temperature, and a subspace from naturally sampled programs could not be estimated because 399 of 400 programs per task were the default.

**Table 33.** Variance of trajectory states (first 64 generated tokens) inside the mode subspace estimated from generated tokens, per temperature (N = 50 per cell).

| Task (layer) | T | Share in mode subspace | Excess over null matched to the data | Mode subspace variance | Complement variance |
|---|---|---|---|---|---|
| Grid (27) | 0.3 | 0.062 | -0.023 | 2,733 | 36,267 |
| Grid (27) | 0.7 | 0.062 | -0.022 | 2,970 | 39,928 |
| Grid (27) | 1.0 | 0.061 | -0.020 | 3,334 | 46,586 |
| Grid (27) | 1.3 | 0.062 | -0.016 | 3,375 | 46,308 |
| Shortest path (22) | 0.3 | 0.018 | -0.050 | 9 | 422 |
| Shortest path (22) | 0.7 | 0.035 | -0.032 | 113 | 2,808 |
| Shortest path (22) | 1.0 | 0.041 | -0.025 | 194 | 4,206 |
| Shortest path (22) | 1.3 | 0.030 | -0.037 | 205 | 5,825 |

**The forecast record.** Eight predictions were committed before generation; six hit. The two misses were the predictions that random seeds stay within 1.5 times the paraphrase null at the read position (the grid word seed exceeded it) and that the excess mode subspace share has a

nonpositive slope in temperature. The preregistered verdict was MIXED.

**What this experiment settled.** Temperature and the sampler are phrasing knobs on these tasks: they never changed the algorithm from 0.3 to 1.3, with or without thinking, while the number of distinct phrasings rose severalfold. Random seeds do not change the output, and apart from the fixed grid word seed their effect on the read position state is at most that of an arbitrary sentence. Context with meaning does move the algorithm, but not reliably: the same semantic seed moved the shortest path task by 66.3 points and the grid task by 2.2 (among correct programs; corrected in iteration 2 from 67 and 2). Explicit novelty instructions move little, and generate then select reaches three to four algorithms at a 24-42-point pass-rate cost.

## 1.11 Comparisons across the six experiments

The strategy deferred the scorecard that combines all experiments to an evaluation artifact in the next iteration. The comparisons below were compiled for this report from each experiment's per-sample rows and result files; the replication table was recomputed with `work/replication.py` in this workspace.

**The same ban, five draws.** Five experiments ran the same modal ban under the shared protocol, each with its own ten paraphrases of `Do not use X`, its own test suite and, in one case, its own engine. Where the ban had compliant samples, they landed on Bellman-Ford/SPFA on the shortest path task in all five draws (272 of 272 compliant programs), and on DFS on the grid task in four of five draws; in the geometry draw, sampled with HF `generate` and with the highest banned share (0.62), the 18 compliant programs split into DFS 10 and unit weight Dijkstra 8, with no clear mode (Table 34). *(Corrected in iteration 2; this sentence previously said that the compliant samples landed on the same substitute every time, DFS on the grid task with a few unit weight Dijkstra programs, which Tables 13 and 34 contradict. Section 2.3.3 gives the per draw counts.)* How often the ban was obeyed did not replicate. The banned mode share ranged from 0.10 to 0.62 on the grid task and from 0.09 to 0.72 on the shortest path task. Within each experiment it depended on the wording: on the grid task the banned counts per paraphrase ranged from 0 of 10 to 9 or 10 of 10 in four of the five experiments (Figure 5). The association experiment's paraphrases were the most effective (eight of ten grid wordings gave 0 of 10 BFS), which is why its compliance looked high. Constraints that oppose a model's default pattern are known to fail at rates between 10% and 100% [38]; here the spread occurs within one model, one task and one ban. Compliance is therefore a property of the ban sentence at least as much as of the banned algorithm, and any analysis that relates ban failure to properties of the algorithm (the ladder limit test) has to hold the wording fixed or model it.

**Table 34.** The modal ban across experiments (N = 100 each; recomputed from per-sample rows; compliant = correct and not the banned algorithm).

| Experiment (engine) | Grid ban BFS: labels | Banned share | Compliant (labels) | Shortest path ban Dijkstra: labels | Banned share | Compliant (labels) |
|---|---|---|---|---|---|---|
| Association (vLLM) | DFS 90, BFS 10 | 0.10 | 38 (DFS 38) | Bellman-Ford/SPFA 90, Dijkstra 9, other 1 | 0.09 | 86 (Bellman-Ford/SPFA 86) |

| | | | | | | |
|---|---|---|---|---|---|---|
| Geometry (HF) | BFS 62, DFS 29, unit weight Dijkstra 9 | 0.62 | 18 (DFS 10, unit weight Dijkstra 8) | Dijkstra 72, Bellman-Ford/SPFA 27, other 1 | 0.72 | 22 (Bellman-Ford/SPFA 22) |
| Forking token (vLLM) | DFS 78, BFS 21, unit weight Dijkstra 1 | 0.21 | 41 (DFS 40, unit weight Dijkstra 1) | Bellman-Ford/SPFA 80, Dijkstra 20 | 0.20 | 80 (Bellman-Ford/SPFA 80) |
| Decommitment (vLLM) | BFS 49, DFS 48, unit weight Dijkstra 2, other 1 | 0.49 | 18 (DFS 16, unit weight Dijkstra 2) | Dijkstra 54, Bellman-Ford/SPFA 40, other 6 | 0.54 | 38 (Bellman-Ford/SPFA 38) |
| Instruction tuning (vLLM) | BFS 62, DFS 36, unit weight Dijkstra 2 | 0.62 | 13 (DFS 12, unit weight Dijkstra 1) | Dijkstra 54, Bellman-Ford/SPFA 46 | 0.54 | 46 (Bellman-Ford/SPFA 46) |
| Pooled, five draws (row added in iteration 2) | n/a | 0.41 | 128 (DFS 116, unit weight Dijkstra 12); DFS share 0.906, Wilson [0.843, 0.946] | n/a | 0.42 | 272 (Bellman-Ford/SPFA 272); share 1.0, Wilson [0.986, 1.000] |

*[Figure fig_replication: generated in the report phase, not in this draft]*

**Two elicitations of the same association map disagree.** The association and geometry experiments asked the same model the same question with different prompt wordings and got different maps (Table 35). For BFS one map put unit weight Dijkstra first with clarity 0.20 and the other put A* first with clarity 0.80; for Dijkstra one put Bellman-Ford/SPFA first with clarity 1.0 and the other put Dial first with clarity 0.579. The realised substitutes (DFS, Bellman-Ford/SPFA) were ranked last for BFS and third for Dijkstra in the second map. The ranking instrument is not stable under rewording, and where its two versions disagree, neither matches the free-form answer the model gives when asked directly.

**Table 35.** The association map of the two default algorithms in the two experiments that elicited it (20 sampled rankings each; 19 for Dijkstra in the geometry experiment).

| X | Elicitation | Borda shares of the alternatives | Winner | Clarity | Kendall's W |
|---|---|---|---|---|---|
| BFS | association experiment | unit weight Dijkstra 0.293, bidirectional BFS 0.283, A* 0.263, IDDFS 0.107, DFS 0.054 | unit weight Dijkstra | 0.20 | 0.52 |
| BFS | geometry experiment | A* 0.371, unit weight Dijkstra 0.288, bidirectional BFS 0.229, IDDFS 0.107, DFS 0.005 | A* | 0.80 | 0.886 |
| Dijkstra | association experiment | Bellman-Ford/SPFA 0.492, Floyd-Warshall 0.226, bidirectional Dijkstra 0.153, Dial 0.129 | Bellman-Ford/SPFA | 1.0 | 0.639 |
| Dijkstra | geometry experiment | Dial 0.424, bidirectional Dijkstra 0.314, Bellman-Ford/SPFA 0.212, Floyd-Warshall 0.051 | Dial | 0.579 | 0.581 |

**The PREDICT column.** Every forecasting family was scored by the experiment that owned it, on that experiment's own draw of the two modal ban events, and all forecasts were committed before the samples (Table 36). The selection rule needs at least three events on which the families' top-1 nominations disagree; the screen produced two shared events (the step-2 events had too few compliant samples or were exploratory), so the rule cannot name a survivor. *(Correction, iteration 2:*

*the rule also gates events on compliance of at least 50%. Under that gate the geometry, instruction tuning and decommitment shortest path events in Table 36 are excluded, the base model, decommitted and geodesic forecasts were scored only on excluded events, and the ranking that follows rests largely on them. On gated events no family can advance; Section 2.3.7.)* Descriptively, the highest scoring forecasts on both events were those that already put most of their mass on the realised substitute before the ban: the model's free-form answer, the listing control distribution, the base model's own distribution and, on the shortest path task, the decommitted distribution. All of them rank the realised substitute second in the preferences the model shows before any ban. The preregistered association map was the worst or second worst forecast in both experiments that elicited it on the grid event. The geodesic kernel beat uniform by 0.29 nats on the grid event and lost to the uniform forecast based on frequency on the shortest path event.

**Table 36.** PREDICT column: raw log score of every committed forecast on the modal ban events (each experiment scored on its own samples; higher is better; uniform over survivors is -1.630 on the grid task and -1.406 on the shortest path task under the shared mix and floor). The geometry experiment scored every sample not labelled as the banned algorithm rather than only compliant ones.

| Forecast family (experiment) | Grid ban BFS | Shortest path ban Dijkstra |
|---|---|---|
| Association map, Borda (association) | -2.817 | -0.755 |
| Association map, first rank (association) | -2.474 | -0.190 |
| External association map, Llama-3.1-8B (association) | -1.412 | -0.928 |
| Free-form answer about the standard alternative (association, not preregistered as a contest predictor) | -0.112 | -0.058 |
| Listing control distribution (association, not preregistered as a contest predictor) | -0.061 | -0.058 |
| Latent frequency, uniform under collapse (association) | -1.630 | -1.406 |
| Association map, Borda (geometry) | -3.543 | -1.647 |
| Geodesic kernel (geometry) | -1.337 | -1.530 |
| Read position decoder (geometry) | -1.287 | -1.587 |
| Latent frequency (geometry) | -1.630 | -1.496 |
| Decommitted survivor distribution (decommitment) | -1.630, degenerate | -0.058 |
| vLLM frequency forecast (decommitment) | -1.630 | -0.219 |
| Base model, plain completion, all samples (instruction tuning, primary) | -0.470 | -0.220 |
| Base model, chat template, correct samples (instruction tuning) | -0.412 | -0.058 |
| Instruct model, chat template, correct samples (instruction tuning) | -1.630 | -0.058 |

**The BUY column.** The rule asks for the highest correctness conditioned Hill number at the strongest setting that keeps at least 80% of the baseline pass rate. Table 37 lists every knob that reached that pass-rate floor with a Hill gain, and the main ones that did not; Figure 6 plots the shortest path knobs as Hill gain against pass-rate points lost. On the grid task the cumulative ban of BFS and DFS kept 88% of its reference pass rate in the decommitment draw and produced 16 compliant programs over two algorithms (unit weight Dijkstra 13, A* 3), but it does not replicate: the geometry draw (pass 0.96 against 0.99) put all 18 of its compliant programs on unit weight Dijkstra, and the association draw (pass 0.53 against 0.933) left 5 compliant programs; the novelty instruction raised Hill_cc to 1.27. *(Corrected in iteration 2; this sentence previously said that no knob keeping 80% of the pass rate produced a second algorithm among compliant programs and that the gain*

*of the cumulative ban came from programs that still used BFS, which the decommitment row of Table 37 contradicts. Section 2.3.4.)* On the shortest path task the prompt ban and the semantic seed reached about 2 effective algorithms at no or small pass cost, while the activation knobs (decommitment, a random direction) reached about +0.6 at 17 points. By the rule's 0.5 margin the prompt ban and the semantic seed tie, so both would advance. *(Correction, iteration 2: the BUY rule names a winner only on the pooled tasks and in direction on each task, and this comparison used the shortest path task alone, with Hill numbers from different statistics. Recomputed with one statistic, rarefied Hill-1 over compliant programs, no knob advances on either task, the prompt ban scores 1.0 on the shortest path task, and the positive target of a second correct algorithm at under 20 pass points is met on neither task; Section 2.3.6.)* The table also exposes a flaw in the metric: Hill_cc counts correct programs of the banned algorithm, so a ban that half fails scores as two algorithms. The prompt-ban rows on the shortest path task are Dijkstra mixed with Bellman-Ford/SPFA; only the semantic seed and generate then select produced diversity without a banned algorithm in the mix.

**Table 37.** BUY column: knobs on the two tasks, with pass rate relative to their own reference, and the Hill number or its change as each artifact reported it.

| Task | Knob (experiment) | Pass / reference pass | Reaches 80% floor | Hill_cc (or Delta Hill) | Correct program composition |
|---|---|---|---|---|---|
| Shortest path | prompt ban (instruction tuning) | 1.00 / 1.00 | yes | 1.99 | Dijkstra 54, Bellman-Ford/SPFA 46 |
| Shortest path | semantic seed (decoding) | 0.92 / 0.99 | yes | 1.988 | Bellman-Ford/SPFA 61, Dijkstra 30, Floyd-Warshall 1 |
| Shortest path | prompt ban (geometry) | 0.94 / 1.00 | yes | 1.723 | Dijkstra 72 correct banned plus Bellman-Ford/SPFA 22 |
| Shortest path | prompt ban (forking token) | 1.00 / 0.995 | yes | 1.649 | Bellman-Ford/SPFA 80, Dijkstra 20 |
| Shortest path | novelty instruction (decoding) | 0.96 / 0.99 | yes | 1.515 | Dijkstra 82, Bellman-Ford/SPFA 14 |
| Shortest path | prompt ban (association) | 0.95 / 1.00 | yes | 1.368 | Bellman-Ford/SPFA 86, Dijkstra 9 |
| Shortest path | prompt ban (decommitment) | 0.87 / 0.89 | yes | Delta +0.79 | Dijkstra 49, Bellman-Ford/SPFA 38 |
| Shortest path | decommitment direction, c = 0.25 | 0.78 / 0.95 | yes (82%) | Delta +0.65 | Bellman-Ford/SPFA 54, Dijkstra 23, other 1 |
| Shortest path | random direction, c = 0.25 (decommitment) | 0.79 / 0.95 | yes (83%) | Delta +0.59 | Dijkstra 53, Bellman-Ford/SPFA 26 |
| Shortest path | generate then select (decoding) | 0.75 / 0.99 | no (76%) | 2.87 | Bellman-Ford/SPFA 31, Dijkstra 29, Floyd-Warshall 15 |
| Shortest path | enumeration prompt (decoding) | 0.75 / 0.99 | no | 1.404 | Dijkstra 67, Bellman-Ford/SPFA 8 |
| Shortest path | temperature 1.0-1.3, top-k = 10 (decoding) | 0.98-1.00 / 0.99 | yes | 1.058-1.059 | Dijkstra with one Bellman-Ford/SPFA |
| Shortest path | sharpening subtraction, any variant (instruction tuning) | 0.00-0.98 / 1.00 | only at layer 18, c of 0.5 or less | 1.00-1.10 | Dijkstra |

| | | | | | |
|---|---|---|---|---|---|
| Grid | cumulative ban of BFS and DFS (decommitment) | 0.88 / 1.00 | yes | Delta +1.27 | BFS 65, unit weight Dijkstra 13, DFS 7, A* 3 |
| Grid | novelty instruction (decoding) | 0.85 / 0.97 | yes | 1.27 | BFS 81, unit weight Dijkstra 2, DFS 1, A* 1 |
| Grid | prompt ban (geometry) | 0.76 / 0.99 | no (77%) | 2.034 | BFS 58 correct banned, DFS 10, unit weight Dijkstra 8 |
| Grid | prompt ban (instruction tuning) | 0.70 / 1.00 | no | 1.70 | BFS 57, DFS 12, unit weight Dijkstra 1 |
| Grid | prompt ban (association) | 0.46 / 0.933 | no | 1.587 | DFS 38, BFS 8 |
| Grid | generate then select (decoding) | 0.55 / 0.97 | no | 3.457 | BFS 26, A* 14, unit weight Dijkstra 9, DFS 6 |
| Grid | decommitment direction, any c | 0.98 at $c = 0.1$ | yes at $c = 0.1$ | 1.0 | BFS |
| Grid | temperature, any setting (decoding) | 0.95-1.00 | yes | 1.0-1.06 | BFS |

*[Figure fig_knobs: generated in the report phase, not in this draft]*

**The locus diagnostic.** The test that would move the geometric analyses from the read position to generated tokens needed a probe at the forking token and commitment positions, and that probe produced no output. The read position evidence is split: it encodes the shortest path default and fails to decode the grid default, while the forking token screen found every baseline path fixed at the read position on both tasks. The diagnostic is undetermined.

## 1.12 What iteration 1 established, and what it abandoned

Everything below is descriptive: one model, two tasks, two informative ban events per experiment, and labellers of which two failed their strict gates.

**Established.**

1. The collapse is total and robust to decoding. Across six independent baselines, 1,199 of 1,200 grid programs were BFS and 1,174 of 1,200 shortest path programs were Dijkstra. Temperature from 0.3 to 1.3, top-k = 10 or top-p = 0.95, and thinking on or off changed the algorithm in at most 3 of 800 programs per task.
2. Temperature is a phrasing knob. At the tightest setting the model writes one idea in a handful of phrasings; loosening temperature multiplies phrasings (4 to 36 clusters on the grid task with top-k = 10) and leaves ideas at one or two. This is the answer this run gives to the request's question 1a-c on these two tasks.
3. Random context seeds change nothing in the output. Apart from the fixed word seed on the grid task (1.56 times the paraphrase noise, against 1.06 for word seeds drawn fresh per paraphrase), they displace the read position state no more than an arbitrary sentence does. Context with meaning moves the state three to four times the paraphrase noise and can move the output, by 66.3 points on the shortest path task and by 2.2 points on the grid task (among correct programs). *(Corrected in iteration 2; the item previously gave 67 and 2 points, 67 being the fall in the Dijkstra share. It also omitted that read position displacement did not predict which algorithm the output moved to on either task: the grid semantic seed pointed at its implied algorithm and behaviour barely moved, the shortest path seed did not point at Bellman-Ford/SPFA and behaviour moved two thirds of the way; and the shortest path entropy seed's interval reaches 1.596, above the equivalence bound. Section 2.3.8.)*

4. When a prompt ban is obeyed, the compliant programs land mostly on one substitute. On the shortest path task it was Bellman-Ford/SPFA in five of five draws (272 of 272 compliant programs); on the grid task DFS was the realised mode in four of five draws (116 of 128 pooled, 0.906, Wilson [0.843, 0.946]), and the HF engine geometry draw, with the highest banned share (0.62), had no clear mode (DFS 10, unit weight Dijkstra 8). A single ban collapses the output again; a second step grid ban can spread compliant output over unit weight Dijkstra and A* (decommitment draw: 13 and 3) or put all of it on unit weight Dijkstra (geometry draw: 18). *(Corrected in iteration 2; the item previously said that the substitute replicated in five independent draws and that bans collapse the output again without spreading it, which the geometry draw and the second step bans contradict. Section 2.3.3.)* How often a ban is obeyed did not replicate (banned mode share 0.09-0.72) and depends on the wording of the ban sentence. Banning the substitute as well fails, and the model then reimplements the banned mechanism under another description.
5. The substitute is the model's second preference, visible before any ban in its free-form answer, in its output when every algorithm is listed, and in the base model's distribution. The preregistered ranked association map did not identify it on the grid task and gave different answers under two elicitation wordings. *(Note added in iteration 2: on the shortest path task random activation directions and every other perturbation that moved the model off Dijkstra also landed on Bellman-Ford/SPFA (Section 2.3.5), so there the second preference of the model cannot be told apart from wherever any perturbation lands.)*
6. The read position carries a low-dimensional map of the catalogue (intrinsic dimension 4-5 against 11-12 for shuffled nulls). On the grid task the mass displaced by bans moved to BFS's geodesic neighbours and never to the far side, *(clause inserted in iteration 2, marked in iteration 3)* which is consistent with geodesic proximity and with typicality, since those neighbours are also second choices of the model; the two are not separated on these events. *(Corrected in iteration 2: the geodesic order is unstable at layer 24 under paraphrase resampling, and on the shortest path task Floyd-Warshall, not the realised Bellman-Ford/SPFA, is the nearest survivor in every subsample. Section 2.3.2.)* The ban shift itself is almost entirely orthogonal to the algorithm directions, so the substitution versus subtraction share test was inconclusive for every algorithm.
7. No activation edit beat the prompt ban. Affine ablation of the modal algorithm did nothing at alpha up to 2 and broke generation at 4, as random directions did; the decommitment direction acted like a random direction; subtracting the instruct minus base direction destroyed generation.
8. Post training sharpened an ordering the base model already had, rather than installing a new attractor; the chat format sharpens it further on the shortest path task. *(Correction, iteration 2: this item replaced the preregistered verdicts with an interpretation. The verdicts are INSTALLED_BY_POST_TRAINING on the shortest path task and MIXED on the grid task under the correctness conditioned Hill number (INSTALLED_BY_POST_TRAINING on both under the Hill number over all programs), and MIXED on both at the planned sample size. The base ordering matches the instruct ordering only under the chat template, where the base model parsed 0.58 and 0.66 of samples and wrote 57 and 90 correct programs, labelled with runtime against AST kappa of 0.20 to 0.50 (Table 24b); in plain completion its shortest path mode is Bellman-Ford/SPFA, 16 of 25 correct programs.)*
9. Without a ban the algorithm is fixed at the read position on both tasks. Under a ban the outcome stays open for dozens of tokens, a few forking tokens exist on the shortest path task, and writing the first token that identifies the algorithm does not lock the algorithm in.

**Abandoned or failed, with the evidence.**

- *Affine directional ablation of the modal algorithm as a removal knob.* 0 of 50 samples moved at alpha 0.5, 1 and 2 on both tasks, with 78-90% of outputs byte-identical to the unedited run; at alpha 4 the edit removed all code. The direction checks had failed beforehand (AUC 0.79-0.89). The prompt-versus-removal comparison that the hypothesis treats as a headline was therefore untestable in this form.
- *Subtracting the instruct minus base direction.* 0 of 100 parseable programs at every band coefficient; the direction promotes format tokens under the logit lens.
- *Forking window temperature.* Every arm was 100% the default algorithm; its phrasing count matched $T = 0.3$.
- *The decommitment direction as a diversity knob.* Its shortest path gain equalled a random direction's (difference $+0.07$, interval -0.26 to 0.39) and the grid task never moved; the enumeration prompt that states the same framing did nothing.
- *The ranked (Borda) association map as the association instrument.* It missed the grid event, ranked the realised substitute last, and changed its winner for both default algorithms under a second elicitation.
- *Baseline frequency as the typicality predictor.* Under a full collapse every survivor has zero frequency, so the forecast is uniform by construction and the discordance strata that the design defined by stability are empty.
- *The mention twin as a placebo, and the sham ban as a negative control.* Mentioning an algorithm recommends it ($+56$ to $+100$ points), so sham bans cannot be tested for equivalence with their mention twins.
- *The locus probes.* The forking token experiment extracted the states but its probe stage produced no output; this is a technical failure, not a negative result, and the measurement is still missing. *(Correction, iteration 2: the probe stage ran before the states were written (Section 1.7); the audit ran it in iteration 2 and found no decision locus among the saved positions, Section 2.3.1.)*
- *The reserved bench.* The dataset artifact delivered source data but no frozen specifications, tests, labellers or forecasts, so the planned out-of-sample confirmation cannot start until the bench is built.

**Spend.** OpenRouter: USD 0.27 (association), about 0.09 (geometry), 0.024 (forking token), 0.062 (decommitment), 0.21 (instruction tuning), 0.11 (decoding) and 0 (dataset), about USD 0.77 in total. GPU: about 6 minutes on an RTX 5090 (association), 2.2 hours on a 23 GB NVIDIA GPU (geometry), 2.1 hours on the same model of GPU (instruction tuning); the other experiments did not record GPU hours. *(Correction, iteration 2: from the union of each experiment's GPU phase log spans, an upper bound that includes waits, geometry 2.92, forking token 1.55, decommitment 3.23, instruction tuning 2.47 and decoding 3.31 GPU hours; Section 2.3.10. The answers this screen gives to the objections of the hypothesis review are recorded in Section 2.1.)*

# Iteration 2: the record repaired, a likelihood instrument, the route of a ban, the top five prohibition and a second model

## 2.1 Why this iteration ran what it ran

**The review of the iteration-1 record.** The review was blocking. It scored the report 4 overall (confidence 4), with soundness 1, presentation 3 and contribution 3. The soundness score came from two places where the report's conclusions contradicted its own tables. The report claimed that the substitute a ban produces replicated in five independent draws, but on the grid task the geometry draw had no clear mode (DFS 10 and unit weight Dijkstra 8 of 18 compliant programs). It also claimed that no grid knob above the 80% pass rate floor produced a second algorithm among compliant programs, but the decommitment draw's cumulative ban of BFS and DFS produced 16 compliant programs over unit weight Dijkstra and A*. The review further found that the selection rule had been quoted without its event gate and then applied to events the gate excludes; that the iteration had added forecasting families instead of events; that the report never named the rival its own controls suggested, namely that random directions push the shortest path task to the same substitute as a ban; that the geometric claim outran its evidence; that the cause of the missing locus probe was recorded nowhere; that several items of the request were not covered; that no positive claim was set against its nearest published neighbour; that the read position evidence on seeds was reported by halves; and that the objections of the hypothesis review had not been answered. Eleven items were marked must fix and three minor. The corrections they require are made in place in the iteration-1 section, each marked with what changed and why. Table 38 maps every item to the place where it is answered.

**Table 38.** Review items on the iteration-1 record and where this report answers them.

| Review item | What the review found | Where it is answered |
|---|---|---|
| Substitute claim over five draws | Grid geometry draw had no clear mode; the few unit weight Dijkstra programs were in fact 8 of 18 | Corrections in Sections 1.11 and 1.12; per draw table in Section 2.3.3 |
| Grid knob verdict | The cumulative BFS and DFS ban produced compliant unit weight Dijkstra and A* programs | Correction in Section 1.11; three draws in Section 2.3.4; positive target per task in Section 2.3.6 |
| Selection rule | Event gate omitted; rule applied to events the gate excludes; one Hill statistic not used | Correction in Section 1.1; gated forecast scores in Section 2.3.7; one statistic knob table in Section 2.3.6 |
| Events, not predictors | Fifteen forecasting families scored on two events | Families frozen at four (this section); bench with a presence pilot in Section 2.8; Section 2.12 |
| Generic perturbation rival | Random directions land on Bellman-Ford/SPFA like the ban | Correction to Table 26 in Section 1.9; perturbation table in Section 2.3.5; external ladder profile in Sections 2.4 and 2.7 |
| Geometric overreach | Proximity and typicality confounded; stability from point resampling | Correction to Section 1.12; layer and k sweep in Section 2.3.2 |
| Locus probe dead end | Cause unrecorded; inputs exist on disk | Correction in Section 1.7; probe re-run in Section 2.3.1 |

| | | |
|---|---|---|
| Request coverage | Top five ban, natural run internals, SAE, thought anchors, same seed, lenses, NLA missing | Sections 2.3.1, 2.5, 2.6 and 2.9 |
| Novelty positioning | No claim set against its nearest neighbour | Section 2.3.11 |
| Seeds half reported | Shortest path cosine, word seed probe shift and entropy seed interval omitted | Corrections in Sections 1.10 and 1.12; Section 2.3.8 |
| Objections of the hypothesis review | Sham bans, simulation of own behaviour, point resampling, missing precedent | This section, below |
| Minor: missing outputs | Executed conditions and diagnostics absent from the record | Section 2.3.10 and notes in Sections 1.5 to 1.10 |
| Minor: post training verdict | Preregistered label replaced by an interpretation | Corrections in Sections 1.9 and 1.12 |
| Minor: bookkeeping | GPU hours, source files, grid failure taxonomy | Section 2.3.10; correction to the spend line in Section 1.12 |

**What the hypothesis review asked for, and what the iteration-1 screen showed.** The review of the hypothesis before iteration 1 raised four objections that the screen can now answer. First, the review argued that if processing a negation promotes an associate [4], a ban of an algorithm the model never writes should still move mass toward a related algorithm, so the preregistered equivalence between a sham ban and its mention twin was inconsistent with the mechanism. Two of nine sham bans did move mass: banning Floyd-Warshall gave Bellman-Ford/SPFA 23% of attempted programs and banning IDDFS gave DFS 12% (Section 2.3.9). Both promoted algorithms are content neighbours of the banned one and also the second choice of the model, so this is weak and confounded evidence for promotion triggered by negation. Second, the review named simulation of its own behaviour [39] as a rival reason why the stated alternative of the model predicts its behaviour: a model can forecast what it will do without any privileged access to a decision. That is exactly the free-form predictor's success in Table 7. Such simulation fits either branch of the substitution test below, so the free-form result says nothing about mechanism. Third, the review warned that resampling points within a fixed graph would make geodesic stability uninformative. The iteration-1 stability of 0.985 came from that procedure, and resampling at the level of paraphrases changes the shortest path picture (Section 2.3.2). Fourth, the review flagged an omitted precedent on approach organised internal states [40] and an unmapped lens named in the request, which we refer to as the j lens. The precedent is positioned in Section 2.3.11, and the lens, a Jacobian lens, was tested in Section 2.9.

**The hypothesis update.** The update kept the claim and changed the instrument. The claim is that a prohibition both subtracts the banned algorithm X and substitutes: it raises particular survivors by more than renormalisation would, and which survivors it raises depends on X. We call the rival the *latent ladder*: a fixed ordering of the survivors that any perturbation of the model reveals, under which a ban only subtracts and exposes the second choice of the model. Iteration 1 could not separate the two accounts, because every substitute it observed was also the second choice of the model and because on the shortest path task random directions reached the same second choice as the ban. The update made four changes. It replaced sampled frequencies, which carry no information under a fully collapsed baseline, with a likelihood instrument read from forward passes (Section 2.2), so that every single ban, sham ban and ban list becomes a measured event. It added a causal test of where a ban acts: through the prompt states before the first generated token, or by generated tokens reading the prohibition directly. It froze the forecasting families at four (ladder, association, prime and veto, proximity), fixed the ban wording at five templates, and moved the

budget from predictors to events. And it ordered the iteration-1 record to be repaired before any new generation. A second claim was added: the read position encodes how strongly a ban will be obeyed, not what replaces X.

**The strategy.** Seven artifacts ran, one audit without generation and six experiments (Table 39). The audit repaired the record and ran every diagnostic that needed no GPU, including the locus probe that had never run. The likelihood screen and the Llama replication tested substitution against the latent ladder on the two path search tasks, in Qwen3-8B [5] and in Llama-3.1-8B-Instruct [22]. The route experiment patched key and value (KV) caches between a ban prompt and its neutral twin. The ban list experiment ran the request's literal prohibition of the top five algorithms. The bench build froze a confirmation bench of 12 new tasks behind a cheap presence pilot, with no ban prompt run on it. The readout experiment looked at the ban with sparse autoencoder features, attention, thought anchors, a same seed temperature test and lenses. The strategy fixed one rule for every experiment: the screen tasks (grid and shortest path) are never pooled into confirmation statistics, so the bench stays untouched until forecasts are committed in the next iteration.

**Table 39.** The seven artifacts of iteration 2, the question each was asked, and its hardware and spend.

| Artifact | Question | Model and hardware | OpenRouter spend |
|---|---|---|---|
| Audit | Repair the iteration-1 record; locus probe; geometry sweep; perturbation table; one statistic knob table; gated forecasts | No generation; probes on one RTX 4090 (about 8 min) | USD 0 |
| Likelihood screen | Substitution or latent ladder in Qwen3-8B (validity gate, gain matrices, X specific share) | Qwen3-8B; one 23 GB NVIDIA GPU, 3.38 GPU hours of scoring plus about 0.24 hours of generation | USD 0.30 |
| Route experiment | Does the ban act through the prompt states or directly at generation time; does the read position know how hard a ban will be obeyed | Qwen3-8B bf16; one RTX 4090, about 1.2 GPU hours | USD 0.32 |
| Ban list experiment | Where the mass lands when four or five algorithms are banned at once; bans that leave one survivor | Qwen3-8B; one RTX 4090, about 1 GPU hour | USD 0.42 |
| Llama replication | The likelihood screen and route readout in a second model family | Llama-3.1-8B-Instruct bf16; one RTX 4090, 0.64 GPU hours logged | USD 0.49 |
| Bench build | A frozen 12-task confirmation bench with presence pilot, labeller gate and baseline likelihoods | Qwen3-8B, official 8-bit floating point checkpoint (and a Llama pilot); one RTX 2000 Ada (16 GB), about 1.4 GPU hours | USD 0.14 |
| Readout experiment | SAE, attention and prime and veto readouts of the ban; thought anchors; same seed temperature test; Jacobian lens and NLA | Qwen3-8B bf16 states, 8-bit twin for generation; one 16 GB GPU, about 1.05 GPU hours | USD 0.04 |

## 2.2 Shared protocol v2 and the likelihood instrument

**Canonical suite and labels.** Every experiment on the path search tasks used the association experiment's task text, prompt scaffold, hidden tests (48 grid cases, 45 shortest path cases), 22

references and labeller, copied byte for byte. Each rechecked the labeller before use: the combined label agreed with the adjudicated gold set on 0.965 or 0.966 of 141 programs in four experiments and on 135 of 144 in the ban list experiment, and all 22 references were labelled correctly. Labels are at the level of mechanism: the runtime signature decides, so a renamed BFS counts as BFS.

**Templates and twins.** Ban wording was frozen at five templates (Table 40), each with a mention twin and a neutral twin. We call a ban's *neutral twin* the same sentence with the algorithm's display name replaced by frequent concrete nouns from the dataset artifact's 718-noun bank (`man`, `guy`, `house` and so on) and padded with filler tokens until the prompt has the ban prompt's exact token length; from the task text onward the two prompts have identical token ids. Two generic perturbations were also frozen: a nonce ban (`Do not use the Zorblax method`, in the five templates) and a novelty instruction (`Write a novel algorithm for this task, one you have not seen before.`). We refer to the templates by their order in Table 40, from the first to the fifth.

**Table 40.** The five frozen ban templates of shared protocol v2 (X is the algorithm's display name, N the neutral nouns).

| Template | Ban | Mention twin | Neutral twin |
|---|---|---|---|
| first | Do not use X for this task. | You could use X for this task. | Do not use N for this task. |
| second | Your solution must not rely on X. | Your solution may rely on X. | Your solution must not rely on N. |
| third | Avoid X entirely. | Consider X first. | Avoid N entirely. |
| fourth | X is not allowed here. | X is allowed here. | N is not allowed here. |
| fifth | Write a correct solution that does not use X. | Write a correct solution that may use X. | Write a correct solution that does not use N. |

**Event gate.** One definition replaces the two that iteration 1 mixed. The *avoidance* of a ban is one minus the share of all sampled programs labelled as a banned mechanism. An event enters a forecast contest only if avoidance is at least 0.5 and at least 20 attempted programs are not banned. The *landing distribution* is the label distribution of those attempted programs not banned. The compliant share (correct and not banned) is reported beside it and drives the knob table only, because most grid programs that fail do so through implementation bugs, not through the tests (Section 2.3.10).

**The likelihood instrument.** Under a fully collapsed baseline every survivor has sampled frequency zero, which made the typicality forecast uniform and the sham bans uninformative in iteration 1. We define the *forced ratio latent probability* (FRLP) of algorithm j under prompt c as an importance sampling estimate of the log probability that the model writes an algorithm j program when given c. The model first writes programs under forced prompts F_j (`Implement j for this task` and nine paraphrases). Programs labelled j are kept; names of catalogue algorithms are scrubbed from identifiers, comments and docstrings, and a scrubbed program is kept only if it passes the same tests and keeps the same runtime label. Each kept program p_m is then scored by teacher forcing under any prompt c, giving its summed log probability l(p_m | c). With the mixture of forced prompts q(p) = (1/K) Σ_k P(p | F_j^k) as the proposal, the balance heuristic of multiple importance sampling [41], and y_j the share of forced attempts labelled j,

$$L_j(c) = \log y_j + \log \operatorname{mean}_m \exp[\, l(p_m \mid c) - \log q(p_m) \,].$$

The same program text is scored under c and under its proposal, so phrasing and length cancel. Each cell reports the Kish effective sample size (ESS) of its importance weights. The *gain* of survivor j under a ban of X with template w is G[X, j, w] = L_j(ban_w X) − L_j(neutral_w X). The mention twin splits a ban into a *prime* (mention minus neutral) and a *veto* (ban minus mention).

**The substitution test.** For each single ban row, the gains are centred over the survivors and regressed on the centred *external ladder profile* u, the mean gain over generic perturbations of matched size: the nonce ban, six random edits of the prompt states in layers 12 to 24 whose norm is pinned per layer and position to the ban's own displacement, and the novelty instruction. The fitted part is the ladder component of the row and the residual its X specific component. Summed over rows and corrected for template and program replicate noise, they give V_u and V_s, and we call SHARE = V_s / (V_s + V_u) the *X specific share*. A ladder exists on a task if the generic profiles agree with one another (mean pairwise cosine over survivors other than the default of at least 0.5). The aim of the X specific residual is its Spearman correlation with the *prime and veto profile* (the mention twin's gains over its neutral twin, with X removed), computed on disjoint program halves and different nouns, against an exact within row permutation null. The branches were fixed before data: substitution (share interval above 0.5 and aim above the null's 95th percentile), X specific but unaimed (share interval above 0.5, aim not above the null), ladder (share interval below 0.5 and a ladder exists), pure subtraction (no between survivor variance beyond replicate noise), and undetermined (any other case). A row needs at least three survivors. Both references (neutral twin primary, mention twin secondary) are reported, and a branch is called robust only if they agree.

## 2.3 The audit: the iteration-1 record repaired, without new generation

The audit read the six iteration-1 experiments in place, generated nothing, made no LLM calls and spent nothing on OpenRouter. It ran first because several of this iteration's analyses depended on what it found: where the algorithm can be read out, whether the geometric neighbours survive resampling, and which iteration-1 events are informative at all. Every number it produced names its source file. Its own checks recomputed 40 numbers quoted in the plan and matched 38; the two mismatches are explained below (the count of gated events and the rarefied Hill number of the semantic seed). An independent script rederived 20 headline numbers with no mismatch, a second one reran five headline tests, and every placebo failed as it should.

### 2.3.1 The locus probe that never ran

**Cause.** The forking token experiment's probe file is 28 bytes, an empty summary written at 18:29:22. Its state log records the start of the state phase at 18:31:55 and the grid and shortest path states saved at 18:34:34 and 18:34:50. The probe stage ran before its inputs existed and found an empty folder. The measurement was therefore a CPU job on files already on disk, not a missing experiment.

**Design.** The saved states hold 300 generations per task (200 baseline, 100 under the modal ban) at eight positions (the read position, generated tokens 8, 16, 32 and 64, and the tokens before, at and after the first algorithm identifying token, which we call the commit token) and at layers 8 to 28. Within one ban prompt the model wrote BFS 21 times and DFS 78 times on the grid task (one unit weight Dijkstra) and Dijkstra 20 times and Bellman-Ford/SPFA 80 times on the shortest path task. These contrasts within one ban prompt are the natural run internals the request asks for: same prompts, different algorithms written. The probes are ridge regularised logistic regression probes fitted with folds that hold out one paraphrase at a time, which is the honest test because the read position is identical within a paraphrase. Each probe is compared with 200 refits on globally shuffled labels and 200 refits on labels shuffled within paraphrase (the null for a probe that knows no more than the base rate of each paraphrase), with a familywise null over layers. A written text control fits the same probe to features of the text written so far. A batched solver was unit tested against sklearn (identical predictions; largest relative coefficient difference 3.9e-5). The preregistered rule moves the locus to the first generated position whose decoder beats the read

position by at least 20 points of balanced accuracy and beats both nulls; a precommit qualifier asks whether the state carries more than the written text.

**Table 41.** Decoder of the algorithm written, within one ban prompt (grid: BFS against DFS; shortest path: Dijkstra against Bellman-Ford/SPFA), balanced accuracy at the best layer per position, with the read position accuracy at that layer, both nulls and the written text control (source: `results/locus.json`).

| Task | Position (median token) | Best layer | Balanced accuracy | Read position at that layer | Difference [95% CI] | Global null p95 | Within paraphrase null p95 | Text control | Passes rule |
|---|---|---|---|---|---|---|---|---|---|
| Grid | read position | 16 | 0.855 | n/a | n/a | n/a | n/a | 0.5 | n/a |
| Grid | before commit token (6) | 20 | 0.855 | 0.791 | +0.064 [0.0, 0.201] | 0.639 | 0.855 | 0.673 | no |
| Grid | after commit token (7) | 23 | 0.855 | 0.791 | +0.064 [0.0, 0.201] | 0.652 | 0.855 | 0.673 | no |
| Grid | token 8 | 26 | 0.855 | 0.791 | +0.064 [0.0, 0.201] | 0.659 | 0.855 | 0.673 | no |
| Grid | token 16 | 9 | 0.798 | 0.402 | +0.396 [-0.065, 0.537] | 0.658 | 0.879 | 0.673 | no |
| Grid | token 32 | 23 | 0.798 | 0.791 | +0.006 [-0.16, 0.229] | 0.653 | 0.850 | 0.542 | no |
| Grid | token 64 | 27 | 0.739 | 0.791 | -0.052 [-0.203, 0.239] | 0.657 | 0.812 | 0.598 | no |
| Shortest path | read position | 13 | 0.625 | n/a | n/a | n/a | n/a | 0.5 | n/a |
| Shortest path | before commit token (6) | 8 | 0.762 | 0.481 | +0.281 [0.145, 0.388] | 0.669 | 0.681 | 0.756 | yes (mechanical) |
| Shortest path | forking token (7) | 17 | 0.731 | 0.550 | +0.181 [0.046, 0.261] | 0.650 | 0.644 | 0.675 | no |
| Shortest path | after commit token (7) | 10 | 0.762 | 0.456 | +0.306 [0.12, 0.463] | 0.656 | 0.675 | 0.762 | yes |
| Shortest path | token 8 | 8 | 0.831 | 0.481 | +0.350 [0.182, 0.525] | 0.663 | 0.682 | 0.831 | yes |
| Shortest path | token 16 | 8 | 0.837 | 0.481 | +0.356 [0.193, 0.525] | 0.650 | 0.650 | 0.837 | yes |
| Shortest path | token 32 | 15 | 0.837 | 0.550 | +0.287 [0.122, 0.399] | 0.675 | 0.669 | 0.831 | yes |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Shortest path | token | 64 | 8 | 0.831 | 0.481 | +0.350 [0.182, 0.525] | 0.669 | 0.662 | 0.819 | yes |

*[Figure fig_locus_text: generated in the report phase, not in this draft]*

**What the locus probe showed.** On the grid task no position passes the rule, so the mechanical locus stays at the read position. The read position already separates the prompts that will produce DFS from those that will produce BFS (balanced accuracy 0.855 at layer 16), but at every generated position the within paraphrase null reaches the observed accuracy: the signal is the paraphrase's base rate, not anything specific to a sample. BFS and DFS programs also share their commit token (`' deque'`; 51 of 78 DFS programs and all 21 BFS programs commit at token 6), so the commit index does not separate the classes on this task. On the shortest path task the first passing position is the token before the commit token (layer 8, +0.281 over the read position, within paraphrase $p = 0.005$; a second implementation with 40 sklearn refits gave $p = 0.024$ against a placebo p of 0.17). The written text control reaches 0.756 at the same position, so the state adds 0.006 over the text, below the preregistered 0.10 margin. The audit therefore qualified the shortest path locus as a readout after commitment. The text separates the classes from generated token 4 on: Dijkstra programs write `import heapq` and Bellman-Ford/SPFA programs write `from collections import deque`. The deciding token is generated token 3, emitted from states at tokens 1 to 3 that were never saved, and the forking token experiment had independently confirmed a forking token at that index (`import`, total variation 0.80; Table 19). The first token at which the text alone reaches 0.8 is token 8. No decision locus was found on either task.

Three further readings came out of the same states. The difference-in-means direction at the token before the commit token has held-out AUC 0.882 (grid) and 0.749 (shortest path), below the 0.9 bar that would reopen a generation time removal. Per paraphrase, the read position probe's held-out probability of the substitute tracks the share of the substitute that paraphrase produced (Spearman 0.63, exact $p = 0.056$, grid; 0.65, $p = 0.045$, shortest path; ten paraphrases each), a first sign that the read position knows how strongly a wording will be obeyed. The positive controls (baseline against ban states) were decoded at balanced accuracy close to 1.0.

### 2.3.2 Geometry under paraphrase resampling

The geometry experiment's stability figure of 0.985 came from resampling points on a fixed graph, which the hypothesis review had warned is uninformative. The audit rebuilt the principal component projection, the nearest neighbour graph and the centroids on 1,000 paraphrase level subsamples, for k in {5, 10, 15, 20} at every layer of the two gate bands (24 to 28 grid, 21 to 27 shortest path), after reproducing the committed distance matrices exactly (largest absolute difference 0.0000). Table 42 gives the $k = 10$ column.

**Table 42.** Nearest surviving centroid to the default algorithm under 1,000 paraphrase subsamples, $k = 10$ (source: `tables/geometry_stability.csv`; distances on full data).

| Task | Layer | P(DFS nearest) | P(unit weight Dijkstra nearest) | P(both are the two nearest) | Full data order | MDS stress-1 |
|---|---|---|---|---|---|---|
| Grid | 24 | 0.523 | 0.477 | 1.0 | unit weight Dijkstra > DFS > A* > IDDFS > bidirectional BFS | 0.202 |
| Grid | 25 | 0.880 | 0.120 | 1.0 | DFS > unit weight Dijkstra > A* > IDDFS > bidirectional BFS | 0.212 |
| Grid | 26 | 0.954 | 0.046 | 1.0 | same | 0.213 |
| Grid | 27 | 1.0 | 0.0 | 1.0 | same | 0.217 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Grid | 28 | 1.0 | 0.0 | 1.0 | same | | 0.230 |

| Task | Layer | P(Floyd-Warshall nearest) | P(Bellman-Ford/SPFA nearest) | P(Bellman-Ford/SPFA nearer than bidirectional Dijkstra) | Rank of Bellman-Ford/SPFA (full data) | Full data order | MDS stress-1 |
|---|---|---|---|---|---|---|---|
| Shortest path | 21 | 1.0 | 0.0 | 0.555 | 3 | Floyd-Warshall > bidirectional Dijkstra > Bellman-Ford/SPFA > Dial | 0.205 |
| Shortest path | 22 | 1.0 | 0.0 | 0.633 | 3 | same | 0.239 |
| Shortest path | 23 | 1.0 | 0.0 | 0.829 | 2 | Floyd-Warshall > Bellman-Ford/SPFA > bidirectional Dijkstra > Dial | 0.235 |
| Shortest path | 24 | 1.0 | 0.0 | 0.820 | 2 | same | 0.224 |
| Shortest path | 25 | 1.0 | 0.0 | 0.810 | 2 | same | 0.226 |
| Shortest path | 26 | 1.0 | 0.0 | 0.897 | 2 | same | 0.227 |
| Shortest path | 27 | 1.0 | 0.0 | 0.964 | 2 | same | 0.229 |

Over all four values of k the picture is the same. On the shortest path task Floyd-Warshall is the nearest survivor to Dijkstra in 100% of subsamples at all 28 (layer, k) settings, and Bellman-Ford/SPFA, the algorithm every ban landed on, is never nearest; its full data rank is 2 in 75% of settings and 3 in 25%. The geometry experiment's sensitivity matrix, which recomputes the two flagged centroids from correctly labelled forced samples only, puts bidirectional Dijkstra (40.91) closer to Dijkstra than Bellman-Ford/SPFA (42.18), with Floyd-Warshall at 25.49 and Dial at 58.01. On the grid task DFS is nearest to BFS at layers 25 to 28 (probability 0.87 to 1.00 over k) and unit weight Dijkstra overtakes it at layer 24 (DFS 0.52 to 0.60), while DFS and unit weight Dijkstra are always the two nearest. Neighbour orders agree across k and layer with mean Kendall tau 0.933 (grid) and 0.870 (shortest path). Centroids from name masked prompts give different orders (grid top-1 stability 0.18 to 0.815). The grid ban's landing on DFS and unit weight Dijkstra is therefore consistent with geodesic proximity and with typicality, since both are also second choices of the model, and the two are not separated on these events. On the shortest path task geodesic proximity predicts the wrong substitute.

### 2.3.3 The substitute, draw by draw

The audit recomputed the modal ban substitutes from the per-sample rows of every draw (Table 43). On the shortest path task the claim survives as written: all 272 compliant programs in the five draws were Bellman-Ford/SPFA. On the grid task DFS was the realised mode in four of five draws and took 116 of 128 compliant programs; in the geometry draw, the only one sampled with HF `generate` and the one with the highest banned share, 8 of 18 compliant programs were unit weight Dijkstra and the event had no clear mode.

**Table 43.** Compliant programs under the modal ban, per draw (source: `corrections.json`; Wilson 95% intervals for the substitute's share).

| Draw (engine) | Grid: DFS / compliant | Other compliant labels | DFS share [95%] | Shortest path: Bellman-Ford/SPFA / compliant | Share [95%] |
|---|---|---|---|---|---|
| Association (vLLM) | 38 / 38 | none | 1.0 [0.908, 1.000] | 86 / 86 | 1.0 [0.957, 1.000] |
| Geometry (HF) | 10 / 18 | unit weight Dijkstra 8 | 0.556 [0.337, 0.754] | 22 / 22 | 1.0 [0.851, 1.000] |
| Forking token (vLLM) | 40 / 41 | unit weight Dijkstra 1 | 0.976 [0.874, 0.996] | 80 / 80 | 1.0 [0.954, 1.000] |
| Decommitment (vLLM) | 16 / 18 | unit weight Dijkstra 2 | 0.889 [0.672, 0.969] | 38 / 38 | 1.0 [0.908, 1.000] |
| Instruction tuning (vLLM) | 12 / 13 | unit weight Dijkstra 1 | 0.923 [0.667, 0.986] | 46 / 46 | 1.0 [0.923, 1.000] |
| Pooled | 116 / 128 | unit weight Dijkstra 12 | 0.906 [0.843, 0.946]; draw cluster bootstrap [0.75, 0.98] | 272 / 272 | 1.0 [0.986, 1.000] |

### 2.3.4 The cumulative ban of BFS and DFS does not replicate

The three draws that banned BFS and DFS together disagree on everything the knob table depends on (Table 44). The decommitment draw spread its 16 compliant programs over two algorithms, the geometry draw put all 18 on one, and in the association draw the model reimplemented a banned mechanism 94 times out of 100. The pairs the review quoted from Table 37 (0.88 / 1.00, 0.96 / 0.99, 0.53 / 0.933) are each draw's pass rate followed by its reference pass rate.

**Table 44.** The cumulative ban of BFS and DFS in three draws (source: `corrections.json#cumulative_bfs_dfs_ban`).

| Draw | N | Banned share (mechanism) | Pass rate | Compliant programs | Compliant composition |
|---|---|---|---|---|---|
| Decommitment | 100 | 0.81 | 0.88 | 16 | unit weight Dijkstra 13, A* 3 |
| Geometry | 50 | 0.62 | 0.96 | 18 | unit weight Dijkstra 18 |
| Association | 100 | 0.94 | 0.53 | 5 | unit weight Dijkstra 4, A* 1 |

### 2.3.5 Every perturbation, side by side: the generic perturbation rival

The audit assembled one table of every condition in the six draws: 249 condition rows covering 20,270 programs, each with its pass cost against its own reference (Newcombe intervals [42]), the label composition among correct and among attempted programs, and the share of the second choice with a Wilson interval (`tables/perturbation_table.csv`). Tables 45 and 46 give the rows that bear on the generic perturbation rival. On the shortest path task every nonsemantic perturbation that produced at least 10 correct programs other than Dijkstra (8 of 8 such rows) had Bellman-Ford/SPFA as their majority, random directions included, and so did 3 of 4 semantic ones. Band random direction 2 of the instruction tuning experiment turned all 33 of its correct programs into Bellman-Ford/SPFA, against Dijkstra 100 of 100 in its control. On the grid task generic perturbations almost never left BFS: random directions broke programs (pass rates 0.41 and 0.68)

but every correct program stayed BFS, and only two rows other than bans with at least 10 correct programs other than the default exist, both mention twins (A* and DFS). On the shortest path task a ban and an arbitrary perturbation end in the same place; on the grid task only the ban reaches DFS at scale.

**Table 45.** Shortest path perturbations: label composition of correct programs and Bellman-Ford/SPFA share (source: `tables/perturbation_table.csv`).

| Draw | Perturbation | N | Pass (reference) | Correct programs | Bellman-Ford/SPFA share [95%] |
|---|---|---|---|---|---|
| Decommitment | random direction 1, c = 0.25 | 100 | 0.79 (0.95) | Dijkstra 53, Bellman-Ford/SPFA 26 | 0.329 [0.236, 0.439] |
| Decommitment | random direction 2, c = 0.25 | 100 | 0.87 (0.95) | Dijkstra 82, Bellman-Ford/SPFA 5 | 0.057 [0.025, 0.128] |
| Decommitment | decommitment direction, c = 0.25 | 100 | 0.78 (0.95) | Bellman-Ford/SPFA 54, Dijkstra 23, other 1 | 0.692 [0.583, 0.784] |
| Decommitment | decommitment direction, c = 0.375 | 100 | 0.67 (0.95) | Bellman-Ford/SPFA 47, Dijkstra 19, other 1 | 0.701 [0.583, 0.798] |
| Decommitment | decommitment direction, c = 0.5 | 100 | 0.37 (0.95) | Dijkstra 25, Bellman-Ford/SPFA 10, other 2 | 0.270 [0.154, 0.430] |
| Decommitment | breadth direction, c = 0.25 | 100 | 0.60 (0.95) | Bellman-Ford/SPFA 31, Dijkstra 28, Dial 1 | 0.516 [0.393, 0.638] |
| Decommitment | other task's direction, c = 0.25 | 50 | 0.72 (0.95) | Bellman-Ford/SPFA 28, Dijkstra 8 | 0.777 [0.619, 0.883] |
| Decommitment | variable name placebo, c = 0.25 | 100 | 0.66 (0.95) | Dijkstra 60, Dial 3, Bellman-Ford/SPFA 2, other 1 | 0.030 [0.008, 0.104] |
| Decommitment | enumeration prompt | 100 | 0.82 (0.885) | Dijkstra 80, Bellman-Ford/SPFA 2 | 0.024 [0.007, 0.085] |
| Instruction tuning | band random direction 1 | 100 | 0.00 (1.00) | none | n/a |
| Instruction tuning | band random direction 2 | 100 | 0.33 (1.00) | Bellman-Ford/SPFA 33 | 1.0 [0.896, 1.000] |
| Instruction tuning | random direction 1, read position only, c = 1.0 | 100 | 1.00 (1.00) | Dijkstra 90, Bellman-Ford/SPFA 9, other 1 | 0.090 [0.048, 0.162] |
| Instruction tuning | sharpening subtraction, layer 18, c = 0.5 | 50 | 0.98 (1.00) | Dijkstra 48, Bellman-Ford/SPFA 1 | 0.020 [0.004, 0.107] |
| Instruction tuning | sharpening subtraction, read position only, no cutoff | 20 | 0.20 (1.00) | Dijkstra 3, Bellman-Ford/SPFA 1 | 0.25 [0.046, 0.699] |
| Instruction tuning | base model, chat template | 300 | 0.30 (1.00) | Dijkstra 63, Bellman-Ford/SPFA 24, other 2, Floyd-Warshall 1 | 0.266 [0.186, 0.366] |
| Instruction tuning | base model, plain completion | 300 | 0.083 (1.00) | Bellman-Ford/SPFA 16, Dijkstra 8, Floyd-Warshall 1 | 0.64 [0.445, 0.798] |
| Instruction tuning | instruct model, plain completion | 100 | 0.72 (1.00) | Dijkstra 57, Floyd-Warshall 8, Bellman-Ford/SPFA 7 | 0.097 [0.048, 0.187] |

| | | | | | |
|---|---|---|---|---|---|
| Decoding | semantic seed | 100 | 0.92 (0.99) | Bellman-Ford/SPFA 61, Dijkstra 30, Floyd-Warshall 1 | 0.663 [0.562, 0.751] |
| Decoding | novelty instruction | 100 | 0.96 (0.99) | Dijkstra 82, Bellman-Ford/ SPFA 14 | 0.145 [0.089, 0.230] |
| Decoding | new algorithm instruction | 100 | 0.94 (0.99) | Dijkstra 90, Bellman-Ford/ SPFA 4 | 0.042 [0.017, 0.104] |
| Decoding | enumeration prompt | 100 | 0.75 (0.99) | Dijkstra 67, Bellman-Ford/ SPFA 8 | 0.106 [0.055, 0.197] |
| Decoding | T = 1.3, top-k = 10 | 100 | 1.00 (0.98) | Dijkstra 99, Bellman-Ford/ SPFA 1 | 0.010 [0.002, 0.054] |
| Association | listing control | 100 | 1.00 (1.00) | Dijkstra 97, Bellman-Ford/ SPFA 3 | 0.030 [0.010, 0.085] |
| Association | sham ban of Floyd-Warshall | 100 | 0.71 (1.00) | Dijkstra 68, Bellman-Ford/ SPFA 3 | 0.042 [0.014, 0.117] |
| All five draws | modal ban of Dijkstra, pooled | 500 | 0.87 to 1.00 | Bellman-Ford/SPFA 272 (compliant) | 1.0 [0.986, 1.000] |

The sham ban of Floyd-Warshall shows why compliant and attempted programs must be told apart: 23 of its 100 attempted programs were Bellman-Ford/SPFA (Table 8), but only 3 of those passed the tests.

**Table 46.** Grid perturbations: label composition of correct programs and DFS share (source: `tables/perturbation_table.csv`).

| Draw | Perturbation | N | Pass (reference) | Correct programs | DFS share [95%] |
|---|---|---|---|---|---|
| Instruction tuning | band random direction 1 | 100 | 0.41 (1.00) | BFS 41 | 0.0 [0.000, 0.086] |
| Instruction tuning | band random direction 2 | 100 | 0.68 (1.00) | BFS 68 | 0.0 [0.000, 0.053] |
| Instruction tuning | random direction 1, read position only, c = 1.0 | 100 | 1.00 (1.00) | BFS 100 | 0.0 [0.000, 0.037] |
| Decommitment | decommitment direction, c = 0.25 | 100 | 0.57 (1.00) | BFS 57 | 0.0 [0.000, 0.063] |
| Decoding | semantic seed | 100 | 0.93 (0.97) | BFS 91, bidirectional BFS 2 | 0.0 [0.000, 0.040] |
| Decoding | novelty instruction | 100 | 0.85 (0.97) | BFS 81, unit weight Dijkstra 2, DFS 1, A* 1 | 0.011 [0.002, 0.064] |
| Decoding | new algorithm instruction | 100 | 0.94 (0.97) | BFS 92, DFS 2 | 0.021 [0.006, 0.074] |
| Decoding | enumeration prompt | 100 | 0.99 (0.97) | BFS 99 | 0.0 [0.000, 0.037] |
| Decoding | T = 1.3, top-k = 10 | 100 | 0.95 (0.98) | BFS 94, DFS 1 | 0.010 [0.002, 0.057] |
| Instruction tuning | base model, chat template | 300 | 0.19 (1.00) | BFS 49, DFS 7, A* 1 | 0.122 [0.061, 0.232] |

| | | | | |
|---|---|---|---|---|
| Instruction tuning | base model, plain completion | 300 0.05 (1.00) | BFS 12, DFS 3 | 0.20 [0.070, 0.452] |
| Instruction tuning | instruct model, plain completion | 100 0.73 (1.00) | BFS 73 | 0.0 [0.000, 0.050] |
| Association | listing control | 100 0.92 (0.933) | BFS 92 | 0.0 [0.000, 0.040] |
| Association | sham ban of IDDFS | 100 0.80 (0.933) | BFS 76, DFS 4 | 0.05 [0.020, 0.122] |
| Association | sham ban of unit weight Dijkstra | 100 0.71 (0.933) | BFS 69, DFS 2 | 0.028 [0.008, 0.097] |
| All five draws | modal ban of BFS, pooled | 500 0.46 to 0.76 | DFS 116, unit weight Dijkstra 12 (compliant) | 0.906 [0.843, 0.946] |

### 2.3.6 The knob table on one statistic

The iteration-1 knob table mixed unrarefied Hill numbers, rarefied differences, and Hill numbers that counted correct programs of the banned algorithm. The audit recomputed every knob with one statistic: the Hill number of order 1 [9] over compliant programs only (banned labels excluded), rarefied to a common depth of 25 by 1,000 draws without replacement, with a 90% interval from 1,000 paraphrase cluster bootstrap resamples times 200 rarefaction draws; cells with fewer than 25 compliant programs are marked and never extrapolated. Pass points are measured against each condition's own reference. Generate then select is a ceiling row, because its diversity comes from an external random choice among model proposed approaches, which is the mechanism of SimpleStrat [43]. Table 47 gives the rows that matter (all 183 rows are in `tables/knob_table.csv`).

**Table 47.** One statistic knob table: rarefied Hill-1 over compliant programs at depth 25 (source: `tables/knob_table.csv`).

| Task | Knob (draw) | Class | Compliant programs | Composition | Hill-1 [90% CI] | Pass points lost |
|---|---|---|---|---|---|---|
| Grid | generate then select (decoding) | ceiling | 55 | BFS 26, A* 14, unit weight Dijkstra 9, DFS 6 | 3.358 [3.315, 3.381] | 42.0 |
| Grid | mention of A* (association) | named mode cue | 41 | A* 24, BFS 11, unit weight Dijkstra 5, DFS 1 | 2.687 [1.861, 3.186] | 11.3 |
| Grid | base model, chat template (instruction tuning) | model swap | 57 | BFS 49, DFS 7, A* 1 | 1.549 [1.284, 1.814] | 81.0 |
| Grid | modal ban, five draws pooled | ban | 128 | DFS 116, unit weight Dijkstra 12 | 1.357 [1.250, 1.425] | 34.3 |
| Grid | novelty instruction (decoding) | diversity knob | 85 | BFS 81, unit weight Dijkstra 2, DFS 1, A* 1 | 1.227 [1.096, 1.346] | 12.0 |
| Grid | sham ban of IDDFS (association) | ban | 80 | BFS 76, DFS 4 | 1.205 [1.0, 1.542] | 13.3 |
| Grid | sham ban of unit weight Dijkstra (association) | ban | 71 | BFS 69, DFS 2 | 1.119 [1.0, 1.245] | 22.3 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Grid | modal ban (forking token) | ban | 41 | DFS 40, unit weight Dijkstra 1 | 1.111 [1.0, 1.243] | 35.0 |
| Grid | semantic seed (decoding) | diversity knob | 93 | BFS 91, bidirectional BFS 2 | 1.096 [1.0, 1.191] | 4.0 |
| Grid | new algorithm instruction (decoding) | diversity knob | 94 | BFS 92, DFS 2 | 1.093 [1.0, 1.187] | 3.0 |
| Grid | T = 1.3, top-k = 10 (decoding) | diversity knob | 95 | BFS 94, DFS 1 | 1.048 [1.0, 1.133] | 3.0 |
| Grid | modal ban (association) | ban | 38 | DFS 38 | 1.0 [1.0, 1.0] | 47.3 |
| Grid | cumulative ban of BFS and DFS (decommitment) | ban | 16 | unit weight Dijkstra 13, A* 3 | below depth (unrarefied 1.620) | 12.0 |
| Grid | cumulative ban of BFS and DFS (geometry) | ban | 18 | unit weight Dijkstra 18 | below depth (unrarefied 1.0) | 3.0 |
| Grid | cumulative ban of BFS and DFS (association) | ban | 5 | unit weight Dijkstra 4, A* 1 | below depth (unrarefied 1.649) | 40.3 |
| Grid | every steering, removal, forking window, random seed and remaining temperature arm | diversity knob | 3 to 307 | BFS only | 1.0, or below depth | 0 to 97 |
| Shortest path | generate then select (decoding) | ceiling | 75 | Bellman-Ford/SPFA 31, Dijkstra 29, Floyd-Warshall 15 | 2.802 [2.778, 2.812] | 24.0 |
| Shortest path | base model, plain completion (instruction tuning) | model swap | 25 | Bellman-Ford/SPFA 16, Dijkstra 8, Floyd-Warshall 1 | 2.179 [1.704, 2.611] | 91.7 |
| Shortest path | decommitment direction, c = 0.5 | diversity knob | 37 | Dijkstra 25, Bellman-Ford/SPFA 10, other 2 | 2.139 [1.553, 2.495] | 58.0 |
| Shortest path | breadth direction, c = 0.25 | diversity knob | 60 | Bellman-Ford/SPFA 31, Dijkstra 28, Dial 1 | 2.096 [1.920, 2.297] | 35.0 |
| Shortest path | base model, chat template | model swap | 90 | Dijkstra 63, Bellman-Ford/SPFA 24, other 2, Floyd-Warshall 1 | 2.018 [1.737, 2.262] | 70.0 |
| Shortest path | semantic seed (decoding) | diversity knob | 92 | Bellman-Ford/SPFA 61, Dijkstra 30, Floyd-Warshall 1 | 1.935 [1.534, 2.076] | 7.0 |
| Shortest path | decommitment direction, c = 0.25 | diversity knob | 78 | Bellman-Ford/SPFA 54, Dijkstra 23, other 1 | 1.911 [1.735, 2.069] | 17.0 |
| Shortest path | decommitment direction, c = 0.375 | diversity knob | 67 | Bellman-Ford/SPFA 47, Dijkstra 19, other 1 | 1.901 [1.675, 2.079] | 28.0 |
| Shortest path | random direction 1, c = 0.25 (decommitment) | diversity knob | 79 | Dijkstra 53, Bellman-Ford/SPFA 26 | 1.861 [1.657, 1.964] | 16.0 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Shortest path | random directions, four arms pooled | diversity knob | 199 | Dijkstra 135, Bellman-Ford/SPFA 64 | 1.851 [1.774, 1.903] | 47.75 |
| Shortest path | other task's direction, c = 0.25 | diversity knob | 36 | Bellman-Ford/SPFA 28, Dijkstra 8 | 1.688 [1.412, 1.857] | 23.0 |
| Shortest path | novelty instruction (decoding) | diversity knob | 96 | Dijkstra 82, Bellman-Ford/SPFA 14 | 1.500 [1.275, 1.692] | 3.0 |
| Shortest path | variable name placebo, c = 0.25 | diversity knob | 66 | Dijkstra 60, Dial 3, Bellman-Ford/SPFA 2, other 1 | 1.451 [1.129, 1.829] | 29.0 |
| Shortest path | random direction 1, read position only (instruction tuning) | diversity knob | 100 | Dijkstra 90, Bellman-Ford/SPFA 9, other 1 | 1.393 [1.209, 1.593] | 0.0 |
| Shortest path | enumeration prompt (decoding) | diversity knob | 75 | Dijkstra 67, Bellman-Ford/SPFA 8 | 1.382 [1.217, 1.581] | 24.0 |
| Shortest path | random direction 2, c = 0.25 (decommitment) | diversity knob | 87 | Dijkstra 82, Bellman-Ford/SPFA 5 | 1.231 [1.097, 1.371] | 8.0 |
| Shortest path | modal ban, five draws pooled | ban | 272 | Bellman-Ford/SPFA 272 | 1.0 [1.0, 1.0] | 2.4 |

**The positive target and the BUY rule.** The strategy's positive target was a knob that buys a second correct algorithm, a Hill number of at least 2 within its own compliant output, at a cost below 20 pass points. Under the single statistic no diversity knob and no ban meets it on either task. The closest is the semantic seed on the shortest path task (1.935 at 7 points). The mention of A* on the grid task reaches 2.69 at 11 points, but a mention recommends a named algorithm, so it meets only the literal reading of the target. The modal ban, the knob that iteration 1 ranked first on the shortest path task, scores exactly 1.0 there once banned programs no longer count: it replaces Dijkstra with one algorithm. The BUY rule names a best knob only when its 90% interval separates from the next; on the grid task the top rows (novelty instruction 1.227 [1.096, 1.346], sham ban of IDDFS 1.205 [1.0, 1.542]) do not separate, and on the shortest path task neither do the top five (Table 47). No knob advances. *(Correction, iteration 3: the sentences above paraphrase the target past its metric. A rarefied Hill-1 of at least 2 is an even two-algorithm split, which a 2:1 split never reaches. Stated in its own metric: no knob reaches an even split under 20 pass points; the semantic seed (Bellman-Ford/SPFA 61, Dijkstra 30 at 7 points; rebuilt Hill-1 1.940, 90% interval [1.770, 2.094], which includes 2) and the A* mention (2.69 at 11.3 points) are the two conditions that buy a second correct algorithm, and both name or describe it. The sham ban of IDDFS ranked above is a control, not a knob; the class rule is now declared in Section 3.2.5.)* *(Correction, iteration 4: the iteration-3 correction is itself incomplete. Two activation rows of Table 91 also buy a second correct algorithm without naming or describing one: the decommitment direction at c = 0.25 (Bellman-Ford/SPFA 54, Dijkstra 23; Hill-1 1.913, 90% interval [1.704, 2.083], which includes 2, at 17 points) and random direction 1 of the decommitment experiment (Dijkstra 53, Bellman-Ford/SPFA 26; 1.86 at 16 points). On the bench, norm-matched random directions written into the task-specification states in iteration 3 reach Hill-1 above 2 at 0 pass points on kth smallest (2.92) and two sum (2.12). The verdict per task over every iteration-3 arm is in Section 4.2.1.)*

### 2.3.7 The forecast contest on gated events

The rule, quoted in full from the strategy, is that a forecasting family advances if its calibrated mean log score beats the next best family by at least 0.10 nats on at least three events where the families' top-1 nominations disagree, scored on informative events only: banned mode present at

10% or more of the latent baseline, compliance of at least 50% and at least two survivors. The hypothesis update fixed compliance in this rule to mean avoidance and added a floor of 20 attempted programs not banned. Of the 64 events with forecasts in the iteration-1 record, 24 were sampled (Table 48). Three informative events pass the gate: the association draw's two modal bans and the decommitment draw's grid modal ban, whose forecasts are all tied. The gate as written also admits the association draw's nine sham bans (12 events in all), but they land on the default by construction and their own artifact marked them uninformative. The geometry, instruction tuning and decommitment shortest path events fail the gate under either compliance definition, and the forking token draw made no family forecasts. On the gated events the families' mean log scores are: association map (Borda) -0.844 (2 events), free-form association -0.145 (2), listing control typicality -0.120 (2), sampled frequency -1.579 (3), decommitted distribution -1.674 (1) and uniform -1.674 (1). Only two gated events have disagreeing nominations, and the largest margin on them is 0.05 nats. No family can advance, so the iteration-1 ranking of forecasts in Table 36 rests largely on events the rule excludes.

**Table 48.** Every sampled iteration-1 event with forecasts, with both gates and the log scores of the main families (nats per attempted program not banned; 5% uniform mixing and a 2% floor for other labels in every family; top-1 nomination in brackets). Source: `results/tables.json#predict_gated`.

| Draw | Event | N | Attempted, not banned | Avoidance share | Compliant share | Avoidance gate | Compliant gate | Log scores [top-1] |
|---|---|---|---|---|---|---|---|---|
| Association | grid ban A* (sham) | 100 | 100 | 1.0 | 0.72 | pass | pass | Borda -1.022 [BFS]; free-form -0.101 [BFS]; frequency -0.101 [BFS]; listing -0.131 [BFS] |
| Association | grid ban BFS (modal) | 100 | 90 | 0.90 | 0.38 | pass | fail | Borda -2.710 [unit weight Dijkstra]; free-form -0.152 [DFS]; frequency -1.630 [tie]; listing -0.101 [DFS] |
| Association | grid ban bidirectional BFS (sham) | 100 | 100 | 1.0 | 0.92 | pass | pass | Borda -1.063 [BFS]; free-form -1.141 [A*]; frequency -0.140 [BFS]; listing -0.169 [BFS] |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Association | grid ban DFS (sham) | 100 | 96 | 0.96 | 0.86 | pass | pass | Borda -0.998; free-form -0.101; frequency -0.101; listing -0.101 [all BFS] |
| Association | grid ban unit weight Dijkstra (sham) | 100 | 100 | 1.0 | 0.71 | pass | pass | Borda -1.113; free-form -0.410; frequency -0.410; listing -0.368 [all BFS] |
| Association | grid ban IDDFS (sham) | 100 | 100 | 1.0 | 0.80 | pass | pass | Borda -1.114; free-form -0.511; frequency -0.564; listing -0.486 [all BFS] |
| Association | grid ban BFS and DFS | 100 | 6 | 0.06 | 0.05 | fail | fail | Borda -1.398 [bidirectional BFS]; others -1.406 [tie] |
| Association | shortest path ban Bellman-Ford/SPFA (sham) | 100 | 95 | 0.95 | 0.92 | pass | pass | Borda -0.926; free-form, frequency, listing -0.136 [all Dijkstra] |
| Association | shortest path ban bidirectional Dijkstra (sham) | 100 | 100 | 1.0 | 1.0 | pass | pass | Borda -0.791; free-form -0.205; frequency -0.205; listing -0.212 [Dijkstra] |
| Association | shortest path ban Dial (sham) | 100 | 100 | 1.0 | 0.97 | pass | pass | Borda -0.803; free-form -0.278; frequency -0.278; listing -0.269 [Dijkstra] |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Association | shortest path ban Dijkstra (modal) | 100 | 91 | 0.91 | 0.86 | pass | pass | Borda -0.813 [Bellman-Ford/SPFA]; free-form -0.138 [Bellman-Ford/SPFA]; frequency -1.434 [tie]; listing -0.138 [Bellman-Ford/SPFA] |
| Association | shortest path ban Floyd-Warshall (sham) | 100 | 100 | 1.0 | 0.71 | pass | pass | Borda -0.911; free-form -0.933; frequency -0.933; listing -0.784 [Dijkstra] |
| Association | shortest path ban Dijkstra and Bellman-Ford/SPFA | 100 | 0 | 0.0 | 0.0 | fail | fail | not scorable |
| Geometry | grid ban BFS | 100 | 38 | 0.38 | 0.18 | fail | fail | association -3.543 [A*]; frequency -1.630 [tie]; geodesic -1.337 [DFS]; decoder -1.287 [DFS] |
| Geometry | grid ban BFS and DFS | 50 | 19 | 0.38 | 0.36 | fail | fail | association -0.998 [unit weight Dijkstra]; geodesic -1.073 [unit weight Dijkstra]; decoder -4.124 [IDDFS] |
| Geometry | grid ban BFS and unit weight Dijkstra | 50 | 17 | 0.34 | 0.10 | fail | fail | association -3.966 [A*]; geodesic -1.280 [A*]; decoder -1.372 [IDDFS] |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Geometry | shortest path ban Dijkstra | 100 | 28 | 0.28 | 0.22 | fail | fail | association -1.647 [Dial]; frequency -1.496 [tie]; geodesic -1.530 [Floyd-Warshall]; decoder -1.587 [bidirectional Dijkstra] |
| Geometry | shortest path ban Dijkstra and Bellman-Ford/SPFA | 50 | 1 | 0.02 | 0.0 | fail | fail | all -3.912 |
| Decommitment | grid ban BFS | 100 | 51 | 0.51 | 0.18 | pass | fail | every family -1.674 [tie] |
| Decommitment | grid ban BFS and DFS | 100 | 19 | 0.19 | 0.16 | fail | fail | every family -1.538 [tie] |
| Decommitment | shortest path ban Dijkstra | 100 | 46 | 0.46 | 0.38 | fail | fail | decommitted -0.594 [Bellman-Ford/SPFA]; frequency (vLLM) -0.731 [Bellman-Ford/SPFA]; uniform -1.733 |
| Decommitment | shortest path ban Dijkstra and Bellman-Ford/SPFA | 100 | 17 | 0.17 | 0.01 | fail | fail | all -3.912 [tie] |
| Instruction tuning | grid ban BFS | 100 | 38 | 0.38 | 0.13 | fail | fail | base plain completion -0.430 [DFS]; base chat -0.304 [DFS]; instruct chat -1.630 [tie] |
| Instruction tuning | shortest path ban Dijkstra | 100 | 46 | 0.46 | 0.46 | fail | fail | base plain completion -0.255; base chat -0.096; instruct chat -0.096 [all Bellman-Ford/SPFA] |

These scores are computed over attempted programs not banned, so they differ slightly from the

compliant only scores of Tables 7, 14 and 36.

### 2.3.8 Seeds at the read position: both halves of the dissociation

Iteration 1 reported that the grid semantic seed moved the read position state toward its implied algorithm while the output barely moved. The other half was in the same file and was not reported (Table 49). On the shortest path task the semantic seed's displacement did not point toward the Bellman-Ford/SPFA centroid, yet the output moved two thirds of the way to it. The word seed moved the grid probe's decoded distribution toward A* by 0.88 with no change in behaviour. The entropy seed's interval on the shortest path task reaches 1.596, above the 1.5 equivalence bound, so its equivalence with paraphrase noise was not shown. Displacement along the mode subspace built from forced prompts predicted behavioural movement in neither direction.

**Table 49.** Read position displacement against behavioural movement for the seeds (sources: decoding experiment `results/analysis.json`, audit `results/tables.json#seeds`; behaviour shifts with Newcombe 95% intervals).

| Task | Seed | Read position evidence | Behaviour among correct programs | Behaviour among attempted programs |
|---|---|---|---|---|
| Grid | semantic (implies bidirectional search) | cosine 0.493 with the bidirectional BFS centroid | bidirectional BFS +2.15 points [-1.97, 7.51]; BFS -1.12 | bidirectional BFS +5.0 points [0.33, 11.18]; BFS -4.0 |
| Shortest path | semantic (implies relaxation over all edges) | cosine -0.087 with the Bellman-Ford/SPFA centroid | Bellman-Ford/SPFA +66.30 points [55.50, 75.14]; Dijkstra -67.39 | Bellman-Ford/SPFA +61.0 points [50.53, 69.98]; Dijkstra -62.0 |
| Grid | word | ratio 1.555 [1.309, 1.85] of paraphrase noise; probe decoded shift +0.88 toward A* | BFS +1.03 points; no program other than BFS | BFS +1.0 point |
| Shortest path | entropy | ratio 1.165 [1.0, 1.596]; upper bound above the 1.5 bound | Dijkstra share unchanged | unchanged |

### 2.3.9 Sham bans as a weak test of negation triggered promotion

The audit compared each survivor's share under each of the nine bans of absent algorithms with its share in the association draw's baseline (N = 300), and called a survivor promoted if it gained at least 10 points with Fisher p below 0.05. Two promotions passed: banning IDDFS raised DFS by 12 points (p = 3.5e-8) and banning Floyd-Warshall raised Bellman-Ford/SPFA by 23 points (p = 1.7e-15). Two smaller shifts fell below the cut: banning unit weight Dijkstra raised DFS by 8 points (p = 1.2e-5) and banning Dial raised Bellman-Ford/SPFA by 5 points (p = 0.0009). In both promotions the gaining algorithm is a content neighbour of the banned one and also the second choice of the model, so the test cannot say whether the negation promoted an associate or merely disturbed the default toward the ladder's next rung. Shuffling the condition labels gave no promotion, as it should.

### 2.3.10 Executed outputs missing from the iteration-1 record, and bookkeeping

**Forced conditions of the association experiment.** Table 6 was captioned as every condition, but the experiment also ran eleven forced implementation conditions of 10 samples each (Table 50). They match the geometry experiment's manipulation check (Table 11): the model cannot write a passing IDDFS on the grid task or a Dial on the shortest path task, and asked for bidirectional

Dijkstra it writes plain Dijkstra every time.

**Table 50.** Forced implementations in the association experiment (N = 10 each; labels of attempted programs; source: association experiment `data/samples.jsonl`, conditions `forced:*`).

| Task | Requested algorithm | Pass | Labels |
|---|---|---|---|
| Grid | A* | 10/10 | A* 9, unit weight Dijkstra 1 |
| Grid | BFS | 9/10 | BFS 10 |
| Grid | bidirectional BFS | 4/10 | BFS 5, bidirectional BFS 5 |
| Grid | DFS | 3/10 | DFS 10 |
| Grid | unit weight Dijkstra | 10/10 | unit weight Dijkstra 8, BFS 1, DFS 1 |
| Grid | IDDFS | 0/10 | IDDFS 10 |
| Shortest path | Bellman-Ford/SPFA | 10/10 | Bellman-Ford/SPFA 10 |
| Shortest path | bidirectional Dijkstra | 4/10 | Dijkstra 10 |
| Shortest path | Dial | 0/10 | other 5, Dial 4, Dijkstra 1 |
| Shortest path | Dijkstra | 10/10 | Dijkstra 10 |
| Shortest path | Floyd-Warshall | 10/10 | Floyd-Warshall 10 |

**Other executed outputs.** The association experiment's robustness block held an in-sample, sharpness matched sensitivity that the report omitted: with one sharpening power per family fitted on the two events, the Borda association map scored -1.386 and sampled frequency -1.475, the first rank map -0.888, the external Llama map -0.437, and the free-form answer and listing control both -0.059. These are in-sample numbers, not calibrated out-of-sample scores. In the forking token experiment, the next token entropy at the 13 forking token candidates averaged 0.527 bits (maximum 1.494) against 0.017 bits (maximum 0.130) at the 10 control positions. Its change point detector, tested on planted changes, caught a total variation change of 0.2, 0.3 and 0.5 at the right position in 82.5%, 95% and 99% of trials with a 6% false positive rate, and the grid forking token candidate on ban path 0 at token 9 confirms at a total variation threshold of 0.10 and 0.15 but not at the preregistered 0.20. Its engine test found greedy prefixes identical in only 14% of runs at 10 tokens and in all runs at 40 tokens. The decommitment experiment's arms at $c = 1$ in layers 14 and 22 passed 0 of 10 on both tasks. The geometry experiment's Procrustes disparity between the named and name masked centroid sets was 0.18 (grid) and 0.25 (shortest path), and it dropped two preregistered components under its drop order: the external model association elicitation and the distribution decoder that holds out one condition at a time. In the decoding experiment's enumeration arm on the shortest path task, the model wrote Bellman-Ford/SPFA 33 times but only 8 of those passed.

**GPU hours.** Iteration 1 recorded GPU time for two experiments. The audit took the first and last timestamp of every GPU phase log and summed the union of the spans (an upper bound that includes waits): geometry 2.92 GPU hours, forking token 1.55, decommitment 3.23, instruction tuning 2.47 and decoding 3.31. The association experiment has no GPU phase log; its own report gave about 6 minutes.

**Grid failure taxonomy.** Because a compliant program must be correct, low grid compliance could reflect test bias rather than broken programs. Of the 273 DFS programs in the association draw, 161 failed: 127 wrong answers, 31 timeouts and 3 runtime errors. The review's own re-run showed that reading a returned None as an empty path rescues only 2 to 3 programs per condition (under the BFS ban, DFS correct programs rise from 38 to 40); the audit could not repeat it because the stored failure fields do not hold returned values. Table 51 gives the failure kinds per grid condition for the draws that stored them.

**Table 51.** Failing grid programs by kind (source: `results/tables.json#missing_outputs.grid_failure_taxonomy`; the forking token and decommitment draws did not store failure kinds).

| Draw | Condition | Failing | Wrong answer | Timeout | Runtime error |
|---|---|---|---|---|---|
| Association | baseline | 20 | 15 | 4 | 1 |
| Association | baseline, T = 1.3 | 21 | 17 | 3 | 1 |
| Association | ban BFS | 54 | 44 | 8 | 2 |
| Association | ban DFS | 12 | 11 | 1 | 0 |
| Association | ban bidirectional BFS | 8 | 8 | 0 | 0 |
| Association | ban IDDFS | 20 | 19 | 1 | 0 |
| Association | ban A* | 28 | 23 | 4 | 1 |
| Association | ban unit weight Dijkstra | 29 | 20 | 8 | 1 |
| Association | ban BFS and DFS | 47 | 40 | 6 | 1 |
| Association | listing control | 8 | 8 | 0 | 0 |
| Association | mention DFS | 35 | 20 | 15 | 0 |
| Association | mention IDDFS | 41 | 25 | 16 | 0 |
| Association | mention bidirectional BFS | 14 | 10 | 1 | 3 |
| Association | forced DFS | 7 | 5 | 2 | 0 |
| Association | forced IDDFS | 10 | 3 | 7 | 0 |
| Geometry | ban BFS | 24 | 9 | 14 | 1 |
| Geometry | ban BFS and unit weight Dijkstra | 14 | 5 | 9 | 0 |
| Geometry | forced DFS | 21 | 3 | 18 | 0 |
| Geometry | forced IDDFS | 30 | 6 | 23 | 1 |
| Instruction tuning | instruct, chat template, ban BFS | 30 | 10 | 19 | 1 |
| Instruction tuning | base model, plain completion | 277 | 151 | 27 | 99 |
| Decoding | generate then select | 45 | 15 | 27 | 2 |
| Decoding | novelty instruction | 15 | 5 | 7 | 1 |

**Base model caveats.** The instruction tuning experiment's base arms rest on few programs and weak labels. Under the chat template the base model produced parseable code in 0.583 [0.527, 0.638] of grid samples and 0.663 [0.608, 0.714] of shortest path samples, 57 and 90 correct programs out of 300, with runtime against AST kappa of 0.417 and 0.567 on those arms. In plain completion its correct shortest path programs were Bellman-Ford/SPFA 16, Dijkstra 8 and Floyd-Warshall 1, so its mode there is Bellman-Ford/SPFA, first rather than second. The preregistered verdicts stand as recorded: on the correctness conditioned Hill number, MIXED on the grid task (modal share gap 14.04 points against a 15-point bar) and INSTALLED_BY_POST_TRAINING on the shortest path task; INSTALLED_BY_POST_TRAINING on both under the Hill number over all programs; MIXED on both at the planned sample size.

### 2.3.11 Positioning against the nearest published work

The audit checked each positive claim against its nearest neighbours on the arXiv abstract pages on 2026-10-20 and recorded what this run adds (Table 52). All eight arXiv identifiers were verified. The venues of two papers (ICLR 2025 for the paper on introspection, NAACL 2025 for denial prompting) are not shown on their abstract pages and are marked unverified there.

**Table 52.** Each positive claim, its nearest neighbours and what this run adds (source: `results/positioning.json`).

| Claim | Nearest neighbours | What this run adds, or replication |
|---|---|---|
| Temperature and the sampler change phrasing, not the algorithm | Temperature correlates weakly with novelty [37]; NoveltyBench [44]; approach level against surface level diversity in math reasoning [45], whose paper uses temperature only as a sampling setting | An executed, mechanism labelled measurement of algorithm choice in code across temperature, sampler and thinking mode: no change in algorithm beside a severalfold rise in phrasings. None of the three neighbours reports this measurement in code |
| Generate then select reaches several algorithms at a pass rate cost | SimpleStrat [43] | Replication in code algorithm space with the pass rate price; a ceiling row, not a knob |
| The substitute is predictable before the ban from the model's own answers | Verbalized Sampling attributes mode collapse to typicality bias [46]; introspection and simulation of own behaviour [39] | Forecasts of the realised substitute under prohibitions in code; descriptive, and no family advances under the rule |
| Internal states are organised by approach | Math reasoning organised by approach, not topic [40] | Code algorithms, prohibitions and a test of the generation time locus; the locus narrows to a paraphrase level signal (grid) and a readout after commitment (shortest path) |
| Bans recollapse on Bellman-Ford/SPFA | The inspiration repository (the specific recollapse statement was not found in its README); denial prompting [23] | Per draw counts across five independent draws with mechanism labels on an open model, and the generic perturbation rival on the same task |

## 2.4 Likelihood screen: does a ban substitute, or expose a latent ladder? (Qwen3-8B)

**Why.** This experiment carried the title claim. On the two path search tasks it had to validate the likelihood instrument against events that iteration 1 had already sampled, then measure the full single ban gain matrices (6 by 5 on the grid task, 5 by 4 on the shortest path task) under the five templates and decompose them into a ladder share and an X specific share.

**Design.** The model wrote 1,170 forced programs (10 forced paraphrases per algorithm, three adaptive vLLM rounds), labelled with the canonical labeller, name scrubbed and reverified (Table 53). Scoring ran in HF transformers with bf16 weights but fp32 activations and TensorFloat-32 matrix products, because pure bf16 moved per program log likelihoods by up to 2.1 nats between equivalent computation paths; after the change the cached and full paths agreed within 0.074 nats per program. Throughput on that GPU (1,046 tokens per second) projected 242 minutes for 30 programs per algorithm against a 150-minute budget, so the preregistered cut rule fixed $M = 20$ programs per algorithm in two halves of 10, and the gate used three baseline paraphrases. In total 684 scoring units ran (one unit is one prompt condition scored on one program set) and 57 were skipped at the GPU deadline. Free-form association answers were sampled 20 times per algorithm; 7% (grid) and 47% (shortest path) were cut at the frozen 32-token limit, so the association profile is weak.

**Table 53.** Forced programs per algorithm (source: `results/yields.json`).

| Task | Algorithm | Forced family | Attempts | Labelled as requested | Yield | Verified after scrub | Scrub mode |
|---|---|---|---|---|---|---|---|
| Grid | BFS | plain | 40 | 40 | 1.00 | 40 | full |
| Grid | DFS | plain | 40 | 40 | 1.00 | 40 | full |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Grid | bidirectional BFS | plain | 120 | 30 | 0.25 | 30 | full |
| Grid | IDDFS | plain | 40 | 40 | 1.00 | 40 | full |
| Grid | A* | plain | 90 | 32 | 0.36 | 32 | full |
| Grid | unit weight Dijkstra | descriptive | 120 | 92 | 0.77 | 92 | full |
| Shortest path | Dijkstra | plain | 40 | 40 | 1.00 | 40 | full |
| Shortest path | Bellman-Ford/ SPFA | plain | 40 | 40 | 1.00 | 40 | full |
| Shortest path | bidirectional Dijkstra | descriptive | 120 | 14 | 0.12 | 11 | comments only (71% of full scrubs failed) |
| Shortest path | Floyd-Warshall | plain | 40 | 40 | 1.00 | 35 | full |
| Shortest path | Dial | descriptive | 120 | 106 | 0.88 | 106 | full |

**The validity gate failed on calibration.** The gate asked three things of the likelihood instrument under the association experiment's own prompts: that the implied probabilities sum to between 0.3 and 3 with at least 0.9 of the mass on the sampled default (calibration), that the implied distribution beat uniform by at least 0.3 nats per sampled program on 19 sampled events (discrimination), and that its argmax match the realised mode on at least 80% of events with a clear mode. Discrimination passed (0.92 nats) and so did the argmax (16 of 19). Calibration failed: the instrument gave DFS a large share of the grid baseline (default share 0.83 on average and 0.44 on one paraphrase) although the association experiment sampled DFS 0 times in 300. Of the 303 gate cells, 298 had an effective sample size below 5, so one or two programs dominated each estimate. The diagnosis is name scrubbing: removing `bfs` and `dfs` identifiers and comments made a forced program much less likely under its own forced prompts than under a neutral prompt, which inflated the weights of minority algorithms. Scored on the unscrubbed programs, with their own proposal denominators, the instrument passed all three criteria (Table 55). By the frozen rule the scrubbed instrument is primary, so every likelihood result of this experiment is descriptive.

**Table 54.** The 19 gate events: realised mode against the likelihood instrument's implied distribution (source: `results/gate.json`; d is the log score over uniform in nats per sampled program).

| Event | Realised mode (allowed labels) | Implied argmax | Hit | d (nats) | Implied shares above 0.01 |
|---|---|---|---|---|---|
| grid baseline | BFS | BFS | yes | 1.31 | BFS 0.66, DFS 0.34 |
| grid ban BFS | DFS | DFS | yes | 1.57 | DFS 1.00 |
| grid mention BFS | BFS | BFS | yes | 1.75 | BFS 1.00 |
| grid mention DFS | DFS | DFS | yes | 1.74 | DFS 1.00 |
| grid mention bidirectional BFS | BFS | bidirectional BFS | no | -1.83 | bidirectional BFS 1.00 |
| grid mention IDDFS | IDDFS | IDDFS | yes | 1.12 | IDDFS 1.00 |
| grid mention A* | A* | BFS | no | -1.75 | BFS 0.78, A* 0.22 |
| grid mention unit weight Dijkstra | BFS | BFS | yes | 1.64 | BFS 0.90, DFS 0.10 |
| grid listing control | BFS | BFS | yes | 1.37 | BFS 0.76, DFS 0.24 |
| grid ban IDDFS | BFS | DFS | no | -2.42 | DFS 0.99, BFS 0.01 |

| | | | | |
|---|---|---|---|---|
| shortest path baseline | Dijkstra | Dijkstra | yes 1.57 | Dijkstra 1.00 |
| shortest path ban Dijkstra | Bellman-Ford/SPFA | Bellman-Ford/ SPFA | yes 1.35 | Bellman-Ford/SPFA 1.00 |
| shortest path mention Dijkstra | Dijkstra | Dijkstra | yes 1.57 | Dijkstra 1.00 |
| shortest path mention Bellman-Ford/SPFA | Bellman-Ford/SPFA | Bellman-Ford/ SPFA | yes 1.57 | Bellman-Ford/SPFA 1.00 |
| shortest path mention bidirectional Dijkstra | Dijkstra | Dijkstra | yes 1.57 | Dijkstra 1.00 |
| shortest path mention Floyd-Warshall | Floyd-Warshall | Floyd-Warshall | yes 1.57 | Floyd-Warshall 1.00 |
| shortest path mention Dial | Dijkstra | Dijkstra | yes 1.57 | Dijkstra 1.00 |
| shortest path listing control | Dijkstra | Dijkstra | yes 1.57 | Dijkstra 1.00 |
| shortest path ban Floyd-Warshall | Dijkstra | Dijkstra | yes 0.64 | Dijkstra 0.98, Bellman-Ford/SPFA 0.02 |

Per baseline paraphrase, the calibration sum S and default share m were 1.55 and 0.70, 2.56 and 0.44, and 1.23 and 0.84 on the grid task, and 0.20 and 1.00, 0.36 and 1.00, and 0.20 and 1.00 on the shortest path task.

**Table 55.** Gate under sensitivity variants (descriptive; the primary verdict is not rechosen; source: `results/gate.json`).

| Variant | Mean S | Mean m | Calibration | Mean d (nats) | Discrimination | Argmax rate | Argmax | All three |
|---|---|---|---|---|---|---|---|---|
| primary (scrubbed, mixture proposal) | 1.02 | 0.83 | fail | 0.92 | pass | 0.84 | pass | fail |
| code body only | 1.02 | 0.83 | fail | 0.92 | pass | 0.84 | pass | fail |
| correct programs only | 0.66 | 1.00 | pass | 0.76 | pass | 0.79 | fail | fail |
| single forced prompt as proposal | 0.99 | 0.94 | pass | 0.99 | pass | 0.84 | pass | pass |
| truncated to 256 tokens | 1.06 | 0.88 | fail | 0.92 | pass | 0.84 | pass | fail |
| unscrubbed | 0.95 | 1.00 | pass | 1.18 | pass | 0.89 | pass | pass |
| unscrubbed, body only, single proposal | 1.01 | 1.00 | pass | 1.11 | pass | 0.89 | pass | pass |

**The substitution test: undetermined.** The between survivor variance of the ban gains is far above replicate noise (F between 16 and 22 under the neutral reference), so the model does not merely subtract. But the pooled X specific share is 0.55 with a 95% interval of [0.25, 0.67] under the neutral twin reference and 0.66 [0.49, 0.87] under the mention twin: both intervals straddle 0.5 and no ladder exists, so the preregistered branch is undetermined, and the same holds under effective sample size weighting (Table 56). The grid task alone reads X specific but unaimed under the mention reference and the weighted variant. Neither rival account fits. The X specific residuals are not aimed at the prime and veto profile (mean Spearman -0.10 against a null 95th percentile of 0.20, $p = 0.80$) or at the free-form association profile (-0.12, $p = 0.77$). And the eight generic perturbation profiles do not agree with one another, so there is no coherent ladder to be exposed (Table 57). *(Correction, iteration 3: the three readings above, that the gains are not pure subtraction, that the residuals are not aimed, and that the generic profiles form no coherent ladder, are not detectable with the current instrument; they are not findings. On the rows whose behaviour is known the X specific residual points away from the realised substitute (Dijkstra ban: Bellman-Ford/SPFA -1.84 against Dial +6.07 and Floyd-Warshall +3.68; BFS ban: DFS +0.83 against IDDFS +7.99), the argmax of the gain*

*hits the realised landing in 0 of 17 sampled Qwen3-8B single bans while the argmax of the implied mass hits 14 of 17, no row keeps three survivors with at least 1% implied mass and an ESS of at least 5, and the bootstrap variance of the mean gain is a median 2.7 times the replicate noise that the frozen correction subtracts, which inflates both F and the share. Section 3.2.1.)*

**Table 56.** Substitution test: noise corrected ladder and X specific sums of squares and the X specific share (source: `results/c1.json`; 95% intervals by bootstrap over templates, programs and random directions).

| Reference | Scope | Rows | V_u (ladder) | V_s (X specific) | Share | Share 95% CI | V_u + V_s 95% CI | F | Branch |
|---|---|---|---|---|---|---|---|---|---|
| neutral twin (primary) | grid | 6 | 239.28 | 412.46 | 0.63 | [0.28, 0.83] | [501.86, 1505.29] | 15.91 | undetermined |
| neutral twin (primary) | shortest path | 5 | 244.64 | 173.12 | 0.41 | [0.10, 0.61] | [316.25, 947.02] | 22.48 | undetermined |
| neutral twin (primary) | pooled | 11 | 483.93 | 585.58 | 0.55 | [0.25, 0.67] | [924.80, 2106.68] | 17.93 | undetermined |
| mention twin | grid | 6 | 1282.94 | 2525.02 | 0.66 | [0.51, 0.98] | [2573.47, 5471.38] | 57.44 | X specific, unaimed |
| mention twin | shortest path | 5 | 522.29 | 979.57 | 0.65 | [0.26, 0.79] | [1182.96, 2115.91] | 37.69 | undetermined |
| mention twin | pooled | 11 | 1805.23 | 3504.59 | 0.66 | [0.49, 0.87] | [4045.59, 7138.92] | 49.98 | undetermined |
| ESS weighted | grid | 6 | -35.45 | 181.32 | 1.24 | [0.57, 1.62] | [116.32, 331.51] | 15.92 | X specific, unaimed |
| ESS weighted | shortest path | 5 | 48.41 | 50.61 | 0.51 | [-1.09, 2.40] | [74.07, 220.58] | 23.24 | undetermined |
| ESS weighted | pooled | 11 | 12.96 | 231.93 | 0.95 | [0.11, 1.70] | [218.16, 478.69] | 18.21 | undetermined |

*(Added in iteration 3: the ESS weighted grid share of 1.24 [0.57, 1.62] is impossible as a proportion. The noise corrected ladder term went negative (V_u = -35.4) once the weights shrank the ladder projection below its noise allowance, so V_s / (V_u + V_s) = 181.3 / 145.9 exceeds 1. The basic bootstrap intervals, which correct for the upward skew of these sums, are given in Section 3.2.1.)*

**Table 57.** Does a latent ladder exist? Agreement among the eight generic perturbation profiles and the aim of the X specific residual (source: `results/c1.json`).

| Task | Ladder exists (mean centred cosine at least 0.5) | Mean centred cosine | Mean raw cosine | Random edits, seed family A against B (r) | Nonce ban, half 1 against half 2 (r) | Aim at prime and veto | Aim p |
|---|---|---|---|---|---|---|---|
| Grid | no | 0.34 | 0.21 | 0.53 | 0.37 | -0.07 | 0.682 |
| Shortest path | no | -0.02 | -0.07 | -0.26 | 0.82 | -0.14 | 0.787 |

*(Added in iteration 3: Tables 56 and 57 omit the aim under the mention twin reference, -0.595 in Qwen3-8B. That aim is negative by construction: the mention twin enters the gain with a minus sign and the prime and veto profile with a plus sign, so shared noise in the mention cell drives the correlation negative. Simulated under the split half design the null has mean -0.27, and -0.595 lies below its 1st percentile. Section 3.2.1.)*

The per row decomposition (Table 58) shows where the X specific variance sits. The modal bans have large X specific residuals: under the BFS ban the largest residual belongs to IDDFS, under the

Dijkstra ban to Dial. The bans of DFS, Bellman-Ford/SPFA, Floyd-Warshall and Dial are mostly close to the ladder (cosine of the row with the ladder profile 0.87 to 0.96). The penultimate column measures each ban's effect on its own label, and it carries the experiment's clearest result: banning an algorithm the model never writes raises that algorithm's likelihood, by +2.5 to +26.9 nats in 8 of the 9 bans of absent algorithms (IDDFS +23.5, Floyd-Warshall +15.4, Dial +26.9). Only the two modal bans (BFS -8.25, Dijkstra -4.95) and the ban of A* (-5.93) lower their own label. Naming an algorithm primes it, even inside a prohibition [7].

**Table 58.** Per row decomposition under the neutral twin reference (source: `results/c1.json`; a_B^2 is the ladder part and ||s_B||^2 the X specific part of the row, N_u and N_s their noise expectations; D_B is the ban's effect on its own label, L_B(ban) - L_B(neutral), in nats; rho(s, PV) is the row's aim).

| Task | Banned | a_B^2 | N_u | s_B^2 | N_s | cos(row, u) | Replicate SD | Between survivor SD | D_B | rho(s, PV) | Largest residual |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Grid | BFS | 139.75 | 2.04 | 163.48 | 6.13 | 0.68 | 4.04 | 7.79 | -8.25 | -0.25 | IDDFS |
| Grid | DFS | 45.20 | 1.53 | 4.32 | 4.60 | 0.96 | 3.50 | 3.15 | 8.43 | -0.10 | IDDFS |
| Grid | bidirectional BFS | 1.03 | 2.04 | 33.27 | 6.13 | -0.17 | 4.04 | 2.62 | 15.20 | 0.75 | IDDFS |
| Grid | IDDFS | 0.47 | 1.98 | 36.72 | 5.95 | 0.11 | 3.98 | 2.73 | 23.53 | 0.25 | bidirectional BFS |
| Grid | A* | 24.13 | 1.97 | 130.22 | 5.92 | 0.40 | 3.97 | 5.56 | -5.93 | -0.50 | IDDFS |
| Grid | unit weight Dijkstra | 39.63 | 1.36 | 77.23 | 4.07 | 0.58 | 3.29 | 4.83 | 2.51 | -0.60 | IDDFS |
| Shortest path | Dijkstra | 9.02 | 0.92 | 116.45 | 1.84 | 0.27 | 2.63 | 5.60 | -4.95 | 0.00 | Dial |
| Shortest path | Bellman-Ford/SPFA | 119.68 | 1.42 | 27.77 | 2.84 | 0.90 | 3.26 | 6.07 | 10.59 | 0.30 | Floyd-Warshall |
| Shortest path | bidirectional Dijkstra | 41.38 | 2.02 | 33.55 | 4.04 | 0.74 | 3.89 | 4.33 | 3.54 | 0.20 | Dial |
| Shortest path | Floyd-Warshall | 73.89 | 1.37 | 5.98 | 2.74 | 0.96 | 3.21 | 4.47 | 15.40 | -0.60 | Dial |
| Shortest path | Dial | 7.16 | 0.75 | 2.33 | 1.51 | 0.87 | 2.38 | 1.54 | 26.88 | -0.60 | Bellman-Ford/SPFA |

**The iteration-1 sham observations are absorbed by the shared component.** In the Floyd-Warshall ban row the Bellman-Ford/SPFA gain (2.73 nats) equals its fitted ladder part (2.79); its X specific residual is -0.06 [-2.76, 3.67]. The IDDFS to DFS residual is -0.77 [-7.20, 4.02], with a mean gain of -0.45 against a fitted ladder part of 0.33. The two sham promotions of Section 2.3.9 therefore look like the ladder component, not like an X specific promotion.

**Sensitivity.** Every variant keeps the pooled branch undetermined except the gate passing unscrubbed programs (Table 59). Scored raw, with the first template only (the only one scored on raw programs), the pooled X specific share is 0.69 [0.51, 0.84], X specific but unaimed on the grid task and undetermined on the shortest path task; the matched scrubbed value is 0.55 [0.18, 0.78]. The aim cannot be tested in that variant, because mention twins were not scored on raw programs.

**Table 59.** Pooled X specific share under sensitivity variants (source: `results/sensitivity.json`).

| Variant | Pooled share | 95% CI | Pooled branch | Aim |
|---|---|---|---|---|
| code body only | 0.55 | [0.25, 0.68] | undetermined | -0.10 |

| | | | | |
|---|---|---|---|---|
| correct programs only | 0.42 | [0.18, 0.57] | undetermined | -0.17 |
| single forced prompt as proposal | 0.54 | [0.26, 0.68] | undetermined | -0.09 |
| truncated to 256 tokens | 0.45 | [0.25, 0.64] | undetermined | -0.11 |
| ladder profile with the iteration-1 steering setting added | 0.55 | [0.25, 0.68] | undetermined | -0.10 |
| ladder profile, unit normalised | 0.42 | [0.25, 0.66] | undetermined | -0.08 |
| ladder profile from the nonce ban only | 0.33 | [0.24, 0.77] | undetermined | 0.10 |
| ladder profile from random edits only | 0.64 | [0.37, 0.89] | undetermined | 0.04 |
| ladder profile from the novelty instruction only | 0.72 | [0.30, 0.81] | undetermined | -0.09 |
| unscrubbed, first template only | 0.69 | [0.51, 0.84] | X specific, unaimed | n/a |
| unscrubbed, body only, single proposal, first template | 0.67 | [0.51, 0.84] | X specific, unaimed | n/a |
| primary, first template only | 0.55 | [0.18, 0.78] | undetermined | -0.16 |

**Double bans.** The instrument put essentially all mass on the banned pair under the cumulative bans: 1.00 on both tasks with the frozen templates, against the association experiment's sampled banned shares of 0.94 (grid, mechanism level) and 1.00 (shortest path). It predicted the failure the succession hit in iteration 1. It placed the remaining mass on IDDFS 0.73 and A* 0.27 (grid) and on Floyd-Warshall 1.00 (shortest path) under the frozen templates, and on IDDFS 0.43, A* 0.33 and unit weight Dijkstra 0.24 (grid) and Floyd-Warshall 0.987 (shortest path) under the association experiment's own step-2 prompts. The six not banned grid programs that experiment sampled were unit weight Dijkstra 4, A* 1 and IDDFS 1. An sham ban outside the catalogue (jump point search, three of five templates) did not single out A*, its declared neighbour, in its residual; the other shams outside the catalogue and the iteration-1 steering setting as a generic perturbation were not run before the GPU deadline, and neither was the planned extension of the gate to the other four iteration-1 draws.

**Coverage and effective sample size.** The forced programs cover the natural ones unevenly (Table 60): well for Dijkstra and Floyd-Warshall (nearest neighbour shingle Jaccard 0.94 to 0.95), poorly for Bellman-Ford/SPFA (0.54) and grid unit weight Dijkstra (0.42). Outside the forced cells, the median effective sample size of a half cell is 1.0 to 1.4 (Table 61).

**Table 60.** Coverage of naturally sampled programs by the forced programs (source: `results/coverage.json`; Jaccard over identifier normalised AST shingles).

| Task and algorithm | Natural programs | Share with nearest forced Jaccard at least 0.5 | Mean nearest Jaccard |
|---|---|---|---|
| Grid BFS | 1102 | 0.69 | 0.66 |
| Grid DFS | 262 | 0.56 | 0.59 |
| Grid bidirectional BFS | 5 | 0.60 | 0.49 |
| Grid IDDFS | 37 | 0.73 | 0.58 |
| Grid A* | 29 | 0.48 | 0.57 |
| Grid unit weight Dijkstra | 12 | 0.17 | 0.42 |
| Shortest path Dijkstra | 1004 | 0.96 | 0.94 |
| Shortest path Bellman-Ford/SPFA | 251 | 0.40 | 0.54 |
| Shortest path Floyd-Warshall | 43 | 1.00 | 0.95 |

**Table 61.** Effective sample size of the importance weights per condition family (source: `results/ess_table.json`).

| Task | Condition family | Half cells | Median ESS | Share below 5 |
|---|---|---|---|---|
| Grid | ban | 516 | 1.13 | 0.99 |

| | | | | |
|---|---|---|---|---|
| Grid | baseline | 12 | 1.07 | 1.00 |
| Grid | random edits | 72 | 1.19 | 0.99 |
| Grid | iteration-1 gate prompts | 228 | 1.21 | 0.98 |
| Grid | forced prompts | 120 | 5.61 | 0.43 |
| Grid | mention | 360 | 1.20 | 0.96 |
| Grid | neutral | 516 | 1.08 | 0.98 |
| Grid | novelty | 12 | 1.10 | 1.00 |
| Shortest path | ban | 350 | 1.17 | 0.97 |
| Shortest path | baseline | 10 | 1.40 | 1.00 |
| Shortest path | random edits | 60 | 1.08 | 0.97 |
| Shortest path | iteration-1 gate prompts | 175 | 1.28 | 0.98 |
| Shortest path | forced prompts | 100 | 3.69 | 0.70 |
| Shortest path | mention | 250 | 1.11 | 0.94 |
| Shortest path | neutral | 350 | 1.27 | 1.00 |
| Shortest path | novelty | 10 | 1.02 | 1.00 |

**Checks and what was frozen.** Synthetic tests of the analysis code recovered a planted ladder (share 0.00), a planted X specific pattern (share 1.00), unbiased noise correction and a planted aim ($p = 0.00025$). An independent rederivation matched the gate, the shares, the aim and every cell to within 1e-15, and a placebo aim sat at its null (mean 0.0007, SD 0.122, against the observed -0.105). The aim profiles, the ladder profile, the templates, nouns and padding, and the code hashes were frozen in a hash chain for the confirmation bench. The experiment's own recommendation, which the Llama replication and the bench inherit, is to score unscrubbed or name matched programs.

## 2.5 Route experiment: does a ban act through the prompt states or at generation time?

**Why.** The hypothesis update predicted that the algorithm specific part of a ban travels by a direct route, with generated tokens reading the prohibition and vetoing X at its signature tokens, and that the read position carries only how strongly the ban will be obeyed. This experiment tested the route causally on the modal bans of both tasks (BFS on the grid task, Dijkstra on the shortest path task), in Qwen3-8B in bf16.

**Design.** Because each ban prompt and its neutral twin have the same token length and identical tokens from the task text onward, their key and value (KV) caches can be spliced position by position, an interchange intervention on cached states in the sense of activation patching [47]. One arm is the full ban and one the full neutral twin. In the *prompt route* arm, generated tokens see the neutral condition segment but the task specification and read position cache entries computed under the ban, so they can learn about the ban only through states that read it. In the *direct route* arm, generated tokens see the ban segment but task specification and read position entries computed under the neutral twin, so they can learn about the ban only by attending to it directly. Two further arms splice only the read position or only the task specification states. A placebo arm splices between two neutral nouns. Attention knockouts [48] block attention from generated tokens to the condition segment for all generated tokens, for the first 16 only or after the first 16, and in three bands of 12 layers; the same full knockout applied to the neutral prompt measures generic damage. The primary readout is the likelihood drop of the banned algorithm X, $D_X(\text{arm}) = L_X(\text{arm}) - L_X(\text{neutral twin})$, over 30 self written, name scrubbed programs per algorithm (323 forced programs in all). The route fractions are $F_{\text{prompt}} = D_X(\text{prompt route}) / D_X(\text{full ban})$ and

F_direct = D_X(direct route) / D_X(full ban), with F_R and F_spec defined the same way for the read position only and task specification only arms, all bootstrapped over templates and programs. The decision rule was hash committed before any arm was scored: the ban acts directly if F_direct is at least 2/3 with an interval above 1/2, and is decided before writing if F_prompt meets the same bar. Behaviour was sampled 100 times per arm and task (five templates by 20) under common random numbers: one uniform stream per sample shared by every arm, inverse CDF sampling in vocabulary order and a constant batch shape.

**The ban is carried by the prompt states.** Under the preregistered readout the ban is decided before writing on both tasks, and the decision is robust (Table 62). Pooled over tasks, the prompt route carries 0.85 [0.61, 1.20] of the likelihood drop and the direct route 0.00 [-0.08, 0.12]. The read position alone carries nothing (F_R about 0); the task specification states carry it (0.86 and 0.81). The placebo splice moved the banned algorithm by 1.2% (grid) and 0.7% (shortest path) of the ban effect, and the full knockout on the neutral prompt cost only -0.21 and -0.28 nats. Leaving out one template at a time kept F_prompt between 0.68 and 0.99 on the grid task and between 0.74 and 0.91 on the shortest path task, and all four preregistered sensitivity analyses gave the same branch (unweighted mean log likelihood: F_prompt 0.70 and 0.82; correct programs only; the generating paraphrase as denominator; no ESS weighting).

**Table 62.** Likelihood drop of the banned algorithm per arm, in nats, with 95% bootstrap intervals over templates and programs (source: `results/analysis.json#route_frlp`).

| Arm | Grid (ban BFS) | Shortest path (ban Dijkstra) |
|---|---|---|
| a, full ban | -12.25 [-20.9, -6.9] | -5.78 [-7.9, -3.1] |
| c, prompt route | -10.65 [-12.3, -9.6] | -4.73 [-6.2, -3.1] |
| d, direct route | -0.23 [-3.0, 0.8] | 0.22 [-0.3, 0.8] |
| e, read position only | 0.00 [-0.4, 0.2] | -0.11 [-0.3, 0.0] |
| f, task specification only | -10.48 [-12.5, -9.3] | -4.66 [-6.1, -3.2] |
| full knockout of the condition segment | -11.65 [-13.6, -9.9] | -5.42 [-6.6, -3.9] |
| knockout, first 16 generated tokens | -10.74 [-15.5, -9.5] | -5.31 [-6.7, -3.5] |
| knockout, after 16 generated tokens | -11.56 [-15.7, -6.8] | -5.91 [-7.8, -3.5] |
| knockout, layers 0 to 11 | -13.88 [-22.3, -7.7] | -5.88 [-8.0, -3.1] |
| knockout, layers 12 to 23 | -8.29 [-11.4, -4.2] | -4.93 [-6.3, -3.3] |
| knockout, layers 24 to 35 | -14.38 [-20.1, -11.5] | -5.24 [-7.3, -2.9] |
| full knockout on the neutral prompt (generic damage) | -0.21 [-0.6, 0.6] | -0.28 [-0.6, -0.0] |
| mention twin | 0.55 [-0.3, 2.1] | -0.17 [-0.7, 0.4] |
| F_prompt [95% CI] | 0.87 [0.55, 1.50] | 0.82 [0.62, 1.20] |
| F_direct [95% CI] | 0.02 [-0.10, 0.18] | -0.04 [-0.25, 0.05] |
| F_spec / F_R | 0.86 / 0.00 | 0.81 / 0.02 |
| interaction D(a) - D(c) - D(d) | -1.37 [-7.97, 3.86] | -1.27 [-2.55, -0.00] |
| Pooled (ratio of sums) | F_prompt 0.85 [0.61, 1.20]; F_direct 0.00 [-0.08, 0.12]; branch: decided before writing, robust | |

*[Figure fig_route: generated in the report phase, not in this draft]*

The knockouts agree. Blocking the condition segment for every generated token restores 5% (grid)

and 6% (shortest path) of the likelihood gap, with intervals that include zero. By layer band, blocking layers 12 to 23 restores 32% [16%, 56%] on the grid task and 15% on the shortest path task, layers 0 to 11 restore about nothing, and layers 24 to 35 restore -17% (interval includes zero) on the grid task and 9% [2%, 16%] on the shortest path task. Whether the veto is read early or late in generation could not be determined (first-16 minus after-16 restoration 0.07 [-0.47, 0.20] and 0.10 [-0.03, 0.18]).

**Behaviour qualifies the grid result.** On the shortest path task behaviour matches the likelihood readout exactly: the prompt route arm reproduces the ban (banned share 0.13 against 0.11) and the direct route arm is the baseline (0.95 against 0.99). On the grid task the prompt route reproduces only part of the ban (0.57 against 0.14), and the direct route also moves it (0.70). A supplementary arm that hides the condition segment from every generated token brought BFS back to 0.47 [0.38, 0.57], close to the prompt route arm (McNemar $p = 0.13$), so on the grid task generated tokens do read the ban directly as well. That reading is redundant over time: blocking only the first 16 generated tokens or only the rest leaves the ban intact (banned shares 0.16 and 0.19). The behavioural route fractions are 0.50 [0.39, 0.62] (prompt) and 0.35 [0.25, 0.45] (direct) on the grid task, and 0.98 [0.89, 1.07] and 0.05 [0.01, 0.09] on the shortest path task. Matched pairs are clean: no sample was banned under the ban and not banned under its neutral twin (0 of 86 and 0 of 88 discordant pairs in that direction).

**Table 63.** Behaviour per arm under common random numbers ($N = 100$ per arm; mechanism labels of attempted programs; banned share with Wilson 95% interval; source: `results/analysis.json#behaviour`, `results/supplementary.json`).

| Task | Arm | Banned share [95%] | Pass rate | Labels | Banned programs per template (first to fifth, 20 each) |
|---|---|---|---|---|---|
| Grid | b, neutral twin | 1.00 [0.96, 1.00] | 0.92 | BFS 100 | 20, 20, 20, 20, 20 |
| Grid | d, direct route | 0.70 | 0.81 | BFS 70, DFS 30 | 15, 12, 17, 20, 6 |
| Grid | c, prompt route | 0.57 [0.47, 0.66] | 0.61 | BFS 57, DFS 43 | 10, 15, 6, 12, 14 |
| Grid | a, full ban | 0.14 [0.09, 0.22] | 0.41 | DFS 86, BFS 14 | 0, 0, 0, 14, 0 |
| Grid | full knockout (supplementary) | 0.47 [0.38, 0.57] | n/a | BFS 47, rest not BFS | n/a |
| Grid | knockout, first 16 tokens | 0.16 | 0.46 | DFS 84, BFS 16 | 1, 0, 1, 14, 0 |
| Grid | knockout, after 16 tokens | 0.19 | 0.49 | DFS 81, BFS 19 | 1, 2, 1, 11, 4 |
| Shortest path | b, neutral twin | 0.99 | 0.99 | Dijkstra 99, Bellman-Ford/SPFA 1 | 20, 20, 20, 19, 20 |
| Shortest path | d, direct route | 0.95 | 0.99 | Dijkstra 95, Bellman-Ford/SPFA 5 | 19, 18, 20, 19, 19 |
| Shortest path | c, prompt route | 0.13 | 0.91 | Bellman-Ford/SPFA 87, Dijkstra 13 | 6, 1, 3, 1, 2 |
| Shortest path | a, full ban | 0.11 | 0.91 | Bellman-Ford/SPFA 86, Dijkstra 11, other 3 | 1, 0, 4, 5, 1 |

| | | | | | |
|---|---|---|---|---|---|
| Shortest path | full knockout (supplementary) | 0.06 | n/a | Dijkstra 6, rest not Dijkstra | n/a |
| Shortest path | knockout, first 16 tokens | 0.11 | 0.94 | Bellman-Ford/SPFA 88, Dijkstra 11, other 1 | 2, 1, 5, 3, 0 |
| Shortest path | knockout, after 16 tokens | 0.11 | 0.96 | Bellman-Ford/SPFA 89, Dijkstra 11 | 3, 0, 4, 3, 1 |

On the grid task, all 14 BFS programs that survived the full ban came from the fourth template (`X is not allowed here`, with BFS as X), which gave 14 of 20 BFS while the other four templates gave 0 of 20.

**Where the ban acts: the first line.** Under the neutral twin, 99% of programs open with an import line (`from collections import deque` on the grid task, `import heapq` on the shortest path task). Under the grid ban, none of 100 programs opens with an import line, including the 14 BFS programs that survive; they open with `def solve(...)` (Table 64). The prompt route arm and every knockout also open without an import, while the direct route arm still opens with one 85% of the time. On the shortest path task the Bellman-Ford/SPFA programs written under the ban mostly open with `from collections import deque`, so the share of import openings stays at 61 to 67% under the ban, the prompt route and the knockouts, and 99% under the direct route. Per-token localisation agrees. On the grid task 63% of the likelihood drop under the full ban, 90% under the prompt route and 95% under the full knockout sits in the first 16 tokens; on the shortest path task almost all of it does. The signature tokens themselves (`deque`, `heapq` and their neighbours) carry only 2.4% (grid) and 11.6% (shortest path) of it. The small direct route drop on the grid task lies later in the program (6% of it in the first 16 tokens). We call this the *opening line veto*: the prompt states suppress the default program's first line before any algorithm specific token is written. On the grid task, the algorithm chosen after `def solve` also depends on direct reads of the ban, and the likelihood readout, dominated by the opening fork, calls the grid task prompt route while its behaviour is mixed.

**Table 64.** Share of programs whose first line is an import (source: `results/supplementary.json#opening_lines`; N = 100 per arm).

| Task | b, neutral | d, direct route | c, prompt route | a, full ban | full knockout | knockout, first 16 | knockout, after 16 |
|---|---|---|---|---|---|---|---|
| Grid | 0.99 | 0.85 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| Shortest path | 0.99 | 0.99 | 0.61 | 0.62 | 0.67 | 0.67 | 0.60 |

**A supplementary readout with natural proposals.** Rescoring, under every arm, the BFS and Dijkstra programs sampled in the neutral arm (exact draws from the neutral prompt's distribution) gives the same ordering with a larger direct share: pooled F_prompt 0.57 [0.41, 0.85] and F_direct 0.16 [0.11, 0.21] (grid 0.58 and 0.18, shortest path 0.56 and 0.11), and the two routes are superadditive on both tasks. Neither likelihood readout is a calibrated probability ratio. The importance weight ESS is about 1 to 3 of 30 programs for the grid full ban arm (8 to 20 on the shortest path task), 858 of the 1,095 cells have ESS below 5, and exp(D) under the natural readout is 3e-7 on the grid task against a behavioural ratio of 0.14. The readouts measure how strongly typical banned algorithm programs, which begin with the import line, are vetoed.

**Where the substitute rises.** Under the full ban the survivors' likelihood gains on the grid task were DFS +9.46, IDDFS +18.40, bidirectional BFS -0.22, A* -0.80 and unit weight Dijkstra -3.53 nats; on the shortest path task Bellman-Ford/SPFA +2.53, bidirectional Dijkstra +30.64, Floyd-Warshall

+12.64 and Dial +6.79. The prompt route arm reproduces this profile (cosine 0.94 [0.85, 0.99] grid, 0.95 [0.90, 0.99] shortest path); the direct route arm reproduces it on the shortest path task (0.98 [0.80, 1.00]) but not reliably on the grid task (0.71 [-0.18, 0.94]). The logit lens and a tuned lens [49] at the planned commitment token (the token after `from` or `import`) found no veto layer: the banned set's rank there is unchanged by the ban (0 of 59 grid and 0 of 60 shortest path programs), because the fork has already happened at the first token. The grid substitute set (DFS) rises from layer 28 (59 of 59 programs; tuned lens median 27). In substitute programs written under the full ban, the banned set is vetoed late (median layer 30 on the grid task, 36 on the shortest path task). The tuned lens beat the logit lens on held-out KL at all 11 probed layers (0.29 against 10.6 at layer 21).

**Does the read position know how hard a ban will be obeyed?** The compliance test regressed the banned share of each wording on the projection of its read position state onto a shared ban direction, the mean ban minus neutral shift estimated leaving one experiment out. It used 110 wordings in 12 experiment by task cells: ten modal ban paraphrases from each of the five iteration-1 draws per task, plus this experiment's five templates per task. The combined within cell Spearman correlation is -0.41 [-0.69, -0.17], one sided stratified permutation $p = 0.0003$ (Holm adjusted over the two axes tested, 0.0006). The preregistered bar was -0.5, so the result is partial: wordings that move the read position further along the ban direction are obeyed more, but less tightly than predicted. The correlation beats 98.5% of 200 random directions (null mean -0.01), and the split half noise ceiling is about 0.77. The decommitment axis of iteration 1 carries nothing (rho 0.05, $p = 0.68$). The DFS share among programs not banned is not decodable beyond the projection (change in R squared below zero, permutation $p = 0.74$ on the grid task; not testable on the shortest path task). The recomputed read position states matched the iteration-1 caches (cosine at least 0.9994 for 40 geometry and 60 decommitment prompts).

**Table 65.** Within cell Spearman correlation between a wording's banned share and its read position projection (source: `results/c5.json`; ten wordings per cell, five for this experiment's templates).

| Cell | Shared ban direction | Decommitment axis |
|---|---|---|
| Association, grid | -0.701 | 0.234 |
| Association, shortest path | -0.811 | 0.416 |
| Geometry, grid | -0.214 | -0.082 |
| Geometry, shortest path | -0.673 | -0.132 |
| Forking token, grid | -0.437 | 0.212 |
| Forking token, shortest path | 0.271 | 0.516 |
| Decommitment, grid | -0.074 | 0.309 |
| Decommitment, shortest path | -0.605 | -0.368 |
| Instruction tuning, grid | 0.036 | 0.207 |
| Instruction tuning, shortest path | -0.171 | -0.734 |
| This experiment's templates, grid | -0.707 | 0.0 |
| This experiment's templates, shortest path | -0.667 | 0.308 |
| Combined [95% CI]; one sided p | -0.41 [-0.69, -0.17]; $p = 0.0003$ | 0.05 [-0.30, 0.36]; $p = 0.68$ |

**Checks.** Spliced caches reproduced the full ban scores bit for bit; the knockout attention with nothing blocked equalled stock attention (largest difference 0); attention to blocked columns was exactly 0. Repeated sampling was 100% identical at a fixed batch shape and only 40% when the shape changed, which is why every arm shares one shape. Two independent rederivations reproduced 49 and 59 headline numbers with no mismatch; McNemar on randomly swapped arm labels rejected 4.0 to 4.4% of the time, the likelihood ban effect's interval included zero for the

placebo splice and for label shuffled data, and the compliance test rejected 0 of 40 shuffled data sets. From the declared dataset dependency, other models' other than Dijkstra choices for weighted shortest path in the inspiration repository spread over Bellman-Ford/SPFA (37%), bidirectional search (32%) and other algorithms (23%), while Qwen3-8B under the Dijkstra ban lands on Bellman-Ford/SPFA 97% of the time; an import opening is the norm in LLM written solutions that name these algorithms (KodCode: BFS 94% of 17, Dijkstra 100% of 40) but not in human library code (TheAlgorithms: 29% of 7 and 17% of 12).

## 2.6 Ban list experiment: the top five prohibition and bans that leave one survivor

**Why.** The request asks what happens when the top five algorithms are prohibited in one prompt: does the implemented algorithm move to the edge of the banned region, or to its opposite side? Iteration 1 could not reach that test through its greedy succession, because the model stopped complying after two bans. This experiment ran the prohibition as single prompt conditions, with a control for each reading of where the mass goes, and asked on the shortest path task whether the model can and will write each algorithm when it is the only one allowed.

**Catalogue extension.** The grid catalogue was extended toward the ten algorithms the request imagines: greedy best first search was split from A*, and IDA*, a 4-neighbour jump point search and bidirectional A* were added, with new references that both labellers got right (25 of 25) and targeted unit probes (4 of 4). The canonical labeller recheck gave 135 of 144 gold programs and 22 of 22 references. The 40-program gold gate of the extended labeller was missed: 33 of 40 at round 0 and 32 of 40 after the one allowed revision, runtime against AST kappa 0.53; unit weight Dijkstra, jump point search and bidirectional A* fell below three of four gold programs and were excluded. An algorithm was kept if at least 70% of its 40 forced programs were labelled as requested (Table 66). Only BFS, DFS, IDDFS, greedy best first and IDA* passed, so the extension failed. The model mostly writes A* as a cost only heap (unit weight Dijkstra 25 of 40), unit weight Dijkstra as BFS (18 of 40), and jump point search and bidirectional A* as BFS and A*. The analysis catalogue (seven algorithms) kept A* and bidirectional BFS, flagged, because each has at least 10 programs and passes the gold set. With seven algorithms the frozen set rule gives four name sets: TOP (the four with the highest baseline likelihood: BFS, DFS, greedy, IDDFS), ARC (four contiguous on the manifold: BFS, bidirectional BFS, A*, greedy) and SCATTER (four spread out: BFS, DFS, greedy, IDA*). The literal ban of five names, TOP5 (BFS, DFS, greedy, IDDFS, A*), was added. Six landing forecasts per set (ladder, prime and veto, proximity, likelihood under the ban, generic perturbation, corpus frequency) were hash chained before the ban lists were sampled, and the chain verifies.

**Table 66.** Keep rule for the extended catalogues: forced programs labelled as requested (source: `results/catalogue_keep.json`; 40 forced programs each).

| Task | Algorithm | Forced prompt | Labelled as requested | Of those, passing | Other labels | Kept |
|---|---|---|---|---|---|---|
| Grid | BFS | plain | 40 | 38 | none | yes |
| Grid | DFS | plain | 40 | 9 | none | yes |
| Grid | IDDFS | plain | 37 | 0 | DFS 3 | yes |
| Grid | greedy best first | plain | 35 | 25 | BFS 2, other 2, A* 1 | yes |
| Grid | IDA* | descriptive | 36 | 0 | A* 3, BFS 1 | yes |

| | | | | | | |
|---|---|---|---|---|---|---|
| Grid | bidirectional BFS | plain | 21 | 5 | BFS 16, other 3 | no (kept in analysis, flagged) |
| Grid | A* | plain | 11 | 11 | unit weight Dijkstra 25, other 3, BFS 1 | no (kept in analysis, flagged) |
| Grid | unit weight Dijkstra | plain | 13 | 11 | BFS 18, other 9 | no |
| Grid | jump point search | descriptive | 7 | 0 | BFS 21, other 11, unit weight Dijkstra 1 | no |
| Grid | bidirectional A* | descriptive | 5 | 0 | A* 28, BFS 3, other 3, bidirectional BFS 1 | no |
| Shortest path | Dijkstra | plain | 40 | 40 | none | always run |
| Shortest path | Bellman-Ford/SPFA | plain | 40 | 34 | none | always run |
| Shortest path | bidirectional Dijkstra | descriptive (corrected in iteration 3 from plain; the plain prompt gave 10 programs, Dijkstra 9) | 14 | 9 | Dijkstra 22, other 4 | always run |
| Shortest path | Floyd-Warshall | plain | 40 | 40 | none | always run |
| Shortest path | Dial | descriptive (corrected in iteration 3 from plain; the plain prompt gave 10 programs) | 36 | 7 | other 4 | always run |

**Every ban list fails.** With one name banned the model complies (avoidance 0.86 on the grid task, 0.84 on the shortest path task). With four or five names it mostly writes a banned algorithm anyway (Table 67): avoidance is 0.13 for TOP, 0.37 for ARC, 0.04 for SCATTER and 0.00 for the literal ban of five names, so every ban list fails the event gate. The banned programs are BFS and DFS reimplemented, often without naming their mechanism (Table 68). We call a banned mechanism program *disguised* when its comments and identifiers never name its own mechanism; 41% to 92% of the banned programs under a ban list are disguised, DFS most often as `backtracking`. The likelihoods committed before sampling foresaw the failure for three of the four sets: the implied avoidance was below 1e-8 for TOP, SCATTER and the five name ban, against realised 0.13, 0.04 and 0.00. For ARC the implied avoidance was 0.9995 against a realised 0.37. A ban prompt raised the likelihood of the banned algorithms' own programs relative to the neutral twin (DFS gained about 10 nats under the first template of TOP), the same priming as in the likelihood screen.

**Table 67.** Grid ban lists and the single name references (N = 100 each, five templates by 20; mechanism labels of attempted programs; source: `results/landing.json`).

| Condition | Banned | Landing (attempted programs) | Avoidance [Wilson 95%] | Template cluster 95% CI | Avoidance per template (first to fifth) | Pass rate |
|---|---|---|---|---|---|---|
| Grid baseline | none | BFS 100 | n/a | n/a | n/a | 0.89 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Grid single ban | BFS | DFS 86, BFS 14 | 0.86 [0.78, 0.91] | [0.58, 1.00] | 1.0, 1.0, 1.0, 0.30, 1.0 | 0.41 |
| TOP | BFS, DFS, greedy, IDDFS | BFS 59, DFS 26, greedy 2, other 13 | 0.13 [0.08, 0.21] | [0.01, 0.29] | 0.15, 0.05, 0.0, 0.0, 0.45 | 0.40 |
| ARC | BFS, bidirectional BFS, A*, greedy | BFS 63, DFS 37 (DFS allowed) | 0.37 [0.28, 0.47] | [0.09, 0.68] | 0.70, 0.15, 0.15, 0.0, 0.85 | 0.60 |
| SCATTER | BFS, DFS, greedy, IDA* | BFS 69, DFS 27, A* 2, other 2 | 0.04 [0.02, 0.10] | [0.00, 0.08] | 0.0, 0.10, 0.0, 0.0, 0.10 | 0.75 |
| TOP5 (literal) | BFS, DFS, greedy, IDDFS, A* | BFS 63, DFS 36, IDDFS 1 | 0.00 [0.00, 0.04] | [0.00, 0.00] | 0.0, 0.0, 0.0, 0.0, 0.0 | 0.68 |
| Shortest path baseline | none | Dijkstra 100 | n/a | n/a | n/a | 1.00 |
| Shortest path single ban | Dijkstra | Bellman-Ford/SPFA 83, Dijkstra 16, other 1 | 0.84 [0.76, 0.90] | [0.72, 0.95] | 0.90, 0.95, 0.70, 0.65, 1.0 | 0.95 |

*(Correction, iteration 3: TOP's avoidance of 0.13 is labeller residue, not avoidance into a catalogue algorithm. Its 13 programs not banned are all labelled other: 12 failing programs that both blind judges classed as degenerate BFS or DFS variants, and 1 correct unit weight Dijkstra outside the $K = 7$ analysis catalogue. Avoidance into a non-banned catalogue algorithm is 0.00 for TOP, 0.37 for ARC, 0.02 for SCATTER and 0.00 for TOP5. The share labelled other is reported separately from avoidance in Section 3.2.3.)*

**Table 68.** Disguise: banned mechanism programs that never name their own mechanism (source: `results/landing.json`).

| Condition | Banned mechanism programs | Disguised | Disguise rate | By mechanism (disguised / programs) | Most frequent self descriptions |
|---|---|---|---|---|---|
| Grid single ban of BFS | 14 | 14 | 1.00 | BFS 14/14 | none named |
| Shortest path single ban of Dijkstra | 16 | 2 | 0.125 | Dijkstra 2/16 | none named |
| ARC | 63 | 58 | 0.92 | BFS 58/63 | `is_valid` 2 |
| SCATTER | 96 | 49 | 0.51 | BFS 43/69, DFS 6/27 | `backtrack` 10, `is_valid` 7, `depth-limited` 1 |
| TOP | 87 | 44 | 0.51 | BFS 37/59, DFS 6/26, greedy 1/2 | `backtrack` 10, `is_valid` 9 |
| TOP5 | 100 | 41 | 0.41 | BFS 34/63, DFS 6/36, IDDFS 1/1 | `is_valid` 10, `backtrack` 8, `find_path` 2 |

*[Figure fig_multiban: generated in the report phase, not in this draft]*

**The list's content barely matters (post hoc).** Across the four ban lists the model writes the same mix of BFS and DFS: the BFS share among BFS and DFS programs is 0.69 (TOP), 0.63 (ARC), 0.72 (SCATTER) and 0.64 (TOP5), chi squared $p = 0.48$ and permutation $p = 0.57$. This holds whether DFS is on the list (TOP, SCATTER, TOP5) or allowed (ARC). Avoidance differences therefore come from whether DFS happens to be banned, not from a list specific landing. A single BFS ban moves the model to DFS 86%; a list of four or five names moves it to about one third DFS, which looks more like a generic perturbation of the default than a veto. This comparison was not preregistered.

**Landing tests are degenerate.** Only ARC (37 DFS programs) and SCATTER (2 A* programs) put any mass on a survivor. On ARC the ladder, prime and veto, likelihood, generic perturbation and corpus forecasts all named DFS (log score +1.06 nats per program over uniform), and proximity named IDA* and missed (-0.02). With three survivors the exact permutation floor is 1/6, every per set p is at least 0.33, and every pooled p is at least 0.11; the edge against opposite statistic was 0 on both sets with mass ($p = 0.67$). The request's literal question therefore has a behavioural answer that is neither of its two options: under a four or five name ban the model does not move to the edge or to the opposite side of the banned region, it reimplements the banned defaults. Over all programs, predicting no change from the default algorithm was right for 64% of programs under a ban list, against at most 10% for any committed survivor forecast; on single bans the highest ranked not banned algorithm predicted 85% and the default 15%; under forced prompts the requested algorithm was written 66% of the time.

**Bans that leave one survivor: can against will.** On the shortest path task each of the five algorithms was left as the only one allowed, by banning the other four (Table 69). The model writes Dijkstra when it is the survivor 95% of the time and Bellman-Ford/SPFA 51% of the time. It writes bidirectional Dijkstra, Floyd-Warshall and Dial 0% of the time, falling back to the banned Dijkstra or Bellman-Ford/SPFA instead, although when forced it writes Floyd-Warshall correctly 40 of 40 times and Dial 36 of 40 times. The likelihood gains follow the same split: banning the other four lowers Dijkstra by 3.35 nats and Bellman-Ford/SPFA by 0.21 when each is the survivor, and raises the three algorithms the model never writes by 7.6 to 24.3 nats without moving them into the sample.

**Table 69.** Shortest path bans that leave one survivor (N = 100 each; will is the share of programs implementing the survivor and can the share of 40 forced programs labelled as the survivor; source: `results/los.json`).

| Survivor (other four banned) | Will [Wilson 95%] | Banned mechanisms written | Other | Compliant share | Can (forced labelled as requested; forced pass) | Mean likelihood gain of the survivor (nats) | Disguised / banned programs |
|---|---|---|---|---|---|---|---|
| Dijkstra | 0.95 [0.89, 0.98] | Bellman-Ford/ SPFA 0.05 | 0.0 | 0.95 | 1.00; 1.00 | -3.35 | 4 / 5 |
| Bellman-Ford/ SPFA | 0.51 [0.41, 0.61] | Dijkstra 0.48 | 0.01 | 0.39 | 1.00; 0.85 | -0.21 | 8 / 48 |
| bidirectional Dijkstra | 0.00 [0.00, 0.04] | Dijkstra 0.52, Bellman-Ford/ SPFA 0.48 | 0.0 | 0.0 | 0.35; 0.225 | +24.26 | 51 / 100 |
| Floyd-Warshall | 0.00 [0.00, 0.04] | Dijkstra 0.59, Bellman-Ford/ SPFA 0.40 | 0.01 | 0.0 | 1.00; 1.00 | +7.63 | 46 / 99 |
| Dial | 0.00 [0.00, 0.04] | Dijkstra 0.52, Bellman-Ford/ SPFA 0.44 | 0.04 | 0.0 | 0.90; 0.175 | +21.20 | 42 / 96 |

**Geometry and natural run internals.** In the teacher forced residual stream, centroid MDS stress-1 was 0.025 (grid) and 0.006 (shortest path). The nearest neighbours were BFS with bidirectional BFS, DFS with IDDFS, and A* with greedy best first; nearest neighbour stability was 0.70 across

settings and 0.96 under a program bootstrap. Persistent homology, which tracks loops in a point cloud across distance scales, found no persistent loop on either task. *(Correction, iteration 3: this sentence originally read the result as a negative answer to the request's question about a circle. It is no evidence of a loop, at weak power, at a label-dependent position. These clouds sit at the token before the commit token, a position that depends on the label; there a planted loop is found with power 1.00, while at the label-independent generated token 16 the power is 0.04. The nearest pair at that position, BFS and bidirectional BFS (centroid distance 0.0147), does not survive at token 16: permuted labels reproduce the adjacency with $p = 0.19$ and the two clouds are not separable. The experiment drew 13 Isomap figures with centroids sized by implemented count, which this section did not show; they are reported with their counts in Section 3.2.2.)* Probes within one ban prompt at generated token 16, a position that does not depend on the label, were near chance on every grid condition and reached balanced accuracy 0.69 ($p = 0.03$) on the shortest path ban that leaves only Bellman-Ford/SPFA. Probes at the token before the commit token were confounded, because that position depends on the label. The difference-in-means direction from forced programs aligned with the natural within one ban direction at that position (cosine 0.6 to 0.95) and on the shortest path task at token 16 (0.3 to 0.99), but not for the grid BFS ban at token 16 (-0.13 to -0.01).

**Deviations and cost.** The experiment ran on an RTX 4090 in eager vLLM mode (no C compiler), replaced one duplicated forced paraphrase, missed the gold gate after its revision, and recorded bf16 scoring path noise of about 1 nat per program; the shortest path likelihoods sum to 0.17, below the 0.3 calibration floor; the second noun placebo ran after the ban lists were sampled. It used about 1 GPU hour and USD 0.42 of judge calls (857 calls).

## 2.7 Llama replication: the same protocol in a second model family

**Why.** A claim about what a ban does inside one model needs a second family. This experiment ran shared protocol v2 on Llama-3.1-8B-Instruct [22] in bf16 with its default chat template, on the canonical suite, retokenising the neutral noun bank under Llama. It drew 1,780 seeded vLLM generations: baselines, modal bans, mention twins, listing controls, two bans of algorithms other than the default and two double bans per task (14 events), 880 forced programs and 220 free-form answers. It scored 248 name scrubbed forced programs in 39,768 program scores (11.3 million tokens). Nothing was tuned to agree with Qwen3-8B.

**Llama's baseline is not collapsed on the grid task.** *(Correction, iteration 3: true of attempted programs only. All 20 correct grid baseline programs are BFS, a correctness conditioned Hill number of 1.00 against 1.87 over attempts, so Llama's correct grid output is as collapsed as Qwen3-8B's; correct-only compositions of every Llama event are in Section 3.2.4.)* It writes BFS 77 times and DFS 20 times in 100, with a low pass rate (0.20). On the shortest path task it is collapsed (Dijkstra 92, pass rate 0.59). The BFS ban sends it to DFS 63 of 100; the Dijkstra ban sends it to Dijkstra 43, other labels 37 and Bellman-Ford/SPFA 18. Both double bans fail the gate (Table 70). Bidirectional Dijkstra and Dial could not be written (2 and 1 forced programs kept after up to 120 attempts), so the shortest path catalogue has three algorithms, no row there has three survivors, and the substitution test runs on the grid task only.

**Table 70.** The 14 sampled Llama events (mechanism labels of attempted programs; source: `results/events.json`).

| Task | Event | N | Labels | Banned share | Avoidance | Compliant share | Clear mode | Gate |
|---|---|---|---|---|---|---|---|---|
| Grid | baseline | 100 | BFS 77, DFS 20, other 3 | 0.0 | 1.0 | 0.20 | BFS | pass |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Grid | ban BFS (modal) | 100 | DFS 63, BFS 29, other 6, A* 1, unit weight Dijkstra 1 | 0.29 | 0.71 | 0.08 | DFS | pass |
| Grid | mention BFS | 50 | BFS 47, other 2, DFS 1 | 0.0 | 1.0 | 0.28 | BFS | pass |
| Grid | listing control | 50 | BFS 36, other 7, DFS 6, unit weight Dijkstra 1 | 0.0 | 1.0 | 0.20 | BFS | pass |
| Grid | ban DFS | 50 | BFS 39, DFS 7, other 4 | 0.14 | 0.86 | 0.14 | BFS | pass |
| Grid | ban unit weight Dijkstra (absent) | 50 | BFS 30, DFS 18, unit weight Dijkstra 1, other 1 | 0.02 | 0.98 | 0.26 | BFS | pass |
| Grid | ban BFS and DFS | 50 | BFS 21, DFS 21, other 6, unit weight Dijkstra 2 | 0.84 | 0.16 | 0.0 | none | fail |
| Shortest path | baseline | 100 | Dijkstra 92, other 7, Bellman-Ford/SPFA 1 | 0.0 | 1.0 | 0.59 | Dijkstra | pass |
| Shortest path | ban Dijkstra (modal) | 100 | Dijkstra 43, other 37, Bellman-Ford/SPFA 18, Floyd-Warshall 2 | 0.43 | 0.57 | 0.05 | none | pass |
| Shortest path | mention Dijkstra | 50 | Dijkstra 39, other 11 | 0.0 | 1.0 | 0.46 | Dijkstra | pass |
| Shortest path | listing control | 50 | Dijkstra 19, Bellman-Ford/SPFA 15, other 14, Floyd-Warshall 2 | 0.0 | 1.0 | 0.24 | none | pass |
| Shortest path | ban Bellman-Ford/SPFA | 50 | Dijkstra 33, other 14, Bellman-Ford/SPFA 2, Floyd-Warshall 1 | 0.04 | 0.96 | 0.26 | Dijkstra | pass |
| Shortest path | ban Floyd-Warshall (absent) | 50 | Dijkstra 25, other 11, Bellman-Ford/SPFA 9, Floyd-Warshall 5 | 0.10 | 0.90 | 0.14 | Dijkstra | pass |
| Shortest path | ban Dijkstra and Bellman-Ford/SPFA | 50 | Dijkstra 22, other 21, Bellman-Ford/SPFA 6, Floyd-Warshall 1 | 0.56 | 0.44 | 0.0 | none | fail |

*(Added in iteration 3: Table 70 lists attempted programs only. Correct programs: grid baseline BFS 20; ban BFS DFS 7, other 1; mention BFS BFS 14; listing control BFS 7, DFS 3; ban DFS BFS 7; ban unit weight Dijkstra BFS 12, DFS 1; shortest path baseline Dijkstra 59; ban Dijkstra Bellman-Ford/SPFA 4, other 1; mention Dijkstra Dijkstra 22, other 1; listing control Dijkstra 9, Bellman-Ford/SPFA 3; ban Bellman-Ford/SPFA Dijkstra 13; ban Floyd-Warshall Dijkstra 6, Bellman-Ford/SPFA 1; both double bans 0. Table 89, Section 3.2.4.)*

**The validity gate fails, and the substitution test is unresolved.** Calibration fails on the grid task: the instrument implies BFS 0.43 and DFS 0.55 against a sampled BFS share of 0.77 (Wilson [0.68, 0.84]); the shortest path task passes (Dijkstra 0.97, sum 0.87). Discrimination passes at 0.59 nats per program over uniform across 13 events (1.0 to 1.6 nats on the grid ban and mention events, 0.1 to 0.8 on the shortest path ones, negative for both listing controls). The argmax matches the realised mode in 7 of 10 clear mode events, below the 0.8 bar; all three misses are on the grid task, where the instrument predicts DFS and sampling lands on BFS. The median cell ESS is 2.5 and 76% of cells are below 5. Under the neutral reference the X specific share is 0.80 with a 95% percentile interval of [0.28, 0.94], with no aim and no ladder, so the result is not resolved at this sample size; under the mention reference it reads X specific but unaimed (Table 71). The branch is not robust. Outside the frozen rule, the program level importance sampling noise dominates: the

bootstrap variance of the mean gain is 5 to 11 times the replicate variance the frozen correction subtracts, the basic interval for the neutral V_u + V_s is [-15.1, 10.2] and covers zero, and the share falls to 0.41 without the five forced programs that the label audit disputed and to 0.44 with a raw mean ladder profile.

**Table 71.** Llama substitution test, grid task only (source: `results/c1.json`).

| Reference | X specific share (95% percentile CI) | V_u / V_s / V_u + V_s | Aim at prime and veto (null 95th pct; p) | Ladder exists (mean cosine) | Branch |
|---|---|---|---|---|---|
| neutral twin (primary) | 0.80 (0.28, 0.94) | 2.0 / 8.2 / 10.1 | 0.067 (0.258; p = 0.35) | no (0.14) | not resolved at this N |
| mention twin (secondary) | 0.78 (0.61, 0.91) | 7.1 / 25.5 / 32.6 | -0.28 (p = 0.96) | no | X specific but unaimed |

The association aim did not pass either (-0.08 against a null 95th percentile of 0.30).

**Llama, like Qwen3-8B, primes the banned name.** Naming an absent algorithm in a ban raised its own likelihood in 5 of 6 cases: bidirectional BFS +3.9 nats, IDDFS +7.6, unit weight Dijkstra +1.8, Bellman-Ford/SPFA +4.1 and Floyd-Warshall +4.8; A* was the exception (-1.2). The mention twin primes these algorithms by +6.7 to +14.4 nats and the ban's veto (-4.9 to -10.1) cancels only part of the prime. Bans of present algorithms do lower them: BFS -1.7, DFS -3.0 and Dijkstra -1.0. Within a ban row, the X specific residual is negatively correlated with how much a survivor's code resembles X (Spearman -0.6 to -0.9 in 5 of 6 grid rows): a ban lowers the survivors that look like the banned algorithm.

**The route, read by likelihood only.** On the two modal bans (20 cells), the ban's drop on the default (-27.2 nats summed) travels mostly by the prompt route: F_prompt 0.84 (bootstrap interval 0.62 to 1.46) and F_direct -0.21 (-1.03 to 0.09). On the grid task the prompt route arm also reproduces the full ban's redistribution over survivors (cosine 0.98); the direct route arm does not (0.20).

**Which forecast predicts the program written?** Over 743 attempted samples not banned the empirical sampled ladder predicted the realised mechanism best (Table 72). The likelihood forecast lost to it overall (difference -0.11, paired sign flip $p < 1e\text{-}4$) and tied it on single bans (+0.01, $p = 0.71$). Asked for the standard alternative, Llama answers DFS 20 of 20 times for BFS and Bellman-Ford/SPFA 19 of 20 times for Dijkstra; its top-1 answers match the OpenRouter Llama map of iteration 1 for 9 of 11 algorithms.

**Table 72.** Per-sample top-1 accuracy of each predictor against the realised mechanism label (source: `results/independent_rederivation_headlines.json`).

| Predictor | All 743 samples | Single bans (313 samples) |
|---|---|---|
| likelihood (FRLP) top-1 | 0.59 | 0.67 |
| empirical sampled ladder | 0.70 | 0.67 |
| free-form association | 0.52 | 0.67 |
| latent ladder from baseline likelihoods | 0.51 | 0.63 |

**Checks.** The canonical labeller recheck reproduced the association experiment exactly (22 of 22 references, combined against adjudicated 0.966, runtime against AST kappa 0.753). A blind audit of 50 Llama programs by a second judge agreed with the final labels at 0.78 (grid 0.74, shortest path 0.87), below 0.9 and flagged; most disputes were flawed attempts the audit judge labelled as other. HF and vLLM scores agreed within 0.003 to 0.007 nats per token. KV cache reuse failed its 0.02-nat tolerance (mean 0.16, maximum 0.40 nats per program, bf16 batch shape numerics), so an

fp32 language model head was adopted. For 91% of forced prompts an algorithm's own likelihood lay in [-0.7, 0.2]; the default's likelihood matched its log sampled share within 0.11 nats on the shortest path task but was 0.59 nats off on the grid task, where the instrument under weights BFS. Independent rederivations matched 34 of 34 quantities and 63 of 63 headline numbers, and a random predictor placebo never beat the ladder. Cost: USD 0.49 of OpenRouter calls and 0.64 logged GPU hours on one RTX 4090. Table 72b lists every test the replication ran.

**Table 72b.** Tests of the Llama replication (source: the artifact's `README.md` test table and the `results/*.json` files it names).

| Test | Result |
|---|---|
| Labeller recheck and label audit | pass (22/22 references, combined against adjudicated 0.966); blind audit of 50 Llama programs 0.78, flagged |
| Prompts | pass: 71/71 task texts equal the association experiment's; one BOS token; identical task suffix; ban and neutral twins of equal length (at most 13 filler tokens, no token level insertion); sampled token ids equal scored ids for 140 conditions |
| Engine agreement | pass: HF against vLLM 0.003 to 0.007 nats per token, program sums within 0.14 to 0.49 nats; 97% of outputs a single fenced code block, 94% exact retokenisation |
| KV cache reuse | fails the 0.02-nat tolerance: mean 0.16, maximum 0.40 nats per program (bf16 batch shape numerics; fp32 language model head adopted) |
| Scrubber | pass: 244 of 248 forced programs keep pass status and label after the full scrub, 4 after the conservative scrub, 8 unparsable originals dropped; on about 1,400 external programs every residual name is an attribute name |
| Likelihood sanity | an algorithm's own likelihood under its forced prompts lies in [-0.7, 0.2] for 91% of forced prompts; the default's likelihood matches its log sampled share within 0.11 nats on the shortest path task and is 0.59 nats off on the grid task |
| Edit hooks | zero coefficient leaves states exactly unchanged; no change outside the edited span or before the layer band; displacement within 4.3% of target (bf16), not 1e-3 |
| Hybrid caches | pass (difference 0.0) |
| Analysis on synthetic data | pass: ladder, substitution and pure subtraction recovered; uniform aim p under the null; unbiased under exchangeable noise, biased under crossed noise (hence a template mean sensitivity) |
| Independent rederivation | 34 of 34 quantities match to 1e-9; placebo p uniform (KS $p = 0.61$) |
| Headline rederivation | 63 of 63 headline numbers match by a separate code path; the sampled ladder beats the likelihood forecast overall (-0.112, $p < 1e\text{-}4$) and ties it on single bans ($+0.010$, $p = 0.71$); a random predictor placebo never beats the ladder |

## 2.8 Bench build: a frozen 12-task confirmation bench behind a presence pilot

**Why.** The review's arithmetic called for events, not predictors, and the hypothesis update needed tasks with a present minor algorithm so that present minor bans and a nondegenerate typicality forecast could exist. This artifact built the bench and ran a cheap baseline pilot before freezing any task. No ban prompt was rendered anywhere in it: an audit of 4,310 rendered prompts against a superset ban lexicon found no hit.

**Design.** Thirteen tasks derived from LeetCode [12] received a frozen specification in new wording on the canonical scaffold, a catalogue of three to six algorithms (54 in all), 684 hidden tests in total (parsed dataset cases filtered to the specification's bounds, plus generated edge and stress cases), order free checkers where the answer is a set, and one verified reference per algorithm. Every

reference passes at no more than a quarter of the per case limit, which is the larger of 1 s and ten times the fastest reference; all 39 planted mutants fail. Each task has a runtime signature labeller backed by an AST labeller, with a blind judge only on disagreement, gated against a gold set adjudicated by two judges and a tiebreaker (Cohen's kappa of at least 0.8, all references correct, runtime against adjudicated of at least 0.90). Qwen3-8B ran from its official 8-bit floating point checkpoint through the 8-bit Marlin kernels of vLLM, because the 16 GB GPU could not hold bf16 weights; decoding was T = 1.0, top-p 1.0, thinking off. Each task received a 50-sample pilot, 20 free-form answers per algorithm to the question `What is the standard alternative to X for this task?`, and at least 40 forced, name scrubbed programs per algorithm, scored by vLLM prompt log probabilities to give baseline likelihoods (corrected for adherence: L_corr = L + log adherence). Tasks were ordered by how much the ladder, association and proximity nominations disagree (Stage A, committed before gating) and the first 12 gated tasks were frozen with sha256 commitments.

**Table 73.** The 13 candidate tasks (source: `README.md` main table, `results/gate_summary.json`, `results/pilot_stats.json`, `results/frlp_validity.json`, `results/stageA_nominations.json`).

| Task | K | Tests | Gate: kappa runtime vs AST [95% CI] / PABAK / runtime vs adjudicated | Qwen pilot: modal algorithm (share, pass) | Baseline likelihood of the modal algorithm | Best alternative (likelihood) | Sum of exp(likelihood) | Association / proximity nomination | Llama-3.1-8B pilot modal (share) | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| kth smallest | 6 | 60 | 0.95 [0.84, 1.00] / 0.96 / 0.96 | full sort (1.00, 1.00) | -0.18 | quickselect (-1.5) | 1.05 | quickselect / heapify and pop | heapify and pop (0.34) | frozen |
| two sum | 4 | 44 | 0.92 [0.75, 1.00] / 0.94 / 0.95 | hash map (1.00, 1.00) | -0.40 | sort and two pointers (-15.5) | 0.67 | sort and two pointers / brute force | hash map (0.96) | frozen |
| find duplicate | 5 | 54 | 1.00 [1.00, 1.00] / 1.00 / 1.00 | hash set (1.00, 1.00) | 0.02 | sort, adjacent check (-12.4) | 1.02 | Floyd cycle detection / sort, adjacent check | hash set (0.43) | frozen |
| top-k frequent | 4 | 38 | 0.84 [0.60, 1.00] / 0.88 / 0.91 | sort by count (1.00, 0.94) | -0.10 | bucket sort (-15.4) | 0.91 | heap / heap | heap (0.64) | frozen |
| connected components | 4 | 57 | 1.00 [1.00, 1.00] / 1.00 / 1.00 | union find (1.00, 1.00) | -0.15 | DFS (-21.8) | 0.86 | DFS / label propagation | union find (0.77) | frozen |
| Fibonacci | 4 | 36 | 0.85 [0.65, 1.00] / 0.89 / 0.91 | iteration (1.00, 1.00) | -0.02 | fast doubling (-11.4) | 0.98 | matrix power / fast doubling | matrix power (0.79) | frozen |
| substring search | 6 | 58 | 0.95 [0.82, 1.00] / 0.96 / 1.00 | naive (1.00, 1.00) | 0.24 | KMP (-14.3) | 1.28 | KMP / Horspool | naive (0.53) | frozen |
| longest increasing subsequence | 3 | 57 | 1.00 [1.00, 1.00] / 1.00 / 1.00 | quadratic DP (1.00, 1.00) | -0.86 | patience sorting with bisect (-14.4) | 0.42 | patience sorting / patience sorting | quadratic DP (0.90) | frozen |
| merge intervals | 3 | 59 | 1.00 [1.00, 1.00] / 1.00 / 0.95 | sort and sweep (1.00, 1.00) | 0.74 | union find (-35.5) | 2.09 | interval tree / union find | sort and sweep (1.00) | frozen |
| nonoverlapping intervals | 4 | 53 | 0.89 [0.63, 1.00] / 0.94 / 0.91 | greedy by end (1.00, 1.00) | -0.83 | DP with bisect (-28.1) | 0.44 | DP with bisect / DAG longest path | greedy by end (0.56) | frozen |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| cycle detection | 4 | 59 | 0.92 [0.73, 1.00] / 0.94 / 0.95 | DFS with parent (1.00, 0.92) | 2.39 | union find (-14.2) | 10.9 | union find / union find | DFS with parent (0.59) | excluded (13th gated; 12-task cap) |
| topological order | 4 | 53 | 1.00 [1.00, 1.00] / 1.00 / 0.95 | Kahn (1.00, 1.00) | 2.44 | DFS postorder (-27.2) | 11.4 | DFS postorder / source removal | Kahn (0.98) | frozen |
| first and last position | 3 | 56 | 0.90 [0.61, 1.00] / 0.93 / 0.95 | binary search (0.98, 0.98) | 0.80 | bisect module (-12.6) | 2.21 | bisect module / exponential search | binary search (1.00) | frozen |

**Qwen3-8B is fully collapsed on all 13 tasks.** The modal share is 50 of 50 everywhere except first and last position (49 of 50), Wilson [0.929, 1.0], and the correctness conditioned Hill number is 1.0 on every task. No task has a minor algorithm at 10% or more, although the presence check is weak at N = 50 (a true share of 0.10 reaches 5 of 50 with probability about 0.57). The pilot pass rate is at least 0.92 on every task. The collapse is specific to the model: Llama-3.1-8B-Instruct, run with the same specifications and decoding, splits over several algorithms on kth smallest (heapify and pop 0.34, size k heap, full sort and quickselect), substring search (naive 0.53) and find duplicate (hash set 0.43), and passes far fewer tests. *(Correction, iteration 3: these Llama shares are of attempted programs. Its 10 correct kth smallest programs are full sort 6, heapify and pop 2, quickselect 1 and size k heap 1, so its correct mode equals Qwen3-8B's; correct-only diversity appears on substring search (naive 18, KMP 5, Rabin-Karp 1), connected components (union find 30, DFS 5), cycle detection (union find 7, DFS with parent 4) and top-k frequent (heap 8, sort by count 4), at small N. Rarefied to 10 correct programs, Llama is more diverse than Qwen3-8B on 9 of 13 tasks. Table 90, Section 3.2.4.)* The presence pilot therefore removes the present minor stratum for Qwen3-8B on every candidate task: confirmation on this bench must rest on likelihood matrices and on modal bans for Qwen3-8B, and Llama is where sampled present minor bans exist.

**Labeller, likelihoods and nominations.** All 13 tasks passed the labeller gate, 11 at round 0 and two (kth smallest, merge intervals) after one revision round with a fresh holdout (5 of 6 and 6 of 6). Both labellers got all 54 references right; runtime and AST labels agreed on 0.91 of 3,230 Qwen attempts, and the judge was needed on 8.4%. The baseline likelihood's argmax equals the pilot mode on 13 of 13 tasks, its log score beats uniform on every task (for example -0.06 against -1.61 on two sum), and the sum of exp(likelihood) lies in [0.3, 3] on 11 of 13 (cycle detection and topological order overshoot, an importance sampling overestimate under low ESS). Of the 54 cells, 34 have Kish ESS below 5. One cell stands out: quickselect on kth smallest has likelihood -1.5 (about 0.21 latent mass) while the pilot drew it 0 of 50 times. The free-form association answers are close to deterministic (the alternative to a full sort is quickselect 20 of 20 times); proximity nominates a different algorithm from association on 10 of 13 modal events; and under total collapse the sampled ladder ties over all survivors. The order recomputed after the likelihoods (Stage B) agrees with Stage A at Kendall tau 0.54. Forced adherence limits the exotic algorithms: 45 of 54 have adherence of at least 0.5, and the lowest are topological layering (0.04) and source removal (0.10), the interval tree (0.24), the DP variants of nonoverlapping intervals (0.28), index marking (0.28) and label propagation (0.29).

**Deviations and what was not done.** Scoring used vLLM prompt log probabilities instead of HF; single request and batched scoring disagreed by 0.62 nats on 1 of 5 checked programs (the others within 0.015). The forced proposal used five paraphrases over eight rounds instead of ten over four. Not produced: the unscrubbed scoring sensitivity that the likelihood screen recommended, Llama forced programs and likelihoods, ban, mention and neutral likelihood matrices, and any sampled

ban event. Cost: about 1.4 GPU hours on one RTX 2000 Ada and USD 0.14 of judge calls (1,124 calls). *(Added in iteration 3: the frozen baseline likelihoods are therefore not commensurable with the screen's, which had to move to fp32 activations because bf16 moved per program log likelihoods by up to 2.1 nats; and the recommendation to score unscrubbed programs is validated only on baseline, mention and listing events, not on ban rows, where names left in identifiers interact with the ban's own naming. Rescored on the fp32 path in iteration 3, the bench cells moved by a median 0.88 nats with the argmax unchanged on 12 of 12 tasks; Section 3.3.)*

## 2.9 Readout experiment: the ban inside the model

**Why.** The request asks for internals read with sparse autoencoders, thought anchors and lenses, and for the literal same seed test of temperature. This experiment ran those readouts on the modal bans of the two path search tasks, each beside a matched baseline on identical rows and folds. Everything in it is descriptive.

**A fresh ban pool.** Under protocol v2 (five templates by 20 samples per task), the banned share was 0.15 on the grid task (Wilson [0.09, 0.23]) and 0.13 on the shortest path task ([0.08, 0.21]); both events pass the gate. On the grid task the fourth template gave BFS 15 of 20 times and every other template 0 of 20, the third independent draw in which that one template fails (Sections 2.5 and 2.6).

**The SAE transfer test failed.** The sparse autoencoders used were Qwen-Scope's, trained on Qwen3-8B-Base and covering all 36 layers [50] (64K latents, top-k activation with $k = 50$ [51]). Before using them on the instruct model, the experiment spliced their reconstructions into the residual stream at layers 15, 20 and 27 and compared the increase in cross-entropy over 200 instruct model sequences with the increase on the base model; the preregistered bar was a ratio of at most 1.5. It failed at all three layers (Table 74). Reconstruction error was similar in the two models, but the instruct model's clean cross-entropy on its own code is far lower (0.030 against 0.260 nats), so the same error costs more. Every latent result below is labelled as computed on out-of-distribution activations, and the difference-in-means baselines lead.

**Table 74.** SAE transfer test: increase in cross-entropy when SAE reconstructions are spliced in (source: `results/transfer_summary.json`; ratio intervals 95%).

| Layer | Increase, instruct | Increase, base | Ratio [95% CI] | Pass (ratio at most 1.5) | Relative MSE instruct / base | Fraction of variance unexplained instruct / base | Cosine instruct / base |
|---|---|---|---|---|---|---|---|
| 15 | 0.173 | 0.053 | 3.29 [3.14, 3.45] | no | 0.196 / 0.165 | 0.366 / 0.288 | 0.896 / 0.914 |
| 20 | 0.150 | 0.077 | 1.95 [1.84, 2.06] | no | 0.230 / 0.203 | 0.094 / 0.243 | 0.877 / 0.893 |
| 27 | 0.097 | 0.061 | 1.59 [1.50, 1.67] | no | 0.229 / 0.197 | 0.153 / 0.325 | 0.877 / 0.896 |

**Readouts of the algorithm written.** On the forking token experiment's saved states, with nested selection that holds out one paraphrase at a time, 200 within paraphrase permutations and a maximum statistic null, the shortest path contrast (Dijkstra against Bellman-Ford/SPFA, same prompts) is decodable at the token before the commit token and at token 16 by every method, with within paraphrase held-out AUC 0.67 to 0.86 and permutation p at most 0.025 (Table 75). The grid

contrast (BFS against DFS) is not decodable within paraphrase at the commitment point: most p values exceed 0.05, no latent passes the maximum statistic null, and only 57 within paraphrase pairs exist. SAE latents never beat the difference-in-means direction or the top raw dimensions by a margin outside the intervals, which matches AxBench [34] and the MIB benchmark [52]. The positive control was weak (within paraphrase AUC at the token after the commit token peaks at 0.75 on the grid task and 0.79 on the shortest path task), because the commit token is the `deque` import for most programs of both classes (DFS 55 of 78, Bellman-Ford/SPFA 78 of 80), so the state at the commit token is the state at the first lexicon match, not at the algorithm decision. The read position, identical within a paraphrase, gave within paraphrase AUC of 0.37 to 0.62, as it must.

**Table 75.** Within paraphrase held-out AUC of each readout of the algorithm written (permutation p in brackets; source: `results/readout_i_discovery.json`). DiM is the difference-in-means direction on raw states.

| Task, layer, position | Top-1 SAE latent | Top-20 SAE latents (logistic) | DiM | Raw logistic probe | Top-1 neuron | Top-20 neurons (logistic) | DiM on SAE codes |
|---|---|---|---|---|---|---|---|
| Grid, 15, before commit | 0.482 (0.881) | 0.588 (0.234) | 0.614 (0.199) | 0.596 (0.184) | 0.535 (0.388) | 0.570 (0.343) | 0.596 (0.249) |
| Grid, 15, token 16 | 0.640 (0.114) | 0.693 (0.040) | 0.544 (0.308) | 0.649 (0.114) | 0.588 (0.149) | 0.518 (0.483) | 0.649 (0.075) |
| Grid, 20, before commit | 0.570 (0.284) | 0.649 (0.124) | 0.719 (0.025) | 0.544 (0.378) | 0.640 (0.060) | 0.649 (0.080) | 0.535 (0.388) |
| Grid, 20, token 16 | 0.640 (0.065) | 0.675 (0.030) | 0.456 (0.682) | 0.667 (0.060) | 0.658 (0.100) | 0.491 (0.527) | 0.518 (0.448) |
| Grid, 27, before commit | 0.535 (0.308) | 0.649 (0.090) | 0.614 (0.179) | 0.526 (0.378) | 0.535 (0.284) | 0.579 (0.244) | 0.711 (0.020) |
| Grid, 27, token 16 | 0.640 (0.124) | 0.588 (0.279) | 0.684 (0.065) | 0.579 (0.294) | 0.675 (0.070) | 0.596 (0.229) | 0.684 (0.075) |
| Shortest path, 15, before commit | 0.761 (0.005) | 0.676 (0.015) | 0.768 (0.005) | 0.778 (0.005) | 0.782 (0.005) | 0.771 (0.005) | 0.757 (0.005) |
| Shortest path, 15, forking token | 0.613 (0.030) | 0.634 (0.025) | 0.553 (0.229) | 0.609 (0.144) | 0.581 (0.119) | 0.683 (0.010) | 0.630 (0.020) |
| Shortest path, 15, token 16 | 0.856 (0.005) | 0.845 (0.005) | 0.856 (0.005) | 0.852 (0.005) | 0.863 (0.005) | 0.856 (0.005) | 0.838 (0.005) |
| Shortest path, 20, before commit | 0.732 (0.005) | 0.665 (0.025) | 0.736 (0.005) | 0.792 (0.005) | 0.810 (0.005) | 0.746 (0.005) | 0.708 (0.005) |
| Shortest path, 20, forking token | 0.697 (0.005) | 0.616 (0.095) | 0.606 (0.104) | 0.581 (0.139) | 0.581 (0.169) | 0.570 (0.219) | 0.708 (0.005) |
| Shortest path, 20, token 16 | 0.863 (0.005) | 0.845 (0.005) | 0.810 (0.005) | 0.817 (0.005) | 0.835 (0.005) | 0.845 (0.005) | 0.856 (0.005) |
| Shortest path, 27, before commit | 0.789 (0.005) | 0.725 (0.005) | 0.764 (0.005) | 0.694 (0.005) | 0.711 (0.005) | 0.789 (0.005) | 0.764 (0.005) |
| Shortest path, 27, forking token | 0.574 (0.095) | 0.560 (0.249) | 0.641 (0.025) | 0.694 (0.020) | 0.623 (0.060) | 0.697 (0.005) | 0.676 (0.005) |
| Shortest path, 27, token 16 | 0.856 (0.005) | 0.803 (0.005) | 0.810 (0.005) | 0.838 (0.005) | 0.827 (0.005) | 0.806 (0.005) | 0.824 (0.005) |

Applied without refitting to the fresh pool's states at the token before the commit token, the difference-in-means direction transferred best. On the grid task its within template AUC was 0.93, 0.97 and 1.00 at layers 15, 20 and 27, but only the fourth template produced both classes (15 BFS against 5 DFS, 75 pairs); the single top latent reached 0.27, 0.80 and 0.52. On the shortest path task the difference-in-means direction reached 0.66, 0.66 and 0.76 and the top-20-latent probe 0.65, 0.66 and 0.77. The top-20 latent lists overlapped above chance on the grid task (8 shared against 1.8 expected at layer 15) and at chance on the shortest path task, and shuffled labels gave AUC about 0.50.

**Attention to the ban.** The same three heads carry ban specific attention on both tasks: layer 21 head 17, layer 26 head 21 and layer 26 head 25. From the query at the token before the commit token, head 21.17 sends 0.207 (grid) and 0.098 (shortest path) more of its attention mass to the ban segment than to the neutral segment of the token matched twin, concentrated on the banned name's tokens (`breadth`, `-first`, `ijkstra`). On the grid task, runs that complied put more top head attention on the ban than runs that did not (0.135 against 0.105, within template permutation $p = 0.0025$); all noncompliant runs came from the fourth template, so this is a contrast within that template. The shortest path task shows no difference ($p = 0.96$). Attention weights are not a causal test, and the route experiment found the generated tokens' direct reads of the ban to matter only on the grid task.

**Prime and veto in the latents.** The preregistered pattern in which a forbidden concept is first activated so that it can be suppressed [53] appeared in none of the six task by layer configurations. The latents that fire on the banned algorithm's name were silent at the read position. At layer 27, both the name latents and the implementation latents of X were lower at the token before the commit token under the ban than under both twins for the same text (grid implementation latents 124 against 169 under the neutral twin; name latents 94 against 129 under the mention twin; implementation, ban against neutral, sign test $p = 2e\text{-}24$). The difference-in-means projection agreed in sign in only 2 of the 6 configurations, so most latent results here are flagged as not reproduced by the baseline.

**Thought anchors.** With thinking on and an 8,192-token cap, ten first template ban traces were sampled (five per task), and four traces per task were resampled at nine sentence boundaries with six continuations each (432 continuations), following the thought anchors method [54]. Grid traces think for 8.4k to 9.3k tokens and none completed within the cap, though 4 of 5 still emitted DFS code. In 4 of 4 resampled grid traces DFS was fixed at the first resample point (every continuation DFS from the first boundary on) and no anchor sentence exists; the traces open by restating the ban and proposing DFS within their first four sentences. Shortest path traces completed in 2.5k to 3.7k tokens (5 of 5) and showed one anchor among 32 sentences, a problem restatement. The early label detector was valid on 13 of 16 checks (0.81).

**The same seed test of temperature.** The request asked whether, with the seed held fixed, raising the temperature changes the idea or only the phrasing. Thirty seeds were run at T in {0.3, 0.7, 1.0, 1.3} with top-k = 10, for the baseline and the first template ban on both tasks (480 rows), coupling samples by inverse CDF sampling over the sorted top 10 tokens with one uniform stream per seed shared across temperatures. Under this coupling the idea never moved: the label changed against $T = 0.3$ in 0 of 30 seeds in 10 of 12 cells and in 1 of 30 in the other two, for same seed and different seed pairs alike (Table 76). With no idea level variation left, the same seed and different seed agreement on ideas is identical. Phrasing (identifier normalised AST 4-gram Jaccard below 0.8) changed less within a seed than across seeds, and the first token where a program diverged from its $T = 0.3$ twin fell at a median of 13 to 117 tokens. vLLM reproduced 5 of 8 rows byte for

byte when rerun.

**Table 76.** Same seed temperature test: rate of change against T = 0.3 for the same seed and for different seeds (30 seeds per cell; phrasing at Jaccard threshold 0.8; source: `results/same_seed.json`).

| Task | Condition | T | Idea change, same seed | Idea change, different seeds | Phrasing change, same seed | Phrasing change, different seeds | Median divergence token | Identical to T = 0.3 |
|---|---|---|---|---|---|---|---|---|
| Grid | baseline (all BFS) | 0.7 | 0/30 | 0.0 | 0.60 | 0.84 | 23 | 7 |
| Grid | baseline | 1.0 | 0/30 | 0.0 | 0.77 | 0.86 | 23 | 4 |
| Grid | baseline | 1.3 | 0/30 | 0.0 | 0.83 | 0.90 | 21 | 3 |
| Grid | ban (all DFS) | 0.7 | 0/30 | 0.0 | 0.87 | 0.94 | 27 | 3 |
| Grid | ban | 1.0 | 0/30 | 0.0 | 0.93 | 0.97 | 13 | 1 |
| Grid | ban | 1.3 | 0/30 | 0.0 | 1.00 | 1.00 | 13 | 0 |
| Shortest path | baseline (all Dijkstra) | 0.7 | 0/30 | 0.0 | 0.07 | 0.06 | 117 | 21 |
| Shortest path | baseline | 1.0 | 0/30 | 0.0 | 0.13 | 0.12 | 91 | 13 |
| Shortest path | baseline | 1.3 | 0/30 | 0.0 | 0.10 | 0.16 | 28 | 7 |
| Shortest path | ban (Bellman-Ford/SPFA) | 0.7 | 0/30 | 0.0 | 0.53 | 0.63 | 74 | 5 |
| Shortest path | ban | 1.0 | 1/30 | 0.033 | 0.59 | 0.68 | 28.5 | 2 |
| Shortest path | ban | 1.3 | 1/30 | 0.033 | 0.55 | 0.76 | 25 | 1 |

Across the four temperatures, each grid baseline seed produced 1.0 distinct idea and 2.37 phrasing clusters on average (3.17 under the ban), and each shortest path seed 1.0 idea and 1.13 phrasing clusters (1.03 ideas and 2.17 clusters under the ban). Under the ban the grid pass rate per temperature was 0.23, 0.57, 0.43 and 0.43: every program was DFS.

**Lenses: NLA and the Jacobian lens.** A dated search in this experiment resolved the two unmapped terms of the request. NLA denotes natural language autoencoders, an activation verbalizer and reconstructor trained jointly with reinforcement learning; released checkpoints cover Qwen2.5-7B-Instruct, two Gemma-3 models and Llama-3.3-70B-Instruct but not Qwen3-8B as of 2026-10-20, and training one is far outside a 16 GB budget, so NLA was not run. We refer to the request's j lens as the Jacobian lens, which reads the state of a layer through the unembedding after multiplying it by the mean Jacobian of the final state with respect to that layer. A community Jacobian lens for Qwen3-8B exists and was gated against the logit lens on 2,000 positions: its top-1 agreement with the model's own next token was below the logit lens at every source layer (layer 27: 0.104 against 0.291), so it failed and its readouts are descriptive only.

**Checks and cost.** The labeller recheck gave 0.965 on 141 gold programs. An independent script rederived 69 headline numbers with no mismatch. Not run: a same seed arm with thinking on, a Gumbel coupling sensitivity, NLA, an adaptive top up and OpenRouter descriptions of latents. Cost: about 1.05 GPU hours and USD 0.04 (112 judge calls).

## 2.10 Comparisons across this iteration's experiments

**The same modal ban, three draws under fixed wording.** Three experiments sampled the modal bans with the five frozen templates and 20 samples each in Qwen3-8B. With the wording fixed, the variation that iteration 1 saw across draws (banned shares from 0.09 to 0.72) disappears: the grid banned share was 0.14, 0.14 and 0.15 and the shortest path banned share 0.11, 0.16 and 0.13 (Table 77). What remains is concentrated in one template. On the grid task every surviving BFS program in all three draws came from the fourth template (`X is not allowed here`, with BFS as X), which produced BFS 43 times in 60 against 0 in 240 for the other four templates. On the shortest path task the third and fourth templates carry most of the failures (15 and 18 of 60, against 1 to 4 for the others). Every attempted grid program not banned in the three draws was DFS (257 of 257), and 255 of 260 on the shortest path task were Bellman-Ford/SPFA (5 other labels). Compliance is a property of the wording, and here of one wording in particular.

**Table 77.** Banned programs per template (of 20) under the modal ban in the three protocol v2 draws of Qwen3-8B (sources: route experiment `results/analysis.json#behaviour`, ban list experiment `results/landing.json`, readout experiment `results/fresh_pool_summary.json`).

| Task | Draw | First template | Second | Third | Fourth | Fifth | Banned share | Programs not banned | Pass rate |
|---|---|---|---|---|---|---|---|---|---|
| Grid | route experiment, arm a | 0 | 0 | 0 | 14 | 0 | 0.14 | DFS 86 | 0.41 |
| Grid | ban list experiment, single ban | 0 | 0 | 0 | 14 | 0 | 0.14 | DFS 86 | 0.41 |
| Grid | readout experiment, fresh pool | 0 | 0 | 0 | 15 | 0 | 0.15 | DFS 85 | 0.34 |
| Shortest path | route experiment, arm a | 1 | 0 | 4 | 5 | 1 | 0.11 | Bellman-Ford/ SPFA 86, other 3 | 0.91 |
| Shortest path | ban list experiment, single ban | 2 | 1 | 6 | 7 | 0 | 0.16 | Bellman-Ford/ SPFA 83, other 1 | 0.95 |
| Shortest path | readout experiment, fresh pool | 1 | 0 | 5 | 6 | 1 | 0.13 | Bellman-Ford/ SPFA 86, other 1 | 0.96 |

**The substitution test in two models.** Neither model gave a robust branch (Table 78). In Qwen3-8B the pooled X specific share sits near 0.55 to 0.66 with intervals that straddle 0.5, and only the unscrubbed programs, which pass the validity gate, place the interval above 0.5. In Llama the point estimate is higher (0.80) but its interval is wider and the post hoc noise analysis shows the importance sampling noise has not been removed. Three things replicate across the families: the gains are far from pure subtraction, the X specific residual is not aimed at the prime and veto profile in either model, and the generic perturbations do not form a coherent ladder in either (mean cosines 0.34 and -0.02 in Qwen3-8B, 0.14 in Llama).

**Table 78.** X specific share in the two models (sources: likelihood screen `results/c1.json`, `results/sensitivity.json`; Llama replication `results/c1.json`).

| Model | Reference | Scope | X specific share | 95% CI | Branch |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| Qwen3-8B | neutral twin | grid | 0.63 | [0.28, 0.83] | undetermined |
| Qwen3-8B | neutral twin | shortest path | 0.41 | [0.10, 0.61] | undetermined |
| Qwen3-8B | neutral twin | pooled | 0.55 | [0.25, 0.67] | undetermined |
| Qwen3-8B | mention twin | grid | 0.66 | [0.51, 0.98] | X specific, unaimed |
| Qwen3-8B | mention twin | shortest path | 0.65 | [0.26, 0.79] | undetermined |
| Qwen3-8B | mention twin | pooled | 0.66 | [0.49, 0.87] | undetermined |
| Qwen3-8B | neutral twin, unscrubbed programs, first template | pooled | 0.69 | [0.51, 0.84] | X specific, unaimed |
| Llama-3.1-8B-Instruct | neutral twin | grid | 0.80 | [0.28, 0.94] | not resolved at this N |
| Llama-3.1-8B-Instruct | mention twin | grid | 0.78 | [0.61, 0.91] | X specific, unaimed |

*[Figure fig_c1share: generated in the report phase, not in this draft]*

**Priming and the route agree across models.** A ban of an algorithm the model does not write raises that algorithm's likelihood in 8 of 9 cases in Qwen3-8B (+2.5 to +26.9 nats) and 5 of 6 in Llama (+1.8 to +7.6), and in both models the bans that lower their own label are those of algorithms the model does write, plus A*. The likelihood drop of the banned default travels by the prompt route in both models: F_prompt 0.85 [0.61, 1.20] in Qwen3-8B and 0.84 [0.62, 1.46] in Llama, with the direct route near zero (0.00 and -0.21). Only the Qwen3-8B grid behaviour shows a second, direct component. *(Correction, iteration 3: both numbers are the likelihood readout, which on the grid task is dominated by the opening fork (ESS 1 to 3 of 30), and in Llama it is the only readout, on an instrument that failed its gate there. The route is restated per task and per readout in Section 3.1, and on the bench the behavioural route is task dependent, with a pooled direct share of 0.74; Section 3.6.)*

## 2.11 What iteration 2 established, what it ruled out, and what it abandoned

Everything below is still descriptive: two screen tasks, two models, one modal ban event per task and model for the route test, and a likelihood instrument whose validity gate failed in both models in its preregistered (name scrubbed) form.

**Established.**

1. *The iteration-1 record, corrected.* On the shortest path task the ban's substitute is Bellman-Ford/SPFA in five of five draws (272 of 272 compliant programs, Wilson [0.986, 1.000]); on the grid task DFS is the realised mode in four of five draws (116 of 128, [0.843, 0.946]), with no clear mode in the HF engine geometry draw. Under fixed wording three new Qwen3-8B draws landed on DFS (257 of 257 programs not banned) and Bellman-Ford/SPFA (255 of 260). The cumulative BFS and DFS ban does not replicate. Under one statistic, no ban and no diversity knob reaches a second correct algorithm (rarefied Hill-1 of at least 2) for less than 20 pass points on either task; the semantic seed on the shortest path task comes closest (1.935 at 7 points). On gated events, no forecasting family can advance (three informative events, largest margin 0.05 nats).

2. *The generic perturbation rival holds on the shortest path task and fails on the grid task.* Every nonsemantic perturbation that left Dijkstra (8 of 8 rows) landed mostly on Bellman-Ford/SPFA, random directions included; on the grid task no nonsemantic perturbation left BFS among correct programs *(corrected in iteration 3: one did, barely; temperature 1.3 with top-k = 10 gave DFS in 1 of 95 correct programs)*, while every ban reached DFS. On the shortest path task a ban and an arbitrary push end in the same place.
3. *The ban acts through the prompt states, before the first generated token.* In cache patching the prompt route carried 0.85 [0.61, 1.20] of the banned algorithm's likelihood drop and the direct route 0.00 [-0.08, 0.12], with the task specification states, not the read position alone, carrying it. The mechanism is an opening line veto: under the grid ban 0 of 100 programs open with the default's import line against 99 of 100 under the neutral twin, and the prompt route arm reproduces this exactly. On the shortest path task behaviour follows the prompt route completely; on the grid task generated tokens also read the ban directly (hiding the ban from them restores BFS to 0.47). Llama's likelihood readout agrees (F_prompt 0.84). *(Correction, iteration 3: the heading and first sentence state the route more strongly than its evidence. Per task and per readout: on the shortest path task the prompt route carries the ban in likelihood (0.82) and in behaviour (0.98 [0.89, 1.07], direct 0.05 [0.01, 0.09]); on the grid task the likelihood readout is dominated by the opening fork (0.87), behaviour is mixed (prompt 0.50 [0.39, 0.62], direct 0.35 [0.25, 0.45]) and the natural proposal readout puts the direct route at 0.16 [0.11, 0.21], an interval that excludes 0; Llama has the likelihood readout only (0.84 [0.62, 1.46]). The direct route is not zero. Splicing the read position alone carries none of the effect (F_R about 0).)*
4. *The read position knows, in part, how hard a wording will be obeyed, and not what replaces X.* Across 110 wordings the projection on a shared ban direction correlates with the banned share at -0.41 [-0.69, -0.17] ($p = 0.0003$), short of the -0.5 bar; the substitute is not decodable beyond that projection. *(Added in iteration 3: splicing the read position alone carries none of the ban's effect (F_R about 0), so this site is decodable but causally inert; a causal test of a shared prohibition direction at the task specification positions was run in iteration 3 and failed, Section 3.7.)*
5. *Naming primes.* A ban of an absent algorithm raises that algorithm's likelihood in both models (8 of 9 and 5 of 6), and the mention twin raises it more. The iteration-1 sham promotions are reproduced by the shared (ladder) component of the gains and are not X specific.
6. *Ban lists fail by reimplementation.* Every four or five name grid ban failed (avoidance 0.00 to 0.37); the model wrote BFS and DFS in the same mix whatever the list contained (BFS share 0.63 to 0.72, chi squared $p = 0.48$), disguising 41% to 92% of the banned programs. On the shortest path task the model will not write bidirectional Dijkstra, Floyd-Warshall or Dial even when each is the only algorithm allowed, though it can write Floyd-Warshall and Dial when told to *(qualified in iteration 3: forced Floyd-Warshall is labelled as requested and correct 40 of 40 times, but forced Dial, from the descriptive prompt, is labelled as requested 36 of 40 times and passes only 7 of 40, so only Floyd-Warshall is written cleanly on request)*.
7. *No decision locus among the saved states.* On the grid task the within one ban signal is fixed per paraphrase; on the shortest path task the first state that beats the read position by 20 points is a readout of text already written (`import heapq` against `from collections import deque`, decided at generated token 3). The forking token, the opening line veto and the locus probe all point at the first three generated tokens, which no artifact has saved.
8. *Temperature does not change the algorithm even for a fixed seed.* With seeds coupled across temperatures, the algorithm changed in 0 of 30 seeds in 10 of 12 cells and 1 of 30 in the

other two, while phrasing changed in 7% to 100% of seeds depending on task and temperature, and less within a seed than across seeds.

9. *Qwen3-8B is collapsed on 13 new tasks as well.* The pilot found a modal share of 1.00 on 12 tasks and 0.98 on one; Llama-3.1-8B-Instruct is more diverse on the same tasks *(qualified in iteration 3: in attempted programs; counted over correct programs and rarefied to 10, Llama is more diverse on 9 of 13 tasks, but much of its attempted diversity is failing code, Section 3.2.4)*. The confirmation bench is frozen (12 tasks, labellers gated) with no ban prompt run on it.

**Ruled out, or not supported.**

- *Geodesic proximity on the read position manifold as the predictor of the shortest path substitute.* Floyd-Warshall is the nearest survivor to Dijkstra in 100% of paraphrase subsamples at every layer and k tested, Bellman-Ford/SPFA is never nearest, and no ban landed on Floyd-Warshall. On the grid task proximity and typicality coincide and cannot be separated.
- *An aimed substitution.* In neither model is the X specific part of the ban gains aimed at the prime and veto profile (Qwen3-8B -0.10, $p = 0.80$; Llama 0.067, $p = 0.35$) or at the stated alternative. *(Correction, iteration 3: not detectable with the current instrument, rather than ruled out; Section 3.2.1.)*
- *A coherent latent ladder.* The generic perturbation profiles do not agree with each other in either model, so the rival in its preregistered form, a fixed ordering any perturbation reveals, was not found either. *(Correction, iteration 3: this item belongs among the effects that an uncalibrated instrument could not detect, not among the accounts ruled out, and it conflicts with item 2 above. Reconciled: behaviour shows a shared top rung on the shortest path task, where 8 of 8 nonsemantic perturbations that left Dijkstra landed mostly on Bellman-Ford/SPFA, and on the grid task only bans reach DFS. The order of the survivors below the top rung is unmeasured: it is unobservable in behaviour under the collapse, and the likelihood cosines that this item rests on come mostly from survivors with an ESS near 1.)*

*(Added in iteration 3: the iteration-2 preregistered predictions and their outcomes, and the outcome of the iteration-2 forecast contest of the four frozen families (NOT RUN on any new event), are listed in Section 3.1.)*

- *A veto at the signature token.* The lens found no layer at which the ban lowers the banned algorithm at the commitment token, because the fork is at the first token; the signature tokens carry 2.4% and 11.6% of the likelihood drop.
- *Activation of a forbidden concept in order to suppress it.* The prime and veto latent pattern appeared in 0 of 6 configurations; the name latents of X were lower under the ban than under both twins.

**Dead ends, with the evidence that closed them.**

- *The likelihood instrument on name scrubbed programs as a calibrated probability.* It failed calibration in Qwen3-8B (default share 0.83, DFS mass under a baseline that never samples DFS) and in Llama (implied BFS 0.43 against sampled 0.77), with effective sample sizes mostly below 5 in every experiment that used it (98% of gate cells, 858 of 1,095 route cells, 97% of ban list cells, 76% of Llama cells). Unscrubbed programs pass the Qwen3-8B gate (calibration 0.95 and 1.00, 1.18 nats, 17 of 19), so the scrubbing, not the instrument, is the closed part.
- *Extending the grid catalogue toward ten algorithms.* Only five of ten algorithms passed the keep

rule; the model writes A* as unit weight Dijkstra, unit weight Dijkstra as BFS, and jump point search and bidirectional A* as BFS and A*.

- *The literal top five landing test as a test of edge against opposite side.* The model does not comply with a five name ban (avoidance 0.00), so no survivor mass exists to place on either side; the question is answered behaviourally (reimplementation of the defaults) rather than geometrically.
- *Qwen-Scope base model SAEs on the instruct model.* The spliced in cross-entropy ratio was 3.29, 1.95 and 1.59 at layers 15, 20 and 27 against a 1.5 bar; latents never beat the difference-in-means direction.
- *The community Jacobian lens for Qwen3-8B.* Its next token agreement was below the logit lens at every layer (0.104 against 0.291 at layer 27).
- *NLA.* No natural language autoencoder exists for Qwen3-8B as of 2026-10-20.
- *A presence piloted present minor stratum for Qwen3-8B.* All 13 bench tasks are fully collapsed in the pilot, so present minor bans and a sampled typicality forecast do not exist for this model on this bench.
- *The free-form association answer as evidence about mechanism.* It predicts the substitute, but simulation of its own behaviour explains that success under either branch of the substitution test.

## 2.12 What the next iteration takes from this one

The title claim is still open, and the reason has changed. Iteration 1 could not separate substitution from a latent ladder because its events were too few and its typicality forecast was degenerate. This iteration measured every single ban in two models with a likelihood instrument and found neither account clean: the gains are not pure subtraction, they are not a coherent ladder, and their X specific part is not aimed where association or prime and veto predict. *(Correction, iteration 3: none of these three is detectable with the current instrument; Section 3.2.1.)* The confirmation bench exists to settle the share with task level clusters. Its inputs are now fixed, and so are its limits. Qwen3-8B is collapsed on all 12 frozen tasks, so its confirmation must use likelihood matrices on unscrubbed programs, which the gate supports, plus modal bans; Llama supplies sampled present minor bans. The ban, mention, neutral and nonce matrices, the Llama forced programs and every sampled ban event on the bench remain to be run, with forecasts hash committed first.

The route result redirects the mechanistic work. A ban is applied by the prompt states as a veto on the default's first line, and on the grid task also by direct reads during generation. The next measurements are therefore at generated tokens 1 to 3, the states no artifact has saved: a decoder and an activation patch there would locate the decision the locus probe could not find, and would say whether the opening line veto is the whole of the algorithm choice on the shortest path task. On the grid task the direct component needs its own localisation (which layers and which tokens read the ban after `def solve`), since the three attention heads that attend to the banned name are a descriptive lead, not a causal one. The template effect must enter every analysis as a random effect; one wording, the fourth template (`X is not allowed here`), accounts for every grid failure in three draws.

Request items still open after this iteration: the activation space manifold plots with centroids coloured by implemented count at a validated locus (no locus validated yet; *corrected in iteration 3: the plots themselves exist, 13 Isomap figures drawn by the ban list experiment at layer 24 and the token before the commit token, Section 3.2.2, but not at a validated locus*), a working SAE readout (the

transfer test failed; a short fine-tuning on instruct activations, or a dictionary trained on them, is the alternative), and a same seed arm with thinking on. The top five prohibition, natural BFS against DFS internals, thought anchors, the same seed test, the logit and tuned lens, and the mapping of NLA and the Jacobian lens were run here. *(Added in iteration 3: the readout experiment did not save generated tokens 1 to 3 because the artifacts of one iteration run in parallel, and its plan was frozen before the audit named those positions; its states stop at the token before the commit token and token 16.)*

# Iteration 3: ban-free forecasts on the bench, the route of a ban across twelve tasks, and how far any knob reaches

## 3.1 Why this iteration ran what it ran

**The review of the iteration-2 record.** The review raised nine major and three minor items. Five concerned evidence the record held but read too strongly. The likelihood substitution test could not settle where a ban sends the mass: on the only rows with known behaviour, its algorithm specific residual pointed away from the substitute the model actually wrote, and its noise correction was never checked on Qwen3-8B. The record also carried two verdicts on the rival account that contradicted each other (a shared second choice on the shortest path task, and the absence of a coherent ladder listed among the accounts ruled out). The closing summary stated that the direct route carried none of the ban, although two of the three readouts in the same artifact showed a direct route. The cross-model contrast rested on failing programs: Llama's correct grid programs were all BFS. The knob verdict was paraphrased past its own metric. Four items concerned work that existed but was missing from the record: the ban list experiment's 13 manifold figures with centroids coloured by implemented count, the out-of-catalogue programs that no one had inspected, positioning rows for the iteration-2 claims, and a statement that the forecast contest of the four frozen families had never been run. The remaining items were the preregistered predictions and their outcomes, the inherited bench instrument, and five factual slips. The corrections are made in place in Sections 1.12 and 2.3 to 2.12, each marked; Table 80 maps every item to where it is answered.

**The hypothesis update.** The update flipped the predicted branch to the side the evidence leaned toward and moved the decision from likelihood cells to sampled programs. Its title claim is that a prompt prohibition removes the named default and does not choose the replacement: the mass that leaves the default lands on what we call the *next rung*, the survivor that the model's own distribution puts next once the default is gone. The rival is substitution specific to X, under which the landing follows the model's stated alternative to X (association) or the survivor whose programs look most like X's (proximity). Under total collapse the next rung cannot be read from sampled frequencies, so the update introduced ban-free counterfactuals that expose it without any ban text.

- We call the *signature veto* a decoding-time mask on the default's distinctive token n-gram: the shortest n-gram (up to 6 tokens) present in at least 80% of the default's baseline programs that minimises its largest share in the other algorithms' forced programs. The one-shot form masks the completing token the first time it comes up; the persistent form masks it at every occurrence; the iterated form adds the n-grams of re-emerged default programs for up to three rounds. We call the landing of the last veto round that leaves at least 20 attempts off the default the *subtraction forecast* of the event.
- We call the nonce ban, the novelty instruction and three random directions added at the task

specification positions at the ban's own displacement norm generic pushes, and the pooled landing of the generic pushes that move at least 20 attempts off the default the *ladder forecast*.

- We call the model's frozen free-form answer when asked for the standard alternative to X the *association forecast*, and the survivor whose forced programs are nearest to the default's by tree edit distance on identifier-normalised syntax trees [64] the *proximity forecast*.
- We call an *exposure ban* a prompt that shows a correct program of a non-default algorithm Y and then asks for a correct solution that does not use Y. Showing Y makes the model copy it, so Y becomes the default to ban, and association and ladder disagree wherever Y's stated alternative is not the model's context-free default.

The update also added a causal test of what we call the shared prohibition direction, the mean displacement that the bans of the other tasks cause at the task specification positions (Section 3.7 gives its construction), a locus probe at generated tokens 1 to 8, the request's top-five prohibition asked again as a cumulative token veto, and the request's own long-form novelty instruction, an instruction to invent an algorithm that asks for a design rationale of about 2,000 words before the code. The event gate stayed as in Section 2.2 (avoidance at least 0.5 and at least 20 attempts not banned). The frozen contest rule names its outcomes, and we call these outcomes branches, written as the rule prints them: SUBTRACTION/LADDER when a ban-free family (subtraction or ladder) advances, followed by the family that advanced; SUBSTITUTION when association or proximity advances; NO FAMILY ADVANCES when none does, which we call aimed but unresolved when some family's top-1 still names the landing more often than chance; and, when no family's top-1 beats chance, the branch we call X specific but unaimed, the name Section 2.2 gave the same outcome. In the n-gram code spans of this section, ␣ marks a space and \n a newline. The likelihood instrument was demoted to a descriptive forecast of mass. The two path search tasks became the screen, and the 12 frozen bench tasks in two models became the confirmation; the two are never pooled.

**The iteration-2 preregistered predictions and their outcomes.** These were not listed in the iteration-2 record; they are recorded here as the hypothesis update recorded them.

1. A gain for DFS under the BFS ban specific to X, larger than the ladder can match: failed. The largest residual of that row belongs to IDDFS ($+7.99$) and DFS's is $+0.83$.
2. The direct route at generation time carries at least two thirds of the ban: failed. The direct share was 0.00 in likelihood, 0.35 in grid behaviour and 0.05 in shortest path behaviour; the preregistered alternative, decided before writing, was met on the likelihood readout.
3. A compliance correlation at the read position of -0.5 or below: missed (-0.41 [-0.69, -0.17], $p = 0.0003$).
4. The validity gate of the likelihood instrument: failed in both models in its preregistered, name scrubbed form.
5. The locus rule: no position passed on the grid task, and the shortest path pass was a readout after commitment.
6. The positive knob target: not met.
7. The forecast contest of the four frozen families (ladder, association, prime and veto, proximity) under the preregistered rule: not run on any new event. Only the three gated iteration-1 events existed, with a largest margin of 0.05 nats, and the ban list landing forecasts were degenerate.

**The route, restated per task and per readout.** The closing summary of iteration 2 said the direct route carried none of the ban. Per readout, the record holds this. On the shortest path task the prompt route carries the ban in likelihood (0.82) and in behaviour (0.98 [0.89, 1.07], against a direct share of 0.05 [0.01, 0.09]). On the grid task the likelihood readout is dominated by the opening fork (0.87), while behaviour is mixed (prompt 0.50 [0.39, 0.62], direct 0.35 [0.25, 0.45]) and the natural proposal readout puts the direct route at 0.16 [0.11, 0.21]. Llama has the likelihood readout only (0.84 [0.62, 1.46]). Splicing the read position alone carries none of the effect ($F_R$ about 0), even though the read position projection correlates with compliance across wordings. Decodable there is not actionable there [55].

**The strategy.** Seven artifacts ran (Table 79). One audit without generation repaired the record. Three experiments sampled the bench bans that decide the title claim, each committing its forecasts by sha256 hash before its own first ban sample, because artifacts of one iteration run in parallel and cannot share commitments: modal bans in Qwen3-8B [5], exposure bans in Qwen3-8B, and modal and present minor bans in Llama-3.1-8B-Instruct [22]. Three experiments asked where and how far: the behavioural route on the bench, the internal locus with the shared prohibition direction, and the reach of prompt bans and token vetoes with the novelty question. The budget went to sampled events rather than likelihood variants, as the review asked.

**Table 79.** The seven artifacts of iteration 3, the question each was asked, and its hardware and spend.

| Artifact | Question | Model and hardware | OpenRouter spend |
|---|---|---|---|
| Record audit | Repair the iteration-2 record; likelihood diagnostics; manifold figures; novelty audit of every correct program outside the catalogue; correct-only Llama compositions; knob table with a class rule; positioning | No generation; CPU only | USD 0.42 (blind judges) |
| Bench forecast experiment | Do ban-free forecasts, sealed before the bans, predict where the 12 modal bench bans land in Qwen3-8B? | Qwen3-8B bf16; one RTX 4090, about 45 GPU minutes | USD 0.024 |
| Exposure experiment | Shown a correct program of Y and told not to use Y, does the model go to Y's stated alternative or back to its default? | Qwen3-8B bf16; one 23 GB NVIDIA GPU, about 1.0 GPU hour | USD 0.096 |
| Llama bench experiment | The same forecast contest in Llama-3.1-8B-Instruct, with present minor bans and a path task screen | Llama-3.1-8B-Instruct bf16; one 23 GB NVIDIA GPU, 0.98 GPU hours of logged generation (about 1.8 hours held) | USD 0.48 |
| Bench route experiment | Does a ban act through the prompt states or by generated tokens reading it, on the 12 bench tasks, in behaviour? | Qwen3-8B bf16 (HF); one 23 GB NVIDIA GPU, about 1.8 GPU hours | USD 0.105 |
| Locus experiment | Is there a shared prohibition direction; is the algorithm decided at generated tokens 1 to 8; do the ban-attending heads matter; do the base model SAEs transfer under a fairer criterion? | Qwen3-8B with 8-bit floating point weight and activation linears (HF); one 16 GB RTX 2000 Ada GPU | USD 0.22 |
| Reach experiment | How far do cumulative token vetoes and bans that leave one survivor push the model; does the request's long-form novelty instruction produce a correct novel algorithm? | Qwen3-8B bf16 (path tasks) and the official 8-bit floating point checkpoint (bench tasks); one 23 GB NVIDIA GPU | USD 0.49 |

**Table 80.** Review items on the iteration-2 record and where this report answers them.

| Review item | What the review found | Where it is answered |
|---|---|---|
| Likelihood substitution test | Residuals point away from the realised substitute; tail cells dominate; noise correction unchecked on Qwen3-8B; impossible weighted share; mention aim omitted | Corrections in Sections 2.4, 2.11 and 2.12; Section 3.2.1 |
| Two verdicts on the rival | A shared second choice in behaviour against a claimed absence of any coherent ladder | Correction in Section 2.11; Section 3.2.1; the bench tests in Sections 3.3 to 3.5 |
| Missing manifold figures | 13 Isomap figures with implemented counts existed; circle result overstated | Corrections in Sections 2.6 and 2.12; Section 3.2.2 |
| Cross-model contrast on failing programs | Llama's correct grid programs are BFS 20 of 20 | Corrections in Sections 2.7, 2.8 and 2.11; Section 3.2.4 |
| Samples, not variants; forecast contest outcome | The contest of the four frozen families never ran; budget arithmetic missing | Section 3.1 (outcome NOT RUN); Section 3.2.1 (arithmetic); Sections 3.3 to 3.5 (sampled bench events) |
| Positioning of iteration-2 claims | No neighbour rows for compliance at the read position, the route, priming, ban lists or SAE transfer | Section 3.2.6 |
| Route stated too strongly | A direct route of zero claimed against two readouts with a direct route | Corrections in Sections 2.10 and 2.11; this section; Section 3.6 |
| Novelty never inspected | About 25 correct programs outside the catalogue unread; the request's 2,000-word novelty instruction never run | Section 3.2.3; Section 3.8 |
| Knob verdict past its metric | A 2:1 split cannot reach Hill-1 of 2; class rule undeclared | Correction in Section 2.3.6; Section 3.2.5 |
| Minor: preregistered predictions; tokens 1 to 3 | Failed targets not listed; why the states were not saved | This section; correction in Section 2.12 |
| Minor: bench instrument | 8-bit prompt log probabilities; unscrubbed variant unvalidated on ban rows | Correction in Section 2.8; Section 3.3 (fp32 rescoring) |
| Minor: five factual slips | Table 66 prompt labels; DFS 1 at T = 1.3; forced Dial; TOP residue; an unmarked insertion in Section 1.12 | Corrections in Sections 1.12, 2.6 and 2.11 |

## 3.2 The record audit: likelihood diagnostics, manifold figures, novelty, correct-only Llama, knob classes

The record audit ran first and on CPU only. It read the iteration-1 and iteration-2 artifacts in place, generated nothing, and spent USD 0.42 on blind LLM judges for the novelty audit. Every number it produced names its source file. An independent script rederived 72 of 72 headline checks from the raw files, and all 7 placebo tests failed as they should.

### 3.2.1 The likelihood substitution test cannot decide where the mass goes

The review's central objection was that the substitution test of Section 2.4 rests on cells that are behaviourally irrelevant and close to unestimable. The audit tested this directly on every sampled single ban in both models: if the likelihood instrument measured where a ban sends the mass, the survivor with the largest gain should be the one the model writes. Table 81 gives every event.

**Table 81.** Every sampled single-ban event with likelihood cells: the realised landing and the argmax of three likelihood readouts (gain against the neutral twin, its algorithm specific residual after the ladder fit, and the implied probability mass under the ban). An argmax equal to the modal landing is marked as a hit. Source: record audit `tables/c1_argmax_vs_landing.csv`.

| Model | Task | Draw | Banned | Kind | Attempted, not banned | Correct, not banned | Argmax of gain | Argmax of residual | Argmax of implied mass |
|---|---|---|---|---|---|---|---|---|---|
| Qwen3-8B | Grid | readout experiment, fresh pool | BFS | modal | DFS 85 | DFS 28 | IDDFS | IDDFS | DFS (hit) |
| Qwen3-8B | Shortest path | readout experiment, fresh pool | Dijkstra | modal | Bellman-Ford/SPFA 86, other 1 | Bellman-Ford/ SPFA 83 | Dial | Dial | Bellman-Ford/ SPFA (hit) |
| Qwen3-8B | Grid | association experiment | A* | absent (sham) | BFS 100 | BFS 72 | IDDFS | IDDFS | DFS |
| Qwen3-8B | Grid | association experiment | BFS | modal | DFS 90 | DFS 38 | IDDFS | IDDFS | DFS (hit) |
| Qwen3-8B | Grid | association experiment | bidirectional BFS | absent (sham) | BFS 99, other 1 | BFS 92 | IDDFS | IDDFS | BFS (hit) |
| Qwen3-8B | Grid | association experiment | DFS | absent (sham) | BFS 96 | BFS 86 | IDDFS | IDDFS | BFS (hit) |
| Qwen3-8B | Grid | association experiment | unit weight Dijkstra | absent (sham) | BFS 92, DFS 8 | BFS 69, DFS 2 | IDDFS | IDDFS | DFS |
| Qwen3-8B | Grid | association experiment | IDDFS | absent (sham) | BFS 88, DFS 12 | BFS 76, DFS 4 | bidirectional BFS | bidirectional BFS | DFS |
| Qwen3-8B | Shortest path | association experiment | Bellman-Ford/SPFA | absent (sham) | Dijkstra 94, other 1 | Dijkstra 92 | Dial | Floyd-Warshall | Dijkstra (hit) |
| Qwen3-8B | Shortest path | association experiment | bidirectional Dijkstra | absent (sham) | Dijkstra 97, Bellman-Ford/SPFA 3 | Dijkstra 97, Bellman-Ford/ SPFA 3 | Dial | Dial | Dijkstra (hit) |
| Qwen3-8B | Shortest path | association experiment | Dial | absent (sham) | Dijkstra 95, Bellman-Ford/SPFA 5 | Dijkstra 95, Bellman-Ford/ SPFA 2 | Bellman-Ford/SPFA | Bellman-Ford/SPFA | Dijkstra (hit) |
| Qwen3-8B | Shortest path | association experiment | Dijkstra | modal | Bellman-Ford/SPFA 90, other 1 | Bellman-Ford/ SPFA 86 | Dial | Dial | Bellman-Ford/ SPFA (hit) |
| Qwen3-8B | Shortest path | association experiment | Floyd-Warshall | absent (sham) | Dijkstra 77, Bellman-Ford/SPFA 23 | Dijkstra 68, Bellman-Ford/ SPFA 3 | Dial | Dial | Dijkstra (hit) |
| Qwen3-8B | Grid | route experiment, full ban | BFS | modal | DFS 86 | DFS 31 | IDDFS | IDDFS | DFS (hit) |
| Qwen3-8B | Shortest path | route experiment, full ban | Dijkstra | modal | Bellman-Ford/SPFA 86, other 3 | Bellman-Ford/ SPFA 80 | Dial | Dial | Bellman-Ford/ SPFA (hit) |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Qwen3-8B | Grid | ban list experiment, single ban | BFS | modal | DFS 86 | DFS 30 | IDDFS | IDDFS | DFS (hit) |
| Qwen3-8B | Shortest path | ban list experiment, single ban | Dijkstra | modal | Bellman-Ford/SPFA 83, other 1 | Bellman-Ford/ SPFA 79 | Dial | Dial | Bellman-Ford/ SPFA (hit) |
| Llama-3.1-8B | Grid | Llama replication | BFS | modal | DFS 63, other 6, A* 1, unit weight Dijkstra 1 | DFS 7, other 1 | A* | A* | DFS (hit) |
| Llama-3.1-8B | Grid | Llama replication | DFS | absent (sham) | BFS 39, other 4 | BFS 7 | A* | A* | BFS (hit) |
| Llama-3.1-8B | Grid | Llama replication | unit weight Dijkstra | absent (sham) | BFS 30, DFS 18, other 1 | BFS 12, DFS 1 | A* | A* | DFS |
| Llama-3.1-8B | Shortest path | Llama replication | Dijkstra | modal | other 37, Bellman-Ford/SPFA 18, Floyd-Warshall 2 | Bellman-Ford/ SPFA 4, other 1 | n/a (three algorithms) | n/a | Bellman-Ford/ SPFA (hit) |
| Llama-3.1-8B | Shortest path | Llama replication | Bellman-Ford/SPFA | absent (sham) | Dijkstra 33, other 14, Floyd-Warshall 1 | Dijkstra 13 | n/a | n/a | Dijkstra (hit) |
| Llama-3.1-8B | Shortest path | Llama replication | Floyd-Warshall | absent (sham) | Dijkstra 25, other 11, Bellman-Ford/SPFA 9 | Dijkstra 6, Bellman-Ford/ SPFA 1 | n/a | n/a | Dijkstra (hit) |

The gains never point where the model writes (Table 82). Over the 17 Qwen3-8B events the argmax of the gain and the argmax of the residual hit the realised landing 0 times, against a chance rate of 0.22; under the BFS ban the largest residual is always IDDFS and under the Dijkstra ban always Dial. The implied mass, which weights each survivor by how much probability it has rather than by how much its log probability moved, hits 14 of 17, and 8 of 8 modal bans. The same split holds in Llama. A log gain of 18 nats on a survivor with negligible mass counts as much as the realised substitute's gain, and those tail cells are the ones with effective sample sizes near 1.

**Table 82.** Hit rate of each likelihood argmax on the realised landing (Wilson 95% intervals; chance is one over the number of survivors, averaged over events). Source: record audit `results/c1_diagnostics.json`.

| Model | Events | Gain (neutral twin) | Residual (neutral twin) | Gain (mention twin) | Residual (mention twin) | Implied mass | Chance |
|---|---|---|---|---|---|---|---|
| Qwen3-8B, all single bans | 17 | 0/17 [0.00, 0.18] | 0/17 [0.00, 0.18] | 1/17 [0.01, 0.27] | 0/17 [0.00, 0.18] | 14/17 [0.59, 0.94] | 0.22 |
| Qwen3-8B, modal bans | 8 | 0/8 [0.00, 0.32] | 0/8 [0.00, 0.32] | 0/8 | 0/8 | 8/8 [0.68, 1.00] | 0.23 |
| Llama-3.1-8B, all single bans | 6 (3 with gains) | 0/3 | 0/3 | 2/3 | 0/3 | 5/6 [0.44, 0.97] | 0.22 |
| Pooled | 23 (20 with gains) | 0/20 [0.00, 0.16] | 0/20 [0.00, 0.16] | 3/20 [0.05, 0.36] | 0/20 [0.00, 0.16] | 19/23 [0.63, 0.93] | 0.22 |

*(Correction, iteration 4: Table 82 treats behavioural replicates as independent events. The 17 Qwen3-8B events are 11 distinct likelihood rows, and the 8 modal bans are 2 rows (the BFS and Dijkstra bans) each scored four times against identical cells, so 8 of 8 is 2 of 2. Per distinct row, the gain and the residual argmax hit 0 of 11 and the implied mass 8 of 11, and 6 of those 8 are bans of absent algorithms, where the landing is the default and the argmax is trivially right. The Wilson intervals computed over replicates are withdrawn; the per-row table is in Section 4.2.6.)*

Restricting the test to estimable mass removes every row. The review asked for the test recomputed on survivors with at least 1% implied mass and an ESS of at least 5; the audit predicted before computing that few or no rows would survive, and none did (0 of 11 Qwen3-8B rows keep three such survivors). The substitution test is not decidable on estimable mass.

The noise correction was too small. Resampling templates and then programs within algorithm halves, the bootstrap variance of the mean gain is a median 2.68 times the replicate noise term that the frozen correction subtracted (range 1.22 to 4.26 over the 11 rows; 1.59 for a one stage bootstrap that resamples programs only). The Llama ratio was 4.8 to 11.3. Because the correction under-subtracts, both the F statistic behind the reading that the gains are not pure subtraction and the X specific share are inflated. The sums are also skewed upward, which the percentile intervals of Section 2.4 hid; basic intervals (twice the point estimate minus the bootstrap quantiles), which correct for that skew, are given in Table 83. The ladder term's basic interval crosses zero on both tasks.

**Table 83.** Qwen3-8B ladder and X specific sums of squares and the X specific share under the neutral twin reference: point estimate, bootstrap mean, and percentile and basic 95% intervals (2,000 two stage draws). Source: record audit `results/c1_diagnostics.json#intervals`.

| Scope | Quantity | Point | Bootstrap mean | Percentile 95% | Basic 95% |
|---|---|---|---|---|---|
| Grid | V_u (ladder) | 239.3 | 396.4 | [135.4, 873.0] | [-394.4, 343.1] |
| Grid | V_s (X specific) | 412.5 | 441.5 | [207.0, 729.0] | [95.9, 617.9] |
| Grid | share | 0.633 | 0.541 | [0.270, 0.781] | [0.484, 0.995] |
| Shortest path | V_u | 244.6 | 364.9 | [146.5, 769.7] | [-280.4, 342.8] |
| Shortest path | V_s | 173.1 | 162.4 | [83.8, 267.2] | [79.1, 262.4] |
| Shortest path | share | 0.414 | 0.335 | [0.121, 0.592] | [0.237, 0.708] |
| Pooled | V_u | 483.9 | 761.3 | [363.7, 1426.3] | [-458.4, 604.1] |
| Pooled | V_s | 585.6 | 603.9 | [340.2, 908.3] | [262.9, 831.0] |
| Pooled | share | 0.548 | 0.453 | [0.252, 0.659] | [0.436, 0.843] |

Two reported numbers were artefacts of construction. The aim under the mention twin reference is biased negative: the gain is the ban minus the mention cell and the prime and veto profile is the mention minus the neutral cell, so noise in the shared mention cell makes their correlation negative even when nothing is aimed. With equal cell variances the expected null correlation is -0.5; simulated under the split half design with the estimated variance components the null has mean -0.27 (SD 0.14, 5th to 95th percentile -0.49 to -0.04), and a placebo with an independent mention cell has mean 0.00. Llama's mention aim of -0.28 sits at the null's 66th percentile, so the bias explains it fully. Qwen3-8B's -0.595, omitted from Tables 56, 57 and 78, still lies below the null's 1st percentile (quantile 0.004). The ESS weighted grid share of 1.24 is impossible as a proportion: the noise corrected ladder term went negative (V_u = -35.4) when the weights shrank the ladder projection below its noise allowance, so V_s / (V_u + V_s) = 181.3 / 145.9.

**What a decision on mass would cost.** With Kish ESS approximately M / (1 + CV^2) for M proposal programs, the number of programs per algorithm needed for a median ESS of 5 is 5(1 +

CV^2). The plug-in estimate of CV^2 from 20 programs cannot exceed 19, so the plug-in M is bounded by 100 and is a lower bound; a lognormal fit to the log weights gives 5 exp(sigma^2), the upper end. Pooled over algorithms, the plug-in median is 50 programs, the lognormal median about 10^31.7, and 22% of the Pareto smoothed importance sampling shape estimates [66] exceed 0.7, the value above which an importance sampling estimate is not reliable. At the plug-in lower bound one bench event costs about 1.4 GPU hours of scoring. The spread of per-event shares (SD 0.41 in Qwen3-8B, 0.23 in Llama) means that separating the share from 0.5 by 0.10 needs about 129 Qwen3-8B events or 43 Llama events (32 and 11 for a margin of 0.20), against 11 and 6 rows measured. The decision therefore moved to sampled landings on the bench (Sections 3.3 to 3.5).

**Table 84.** Programs per algorithm needed for a median Kish ESS of 5, from the observed weight dispersion of the likelihood cells. Source: record audit `tables/m_arithmetic.csv`.

| Model | Algorithm | Cells | Median ESS now (M = 20) | Median SD of log weights | Plug-in programs needed, median | Plug-in programs needed, 90th percentile | Lognormal programs needed, median |
|---|---|---|---|---|---|---|---|
| Qwen3-8B | A* | 202 | 1.03 | 5.99 | 55.2 | 99.9 | 1.9e16 |
| Qwen3-8B | Bellman-Ford/ SPFA | 165 | 1.82 | 18.55 | 40.1 | 82.1 | 1.2e150 |
| Qwen3-8B | BFS | 202 | 4.27 | 2.43 | 15.0 | 64.1 | 1813.9 |
| Qwen3-8B | bidirectional BFS | 202 | 1.00 | 29.50 | 100.0 | 100.0 | 5.1e304 |
| Qwen3-8B | bidirectional Dijkstra | 165 | 1.01 | 26.56 | 36.8 | 55.0 | 5.1e304 |
| Qwen3-8B | DFS | 202 | 1.12 | 7.81 | 50.0 | 98.7 | 1.4e27 |
| Qwen3-8B | Dial | 165 | 1.00 | 13.36 | 73.5 | 100.0 | 1.7e78 |
| Qwen3-8B | Dijkstra and unit weight Dijkstra | 367 | 1.86 | 5.75 | 32.6 | 96.7 | 1.2e15 |
| Qwen3-8B | Floyd-Warshall | 165 | 1.07 | 7.39 | 64.3 | 98.3 | 2.5e24 |
| Qwen3-8B | IDDFS | 202 | 1.00 | 12.81 | 79.7 | 100.0 | 8.5e71 |
| Llama-3.1-8B | A* | 120 | 2.51 | 3.03 | 35.3 | 84.6 | 4.9e4 |
| Llama-3.1-8B | Bellman-Ford/ SPFA | 100 | 1.02 | 5.23 | 70.0 | 144.7 | 3.7e12 |
| Llama-3.1-8B | BFS | 120 | 8.61 | 1.18 | 9.6 | 39.7 | 20.0 |
| Llama-3.1-8B | bidirectional BFS | 120 | 1.33 | 4.39 | 28.1 | 50.5 | 1.2e9 |
| Llama-3.1-8B | DFS | 120 | 3.97 | 2.09 | 22.3 | 53.3 | 390.2 |
| Llama-3.1-8B | Dijkstra and unit weight Dijkstra | 220 | 3.37 | 2.45 | 28.9 | 59.3 | 2050.0 |
| Llama-3.1-8B | Floyd-Warshall | 100 | 1.34 | 7.05 | 60.6 | 114.9 | 1.9e22 |
| Llama-3.1-8B | IDDFS | 120 | 1.57 | 6.37 | 51.0 | 94.7 | 2.2e18 |

The ladder verdicts of Section 2.11 are reconciled as the review asked (correction in place there). We call a survivor that the ban and every generic perturbation reach a shared top rung. Behaviour shows a shared top rung on the shortest path task: Bellman-Ford/SPFA took 272 of 272 compliant programs in the five iteration-1 draws and 255 of 260 attempts not banned in the three iteration-2 draws, and 8 of 8 nonsemantic perturbations that left Dijkstra landed there. On the grid task only

bans reach DFS (116 of 128 compliant iteration-1 programs, 257 of 257 iteration-2 attempts not banned). The order below the top rung is unmeasured, and the latent ladder of Section 2.1 moves from the accounts ruled out to the effects that an uncalibrated instrument could not detect.

### 3.2.2 The ban list manifold figures

The ban list experiment had drawn 13 Isomap figures, not the 14 its summary claimed, plus two figures of neighbour stability over k. *(Added in iteration 4: four of these Isomaps, the grid baseline, the single BFS ban, the five-name ban and the shortest path ban that leaves only Bellman-Ford/SPFA, are now plotted from their centroid coordinates in Section 4.2.5, with the position caveat in the caption; the token-16 recomputation is tabulated there, because it yields neighbour orders rather than coordinates.)* Each figure embeds teacher forced states of forced programs at layer 24 and the token before the commit token (k = 10). Its centroids are sized and coloured by the number of programs implementing each algorithm under the condition, banned centroids are crossed out, and natural programs are overlaid. The audit recomputed the geometry from the saved states and reproduced the experiment's centroid matrix within 1e-7 and its nearest neighbour maps in 24 of 24 settings. Table 85 gives the implemented counts that the figures encode. Under the literal five name ban the counts stay on the banned BFS and DFS centroids (63 and 36 attempted, 53 and 15 correct).

**Table 85.** Implemented counts per centroid in the 13 manifold figures (attempted / correct; N = 100 per condition; banned algorithms marked †). Grid columns: BFS, DFS, bidirectional BFS, IDDFS, A*, greedy best first, IDA*, other. Shortest path columns: Dijkstra, Bellman-Ford/SPFA, bidirectional Dijkstra, Floyd-Warshall, Dial, other. Source: record audit `tables/manifold_implemented_counts.csv`.

| Condition | Banned | Counts |
|---|---|---|
| Grid baseline | none | BFS 100/89; all others 0 |
| Grid single ban | BFS | BFS† 14/11, DFS 86/30 |
| Grid TOP | BFS, DFS, greedy, IDDFS | BFS† 59/38, DFS† 26/0, greedy† 2/1, other 12/0, unit weight Dijkstra (outside the K = 7 catalogue) 1/1 |
| Grid ARC | BFS, bidirectional BFS, A*, greedy | BFS† 63/40, DFS 37/20 |
| Grid SCATTER | BFS, DFS, greedy, IDA* | BFS† 69/53, DFS† 27/18, A* 2/2, unit weight Dijkstra 2/2 |
| Grid TOP5 | BFS, DFS, greedy, IDDFS, A* | BFS† 63/53, DFS† 36/15, IDDFS† 1/0 |
| Shortest path baseline | none | Dijkstra 100/100 |
| Shortest path single ban | Dijkstra | Dijkstra† 16/16, Bellman-Ford/SPFA 83/79, other 1/0 |
| Shortest path, only Dijkstra allowed | the other four | Dijkstra 95/95, Bellman-Ford/SPFA† 5/5 |
| Shortest path, only Bellman-Ford/SPFA allowed | the other four | Dijkstra† 48/47, Bellman-Ford/SPFA 51/39, other 1/0 |
| Shortest path, only bidirectional Dijkstra allowed | the other four | Dijkstra† 52/51, Bellman-Ford/SPFA† 48/36 |
| Shortest path, only Floyd-Warshall allowed | the other four | Dijkstra† 59/59, Bellman-Ford/SPFA† 40/28, other 1/0 |
| Shortest path, only Dial allowed | the other four | Dijkstra† 52/50, Bellman-Ford/SPFA† 44/28, other 4/0 |

The neighbourhood that the iteration-2 record reported (BFS next to bidirectional BFS, centroid distance 0.0147 against a within-BFS spread of 0.0154) does not survive at a position independent

of the label. At generated token 16, BFS and bidirectional BFS are still the nearest pair, but permuting the algorithm labels reproduces that identity with $p = 0.19$, and the two clouds are not separable (centroid distance 0.96 of the within-class spread; exchangeability test $p = 0.48$). Table 86 gives the nearest neighbour maps at layer 24 and $k = 10$ for both positions. Persistent homology found no loop at either position. At the token before the commit token a planted loop of matched size is detected with power 1.00 (grid) and 0.975 (shortest path), but that position partly encodes the label; at token 16 the power is 0.035 and 0.045. The absence of a loop is therefore established with power only where the label confounds the geometry.

**Table 86.** Nearest centroid of each algorithm at layer 24, $k = 10$, at the label-dependent position (token before the commit token) and at generated token 16, with MDS stress-1. Source: record audit `tables/manifold_grid.csv`.

| Task | Position | Nearest neighbour map | Stress-1 | BFS and bidirectional BFS: distance / separability |
|---|---|---|---|---|
| Grid | before commit token | BFS: bidirectional BFS; DFS: IDDFS; bidirectional BFS: BFS; IDDFS: DFS; A*: greedy; greedy: A*; IDA*: greedy | 0.025 | 0.0147 / 0.954 |
| Grid | token 16 | BFS: bidirectional BFS; DFS: BFS; bidirectional BFS: BFS; IDDFS: bidirectional BFS; A*: greedy; greedy: A*; IDA*: bidirectional BFS | 0.104 | 0.0272 / 0.960 |
| Shortest path | before commit token | Dijkstra: bidirectional Dijkstra; Bellman-Ford/SPFA: bidirectional Dijkstra; bidirectional Dijkstra: Dijkstra; Floyd-Warshall: Dijkstra; Dial: Floyd-Warshall | 0.006 | n/a |
| Shortest path | token 16 | Dijkstra: bidirectional Dijkstra; Bellman-Ford/SPFA: Dial; bidirectional Dijkstra: Dijkstra; Floyd-Warshall: Dial; Dial: Floyd-Warshall | 0.005 | n/a |

**Table 86b.** The full recomputed grid behind Table 86: every layer, position and k (nearest neighbour of the default and of its realised substitute; the BFS and bidirectional BFS distance and separability on the grid task). Source: record audit `tables/manifold_grid.csv`.

| Task | Layer | Position | k | Stress-1 | Nearest to the default | Nearest to the substitute | BFS to bidirectional BFS distance | Separability |
|---|---|---|---|---|---|---|---|---|
| Grid | 20 | before commit token | 5 | 0.0273 | BFS: bidirectional BFS | DFS: IDDFS | 0.0150 | 0.978 |
| Grid | 20 | before commit token | 10 | 0.0427 | BFS: bidirectional BFS | DFS: IDDFS | 0.0143 | 0.962 |
| Grid | 20 | before commit token | 15 | 0.0360 | BFS: bidirectional BFS | DFS: IDDFS | 0.0159 | 0.972 |
| Grid | 20 | before commit token | 20 | 0.0286 | BFS: bidirectional BFS | DFS: IDDFS | 0.0146 | 0.967 |
| Grid | 20 | token 16 | 5 | 0.0649 | BFS: bidirectional BFS | DFS: A* | 0.0169 | 0.971 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Grid | 20 | token 16 | 10 | 0.1276 | BFS: bidirectional BFS | DFS: A* | 0.0151 | 0.938 |
| Grid | 20 | token 16 | 15 | 0.1332 | BFS: bidirectional BFS | DFS: A* | 0.0135 | 0.990 |
| Grid | 20 | token 16 | 20 | 0.1325 | BFS: bidirectional BFS | DFS: A* | 0.0096 | 0.938 |
| Grid | 24 | before commit token | 5 | 0.0908 | BFS: bidirectional BFS | DFS: IDDFS | 0.0147 | 0.974 |
| Grid | 24 | before commit token | 10 | 0.0253 | BFS: bidirectional BFS | DFS: IDDFS | 0.0147 | 0.954 |
| Grid | 24 | before commit token | 15 | 0.0178 | BFS: bidirectional BFS | DFS: IDDFS | 0.0158 | 0.959 |
| Grid | 24 | before commit token | 20 | 0.0173 | BFS: bidirectional BFS | DFS: IDDFS | 0.0144 | 0.968 |
| Grid | 24 | token 16 | 5 | 0.1040 | BFS: bidirectional BFS | DFS: A* | 0.0178 | 0.860 |
| Grid | 24 | token 16 | 10 | 0.1038 | BFS: bidirectional BFS | DFS: BFS | 0.0272 | 0.960 |
| Grid | 24 | token 16 | 15 | 0.1542 | BFS: bidirectional BFS | DFS: A* | 0.0172 | 0.997 |
| Grid | 24 | token 16 | 20 | 0.1415 | BFS: bidirectional BFS | DFS: A* | 0.0127 | 0.999 |
| Grid | 28 | before commit token | 5 | 0.0211 | BFS: bidirectional BFS | DFS: IDDFS | 0.0257 | 0.974 |
| Grid | 28 | before commit token | 10 | 0.0245 | BFS: bidirectional BFS | DFS: IDDFS | 0.0232 | 0.984 |
| Grid | 28 | before commit token | 15 | 0.0234 | BFS: bidirectional BFS | DFS: IDDFS | 0.0253 | 0.976 |
| Grid | 28 | before commit token | 20 | 0.0234 | BFS: bidirectional BFS | DFS: IDDFS | 0.0198 | 0.975 |
| Grid | 28 | token 16 | 5 | 0.0995 | BFS: bidirectional BFS | DFS: A* | 0.0211 | 0.951 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Grid | 28 | token 16 | 10 | 0.0598 | BFS: bidirectional BFS | DFS: IDDFS | 0.0229 | 0.992 |
| Grid | 28 | token 16 | 15 | 0.0737 | BFS: bidirectional BFS | DFS: IDDFS | 0.0223 | 0.995 |
| Grid | 28 | token 16 | 20 | 0.0774 | BFS: bidirectional BFS | DFS: IDDFS | 0.0157 | 0.956 |
| Shortest path | 20 | before commit token | 5 | 0.0047 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: bidirectional Dijkstra | n/a | n/a |
| Shortest path | 20 | before commit token | 10 | 0.0039 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: bidirectional Dijkstra | n/a | n/a |
| Shortest path | 20 | before commit token | 15 | 0.0039 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: bidirectional Dijkstra | n/a | n/a |
| Shortest path | 20 | before commit token | 20 | 0.0032 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: bidirectional Dijkstra | n/a | n/a |
| Shortest path | 20 | token 16 | 5 | 0.0066 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: bidirectional Dijkstra | n/a | n/a |
| Shortest path | 20 | token 16 | 10 | 0.0054 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: Dijkstra | n/a | n/a |
| Shortest path | 20 | token 16 | 15 | 0.0048 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: bidirectional Dijkstra | n/a | n/a |
| Shortest path | 20 | token 16 | 20 | 0.0035 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: Dijkstra | n/a | n/a |
| Shortest path | 24 | before commit token | 5 | 0.0075 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: bidirectional Dijkstra | n/a | n/a |
| Shortest path | 24 | before commit token | 10 | 0.0060 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: bidirectional Dijkstra | n/a | n/a |
| Shortest path | 24 | before commit token | 15 | 0.0056 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: bidirectional Dijkstra | n/a | n/a |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Shortest path | 24 | before commit token | 20 | 0.0061 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: bidirectional Dijkstra | n/a | n/a |
| Shortest path | 24 | token 16 | 5 | 0.0062 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: Dial | n/a | n/a |
| Shortest path | 24 | token 16 | 10 | 0.0050 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: Dial | n/a | n/a |
| Shortest path | 24 | token 16 | 15 | 0.0050 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: bidirectional Dijkstra | n/a | n/a |
| Shortest path | 24 | token 16 | 20 | 0.0042 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: Dijkstra | n/a | n/a |
| Shortest path | 28 | before commit token | 5 | 0.0042 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: bidirectional Dijkstra | n/a | n/a |
| Shortest path | 28 | before commit token | 10 | 0.0061 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: bidirectional Dijkstra | n/a | n/a |
| Shortest path | 28 | before commit token | 15 | 0.0075 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: bidirectional Dijkstra | n/a | n/a |
| Shortest path | 28 | before commit token | 20 | 0.0073 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: bidirectional Dijkstra | n/a | n/a |
| Shortest path | 28 | token 16 | 5 | 0.0038 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: Dial | n/a | n/a |
| Shortest path | 28 | token 16 | 10 | 0.0039 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: Dial | n/a | n/a |
| Shortest path | 28 | token 16 | 15 | 0.0039 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: Dial | n/a | n/a |
| Shortest path | 28 | token 16 | 20 | 0.0034 | Dijkstra: bidirectional Dijkstra | Bellman-Ford/ SPFA: Dial | n/a | n/a |

### 3.2.3 Novelty: every correct program outside the catalogue, read

The request asks whether a model can be made to give a truly novel solution. The only place such a solution could have appeared in the run's data is among programs with the label other. The audit collected every correct program labelled other across the eleven declared artifacts and the 1,200-

program copy of the forking token pools: 78 unique programs (Set A). It added 138 unique failing programs labelled other and written under bans (Set B), and the 12 failing programs labelled other under the TOP ban list (the selection was hash committed before any judge call). Two blind judges from different families (gpt-4.1-mini and gemini-2.5-flash, tie-break deepseek-chat) classed each program as a novel algorithm, a variant of a catalogue algorithm, a labeller miss, or degenerate or hard-coded, and the executor read all 78 Set A programs and 112 Set B programs by hand.

None of the 78 correct programs is a novel algorithm (Table 87). Sixty-five are labeller misses (catalogue algorithms the runtime and AST labellers failed to name) and 13 are variants of catalogue algorithms; the manual pass, which read every program, agreed on the exact class for 60 of 78 and on all 78 once the catalogue variant and labeller miss classes are merged. Sixty-two of the 78 came from forced prompts and 4 from the decommitment steering arms; none came from a creativity, new algorithm or novelty instruction (Table 87b). The rule of three [67] bounds the novel rate at $3/78 = 0.038$ among correct programs outside the catalogue and 0.00026 over all 11,763 unique correct programs audited. About 9,000 other programs of the forking token experiment were not audited.

**Table 87.** Final classes of the audited programs. Source: record audit `results/novelty_audit.json`.

| Set | Programs | Labeller miss | Catalogue variant | Degenerate or hard-coded | Novel algorithm |
|---|---|---|---|---|---|
| A: correct, labelled other | 78 | 65 | 13 | 0 | 0 |
| B: failing, labelled other, under bans (final) | 138 | 53 | 41 | 44 | 0 |
| B: judges only, before the tie-break and manual pass | 138 | 53 | 28 | 40 | 17 |
| TOP ban list residue | 12 | 0 | 0 | 12 | 0 |

**Table 87b.** The 78 correct programs labelled other, by model, task and condition, with their final class (Set A of Table 87). Source: record audit `tables/novelty_class_counts.csv`.

| Model | Task | Condition | Labeller miss | Catalogue variant | Novel |
|---|---|---|---|---|---|
| Qwen3-8B | kth smallest | forced prompts | 13 | 0 | 0 |
| Qwen3-8B | merge intervals | forced prompts | 0 | 1 | 0 |
| Qwen3-8B | Grid | forced A* | 21 | 0 | 0 |
| Qwen3-8B | Grid | forced bidirectional BFS | 1 | 1 | 0 |
| Qwen3-8B | Grid | forced unit weight Dijkstra | 15 | 0 | 0 |
| Qwen3-8B | Grid | forced greedy best first | 2 | 0 | 0 |
| Qwen3-8B | Grid | forced IDA* | 1 | 0 | 0 |
| Qwen3-8B | Grid | forced jump point search | 0 | 1 | 0 |
| Qwen3-8B | Grid | forced jump point search (descriptive) | 1 | 1 | 0 |
| Qwen3-8B | Grid | mention of bidirectional BFS | 1 | 0 | 0 |
| Qwen3-8B | Grid | SCATTER ban list | 2 | 0 | 0 |
| Qwen3-8B | Grid | TOP ban list | 1 | 0 | 0 |
| Llama-3.1-8B-Instruct | Grid | ban BFS, first template | 0 | 1 | 0 |

| | | | | | |
|---|---|---|---|---|---|
| Llama-3.1-8B-Instruct | Grid | forced A* (paraphrase 5) | 1 | 0 | 0 |
| Llama-3.1-8B-Instruct | Grid | forced A* (paraphrase 9) | 0 | 1 | 0 |
| Qwen3-8B | Shortest path | baseline | 0 | 1 | 0 |
| Qwen3-8B | Shortest path | forced bidirectional Dijkstra (descriptive) | 1 | 1 | 0 |
| Qwen3-8B | Shortest path | decommitment direction, c = 0.25 | 1 | 0 | 0 |
| Qwen3-8B | Shortest path | decommitment direction, c = 0.375 | 0 | 1 | 0 |
| Qwen3-8B | Shortest path | decommitment direction, c = 0.5 | 1 | 1 | 0 |
| Qwen3-8B | Shortest path | variable name placebo direction, c = 0.25 | 0 | 1 | 0 |
| Qwen3-8B | Shortest path | second succession step | 0 | 1 | 0 |
| Qwen3-8B-Base | Shortest path | baseline | 1 | 1 | 0 |
| Llama-3.1-8B-Instruct | Shortest path | ban Dijkstra, fourth template | 1 | 0 | 0 |
| Llama-3.1-8B-Instruct | Shortest path | mention of Dijkstra, fifth template | 1 | 0 | 0 |
| All | | | 65 | 13 | 0 |

The judges agreed only fairly (Cohen's kappa 0.38 [0.31, 0.45]; 0.44 when known, novel and degenerate are collapsed), and gpt-4.1-mini called failing, broken code novel. The positive controls (catalogue references presented as unknown) failed the strict class gate at 7 of 10, but both judges named the right algorithm in 10 of 10; the negative controls passed 5 of 5, and no control was called novel.

The TOP ban list's avoidance of 0.13 is residue (Table 88): its 13 programs not banned are 12 failing degenerate BFS or DFS variants and 1 correct unit weight Dijkstra outside the K = 7 catalogue. Counted over catalogue algorithms only, avoidance is 0.00 for TOP and TOP5, 0.02 for SCATTER and 0.37 for ARC.

**Table 88.** Ban lists with the share labelled other reported separately from avoidance (N = 100 each; Wilson 95% intervals). Source: record audit `tables/multiban_avoidance_corrected.csv`.

| Ban list | Banned share | Share in a catalogue algorithm not banned | Share labelled other | Avoidance as reported | Avoidance into the catalogue |
|---|---|---|---|---|---|
| TOP | 0.87 [0.79, 0.92] | 0.00 [0.00, 0.04] | 0.13 [0.08, 0.21] | 0.13 | 0.00 |
| ARC | 0.63 [0.53, 0.72] | 0.37 [0.28, 0.47] | 0.00 | 0.37 | 0.37 |
| SCATTER | 0.96 [0.90, 0.98] | 0.02 [0.01, 0.07] | 0.02 [0.01, 0.07] | 0.04 | 0.02 |
| TOP5 | 1.00 [0.96, 1.00] | 0.00 | 0.00 | 0.00 | 0.00 |

### 3.2.4 Llama's compositions over correct programs

Every composition quoted in the review reproduced exactly. Llama's apparent diversity on the grid task is failing code: its 20 correct baseline programs are all BFS (correctness conditioned Hill number 1.00, against 1.87 over attempts). Tables 89 and 90 give every Llama event and the bench pilot with correct programs beside attempted ones.

**Table 89.** Llama-3.1-8B-Instruct path task events: correct compliant programs against attempted programs, with the correctness conditioned Hill number and the Qwen3-8B value for the matched condition. Source: record audit `tables/llama_correct_only.csv`.

| Task | Event | N | Pass rate | Correct, compliant | Attempted | Hill (attempts) | Hill_cc | Qwen3-8B Hill_cc |
|---|---|---|---|---|---|---|---|---|
| Grid | baseline | 100 | 0.20 | BFS 20 | BFS 77, DFS 20, other 3 | 1.87 | 1.00 | 1.00 |
| Grid | ban BFS | 100 | 0.12 | DFS 7, other 1 | DFS 63, BFS 29, other 6, A* 1, unit weight Dijkstra 1 | 2.49 | 1.46 (8 programs) | 1.00 |
| Grid | mention BFS | 50 | 0.28 | BFS 14 | BFS 47, other 2, DFS 1 | 1.30 | 1.00 | 1.00 |
| Grid | listing control | 50 | 0.20 | BFS 7, DFS 3 | BFS 36, other 7, DFS 6, unit weight Dijkstra 1 | 2.33 | 1.84 | 1.00 |
| Grid | ban DFS | 50 | 0.16 | BFS 7 | BFS 39, DFS 7, other 4 | 1.96 | 1.00 (7 programs) | 1.00 |
| Grid | ban unit weight Dijkstra | 50 | 0.26 | BFS 12, DFS 1 | BFS 30, DFS 18, unit weight Dijkstra 1, other 1 | 2.30 | 1.31 | 1.14 |
| Grid | ban BFS and DFS | 50 | 0.06 | none | BFS 21, DFS 21, other 6, unit weight Dijkstra 2 | 3.04 | n/a | 1.65 |
| Shortest path | baseline | 100 | 0.59 | Dijkstra 59 | Dijkstra 92, other 7, Bellman-Ford/SPFA 1 | 1.36 | 1.00 | 1.00 |
| Shortest path | ban Dijkstra | 100 | 0.25 | Bellman-Ford/ SPFA 4, other 1 | Dijkstra 43, other 37, Bellman-Ford/ SPFA 18, Floyd-Warshall 2 | 3.06 | 1.65 (5 programs) | 1.00 |
| Shortest path | mention Dijkstra | 50 | 0.46 | Dijkstra 22, other 1 | Dijkstra 39, other 11 | 1.69 | 1.20 | 1.00 |
| Shortest path | listing control | 50 | 0.24 | Dijkstra 9, Bellman-Ford/ SPFA 3 | Dijkstra 19, Bellman-Ford/SPFA 15, other 14, Floyd-Warshall 2 | 3.37 | 1.75 | 1.14 |
| Shortest path | ban Bellman-Ford/SPFA | 50 | 0.26 | Dijkstra 13 | Dijkstra 33, other 14, Bellman-Ford/ SPFA 2, Floyd-Warshall 1 | 2.31 | 1.00 | 1.00 |
| Shortest path | ban Floyd-Warshall | 50 | 0.18 | Dijkstra 6, Bellman-Ford/ SPFA 1 | Dijkstra 25, other 11, Bellman-Ford/ SPFA 9, Floyd-Warshall 5 | 3.38 | 1.51 (7 programs) | 1.19 |
| Shortest path | ban Dijkstra and Bellman-Ford/SPFA | 50 | 0.30 | none | Dijkstra 22, other 21, Bellman-Ford/ SPFA 6, Floyd-Warshall 1 | 2.88 | n/a | n/a |

**Table 90.** Bench pilot (N = 50 per task), Llama-3.1-8B-Instruct correct programs against attempted programs; Qwen3-8B's correctness conditioned Hill number was 1.00 on every task. Source: record

audit `tables/llama_correct_only.csv`.

| Task | Pass rate | Correct programs | Attempted programs | Hill (attempts) | Hill_cc | Hill_cc rarefied to 10 |
|---|---|---|---|---|---|---|
| kth smallest | 0.20 | full sort 6, heapify and pop 2, quickselect 1, size-k heap 1 | heapify and pop 16, size-k heap 11, full sort 7, other 6, quickselect 5, counting sort 2 | 5.08 | 2.97 | 2.97 |
| substring search | 0.48 | naive 18, KMP 5, Rabin-Karp 1 | naive 24, KMP 14, Rabin-Karp 6, other 1 | 2.86 | 1.96 | 1.87 |
| two sum | 0.80 | hash map 39, brute force 1 | hash map 48, other 1, brute force 1 | 1.22 | 1.12 | 1.10 |
| find duplicate | 0.34 | hash set 16, Floyd cycle detection 1 | hash set 20, other 15, Floyd 4, sort and adjacent check 3, index marking 3, binary search on values 1 | 3.97 | 1.25 | 1.23 |
| Fibonacci | 0.44 | matrix power 19, iteration 2, memoised recursion 1 | matrix power 37, iteration 4, other 4, memoised recursion 2 | 2.10 | 1.62 | 1.54 |
| connected components | 0.70 | union find 30, DFS 5 | union find 37, DFS 11 | 1.71 | 1.51 | 1.47 |
| longest increasing subsequence | 0.76 | quadratic DP 38 | quadratic DP 45, other 3, patience sorting 2 | 1.48 | 1.00 | 1.00 |
| merge intervals | 0.56 | sort and sweep 28 | sort and sweep 42 | 1.00 | 1.00 | 1.00 |
| top-k frequent | 0.24 | heap 8, sort by count 4 | heap 28, sort by count 14, other 1, quickselect on counts 1 | 2.28 | 1.89 | 1.87 |
| nonoverlapping intervals | 0.04 | DP with bisect 1, greedy by end 1 | greedy by end 23, other 14, DP with bisect 3, DAG longest path 1 | 2.65 | 2.00 (2 programs) | 2.00 |
| cycle detection | 0.22 | union find 7, DFS with parent 4 | DFS with parent 29, union find 16, other 4 | 2.41 | 1.93 | 1.92 |
| topological order | 0.82 | Kahn 41 | Kahn 48, DFS postorder 1 | 1.10 | 1.00 | 1.00 |
| first and last position | 0.56 | binary search 28 | binary search 47 | 1.00 | 1.00 | 1.00 |

Counted over correct programs and rarefied to 10, Llama is more diverse than Qwen3-8B on 9 of 13 tasks (median 1.47 against 1.00). The cross-model sentence of iteration 2 therefore becomes: Llama attempts more algorithms, but its correct grid output is as collapsed as Qwen3-8B's. Correct-only diversity appears on substring search, connected components, cycle detection and top-k frequent, at small N.

### 3.2.5 The knob table under a declared class rule

The audit rebuilt the knob table with one statistic (rarefied Hill-1 over correct compliant programs at depth 25) and a class rule declared before ranking. Knobs are prompt bans, seeds, instructions, decoding settings, activation interventions and token vetoes. Controls (mention twins, sham bans of absent algorithms, forced prompts, and format or weight arms) are listed but never ranked. Generate then select is a ceiling, and baselines are references. *(Correction, iteration 5: the class rule omitted a class. Exposure arms, any arm whose prompt shows a correct program of an algorithm (the*

*copy twin, the named and unnamed exposure bans and the exposure vetoes), supply the text of the alternative and are therefore controls, like the mention twin; their Hill-1 is reported beside the knob verdict and never as the best knob. The per-task verdict with exposure arms as controls is in Table 172, Section 5.2.)* The rebuilt table has 306 rows (276 without the instruction tuning experiment's arms), against the 249 of iteration 2, because steering arms, instruct arms and temperature cells are kept as separate units: 185 knobs, 103 controls, 16 references and 2 ceilings, with 108 rankable knob rows under 20 pass points.

No knob row under 20 pass points reaches a rarefied Hill-1 of 2 (Table 91). The best is the semantic seed on the shortest path task, 1.940 with a 90% interval of [1.770, 2.094] that includes 2. The A* mention reaches 2.69 at 11.3 points, but it is a control: it recommends a named algorithm. Stated in its own metric: no knob reaches an even two-algorithm split under 20 points; the semantic seed (Bellman-Ford/SPFA 61, Dijkstra 30 at 7 points) and the A* mention are the two conditions that buy a second correct algorithm, and both name or describe it. *(Correction, iteration 4: the claim that these are the only two such conditions is wrong. The decommitment direction (1.913, 90% interval [1.704, 2.083], at 17 points) and random direction 1 (1.86 at 16 points) of Table 91 also buy a second correct algorithm and name nothing, and the decommitment interval includes 2 by the same criterion used to single out the semantic seed. The iteration-3 arms were never entered into this table; with them, norm-matched random directions at the task-specification positions are the cheapest diversity knob on kth smallest (2.92 at 0 points) and two sum (2.12 at 0 points), 7 of 21 random direction and dose cells qualify on kth smallest, and the rebuilt verdict per task is in Section 4.2.1.)*

**Table 91.** The twelve best knob rows under 20 pass points (rarefied Hill-1 at depth 25 over correct compliant programs, 90% interval; pass points against each row's own reference). Source: record audit `results/knob_table.json#ranking`.

| Task | Knob (experiment) | Pass rate | Correct compliant programs | Hill-1 [90%] | Pass points lost | Second algorithm (share) |
|---|---|---|---|---|---|---|
| Shortest path | semantic seed (decoding) | 0.92 | Bellman-Ford/SPFA 61, Dijkstra 30, Floyd-Warshall 1 | 1.940 [1.770, 2.094] | 7.0 | Dijkstra (0.33) |
| Shortest path | decommitment direction, c = 0.25 | 0.78 | Bellman-Ford/SPFA 54, Dijkstra 23, other 1 | 1.913 [1.705, 2.088] | 17.0 | Dijkstra (0.29) |
| Shortest path | random direction 1, c = 0.25 (decommitment) | 0.79 | Dijkstra 53, Bellman-Ford/SPFA 26 | 1.858 [1.728, 1.947] | 16.0 | Bellman-Ford/SPFA (0.33) |
| Shortest path | novelty instruction (decoding) | 0.96 | Dijkstra 82, Bellman-Ford/SPFA 14 | 1.503 [1.324, 1.649] | 3.0 | Bellman-Ford/SPFA (0.15) |
| Shortest path | random direction 1 at the read position only, c = 1.0 (instruction tuning) | 1.00 | Dijkstra 90, Bellman-Ford/SPFA 9, other 1 | 1.403 [1.210, 1.566] | 0.0 | Bellman-Ford/SPFA (0.09) |
| Shortest path | random direction 2, c = 0.25 (decommitment) | 0.87 | Dijkstra 82, Bellman-Ford/SPFA 5 | 1.229 [1.091, 1.376] | 8.0 | Bellman-Ford/SPFA (0.06) |
| Grid | novelty instruction (decoding) | 0.85 | BFS 81, unit weight Dijkstra 2, DFS 1, A* 1 | 1.228 [1.053, 1.401] | 12.0 | unit weight Dijkstra (0.02) |

| | | | | | | |
|---|---|---|---|---|---|---|
| Shortest path | new algorithm instruction (decoding) | 0.94 | Dijkstra 90, Bellman-Ford/SPFA 4 | 1.178 [1.048, 1.308] | 5.0 | Bellman-Ford/ SPFA (0.04) |
| Shortest path | enumeration prompt (decommitment) | 0.82 | Dijkstra 80, Bellman-Ford/SPFA 2 | 1.109 [1.000, 1.215] | 6.5 | Bellman-Ford/ SPFA (0.02) |
| Shortest path | enumeration prompt, 1,024 tokens (decommitment) | 0.82 | Dijkstra 80, Bellman-Ford/SPFA 2 | 1.108 [1.000, 1.236] | 6.5 | Bellman-Ford/ SPFA (0.02) |
| Grid | new algorithm instruction (decoding) | 0.94 | BFS 92, DFS 2 | 1.093 [1.000, 1.191] | 3.0 | DFS (0.02) |
| Grid | semantic seed (decoding) | 0.93 | BFS 91, bidirectional BFS 2 | 1.092 [1.000, 1.192] | 4.0 | bidirectional BFS (0.02) |

Four of the nine shortest path rows in Table 91 are activation edits: the decommitment direction, which iteration 1 found no better than a random direction, and three random directions. In every one of them the only catalogue algorithm besides Dijkstra is Bellman-Ford/SPFA, the same second choice the ban reaches.

### 3.2.6 Positioning of the iteration-2 claims

The audit checked each iteration-2 claim against its nearest published neighbour with verbatim quotes from the abstract pages, all 12 verified on 2026-10-20 (Table 92). Two contrasts matter for interpretation. Heo et al. found a dimension in input embedding space that predicts instruction following, is tied to prompt phrasing, and raises success when intervened on [58]; this run replicates the phrasing link for prohibitions in code but finds the read position causally inert when spliced alone. Sia, Mueller and Duh found a task recognition point after which attention to the instruction is no longer needed [59]; this run replicates that on the shortest path task and finds the opposite on the grid task. The base to chat SAE transfer study of Kissane and colleagues (an Alignment Forum post that the bibliography tool could not retrieve, so it carries no number here) is the source of the fine-tuning remedy that Section 2.12 proposed without citation.

**Table 92.** The iteration-2 claims, their nearest neighbours, and what this run adds. Source: record audit `tables/positioning.csv`.

| Claim in this run | Neighbour | Neighbour's finding (paraphrased; the audit verified each against a verbatim quote) | Relation | What this run adds |
|---|---|---|---|---|
| Compliance is decodable at the read position (-0.41 across wordings) yet splicing the read position carries nothing | Heo et al. [58] | a direction in input embedding space predicts instruction following, tracks prompt phrasing more than task difficulty, and raises success when intervened along | replication (phrasing) and contrast (decodable but inert under splicing) | a compliance-predictive direction at the read position that is causally inert; the ban travels through the task specification states |
| Attention knockouts and cache splicing place the path task ban on the prompt route | Geva et al. [48] | interventions on attention edges locate two points where information flows to the prediction | extension (method) | edge knockouts applied to a prompt-level prohibition in code, with token-matched neutral twins |

| | | | | |
|---|---|---|---|---|
| Generated tokens stop needing the ban on the shortest path task (direct 0.05) but not on the grid task (direct 0.35) | Sia et al. [59] | after a task recognition point the task is encoded in the input representations and attention to the context is no longer needed | replication (shortest path) and contrast (grid) | the same redundancy for a prohibition on one task and its failure on another |
| Naming an absent algorithm in a ban raises its likelihood (8 of 9 in Qwen3-8B, 5 of 6 in Llama) | Rana [7] | explicitly mentioning a forbidden word activates its representation instead of suppressing it | replication | priming at the level of whole algorithms, in two model families |
| (same) | Mann et al. [53] | suppressing a concept requires activating it internally, which can prime rebound rather than avoidance | replication | rebound measured on implemented mechanisms rather than tokens |
| Ban lists fail by reimplementation, 41% to 92% disguised | Lu et al. [23] | denial prompting adds a new constraint on the previous solution at each step to push the model toward more creative solutions | extension | mechanism labels show the constraint satisfied in name only |
| Qwen-Scope base SAEs fail the cross-entropy ratio on the instruct model | Kissane et al. (blog post, not in the bibliography) | base model SAEs usually transfer to chat models, but not on Gemma v1 2B | contrast and source of the remedy | Qwen3-8B would join Gemma v1 2B as a pair where base SAEs do not transfer under that criterion; Section 3.7 reverses this under a fairer criterion |
| Decoding-time constraints as the alternative to prompt bans | Biagiola et al. [62] | an incomplete constraint at decoding time pushes the model into low-probability regions of program space, causing time-outs and up to 97% lower functional correctness | contrast | single name prompt bans are obeyed at 0.86 and 0.84; vetoes at decoding time carry the cost Biagiola et al. measure (Section 3.8) |
| A single ban suppresses the default opening and promotes one substitute | Yoon et al. [4] | negation suppresses retrieval of the original answer while promoting a favoured candidate in the same answer category | replication at program level | the promoted candidate is the shared top rung, not a substitute specific to X |
| (same) | Zhou et al. [56] | models process negation by both suppression and construction, with construction more prominent | contrast | for a prohibition on code the route is carried by task specification states; construction of the negated phrase was not tested |

| | | | | |
|---|---|---|---|---|
| The ban is carried by task specification states and not by the read position alone | Hendel et al. [60] | functions learned in context often have a very simple structure, a task vector | contrast | a prohibition does not compress into one position's state |
| (same) | Todd et al. [61] | a few attention heads transport a compact representation of the demonstrated task, a function vector | contrast | ban-attending heads exist (layer 21 head 17, layer 26 heads 21 and 25), but the effect is spread over task specification positions; Section 3.7 tests the heads causally |

### 3.2.7 Corrections and reproduction targets

The audit wrote 12 correction items, each with its old wording, the recomputed numbers and a replacement sentence; they are applied in place in Sections 1.12 and 2.3 to 2.12 and marked. It also recomputed 19 numbers quoted in the record; 16 reproduced (Table 93). One mismatch is a difference of definition rather than of data. The audit counted DFS as the grid substitute's mode in 5 of 5 iteration-1 draws by plurality, while the record's 4 of 5 applies the realised mode rule of Section 1.2, which requires a 15-point lead; the geometry draw (DFS 10, unit weight Dijkstra 8 of 18) fails that lead. The record's count stands.

**Table 93.** Numbers quoted in the record and their recomputation. Source: record audit `corrections.json#reproduction_targets`.

| Quantity | Quoted | Recomputed | Match | Cause of a mismatch |
|---|---|---|---|---|
| Shortest path Bellman-Ford/ SPFA, iteration-1 five draws | 272/272 | 272/272 | yes | |
| Grid DFS, iteration-1 five draws | 116/128 (0.906) | 116/128 | yes | |
| Grid draws with DFS as realised mode | 4 of 5 | 5 of 5 | no | plurality count; the record applies the 15-point lead of Section 1.2 |
| Iteration-2 pooled shortest path Bellman-Ford/SPFA (attempted, not banned) | 255/260 | 255/260 | yes | |
| Iteration-2 pooled grid DFS (attempted, not banned) | 257/257 | 257/257 | yes | |
| Residual of IDDFS under the BFS ban | 7.988 | 7.988 | yes | |
| Semantic seed, shortest path Hill-1 | 1.935 [1.534, 2.076] at 7 points | 1.940 [1.770, 2.094] at 7.0 points | yes | point within 0.005; the iteration-2 bootstrap scheme was not available |
| A* mention Hill-1 (control) | 2.69 at 11 points | 2.69 at 11.3 points | yes | |
| Generate then select | 3 to 4 algorithms at 24 to 42 points | grid 4 at 42; shortest path 3 at 24 | yes | |
| Knob table rows | 249 | 306 (276 without the instruction tuning arms) | no | steering arms, instruct arms and temperature cells kept as separate units |
| Ban list manifold figures | 14 | 13 | no | 13 files exist |

| | | | | |
|---|---|---|---|---|
| TOP5 BFS / DFS attempted | 63 / 36 | 63 / 36 | yes | |
| Before the commit token, BFS within spread against BFS to bidirectional BFS distance | 0.0154 against 0.0147 | 0.0154 against 0.0147 | yes | |
| ESS weighted share | 1.24 [0.57, 1.62] | 1.24 [0.57, 1.62] | yes | |
| Qwen3-8B aim, mention reference | -0.595 | -0.595 | yes | |
| Llama aim, mention reference | -0.28 | -0.283 | yes | |
| Llama bootstrap against replicate variance | 5 to 11 times | 4.8 to 11.3 times | yes | |
| Llama grid baseline | BFS 77, DFS 20 attempted; BFS 20 of 20 correct | same | yes | |
| Gated forecast events of iteration 2 | 3 events, 0.05-nat margin | quoted, not recomputed | yes | record fact |

## 3.3 Bench forecast experiment: do ban-free forecasts predict where the modal bench bans land? (Qwen3-8B)

**Why.** This experiment was the confirmation of the title claim in Qwen3-8B. On the 12 frozen bench tasks, which had never seen a ban prompt, it built the four ban-free forecasts of where each task's modal ban would land, sealed them in a sha256 hash chain, and only then sampled the bans. If a ban only removes the default, the subtraction or ladder forecast should win; if it substitutes, association or proximity should.

**Design.** The model ran in bf16 (revision b968826d, thinking off, $T = 1.0$) on one RTX 4090. The bench, its labellers (runtime trace first, AST second, blind judge on disagreement) and its gold sets were copied with all 146 committed file hashes verified, and the labeller recheck matched every gate value on 12 of 12 tasks. Ban prompts used the five frozen templates with token-length matched neutral twins (identical token ids from the task text on). The ban-free arms had 600 samples each over the 12 tasks: the neutral twin, the one-shot, persistent and sham signature vetoes, the nonce ban, the novelty instruction, and up to three iterated veto rounds. Three seeded isotropic random directions were patched into the task specification states at layers 12 to 23 at the ban's own displacement norm (50 samples per task each). The vetoes used exact rejection and a masked redraw coupled to the twin samples by seed. Seventeen forecast files, the n-grams, seeds, norms and the analysis code were sealed at 18:50:57Z; the first ban request followed at 18:55:16Z. Then 1,200 ban samples (5 templates by 20 per task) were labelled, and a band patch control spliced the ban's own task specification states at layers 12 to 23 into the twin (600 samples). Every family's forecast was smoothed the same way (93% forecast, 5% uniform over survivors, 2% on other) and scored by its mean log score per attempt not banned. The frozen rule advances a family that beats the next best by at least 0.10 nats on at least 3 gated events whose top-1 nominations disagree. An undefined forecast is scored as uniform.

**Table 94.** Forecasts committed before any ban sample (top-1 survivor; a forecast marked undefined is scored as uniform). The distinctive n-gram is shown as tokenised (␣ marks a space); its source is the 8-bit pilot unless marked. Source: bench forecast experiment `README.md` and `predictions/`.

| Task | Default | Distinctive n-gram | Subtraction | Ladder | Association | Proximity | Nominations disagree |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| kth smallest | full sort | `␣␣␣␣nums` | undefined | quickselect | quickselect | heapify and pop | yes |
| substring search | naive | `␣needle: \n␣␣␣␣␣␣␣␣␣␣␣` | undefined | undefined | KMP | built-in find | yes |
| find duplicate | hash set | `␣set` | sort and adjacent check | undefined | Floyd cycle detection | sort and adjacent check | yes |
| two sum | hash map | `␣if␣complement` | undefined | undefined | sort and two pointers | brute force | yes |
| connected components | union find | `␣find` | DFS | undefined | DFS | label propagation | yes |
| nonoverlapping intervals | greedy by end | `␣if␣start` (rederived on bf16) | undefined | undefined | DP with bisect | DAG longest path | yes |
| Fibonacci | iteration | `(` | undefined | matrix power | matrix power | memoised recursion | yes |
| topological order | Kahn | `.p` | undefined | undefined | DFS postorder | undefined | no |
| merge intervals | sort and sweep | `␣last` | undefined | undefined | interval tree | union find | yes |
| first and last position | binary search | `_last` | undefined | undefined | bisect module | bisect module | no |
| top-k frequent | sort by count | `_counts` (rederived on bf16) | undefined | undefined | heap | heap | no |
| longest increasing subsequence | quadratic DP | `␣j` | undefined | patience sorting | patience sorting | patience sorting | no |

The subtraction forecast was undefined on 7 tasks because the default kept at least half of the attempts in every veto round, and on 3 more because no round produced 20 attempts off the default. Bans then failed on most tasks (Table 95): only 5 of 12 passed the gate. On nonoverlapping intervals, topological order, first and last position and longest increasing subsequence the model kept its default on almost every sample whatever the wording, and 88% to 100% of the programs that kept the banned mechanism named neither its alias nor its display name.

**Table 95.** What the named ban did (N = 100 per task; landing = attempted programs not using the default), against the ban-free arms. Avoidance is one minus the default's share of attempts (Wilson 95%). Source: bench forecast experiment `README.md`, `results/c1_events.jsonl`.

| Task | Ban avoidance | Gate | Ban landing | Twin default share | One-shot veto avoidance | Best ban-free push (avoidance) | Band patch avoidance | Frozen rule winner |
|---|---|---|---|---|---|---|---|---|
| kth smallest | 0.97 [0.92, 0.99] | pass | quickselect 59, size-k heap 38 | 0.96 | 0.38 | novelty 1.00 | 1.00 | ladder |
| substring search | 0.40 [0.31, 0.50] | fail | KMP 21, built-in find 16, Rabin-Karp 2, Z algorithm 1 | 1.00 | 0.00 | novelty 0.26 | 0.00 | none |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| find duplicate | 0.79 [0.70, 0.86] | pass | index marking 73, Floyd cycle detection 5, sort and adjacent check 1 | 1.00 | 0.00 | novelty 0.02 | 0.04 | ladder (undefined, scored uniform) |
| two sum | 0.86 [0.78, 0.91] | pass | brute force 86 | 1.00 | 0.00 | novelty 0.08 | 0.22 | proximity |
| connected components | 1.00 [0.96, 1.00] | pass | BFS 83, DFS 17 | 1.00 | 0.33 | random direction 3, 0.26 | 1.00 | ladder (undefined, scored uniform) |
| nonoverlapping intervals | 0.00 [0.00, 0.04] | fail | none | 1.00 | 0.00 | none (0.00) | 0.00 | none |
| Fibonacci | 0.34 [0.25, 0.44] | fail | other 33, fast doubling 1 | 1.00 | 0.00 | novelty 0.40 | 0.00 | none |
| topological order | 0.08 [0.04, 0.15] | fail | DFS postorder 8 | 1.00 | 0.00 | none (0.00) | 0.00 | none |
| merge intervals | 0.58 [0.48, 0.67] | pass | other 44, union find 14 | 1.00 | 0.00 | none (0.00) | 0.14 | proximity |
| first and last position | 0.03 [0.01, 0.08] | fail | other 2, bisect module 1 | 0.80 | 0.42 | nonce 0.36 | 0.00 | none |
| top-k frequent | 0.48 [0.38, 0.58] | fail | heap 32, quickselect on counts 15, other 1 | 0.96 | 0.04 | nonce 0.04 | 0.00 | none |
| longest increasing subsequence | 0.02 [0.01, 0.07] | fail | patience sorting 2 | 1.00 | 0.00 | novelty 0.68 | 0.00 | none |

*(Added in iteration 4: avoidance in Table 95 counts failing programs labelled other. Counted over catalogue algorithms only, avoidance is 0.14 on merge intervals, where 44 of the 58 programs off the default are labelled other, and 0.01 on Fibonacci (33 of 34); merge intervals therefore passes the gate only through residue. The other tasks change by at most 0.02. Catalogue-only avoidance for every bench event is in Section 4.2.4.)*

The frozen rule returned the SUBTRACTION/LADDER branch, with the ladder advancing: on the five gated events, all with disagreeing nominations, the ladder won 3 (kth smallest, find duplicate, connected components) and proximity 2 (two sum, merge intervals). Table 96 shows why that label carries little. On find duplicate and connected components the ladder was undefined and scored as uniform, and it won because every defined forecast named the wrong survivor. Counting only wins by defined forecasts (a post hoc check, not preregistered) gives ladder 1 and proximity 2, and no family advances. Against the uniform reference over the gated events, the mean differences are ladder +0.06 (task bootstrap 95% [0.00, 0.18]), subtraction -0.69 [-1.42, 0.00], proximity -1.53 [-2.96, 0.09] and association -1.78 [-2.73, -0.73]: no family forecasts better than uniform on average, and the association forecast is worse than uniform on all five events.

**Table 96.** Mean log score per attempt not banned for each forecast (nats; higher is better; uniform over the survivors is the reference). Gated events are marked †. The 8-bit likelihood column is the

frozen bench likelihood argmax, a descriptive forecast. Source: bench forecast experiment `results/c1_events.jsonl`.

| Task | Subtraction | Ladder | Association | Proximity | 8-bit likelihood | Uniform |
|---|---|---|---|---|---|---|
| kth smallest † | -1.630 | -1.326 | -1.842 | -4.605 | -1.842 | -1.630 |
| substring search | -1.630 | -1.630 | -2.220 | -2.788 | -1.630 | -1.630 |
| find duplicate † | -3.210 | -1.406 | -4.108 | -4.327 | -1.406 | -1.406 |
| two sum † | -1.119 | -1.119 | -4.094 | -0.055 | -1.119 | -1.119 |
| connected components † | -2.789 | -1.119 | -3.408 | -4.094 | -1.119 | -1.119 |
| nonoverlapping intervals | n/a | n/a | n/a | n/a | n/a | n/a |
| Fibonacci | -3.830 | -2.672 | -3.917 | -3.917 | -3.830 | -3.830 |
| topological order | -1.119 | -1.119 | -0.055 | -1.119 | -1.119 | -1.119 |
| merge intervals † | -3.140 | -3.140 | -3.858 | -2.979 | -3.140 | -3.140 |
| first and last position | -2.846 | -2.846 | -2.623 | -2.623 | -2.846 | -2.846 |
| top-k frequent | -1.177 | -1.177 | -1.398 | -1.398 | -1.177 | -1.177 |
| longest increasing subsequence | -0.713 | -2.234 | -0.046 | -0.046 | -0.713 | -0.713 |

*[Figure fig_bench_forecasts: generated in the report phase, not in this draft]*

The landings were specific to the banned task and mostly unforeseen. Find duplicate went to index marking, which places each value at index value minus one: the model wrote it only once in 40 template-forced attempts in the bench build (adherence 0.28 over all forced prompts), and no family nominated it. Connected components went to BFS (83 of 100) while subtraction and association said DFS. Merge intervals went mostly to programs labelled other, which failed every time (0 of 44) *(corrected in iteration 4 from the earlier description as a merge sweep without sorting: a manual read of all 44 found 29 programs that mark coverage over the coordinate range in a boolean array, 12 unsorted sweeps and 3 others; Section 4.2.4)*. On Fibonacci the programs labelled other used Binet's closed form, which also failed (0 of 33). Only the kth smallest landing (quickselect: ladder, association and the 8-bit likelihood) and the two sum landing (brute force: proximity) were hit by a top-1 nomination. On 3 of the 5 gated events no family's top-1 named the realised landing.

**Signature vetoes move the surface, not the algorithm.** The distinctive n-gram fired in 95% of twin samples and the redraw always avoided it, yet the default survived every persistent and iterated round on 9 of 12 tasks. The model respelled the same mechanism: on kth smallest `nums.sort()` became `return sorted(nums)[k-1]`, and on two sum the next round's distinctive token was `␣compliment`, a misspelling that dodges `␣complement`. The vetoed text reappeared under another token split in 23% of one-shot samples and 54% by round 2. Where vetoes did move off the default, most outputs were invalid. The sham veto of a shared token broke programs without moving the algorithm (for example, the model indented with non-breaking spaces after the indentation token was masked). The test of whether the one-shot veto reproduces the ban's landing was therefore undetermined: the one-shot veto never reached 0.5 avoidance on a gated task (kth smallest 0.38, connected components 0.33, others 0). Descriptively, the last iterated round was eligible on find duplicate and connected components, and both landed elsewhere than the ban: sort and adjacent check against the ban's index marking (Jensen-Shannon divergence 0.77 against a template resampling 95th percentile of 0.09), and DFS against the ban's BFS (0.58 against 0.09).

**Generic pushes rarely move this collapsed model.** Nonce bans moved nothing except on first and last position, whose twin was already noisy. The novelty instruction moved kth smallest (all 50 off the default, 45 to quickselect), longest increasing subsequence, Fibonacci, substring search and connected components. One of three random directions moved kth smallest (avoidance 0.82,

mostly quickselect) and connected components a little (0.26, to DFS); the HF twin control kept the default at 0.98 and 1.00 on those tasks, so the movement is not an engine artefact. The positive control of patching the ban's own task specification states into the twin at layers 12 to 23 reproduced the ban's avoidance only on kth smallest and connected components (1.00 each), partly on two sum (0.22) and merge intervals (0.14), and nowhere else. On most tasks the ban's effect is not carried by the task specification states of that band alone, which Section 3.6 confirms behaviourally.

**Wording.** The banned share differed by template far more than by chance (Table 97), with a likelihood ratio of 374 on 4 degrees of freedom (p about 1e-79). A mixed logistic model put the template standard deviation at 2.0 logits [1.1, 3.5] and the template by task deviation at 1.25 [1.05, 1.49]; a shuffled template placebo gave $p = 0.44$ (0.39 in the second audit's own likelihood ratio test). Under the fourth template the default kept at least 70% of samples on 10 of 12 tasks.

**Table 97.** Banned programs per template (of 20) under each modal bench ban. Source: bench forecast experiment `results/template_effect.json`.

| Task | First | Second | Third | Fourth | Fifth |
|---|---|---|---|---|---|
| kth smallest | 0 | 0 | 0 | 3 | 0 |
| substring search | 11 | 6 | 19 | 20 | 4 |
| find duplicate | 0 | 1 | 0 | 20 | 0 |
| two sum | 0 | 0 | 0 | 14 | 0 |
| connected components | 0 | 0 | 0 | 0 | 0 |
| nonoverlapping intervals | 20 | 20 | 20 | 20 | 20 |
| Fibonacci | 20 | 6 | 15 | 20 | 5 |
| topological order | 20 | 16 | 19 | 20 | 17 |
| merge intervals | 12 | 5 | 3 | 20 | 2 |
| first and last position | 20 | 20 | 20 | 20 | 17 |
| top-k frequent | 12 | 2 | 16 | 20 | 2 |
| longest increasing subsequence | 20 | 20 | 20 | 20 | 18 |
| All tasks (of 240) | 135 | 96 | 132 | 197 | 85 |

*[Figure fig_bench_templates: generated in the report phase, not in this draft]*

**The bench likelihoods on the screen's path.** Rescoring the frozen bench likelihood sets with fp32 activations, the path the likelihood screen had validated, moved the adherence corrected likelihood of a cell by a median 0.88 nats (maximum 9.0) from the 8-bit prompt log probabilities. Every shift above 2 nats sat in a cell with Kish ESS below 5, and the argmax algorithm was unchanged on 12 of 12 tasks. The rescoring code first reproduced 5 stored program scores of the likelihood screen within 0.021 nats. The 8-bit likelihood forecast was defined only on kth smallest, where it named quickselect.

**Checks.** An independent rederivation from the raw label files reproduced every headline number with no mismatch; a task shuffled placebo collapsed the family wins (mean ladder wins 0.52 against 3 observed); a template permutation test gave $p = 0.002$ on real data and 0.42 on shuffled data; and the hash chain verified. One file changed after the seal: absolute input paths in a configuration module were replaced by relative paths for portability, with the sealed bytes and the diff kept. GPU time was about 45 minutes, judge spend USD 0.024 over 344 calls.

## 3.4 Exposure experiment: show a solution, ban it, and see what replaces it (Qwen3-8B)

**Why.** Under total collapse, every modal ban's association and ladder nominations are close to each other, and bench bans in Qwen3-8B pass the gate on only five tasks. Exposure bans create events where the two accounts finally disagree: the prompt shows correct programs of a non-default algorithm Y, which the model then copies, and asks for a solution that does not use Y. Subtraction predicts a return to what the model would write without Y in view, which in context is its default; substitution predicts Y's stated alternative.

**Design.** The experiment took every non-default algorithm Y of the 12 frozen bench tasks with at least one correct forced program: 33 Y. We call the 19 whose frozen associate differs from the default priority Y. The prompt shows up to 5 name-scrubbed correct programs of Y and then asks one of three things: write a correct solution, which we call the copy twin; write a correct solution that does not use Y, the named ban (the fifth template); or write a correct solution that uses a different algorithm from the one above, the unnamed ban. Each arm had 50 samples per Y, with seeds shared across arms (Qwen3-8B bf16, vLLM 0.11.0, one 23 GB NVIDIA GPU). Four forecasts were committed by hash before any ban sample: ladder (the context-free default), association (the modal frozen answer for Y), proximity (the survivor whose forced programs are nearest the shown programs by Zhang-Shasha tree edit distance [64]) and the subtraction forecast in context (the landing of an iterated persistent signature veto of Y under the copy twin prompt, with no ban text). GPU smoke tests confirmed the veto: 0 of 20 veto samples contained the vetoed n-gram against 20 of 20 copy samples, an inert veto left 20 of 20 outputs identical, and veto and copy twins diverged exactly at the vetoed position in 20 of 20. Of 10,600 programs, 7,478 were labelled by agreeing runtime and AST labellers, 1,398 by the judge, 147 by runtime alone, and 1,577 had no code block (mostly from veto rounds).

**Showing a solution makes the model copy it, and banning it sends the model home.** Under the copy twin a mean 0.93 of attempts used the shown algorithm (median 1.00; none below 0.2), and the default appeared in 0.05. Under the named ban, avoidance averaged 0.95 and the default's share of attempts rose to 0.72 (Table 98). We call this return to the context-free default *snap-back*. In 24 of the 31 gated named events the modal replacement was the default (14 of 18 for priority Y). Named and unnamed bans picked the same modal replacement for 0.88 of Y, and their landings were consistent under the Jensen-Shannon resampling null for 0.79 of Y: naming Y adds little beyond asking for a different algorithm.

**Table 98.** The default's share of attempts by arm: the mean over the 33 shown algorithms of each one's share (over 29 for the final veto round, where 4 shown algorithms had no valid attempts). Source: exposure experiment `README.md`, `results/other_reports.json`.

| Copy twin | Final veto round under the copy twin | Named ban | Unnamed ban |
|---|---|---|---|
| 0.05 | 0.10 | 0.72 | 0.82 |

*[Figure fig_exposure_snapback: generated in the report phase, not in this draft]*

Table 99 gives every named exposure event. The frozen rule (at least 3 wins by 0.10 nats on gated, disagreeing events) advanced the ladder with 7 wins, against subtraction 2, association 2 and proximity 0, so the frozen branch is SUBTRACTION/LADDER, with the ladder advancing. Gating removed two events (size-k heap on kth smallest and patience sorting on longest increasing subsequence, where Y was kept), and 30 of the 31 gated events had disagreeing nominations.

**Table 99.** Every named exposure event (N = 50 per arm). The winner is the family that beats the

next best by at least 0.10 nats on a gated event with disagreeing nominations (none when the top two tie; n/a when the event is ungated or its nominations do not disagree). An undefined subtraction forecast is scored as uniform. Source: exposure experiment `results/c1_events.jsonl`, `results/arm_stats.json`.

| Task | Shown algorithm Y | Priority | Default | Associate of Y | Copy: share of Y | Named: avoidance | Named landing (attempts not Y) | Default share | Gate | Subtraction top-1 | Proximity top-1 | Winner |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| kth smallest | size-k heap | yes | full sort | quickselect | 1.00 | 0.20 | full sort 10 | 0.20 | fail | undefined | heapify and pop | n/a |
| kth smallest | heapify and pop | yes | full sort | quickselect | 1.00 | 1.00 | quickselect 45, full sort 5 | 0.10 | pass | undefined | full sort | association |
| kth smallest | quickselect | yes | full sort | heapify and pop | 1.00 | 1.00 | full sort 50 | 1.00 | pass | undefined | median of medians | ladder |
| kth smallest | median of medians | yes | full sort | quickselect | 1.00 | 1.00 | quickselect 50 | 0.00 | pass | quickselect | quickselect | none |
| kth smallest | counting sort | yes | full sort | quickselect | 0.80 | 1.00 | full sort 49, quickselect 1 | 0.98 | pass | undefined | full sort | ladder |
| substring search | KMP | no | naive | naive | 1.00 | 1.00 | naive 50 | 1.00 | pass | undefined | Boyer-Moore-Horspool | none |
| substring search | Rabin-Karp | yes | naive | KMP | 1.00 | 1.00 | naive 50 | 1.00 | pass | undefined | Boyer-Moore-Horspool | ladder |
| substring search | Z algorithm | yes | naive | KMP | 1.00 | 1.00 | naive 44, KMP 6 | 0.88 | pass | undefined | Boyer-Moore-Horspool | ladder |
| substring search | Boyer-Moore-Horspool | yes | naive | KMP | 1.00 | 1.00 | naive 50 | 1.00 | pass | undefined | naive | ladder |
| substring search | built-in find | yes | naive | KMP | 1.00 | 1.00 | naive 50 | 1.00 | pass | undefined | naive | none |
| find duplicate | sort and adjacent check | yes | hash set | Floyd cycle detection | 1.00 | 1.00 | hash set 50 | 1.00 | pass | undefined | hash set | none |
| find duplicate | Floyd cycle detection | no | hash set | hash set | 1.00 | 1.00 | hash set 50 | 1.00 | pass | undefined | hash set | none |
| find duplicate | binary search on values | yes | hash set | Floyd cycle detection | 0.98 | 1.00 | hash set 50 | 1.00 | pass | undefined | hash set | ladder |
| find duplicate | index marking | yes | hash set | Floyd cycle detection | 1.00 | 1.00 | hash set 50 | 1.00 | pass | undefined | sort and adjacent check | ladder |
| two sum | sort and two pointers | no | hash map | hash map | 1.00 | 1.00 | hash map 50 | 1.00 | pass | undefined | hash map | none |
| two sum | brute force | yes | hash map | sort and two pointers | 0.36 | 1.00 | hash map 50 | 1.00 | pass | hash map | hash map | none |
| two sum | sort and binary search | yes | hash map | sort and two pointers | 1.00 | 1.00 | hash map 50 | 1.00 | pass | undefined | hash map | none |
| connected components | BFS | yes | union find | DFS | 1.00 | 1.00 | DFS 50 | 0.00 | pass | undefined | DFS | none |
| connected components | DFS | yes | union find | BFS | 1.00 | 1.00 | BFS 50 | 0.00 | pass | undefined | BFS | association |
| connected components | label propagation | yes | union find | DFS | 0.84 | 1.00 | union find 49, DFS 1 | 0.98 | pass | union find | union find | none |
| nonoverlapping intervals | memoised recursion | no | greedy by end | greedy by end | 1.00 | 1.00 | DP with bisect 41, other 7, greedy by end 2 | 0.04 | pass | undefined | greedy by end | subtraction |
| nonoverlapping intervals | DAG longest path | no | greedy by end | greedy by end | 1.00 | 0.94 | greedy by end 47 | 0.94 | pass | undefined | greedy by end | none |
| Fibonacci | memoised recursion | no | iteration | iteration | 1.00 | 1.00 | iteration 50 | 1.00 | pass | undefined | iteration | none |
| Fibonacci | matrix power | no | iteration | iteration | 1.00 | 1.00 | iteration 50 | 1.00 | pass | undefined | fast doubling | none |
| Fibonacci | fast doubling | no | iteration | iteration | 1.00 | 1.00 | other 49, iteration 1 | 0.02 | pass | undefined | iteration | none |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| merge intervals | union find | no | sort and sweep | sort and sweep | 1.00 | 1.00 | sort and sweep 50 | 1.00 | pass | undefined | sort and sweep | none |
| first and last position | bisect module | no | binary search | binary search | 1.00 | 1.00 | other 49, binary search 1 | 0.02 | pass | undefined | binary search | none |
| first and last position | exponential search | no | binary search | binary search | 0.28 | 1.00 | binary search 50 | 1.00 | pass | binary search | binary search | n/a |
| top-k frequent | heap | no | sort by count | sort by count | 1.00 | 1.00 | sort by count 50 | 1.00 | pass | undefined | sort by count | none |
| top-k frequent | bucket sort | no | sort by count | sort by count | 0.36 | 1.00 | sort by count 50 | 1.00 | pass | undefined | heap | none |
| top-k frequent | quickselect on counts | yes | sort by count | heap | 1.00 | 0.94 | sort by count 31, bucket sort 7, other 6, heap 3 | 0.62 | pass | undefined | bucket sort | none |
| longest increasing subsequence | patience sorting | no | quadratic DP | quadratic DP | 1.00 | 0.16 | quadratic DP 8 | 0.16 | fail | undefined | quadratic DP | n/a |
| longest increasing subsequence | Fenwick tree | yes | quadratic DP | patience sorting | 1.00 | 1.00 | quadratic DP 35, patience sorting 15 | 0.70 | pass | undefined | quadratic DP | subtraction |

The paired tests point the same way but none survives correction (Table 100). Over the 30 gated disagreeing named events, paired Wilcoxon signed-rank tests [65] of the per-event log score differences, Holm corrected over four comparisons, favour the ban-free families over association by about 1 nat per attempt, with task cluster bootstrap intervals above zero, but Holm adjusted p stays at 0.081. Top-1 hit rates on the gated named events were 0.77 for the ladder, 0.61 for proximity, 0.45 for association and 0.36 for subtraction. *(Correction, iteration 4: the subtraction rate is almost entirely tie credit. Subtraction was defined for 4 of 33 shown algorithms, and where defined it named the landing 4 of 4 times in both the named and the unnamed arms (median of medians to quickselect, brute force to hash map, label propagation to union find, exponential search to binary search); on every other event its uniform tie earned fractional credit. Defined-only rates are in Section 4.2.6.)* The unnamed events, which are descriptive, give the same branch, and there the ladder beats association after Holm correction ($+1.42$ nats, adjusted $p = 0.0056$).

**Table 100.** Paired log score differences over the 30 gated, disagreeing named exposure events (positive favours the first family). Source: exposure experiment `results/c1_summary.json`.

| Difference | Mean (nats) | Task cluster 95% CI | p | Holm adjusted p |
|---|---|---|---|---|
| Subtraction minus association | +1.02 | [0.35, 1.56] | 0.020 | 0.081 |
| Ladder minus association | +1.29 | [0.10, 2.38] | 0.027 | 0.081 |
| Ladder minus proximity | +0.50 | [-0.40, 1.16] | not reported | 0.159 |
| Subtraction minus proximity | +0.23 | not reported | not reported | 0.96 |

**How strong the verdict is.** The counting rule is not a significance test. In a unit test with no information, it advanced some family in 85% of 200 simulated draws, while the Holm-corrected Wilcoxon test erred in 2%; on the real events, a label permutation placebo advanced some family in 100% of draws (the ladder in 43%). The branch label is therefore weak evidence, and the paired tests are the informative part. The signature veto rarely removed Y. The subtraction forecast in context was defined for only 4 of 33 Y (median of medians to quickselect, brute force to hash map, label propagation to union find, exponential search to binary search). For 25 of 33 Y, Y still re-emerged in at least 20% of attempts after three rounds. The veto cost a mean 0.70 of pass rate, and 46% of final round outputs were invalid. Because an undefined subtraction forecast is uniform, its tied top-1 is what makes almost every event disagreeing. Every preregistered sensitivity returned the ladder with 7 to 9 wins (subtraction top-1, excluding co-vetoed survivors, priority Y only with 18 events, token distance proximity, correct-only landings, and excluding labeller-flagged tasks) except the one that drops events with an undefined subtraction forecast, which leaves 2 events and is undetermined. An exploratory comparison of ladder, association and proximity on the 21 events

where those three disagree gave wins 8, 3 and 0 (ladder minus association p = 0.027, Holm over three 0.081). The secondary exposure test, whether the ban's modal survivor matches the veto's where the associate differs, had 2 qualifying events (brute force and label propagation); both agreed, but both landings are the default, so this is snap-back, not a test of substitution.

**Wording.** A template arm (first to fourth templates, 10 samples for each of the 19 priority Y; descriptive and never in the contest) gave mean avoidance 0.93, 0.94, 0.96 and 0.81: the fourth template is again the weakest. *(Added in iteration 4: wording also changes the landing. Pooled over the first four templates the modal replacement is the default for 13 of 19 priority Y, against 15 of 19 under the fifth template of Table 99; it flips to the associate for the size-k heap (quickselect 13, full sort 2, against full sort 10 under the fifth template) and for the Z algorithm (KMP 29, naive 11, against naive 44, KMP 6). Snap-back as recorded in Table 99 rests on the fifth template.)* A stop token bug in the first round-2 veto run (the end-of-turn token was chosen as the n-gram) was found and fixed; the 750 affected samples are kept apart. Deduplication left 11 of 33 Y with a single shown program. GPU time was about 1.0 GPU hour, judge spend USD 0.096 over 1,229 calls, and an independent rederivation matched all 66 events.

## 3.5 Llama bench experiment: the same forecast contest in a second model family

**Why.** Llama-3.1-8B-Instruct [22] is less collapsed than Qwen3-8B on the bench pilot, so it supplies present minor bans, where a correct minor algorithm already exists and subtraction makes a non-trivial prediction. This experiment replicated the frozen forecast contest on the 12 bench tasks in bf16 (revision 0e9e39f), with every forecast computed without ban text and hash chained before any ban prompt was rendered (a grep audit found 0 ban prompts in the 10,328 rows generated before the commit).

**Design.** The default of each task was frozen from a decision set (the 8-bit Llama pilot pooled with 600 bf16 neutral twin samples after a homogeneity test) as the modal correct label when at least 10 programs were correct, else the modal attempted label (Table 101). Present minor events required at least 10% of correct programs, at least 3 correct, and at least 10 correct programs on the task. The four forecasts were built as in the Qwen3-8B experiment. The subtraction forecast used exact one-shot, persistent and iterated signature vetoes by truncate and redraw; the ladder pooled the nonce ban, the novelty instruction and three matched-norm random directions added at the task specification positions of layers 11 to 21 during prefill; association used Llama's own frozen answers; and proximity used Zhang-Shasha distance [64]. Association and proximity took a form that transfers the default's mass, with point mass forms as sensitivities. Then 1,800 bans were sampled (12 modal and 6 present minor events, 5 templates by 20) and labelled with the bench labellers; references were recovered 50 of 50, and gold agreement was 0.959. A second blind judge audited 60 Llama programs at 0.88 agreement, below 0.85 on four tasks (nonoverlapping intervals, kth smallest, two sum, top-k frequent).

**Table 101.** Llama defaults and events, frozen before any ban (decision set: 8-bit pilot plus bf16 neutral twins). Source: Llama bench experiment `README.md`, `predictions/events.json`.

| Task | Pilot mode | Default | Default from attempts | Valid programs | Correct programs | Default share | Hill_cc |
|---|---|---|---|---|---|---|---|
| kth smallest | heapify and pop | full sort | | 94 | 18 | 0.17 | 3.80 |
| substring search | naive | naive | | 89 | 49 | 0.54 | 2.00 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| find duplicate | hash set | hash set | | 93 | 34 | 0.44 | 1.48 |
| two sum | hash map | hash map | | 99 | 84 | 0.97 | 1.07 |
| connected components | union find | union find | | 97 | 74 | 0.75 | 1.59 |
| nonoverlapping intervals | greedy by end | greedy by end | yes | 89 | 8 | 0.63 | 2.46 |
| Fibonacci | matrix power | matrix power | | 96 | 44 | 0.76 | 1.78 |
| topological order | Kahn | Kahn | | 99 | 88 | 0.98 | 1.06 |
| merge intervals | sort and sweep | sort and sweep | | 89 | 65 | 1.00 | 1.00 |
| first and last position | binary search | binary search | | 93 | 54 | 0.99 | 1.10 |
| top-k frequent | heap | heap | | 86 | 22 | 0.62 | 1.80 |
| longest increasing subsequence | quadratic DP | quadratic DP | | 94 | 68 | 0.87 | 1.08 |

Present minor events: top-k frequent sort by count (6 of 22 correct), kth smallest heapify and pop (4 of 18), kth smallest quickselect (4 of 18), substring search KMP (9 of 49), connected components DFS (13 of 74), kth smallest size-k heap (3 of 18).

**The contest does not resolve in Llama.** Thirteen of 18 events passed the gate and 11 were gated and disagreeing, but only 2 had a winner by 0.10 nats (Table 102). One was the subtraction forecast on two sum, through its undefined forecast scored as uniform, because the ban split between sort and two pointers (42) and brute force (32). The other was association on kth smallest heapify and pop. No family advances; the preregistered branch is NO FAMILY ADVANCES, aimed but unresolved. The landing is aimed: the ladder and association each named the realised landing on 7 of 13 gated events, against a chance rate of about 0.27 (exact binomial $p = 0.033$ each); subtraction hit 4 of 12 ($p = 0.38$) and proximity 3 of 13 ($p = 0.71$). *(Correction, iteration 4: these hits were scored against a label this section did not define. Modal events were scored against the modal landing, but the six present minor events were scored against the survivor with the largest excess over the renormalised ban-free baseline. The ladder also nominated other on 3 of the 13 gated events, and on find duplicate both its nomination and the landing were other (50 of 94 attempts, 71 not banned; 2 correct). With other excluded as nominee and as landing, the ladder hits 6 of 13 (Poisson-binomial $p = 0.100$) and association 7 of 13 ($p = 0.032$), so only association is above chance. On the present minor events Llama returns to its default: union find 0.93, heap 0.72 and naive 0.73 of attempts not banned. Section 4.2.3.)* Five defaults resisted the ban itself (nonoverlapping intervals, topological order, merge intervals, first and last position, longest increasing subsequence; avoidance 0.04 to 0.30).

**Table 102.** Every Llama ban event (N = 100). s_X is the banned algorithm's share of attempts in the decision set before the ban; † marks a present minor event. Each family's top-1 is given with its mean log score per attempt not banned in parentheses; a forecast marked undefined is scored as uniform. Source: Llama bench experiment `README.md`, `results/c1_events.jsonl`.

| Event | s_X | Avoidance | Gate | Attempts not banned | Modal landing | Subtraction | Ladder | Association | Proximity | Renormalised baseline | Winner (margin) | Hill_cc (correct programs) | Other / invalid |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| kth smallest, ban full sort | 0.17 | 0.95 | pass | 87 | quickselect | heapify and pop (-1.71) | other (-1.46) | quickselect (-1.40) | size-k heap (-1.66) | -1.64 | none (0.06) | 1.00 (15) | 0.09 / 0.08 |
| substring search, ban naive | 0.54 | 0.79 | pass | 64 | KMP | Rabin-Karp (-1.20) | KMP (-1.25) | KMP (-1.18) | built-in find (-1.83) | -1.20 | none (0.02) | 1.38 (20) | 0.05 / 0.19 |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| find duplicate, ban hash set | 0.44 | 0.76 | pass | 71 | other | other (-0.98) | other (-1.00) | Floyd cycle detection (-1.53) | sort and adjacent check (-1.37) | -1.02 | none (0.02) | 3.59 (7) | 0.53 / 0.06 |
| two sum, ban hash map | 0.97 | 0.91 | pass | 86 | sort and two pointers | undefined (-1.39) | other (-1.90) | sort and two pointers (-1.94) | brute force (-2.37) | -1.39 | subtraction (0.51) | 1.63 (34) | 0.07 / 0.05 |
| connected components, ban union find | 0.75 | 0.82 | pass | 78 | DFS | DFS (-1.00) | DFS (-0.80) | DFS (-0.98) | DFS (-0.98) | -0.98 | none (0.18) | 1.68 (42) | 0.04 / 0.05 |
| nonoverlapping intervals, ban greedy by end | 0.63 | 0.29 | fail | 26 | other | other (-0.89) | other (-0.97) | DP with bisect (-1.51) | memoised recursion (-1.59) | -0.85 | none (0.08) | n/a (0) | 0.21 / 0.11 |
| Fibonacci, ban matrix power | 0.76 | 0.79 | pass | 72 | iteration | other (-1.13) | iteration (-1.10) | fast doubling (-2.31) | memoised recursion (-1.94) | -1.21 | none (0.03) | 1.72 (44) | 0.13 / 0.09 |
| topological order, ban Kahn | 0.98 | 0.18 | fail | 17 | DFS postorder | undefined (-1.39) | undefined (-1.39) | DFS postorder (-0.92) | source removal (-3.81) | -1.39 | none (0.46) | 1.00 (1) | 0.04 / 0.04 |
| merge intervals, ban sort and sweep | 1.00 | 0.30 | fail | 25 | other | undefined (-1.11) | undefined (-1.11) | union find (-3.13) | interval tree (-3.71) | -1.11 | none (0.00) | n/a (0) | 0.23 / 0.18 |
| first and last position, ban binary search | 0.99 | 0.04 | fail | 2 | other | undefined (-1.11) | undefined (-1.11) | bisect module (-3.76) | bisect module (-3.76) | -1.11 | none (0.00) | 1.00 (1) | 0.02 / 0.03 |
| top-k frequent, ban heap | 0.62 | 0.89 | pass | 84 | sort by count | sort by count (-0.74) | sort by count (-0.68) | sort by count (-0.70) | sort by count (-0.70) | -0.66 | none (0.02) | 1.31 (13) | 0.05 / 0.06 |
| longest increasing subsequence, ban quadratic DP | 0.87 | 0.23 | fail | 22 | patience sorting | undefined (-1.10) | undefined (-1.10) | Fenwick tree (-2.56) | patience sorting (-1.02) | -0.83 | none (0.08) | 1.00 (8) | 0.09 / 0.05 |
| top-k frequent, ban sort by count † | 0.36 | 0.82 | pass | 75 | quickselect on counts | heap (-1.05) | heap (-1.02) | heap (-1.06) | bucket sort (-1.34) | -1.02 | none (0.03) | 1.19 (24) | 0.09 / 0.09 |
| kth smallest, ban heapify and pop † | 0.33 | 0.88 | pass | 77 | quickselect | other (-1.54) | full sort (-1.57) | quickselect (-1.33) | size-k heap (-1.81) | -1.57 | association (0.21) | 2.57 (27) | 0.09 / 0.12 |
| kth smallest, ban quickselect † | 0.12 | 0.76 | pass | 71 | full sort | full sort (-1.65) | full sort (-1.73) | full sort (-1.56) | size-k heap (-1.79) | -1.73 | none (0.09) | 2.12 (34) | 0.10 / 0.06 |
| substring search, ban KMP † | 0.26 | 0.70 | pass | 66 | Boyer-Moore-Horspool | other (-0.99) | naive (-1.01) | Rabin-Karp (-1.22) | naive (-1.01) | -1.01 | none (0.01) | 1.14 (34) | 0.09 / 0.06 |
| connected components, ban DFS † | 0.25 | 0.89 | pass | 85 | BFS | union find (-0.35) | union find (-0.34) | union find (-0.34) | BFS (-0.47) | -0.34 | none (0.00) | 1.15 (65) | 0.02 / 0.04 |
| kth smallest, ban size-k heap † | 0.24 | 0.84 | pass | 78 | full sort | heapify and pop (-1.68) | full sort (-1.72) | quickselect (-1.70) | heapify and pop (-1.88) | -1.72 | none (0.02) | 2.23 (27) | 0.10 / 0.07 |

*(Correction, iteration 4: the modal landing column of Table 102 is mislabelled for three present minor events, where it shows the largest-excess survivor used for scoring. The modal landings are heap (54 of 75 attempts not banned) under the ban of sort by count, naive (48 of 66) under the ban of KMP and union find (79 of 85) under the ban of DFS, against quickselect on counts (12), Boyer-Moore-Horspool (5) and BFS (4) shown. Both labels are given side by side in Section 4.2.3.)*

The exact Wilcoxon tests over the 11 gated disagreeing events, Holm corrected, separate the ban-free families from proximity but not from association. Subtraction minus proximity (+0.345 nats [0.190, 0.633], Holm 0.0088) and ladder minus proximity (+0.306 [0.204, 0.513], Holm 0.0078) are rejected; subtraction minus association (+0.172 [-0.049, 0.606], $p = 0.52$) and ladder minus association (+0.133 [-0.056, 0.542], $p = 0.77$) are not. In Llama, code proximity is the worst forecast, and association, subtraction and the ladder cannot be told apart at this N, which is below the pooled floor of 16 events.

The verdict is fragile. Of ten preregistered sensitivity rows, three return substitution with association advancing on 3 wins: rescoring on correct programs only, the bf16-only decision set, and dropping the fourth template. Excluding the task whose default came from attempts and the tasks the label audit flagged leaves 5 disagreeing gated events and reads X specific but unaimed. The point mass forms, dropping co-vetoed survivors and dropping any other single template all keep NO FAMILY ADVANCES.

**Signature vetoes rarely remove a Llama default either (Table 103).** Only the heap of top-k frequent collapsed (0.02 once `(k` and `␣heapq` were both masked). Elsewhere the default re-emerged at 0.20 to 0.96 in the last persistent round; hash map (0.89), sort and sweep (0.88), Kahn and binary search (0.96) barely moved.

**Table 103.** Share of the banned algorithm among veto attempts per round (persistent, iterated arm; weighted counts for present minor events; ␣ marks a space in the n-grams). Source: Llama bench experiment `README.md`.

| Event | Round-1 n-gram | Round 1 | Round 2 | Round 3 | Note |
|---|---|---|---|---|---|
| kth smallest, ban full sort | `,` | 0.20 | | | co-vetoes every survivor |
| substring search, ban naive | `1):\n` | 0.44 | 0.42 | 0.54 | not carried by a small set of surface tokens; co-vetoes Rabin-Karp |
| find duplicate, ban hash set | `␣nums:\n␣␣␣␣␣␣␣` | 0.38 | 0.26 | 0.26 | not carried by surface tokens |
| two sum, ban hash map | `␣i]\n` | 0.93 | 0.95 | 0.89 | not carried by surface tokens |
| connected components, ban union find | `␣def␣find` | 0.59 | 0.57 | 0.23 | not carried by surface tokens |
| nonoverlapping intervals, ban greedy by end | `.sort` | 0.60 | 0.50 | 0.36 | not carried by surface tokens; co-vetoes DP with bisect |
| Fibonacci, ban matrix power | `␣matrix` | 0.46 | 0.43 | 0.28 | not carried by surface tokens |
| topological order, ban Kahn | `␣queue` | 0.98 | 0.96 | 0.96 | not carried by surface tokens |
| merge intervals, ban sort and sweep | `\ndef␣solve` | 0.84 | 1.00 | 0.88 | not carried by surface tokens |
| first and last position, ban binary search | `␣=␣0,` | 0.98 | 0.96 | 0.96 | not carried by surface tokens |
| top-k frequent, ban heap | `(k` | 0.57 | 0.02 | | removed |
| longest increasing subsequence, ban quadratic DP | `[j]␣+` | 0.85 | 0.83 | 0.80 | not carried by surface tokens |
| top-k frequent, ban sort by count † | `␣[` | 0.37 | 0.34 | 0.35 | not carried by surface tokens; co-vetoes heap and quickselect on counts |
| kth smallest, ban heapify and pop † | `␣{}\n␣␣␣` | 0.25 | 0.15 | | |
| kth smallest, ban quickselect † | `␣quick` | 0.12 | | | |
| substring search, ban KMP † | `␣␣␣␣␣␣␣␣␣␣␣␣␣␣␣␣␣␣j` | 0.26 | 0.24 | | |

| | | |
|---|---|---|
| connected components, ban DFS † | `␣dfs` | 0.17 |
| kth smallest, ban size-k heap † | `.n` | 0.09 |

Of the two events where the one-shot veto moved at least half of the banned algorithm's attempts, neither shared the ban's modal survivor within the template bootstrap null (0 of 2, against a preregistered bar of two thirds, quoted for reference only). The test that matters most for substitution is the present minor gain (Table 104): if a named ban only removes the minor algorithm, the associate's share under the ban should equal its share under the persistent veto of that algorithm. On two of six events the associate gains more under the named ban (kth smallest heapify and pop to quickselect, +0.25 [0.12, 0.37]; kth smallest quickselect to full sort, +0.24 [0.10, 0.36]); on substring search KMP the associate (Rabin-Karp) is less frequent under the ban (-0.08 [-0.17, -0.00]); on three it is unchanged. Naming the banned algorithm sometimes steers Llama toward its associate beyond what removing it does, but not consistently.

**Table 104.** Present minor events: the associate's share under the named ban against the persistent veto of the same algorithm (Newcombe 95% intervals [42]). Source: Llama bench experiment `README.md`.

| Event | Associate | Ban share | Veto share | Baseline share | Ban minus veto |
|---|---|---|---|---|---|
| top-k frequent, ban sort by count | heap | 0.59 | 0.63 | 0.62 | -0.04 [-0.18, 0.10] |
| kth smallest, ban heapify and pop | quickselect | 0.42 | 0.17 | 0.12 | 0.25 [0.12, 0.37] |
| kth smallest, ban quickselect | full sort | 0.44 | 0.20 | 0.17 | 0.24 [0.10, 0.36] |
| substring search, ban KMP | Rabin-Karp | 0.04 | 0.12 | 0.12 | -0.08 [-0.17, -0.00] |
| connected components, ban DFS | union find | 0.82 | 0.81 | 0.75 | 0.01 [-0.10, 0.12] |
| kth smallest, ban size-k heap | quickselect | 0.22 | 0.18 | 0.12 | 0.04 [-0.08, 0.15] |

**Pushes.** The nonce ban left the default near its baseline on most tasks. Matched-norm random directions moved the default substantially on a few tasks (Fibonacci matrix power 0.76 to 0.16 to 0.51; top-k frequent heap 0.62 to 0.39 to 0.56), and the novelty instruction lowered the default almost everywhere. The realised layer-21 displacement of the random pushes was about three times the ban's, because pushes at layers 11 to 20 propagate, so the ladder inputs are generous pushes. A second blind judge re-audited 187 ban outputs labelled other and called 8 (4%) the banned algorithm and 24 a catalogue survivor; other is never folded into avoidance.

**The path task screen (descriptive, never pooled).** On the iteration-2 Llama path tasks, the same ban-free families were scored against the Llama replication's existing ban landings, using that experiment's own prompt builders (10 of 10 prompts reproduced byte for byte) and labeller (Table 105). The vetoes fired (25 of 50 grid and 40 to 48 of 50 shortest path pass-1 samples) but BFS (0.52 to 0.58) and Dijkstra (0.78 to 0.82) re-emerged through other surface forms in every round. On the grid task all four families named DFS, the realised landing. On the shortest path task, the ban's large other share (37 of 100) was not reproduced by any veto round (8 of 50 other each); the novelty instruction came close (18 of 50 other).

**Table 105.** Path task screen in Llama. Source: Llama bench experiment `README.md`.

| Task | Ban landing (Llama replication) | Modal | Subtraction | Ladder | Association | Proximity | Winner | Veto: banned share by round |
|---|---|---|---|---|---|---|---|---|

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Grid | DFS 63, BFS 29, other 6, A* 1, unit weight Dijkstra 1 | DFS | DFS (-0.54) | DFS (-0.88) | DFS (-0.47) | DFS (-0.47) | none (0.00) | ␣queue:\n␣␣␣␣␣␣␣: 0.56, 0.52, 0.58 |
| Shortest path | Dijkstra 43, Bellman-Ford/SPFA 18, other 37, Floyd-Warshall 2 | other | undefined (-1.59) | undefined (-1.59) | Bellman-Ford/SPFA (-1.79) | Floyd-Warshall (-2.81) | none (0.00) | ␣continue: 0.82, 0.78, 0.80 |

**Checks.** Input integrity, hooks, prompts, sampler, scoring and veto tests passed; every committed file preceded the first ban sample; an independent rederivation matched the gate, landings, log scores, wins and branch, and a family permutation placebo advanced a family in 0.00 of 1,000 permutations; the exact Wilcoxon p values, Holm decisions and binomial tests were rederived without the project code. Not rederived: the sensitivity rows, the bootstrap intervals, the one-shot veto comparison, the present minor gain and the screen scores. GPU time was 0.98 hours of logged generation (about 1.8 hours with the GPU held), judge spend USD 0.48.

## 3.6 Bench route experiment: which internal route carries a ban, task by task (Qwen3-8B)

**Why.** Iteration 2 found on the two path tasks that the ban is carried mostly by the prompt states. The hypothesis update made that the second claim of the title account: if a ban only removes the default, the removal should be applied by the task specification states before the first generated token, with generated tokens reading the ban directly only where the one-shot veto does not suffice. This experiment measured the route in behaviour on the 12 bench tasks, with the iteration-2 engine.

**Design.** Qwen3-8B ran in bf16 under HF with common random numbers (one uniform per row and step, inverse CDF sampling) and cache splicing by position class, on one 23 GB NVIDIA GPU with batches of 40 rows of one shape class. Each slot ran seven kinds of arm. The full ban (a) and the token-length matched neutral twin (b) are the endpoints. The prompt route (c) splices the twin's condition segment with the ban's task specification and read position states; the direct route (d) splices the ban's condition segment with the twin's later states. The full knockout arm removes the condition span from the cache seen by generated tokens. A one-shot veto arm applies the signature veto of the default under the twin, and prefills force the default's opening or a neutral `def` line under the ban and the twin. N was 100 slots per arm (5 templates by 20), and 13,030 samples were labelled by the frozen bench labellers. Route fractions are defined on the default's share of attempts s as F_prompt = (s_b - s_c) / (s_b - s_a), F_direct = (s_b - s_d) / (s_b - s_a), the interaction I = 1 - F_prompt - F_direct, and the knockout fraction F_knockout = (s_b - s_knockout) / (s_b - s_a), where s_knockout is the default's share under the full knockout. The decision rule (prompt route at least two thirds, with the interval above one half; disconfirmed if the direct route reaches one third) was hash committed before any ban sample.

**Only five bench bans take.** Table 106 gives the default's share in every arm for all 12 tasks. The gate passed on kth smallest (avoidance 0.99), connected components (1.00), two sum (0.83), find duplicate (0.79) and merge intervals (0.59), the same five tasks as in the bench forecast experiment. The fourth template was again the weakest (pooled avoidance 0.19 against 0.42 to 0.62 for the others) and produced 29% of all ban failures.

**Table 106.** The default's share of attempts by arm (N attempted in parentheses), and the ban's

landing. Source: bench route experiment `results/shares.json`.

| Task | Full ban (a) | Twin (b) | Prompt route (c) | Direct route (d) | Full knockout | One-shot veto | Ban landing (attempts off the default) |
|---|---|---|---|---|---|---|---|
| kth smallest | 0.01 (100) | 0.95 (100) | 0.01 (100) | 0.25 (100) | 0.01 (100) | 0.59 (100) | quickselect 70, size-k heap 28, heapify and pop 1 |
| connected components | 0.00 (100) | 0.99 (100) | 0.00 (100) | 0.20 (100) | 0.00 (100) | 0.70 (20) | BFS 75, DFS 25 |
| two sum | 0.17 (100) | 1.00 (100) | 0.66 (100) | 0.20 (100) | 0.67 (100) | 1.00 (100) | brute force 83 |
| find duplicate | 0.21 (100) | 1.00 (100) | 0.97 (100) | 0.22 (100) | 0.95 (100) | 1.00 (100) | index marking 77, Floyd cycle detection 1, other 1 |
| merge intervals | 0.41 (100) | 1.00 (100) | 0.77 (100) | 0.99 (100) | 0.74 (100) | 1.00 (69) | other 44, union find 15 |
| substring search | 0.58 (100) | 1.00 (100) | 1.00 (100) | 0.89 (100) | 1.00 (50) | 1.00 (94) | KMP 18, built-in find 17, Boyer-Moore-Horspool 5, Rabin-Karp 2 |
| top-k frequent | 0.59 (100) | 0.93 (100) | 0.97 (100) | 0.77 (100) | 0.94 (50) | 0.93 (100) | heap 30, quickselect on counts 10, other 1 |
| Fibonacci | 0.69 (100) | 1.00 (100) | 0.99 (100) | 0.72 (100) | 0.96 (50) | 1.00 (100) | other 31 |
| topological order | 0.96 (100) | 1.00 (100) | 1.00 (100) | 1.00 (100) | 1.00 (50) | 1.00 (46) | DFS postorder 4 |
| longest increasing subsequence | 0.98 (100) | 1.00 (100) | 1.00 (100) | 1.00 (100) | 1.00 (50) | 1.00 (42) | other 1, patience sorting 1 |
| first and last position | 0.98 (100) | 0.76 (100) | 1.00 (100) | 0.76 (100) | 1.00 (50) | 0.61 (62) | other 1, bisect module 1 |
| nonoverlapping intervals | 1.00 (100) | 1.00 (100) | 1.00 (100) | 1.00 (100) | 1.00 (50) | 1.00 (97) | none |

*(Added in iteration 4: the full-ban avoidance of merge intervals in Table 106 (0.59) includes 44 failing programs labelled other; its catalogue-only avoidance is 0.15, below the gate, so under catalogue-only avoidance the route analysis has four gated tasks. Section 4.2.4.)*

**The route is task dependent, and pooled it is not the prompt route.** On the five gated tasks the mechanisms differ (Table 107). On kth smallest and connected components the information is redundant: the prompt route alone and the knockout (which removes direct reading) each reproduce the ban, and the direct route alone does most of it as well. On two sum and find duplicate the ban acts almost entirely through generated tokens reading the condition segment: the direct route arm reproduces the ban, while the prompt route and the knockout do not (find duplicate: 0 against 76 discordant slots between the full ban and the prompt route; exact McNemar p below 1e-6). On merge intervals neither route suffices alone. Pooled over the five tasks, the prompt route carries 0.61 and the direct route 0.74 (Table 108). The preregistered route claim is therefore disconfirmed under all three intervals: the direct route exceeds one third, and the prompt route falls below two thirds with a lower bound far under one half. The bootstrap probability that F_prompt is at most one half is 0.28, and that F_direct is at least one third, 0.99. Leaving one template out kept F_prompt between 0.55 and 0.66 and F_direct between 0.71 and 0.84; a Bayesian mixed logistic sensitivity gave 0.74 and 0.89.

**Table 107.** Route fractions on the gated bench tasks (template then slot bootstrap 95% intervals).

Source: bench route experiment `results/route.json`.

| Task | F_prompt | F_direct | Interaction | F_knockout | Reading |
|---|---|---|---|---|---|
| kth smallest | 1.00 [1.00, 1.00] | 0.74 [0.36, 0.99] | -0.74 | 1.00 | redundant: either route suffices |
| connected components | 1.00 [1.00, 1.00] | 0.80 [0.40, 1.00] | -0.80 | 1.00 | redundant |
| two sum | 0.41 [0.03, 0.82] | 0.96 [0.80, 1.00] | -0.37 | 0.40 | mostly direct |
| find duplicate | 0.04 [0.00, 0.13] | 0.99 [0.90, 1.05] | -0.03 | 0.06 | direct only |
| merge intervals | 0.39 [0.00, 0.73] | 0.02 [0.00, 0.08] | 0.59 | 0.44 | superadditive: both needed |

**Table 108.** Pooled route fractions over the five gated tasks (ratio of sums; hierarchical task, template and slot bootstrap with 10,000 replicates; the leave-one-task-out jackknife is primary because there are only five clusters). Source: bench route experiment `results/route.json`.

| Quantity | Point | Jackknife 95% (primary) | Percentile 95% | BCa 95% [68] |
|---|---|---|---|---|
| F_prompt | 0.611 | [0.06, 1.17] | [0.23, 0.93] | [0.22, 0.93] |
| F_direct | 0.744 | [0.38, 1.11] | [0.42, 0.97] | [0.27, 0.94] |
| Interaction | -0.355 | [-0.98, 0.27] | [-0.73, 0.14] | [-0.71, 0.17] |
| F_knockout | 0.621 | [0.08, 1.16] | [0.24, 0.93] | [0.24, 0.93] |

*(Correction, iteration 4: the superadditive class of Table 107 rests entirely on merge intervals, which is gated only through residue and has no route class under catalogue-only avoidance. Without it the pooled fractions are F_prompt 0.648 [-0.08, 1.38] and F_direct 0.865 [0.68, 1.05] (jackknife). The two pooled point estimates of Table 108 have overlapping jackknife intervals and a negative interaction, so the direct route carries more than the prompt route only in point estimate. The landing composition of every route arm is in Section 4.2.2: on connected components either route alone lands mostly on DFS, and BFS is chosen only when both routes act.)*

*[Figure fig_bench_route: generated in the report phase, not in this draft]*

An independent second draw with fresh uniforms (slots 20 to 39, arms a to d on the five gated tasks, N = 100 each) reproduced the picture. Pooled F_prompt was 0.64 [0.27, 0.94] and F_direct 0.74 [0.43, 0.94], per-task fractions moved by about 0.1 at most (find duplicate F_direct 0.99 then 0.93; two sum 0.96 then 0.93; merge intervals F_prompt 0.39 then 0.49), and the Spearman rank agreement between the draws was 0.89 and 0.90. The task differences are not sampling noise. A continuous companion readout (the teacher forced log probability of the default's opening up to its distinctive n-gram, under each arm) split the effect across both routes as behaviour does (pooled 0.40 prompt, 0.45 direct).

**Veto sufficiency does not predict the route.** The update predicted that the direct share would be small where the one-shot veto suffices and large where it does not. The frozen n-gram rule picked identifier or structural tokens with zero collateral (two sum `␣if␣complement`, find duplicate `␣set`, Fibonacci `()`), so the one-shot veto rarely changed the mechanism: veto avoidance V_t was 0.41 on kth smallest, 0.39 on first and last position, 0.30 on connected components and at most 0.07 elsewhere. The veto fired in 61% to 100% of rows, and the model renamed or retokenised (two sum: `if compliment in seen`) or broke the code (connected components: 80 of 100 invalid). Every task was classed in the second commit as needing a large direct route (V_t below 0.5, predicting F_direct of at least 0.25). Four of the five gated tasks hit; merge intervals missed with F_direct 0.02. The Spearman correlation between V_t and F_direct was -0.34, with a one-sided

permutation p of 0.30 over five points, which fails the rule. The test is inconclusive, and the comparison of the one-shot veto's landing with the ban's had no eligible task.

**The ban is decided at the opening of the body.** Forcing the default's own opening, up to its distinctive n-gram, undid the ban completely on every gated task (the share of the ban effect undone, P_open, 0.98 to 1.00), while forcing only a neutral `def solve(...):` line undid none of it (-0.12 to 0.01). The depth sweep (Table 109) localises this: the ban holds through the whole `def` line and gives way at the first content token of the default's body, which we call the first body token. On kth smallest, forcing the `def` line and its indentation (10 tokens) leaves the ban intact, and forcing one more token, `␣nums`, undoes it. Under common random numbers, arms a, c and d leave the twin's tokens at the first body token (median token 8 to 10 on kth smallest, two sum, find duplicate and connected components; 19 to 21 on merge intervals, whose shared `if not intervals` guard comes first). Once the first tokens of the default's body are on the page, the model continues the default; the ban is applied at the opening and not re-read in the body, as Qi et al. found for safety refusals [63].

**Table 109.** Share of the ban effect undone by forcing a prefix (P_open), by prefix: a neutral `def` line, or the default's modal program forced to 25%, 50% or 75% of its tokens or up to its distinctive n-gram (forced tokens in parentheses; N = 50 per cell, under the ban and the twin). Source: bench route experiment `README.md`, `results/replication.json`.

| Task | Neutral `def` line | 25% of the program | Up to the distinctive n-gram | 50% | 75% |
|---|---|---|---|---|---|
| kth smallest | -0.01 (9 tokens) | -0.01 (5: `def solve`) | 0.98 (11: `def` line, indent, `␣nums`) | -0.01 (10: `def` line and indent) | 1.00 (16) |
| find duplicate | -0.01 (7) | 1.00 (10: `seen =`) | 1.00 (11: `seen = set`) | 1.00 (20) | 1.00 (29) |
| two sum | 0.01 (9) | 1.00 (13: `seen = {}`) | 1.00 (32) | 1.00 (26) | 1.00 (40) |
| connected components | 0.00 (9) | 1.00 (31, inside `find`) | 1.00 (20: `def find`) | 1.00 (62) | 1.00 (92) |
| merge intervals | -0.12 (8) | 1.00 (30: `intervals.sort(key)`) | 1.00 (58) | 1.00 (59) | 1.00 (88) |

*[Figure fig_prefill_depth: generated in the report phase, not in this draft]*

**The path task screen.** Under the iteration-2 twin prompts, every veto variant failed to move either path default: one-shot, persistent, iterated rounds with the re-emerged programs' new n-grams, and a veto of the import line itself. The share off the default was 0.00 on the grid task and at most 0.01 on the shortest path task, and the default re-emerged in 99% to 100% of attempts after each round. The committed prediction that the grid one-shot veto would not suffice hit; the prediction that the shortest path one-shot veto would suffice and land on Bellman-Ford/SPFA missed (V = 0). The iteration-2 ban landings recomputed exactly from raw rows (grid DFS 257 of 257; shortest path Bellman-Ford/SPFA 255 of 260). A ban moves these tasks; a surface veto does not.

**Controls and checks.** A placebo splice (one twin's condition segment with another twin's later states) moved the default's share by at most 0.04 on all six tasks where it ran, so splicing itself is inert; the knockout applied to the twin left the default at 1.00 on six tasks. GPU unit tests confirmed common random number determinism, splice identities (splicing a prompt with itself reproduces it in 100% of rows) and zero attention mass on the condition span under the knockout in all 36 layers. A reproduction of the iteration-2 shortest path arms gave banned shares of 0 of 20, 20 of 20, 3 of 20 and 19 of 20 for arms a to d, inside their pass bands. Two audit scripts rederived

every headline from raw rows with no mismatch, and shuffling arm labels within slots collapsed the mean gap between twin and ban from 0.83 to -0.001. A worker restart briefly ran three labelling workers on the same rows; duplicates were compacted to the last row per sample, which every reader used. GPU time was about 1.8 GPU hours, judge spend USD 0.105.

## 3.7 Locus experiment: a shared prohibition direction, a locus at generated tokens 1 to 8, the ban-attending heads, and SAE transfer (Qwen3-8B)

**Why.** Four questions about where a prohibition lives inside the model were open after iteration 2. Is there a task-general direction that carries the instruction to abandon the default, and does writing it into a neutral prompt work like a named ban (the causal test of the read position correlation, moved to the positions where the route test located the effect)? Is the within-ban choice of algorithm decodable at generated tokens 1 to 8 beyond the written text? Do the three heads that attend to the banned name matter? Do the Qwen-Scope base model SAEs [50] transfer to the instruct model under a criterion that does not penalise its lower clean cross-entropy? This experiment ran all four beside their controls under common random numbers, on the two path tasks and four bench tasks (kth smallest, substring search, find duplicate, two sum).

**Design.** The machine had a 16 GB RTX 2000 Ada, too small for bf16 weights, so every decoder linear layer was quantised at load time to 8-bit floating point with per-channel weight and per-token activation scales (embeddings, norms and the output head stayed bf16). A validation reproduced the iteration-2 shortest path arms (ban 2 of 20 Dijkstra, twin 20 of 20). The sampler coupled every arm exactly (identical uniforms in different rows give identical tokens at 20 rows per batch), and teacher forcing at batch 1 was bit exact. The shared prohibition direction v is the leave-one-task-out mean, over the 13 other tasks and all five templates, of the ban minus twin displacement at the task specification positions in blocks 12 to 23. The ADD arm wrote v into the neutral twin's prefill with the displacement targeted to alpha times the ban's own per-layer norm (alpha in {1, 2}, plus 0.5 on some tasks); the SUBTRACT arm projected v out of the ban prompt's prefill. Controls were three random directions with the same blocks, positions and norms, the twin, and the named ban ($N = 100$). In all, 5,500 samples followed the frozen design and 1,800 were exploratory. Decision rules were hash committed before sampling.

**The direction exists, but adding it is no better than a random push.** The leave-one-task-out v was reliable (split half cosine 0.95 to 0.99 across blocks 12 to 23). It was aligned with every held-out task's mean ban displacement (cosine 0.29 to 0.86 at block 18, against a 95th percentile of 0.03 for random directions), yet it carried only 2% to 6% of each position's own ban displacement. The preregistered test failed on every gated task (Table 110): ADD beat every random control on 0 of 5 gated tasks, and its avoidance was not above the maximum of the three random directions on 5 of 6 tasks. Where ADD moved behaviour it did not reproduce the ban. On kth smallest it landed on a size-k heap while the ban landed on quickselect (Jensen-Shannon divergence 0.52 against a permutation 95th percentile of 0.03; the committed forecast missed); on the shortest path task ADD and the ban both landed on Bellman-Ford/SPFA, but so did the random directions that moved it. The dose response was not monotone (shortest path avoidance 0.02, 0.35 and 0.12 at alpha 0.5, 1 and 2 over all slots), and the random edits grew more than v downstream (block 35: about 283 against 219 on the shortest path task), so a weak control does not explain the result. SUBTRACT never restored the default: pooled over the gated tasks the banned share changed by -0.026 [-0.07, 0.03] against a required $+0.3$, and on substring search, find duplicate and two sum removing v made the ban slightly stronger. On the path tasks neither plain nor affine subtraction exceeded a random edit of the same norm by the required 0.15, and that random edit was destructive (pass

rate 0.00).

**Table 110.** The shared prohibition direction against matched random directions (avoidance; slots 0 to 9 for ADD and random arms). Source: locus experiment `results/c2d.json`.

| Task (default) | Named ban avoidance and landing | ADD, alpha 1 / 2 | Random directions, alpha 1 (three seeds) | SUBTRACT: change in banned share [95% CI] |
|---|---|---|---|---|
| Shortest path (Dijkstra) | 0.88, to Bellman-Ford/ SPFA | 0.44 / 0.12 | 0.06, 0.50, 0.34 | +0.07 [-0.02, 0.17] |
| Grid (BFS) | 0.87, to DFS (all 13 BFS under the fourth template) | 0.06 / 0.04 | 0.04, 0.00, 0.02 | -0.05 [-0.11, 0.00] |
| kth smallest (full sort) | 1.00, to quickselect 79, size-k heap 21 | 0.80 / 0.70 | 0.24, 0.70, 0.36 | 0.00 (no default to restore) |
| substring search (naive) | 0.49, to KMP (fails the gate) | 0.00 / 0.00 | 0.00, 0.00, 0.00 | -0.12 [-0.22, -0.02] |
| find duplicate (hash set) | 0.74, to index marking | 0.00 / 0.00 | 0.00, 0.00, 0.00 | -0.06 [-0.11, -0.02] |
| two sum (hash map) | 0.80, to brute force | 0.00 / 0.00 | 0.00, 0.10, 0.00 | -0.09 [-0.15, -0.04] |

*(Added in iteration 4: Table 110 reports avoidance only and omits the three random directions at alpha 2. At alpha 2 they moved kth smallest (avoidance 0.96 at pass 0.84, correct quickselect 40, full sort 2; 0.52 at pass 0.98, correct quickselect 24, full sort 24, size-k heap 1; 0.30 at pass 0.98), find duplicate (0.86 at pass 0.82, correct sort and adjacent check 30, hash set 6, other 5) and two sum (0.47 and 0.36 at pass 1.00; correct hash map 26, brute force 23 and hash map 32, brute force 18), where ADD at alpha 2 moved nothing (0.00; on two sum at pass 0.48). On kth smallest and two sum these random directions are the cheapest diversity knob in the record of the run. Every alpha-2 row with its pass rate and composition is in Section 4.2.1.)*

An exploratory run added 12 random directions (15 in all, alpha 1). On kth smallest ADD exceeded all 15 (0.80 against a maximum of 0.70 and a mean of 0.24) but landed on the wrong algorithm. On the shortest path task it sat at the 80th percentile (3 of 15 random directions moved it more; random mean 0.20), and on the grid task it was indistinguishable from random (0.06 against a mean of 0.07). Writing the shared component of the ban displacement into a neutral prompt is a direction-specific push on one task and a generic one elsewhere, and removing it is not necessary for the ban.

**No within-ban locus at generated tokens 1 to 16 (Table 111).** Decoders of default against other were fitted on batch-1 teacher forced states at blocks 8 to 28: the task specification mean, the read position, and generated tokens 1 to 8 and 16. Each was compared with 500 within-template permutations for the path tasks (200 for the bench tasks), taking the maximum over layers, and with three written text controls of matched or greater capacity (token one-hots, layer-0 embeddings reduced to 64 principal components, exact prefix match). No decoder passed the rule on any testable task. On the grid task and three bench tasks the class is set by the template: under the fourth template the model keeps the default in 20 of 20 bench samples and writes all 61 grid BFS programs, so every decoder (state, read position, text) reaches the template-only accuracy and none exceeds it. On the shortest path task the default is spread over templates (2, 0, 4, 3 and 3 of 20), and the choice is made inside the program. The first token at which the two classes diverge comes at a median of about 29 generated tokens, after the frozen window, and neither state nor text decodes the choice within tokens 1 to 16. The committed prediction of a fork at the import line

(tokens 3 to 5) was wrong for this sample. At batch 1, identical prefixes gave bit-identical states (maximum difference 0 over 89 to 95 pairs per position), so a state at position k is a function of the written prefix.

**Table 111.** Within-ban decoders of the algorithm written (default against other) on batch-1 teacher forced states. BA is balanced accuracy. Source: locus experiment `results/c2e.json`.

| Task | Programs (default / other) | Largest state BA, tokens 1 to 8 | Best of read position and text controls at token 8 | Locus |
|---|---|---|---|---|
| Shortest path | 100 (12 / 88) | 0.53 rising to 0.63, never above the null 95th percentile (about 0.60 to 0.70) | 0.55 | none |
| Grid (with a fourth-template top-up) | 160 (61 / 99) | 0.904 at every k | 0.904 (template only) | none |
| substring search | 100 (51 / 49) | 0.832 at every k | 0.832 (template only) | none |
| find duplicate | 100 (26 / 74) | 0.88 at every k | 0.88 (template only) | none |
| two sum | 100 (20 / 80) | 1.00 at every k | 1.00 (template only) | none |
| kth smallest | 100 (0 / 100) | not testable (no default under the ban) | | |

A secondary test pooled the archived ban pools of the route and readout experiments (200 rows per path task): the shortest path state reached balanced accuracy 0.78 to 0.79 at tokens 4 to 8 against 0.75 for the template-only decoder, a margin of 0.03 to 0.05, below the 0.10 rule. The first run on states extracted in 20-row batches, where 8-bit matrix tiling made identical prefixes differ by 4% to 8%, gave the same verdict on every task. An exploratory extension to tokens 24 to 96 found text matching or beating the state on the grid and substring search tasks. On the shortest path task the state curve was noisy (0.57, 0.66, 0.45, 0.50 and 0.75 at tokens 24, 32, 48, 64 and 96). Only token 96 cleared both the null (0.66) and the text controls (at most 0.60), a single unreplicated point with 12 minority samples.

**The ban-attending heads are not special.** Mean ablation of layer 21 head 17 and layer 26 heads 21 and 25 at generated positions under the full ban raised the grid BFS share from 0.12 (the ban, same slots) to 0.24. Every set of three random heads in the same layers raised it to 0.18, so the difference is $+0.06$ (90% interval [0.02, 0.12]; exact paired $p = 0.125$ against each random set), short of the required 0.15. The shortest path task was unchanged (-0.03 [-0.07, 0.03]). Almost all the grid movement sat in the fourth template, which carries every BFS under the ban: BFS went from 6 of 10 to 10 of 10 under the target heads and 9 of 10 under each random set. The fourth-template ban is fragile to any small perturbation of layers 21 and 26.

**Under a fairer criterion the base SAEs pass, but they lose most of the ban displacement.** The iteration-2 transfer test compared increases in cross-entropy, which penalises the instruct model for its lower clean cross-entropy (0.030 against 0.260 nats). This experiment scored each model on its own text in a two by two design (model by text source, 100 sequences each) by the fraction of cross-entropy recovered relative to zero ablation, FR_zero (Table 112). The frozen rule, FR_zero of the instruct model on its own text at least 0.9 times the base model's, passed at layers 15, 20 and 27. As the plan had computed, the zero-ablation denominator of 12 to 15 nats makes this rule lenient. A placebo that swaps the SAE and zero-ablation cross-entropy per sequence drives FR_zero to about 0, yet still passes the 0.9 rule at layer 20; the values themselves (about 0.99, against about 0 on the placebo) carry the information. Stricter companions: the fraction recovered relative to mean ablation (FR_mean) passed the same rule at all three layers; the ratio of cross-entropy

increases on own text (instruct over base) was 1.68, 1.30 and 0.76 at layers 15, 20 and 27 (iteration 2, one shared corpus in bf16: 3.29, 1.95, 1.59). The task-relevant test fails: the SAE reconstructions keep only about a third of the direction of the ban minus twin displacement at the task specification positions (cosine 0.27, 0.30 and 0.32 at layers 15, 20 and 27; norm ratio 0.65 to 0.79), so what the ban changes in the residual stream is largely lost in reconstruction error. The passing fraction recovered agrees with the finding of Kissane and colleagues that base model SAEs usually transfer to chat models; the displacement test, which measures what matters for a ban, is what fails, and latent-level claims about the ban should therefore not rest on these SAEs.

**Table 112.** Qwen-Scope SAEs spliced into each model (8-bit linears) on each text source: FR_zero / FR_mean / increase in cross-entropy (nats) / KL(clean, SAE) / fraction of variance unexplained. Source: locus experiment `results/sae_transfer.json`.

| Model / text | Clean cross-entropy | Layer 15 | Layer 20 | Layer 27 |
|---|---|---|---|---|
| instruct / own text | 0.032 | 0.9890 / 0.984 / 0.168 / 0.171 / 0.365 | 0.9903 / 0.985 / 0.169 / 0.175 / 0.096 | 0.9939 / 0.991 / 0.095 / 0.097 / 0.156 |
| base / own text | 0.926 | 0.9928 / 0.988 / 0.100 / 0.102 / 0.294 | 0.9906 / 0.986 / 0.130 / 0.134 / 0.113 | 0.9907 / 0.985 / 0.125 / 0.127 / 0.228 |
| instruct / base text | 1.861 | 1.0072 / 1.010 / -0.099 / 0.181 / 0.358 | 1.0066 / 1.010 / -0.104 / 0.220 / 0.098 | 1.0170 / 1.026 / -0.245 / 0.169 / 0.156 |
| base / instruct text | 0.251 | 0.9966 / 0.994 / 0.051 / 0.049 / 0.287 | 0.9949 / 0.992 / 0.075 / 0.075 / 0.281 | 0.9961 / 0.994 / 0.054 / 0.057 / 0.334 |

The instruct model on base text has a negative increase (the base SAE lowers its cross-entropy, most at layer 27, from 1.86 to 1.62 nats), consistent with the SAE pulling instruct activations toward base ones. Identity splices reproduced the clean cross-entropy, a zero splice at layer 15 gave 13.1 and 12.3 nats, and the instruct fraction of variance unexplained matched iteration 2 (0.365 against 0.366 at layer 15). As a readout, the latents never beat the difference-in-means direction: over 63 cells (6 tasks, 3 layers, a prohibition-present readout at the task specification mean and a default-plan readout at tokens 1 to 3), a top-1 or top-20 latent beat difference-in-means beyond its interval in 1 cell, a trivial one where every readout is at ceiling, as AxBench found [34].

**Geometry of early generated states (descriptive).** At the mean of generated tokens 1 to 3 and at token 16 (blocks 20 and 23), the k-nearest-neighbour graphs ($k = 10$) over all arms were fragmented (100 to 440 components for 800 to 1,100 points), because early states are fixed by shared boilerplate (the code fence and the `solve` signature) and many points coincide. No cloud had a persistent loop beyond its coordinate permutation null. These arms produced no bidirectional BFS, A* or greedy programs, so the BFS and bidirectional BFS neighbourhood of the ban list experiment could not be re-examined here.

**Table 113.** Committed predictions of the locus experiment and their outcomes. Source: locus experiment `predictions/commit1.json`, `README.md`.

| Prediction | Outcome |
|---|---|
| ADD (alpha 1 or 2) beats every random control on at least two thirds of gated tasks and lands on the ban's modal survivor | no: 0 of 5 gated tasks |
| SUBTRACT raises the banned share by at least 0.3 | no: pooled -0.026 [-0.07, 0.03] |
| No locus; state and text rise together at the first divergent token; the shortest path fork is at the import line (tokens 3 to 5) | no locus: yes; fork position: no (median about 29 tokens) |
| Head ablation raises the grid banned share by at least 0.15 over random heads; shortest path unchanged | grid no (+0.06); shortest path unchanged yes (-0.03) |

| | |
|---|---|
| The SAEs pass the frozen transfer rule, FR_zero about 0.98 to 0.995 for both models | pass; FR_zero instruct 0.989 to 0.994, base 0.991 to 0.993 |

**Checks.** An independent recomputation reproduced every arm share; shuffling arm labels within slots erased the ADD and head contrasts; the hash chain had no breaks and both commits preceded their first samples. A separate script with no shared code (exact McNemar by enumeration, an sklearn decoder) matched the pooled SUBTRACT change, the ADD pass count, the head deltas, FR_zero and the validation counts; its shortest path decoder at token 8 gave 0.638 against the frozen null of 0.695. Blind judge spend was USD 0.22.

## 3.8 Reach experiment: how far bans and token vetoes push, and what the long-form novelty instruction produces (Qwen3-8B)

**Why.** The update's third claim was that prompts move the model exactly one rung. This experiment asked three things. First, do cumulative token vetoes, an operation the model cannot ignore, reach algorithms that prompt bans cannot? We call this the *token walk*: veto the default's distinctive n-gram, then the next landing's, and so on, up to five algorithms. Second, does a ban that leaves exactly one survivor get that survivor written whenever the model can write it on request? We call the share of programs implementing the survivor *will*, and the share of forced programs labelled as the requested algorithm and correct *can*. Third, does the request's own long-form novelty instruction produce a correct novel algorithm? Rungs are numbered by the walk: rung 1 is the default, rung 2 is the walk's first landing (or, where the walk stalled, a flagged fallback committed in advance), and every other catalogue algorithm is at rung 3 or more. Six tasks ran: the two path tasks in bf16, and kth smallest, substring search, find duplicate and connected components on the official 8-bit checkpoint at which their iteration-2 references were made. Every mask, forecast and protocol was hash committed before the samples it governs.

**The token walk stalls at the default (Table 114).** The persistent veto, implemented as a vLLM logits processor, was exact: no banned token sequence occurred in any of 1,450 vetoed generations. The algorithm came back anyway. On the grid task `current_row` and `current_col` replaced `r, c` in 63% of re-emerged BFS programs (7% in references), and on the shortest path task `current_distinct` and `d` replaced `current_dist`. Vetoed strings also leaked through non-canonical spellings, in up to 0.86 of samples on connected components and 0.80 on find duplicate at step 2. The grid code degenerated into near-miss syntax (`queue popleft()`, 66% to 86% invalid). Imports and calls barely changed: the rerouting is lexical, not algorithmic. Only find duplicate moved one rung, to sort and adjacent check, the same single-rung move that prompt bans produce. A sham veto of an algorithm-generic n-gram of matched length and presence kept the default: grid 50 of 50 (`␣start␣==␣goal:\n`), shortest path 50 of 50 (`␣u,␣v,␣w`), substring search 49 of 50 (`␣len`) and connected components 44 of 44 valid (`␣range`). The distinctive veto cut correctness where the sham did not (shortest path pass rate 0.41 against 1.00, Newcombe difference 0.59 [0.43, 0.72]; grid 0.76 against 0.96). A fixed veto of the distinctive n-grams of BFS, DFS, IDDFS and A* at once, the request's top-five prohibition asked as an operation, left BFS 13, DFS 2 and 35 invalid programs on the grid task after two extra rounds; the prompt version gave avoidance 0.00 (Section 2.6). Neither knob reaches the rest of the catalogue.

**Table 114.** The token walk under the neutral twins (no ban text; N = 50 per round, 10 per template; ␣ marks a space in the n-grams). Source: reach experiment `results/walk.json`, `README.md`.

| Task | Step 0 (no veto) | Vetoed n-grams, by round | Final round of the last step | Status |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Grid (bf16) | BFS 50 | `␣c,␣path␣=␣queue.p`, then `␣queue.append(start`, then `None␣for` | BFS 6, DFS 1, invalid 43 | stalled at step 1 |
| Shortest path (bf16) | Dijkstra 50 | `␣heap␣=␣[(0,␣s`, then `␣current_dist,`, then `␣dist[u]:\n` | Dijkstra 27, invalid 23 (pass 0.30) | stalled at step 1 |
| kth smallest (8-bit) | full sort 45, quickselect 5 | `␣nums.sort`, then `_sorted`, then `␣sorted(nums` | full sort 23, counting sort 10, quickselect 6, invalid 10 | stalled at step 1 |
| substring search (8-bit) | naive 50 | `+len`, then `[i:i␣+`, then a leak span | naive 50 | stalled at step 1 |
| find duplicate (8-bit) | hash set 50 | `␣set`, `␣seen` and leak spans; then, at rung 2 (sort and adjacent check, 37 of 50), `.sort` and 5 leak spans | sort and adjacent check 20, other 13, hash set 10 | stalled at step 2 |
| connected components (8-bit) | union find 47, DFS 3 | `␣parent` and 8 leak spans (`(parent`, `parents`, `.parent`, tab-indented `parent`, and others) | union find 44, DFS 6 | stalled at step 1 |

**Prompts reach further than one rung on two tasks (Tables 115 and 116).** The preregistered reach claim was that will is at most 0.10 at every rung of 3 or more where can is at least 0.7, and it is disconfirmed by a will of 0.30 or more. It was disconfirmed on connected components, where the list that leaves only BFS got BFS 50 of 50, and on kth smallest, where the list that leaves only quickselect got quickselect 50 of 50. Both cells were placed at rung 3 or more through the flagged fallback, because the walk stalled at step 1 and the fallback set the second rung (counting sort on kth smallest, DFS on connected components). A post hoc sensitivity that sets the second rung to the association answer keeps connected components disconfirming (association says DFS, so BFS stays at rung 3 or more) and moves quickselect to rung 2. On connected components the two traversals are interchangeable alternatives (BFS 50 of 50 when only BFS is allowed, DFS 49 of 50 when only DFS is), so a single chain of rungs does not describe the repertoire. The claim was confirmed on the shortest path task (Floyd-Warshall, from Section 2.6), on two kth smallest cells and on find duplicate's binary search on values; find duplicate's Floyd cycle detection reached 0.14; substring search had no rung of 3 or more with can of at least 0.7.

**Table 115.** Reach cells: rungs of 3 or more with can of at least 0.7. Source: reach experiment `results/c3.json`.

| Cell | Rung | Can | Will (N = 50) [95% CI] | Verdict |
|---|---|---|---|---|
| Shortest path, Floyd-Warshall (Section 2.6, not rerun) | 3 or more | 1.00 | 0.00 [0.00, 0.04] | confirm |
| kth smallest, quickselect | 3 or more (fallback) | 0.95 | 1.00 [0.93, 1.00] | disconfirm |
| kth smallest, size-k heap | 3 or more | 0.75 | 0.00 [0.00, 0.07] | confirm |
| kth smallest, heapify and pop | 3 or more | 0.70 | 0.00 [0.00, 0.07] | confirm |
| find duplicate, binary search on values | 3 or more | 0.78 | 0.00 [0.00, 0.07] | confirm |
| find duplicate, Floyd cycle detection | 3 or more | 0.70 | 0.14 [0.07, 0.26] | neither |
| connected components, BFS | 3 or more (fallback) | 1.00 | 1.00 [0.93, 1.00] | disconfirm |

**Table 116.** Every ban that leaves one survivor on the bench tasks (N = 50; disguise is the share of banned-mechanism programs that name none of the banned algorithms). Source: reach experiment `results/reach_events.jsonl`.

| Task | Only allowed algorithm | Labels of attempted programs | Will [95% CI] | Pass rate | Disguise rate |
|---|---|---|---|---|---|
| kth smallest | full sort | quickselect 50 | 0.00 [0.00, 0.07] | 0.94 | 0.04 |
| kth smallest | quickselect | quickselect 50 | 1.00 [0.93, 1.00] | 0.98 | n/a |
| kth smallest | size-k heap | quickselect 50 | 0.00 [0.00, 0.07] | 0.90 | 0.08 |
| kth smallest | counting sort | quickselect 44, other 6 | 0.00 [0.00, 0.07] | 1.00 | 0.09 |
| kth smallest | heapify and pop | quickselect 50 | 0.00 [0.00, 0.07] | 0.94 | 0.00 |
| substring search | naive | naive 50 | 1.00 [0.93, 1.00] | 1.00 | n/a |
| substring search | KMP | naive 37, KMP 13 | 0.26 [0.16, 0.40] | 0.98 | 1.00 |
| substring search | Rabin-Karp | naive 36, KMP 8, Rabin-Karp 5, Boyer-Moore-Horspool 1 | 0.10 [0.04, 0.21] | 0.92 | 0.76 |
| substring search | built-in find | naive 49, KMP 1 | 0.00 [0.00, 0.07] | 0.98 | 0.92 |
| substring search | Z algorithm | naive 46, KMP 3, Boyer-Moore-Horspool 1 | 0.00 [0.00, 0.07] | 0.96 | 0.92 |
| find duplicate | hash set | hash set 43, index marking 4, Floyd cycle detection 3 | 0.86 [0.74, 0.93] | 0.88 | 0.57 |
| find duplicate | binary search on values | index marking 40, hash set 10 | 0.00 [0.00, 0.07] | 0.40 | 1.00 |
| find duplicate | Floyd cycle detection | index marking 34, hash set 9, Floyd cycle detection 7 | 0.14 [0.07, 0.26] | 0.38 | 1.00 |
| find duplicate | sort and adjacent check | index marking 33, hash set 11, Floyd cycle detection 3, other 1, binary search on values 1, sort and adjacent check 1 | 0.02 [0.00, 0.10] | 0.36 | 0.96 |
| connected components | BFS | BFS 50 | 1.00 [0.93, 1.00] | 1.00 | n/a |

| | | | | | |
|---|---|---|---|---|---|
| connected components | DFS | DFS 49, union find 1 | 0.98 [0.90, 1.00] | 0.98 | 1.00 |
| connected components | union find | union find 47, DFS 3 | 0.94 [0.84, 0.98] | 1.00 | 0.67 |

Quickselect is kth smallest's universal answer to a ban list: every list that bans the full sort yields quickselect in 44 to 50 of 50 programs, including the lists that ban quickselect itself (only size-k heap allowed, only heapify and pop allowed: 50 of 50 quickselect, disguised in 0% to 8%). The sanity check of the list that leaves only the default gave will 1.00 on substring search, 0.86 on find duplicate and 0.94 on connected components, but 0.00 on kth smallest. There the prompt banned the size-k heap, heapify and pop, quickselect, median of medians and counting sort, and all 50 programs implemented quickselect, 48 of them in a function named `quickselect`, while full sort was 45 of 50 under the twin. This is the starkest priming by a negative constraint in the run [7, 53]. Forced can underestimates some algorithms: find duplicate's index marking has a forced adherence of 0.03, yet the model wrote it in 34 to 40 of 50 programs under lists that forbid it, always disguised. The preregistered list independence test asks whether the default's share among default and second-rung programs is the same across lists that ban both. It failed on substring search: 0.82 under the Rabin-Karp list against 0.98 under the built-in list (permutation chi-squared $p = 0.033$; largest difference 0.16 [0.04, 0.30]; placebos with shuffled membership gave $p = 0.61$, 0.56 and 0.06). It was descriptive on find duplicate and not testable on kth smallest and connected components.

**Two-step bans of the default and its second rung (N = 100) split the tasks (Table 117).** On the path tasks and substring search they failed (compliance 0.02, 0.01 and 0.14, with snap-back to the default on substring search), replicating iteration 1; on kth smallest, find duplicate and connected components they were obeyed (1.00, 0.96 and 1.00). The committed forecasts hit find duplicate (the second walk landing, index marking), kth smallest and connected components (association: quickselect and BFS) and substring search (snap-back).

**Table 117.** Two-step bans of the default and its second rung (N = 100; labels of attempted programs not banned). Source: reach experiment `results/reach_events.jsonl`, `README.md`.

| Task | Banned | Compliance | Attempted programs (banned mechanisms included) |
|---|---|---|---|
| Grid | BFS, DFS | 0.02 [0.01, 0.07] | DFS 82, BFS 16, unit weight Dijkstra 1, other 1 |
| Shortest path | Dijkstra, Bellman-Ford/SPFA | 0.01 | Bellman-Ford/SPFA 67, Dijkstra 32, other 1 |
| kth smallest | full sort, counting sort | 1.00 | quickselect 63, size-k heap 37 |
| substring search | naive, KMP | 0.14 | naive 74, Rabin-Karp 9, built-in find 2, Boyer-Moore-Horspool 2, other 1 (the rest KMP) |
| find duplicate | hash set, sort and adjacent check | 0.96 | index marking 52, Floyd cycle detection 42, other 2 (the rest banned) |
| connected components | union find, DFS | 1.00 | BFS 100 |

**The long-form novelty instruction produces claims of novelty, not new algorithms (Table 118).** The request contains two long-form novelty instructions, which we call the first long-form instruction (asking the model to make a new algorithm) and the second long-form instruction

(asking for a fully novel algorithm never seen before); each asks for a design rationale of about 2,000 words before the code. They were run with thinking off (N = 25 to 50 per cell) and on (N = 10 per prompt and task, 16k thinking cap). All 100 thinking traces completed, at 4.1k to 12.4k thinking tokens. Of 574 outputs with a claim record, 484 (84%) claimed a new algorithm. Of those, 189 (39%) contained no code at all, only a rationale of 1,300 to 1,700 words, and 220 of the 295 with code (75%) were catalogue algorithms by the mechanism labeller, under names that modify a catalogue algorithm, such as a modified Bellman-Ford, a BFS with dynamic programming, or an optimised Dijkstra with early termination. The instructions do diversify the output (correct-only Hill numbers of 1.8 to 4.5 in the thinking-off cells, against a collapsed baseline), at a cost in correctness (pass rates of 0.33 to 0.44 on the grid task and 0.32 to 0.37 on find duplicate). With thinking on, the code stayed in the catalogue (grid mostly DFS, 13 of 20; shortest path Bellman-Ford/SPFA, Dijkstra and Floyd-Warshall only; kth smallest counting sort 13 of 20; substring search naive 14 of 20; find duplicate off-catalogue 11 of 20, all failing arithmetic tricks that assume one duplicate). A bf16 against 8-bit cross-check on the shortest path task under the first instruction gave a label composition Jensen-Shannon divergence of 0.033 bits against a null 95th percentile of 0.068 ($p = 0.35$).

**Table 118.** Long-form novelty instructions: composition of outputs per cell (first and second long-form instructions as defined above). Source: reach experiment `results/longform.json`, `README.md`.

| Cell | N | Default | Other catalogue | Off catalogue | Invalid or no code | Correct |
|---|---|---|---|---|---|---|
| kth smallest, first, thinking off (8-bit) | 50 | 3 | 17 | 12 | 18 | 25 |
| kth smallest, first, thinking on (8-bit) | 10 | 0 | 7 | 1 | 2 | 8 |
| kth smallest, second, thinking off (8-bit) | 50 | 0 | 12 | 8 | 30 | 15 |
| kth smallest, second, thinking on (8-bit) | 10 | 0 | 9 | 0 | 1 | 8 |
| substring search, first, thinking off (8-bit) | 50 | 27 | 7 | 1 | 15 | 32 |
| substring search, first, thinking on (8-bit) | 10 | 7 | 3 | 0 | 0 | 10 |
| substring search, second, thinking off (8-bit) | 50 | 10 | 8 | 4 | 28 | 13 |
| substring search, second, thinking on (8-bit) | 10 | 7 | 1 | 0 | 2 | 8 |
| find duplicate, first, thinking off (8-bit) | 50 | 5 | 13 | 17 | 15 | 13 |
| find duplicate, first, thinking on (8-bit) | 10 | 0 | 4 | 5 | 1 | 4 |
| find duplicate, second, thinking off (8-bit) | 50 | 4 | 14 | 7 | 25 | 8 |
| find duplicate, second, thinking on (8-bit) | 10 | 0 | 3 | 6 | 1 | 3 |
| Grid, first, thinking off (bf16) | 25 | 3 | 12 | 1 | 9 | 7 |
| Grid, first, thinking on (8-bit) | 10 | 0 | 8 | 0 | 2 | 3 |
| Grid, second, thinking off (bf16) | 25 | 3 | 3 | 2 | 17 | 3 |
| Grid, second, thinking on (8-bit) | 10 | 0 | 7 | 2 | 1 | 3 |
| Shortest path, first, thinking off (bf16) | 50 | 17 | 20 | 3 | 10 | 35 |
| Shortest path, first, thinking off (8-bit) | 50 | 18 | 12 | 4 | 16 | 29 |

| | | | | | |
|---|---|---|---|---|---|
| Shortest path, first, thinking on (8-bit) | 10 2 | 6 | 0 | 2 | 8 |
| Shortest path, second, thinking off (bf16) | 25 6 | 6 | 5 | 8 | 9 |
| Shortest path, second, thinking on (8-bit) | 10 6 | 4 | 0 | 0 | 9 |

**No novel algorithm in any cell.** Every correct program outside the catalogue from every arm of this experiment went through an audit: 38 correct programs (23 from the long-form cells, 10 from the walk, 5 from the lists) plus 15 seeded failing ones. Two blind judges (gemini-2.5-flash-lite and gpt-4.1-mini; each passed a 6-program control gate at 5 of 6 after a prompt fix) compared each program with its three nearest known solutions from the declared dataset's 472 human and textbook programs and its three nearest catalogue references, and the executor read each one and ran it on three extra random inputs. The 38 correct programs were 30 labeller misses (mostly counting sort on kth smallest written with a dictionary or Counter), 7 catalogue variants (counting plus binary search on cumulative counts; a last-in-first-out label correcting search on the shortest path task) and 1 degenerate quadratic brute force. Judge A called 10 programs novel and judge B none (kappa 0.27); the manual pass found 5 of those 10 degenerate (XOR or sum tricks, broken loops), 3 labeller misses and 2 catalogue variants. The exact one-sided 95% upper bound on the novel rate is 0.058 per cell of 50, 0.113 per cell of 25 and 0.26 per thinking-on cell of 10. Together with the record audit (0 of 78), the run's direct answer to the novelty question is that no instruction or knob produced a correct algorithm outside the catalogue and its known variants.

**Deviations and checks.** Distinctive n-grams had to be lexical (a veto of an indentation-tailed n-gram was bypassed by re-indenting in 20 of 20 unit tests), and shams could not be constructed for kth smallest and find duplicate. The analysis code was edited after its hash was committed (corrected compositions, a post hoc sensitivity, a claim mechanism check); an unchanged rederivation script reproduced every preregistered number from raw rows with no mismatch. Dropped: placement of the walk landings on the manifold, the thinking-on temperature sweep, and extending the path task thinking cells to $N = 25$. Judge spend was USD 0.49.

## 3.9 Comparisons across this iteration's experiments

**The same modal bench ban, drawn three times in Qwen3-8B and once in Llama.** Three Qwen3-8B artifacts sampled the modal bench bans with the five frozen templates and 20 samples each, on different engines: vLLM in bf16 (bench forecast experiment), HF in bf16 with common random numbers (bench route experiment, full ban arm) and HF with 8-bit linears (locus experiment, four bench tasks). Table 119 sets them side by side with Llama. With the wording fixed, avoidance replicates within a few points and the landing replicates in kind. The same five tasks pass the gate in both artifacts that ran all twelve. Every Qwen3-8B draw sends kth smallest to quickselect and size-k heap, two sum to brute force, find duplicate to index marking, connected components to BFS, and merge intervals to a failing merge without sorting. Llama lands elsewhere on two of the shared tasks (two sum to sort and two pointers, connected components to DFS), on both of which Llama's own association answer matches its landing.

**Table 119.** The modal bench bans across draws (avoidance; landing among attempts off the default). Sources: bench forecast experiment `results/c1_events.jsonl`; bench route experiment `results/shares.json`; locus experiment `README.md`; Llama bench experiment `README.md`.

| Task | Qwen3-8B, vLLM bf16 | Qwen3-8B, HF bf16 (full ban arm) | Qwen3-8B, HF 8-bit linears | Llama-3.1-8B-Instruct, vLLM bf16 |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| kth smallest | 0.97: quickselect 59, size-k heap 38 | 0.99: quickselect 70, size-k heap 28, heapify and pop 1 | 1.00: quickselect 79, size-k heap 21 | 0.95: quickselect (modal of 87) |
| two sum | 0.86: brute force 86 | 0.83: brute force 83 | 0.80: brute force | 0.91: sort and two pointers 42, brute force 32 |
| find duplicate | 0.79: index marking 73, Floyd 5, sort and adjacent check 1 | 0.79: index marking 77, Floyd 1, other 1 | 0.74: index marking | 0.76: other (0.53 of attempts) |
| connected components | 1.00: BFS 83, DFS 17 | 1.00: BFS 75, DFS 25 | not run | 0.82: DFS (modal of 78) |
| merge intervals | 0.58: other 44, union find 14 | 0.59: other 44, union find 15 | not run | 0.30 (fails the gate) |
| substring search | 0.40: KMP 21, built-in find 16, Rabin-Karp 2, Z algorithm 1 | 0.42: KMP 18, built-in find 17, Boyer-Moore-Horspool 5, Rabin-Karp 2 | 0.49: KMP | 0.79: KMP (modal of 64) |
| top-k frequent | 0.48 | 0.41 | not run | 0.89: sort by count |
| Fibonacci | 0.34 | 0.31 | not run | 0.79: iteration |
| topological order | 0.08 | 0.04 | not run | 0.18 |
| first and last position | 0.03 | 0.02 | not run | 0.04 |
| longest increasing subsequence | 0.02 | 0.02 | not run | 0.23 |
| nonoverlapping intervals | 0.00 | 0.00 | not run | 0.29 |

**One wording, everywhere.** The fourth template of Table 40, which states only that X is not allowed here, is the weakest wording in every artifact that measured it. In the bench forecast experiment it accounted for 197 of 240 banned programs, against 85 to 135 for the other templates. In the bench route experiment its pooled avoidance was 0.19, against 0.42 to 0.62, and it produced 29% of all ban failures. In the exposure template arm its mean avoidance was 0.81, against 0.93 to 0.96. In the locus experiment it kept the default in 20 of 20 bench samples and carried all 61 grid BFS programs under the ban, which is why every within-ban decoder there reduced to a template decoder. In Llama, dropping the fourth template alone flips the frozen branch from no family advancing to substitution. Wording is the largest measured source of variance in compliance, and any analysis that pools templates without modelling them pools different experiments.

**Signature vetoes, five times.** Every artifact that vetoed a default's distinctive n-gram found the default carried by more than its surface tokens. The default survived every round on 9 of 12 bench tasks in the bench forecast experiment, and Y re-emerged after three rounds for 25 of 33 shown algorithms in the exposure experiment. Only one Llama default (the heap of top-k frequent) was removed, the one-shot veto's avoidance was at most 0.41 in the bench route experiment, and the token walk stalled on all six tasks of the reach experiment although the veto was exact. On both path tasks no veto variant moved the default at all, in Qwen3-8B or in Llama, while a single prompt ban moves them by about 85 points. The subtraction forecast was therefore undefined on most events, and the test of whether a ban-free veto reproduces a ban's landing was undetermined everywhere it was planned.

**Three forecast contests.** Table 120 summarises the contests that decide the title claim, each under

the same frozen rule. On the Qwen3-8B modal bans no family's forecast beats uniform by a decisive margin, and in neither model does a family advance once undefined forecasts are not counted as wins. On the exposure bans, where the default is still available, the ladder (that is, the default) wins: snap-back. Association is never better than the ban-free families in Qwen3-8B, and in Llama it ties them.

**Table 120.** The three forecast contests of iteration 3 (gated events; top-1 hit rates against the realised modal landing). Sources: bench forecast, exposure and Llama bench experiments.

| Contest | Gated events (disagreeing) | Frozen branch | Wins by 0.10 nats | Top-1 hit rates | Paired tests |
|---|---|---|---|---|---|
| Qwen3-8B modal bench bans | 5 (5) | SUBTRACTION/ LADDER, ladder advancing; NO FAMILY ADVANCES if undefined forecasts do not count as wins | ladder 3 (2 as uniform), proximity 2 | ladder 1 of 5 and association 1 of 5 (both kth smallest), proximity 1 of 5 (two sum), subtraction 0 of 5 | not run (5 events); mean log score minus uniform: ladder +0.06, subtraction -0.69, proximity -1.53, association -1.78 |
| Qwen3-8B exposure bans, named | 31 (30) | SUBTRACTION/ LADDER, ladder advancing | ladder 7, subtraction 2, association 2, proximity 0 | ladder 0.77, proximity 0.61, association 0.45, subtraction 0.36 | ladder minus association +1.29 nats [0.10, 2.38], Holm 0.081; subtraction minus association +1.02 [0.35, 1.56], Holm 0.081 |
| Llama modal and present minor bench bans | 13 (11) | NO FAMILY ADVANCES, aimed but unresolved | subtraction 1 (as uniform), association 1 | ladder 7 of 13, association 7 of 13 (binomial p 0.033 each), subtraction 4 of 12, proximity 3 of 13 | both ban-free families beat proximity (Holm 0.0088 and 0.0078); neither separates from association (p 0.52 and 0.77) |

*(Correction, iteration 4: the top-1 rates of Table 120 use three different rules for undefined forecasts: a miss in the bench contest, fractional tie credit in the exposure contest and a dropped event in the Llama contest. Defined-only rates with the number of defined events are in Section 4.2.6. The exposure subtraction rate of 0.36 is tie credit; where defined (4 of 33 shown algorithms) subtraction named the landing every time. The figure below now plots the defined-only rates.)*

*[Figure fig_forecast_hits: generated in the report phase, not in this draft]*

## 3.10 What iteration 3 established, what it ruled out, and what it abandoned

Everything below is still one model per contest, at most 31 gated events per contest, and labels from labellers that passed their gates; the bench is memorised LeetCode material, and the path tasks are a screen that is never pooled.

**Established.**

1. *The likelihood substitution test of iteration 2 cannot decide the title claim.* Its gain and residual argmax hit the realised landing in 0 of 17 Qwen3-8B single bans, against 14 of 17 for the

implied mass. No row keeps three estimable survivors, the bootstrap variance of the mean gain is 2.7 times the noise term it subtracted, and a decision on mass would need about 50 programs per algorithm at the lower bound and 129 Qwen3-8B or 43 Llama events. Whether the gains are pure subtraction, whether their specific part is aimed, and whether the generic profiles form a ladder are all undetectable with that instrument.

2. *Shown a correct program of a non-default algorithm and told not to use it, Qwen3-8B snaps back to its default.* The default's share of attempts rises from 0.05 under the copy twin to 0.72 under the named ban and 0.82 under the unnamed one. The modal replacement is the default in 24 of 31 gated events, and the ladder forecast wins the frozen contest (7 wins; top-1 0.77). Naming the banned algorithm adds little beyond asking for something else (same modal replacement for 0.88 of Y). *(Correction, iteration 4: the frozen counting rule advances some family in 85% of no-information simulations and in 100% of label-permutation placebo draws (the ladder in 43%), and no paired test survives Holm on the named events, so the 7 wins of the ladder are weak evidence. Wording also changes the landing: under templates one to four the default is modal for 13 of 19 priority algorithms, against 15 of 19 under the fifth, and the replacement flips to the associate for the size-k heap and the Z algorithm. Snap-back rests on one wording. It replicates in the present minor bans of Llama, which return to the default in 0.72 to 0.93 of attempts not banned; Section 4.2.3.)*
3. *Where the default is not available, the landing is specific to the task and mostly unforeseen.* On the five gated Qwen3-8B modal bench bans, no ban-free or association forecast beats uniform on average, and three of five landings (index marking, BFS, a failing merge without sorting) were named by no family. In Llama the landing is predictable above chance (ladder and association 7 of 13 each), but the families do not separate. *(Correction, iteration 4: the merge intervals landing is residue, failing programs labelled other (29 coverage arrays, 12 unsorted sweeps, 3 others), and that event passes the gate only through them (catalogue-only avoidance 0.14). In Llama, with other excluded, association hits 7 of 13 ($p = 0.032$) and the ladder 6 of 13 ($p = 0.100$), so only association is above chance.)*
4. *A ban's route is task dependent.* On the bench, pooled over five gated tasks, generated tokens reading the ban carry 0.74 of the effect and the prompt states 0.61. The mechanism is redundant on kth smallest and connected components, direct on two sum and find duplicate, and superadditive on merge intervals, and a second draw reproduces every task. The prompt route story of the path tasks does not generalise. *(Correction, iteration 4: the claim that generated tokens carry more of the effect rested on a comparison of point estimates. With their jackknife intervals the fractions are F_direct 0.744 [0.38, 1.11] and F_prompt 0.611 [0.06, 1.17], which overlap, with interaction -0.355. The superadditive class rests on merge intervals, which is gated only through residue; without it F_prompt is 0.648 [-0.08, 1.38] and F_direct 0.865 [0.68, 1.05].)*
5. *The ban is applied at the opening of the body and not re-read.* Forcing the default's first body token undoes the ban completely on every gated task, while forcing the `def` line undoes none of it. *(Correction, iteration 4: the locus is resolved to a single token only on kth smallest, where one token after the `def` line and its indentation undoes the ban; on the other four gated tasks the shortest forced prefix in the depth sweep that undoes it is 10 to 30 tokens long. Restated: the ban gives way at the opening of the body, to a single token on kth smallest.)*
6. *Wording dominates compliance.* The fourth template is the weakest in every artifact (template SD about 2.0 logits; likelihood ratio 374), and dropping it flips the Llama branch.
7. *The default is not carried by any small set of surface tokens.* Exact signature vetoes, one-shot, persistent, iterated or cumulative, almost never remove it; the model renames, respells or

breaks the code. A single prompt ban moves the model further than any veto.

8. *Prompts reach past one rung where the alternatives are interchangeable.* Bans that leave one survivor got BFS 50 of 50 on connected components and quickselect 50 of 50 on kth smallest; on kth smallest quickselect is the answer to every ban list, including lists that ban it and the list that leaves only the default. Elsewhere (Floyd-Warshall, size-k heap, heapify and pop, binary search on values) the model will not write a survivor it can write on request. *(Correction, iteration 4: the kth smallest disconfirmation depends on the rung definition. It uses the fallback rung of the token walk; three bench draws put the landing of the ban on quickselect (59 to 79%), which makes quickselect the behavioural second rung, and under that definition the cell confirms the claim (size-k heap, will 0.00). Connected components disconfirms under either definition.)*
9. *No instruction or knob produced a correct novel algorithm.* Zero of 78 correct out-of-catalogue programs in the iteration-1 and iteration-2 data and 0 of 38 in iteration 3 are novel; the request's long-form instruction makes 84% of outputs claim novelty, 39% of those contain no code, and 75% of the rest are catalogue algorithms.
10. *Llama's correct output is as collapsed as Qwen3-8B's on the grid task* (BFS 20 of 20 correct), and more diverse on 9 of 13 bench tasks counted over correct programs.

**Ruled out, or not supported.**

- *Substitution by association in Qwen3-8B.* In none of the three contests did association win; it was worse than uniform on all five gated modal bans and lost to the ladder by about 1.3 nats on exposure bans (Holm 0.081 named, 0.0056 unnamed).
- *A shared prohibition direction as the carrier of the ban.* Adding the leave-one-task-out ban displacement to a neutral prompt passed on 0 of 5 gated tasks against matched random directions, and removing it from a ban never restored the default (pooled -0.026 [-0.07, 0.03]).
- *A ban-free signature veto as the forecast of the next rung.* It was undefined on 10 of 12 Qwen3-8B modal events and 29 of 33 exposure events, and where defined on the path tasks or the bench it landed elsewhere than the ban.
- *The prompt route as the general mechanism of a ban.* Disconfirmed on the bench (pooled direct share 0.74, jackknife [0.38, 1.11]).
- *Prompts move the model exactly one rung.* Disconfirmed on connected components (BFS will 1.00 at can 1.00) and, through the fallback rung, on kth smallest.
- *The three ban-attending heads as a causal site.* Ablating them moved the grid ban by $+0.06$ over random heads in the same layers, inside the range of any perturbation of the fragile fourth-template ban.
- *A locus at generated tokens 1 to 16.* No decoder beat the template-only or written text controls on any testable task; on the shortest path task the choice diverges at a median of about 29 tokens.

**Dead ends, with the evidence that closed them.**

- *The forced ratio latent probability as the decision instrument for where a ban sends the mass.* Closed by Table 82 (argmax 0 of 17 against the realised landing) and Table 84 (programs and events needed far beyond the budget).
- *Signature vetoes and the token walk as a way past the default.* Closed by the bench forecast,

exposure, Llama, route and reach experiments: the default survives exact vetoes by renaming and respelling, and the walk stalled at rung 1 or 2 on all six tasks.

- *The shared prohibition direction.* Closed by Table 110.
- *Head ablation as a test of the attention heads.* Closed by the random-head control.
- *Qwen-Scope base SAEs for readouts of the ban.* They now pass a fairer transfer criterion (fraction of cross-entropy recovered 0.989 to 0.994), but their reconstructions keep only a cosine of about 0.3 of the ban displacement, and their latents never beat difference-in-means.
- *Long-form novelty instructions as a source of novel algorithms.* Closed by Table 118 and the audit (0 novel of 38 correct programs outside the catalogue).

## 3.11 What the next iteration takes from this one

The title claim of the update, that a ban removes the default and the model falls to a next rung that ban-free counterfactuals can read in advance, did not survive in that form. The part that survived is snap-back: with the default available, a ban on a shown algorithm returns the model to its default, which is subtraction in its simplest form. Where the default itself is banned, the counterfactuals meant to expose the next rung either fail to move the model (signature vetoes) or move it somewhere else (random pushes, the shared prohibition direction), and the landing is specific to the task in a way no frozen family foresaw in Qwen3-8B. Substitution by association is not supported in Qwen3-8B and is tied with the ladder in Llama. What decides the replacement under a modal ban remains open, and the run now has two concrete leads for it: the opening of the body, where the prefill sweep localises the ban to the first body token, and the template, which carries more of the variance in compliance than anything else measured. *(Added in iteration 4: a third lead was already in the iteration-3 data, the landing composition per route arm. On connected components either route alone lands mostly on DFS, which is the generic landing of random pushes and the association answer of Qwen3-8B, and BFS is chosen only when both routes act; Section 4.2.2.)*

Three things follow for the next iteration. First, the contest needs events on which the default is unavailable and the families disagree, in numbers the paired tests can use: the modal bench bans give 5 gated events in Qwen3-8B, so pooling with Llama (11) and adding bench tasks where bans take is the arithmetic, with the template entered as a random effect rather than pooled over. Second, the first body token is now a measured locus in behaviour; a decoder or patch there, in states saved at that position for each task, would ask what is decided when the ban gives way. Third, the generic perturbation rival needs a behavioural test where its runner-up and the associate differ, which the Qwen3-8B bench rarely provides; Llama's present minor events are where the named ban gained the associate about 0.25 of attempts beyond the veto on two of six events, and that is the next place to look.

Request items still open: a same seed temperature arm with thinking on (dropped again), manifold placement of the walk landings (dropped; the walk stalled), and an SAE readout that keeps the ban displacement. *(Added in iteration 4: the same seed arm with thinking on ran in iteration 4, Section 4.5.)*

# Iteration 4: plan noise priced against temperature, where the noise must enter, the plan subspace, and a held-out model family

## 4.1 Why this iteration ran what it ran

**The review of the iteration-3 record.** The review was blocking again. It raised six major items, all marked must fix, and three minor ones. The first item went to the answer the record gave to the second question of the request, whether the algorithm a model writes can be influenced. The closing section and Section 3.2.5 said that only the semantic seed and the A* mention buy a second correct algorithm, and that both name or describe it. Table 91 itself held two activation rows that buy one without naming anything, and none of the iteration-3 arms had been entered into the knob table, although random directions written into the task-specification states had reached a Hill-1 near 3 on kth smallest at no pass cost. The second item was an executed result missing from the record: the landing composition of every route arm, which bears directly on what picks the replacement under a ban. The third concerned the Llama contest, which scored its present minor events against a label the record never defined and mislabelled three rows of Table 102; with other excluded, only association hit above chance. The fourth was that the event gate counted failing programs labelled other as avoidance, the defect the record had already corrected for the TOP ban list, and merge intervals passed the gate only through such residue. The fifth was coverage of the request: the activation-space plots with centroids coloured by implemented count existed but had no figure marker, the same seed temperature arm with thinking on had been dropped twice, and the run had located where a ban acts but not where the choice of algorithm itself is represented. The sixth asked for the iteration-3 claims to be set against their nearest published neighbours. The minor items asked for top-1 hit rates under one rule for undefined forecasts, Table 82 per distinct likelihood row, and four sentences of Section 3.10 that said more than their tables. Table 121 maps every item to its answer. The corrections are made in place in Sections 2.3.6 and 3.2 to 3.11, each marked as an iteration-4 correction.

**Table 121.** Review items on the iteration-3 record and where this report answers them.

| Review item | What the review found | Where it is answered |
|---|---|---|
| Answer to the question whether the algorithm can be influenced | Decommitment and random directions also buy a second algorithm; iteration-3 arms absent from the knob table; alpha-2 random rows hidden by Table 110 | Corrections in Sections 2.3.6, 3.2.5 and 3.7; Section 4.2.1; the plan noise experiment, Section 4.3 |
| Landing by route | Compositions of every route arm unreported | Correction in Section 3.6; Section 4.2.2 |
| Llama hit rule | Present minor events scored against the largest-excess survivor; three rows of Table 102 mislabelled; hits on other | Corrections in Sections 3.5 and 3.10; Section 4.2.3 |
| Residue in the event gate | Failing other programs counted as avoidance; merge intervals gated through residue; other landings never audited | Corrections in Sections 3.3, 3.6 and 3.10; Section 4.2.4 |
| Request coverage | No geometry figure marker; same seed arm with thinking on dropped twice; no internal account of where the choice lives | Section 4.2.5 (figure); Section 4.5 (thinking on); Sections 4.4 and 4.6 (site and subspace) |

| | | |
|---|---|---|
| Positioning of iteration-3 claims | No neighbour rows for snap-back, wording, body-opening enforcement, vetoes, novelty, random pushes | Table 92b and Table 92c, Section 4.8 |
| Minor: hit-rate rules | Three rules for undefined forecasts across the contests | Correction in Section 3.9; Section 4.2.6; corrected figure |
| Minor: Table 82 | Replicates treated as independent events | Correction in Section 3.2.1; Section 4.2.6 |
| Minor: Section 3.10 wording | Calibration caveat, single-token locus, rung definition, the claim that one route carries more | Corrections in Section 3.10 items 2 to 5 and 8 |

**The hypothesis update.** The update reframed the run around a result that had sat in the control arms of the iteration-3 locus experiment. Isotropic random directions added to the task-specification states at blocks 12 to 23, at the size of the displacement a ban itself causes, had made Qwen3-8B [5] write two or three different correct algorithms for one prompt at no loss of correctness. On kth smallest, 7 of 15 directions moved at least 20% of programs off the default, and five reached a correct-only Hill-1 of 2 or more. The update read this as evidence that the algorithm is chosen in those states before any code is written. We call that state the *plan* below, with one caveat stated now: the word names a site in the network, and whether a plan representation sits there is what this iteration tested. Temperature never touches the plan, which would explain why it changes wording and not ideas, while a nudge to the plan changes ideas. The question that ran through iterations 1 to 3, substitution against subtraction, becomes a special case: a ban is one large displacement of the plan, plus generated tokens reading the ban. Three claims followed, each with a rule committed before its first decisive sample.

- *The price claim.* Plan noise moves the algorithm while temperature moves only the wording, and on tasks where the plan is movable it buys correct algorithmic diversity more cheaply than every token-level and prompt-level knob tested. Two sub-claims: the effect is specific to the task-specification states at blocks 12 to 23 (the site claim), and with the token seed held fixed, temperature changes the algorithm in at most 1 of 20 seeds while a change of plan noise draw changes it often (the same seed claim, which is question 1a-c of the request).
- *The subspace claim.* The choice is represented in a low-dimensional subspace of the task-specification states, spanned by the differences between per-algorithm centroids under forced prompts, and a push changes the algorithm to the extent that it moves the state along that subspace.
- *The movability claim.* Whether a ban on the default of a task is obeyed is predicted, without any ban sample, by whether random pushes can move that task, and the ban lands where the pushes land unless generated tokens read the ban. The claim was to be confirmed in a model family never run in this study. Two further tests rode along: a norm sweep to the point of breakage with a full audit for novel algorithms, and the top-five prohibition of the request applied as a push at the plan.

We define the instruments once. A *ban unit* is the norm of the ban minus neutral twin displacement per layer and per task-specification position at blocks 12 to 23, averaged over the five modal ban templates; for a model never banned it is estimated without ban text from the residual norm times the median ratio, in Qwen3-8B, of ban displacement to residual norm at the same relative depth. *Plan noise* at dose alpha is a fresh isotropic random direction per sample, added during prefill at every task-specification position of blocks 12 to 23, with the displacement along it set to alpha ban units at each block, so the edits do not compound. A fixed direction *escapes* when, over its first 10 samples, the share of attempts off the default exceeds that of the twin by at least 0.20 at a pass rate

of at least 0.8 times that of the twin; two experiments added a confirmation at N = 30 by an exact binomial test. The *escape fraction* of a task is the share of its fixed directions that escape, and we call a task *movable* when its escape fraction is above zero at some dose of at most 2. The *plan subspace* of a task is the span of the differences between per-algorithm centroids of the task-specification states under 30 forced paraphrases per algorithm, with a name-masked twin built from descriptions. The *push-landing forecast* names the modal catalogue survivor among off-default attempts of escaping directions, and the *opening ladder* ranks survivors by the teacher-forced log probability of their modal forced-program body opening under the neutral twin. We price a knob by its *5-point Hill-1*, the highest correct-only rarefied Hill-1 (depth 25) it reaches at a cost of at most 5 pass points, and by its *price of 1.5* and *price of 2*, the smallest pass cost at which it reaches a Hill-1 of 1.5 and of 2. The plan noise and site experiments compute them on the *randomisation hull* of each knob, the set of mixtures of the reference arm with the settings of the knob, which turns the dial of a knob into a continuous price curve.

**The strategy.** Seven artifacts ran (Table 122). One evaluation repaired the record from rows on disk and froze the push screen by hash, so that every later analysis cites one screen. Four Qwen3-8B experiments tested the price, site, same seed and subspace claims, and one experiment ran the movability claim in a held-out family. A research artifact positioned the iteration-3 and iteration-4 claims against published work. Two departures from the plan shaped what could be claimed. The held-out family was planned as gemma-2-9b-it, but the only GPU available held 16 GB, so the experiment ran gemma-3-4b-it [76], another family never run in this study. Every Qwen3-8B GPU artifact ran on the same 16 GB card with 8-bit floating point weight and activation linear layers, the engine of the iteration-3 screen; each reproduced the rows of the screen byte for byte, so every plan noise result below is a result for that engine, with no measured gap to bf16. The planned second screen body in Llama-3.1-8B-Instruct was first in the drop order and did not run.

**Table 122.** The seven artifacts of iteration 4, the question each was asked, and its hardware and spend.

| Artifact | Question | Model and hardware | OpenRouter spend |
|---|---|---|---|
| Record audit | Repair the iteration-3 record (knob table, route landings, Llama hit rule, other audit, figures, rigor items); freeze the push screen | No generation; CPU only | USD 0.22 (blind judges) |
| Plan noise experiment | Is fresh plan noise the cheapest knob for correct algorithmic diversity? | Qwen3-8B, 8-bit linears, one 16 GB RTX 2000 Ada; 10,100 programs; 2.53 GPU hours | USD 0.23 |
| Site experiment | Where must the noise enter to change the algorithm; what does a top-five push at the plan do? | Qwen3-8B, 8-bit linears, same card; 4,510 programs; 1.74 GPU hours | USD 0.19 |
| Same seed experiment | With the token seed fixed, does temperature or plan noise change the algorithm, thinking off and on? | Qwen3-8B, 8-bit linears (HF) and the official 8-bit checkpoint (vLLM), same card; 3,140 programs, thinking-on traces included; 2.74 GPU hours (HF 3,297 s, vLLM 6,580 s; added in iteration 5) | under USD 0.10 |
| Plan subspace experiment | Does a subspace read from forced prompts predict which random pushes move the algorithm? | Qwen3-8B, 8-bit linears, same card; 13,500 programs; 2.51 GPU hours | USD 0.20 |

| | | | |
|---|---|---|---|
| Held-out family experiment | Does push movability, measured before any ban, predict which bans gemma-3-4b-it obeys? | gemma-3-4b-it bf16, same card; 10,730 programs; 1.28 GPU hours | USD 0.28 |
| Positioning audit | Set the iteration-3 and iteration-4 claims against their nearest published findings | Web search and full-text quote checks | none recorded |

## 4.2 The record audit: the iteration-3 record repaired from rows on disk

The record audit generated no model text. It recomputed every number from the labelled rows of eleven earlier artifacts, citing each source file, and spent USD 0.22 on the blind judges of its audit of other programs. A stdlib-only reimplementation rederived 29 headline checks with no mismatch and reproduced the frozen screen file byte for byte, a second script added 12 checks with no mismatch, and all 7 placebo tests failed as they should.

### 4.2.1 One knob table for every iteration-3 arm

The audit rebuilt the knob table over every iteration-3 arm, 1,120 cells of artifact, task, arm and dial, each priced against its own twin with a Newcombe interval on the pass cost [42], the correct-only rarefied Hill-1 at depth 25 with template and independent-sample 90% intervals, the Good-Turing coverage estimate, catalogue-only avoidance and two flags. A cell *escapes* by the rule of Section 4.1, and we say it *qualifies* when it escapes and reaches a Hill-1 of at least 2 within 20 pass points. Of the 1,120 cells, 752 are knobs, 348 controls, 18 ablations on a ban and 2 ceilings; 945 come from Qwen3-8B and 175 from Llama-3.1-8B-Instruct. Table 123 gives the verdict per task that the review asked for, read from the table under the class rule of Section 3.2.5. The band patch is listed separately, because it writes the states of the ban itself into the twin and so carries the ban; exposure arms are priced against their own copy twin, in which the shown algorithm is the default.

**Table 123.** Best non-control knob per task (Qwen3-8B; correct-only rarefied Hill-1 at depth 25; pass points lost against the twin of the same row) within 5 and within 20 pass points, the band patch for reference, and the number of random direction and dose cells that qualify. Source: record audit `tables/per_task_verdict.csv`.

| Task (default) | Best within 5 points (class) | Best within 20 points (class) | Band patch | Random cells qualifying | A non-activation knob reaches 1.5 within 20 points |
|---|---|---|---|---|---|
| kth smallest (full sort) | random direction 9002, alpha 1: 2.92 at 0 (random) | same | 1.97 | 7 of 21 | yes (modal ban, 2.17 at 8) |
| two sum (hash map) | random direction 9002, alpha 2: 2.12 at 0 (random) | same | 1.68 | 1 of 9 | yes (modal ban of the locus experiment, 1.62 at 0) |
| find duplicate (hash set) | first iterated veto round: 2.51 at 2 (veto) | same | 1.17 | 1 of 9 | yes (iterated veto) |
| connected components (union find) | unnamed exposure ban of BFS: 1.85 at 0 (exposure) | same | 1.88 | 0 of 3 | yes (exposure ban) |
| substring search (naive) | novelty instruction: 1.98 at 0 (prompt) | modal ban of the locus experiment: 2.62 at 8 (ban) | 1.00 | 0 of 9 | yes |

| | | | | | |
|---|---|---|---|---|---|
| Fibonacci (iteration) | random direction 3: 1.09 at 0 (random) | same | 1.00 | 0 of 3 | no (novelty 1.90, but at 24 points) |
| longest increasing subsequence (quadratic DP) | named exposure ban of the Fenwick tree: 1.78 at 4 (exposure) | same | 1.00 | 0 of 3 | yes (exposure ban) |
| merge intervals (sort and sweep) | nonce ban: 1.00 at 0 | same | 1.00 | 0 of 3 | no |
| top-k frequent (sort by count) | novelty instruction: 1.19 at -12 (prompt) | modal ban: 1.78 at 17 (ban) | 1.00 | 0 of 3 | yes |
| nonoverlapping intervals (greedy by end) | novelty instruction: 1.00 at 0 | named exposure ban of the DAG longest path: 1.26 at 6 | 1.00 | 0 of 3 | no |
| topological order (Kahn) | nonce ban: 1.00 at 0 | modal ban: 1.05 at 7 | 1.00 | 0 of 3 | no |
| first and last position (binary search) | exposure veto of exponential search, round 1: 1.78 at 0 (veto) | same, round 3: 1.84 at 8 | 1.00 | 0 of 3 | no (the veto is an exposure arm) |
| grid (BFS) | random direction 9013, alpha 1: 1.51 at -13 (random) | second succession step: 2.19 at 12 (ban) | not run | 0 of 18 | yes |
| shortest path (Dijkstra) | random direction 9004, alpha 1: 1.975 at 4 (random) | same | not run | 0 of 18 | yes (leave-one-survivor list, 1.96 at 14; semantic seed, 1.95 at 7) |

Read per task, the verdict the record should have carried is this. On kth smallest and two sum, matched-norm random pushes at the task-specification positions reach a Hill-1 above 2 at no pass cost, ahead of every prompt knob; on kth smallest the three signature vetoes and the band patch come close (1.89 to 1.97 at 2 points or less). On connected components a random direction, the novelty instruction and the band patch all pass 1.5 at no cost, and none reaches 2. Substring search moves with words: the novelty instruction reaches 1.98 at no cost, and no random direction moves it. Find duplicate moves with an iterated veto (2.51 at 2 points; sort and adjacent check 32, Floyd cycle detection 9, hash set 7, index marking 1). On the path tasks the best rows, a random direction, the semantic seed and the decommitment direction, cluster just under 2. Merge intervals, topological order and nonoverlapping intervals do not reach 1.5 within 20 points under any knob, and Fibonacci reaches it only at 24 points. The sentence of the review that nothing reaches 1.5 within 20 points on six tasks did not reproduce, because the modal ban of top-k frequent (1.78 at 17 points) and exposure arms on longest increasing subsequence and first and last position do. Of the 27 numbers the review quoted, 25 reproduced (Table 126b); the other mismatch was the two sum avoidance at alpha 2, 0.469 on the attempted denominator (one row had no program) against the quoted 0.48. *(Correction, iteration 5: three qualifications. First, the numbers for random directions above are maxima over directions, not the effect of a typical direction: on kth smallest 2.92 is the best of 21 direction and dose cells, of which 7 qualify, and on two sum 2.12 is the best of 9 directions, of which 1 qualifies; fresh per-sample plan noise reaches 1.29 on two sum, and 0 of 40 fresh directions escape there at alpha 1 and 5 of 40 at alpha 2 (Sections 4.3 and 4.6; distributions per direction in Table 175). Second, under the exposure class now declared in Section 3.2.5, exposure arms are controls. With them set aside, in the knob rows of the plan noise experiment a knob outside the exposure class reaches 1.5 within 20 pass points on the grid, shortest path, kth smallest, substring search, two sum, find duplicate and*

*connected components tasks; longest increasing subsequence and first and last position join the tasks where nothing does, and the modal ban of top-k frequent (1.78 at 17 points) is carried from the record because ban arms are not among those rows (Table 172). Third, the record omitted the Llama block of the same table (Table 126c, below Table 126b).)*

Table 124 counts how many random directions qualify on each task, so that the result reads as a property of pushes rather than of one fortunate seed. Most bench tasks saw only three directions at alpha 1, which leaves the immovable side weakly tested: with 3 directions, a true escape rate of 0.54 would still produce no escape 10% of the time.

**Table 124.** Random directions per task and dose: run, escaping, qualifying, and the split by engine (8-bit linears in the locus experiment, bf16 in the bench forecast experiment). Source: record audit `tables/random_direction_counts.csv`.

| Task | Alpha | Run | Escape | Qualify | Escaping directions | 8-bit: run / escape / qualify | bf16: run / escape / qualify |
|---|---|---|---|---|---|---|---|
| kth smallest | 1 | 18 | 8 | 5 | 9001, 9002, 9003, 9006, 9010, 9011, 9014 and bench direction 3 | 15 / 7 / 5 | 3 / 1 / 0 |
| kth smallest | 2 | 3 | 3 | 2 | 9001, 9002, 9003 | 3 / 3 / 2 | 0 |
| two sum | 1 | 6 | 0 | 0 | none | 3 / 0 / 0 | 3 / 0 / 0 |
| two sum | 2 | 3 | 2 | 1 | 9002, 9003 | 3 / 2 / 1 | 0 |
| find duplicate | 1 | 6 | 0 | 0 | none | 3 / 0 / 0 | 3 / 0 / 0 |
| find duplicate | 2 | 3 | 1 | 1 | 9003 | 3 / 1 / 1 | 0 |
| connected components | 1 | 3 | 1 | 0 | bench direction 3 | 0 | 3 / 1 / 0 |
| substring search | 1 | 6 | 0 | 0 | none | 3 / 0 / 0 | 3 / 0 / 0 |
| substring search | 2 | 3 | 0 | 0 | none | 3 / 0 / 0 | 0 |
| grid | 1 | 15 | 1 | 0 | 9014 | 15 / 1 / 0 | 0 |
| grid | 2 | 3 | 0 | 0 | none | 3 / 0 / 0 | 0 |
| shortest path | 1 | 15 | 3 | 0 | 9003, 9004, 9007 | 15 / 3 / 0 | 0 |
| shortest path | 2 | 3 | 0 | 0 | none | 3 / 0 / 0 | 0 |
| Fibonacci, longest increasing subsequence, merge intervals, top-k frequent, nonoverlapping intervals, topological order, first and last position | 1 | 3 each | 0 | 0 | none | 0 | 3 / 0 / 0 each |

Table 110 of the locus experiment reported avoidance alone and left out the three random directions at alpha 2. Table 125 gives every alpha-2 row with its pass rate and correct composition (the correction is marked in Section 3.7). On kth smallest and two sum the random directions buy two correct algorithms at almost no cost, and on find duplicate one of them buys sort and adjacent check at 18 points, where the structured edit ADD moved nothing.

**Table 125.** The alpha-2 rows of the locus experiment (N = 50 each): avoidance, pass rate and cost against the twin, rarefied Hill-1, and the composition of correct programs. Source: record audit `tables/table110_alpha2.csv`.

| Task | Arm | Avoidance (catalogue only) | Pass rate (points lost) | Hill-1 | Correct programs |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| kth smallest | ADD | 0.70 (0.70) | 1.00 (0) | 2.32 | size-k heap 31, full sort 15, quickselect 4 |
| kth smallest | random 9001 | 0.96 (0.96) | 0.84 (16) | 1.20 | quickselect 40, full sort 2 |
| kth smallest | random 9002 | 0.52 (0.52) | 0.98 (2) | 2.13 | quickselect 24, full sort 24, size-k heap 1 |
| kth smallest | random 9003 | 0.30 (0.30) | 0.98 (2) | 2.29 | full sort 35, quickselect 7, size-k heap 6, heapify and pop 1 |
| two sum | ADD | 0.00 (0.00) | 0.48 (52) | not rarefiable | hash map 24 |
| two sum | random 9001 | 0.00 (0.00) | 1.00 (0) | 1.00 | hash map 50 |
| two sum | random 9002 | 0.47 (0.47) | 1.00 (0) | 2.12 | hash map 26, brute force 23, disputed 1 |
| two sum | random 9003 | 0.36 (0.36) | 1.00 (0) | 1.90 | hash map 32, brute force 18 |
| find duplicate | ADD | 0.00 (0.00) | 1.00 (0) | 1.00 | hash set 50 |
| find duplicate | random 9001 | 0.00 (0.00) | 1.00 (0) | 1.00 | hash set 50 |
| find duplicate | random 9002 | 0.00 (0.00) | 1.00 (0) | 1.00 | hash set 50 |
| find duplicate | random 9003 | 0.86 (0.76) | 0.82 (18) | 2.13 | sort and adjacent check 30, hash set 6, other 5 |
| substring search | ADD and random 9001 to 9003 | 0.00 each | 1.00 (0) each | 1.00 each | naive 50 each |
| grid | ADD | 0.04 (0.04) | 0.86 (-5) | 1.10 | BFS 42, DFS 1 |
| grid | random 9001 | 0.04 (0.04) | 0.44 (37) | not rarefiable | BFS 20, DFS 2 |
| grid | random 9002 | 0.10 (0.10) | 0.58 (23) | 1.00 | BFS 29 |
| grid | random 9003 | 0.18 (0.16) | 0.32 (49) | not rarefiable | BFS 16 |
| shortest path | ADD | 0.12 (0.10) | 0.84 (16) | 1.00 | Dijkstra 42 |
| shortest path | random 9001 | 0.10 (0.10) | 0.82 (18) | 1.00 | Dijkstra 41 |
| shortest path | random 9002 | 0.74 (0.74) | 0.56 (44) | 1.99 | Bellman-Ford/SPFA 15, Dijkstra 13 |
| shortest path | random 9003 | 0.06 (0.06) | 0.98 (2) | 1.18 | Dijkstra 47, Bellman-Ford/SPFA 2 |

Table 126 condenses the knob cells of Qwen3-8B into the best row per task and knob family within 20 points, with the number of cells in each family, so that the whole table can be read at a glance. The pattern across families is the starting point of the iteration: random directions lead on kth smallest, two sum and the shortest path task, prompt bans and exposure bans lead on substring search and the grid task, and on the tasks where bans fail no family reaches 1.5 within 20 points except through exposure arms.

**Table 126.** Best Hill-1 within 20 pass points per task and knob family (Qwen3-8B knob cells; cells in the family in brackets; a dash means the family was not run on the task; the entry none means that no cell within 20 points was rarefiable). Source: record audit `tables/knob_table.csv`.

| Task | Random directions | Structured activation edits | Band patch | Prompt ban | Exposure ban | Ban list | Instruction | Novelty instruction | Nonce ban | Seed | Decoding | Veto |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| kth smallest | 2.92 (21) | 2.32 (3) | 1.97 (1) | 2.17 (3) | 1.96 (30) | 1.43 (5) | none (4) | 1.11 (1) | 1.00 (1) | - | - | 1.98 (23) |
| two sum | 2.12 (9) | 1.00 (3) | 1.68 (1) | 1.62 (2) | 1.00 (14) | - | - | 1.18 (1) | 1.00 (1) | - | - | 1.92 (14) |
| find duplicate | 2.13 (9) | 1.00 (3) | 1.17 (1) | 2.01 (3) | 1.00 (20) | 1.10 (4) | none (4) | 1.09 (1) | 1.00 (1) | - | - | 2.51 (21) |
| connected components | 1.76 (3) | - | 1.88 (1) | 1.56 (2) | 1.85 (18) | 1.25 (3) | - | 1.68 (1) | 1.00 (1) | - | - | 1.51 (10) |
| substring search | 1.00 (9) | 1.00 (3) | 1.00 (1) | 2.62 (3) | 1.95 (26) | 2.02 (5) | none (4) | 1.98 (1) | 1.00 (1) | - | - | 1.12 (23) |
| Fibonacci | 1.09 (3) | - | 1.00 (1) | none (1) | 1.00 (6) | - | - | none (1) | 1.00 (1) | - | - | 1.00 (13) |
| longest increasing subsequence | 1.00 (3) | - | 1.00 (1) | 1.09 (1) | 1.78 (8) | - | - | none (1) | 1.00 (1) | - | - | 1.00 (10) |
| merge intervals | 1.00 (3) | - | 1.00 (1) | none (1) | 1.00 (2) | - | - | 1.00 (1) | 1.00 (1) | - | - | 1.00 (8) |
| top-k frequent | 1.00 (3) | - | 1.00 (1) | 1.78 (1) | 1.51 (10) | - | - | 1.19 (1) | 1.00 (1) | - | - | 1.00 (14) |
| nonoverlapping intervals | 1.00 (3) | - | 1.00 (1) | 1.00 (1) | 1.26 (4) | - | - | 1.00 (1) | 1.00 (1) | - | - | 1.00 (11) |
| topological order | 1.00 (3) | - | 1.00 (1) | 1.05 (1) | - | - | - | 1.00 (1) | 1.00 (1) | - | - | none (5) |
| first and last position | 1.21 (3) | - | 1.00 (1) | 1.04 (1) | 1.00 (4) | - | - | 1.09 (1) | 1.48 (1) | - | - | 1.84 (11) |
| grid | 1.75 (20) | 1.10 (13) | - | 2.19 (6) | - | 2.10 (4) | 1.23 (10) | - | - | 1.09 (3) | 1.04 (16) | none (5) |
| shortest path | 1.97 (20) | 1.91 (13) | - | 1.95 (5) | - | 1.96 (5) | 1.51 (11) | - | - | 1.95 (3) | 1.05 (16) | none (3) |

**Table 126b.** The 27 numbers quoted by the review and their recomputation from rows (tolerances from 0.005 to 0.03 in Hill-1 or share). Source: record audit `tables/verification_targets.csv`.

| Quoted number | Quoted | Recomputed | Match |
|---|---|---|---|
| kth smallest, alpha 1: escaping directions of 15 (locus experiment) | 7 of 15 | 7 of 15 | yes |
| kth smallest: Hill-1 of the escaping directions at pass of at least 0.96 | 2.92, 2.42, 2.32, 2.14, 2.10 | 2.923, 2.420, 2.320, 2.134, 2.109 | yes |
| kth smallest: composition of the best direction | size-k heap 19, quickselect 16, full sort 15 | same | yes |
| shortest path, alpha 1: escaping directions of 15 at pass of at least 0.88 | 3 of 15, minimum pass 0.88 | same | yes |
| shortest path: best direction | Dijkstra 26, Bellman-Ford/SPFA 22, pass 0.96, Hill-1 1.97 | same, Hill-1 1.975 | yes |
| shortest path: landing of every escaping direction | Bellman-Ford/SPFA | Bellman-Ford/SPFA | yes |

| | | | |
|---|---|---|---|
| two sum: escapes at alpha 1 and at alpha 2 | 0 of 3; 2 of 3 | same | yes |
| two sum: Hill-1 of the alpha-2 directions | 2.13, 1.90, 1.00, 1.00 | 2.121, 1.899, 1.00, 1.00 | yes |
| two sum: alpha-2 best composition | hash map 26, brute force 23 | hash map 26, brute force 23, disputed 1 (next: hash map 32, brute force 18) | yes |
| find duplicate: alpha-2 escapes, composition, pass | 1 of 3; sort and adjacent check 30 of 41; 0.82 | 1 of 3; sort and adjacent check 30, hash set 6, other 5; 0.82 | yes |
| substring search: escaping direction and dose cells | 0 of 6 | 0 of 6 | yes |
| bench forecast experiment: kth smallest random row | quickselect 34, full sort 9 | same | yes |
| bench forecast experiment: connected components random row | DFS 13 of 50, pass 1.00 | same | yes |
| decommitment direction, c = 0.25 (shortest path) | 1.913 [1.705, 2.088] at 17 points | 1.913 [1.704, 2.083] at 17.0 | yes |
| decommitment experiment, random direction 1 (shortest path) | 1.858 at 16 points | 1.86 at 16.0 | yes |
| semantic seed (shortest path) | 1.940 [1.770, 2.094] | 1.945 [1.780, 2.091] at 7.0 | yes |
| kth smallest vetoes | 1.89 to 1.91 at 2 points | one-shot 1.913, persistent 1.891, first iterated 1.889, each at 2.0 | yes |
| kth smallest band patch | 1.98 | 1.973 | yes |
| connected components: random direction, novelty instruction, band patch | 1.76, 1.68, 1.88, all at 0 points | 1.756, 1.681, 1.882, all at 0.0 | yes |
| substring search: novelty instruction | 1.97 to 1.99 at pass 1.00 | 1.976 at pass 1.00 | yes |
| find duplicate: first iterated veto round | 2.49 at 2 points (sort and adjacent check 32, Floyd cycle detection 9, hash set 7, index marking 1) | 2.513 at 2.0, same composition | yes |
| Fibonacci: novelty instruction | 1.90 at 24 points | 1.898 at 24.0 | yes |
| Table 110: kth smallest alpha-2 avoidance | 0.96 | 0.96 | yes |
| Table 110: find duplicate alpha-2 avoidance and landing | 0.86, sort and adjacent check | 0.86; sort and adjacent check 30, index marking 8, other 5 | yes |

| | | | |
|---|---|---|---|
| Table 110: two sum alpha-2 avoidance (two directions) | 0.48, 0.36 | 0.469, 0.36 | no |
| Table 110: ADD at alpha 2 on find duplicate and two sum | 0.00, 0.00 | 0.00, 0.00 (two sum at pass 0.48) | yes |
| nothing reaches Hill-1 1.5 within 20 points on six immovable tasks | no row | top-k frequent modal ban 1.783 at 17; first and last position exposure vetoes 1.778 at 0, 1.827 at 6, 1.837 at 8; longest increasing subsequence named exposure bans of the Fenwick tree 1.775 at 4 and of patience sorting 1.538 at 0; top-k frequent unnamed exposure ban of quickselect on counts 1.511 at 6 | no |

The two mismatches are explained in the text above: the two sum share uses the attempted denominator (1 - 26/49), and the six-task sentence ignored the top-k frequent modal ban and two exposure arms.

*(Added in iteration 5: the Llama-3.1-8B-Instruct block of the knob table, which this section did not report. Source: iteration-5 record audit* `tables/llama_block.csv`*, carried from the iteration-4 record audit's knob table.)*

**Table 126c.** Llama-3.1-8B-Instruct cells of the knob table (175 cells): random directions against the knob cells that qualify (Hill-1 of at least 2 within 20 pass points).

| Row | Value | Note |
|---|---|---|
| cells of random directions | 0 of 36 qualify | pushed at about 3 times the ban displacement at layer 21; the dose was never swept down |
| qualifying knob cells | 5 | prompt bans and one veto |
| Fibonacci modal ban | 2.43 | prompt ban |
| substring search one-shot veto | 2.54 | signature veto |
| kth smallest present-minor ban of heapify and pop (K0522) | 2.87 (90% [2.1845, 3.3413]) | prompt ban; pass 0.28 against 0.16, cost -12 points (this row and the next two completed in iteration 6) |
| kth smallest present-minor ban of quickselect (K0523) | 2.96 (90% [2.406, 3.1827]) | prompt ban; pass 0.43 against 0.16, cost -27 points |
| kth smallest present-minor ban of size-k heap (K0524) | 2.92 (90% [2.2306, 3.3258]) | prompt ban; pass 0.31 against 0.16, cost -15 points |

In Llama the cheapest diversity came from prompt bans, not from random pushes, and the pushes ran only at a dose about three times the ban's own displacement, where they broke programs rather than redirected them.

### 4.2.2 Where each route lands

The bench route experiment had written the composition of every route arm to disk, and the record had reported only the share of the default. Table 127 gives the programs that left the default under the full ban, the prompt route, the direct route, the condition-span knockout and the one-shot veto on the five gated tasks, with the Jensen-Shannon divergence between the landing of each arm and that of the full ban, and the 95th percentile of its resampling null. All seven readings the review quoted reproduced, and the divergences matched those of the experiment itself to 1e-6 on 17 of 17 arms.

**Table 127.** Landing of the attempts that left the default under each route arm (N = 100 slots per arm; the one-shot veto on connected components produced 20 valid attempts), the divergence from the landing of the full ban against its null, and the frozen family nominations for the task. Source:

record audit `tables/route_landing.csv`.

| Task | Arm | Attempts off the default (of attempted) | JS from full ban (null p95) | Association / likelihood runner-up / proximity / push top-1 |
|---|---|---|---|---|
| kth smallest | full ban | quickselect 70, size-k heap 28, heapify and pop 1 (99 of 100) | n/a | quickselect / quickselect / heapify and pop / quickselect |
| kth smallest | prompt route | quickselect 57, size-k heap 41, heapify and pop 1 (99) | 0.0133 (0.0084), outside | |
| kth smallest | direct route | quickselect 74, size-k heap 1 (75) | 0.1146 (0.0107), outside | |
| kth smallest | knockout | quickselect 62, size-k heap 36, heapify and pop 1 (99) | 0.0052 (0.0084), within | |
| kth smallest | one-shot veto | quickselect 41 (41) | 0.1279 (0.0211), outside | |
| two sum | full ban | brute force 83 (83) | n/a | sort and two pointers / sort and two pointers / brute force / brute force |
| two sum | prompt route | brute force 34 (34) | degenerate (single label) | |
| two sum | direct route | brute force 80 (80) | degenerate | |
| two sum | knockout | brute force 33 (33) | degenerate | |
| two sum | one-shot veto | none (0) | n/a | |
| find duplicate | full ban | index marking 77, Floyd cycle detection 1, other 1 (79) | n/a | Floyd cycle detection / sort and adjacent check / sort and adjacent check / sort and adjacent check |
| find duplicate | prompt route | index marking 3 (3) | 0.1245 (0.2436), within | |
| find duplicate | direct route | index marking 74, other 3, sort and adjacent check 1 (78) | 0.0086 (0.0061), outside | |
| find duplicate | knockout | Floyd cycle detection 3, index marking 2 (5) | 0.3507 (0.1455), outside | |
| find duplicate | one-shot veto | none (0) | n/a | |
| connected components | full ban | BFS 75, DFS 25 (100) | n/a | DFS / DFS / label propagation / DFS |
| connected components | prompt route | DFS 56, BFS 44 (100) | 0.0711 (0.0067), outside | |
| connected components | direct route | DFS 70, BFS 10 (80; union find 20) | 0.3000 (0.0085), outside | |

| | | | | |
|---|---|---|---|---|
| connected components | knockout | DFS 57, BFS 43 (100) | 0.0756 (0.0067), outside | |
| connected components | one-shot veto | DFS 6 (6 of 20) | 0.3981 (0.0891), outside | |
| merge intervals | full ban | other 44, union find 15 (59) | n/a | interval tree / union find / union find / none |
| merge intervals | prompt route | other 19, union find 4 (23) | 0.0071 (0.0312), within | |
| merge intervals | direct route | other 1 (1) | 0.0865 (0.2479), within | |
| merge intervals | knockout | other 22, union find 4 (26) | 0.0104 (0.0341), within | |
| merge intervals | one-shot veto | none (0) | n/a | |

The table answers the question of the review in the direction it pointed. On connected components the two routes together choose the substitute, and neither chooses it alone. Each single route removes union find, but the prompt route alone, the direct route alone and the knockout all land mostly on DFS, which is the association answer of Qwen3-8B, the runner-up of the baseline likelihood order and the modal landing of random pushes on the task (DFS 0.83 and BFS 0.17 of the excess-weighted push landing in Section 4.6); BFS becomes the modal landing only when both routes act. On kth smallest the direct route alone almost eliminates the size-k heap (1 against 28 under the full ban), so the size-k heap share travels by the prompt states, and the random pushes of Section 4.6 land on the same two algorithms. On find duplicate the direct route reproduces the index marking of the full ban, and the five programs of the knockout land on Floyd cycle detection 3 times, the association answer. Two sum lands on brute force under every arm, and merge intervals lands mostly on residue. We carry this forward as the route-landing lead: where a single route suffices to remove the default, it lands on the generic neighbour that pushes and the stated alternative of the model agree on, and departures from that neighbour appear when both routes act. *(Correction, iteration 5: this generalises from one task. It is an observation on a single task, connected components, where either route alone lands mostly on DFS, the neighbour that pushes and association agree on, and both routes together choose BFS. Find duplicate is the counterexample in the same table: the direct route alone removes the default and lands on index marking (74 of 78), which is neither the push landing (sort and adjacent check) nor the association answer (Floyd cycle detection). On two sum pushes and association disagree, so there is no agreed neighbour to land on.)*

Table 128 restates the pooled route fractions with merge intervals and without it, and under catalogue-only shares. Removing merge intervals raises both fractions, and the jackknife interval of the direct route then lies above one half; the two pooled point estimates still have overlapping intervals and a negative interaction.

**Table 128.** Pooled route fractions over the gated tasks (ratio of sums; leave-one-task-out jackknife 95% intervals), with and without merge intervals, using the share of attempts in the default (plain) or catalogue-only shares. Source: record audit `results/route.json`.

| Tasks | Shares | F_prompt [jackknife] | F_direct [jackknife] | Interaction [jackknife] | F_knockout [jackknife] |
|---|---|---|---|---|---|
| five gated tasks | plain | 0.611 [0.06, 1.17] | 0.744 [0.38, 1.11] | -0.355 [-0.98, 0.27] | 0.621 [0.08, 1.16] |
| without merge intervals | plain | 0.648 [-0.08, 1.38] | 0.865 [0.68, 1.05] | -0.513 [-1.07, 0.04] | 0.651 [-0.07, 1.37] |
| five gated tasks | catalogue only | 0.634 [0.01, 1.26] | 0.824 [0.65, 1.00] | -0.458 [-0.97, 0.06] | 0.637 [0.02, 1.25] |
| without merge intervals | catalogue only | 0.650 [-0.08, 1.38] | 0.859 [0.68, 1.04] | -0.508 [-1.07, 0.06] | 0.653 [-0.06, 1.37] |

Merge intervals has no route class under catalogue-only avoidance: its full-ban avoidance into the catalogue is 0.15, below the gate. The superadditive class of Table 107 therefore rests on residue.

### 4.2.3 The Llama hit rule

The Llama bench experiment scored its 12 modal events against the modal landing and its 6 present minor events against the survivor with the largest excess over the renormalised ban-free baseline, a rule written in its scoring code and never stated in the record. Table 129 recomputes the top-1 hits of every family on the 13 gated events under each candidate rule, with Poisson-binomial p values against the per-event chance rates. The ladder nominated other on 3 of the 13 gated events, and on find duplicate both its nomination and the landing were other.

**Table 129.** Top-1 hits of each frozen family on the 13 gated Llama events under five labelling rules (hits of 13; Poisson-binomial p against per-event chance). Source: record audit `results/llama.json`.

| Rule | Subtraction | Ladder | Association | Proximity |
|---|---|---|---|---|
| As recorded (modal events: modal landing; present minor events: largest excess) | 4 (p 0.468) | 7 (p 0.032) | 7 (p 0.032) | 3 (p 0.715) |
| Modal landing on all 13 events (label corrected in iteration 6) | 6 (p 0.100) | 10 (p 0.00018) | 9 (p 0.0014) | 3 (p 0.715) |
| Largest excess on all 13 events (label corrected in iteration 6) | 1 (p 0.983) | 4 (p 0.468) | 5 (p 0.244) | 1 (p 0.983) |
| As recorded, other excluded as nominee and landing | 3 (p 0.715) | 6 (p 0.100) | 7 (p 0.032) | 3 (p 0.715) |
| Modal landing, other excluded | 8 (p 0.0077) | 11 (p 0.000016) | 9 (p 0.0014) | 3 (p 0.715) |

Under the recorded rule with other excluded, only association hits above chance (7 of 13, $p = 0.032$); the ladder hits 6 of 13 ($p = 0.100$). The verdict depends on the rule, and the rule decides it in an informative way. On present minor events the model mostly writes its default, which the largest-excess rule never counts as a landing, so the ladder, which nominates the default, is scored a miss on exactly the events where it names what the model writes. Under the modal rule the ladder hits 10 of 13. Table 130 gives the present minor events with both labels side by side, and the share of the default among attempts not banned. On five of the six events the modal landing is the default, which is the snap-back of the Qwen3-8B exposure experiment replicated in a second family: the default takes 0.93, 0.73 and 0.72 of attempts not banned under the bans of DFS, KMP and sort by count, and 0.58 and 0.35 under two of the kth smallest bans. The ban of heapify and pop is the exception; it moved Llama to quickselect, and the full sort kept 0.23.

**Table 130.** Llama present minor events: modal landing (with and without other), largest-excess landing, and the share of the default among attempts not banned (Wilson 95%). Source: record

audit `results/llama.json`.

| Event | Default | Modal landing | Largest-excess landing | Default among attempts not banned [95%] |
|---|---|---|---|---|
| top-k frequent, ban of sort by count | heap | heap | quickselect on counts | 54 of 75 = 0.72 [0.61, 0.81] |
| kth smallest, ban of heapify and pop | full sort | quickselect | quickselect | 18 of 77 = 0.23 [0.15, 0.34] |
| kth smallest, ban of quickselect | full sort | full sort | full sort | 41 of 71 = 0.58 [0.46, 0.69] |
| substring search, ban of KMP | naive | naive | Boyer-Moore-Horspool | 48 of 66 = 0.73 [0.61, 0.82] |
| connected components, ban of DFS | union find | union find | BFS | 79 of 85 = 0.93 [0.85, 0.97] |
| kth smallest, ban of size-k heap | full sort | full sort | full sort | 27 of 78 = 0.35 [0.25, 0.46] |

### 4.2.4 The audit of other programs, and catalogue-only avoidance

The audit pooled every program labelled other that iteration 3 produced under a ban: 21 events and 226 unique programs, of which 6 passed the hidden tests (Table 131). Two blind judges from different families classified each as a labeller miss, a variant of a catalogue algorithm, a degenerate or hard-coded program, or a novel algorithm, a third judge broke ties, and the executor read 73 programs by hand: every passing program, every program a judge called novel, all 44 merge intervals programs and a seeded sample of the rest. Judge A (gpt-4.1-mini) fell below the control gate (4 of 6 positive controls in the strict class) and received no weight; judge B (gemini-2.5-flash) passed at 5 of 6 and was weighted; the two agreed at Cohen's kappa [11] 0.77 [0.67, 0.86]. The final classes were 68 catalogue variants, 156 degenerate programs, 2 labeller misses and no novel algorithm. Judges flagged 7 programs as novel and the manual pass confirmed none; the one passing program a judge called novel reran 60 of 60 hidden tests and 3 of 3 stress tests and is a textbook binary search over values after a sort.

**Table 131.** Every event audited for programs labelled other (unique programs; sampled up to 25 per event; final classes; a program shared by two events is counted in both, so the rows sum to 231 sampled programs and the pool holds 226 unique ones). Source: record audit `tables/other_audit_events.csv`.

| Event | Programs labelled other (unique) | Sampled | Passing | Labeller miss | Catalogue variant | Degenerate | Novel |
|---|---|---|---|---|---|---|---|
| Qwen3-8B bench ban, Fibonacci | 33 (10) | 10 | 0 | 0 | 10 | 0 | 0 |
| Qwen3-8B bench ban, merge intervals | 44 (44) | 25 | 0 | 0 | 23 | 2 | 0 |
| Qwen3-8B route full ban, Fibonacci | 31 (6) | 6 | 0 | 0 | 6 | 0 | 0 |
| Qwen3-8B route full ban, merge intervals | 44 (44) | 25 | 0 | 0 | 17 | 8 | 0 |
| Llama modal ban, nonoverlapping intervals | 19 (19) | 19 | 0 | 0 | 1 | 18 | 0 |
| Llama modal ban, kth smallest | 8 (8) | 8 | 0 | 0 | 0 | 8 | 0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Llama present minor ban of heapify and pop | 8 (8) | 8 | 0 | 0 | 0 | 8 | 0 |
| Llama present minor ban of quickselect | 9 (9) | 9 | 2 | 1 | 1 | 7 | 0 |
| Llama present minor ban of size-k heap | 9 (9) | 9 | 0 | 0 | 0 | 9 | 0 |
| Llama present minor ban of KMP | 8 (8) | 8 | 0 | 0 | 0 | 8 | 0 |
| Llama modal ban, two sum | 7 (7) | 7 | 0 | 0 | 0 | 7 | 0 |
| Llama modal ban, find duplicate | 50 (50) | 25 | 1 | 0 | 1 | 24 | 0 |
| Llama modal ban, Fibonacci | 12 (12) | 12 | 1 | 1 | 5 | 6 | 0 |
| Llama modal ban, longest increasing subsequence | 9 (9) | 9 | 0 | 0 | 0 | 9 | 0 |
| Llama modal ban, merge intervals | 19 (19) | 19 | 0 | 0 | 0 | 19 | 0 |
| Llama modal ban, top-k frequent | 5 (5) | 5 | 0 | 0 | 0 | 5 | 0 |
| Llama present minor ban of sort by count | 8 (8) | 8 | 0 | 0 | 0 | 8 | 0 |
| Qwen3-8B exposure, named ban of memoised recursion (nonoverlapping intervals) | 7 (7) | 7 | 0 | 0 | 0 | 7 | 0 |
| Qwen3-8B exposure, unnamed ban of the bisect module (first and last position) | 23 (1) | 1 | 0 | 0 | 1 | 0 | 0 |
| Qwen3-8B exposure, named ban of fast doubling (Fibonacci) | 49 (6) | 6 | 0 | 0 | 6 | 0 | 0 |
| Qwen3-8B exposure, named ban of quickselect on counts (top-k frequent) | 6 (5) | 5 | 2 | 0 | 2 | 3 | 0 |

The merge intervals landing is not the merge sweep without sorting that the record described. All 44 programs fail; the manual read found 29 that mark coverage over the coordinate range in a boolean array, 12 sweeps over unsorted intervals and 3 others (a regular expression pass gave 29, 14 and 1). The Fibonacci programs use the closed-form Binet formula, which sits in the extended catalogue and fails beyond n of about 70. We call an event *gated through residue* when it passes the avoidance gate only because of programs labelled other. Three events are: the Qwen3-8B merge intervals ban in the bench forecast experiment, the full-ban arm of the same ban in the route experiment, and the Llama find duplicate ban (catalogue-only avoidance 0.22, with 50 of 94 attempts labelled other). Table 132 gives catalogue-only avoidance beside avoidance for every modal bench event in both models and for the full-ban arm of the route experiment.

**Table 132.** Avoidance and catalogue-only avoidance (share of attempts in a catalogue algorithm that is not the default) per bench task, with the share labelled other; events gated through residue are marked †. Source: record audit `tables/catalogue_only_avoidance.csv`.

| Task | Qwen3-8B bench ban: avoidance / catalogue only / other | Qwen3-8B route full ban: avoidance / catalogue only / other | Llama modal ban: avoidance / catalogue only / other |
|---|---|---|---|

| | | | |
|---|---|---|---|
| kth smallest | 0.97 / 0.97 / 0.00 | 0.99 / 0.99 / 0.00 | 0.95 / 0.86 / 0.09 |
| substring search | 0.40 / 0.40 / 0.00 | 0.42 / 0.42 / 0.00 | 0.79 / 0.74 / 0.05 |
| find duplicate | 0.79 / 0.79 / 0.00 | 0.79 / 0.78 / 0.01 | 0.76 / 0.22 / 0.53 † |
| two sum | 0.86 / 0.86 / 0.00 | 0.83 / 0.83 / 0.00 | 0.91 / 0.83 / 0.07 |
| connected components | 1.00 / 1.00 / 0.00 | 1.00 / 1.00 / 0.00 | 0.82 / 0.78 / 0.04 |
| nonoverlapping intervals | 0.00 / 0.00 / 0.00 | 0.00 / 0.00 / 0.00 | 0.29 / 0.08 / 0.21 |
| Fibonacci | 0.34 / 0.01 / 0.33 | 0.31 / 0.00 / 0.31 | 0.79 / 0.66 / 0.13 |
| topological order | 0.08 / 0.08 / 0.00 | 0.04 / 0.04 / 0.00 | 0.18 / 0.14 / 0.04 |
| merge intervals | 0.58 / 0.14 / 0.44 † | 0.59 / 0.15 / 0.44 † | 0.30 / 0.07 / 0.23 |
| first and last position | 0.03 / 0.01 / 0.01 | 0.02 / 0.01 / 0.01 | 0.04 / 0.00 / 0.02 |
| top-k frequent | 0.48 / 0.47 / 0.01 | 0.41 / 0.40 / 0.01 | 0.89 / 0.84 / 0.05 |
| longest increasing subsequence | 0.02 / 0.02 / 0.00 | 0.02 / 0.01 / 0.01 | 0.23 / 0.14 / 0.09 |

**Table 132b.** Route experiment arms whose catalogue-only avoidance differs from their avoidance (all other arms of the 12 bench tasks are identical in the two measures). Source: record audit `tables/catalogue_only_avoidance.csv`.

| Task | Arm | Attempted | Avoidance | Catalogue only | Other |
|---|---|---|---|---|---|
| merge intervals | full ban | 100 | 0.59 | 0.15 | 0.44 |
| merge intervals | prompt route | 100 | 0.23 | 0.04 | 0.19 |
| merge intervals | knockout | 100 | 0.26 | 0.04 | 0.22 |
| merge intervals | direct route | 100 | 0.01 | 0.00 | 0.01 |
| Fibonacci | full ban | 100 | 0.31 | 0.00 | 0.31 |
| Fibonacci | direct route | 100 | 0.28 | 0.00 | 0.28 |
| Fibonacci | knockout | 50 | 0.04 | 0.00 | 0.04 |
| Fibonacci | prompt route | 100 | 0.01 | 0.00 | 0.01 |
| find duplicate | full ban | 100 | 0.79 | 0.78 | 0.01 |
| find duplicate | direct route | 100 | 0.78 | 0.75 | 0.03 |
| top-k frequent | full ban | 100 | 0.41 | 0.40 | 0.01 |
| top-k frequent | direct route | 100 | 0.23 | 0.19 | 0.04 |
| top-k frequent | knockout | 50 | 0.06 | 0.04 | 0.02 |
| top-k frequent | twin and one-shot veto | 100 each | 0.07 | 0.05 | 0.02 |
| first and last position | one-shot veto | 62 | 0.39 | 0.00 | 0.37 |
| first and last position | twin and direct route | 100 each | 0.24 | 0.00 | 0.21 to 0.23 |
| first and last position | full ban | 100 | 0.02 | 0.01 | 0.01 |
| longest increasing subsequence | full ban | 100 | 0.02 | 0.01 | 0.01 |

The Llama present minor events keep their gates under catalogue-only avoidance (0.62 to 0.86). Without the residue-gated events, the summaries change as follows. In the Qwen3-8B bench contest, recomputed from Table 96 on the four remaining gated events, the mean log score minus uniform is ladder +0.08, subtraction -0.87, proximity -1.95 and association -2.04, against +0.06, -0.69, -1.53 and -1.78 with merge intervals; the frozen counting rule still gives the ladder 3 wins, 2 of them through an undefined forecast scored as uniform, and proximity loses its merge intervals win. In the Llama contest without find duplicate, the ladder hits 6 of 12 and association 7 of 12 under the recorded rule. The route fractions without merge intervals are in Table 128.

### 4.2.5 Geometry in activation coordinates

The request asks for actual geometry plots in activation space, with centroids coloured by implemented count, before and after prohibiting the top five algorithms. The ban list experiment had drawn them in iteration 2, and the record had only tabulated their counts (Table 85). The figure below plots four of them from the saved centroid coordinates: the Isomap embedding ($k = 10$) of teacher-forced residual stream states at layer 24, read at the token before the commit token, with each centroid sized by the number of programs implementing it under the condition. The caveat travels with the figure: the token before the commit token is a position whose index depends on the label of the program itself, so part of the separation between centroids can reflect position rather than algorithm. Under the five-name ban, every implemented program sits on a banned centroid (BFS 63, DFS 36, IDDFS 1); under the single BFS ban the mass moves to the DFS centroid, which is far from BFS in this embedding; and on the shortest path task the ban that leaves only Bellman-Ford/SPFA splits between the allowed Bellman-Ford/SPFA centroid (51) and the banned Dijkstra centroid (48). The MDS stress-1 of the centroid geodesic matrix is 0.025 on the grid task and 0.006 on the shortest path task. BFS and bidirectional BFS coincide in this embedding (centroid distance 0.0147 in MDS units).

*[Figure fig_isomap_bans: generated in the report phase, not in this draft]*

The label-independent recomputation at generated token 16 gives neighbour orders, not coordinates, so it is tabulated rather than plotted (Table 133). At the label-dependent position the nearest centroid to BFS is bidirectional BFS, and a label permutation reproduces that identity in 1% of draws; at token 16 the nearest centroid to BFS is IDA*, and bidirectional BFS falls to fourth.

**Table 133.** The neighbour order of BFS among grid centroids (forced programs under the baseline prompt, layer 24; 782 programs) at the label-dependent position and at the fixed generated token 16, with the share of 500 label permutations in which the nearest neighbour is bidirectional BFS. Source: record audit `results/figures.json`.

| Position | Neighbour order from BFS | Nearest | Distance BFS to bidirectional BFS | Permutation share with bidirectional BFS nearest |
|---|---|---|---|---|
| token before the commit token | bidirectional BFS, greedy best first, DFS, A*, unit weight Dijkstra, other, IDA*, IDDFS | bidirectional BFS | 81.25 | 0.010 |
| generated token 16 | IDA*, IDDFS, DFS, bidirectional BFS, unit weight Dijkstra, other, greedy best first, A* | IDA* | 99.11 | 0.022 |

### 4.2.6 Hit rates under one rule, Table 82 per row, rival predictors and the frozen screen

The three contests of iteration 3 used three rules for undefined forecasts. Table 134 reports defined-only top-1 hit rates with the number of defined events in each contest. Subtraction, which looked worst in the exposure contest (0.36, almost all of it fractional credit for its uniform tie on events where it was undefined), was defined there for 8 of 64 gated events (the named and unnamed arms of 4 shown algorithms) and named the landing in all 8.

**Table 134.** Defined-only top-1 hit rates in the three iteration-3 contests (events gated; defined events; hits among defined events). Source: record audit `tables/defined_only_hits.csv`.

| Contest | Family | Gated events | Defined | Hits (defined only) | Rate |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| Qwen3-8B modal bench bans | subtraction | 5 | 2 | 0 | 0.00 |
| Qwen3-8B modal bench bans | ladder | 5 | 1 | 1 | 1.00 |
| Qwen3-8B modal bench bans | association | 5 | 5 | 1 | 0.20 |
| Qwen3-8B modal bench bans | proximity | 5 | 5 | 1 | 0.20 |
| Qwen3-8B exposure bans, named and unnamed | subtraction | 64 | 8 | 8 | 1.00 |
| Qwen3-8B exposure bans, named and unnamed | ladder | 64 | 64 | 52 | 0.81 |
| Qwen3-8B exposure bans, named and unnamed | association | 64 | 64 | 31 | 0.48 |
| Qwen3-8B exposure bans, named and unnamed | proximity | 64 | 64 | 42 | 0.66 |
| Llama modal and present minor bans | subtraction | 13 | 12 | 4 | 0.33 |
| Llama modal and present minor bans | ladder | 13 | 13 | 7 | 0.54 |
| Llama modal and present minor bans | association | 13 | 13 | 7 | 0.54 |
| Llama modal and present minor bans | proximity | 13 | 13 | 3 | 0.23 |

Table 135 restates Table 82 per distinct likelihood row. The gains never point at the landing, and the implied mass hits 8 of 11 rows, 6 of them bans of absent algorithms where the landing is the default.

**Table 135.** The 11 distinct Qwen3-8B likelihood rows: realised landing and the argmax of the gain, of the residual and of the implied mass among survivors. Source: record audit `tables/table82_distinct_rows.csv`.

| Task | Banned | Row type | Realised landing | Gain argmax | Residual argmax | Implied-mass argmax | Implied mass hit |
|---|---|---|---|---|---|---|---|
| grid | A* | sham | BFS | bidirectional BFS | bidirectional BFS | DFS | no |
| grid | BFS | modal | DFS | IDDFS | IDDFS | DFS | yes |
| grid | bidirectional BFS | sham | BFS | IDDFS | IDDFS | BFS | yes |
| grid | DFS | sham | BFS | bidirectional BFS | bidirectional BFS | BFS | yes |
| grid | unit weight Dijkstra | sham | BFS | IDDFS | bidirectional BFS | DFS | no |
| grid | IDDFS | sham | BFS | bidirectional BFS | bidirectional BFS | DFS | no |
| shortest path | Bellman-Ford/SPFA | sham | Dijkstra | Dial | Floyd-Warshall | Dijkstra | yes |
| shortest path | bidirectional Dijkstra | sham | Dijkstra | Dial | Dial | Dijkstra | yes |
| shortest path | Dial | sham | Dijkstra | bidirectional Dijkstra | bidirectional Dijkstra | Dijkstra | yes |
| shortest path | Dijkstra | modal | Bellman-Ford/SPFA | Dial | Dial | Bellman-Ford/SPFA | yes |
| shortest path | Floyd-Warshall | sham | Dijkstra | Dial | Dial | Dijkstra | yes |

The audit also scored, on the iteration-3 screen, the predictors of ban compliance that the movability claim competes with (Table 136). Movability and the maximum escape fraction rank the catalogue-only ban avoidance of the 14 tasks far better than the forced-prompt success rate, the clarity of the association, the likelihood mass of the runner-up, the nonce ban or the novelty instruction. These are screen values: the same data produced the movable set.

**Table 136.** Predictors of catalogue-only modal ban avoidance across the 14 Qwen3-8B tasks

(Spearman rho with Monte Carlo permutation p; balanced accuracy at a median cut). Source: record audit `tables/rival_predictors.csv`.

| Predictor | Tasks | Spearman rho | Permutation p | Kendall tau-b | Balanced accuracy | Fisher p |
|---|---|---|---|---|---|---|
| movable (escape fraction above zero) | 14 | 0.86 | 0.00066 | 0.73 | 1.00 | 0.00033 |
| maximum escape fraction | 14 | 0.85 | 0.00022 | 0.69 | 1.00 | 0.00033 |
| forced-prompt success rate (can) | 14 | 0.16 | 0.58 | 0.15 | 0.50 | 1.00 |
| clarity of the association answer | 14 | -0.07 | 0.82 | -0.06 | 0.50 | 1.00 |
| likelihood mass of the runner-up | 14 | 0.19 | 0.52 | 0.08 | 0.50 | 1.00 |
| nonce ban off-default share | 12 | -0.19 | 0.56 | -0.16 | 0.38 | 0.52 |
| novelty instruction off-default share | 14 | 0.35 | 0.21 | 0.28 | 0.65 | 0.59 |

The frozen screen (sha256 d6a2ca9711f461973c39482314a523d448d0590cb40e32432826dec1da4846e8, reproduced byte for byte by a stdlib-only rule from its input file) names six movable tasks: kth smallest, two sum, find duplicate, connected components, and the grid and shortest path tasks, the set the price claim froze. Against catalogue-only ban avoidance of at least 0.5 the table of the screen is 6, 0, 0, 8 (one-sided Fisher $p = 1/3003$), a description of the screen and not a confirmation. Connected components escapes only on the bf16 engine of the bench forecast experiment; the other movable tasks escape on the 8-bit directions of the locus experiment. *(Correction, iteration 6: the escapes of the screen are properties of the model, not of the 8-bit engine. The 40-direction escape fractions of iteration 4 reproduce in bf16 at the same absolute displacement, and the shortest path positive-control direction 9004 of the screen moves that task in bf16 as well (Section 6.3); the shrinkage of the screen under the ban-free unit of iteration 5 belongs to that unit and its direction seeds.)*

## 4.3 Plan noise priced against every other knob (Qwen3-8B)

**Why.** The price claim was the primary claim of the iteration and the question of the request in its plainest form: which knob buys correct algorithmic diversity most cheaply. The iteration-3 evidence for random pushes came from fixed directions, three per task on most of the bench, chosen by nobody. This experiment drew a fresh direction for every sample, which is the form a practitioner would use, and priced it on the six movable tasks of the frozen screen against every token-level and prompt-level knob the run had tried, all under one statistic and one price convention.

**Design.** Qwen3-8B ran with 8-bit linear layers on the 16 GB card, thinking off, and every arm of a task shared the same uniform stream per template and slot, so arms differ only by the knob. Plan noise ran at alpha 0.5, 1, 2 and 4 (and 8 in the dose sweep). The comparison knobs were temperature 1.3 and 1.6 without truncation, top-k 10, the nonce ban, the novelty instruction, the semantic seed on the path tasks and the iteration-3 signature vetoes (one-shot, persistent and iterated, reused rows); min-p 0.1 at temperature 1.6 and 2.0 and a verbalized sampling prompt [46] were preregistered sensitivities. Every cell had $N = 50$ (5 templates by 10 slots): 202 cells and 10,100 labelled programs, with no cell dropped. The 5-point Hill-1 and the prices of 1.5 and 2 were computed on the randomisation hull of each knob with 2,000 bootstrap replicates, and the choice of the best setting of a knob was made inside every replicate. The rule, committed by hash before

the first sample beyond stage 10, confirmed the price claim if plan noise reached a 5-point Hill-1 of at least 1.5 on at least 3 of the 6 tasks, beat the 5-point Hill-1 of the best non-activation knob on at least 4 of 6, and had a pooled margin with a task-bootstrap 95% interval above 0.

**The rule returned PARTIAL.** Plan noise is not the uniformly cheapest knob (Table 137). It reaches a 5-point Hill-1 of at least 1.5 on only 2 of 6 tasks, against the 3 required, and beats the best non-activation knob on 4 of 6, but the pooled margin is -0.22 with an interval from -0.89 to 0.30 (one-sided $p = 0.76$; exact sign test over the six margins $p = 0.34$). Plan noise wins clearly on kth smallest (2.72 against 1.96 for the novelty instruction, ahead in every bootstrap replicate) and on connected components (2.09 against 1.96 for the iterated veto, ahead in 91%), and on both it reaches Hill-1 of 1.5 and of 2 at no pass cost. It edges two sum and the grid task by margins inside their intervals. It loses decisively on find duplicate, where it does not move the hash set at alpha 2 or less while the reused iterated veto reaches 2.73, and on the shortest path task, where the semantic seed reaches 1.93 against 1.35 for plan noise because pushed Bellman-Ford/SPFA programs cost 14 to 26 pass points.

**Table 137.** The preregistered price comparison on the six movable tasks: 5-point Hill-1 of plan noise, the best non-activation knob and its 5-point Hill-1, the margin with its bootstrap 95% interval, the bootstrap probability that plan noise is ahead, and the price of 1.5 and price of 2 for plan noise (pass points; a dash means not reached). Source: plan noise experiment `results/c1.json`, `results/summary.md`.

| Task (default) | 5-point Hill-1, plan noise | Best non-activation knob | Its 5-point Hill-1 | Margin [95% CI] | P(plan noise ahead) | Price of 1.5, plan noise | Price of 2, plan noise |
|---|---|---|---|---|---|---|---|
| kth smallest (full sort) | 2.72 | novelty instruction | 1.96 | 0.76 [0.21, 1.16] | 1.00 | 0.0 | 0.0 |
| two sum (hash map) | 1.29 | novelty instruction | 1.24 | 0.05 [-0.20, 0.37] | 0.67 | 11.1 | - |
| find duplicate (hash set) | 1.00 | iterated veto | 2.73 | -1.73 [-2.06, -1.39] | 0.00 | - | - |
| connected components (union find) | 2.09 | iterated veto | 1.96 | 0.13 [-0.00, 0.37] | 0.91 | 0.0 | 0.0 |
| grid (BFS) | 1.42 | novelty instruction | 1.34 | 0.07 [-0.57, 0.41] | 0.49 | 18.0 | 49.8 |
| shortest path (Dijkstra) | 1.35 | semantic seed | 1.93 | -0.58 [-0.69, -0.14] | 0.00 | 9.3 | - |

Table 138 gives 5-point Hill-1 for every knob class on the six tasks. Temperature, top-k 10 and the nonce ban buy no correct algorithmic diversity on any of them (5-point Hill-1 at most 1.09), and min-p reaches at most 1.17. The novelty instruction switches kth smallest wholesale to quickselect, a Hill-1 of 1.0 on its own; its 1.96 comes from mixing it with the twin on the hull. Top-k 10 never changed a single token sequence, because the top 10 tokens of the model carry essentially all the probability mass on these prompts. The verbalized sampling prompt is a measurement artifact rather than a knob: 46 to 49 of 50 of its programs on every bench task and on the grid task call a random choice inside the solution function, so their mechanism label is whichever branch the trace

of the labeller happened to execute.

**Table 138.** 5-point Hill-1 per knob class on the six movable tasks (randomisation hull, cost at most 5 pass points; a dash means the class was not run on the task). Source: plan noise experiment `results/c1.json`.

| Task | Plan noise | Temperature 1.3/1.6 | Top-k 10 | Nonce ban | Novelty instruction | Semantic seed | One-shot veto | Persistent veto | Iterated veto | Min-p | Verbalized sampling |
|---|---|---|---|---|---|---|---|---|---|---|---|
| kth smallest | 2.72 | 1.09 | 1.00 | 1.00 | 1.96 | - | 1.90 | 1.88 | 1.89 | 1.16 | 1.29 |
| two sum | 1.29 | 1.00 | 1.00 | 1.00 | 1.24 | - | 1.00 | 1.00 | 1.00 | 1.00 | 1.04 |
| find duplicate | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | - | 1.00 | 1.00 | 2.73 | 1.00 | 1.01 |
| connected components | 2.09 | 1.00 | 1.00 | 1.00 | 1.17 | - | 1.02 | 1.03 | 1.96 | 1.09 | 1.02 |
| grid | 1.42 | 1.00 | 1.00 | 1.00 | 1.34 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| shortest path | 1.35 | 1.09 | 1.00 | 1.00 | 1.24 | 1.93 | 1.00 | 1.00 | 1.00 | 1.17 | 1.00 |

Every preregistered sensitivity returns the same verdict (Table 139). Excluding the reused vetoes, which came from another engine, turns the pooled margin positive (+0.20, interval -0.19 to 0.60); plan noise then loses only to the semantic seed on the shortest path task.

**Table 139.** Sensitivities of the price rule; each re-evaluates the full rule. Source: plan noise experiment `results/c1.json`.

| Sensitivity | Verdict | Tasks with 5-point Hill-1 at least 1.5 | Tasks where plan noise beats the best | Pooled margin [95% CI] |
|---|---|---|---|---|
| hull restricted to the twin segment | PARTIAL | 2 | 3 | -0.266 [-0.905, 0.246] |
| min-p and verbalized sampling added to the comparison set | PARTIAL | 2 | 4 | -0.218 [-0.897, 0.314] |
| min-p only added | PARTIAL | 2 | 4 | -0.218 [-0.863, 0.305] |
| dial interpolation instead of the hull | PARTIAL | 2 | 4 | -0.022 [-0.710, 0.607] |
| template-cluster bootstrap | PARTIAL | 2 | 4 | -0.218 [-0.866, 0.329] |
| slot-paired bootstrap | PARTIAL | 2 | 4 | -0.218 [-0.893, 0.322] |
| other programs excluded | PARTIAL | 2 | 4 | -0.243 [-0.892, 0.263] |
| other programs as one category | PARTIAL | 2 | 4 | -0.219 [-0.892, 0.326] |

| | | | | |
|---|---|---|---|---|
| reused vetoes excluded | PARTIAL | 2 | 4 | 0.202 [-0.188, 0.600] |
| hull vertices from the record audit only | PARTIAL | 2 | 3 | -0.134 [-0.721, 0.512] |
| deviation on hull edges (added in iteration 5): the preregistered hull search is restricted to edges; it matched an optimiser over the full simplex within 0.02 on the unit cases but fell short by up to 0.19 Hill-1 on random 5-vertex Dirichlet(0.5) cases, a shortfall that falls on plan noise (5 hull vertices against 2 or 3 for most comparison knobs) | not evaluated again | n/a | n/a | n/a |
| triangle check on the real data (added in iteration 5): the largest gain of a 3-vertex face over the best edge, over all classes, is +0.008 (connected components, plan noise; 42 entries with a value, 113 null) | verdict unaffected | 2 | 4 | -0.22 |

**Ideas against phrasings.** Under common random numbers the effect of each knob can be read slot by slot (Table 140). Temperature rewrites code without changing the algorithm: on the grid task temperature 1.6 changes the code in 88% of slots and the label in 2%, and it raises the phrasing clusters within BFS from 13.6 to 18.5 (20 programs, Jaccard at least 0.8). Plan noise changes the algorithm: at alpha 2 the label differs from that of the twin in 50% of slots on kth smallest, 60% on connected components and 54% on the shortest path task.

**Table 140.** Share of common-random-number slots in which the code differs from the twin and in which the algorithm label differs (N = 50 per cell). Source: plan noise experiment `results/flips.json`.

| Task | Temperature 1.3 | Temperature 1.6 | Top-k 10 | Nonce ban | Novelty instruction | Plan noise 0.5 | Plan noise 1 | Plan noise 2 | Plan noise 4 |
|---|---|---|---|---|---|---|---|---|---|
| kth smallest | 0.04 / 0.02 | 0.14 / 0.02 | 0.00 / 0.00 | 0.12 / 0.00 | 1.00 / 1.00 | 0.14 / 0.08 | 0.38 / 0.28 | 0.72 / 0.50 | 0.88 / 0.54 |
| two sum | 0.16 / 0.00 | 0.32 / 0.00 | 0.00 / 0.00 | 0.14 / 0.00 | 0.94 / 0.08 | 0.32 / 0.00 | 0.40 / 0.00 | 0.50 / 0.02 | 0.86 / 0.28 |
| find duplicate | 0.08 / 0.00 | 0.10 / 0.00 | 0.00 / 0.00 | 0.28 / 0.00 | 0.82 / 0.02 | 0.20 / 0.00 | 0.50 / 0.00 | 0.82 / 0.02 | 0.98 / 0.56 |
| connected components | 0.22 / 0.00 | 0.48 / 0.02 | 0.00 / 0.00 | 0.66 / 0.00 | 0.94 / 0.04 | 0.52 / 0.00 | 0.84 / 0.12 | 0.90 / 0.60 | 0.98 / 0.70 |
| grid | 0.64 / 0.00 | 0.88 / 0.02 | 0.00 / 0.00 | 0.94 / 0.00 | 1.00 / 0.56 | 0.96 / 0.02 | 1.00 / 0.06 | 1.00 / 0.24 | 1.00 / 0.30 |
| shortest path | 0.36 / 0.00 | 0.66 / 0.02 | 0.00 / 0.00 | 0.56 / 0.00 | 0.98 / 0.06 | 0.66 / 0.08 | 0.92 / 0.26 | 0.98 / 0.54 | 1.00 / 0.66 |

Each cell gives code differs / label differs.

**Dose and breakage.** The share of attempts off the default rises with alpha up to 2 on kth smallest, connected components and both path tasks, while two sum and find duplicate stay on their defaults until alpha 4 (Table 141). Breakage, a pass rate below 0.8 times that of the twin, arrives at alpha 4 on kth smallest, find duplicate, connected components and the grid task, at alpha 2 on the shortest path task and at alpha 8 on two sum; at alpha 8 almost nothing passes. Correct-only diversity peaks at alpha 2 on kth smallest (2.87, pass 0.94) and the shortest path task (1.95, pass 0.74) and at alpha 4 on connected components (2.56, pass 0.42). The realised displacement at block 12 matched

its target within 0.15% on all 3,220 plan noise rows.

**Table 141.** Dose response of plan noise on the six movable tasks (N = 50 per dose): pass rate, share of attempts off the default, correct-only rarefied Hill-1 and the composition of correct programs. Source: plan noise experiment `results/alpha_sweep.json`.

| Task | Alpha | Pass | Off default | Hill-1 | Correct programs |
|---|---|---|---|---|---|
| kth smallest | 0 | 1.00 | 0.00 | 1.00 | full sort 50 |
| kth smallest | 0.5 | 1.00 | 0.08 | 1.30 | full sort 46, quickselect 4 |
| kth smallest | 1 | 1.00 | 0.28 | 2.14 | full sort 36, quickselect 9, size-k heap 5 |
| kth smallest | 2 | 0.94 | 0.50 | 2.87 | full sort 24, quickselect 14, size-k heap 8, other 1 |
| kth smallest | 4 | 0.74 | 0.54 | 2.71 | quickselect 17, full sort 16, size-k heap 3, heapify and pop 1 |
| kth smallest | 8 | 0.04 | 0.38 (invalid 0.54) | 2.00 | quickselect 1, full sort 1 |
| two sum | 0 to 1 | 1.00 | 0.00 | 1.00 | hash map 50 |
| two sum | 2 | 1.00 | 0.02 | 1.09 | hash map 49, brute force 1 |
| two sum | 4 | 0.80 | 0.28 | 1.79 | hash map 29, brute force 11 |
| two sum | 8 | 0.00 | 0.08 (invalid 0.92) | n/a | none |
| find duplicate | 0 to 1 | 1.00 | 0.00 | 1.00 | hash set 50 |
| find duplicate | 2 | 0.98 | 0.02 | 1.00 | hash set 49 |
| find duplicate | 4 | 0.24 | 0.52 | 1.00 | hash set 12 |
| find duplicate | 8 | 0.00 | 0.06 (invalid 0.94) | n/a | none |
| connected components | 0 to 0.5 | 1.00 | 0.00 | 1.00 | union find 50 |
| connected components | 1 | 1.00 | 0.12 | 1.42 | union find 44, DFS 6 |
| connected components | 2 | 1.00 | 0.60 | 2.10 | DFS 29, union find 20, BFS 1 |
| connected components | 4 | 0.42 | 0.64 | 2.56 | union find 10, DFS 9, BFS 2 |
| connected components | 8 | 0.00 | 0.14 (invalid 0.86) | n/a | none |
| grid | 0 | 0.74 | 0.00 | 1.00 | BFS 37 |
| grid | 0.5 | 0.86 | 0.02 | 1.00 | BFS 43 |
| grid | 1 | 0.70 | 0.06 | 1.14 | BFS 34, DFS 1 |
| grid | 2 | 0.72 | 0.24 | 1.42 | BFS 32, DFS 4 |
| grid | 4 | 0.24 | 0.30 | 2.06 | BFS 9, DFS 2, Dijkstra 1 |
| grid | 8 | 0.00 | 0.70 (invalid 0.06) | n/a | none |
| shortest path | 0 | 1.00 | 0.00 | 1.00 | Dijkstra 50 |
| shortest path | 0.5 | 0.98 | 0.08 | 1.26 | Dijkstra 46, Bellman-Ford/SPFA 3 |
| shortest path | 1 | 0.86 | 0.26 | 1.66 | Dijkstra 34, Bellman-Ford/SPFA 9 |
| shortest path | 2 | 0.74 | 0.54 | 1.95 | Dijkstra 22, Bellman-Ford/SPFA 15 |
| shortest path | 4 | 0.04 | 0.66 | 1.00 | Dijkstra 2 |
| shortest path | 8 | 0.00 | 0.90 (invalid 0.04) | n/a | none |

*[Figure fig_pn_dose: generated in the report phase, not in this draft]*

**The held-out task and the immovable tasks.** Cycle detection, the bench task that the 12-task cap had excluded and on which nothing had been run, barely moves under plan noise: an off-default excess of 0.01 at alpha 2 or less, a 5-point Hill-1 of 1.09 and a price of 1.5 of 29 points. On the eight bench tasks the screen called immovable, plan noise at alpha 2 or less moves almost nothing (Table 142), which agrees with the screen. The strongest knob there is the novelty instruction on substring search (5-point Hill-1 2.57).

**Table 142.** Cycle detection (held out) and the eight immovable bench tasks: plan noise off-default excess at alpha 2 or less, and 5-point Hill-1 per knob. Source: plan noise experiment `results/nonmovable.json`.

| Task | Plan noise off-default excess | 5-point Hill-1, plan noise | 5-point Hill-1, temperature | 5-point Hill-1, nonce ban | 5-point Hill-1, novelty instruction | Price of 1.5, plan noise |
|---|---|---|---|---|---|---|
| nonoverlapping intervals | 0.00 | 1.00 | 1.00 | 1.00 | 1.00 | - |
| cycle detection (held out) | 0.01 | 1.09 | 1.00 | 1.00 | 1.37 | 29.0 |
| topological order | 0.00 | 1.00 | 1.00 | 1.00 | 1.00 | - |
| first and last position | -0.08 | 1.37 | 1.42 | 1.21 | 1.20 | - |
| substring search | 0.00 | 1.00 | 1.00 | 1.00 | 2.57 | - |
| Fibonacci | 0.01 | 1.00 | 1.00 | 1.00 | 1.04 | - |
| longest increasing subsequence | 0.03 | 1.01 | 1.00 | 1.00 | 1.05 | - |
| merge intervals | 0.00 | 1.00 | 1.00 | 1.00 | 1.00 | - |
| top-k frequent | 0.04 | 1.14 | 1.00 | 1.00 | 1.00 | - |

**Table 142b.** The screen forecast committed before stage 100, which treated fresh plan noise as an equal mixture of the fixed directions of the screen, against the realised result: 5-point Hill-1 of plan noise, the margin over the best non-activation knob, and the divergence of the forecast alpha-1 composition from the realised one against its resampling null. Source: plan noise experiment `results/forecast_score.json`.

| Task | Forecast 5-point Hill-1 | Realised | Forecast margin | Realised margin | Divergence (null p95) | Within null |
|---|---|---|---|---|---|---|
| kth smallest | 2.38 | 2.72 | 0.45 | 0.76 | 0.0056 (0.0768) | yes |
| two sum | 1.78 | 1.29 | 0.61 | 0.05 | 0.0101 (0.0) | no |
| find duplicate | 1.76 | 1.00 | -0.97 | -1.73 | 0.0 (0.0) | yes |
| connected components | 1.33 | 2.09 | -0.63 | 0.13 | 0.0021 (0.0307) | yes |
| grid | 1.09 | 1.42 | n/a | 0.07 | 0.0087 (0.0301) | yes |
| shortest path | 1.35 | 1.35 | 0.35 | -0.58 | 0.0041 (0.115) | yes |

The forecast implied PARTIAL, and the realised verdict was PARTIAL. Unlike a prompt ban, plan noise moves the model under every template, the fourth included (Table 142c), so its effect does not inherit the wording dependence of bans.

**Table 142c.** Plan noise off-default share per template (N = 10 per cell) on the four tasks it moves, with the novelty instruction for comparison. Source: plan noise experiment `results/per_template.json`.

| Task | Arm | First | Second | Third | Fourth | Fifth |
|---|---|---|---|---|---|---|
| kth smallest | plan noise, alpha 1 | 0.1 | 0.2 | 0.5 | 0.1 | 0.5 |
| kth smallest | plan noise, alpha 2 | 0.4 | 0.5 | 0.6 | 0.5 | 0.5 |
| kth smallest | novelty instruction | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 |
| connected components | plan noise, alpha 1 | 0.2 | 0.2 | 0.0 | 0.0 | 0.2 |
| connected components | plan noise, alpha 2 | 0.8 | 0.6 | 0.4 | 0.6 | 0.6 |
| connected components | novelty instruction | 0.1 | 0.1 | 0.0 | 0.0 | 0.0 |
| grid | plan noise, alpha 1 | 0.1 | 0.0 | 0.1 | 0.1 | 0.0 |
| grid | plan noise, alpha 2 | 0.2 | 0.3 | 0.2 | 0.2 | 0.3 |
| grid | novelty instruction | 0.6 | 0.6 | 0.7 | 0.3 | 0.6 |
| shortest path | plan noise, alpha 1 | 0.2 | 0.1 | 0.1 | 0.3 | 0.6 |
| shortest path | plan noise, alpha 2 | 0.5 | 0.4 | 0.6 | 0.6 | 0.6 |
| shortest path | novelty instruction | 0.1 | 0.0 | 0.0 | 0.2 | 0.0 |

**Controls and checks.** The prohibition frame of the neutral twin is inert for the algorithm: a plain prompt with the specification alone and the twin have identical off-default shares on five of the six movable tasks ($+0.02$ on the shortest path task), although the frame costs 18 pass points on the grid task and adds 22 on cycle detection. *(Added in iteration 5: plan noise was also run on the plain prompt without the prohibition frame of the twin (alpha 1, 350 programs on the same common-random-number slots), and the record did not report it. Its off-default shares were 0.16 on kth smallest (0.28 under the twin frame), 0.24 on connected components (0.12), 0.26 on the shortest path task (0.26), 0.02 on the grid task (0.06) and 0 on two sum, find duplicate and cycle detection. Plan noise does not need the frame on kth smallest, connected components and the shortest path task; under plan noise at alpha 1 the frame costs 10 pass points on the grid task (interval -6 to 28). Table 173 gives the rows.)* A forecast committed before stage 100, which treated fresh noise as an equal mixture of the fixed directions of the screen, predicted the PARTIAL verdict and the alpha-1 composition within the resampling null on 5 of 6 tasks; it over-predicted find duplicate (1.76 forecast against 1.00) and two sum and under-predicted connected components. Two screen rows rerun with the keys of the screen itself agreed slot for slot (label agreement 1.00 on all four cells). The labellers recovered every reference on all 13 bench tasks. The 12 correct programs labelled other were all labeller misses, and no novel algorithm appeared. A judge agreed with the labeller on 1.00 of audited programs everywhere except kth smallest (0.80), and rescoring kth smallest with the labels of the judge left its margin at $+0.71$ against $+0.76$. Replacing every plan noise setting by the twin gave margins of 0 or below on all six tasks, and under 100 within-task shuffles of the arm labels the full committed rule never returned CONFIRMED; the count criteria alone fired in 8 of 100 shuffles in the independent rederivation, so the interval criterion is what makes the rule specific. The rederivation matched 12 of 12 checks, and the hash chain shows the commitment preceding the first row beyond stage 10.

## 4.4 Where the noise must enter (Qwen3-8B)

**Why.** If the plan lives in the task-specification states, the same random displacement should change the algorithm there and nowhere else. If any perturbation of matched size does as well, the effect is generic sensitivity, and the word plan names nothing. This experiment wrote one random displacement per common-random-number slot into six sites at matched per-position dose in ban units and priced each site on the randomisation hull.

**Design.** The six sites were the task-specification states at blocks 12 to 23 (the plan site of the hypothesis), the read position, the condition sentence, generated tokens 1 to 32 (through teacher-forced shadow rows), the task-specification states at blocks 0 to 11, and the same states at blocks

24 to 35. Each ran at alpha 1, 2 and 4 with N = 30 per cell (twin N = 50) on the six movable tasks, with energy-matched variants of the read position and condition sites and the plan site at alpha 8 for the boundary. One fresh direction and one stream of token uniforms per slot were shared by every arm, so every site contrast is an exact paired flip. In all 4,510 programs ran; 4,460 decisive rows followed the commitment, and none preceded it. The preregistered rule called the site claim confirmed if the plan site had the highest 5-point Hill-1 on at least 4 of 6 tasks.

**The rule returned PARTIAL, and three sites are inert.** The plan site has the best 5-point Hill-1 on 3 of 6 tasks (kth smallest, the grid task and the shortest path task; p = 0.062 against a binomial with chance 1/6), generated-token noise is best on two sum and find duplicate, and early-block noise on connected components (Table 143). Every per-task margin is a bootstrap close call except the shortest path task, where the plan site is ahead in 97% of replicates. The read position and the condition sentence never changed the algorithm at per-position doses of 1 to 4 ban units (corrected in iteration 5 from the phrase *at any dose*; the energy-matched read position below is weaker but not inert): 0 of 526 and 0 of 527 correct programs off the default at alpha 1 to 4. The late blocks are almost as inert (6 of 521). The outputs at these three sites are often byte-identical to those of the twin, 90% to 100% at alpha 1 on kth smallest, two sum and find duplicate for the read position and the late blocks.

**Table 143.** 5-point Hill-1 per site (correct-only rarefied Hill-1 within 0.05 pass cost of the twin; randomisation hull; 2,000 template-stratified bootstrap replicates with the hull maximum recomputed inside each), the best site, the margin of the plan site over the best other site and the probability that the plan site is best. Source: site experiment `results/c1_site.json`.

| Task (default) | Task spec, blocks 12-23 | Read position | Condition sentence | Generated tokens 1-32 | Task spec, blocks 0-11 | Task spec, blocks 24-35 | Best | Margin [95% CI] | P(plan site best) |
|---|---|---|---|---|---|---|---|---|---|
| kth smallest (full sort) | 2.61 | 1.00 | 1.00 | 2.24 | 2.46 | 1.00 | plan site | 0.15 [-0.75, 0.87] | 0.59 |
| two sum (hash map) | 1.28 | 1.00 | 1.00 | 1.28 | 1.14 | 1.00 | generated tokens | -0.00 [-0.49, 0.32] | 0.38 |
| find duplicate (hash set) | 1.16 | 1.00 | 1.00 | 1.24 | 1.00 | 1.00 | generated tokens | -0.08 [-0.41, 0.27] | 0.31 |
| connected components (union find) | 2.23 | 1.00 | 1.00 | 1.24 | 2.32 | 1.38 | blocks 0-11 | -0.08 [-0.78, 0.49] | 0.33 |
| grid (BFS) | 1.29 | 1.00 | 1.00 | 1.18 | 1.13 | 1.00 | plan site | 0.11 [-0.28, 0.57] | 0.67 |
| shortest path (Dijkstra) | 1.66 | 1.00 | 1.00 | 1.12 | 1.22 | 1.15 | plan site | 0.44 [-0.00, 0.69] | 0.97 |

*[Figure fig_site_h5: generated in the report phase, not in this draft]*

The tie on two sum (1.28 each) went to generated-token noise on the price of 1.5 (0.12 against 0.53, Table 144). Early-block noise winning connected components carries a dose-convention confound that the plan had named: a ban unit at blocks 0 to 11 is only 4.5% to 7.2% of the residual

norm, against 12.7% to 18.3% at blocks 12 to 23, so early-block noise at alpha 4 is a relative dose comparable to the plan site at alpha 1. No other site matched or beat the 5-point Hill-1 of the plan site on 4 tasks, so the claim is not disconfirmed either. Appending alpha 8 to the hull of the plan site, or allowing only mixtures of adjacent doses, left every ranking unchanged.

**Table 144.** Prices of 1.5 and 2 per site: the cheapest pass cost (as a fraction) at which Hill-1 reaches 1.5 and 2 on the hull (n/a: not reached). Source: site experiment `results/hull.json`.

| Task | Price of 1.5: blocks 12-23 / read position / condition / generated / blocks 0-11 / blocks 24-35 | Price of 2: same order |
|---|---|---|
| kth smallest | 0.00 / n/a / n/a / 0.01 / 0.00 / n/a | 0.00 / n/a / n/a / 0.04 / 0.00 / n/a |
| two sum | 0.53 / n/a / n/a / 0.12 / n/a / n/a | n/a for every site |
| find duplicate | 0.54 / n/a / n/a / 0.16 / n/a / n/a | 0.75 / n/a / n/a / 0.37 / n/a / n/a |
| connected components | 0.00 / n/a / n/a / 0.11 / 0.00 / n/a | 0.01 / n/a / n/a / 0.19 / 0.02 / n/a |
| grid | 0.14 / n/a / n/a / n/a / n/a / n/a | n/a for every site |
| shortest path | 0.00 / n/a / n/a / n/a / 0.25 / n/a | n/a for every site |

Table 145 classifies the failure mode of each site under the frozen rule. Generated-token noise moves four tasks but pays in breakage: it is the only site whose invalid rate climbs to between 0.67 and 1.00 at alpha 4.

**Table 145.** Failure mode per site and task under the frozen rule († weak movement: a correct-only excess between 0.10 and 0.20 at a pass rate of at least 0.8 times that of the twin, which the rule does not classify). Source: site experiment `results/failure_modes.json`.

| Site | kth smallest | two sum | find duplicate | connected components | grid | shortest path |
|---|---|---|---|---|---|---|
| Task spec, blocks 12-23 | moves | moves with cost | moves with cost | moves | moves with cost | moves |
| Read position | no movement | no movement | no movement | no movement | no movement | no movement |
| Condition sentence | no movement | no movement | no movement | no movement | no movement | no movement |
| Generated tokens 1-32 | moves | moves | moves with cost | moves | breakage first | breakage first |
| Task spec, blocks 0-11 | moves | weak movement † | no movement | moves | breakage first | weak movement † |
| Task spec, blocks 24-35 | no movement | no movement | no movement | weak movement † | no movement | no movement |

The paired flips at alpha 2 make the comparison exact (Table 146). The plan site flips significantly more of the 30 shared slots off the default than the read position, the condition sentence and the late blocks on kth smallest, connected components and both path tasks, and more than the early blocks on kth smallest and connected components. Generated-token noise flips more than the plan site on find duplicate (12 against 0).

**Table 146.** Paired flips at alpha 2 (30 shared slots): slots off the default only under the plan site : only under the other site, with the exact McNemar p and the byte-identical rate of the other site against the twin. Source: site experiment `results/paired_flips.json`.

| Other site | kth smallest | two sum | find duplicate | connected components | grid | shortest path |
|---|---|---|---|---|---|---|
| Read position | 14:0 (p 0.00012; 0.93) | 3:0 (p 0.25; 0.97) | 1:0 (p 1; 0.80) | 18:0 (p 7.6e-06; 0.33) | 6:0 (p 0.031; 0.13) | 8:0 (p 0.0078; 0.57) |
| Condition sentence | 14:0 (p 0.00012; 0.87) | 3:0 (p 0.25; 0.70) | 1:0 (p 1; 0.87) | 18:0 (p 7.6e-06; 0.40) | 6:0 (p 0.031; 0.07) | 8:1 (p 0.039; 0.57) |
| Generated tokens 1-32 | 6:8 (p 0.79; 0.13) | 2:7 (p 0.18; 0.10) | 0:12 (p 0.00049; 0.07) | 10:2 (p 0.039; 0.00) | 5:1 (p 0.22; 0.00) | 6:3 (p 0.51; 0.00) |
| Task spec, blocks 0-11 | 11:1 (p 0.0063; 0.63) | 3:0 (p 0.25; 0.57) | 1:0 (p 1; 0.40) | 13:1 (p 0.0018; 0.27) | 5:2 (p 0.45; 0.00) | 5:3 (p 0.73; 0.20) |
| Task spec, blocks 24-35 | 14:0 (p 0.00012; 0.97) | 3:0 (p 0.25; 0.97) | 1:0 (p 1; 0.93) | 16:1 (p 0.00027; 0.33) | 7:0 (p 0.016; 0.27) | 7:0 (p 0.016; 0.20) |

**The effect is not generic transmission.** Noise at the read position produces as large a displacement at the last block as the plan site does (medians 141, 250 and 474 against 159, 272 and 461 at alpha 1, 2 and 4), yet it flips nothing. A logistic regression of a correct program being off the default on the log of the realised displacement at the read position, with template and task fixed effects, over 2,769 correct programs from the prefill sites, gives the plan site an indicator of +2.15 [1.79, 2.51] beside a displacement slope of +0.88 [0.62, 1.14]. In the matched-dose bins (the upper three quintiles of displacement) the plan site is off the default in 16% to 27% of programs, against 3% to 7% for the other prefill sites. A hand-coded Mantel-Haenszel odds ratio, which pools the odds of an off-default program at the plan site against the other prefill sites across task and displacement quintile strata, gives 5.72 [3.94, 8.30], and its placebo, with plan-site labels shuffled within task, has a median of 0.98. A Bayesian mixed logistic model with random task and template effects puts the site effects relative to the plan site at -3.0 (read position), -2.9 (condition sentence), -3.0 (late blocks), -0.8 (generated tokens) and -1.1 (early blocks), with the slope of the plan site on log2 alpha at +1.11. Concentrating the whole ban energy at the read position (median displacement there 1,002, six times that of the plan site at alpha 1) moved find duplicate by +0.23 and connected components by +0.18 at pass ratios of 0.87 to 0.93; spreading it over the condition span stayed inert. Generated-token noise produces the most phrasing clusters per idea (4.2 and 4.7 at alpha 1 and 2, against 2.8 and 3.4 for the plan site, AST-shingle components at Jaccard 0.8).

**The boundary.** At alpha 8 the task specification is effectively erased (Table 147): the model echoes the condition sentence, writes placeholder code or asks for clarification, and the pass rate is 0.00 on every task. Correct-only diversity peaks at alpha 2 (kth smallest, grid, shortest path) or alpha 4 (two sum, find duplicate, connected components), and alpha 4 already costs 33% to 97% of the passes. Across every arm 3,555 programs were correct, and exactly one was labelled other: a find duplicate program from generated-token noise at alpha 2 that calls a count of each element over the list. The judges called it degenerate and a labeller miss, and the manual read, beside its three nearest known solutions, called it the textbook quadratic brute force. No novel algorithm appeared (rule-of-three bound 3/3,555 = 0.08% per correct program [67]).

**Table 147.** The plan site from alpha 0 to 8: pass rate / invalid rate / correct-only Hill-1. Source: site experiment `results/boundary.json`.

| Task | Alpha 0 | Alpha 1 | Alpha 2 | Alpha 4 | Alpha 8 |
|---|---|---|---|---|---|
| kth smallest | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 2.52 | 0.93 / 0.00 / 2.68 | 0.67 / 0.00 / 2.23 | 0.00 / 0.94 / n/a |
| two sum | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 1.00 | 0.97 / 0.00 / 1.28 | 0.30 / 0.07 / 1.70 | 0.00 / 0.94 / n/a |

| | | | | | |
|---|---|---|---|---|---|
| find duplicate | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 1.15 | 0.17 / 0.03 / 2.59 | 0.00 / 0.94 / n/a |
| connected components | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 1.84 | 0.97 / 0.00 / 2.23 | 0.23 / 0.10 / 2.73 | 0.00 / 0.96 / n/a |
| grid | 0.74 / 0.00 / 1.00 | 0.72 / 0.00 / 1.39 | 0.68 / 0.00 / 1.51 | 0.12 / 0.06 / 1.00 | 0.00 / 0.82 / n/a |
| shortest path | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 1.65 | 0.90 / 0.00 / 1.70 | 0.03 / 0.07 / 1.00 | 0.00 / 0.88 / n/a |

*(Correction, iteration 5: the alpha-4 peaks of this table rest on few correct programs and were rarefied at those depths, not at 25: two sum 9, find duplicate 5 and connected components 7 correct programs (kth smallest 20, grid 6, shortest path 1). The statement that correct-only diversity peaks at alpha 4 on three tasks is therefore a statement about 5 to 9 programs. Depth per cell in Table 178.)*

**The top-five prohibition at the plan.** The request asks what happens when the five most implemented algorithms are prohibited. The prompt version had failed by reimplementation (avoidance 0.00) and the cumulative token veto had stalled. This experiment applied the prohibition as an activation push: the plan states of the grid task were pushed along the per-block direction from the mean of the five banned centroids to the mean of the survivor centroids, estimated from 300 forced prompts in which the centroid of every algorithm was separable from the others (nearest-centroid separability 0.83 to 1.00). The push never moved the model off BFS (Table 148). Correct programs were BFS 30 of 30 at alpha 1 and 19 of 19 at alpha 2, and none passed at alpha 4; fresh random plan noise at the same norm did no better. The test of edge against opposite side could not be computed because no survivor program appeared at any dose, and the plan geometry leaves little to resolve: bidirectional BFS and IDA* sit almost equally far from the banned set (geodesic 0.36 against 0.37; MDS stress-1 0.147; no persistent loop beyond a permutation null).

**Table 148.** The top-five push at the plan states of the grid task (N = 50 per arm; slots 0 to 9): correct programs, avoidance of the five banned algorithms with Wilson 95% interval, and the composition of correct programs. Source: site experiment `results/c3d.json`.

| Arm | Correct | Avoidance of the top five [95%] | Correct programs |
|---|---|---|---|
| top-five push, alpha 1 | 30 | 0.00 [0.00, 0.11] | BFS 30 |
| random plan noise, alpha 1 | 36 | 0.03 [0.00, 0.14] | BFS 33, DFS 2, other 1 |
| top-five push, alpha 2 | 19 | 0.00 [0.00, 0.17] | BFS 19 |
| random plan noise, alpha 2 | 34 | 0.00 [0.00, 0.10] | BFS 29, DFS 5 |
| top-five push, alpha 4 | 0 | n/a | none |
| random plan noise, alpha 4 | 6 | 0.00 [0.00, 0.39] | BFS 6 |
| twin | 37 | 0.00 [0.00, 0.09] | BFS 37 |

**The scorecard and checks.** Table 149 scores the predictions the experiment committed before its first decisive sample. The engine reproduced the kth smallest and shortest path cells of the screen byte for byte (50 of 50 identical programs on each task, divergence 0.000 against null 95th percentiles of 0.029 and 0.025). Alpha 0 reproduced the twin at every site; the realised displacement along the direction equalled its target within 0.26% at every prefill site. One GPU test failed: shadow states for generated-token noise diverged from an independent teacher-forced pass (minimum cosine 0.679 on kth smallest), and a follow-up showed clean decode-path states diverging by the same order, a property of the 8-bit decode path rather than of the shadow coupling. An independent rederivation checked 146 cells with no mismatch, and a second, written only from raw files, reproduced the headline counts. The alpha-4 compositions of this experiment

differ from the alpha-4 rows of the plan noise experiment beyond a resampling null on four tasks, so these cells do not stand in for the knob table.

**Table 149.** Predictions committed before the first decisive sample and their outcomes. Source: site experiment `README.md`, `predictions/predictions.json`.

| Prediction | Outcome |
|---|---|
| The plan site has the highest 5-point Hill-1 on at least 4 of 6 tasks | missed: 3 of 6 (PARTIAL) |
| The read position shows no movement on at least 5 of 6 | hit: 6 of 6 |
| The condition sentence stays below the plan site on at least 4 of 6, and may move two sum or find duplicate | hit: 6 of 6 below (5-point Hill-1 1.00 everywhere); it moves nothing |
| Generated-token noise: more phrasing clusters per idea than the plan site, little idea change, breakage first by alpha 4 | clusters hit; little idea change missed (it moves four tasks); breakage by alpha 4 hit (invalid 0.67 to 1.00) |
| Early-block noise breaks first at alpha 2 or more | missed: it moves kth smallest and connected components at alpha 4 with pass kept |
| Late-block noise shows no movement at alpha 1 to 2 and breaks at alpha 4 | no movement hit (5 of 6; connected components weak); breakage missed (pass at least 0.87) |
| Energy-matched read position and condition variants break (pass below 0.8 times that of the twin) on at least 4 of 6 | missed: 0 of 6 for both |
| Boundary: pass below 0.5 at alpha 8 on at least 4 of 6 | hit: 6 of 6 (pass 0.00 everywhere) |
| Boundary: correct-only Hill peaks at alpha 2 to 4 | hit: alpha 2 on three tasks, alpha 4 on three |
| Boundary: no judge-confirmed novel algorithm | hit: 0 of 3,555 correct programs |
| Top-five push: avoidance below 0.5 at alpha 2 or less | hit: 0.00 at alpha 1 and 2 |
| Top-five push lands at the edge (bidirectional BFS) rather than the opposite side | not testable: no survivor program at any dose |

## 4.5 Same seed: temperature against plan noise (Qwen3-8B, question 1a-c of the request)

**Why.** The request asks three things about temperature: with a tight temperature and top-k 10, does the model produce one idea in many phrasings or many ideas; does loosening it multiply ideas or phrasings; and does the answer hold with the same seed. The run had answered the first two in iteration 1 and the third with thinking off in iteration 2, and had dropped the thinking-on arm twice. This experiment ran the seed-coupled design on the grid task, the shortest path task and kth smallest, with the thinking-on arm made non-droppable, and added plan noise as the second factor so that the same seed could be varied in temperature or in plan.

**Design.** Each task had 20 token seeds, one per template and slot, and each seed kept one noise stream across every condition (inverse-CDF sampling with common random numbers). The temperature block ran each seed at 0.3, 0.7, 1.0 and 1.3 (top-p 1) and with top-k 10 at 0.7. The plan noise block ran 6 fixed isotropic directions at alpha 1 and 2 and temperature 0.7 and 1.0. The change count of a task is the number of seeds whose algorithm differs across the conditions of a block; for plan noise it is averaged over the 20 three-direction subsets of the 6 directions. Thinking on ran in vLLM with per-request seeds (an exponential race coupling, confirmed by recomputing the first token in 32 of 32 requests), a 16k cap and the completion rate reported. *(Correction, iteration 5: the top-k 10 answer to the request with thinking off is Table 76 of iteration 2: with seeds coupled, the label changed in 0 of 30 seeds in 10 of 12 cells. The thinking-on arm of this experiment ran*

*without top-k (top-k off, Gumbel-max coupling through per-request seeds, temperatures 0.3, 0.7, 1.0 and 1.3), so the literal cell of the request, thinking on with top-k 10 and the same seed, was not run in iteration 4. It ran in iteration 5, Section 5.7.)*

**Temperature never changed the algorithm; plan noise changed it in most seeds on the shortest path task.** With thinking off, the algorithm stayed the same across all four temperatures in 20 of 20 seeds on all three tasks, and with thinking on in 20 of 20 seeds on the shortest path task (Table 150); the exact 95% upper bound on the per-seed change rate is 0.168 for each count of 0 in 20. One random plan direction at alpha 1 changed the algorithm in 17.25 of 20 seeds on the shortest path task, 4.55 on kth smallest and 2.5 on the grid task. *(Correction, iteration 5: these are not rates per direction. Each is the mean, over the 20 three-direction subsets of the 6 directions, of the number of seeds whose label differs anywhere among the no-noise arm and the three directions of the subset, a union over three directions. The single-direction counts of changed seeds at alpha 1 and temperature 1.0 are 7, 10, 15, 17, 3 and 8 of 20 on the shortest path task, 2, 0, 0, 7, 2 and 0 on kth smallest, and 0, 0, 1, 1, 3 and 0 on the grid task.)* The paired contrast per seed, plan noise change minus temperature change, pooled over the shortest path task and kth smallest (40 seeds), has a mean of 0.545 [0.41, 0.67] and a one-sided sign-flip p of 1.0e-5. That is the single entry of the experiment in the secondary family of the iteration. A within-seed permutation of conditions puts the observed mean above all 200 placebo means (maximum 0.155; $p = 0.005$).

**Table 150.** Seeds (of 20) whose algorithm changes across temperatures, thinking off and on, and across plan noise draws (mean over 20 three-direction subsets), by dose and temperature. Source: same seed experiment `results/c1seed.json`.

| Task | Temperature, thinking off (4 temperatures) | Temperature, thinking on (4 temperatures) | Plan noise alpha 1, T 0.7 | Plan noise alpha 1, T 1.0 | Plan noise alpha 2, T 0.7 | Plan noise alpha 2, T 1.0 | Correct only, alpha 1, T 1.0 |
|---|---|---|---|---|---|---|---|
| shortest path | 0 | 0 | 17.70 | 17.25 | 18.45 | 19.05 | 10.25 |
| kth smallest | 0 | not run | 4.40 | 4.55 | 10.30 | 10.60 | 4.05 |
| grid | 0 | 0 of 10 seeds (reduced arm) | 1.50 | 2.50 | 8.95 | 8.70 | 1.45 |

*[Figure fig_seed_counts: generated in the report phase, not in this draft]*

The preregistered verdict is PARTIAL: the temperature, phrasing and coupling predictions hold, and the plan noise prediction (at least 5 of 20 seeds) fails on kth smallest, at 4.55. The plan prescribed the unit of the bench forecast experiment for kth smallest, which is 1/1.30 of the unit the screen used, so alpha 1 here is a weaker push than alpha 1 of the screen; at alpha 2 kth smallest reaches 10.6, and under a Gumbel-max coupling 5.8. Counted over correct programs or with invalid outputs as a label, the temperature count stays 0 on every task; with top-k 10 added as a fifth condition it stays 0; with min-p 0.1 at temperature 2.0 added it becomes 1 on the shortest path task and kth smallest (one Bellman-Ford/SPFA and one quickselect program out of 20) and stays 0 on the grid task.

**Temperature changes phrasings, not ideas.** Phrasings per idea grow with temperature (Table 151): the pooled slope of log phrasings per idea on temperature is 0.64, 0.70 and 0.81 at Jaccard thresholds 0.7, 0.8 and 0.9, each 95% interval excluding 0. Almost all of the growth is on the grid task (3 to 13 clusters at threshold 0.8). The shortest path task barely moves (1 to 2) and kth smallest does not move (2 at every temperature). At alpha 1 and temperature 0.7, the rarefied

Hill-1 of correct ideas is 1.87 on the shortest path task, 1.31 on kth smallest and 1.15 on the grid task, while top-k 10 gives 1.00 on all three. Plan noise lands on Bellman-Ford/SPFA on the shortest path task (52 of 60 off-default programs, 8 other), on quickselect on kth smallest (11 of 11) and on DFS on the grid task (5). It costs correctness on the shortest path task: pass 0.63 at alpha 1 and temperature 1.0 and 0.42 at alpha 2, against 1.00 without it. Once a seed has been nudged off the default, temperature can act on the idea: under the escaping direction, temperature 0.7 against 1.0 changes the label in 5 of 20 seeds on the shortest path task and 2 of 20 on kth smallest, against 0 of 20 without plan noise.

**Table 151.** Phrasing clusters per idea at Jaccard threshold 0.8 by temperature (20 programs per cell, thinking off, no plan noise). Source: same seed experiment `results/phrasing.json`.

| Task | T 0.3 | T 0.7 | T 1.0 | T 1.3 |
|---|---|---|---|---|
| shortest path | 1 | 2 | 2 | 2 |
| kth smallest | 2 | 2 | 2 | 2 |
| grid | 3 | 10 | 12 | 13 |

**Thinking on.** All 100 shortest path traces completed within the cap (Table 152). Every final program and every first approach named was Dijkstra at every temperature, and every program passed. A repeat at temperature 0.7 diverged byte for byte in all 20 traces (first divergence at token 34 to 653), because this GPU generation lacks batch invariance, yet the final label and the first approach named agreed in 20 of 20; the temperature count of 0 is read against a noise floor of 0. A judge naming the first approach of 20 traces agreed with the lexicon on 20 of 20. In the reduced grid arm (10 seeds) all 40 traces completed (median 5.2k to 6.2k thinking tokens, maximum 9.0k), and every final program and first approach was BFS, a temperature count of 0 of 10 (exact 95% upper rate 0.31). With thinking on, plan noise at alpha 2 and temperature 0.7 (2,048-token cap, the three directions with the highest thinking-off off-default share) moved the first approach named in 1, 0 and 0 of 10 seeds, short of the committed expectation of at least 3 of 10 for some direction. Once the model thinks, the prefill nudge no longer redirects the first idea it names.

**Table 152.** Thinking on, shortest path task (20 seeds per temperature, 16k cap, vLLM). Source: same seed experiment `results/thinking.json`.

| Temperature | Completed | Median thinking tokens (maximum) | Final label | First approach named |
|---|---|---|---|---|
| 0.3 | 20 of 20 | 4,097 (7,381) | Dijkstra 20 | Dijkstra 20 |
| 0.7 | 20 of 20 | 3,952 (6,327) | Dijkstra 20 | Dijkstra 20 |
| 1.0 | 20 of 20 | 4,320 (8,808) | Dijkstra 20 | Dijkstra 20 |
| 1.3 | 20 of 20 | 4,020 (12,612) | Dijkstra 20 | Dijkstra 20 |
| 0.7, repeat | 20 of 20 | 4,001 (7,427) | Dijkstra 20 | Dijkstra 20 |

**Checks.** The engine check reran the kth smallest and shortest path cells of the screen with their twins and reproduced 200 of 200 programs token for token (size-k heap 19, quickselect 16, full sort 15; Dijkstra 26, Bellman-Ford/SPFA 24). Filler repeats in other batches were byte-identical in 60 of 60, and the engine-check slots matched the rows of the temperature block at 1.0 in 40 of 40. Under a Gumbel-max replicate the temperature count stays 0 on both tasks, plan noise changes 16.05 of 20 seeds on the shortest path task and 5.8 on kth smallest, and the pooled contrast is 0.546 ($p = 1.0e\text{-}5$). With min-p 0.1 at temperature 2.0, the shortest path task wrote Dijkstra 19 and Bellman-Ford/SPFA 1, kth smallest full sort 19 and quickselect 1, and the grid task BFS 20. The condition-shuffle placebo rejected at p below 0.05 in 2% of 200 runs, and the placebo of the independent rederivation in 10.5%; over 1,440 placebo runs the rejection rate is about 6%. The only correct program labelled other was a naive Dijkstra, a labeller miss; no novel algorithm appeared. The rederivation, sharing no code with the analysis, matched 33 of 33 checks, and the commitment

precedes the first decisive row. An 8-bit cross-check of the two engines on the shortest path twin gave Dijkstra 50 of 50 on each, a divergence of 0 bits, which is weak by construction because the cell is all Dijkstra.

## 4.6 The plan subspace: does the geometry of forced prompts explain which pushes move the algorithm? (Qwen3-8B)

**Why.** The site experiment located the choice in the task-specification states at early to middle blocks, but a site is not a representation. The subspace claim gave the internal account the request asks for a testable form: if the algorithm choice is represented along the directions that separate the per-algorithm centroids, a random push should change the algorithm to the extent that it moves the state along them. This experiment built the subspace from forced prompts alone and asked whether it ranks 40 fresh random directions by their effect, whether removing the in-subspace component of a push removes the effect, and whether a push aimed at the runner-up centroid crosses at the predicted distance.

**Design.** For each of 15 tasks the plan subspace at blocks 12 to 35 was the span of the centroid differences of the task-specification states (mean over the specification positions) under 30 forced paraphrases per algorithm (10 forced templates by 3 names), and a masked twin used 30 leak-checked descriptions per algorithm. The projection share of a direction is its squared norm in the subspace over its total, averaged over blocks. Forty never-run directions at alpha 1 and 2, with $N = 10$ each and passers topped up to $N = 30$, ran on the six movable tasks; 8 directions at both doses ran on the other nine, including cycle detection on its ban-free unit. Every predictor was committed by hash before the first population sample. The run sampled 13,500 programs in 675 batches of 20 over 2.51 GPU hours, labelled by runtime tracer and AST rules with a blind judge on 752 disagreements; 4 disputed and 97 invalid rows were excluded from shares.

**The subspace is real and reliable.** At block 18 the split-half principal-angle cosine between two halves of the paraphrases is 0.89 to 0.97 on all 15 tasks, against 0.02 to 0.03 for random subspaces of equal dimension, and every centroid difference has a split-half cosine of at least 0.74. Named and masked differences agree only moderately (cosine 0.12 to 1.00 at block 18), so the subspace carries a lexical component. The iteration-3 ban displacement projects strongly onto it (projection share 0.02 to 0.77; 0.08 to 0.39 on the six movable tasks), while a random direction has a share near k/4096, about 0.001. The committed plausibility ratio, the expected in-subspace part of a random push over the distance from the default to the runner-up centroid, is 0.10 to 0.29 on 14 tasks and 0.51 on the grid task: a linear plan account predicted before sampling that random pushes would almost never cross to the runner-up through the subspace.

**The projection share does not predict which pushes move the algorithm.** The preregistered test, a within-task Spearman correlation of at least 0.4 with p below 0.01 on at least 4 of 6 tasks, failed on all six, and so did the masked variant (Table 153). The pooled within-task correlation is 0.03 [-0.03, 0.09], and the correlation of every task lies inside its random-subspace null. The null is not for want of movers: 34 to 37 of 40 directions moved the algorithm at least once on kth smallest, the shortest path task and connected components. Nor is it noise. Splitting the first 10 samples of each direction by template, the two halves rank the 40 directions alike (Spearman 0.80 on kth smallest, 0.81 on the shortest path task, 0.76 on two sum, 0.90 on connected components; projected to full length by the Spearman-Brown formula, 0.86 to 0.95), and alpha 1 and alpha 2 rank them alike (0.61 to 0.72). Which direction moves the algorithm is a stable property of the direction, with a ceiling near 0.9 for any predictor, and the subspace share explains none of it. The only signal came from the signed component toward the runner-up on connected components (0.32

to 0.33, unadjusted p near 0.02), and the within-class control did as well (0.37).

**Table 153.** Correlation of the off-default share of each direction (first 10 samples, averaged over alpha 1 and 2) with the committed predictors, over 40 directions per task (Spearman rho, permutation p in brackets; κ is the committed plausibility ratio). Source: plan subspace experiment `results/c2a.json`.

| Task | Directions that moved | Projection share | Masked share | Signed component | Signed toward runner-up | Realised share | Wrong subspace | Within-class | Gain | Random-subspace null p95 | Partial, given gain | κ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| kth smallest | 34 of 40 | 0.14 (0.21) | 0.11 (0.26) | -0.22 (0.91) | -0.16 (0.84) | 0.11 | 0.03 | 0.20 | -0.10 | 0.26 | 0.12 | 0.13 |
| shortest path | 37 of 40 | 0.09 (0.29) | -0.05 (0.62) | 0.01 (0.49) | -0.03 (0.57) | 0.15 | -0.30 | -0.05 | -0.04 | 0.25 | 0.08 | 0.20 |
| two sum | 10 of 40 | 0.08 (0.33) | 0.03 (0.42) | 0.09 (0.29) | 0.21 (0.10) | -0.24 | 0.05 | 0.07 | -0.16 | 0.27 | 0.05 | 0.14 |
| connected components | 36 of 40 | -0.05 (0.62) | -0.03 (0.57) | 0.32 (0.026) | 0.33 (0.021) | -0.14 | -0.12 | 0.37 | -0.06 | 0.26 | -0.05 | 0.14 |
| find duplicate | 3 of 40 (underpowered) | -0.01 (0.52) | -0.04 (0.60) | -0.06 (0.63) | -0.05 (0.62) | -0.02 | -0.17 | -0.11 | -0.15 | 0.25 | -0.00 | 0.21 |
| grid | 22 of 40 | -0.06 (0.64) | 0.05 (0.37) | 0.18 (0.13) | 0.00 (0.49) | -0.09 | -0.07 | -0.04 | 0.25 | 0.25 | -0.01 | 0.51 |

**Escape, confirmed on fresh directions.** Random pushes at the plan site buy correct alternative algorithms on four tasks, rarely on the grid task and two sum at alpha 1, and never on the remaining nine (Table 154). Every twin wrote its default in all samples (top-k frequent 90%). The landing is a property of the task, not of the direction, at a given site (qualified in iteration 5: on find duplicate the landing depends on the site; generated-token noise and the energy-matched read position land on Floyd cycle detection, the single plan-site escape of the screen on sort and adjacent check): on every movable task the excess-weighted landing of the 40 fresh directions reproduces the landing of the screen. The single screen escape of find duplicate did not replicate in 40 fresh directions at either dose, so find duplicate is not movable on fresh draws, and cycle detection, pushed on its ban-free unit, did not move in 16 cells. Population programs stayed mostly correct (pass over all population and top-up programs 0.98 on kth smallest, 1.00 on two sum, 0.99 on find duplicate, 0.97 on connected components), with the path tasks the exception (0.84 and 0.79): 255 of 422 pushed Bellman-Ford/SPFA programs and 55 of 123 pushed DFS programs passed.

**Table 154.** Escape fractions on 40 fresh directions per dose (confirmed at N = 30 by an exact binomial test; 95% interval), with the excess-weighted landing of this run and of the iteration-3 screen. Source: plan subspace experiment `results/escape.json`.

| Task (default) | Escape, alpha 1 [95%] | Escape, alpha 2 [95%] | Push landing, this run | Screen landing |
|---|---|---|---|---|
| kth smallest (full sort) | 12/40 = 0.30 [0.18, 0.45] | 25/40 = 0.63 [0.47, 0.76] | quickselect 0.66, size-k heap 0.31, heapify and pop 0.03 | quickselect 0.64, size-k heap 0.35 |
| connected components (union find) | 15/40 = 0.38 [0.24, 0.53] | 29/40 = 0.73 [0.57, 0.84] | DFS 0.83, BFS 0.17 | DFS 1.00 |

| | | | | |
|---|---|---|---|---|
| shortest path (Dijkstra) | 7/40 = 0.18 [0.09, 0.32] | 8/40 = 0.20 [0.10, 0.35] | Bellman-Ford/ SPFA 0.99 | Bellman-Ford/ SPFA 1.00 |
| two sum (hash map) | 0/40 [0, 0.09] | 5/40 = 0.13 [0.05, 0.26] | brute force 1.00 | brute force 1.00 |
| grid (BFS) | 2/40 = 0.05 [0.01, 0.17] | 5/40 = 0.13 [0.05, 0.26] | DFS 1.00 | DFS 1.00 |
| find duplicate (hash set) | 0/40 [0, 0.09] | 0/40 [0, 0.09] | none | sort and adjacent check 0.79 (1 of 3 directions) |
| substring search, Fibonacci, longest increasing subsequence, merge intervals, top-k frequent, nonoverlapping intervals, topological order, first and last position, cycle detection | 0/8 each | 0/8 each (one top-k frequent screen passer not confirmed) | none | none |

**The in-subspace component is not needed.** Up to four confirmed escaping directions per task (20 in all) were resampled on the same slots in three forms: with the subspace component removed, with only that component amplified to the full norm of the push, and with the subspace component of the realised displacement clamped to 0 at every edited block. Table 155 gives the effect ratio of each form against the original push. On every task with clear escapes, removing the subspace component leaves the effect intact (pooled median ratio 1.00 [0.88, 1.08]), and clamping leaves it intact on kth smallest, two sum and connected components while removing about a third of it on the path tasks. The in-subspace component alone, at the same norm, also escapes for 9 of 20 directions, all four on kth smallest. On the most movable tasks a push of this size flips the algorithm along many directions, inside the subspace or outside it. This is the generic-sensitivity account the plan named as the main confound.

**Table 155.** Effect ratio (arm share minus twin share, over original push share minus twin share) of the decomposed pushes on the same common-random-number slots (N = 20 per arm). Source: plan subspace experiment `results/decomposition.json`.

| Task | Committed reading | Median ratio, subspace part removed [CI] | Median ratio, subspace part only | Subspace part escapes alone | Median ratio, realised subspace part clamped |
|---|---|---|---|---|---|
| kth smallest | subspace not needed; sufficient when amplified | 1.00 [1.00, 1.00] | 0.77 | 4 of 4 | 1.00 |
| shortest path | subspace not needed | 1.00 [0.74, 1.40] | 0.09 | 1 of 4 | 0.63 |
| two sum | subspace not needed | 0.81 [0.60, 1.25] | 0.70 | 2 of 4 | 1.04 |
| connected components | subspace not needed | 1.00 [0.90, 1.02] | 1.00 | 2 of 4 | 1.00 |
| grid | mixed | 0.78 [0.27, 1.50] | 0.06 | 0 of 4 | 0.64 |
| pooled (20 directions) | subspace not needed | 1.00 [0.88, 1.08] | 0.32 | 9 of 20 | 1.00 |

**A targeted push works on one task, and there the geometry is exact.** A push along the direction from the default centroid to the runner-up centroid moved the mean specification state by

a fraction s of their distance, with s = 0.5 the predicted midpoint; its control was a random direction with the same per-position magnitude on the same slots. The committed rule (Hill-1 of at least 1.8 within 5 pass points at the midpoint on at least 4 of 6 tasks, at no more than half the median escaping random norm) was met on 1 of 6 tasks, and the targeted push was indistinguishable from the matched random push on 5 of 6 (Table 156). Connected components is the exception. Half the distance to the DFS centroid, 0.15 ban units or a tenth of a random push, turned 48% of programs into correct DFS against 0% for the matched random push (24 slots against 0; McNemar p = 1.2e-7), and the empirical crossing at 0.59 sits at the predicted midpoint. On kth smallest the crossing is at 0.85 of the distance, and the masked axis is stronger than the named one at the midpoint (0.43 against 0.08). On the other four tasks the midpoint does nothing; at 4 ban units the same axis gives 100% brute force on two sum (77% correct), 100% DFS on the grid task (20% correct) and 63% Bellman-Ford/SPFA on the shortest path task (none correct).

**Table 156.** The targeted push along the default-to-runner-up axis: off-default share at s = 0.25, 0.5 and 1.0, the matched random push, paired flips and the share difference, the 50% crossing, the masked axis at the midpoint, and the sweep at alpha 4 and 8 (pass in brackets). Source: plan subspace experiment `results/c2c.json`.

| Task (runner-up) | Off default at s = 0.25 / 0.5 / 1.0 | Random, same magnitude (s = 0.5) | Flips targeted only / random only; difference [95% CI] | 50% crossing s [CI] | Masked axis (s = 0.5) | Sweep alpha 4 / 8 (pass) |
|---|---|---|---|---|---|---|
| connected components (DFS) | 0.07 / 0.48 / 0.93 | 0.00 | 24 / 0; +0.48 [0.34, 0.62] | 0.59 [0.50, 0.70] | 0.63 | 1.00 (0.97) / 0.40 (0.00) |
| kth smallest (quickselect) | 0.03 / 0.08 / 0.73 | 0.02 | 3 / 0; +0.06 [0.00, 0.12] | 0.85 [0.76, 0.95] | 0.43 | 1.00 (1.00) / 1.00 (0.00) |
| shortest path (Bellman-Ford/ SPFA) | 0.00 / 0.02 / 0.07 | 0.00 | 1 / 0; +0.02 [0.00, 0.06] | 1.69 [1.16, 6.7] (extrapolated) | 0.03 | 0.63 (0.00) / 0.00 (0.00) |
| two sum (brute force) | 0 / 0 / 0 | 0 | 0 / 0 | none | 0 | 1.00 (0.77) / 0.08 (0.00) |
| find duplicate (sort and adjacent check) | 0 / 0 / 0 | 0 | 0 / 0 | none | 0 | 0.03 (0.00) / 0.20 (0.00) |
| grid (DFS) | 0.00 / 0.04 / 0.00 | 0.00 | 2 / 0; +0.04 [0.00, 0.10] | not reached (extrapolated 5.7) | 0.03 | 1.00 (0.20) / 0.53 (0.00) |

**Knob displacements do not project in proportion to their effect.** Over 270 knob points on 14 tasks (nonce bans, the novelty instruction, modal bans, band patches, iteration-3 bans and random pushes), the pooled within-task Spearman correlation of the projection share of a knob with its recorded off-default share is 0.27, below the committed 0.5, and its advantage over the displacement norm (0.12) has a paired-bootstrap interval of -0.02 to 0.31. Within task it ranges

from -0.29 (Fibonacci) to 0.84 (connected components). The class means show the dissociation: modal bans project strongly (mean share 0.23, off-default share 0.40), nonce bans weakly (0.07; 0.00) and the novelty instruction in between (0.15; 0.17), while the iteration-3 random pushes barely project (0.015) yet move the algorithm as often as the novelty instruction (0.17). An exploratory analysis, written after sampling, found that the signed progress of the realised displacement toward the runner-up centroid correlates with the effect on most tasks and blocks (up to 0.57 on connected components at block 31, 0.47 on two sum at block 17, 0.30 on the shortest path task at block 16, 0.26 on kth smallest at block 25, 0.48 on the grid task at block 35), while the unsigned share hovers near 0 (-0.35 to 0.21 at block 18). These are 144 uncorrected correlations, and they point at a direction effect that the committed unsigned predictor cannot see.

**Push landings forecast ban landings.** On the five gated iteration-3 bench bans, already seen and so scored descriptively, the push-landing forecast of this run was defined on 3 events and scored a mean log score of -0.76, against -3.46 for association and -3.21 for proximity (Table 157). The opening ladder did better than both frozen families (-2.39) and worse than either push forecast. Connected components is the departure: the ban lands on BFS (83%) while pushes land on DFS, which the route landings of Section 4.2.2 explain. Across 14 tasks, the escape fraction of this run correlates with the avoidance of the bench ban at Spearman 0.86: the five movable tasks have avoidance 0.86 to 1.00 and the nine immovable ones 0.00 to 0.79. The opening margin does not track avoidance (0.19). *(Correction, iteration 5: the comparison above sets the push forecast's mean over its 3 defined events against the association mean over 5. On matched events, on the 3 events where the push forecast is defined (kth smallest, two sum, connected components) it scores -0.76 against -3.11 for association and -2.92 for proximity; over all 5 gated events, with an undefined forecast scored as uniform, it scores -1.37 against -3.46. The -1.37 coincides at two decimals with the route replication cell of the screen push in Table 157 (-1.369), a different quantity. Table 171 gives the matched means.)*

**Table 157.** Forecasts of the five gated iteration-3 bench ban landings (mean log score of the realised attempts not banned under the smoothing of the bench forecast experiment; replication on the full-ban arm of the route experiment; top-1 hits). Source: plan subspace experiment `results/c3b.json`.

| Forecast | Events defined | Mean log score (bench ban) | Route full-ban replication | Top-1 hits |
|---|---|---|---|---|
| push landing of the confirmed escapes of this run | 3 (kth smallest, two sum, connected components) | -0.76 | -0.71 | 2 of 3 |
| push landing of the iteration-3 screen | 4 | -1.48 | -1.37 | 2 of 4 |
| opening ladder | 5 | -2.39 | -2.24 | 2 of 5 |
| association | 5 | -3.46 | -3.35 | 1 of 5 |
| proximity | 5 | -3.21 | -3.20 | 1 of 5 |

**Geometry at the plan site.** At blocks 18 and 23, metric MDS of the geodesic distances among the forced-prompt states (PCA to 50 dimensions, $k = 10$) has stress-1 of 0.06 to 0.25 with trustworthiness 0.96 to 0.99; Isomap separates the algorithms into tight clusters (trustworthiness 0.84 to 0.98) but with high stress-1 (0.58 to 0.83), because the neighbour graph is clustered. Persistent loops exceed their coordinate-permutation null only on the shortest path task at block 18 and two sum at blocks 18 and 23, and a planted circle at matched size and noise was detected in every case. The figure below plots the block-18 Isomap of connected components and kth smallest in activation coordinates, with each forced-prompt centroid sized by its implemented count under the push population, and with the mean projected state of the twins, the bans, the targeted pushes

and the population pushes grouped by the algorithm they produced. The pushed states stay near the twin region, apart from every forced-prompt centroid, and pushes that changed the algorithm sit among pushes that did not (the group means lie within about one unit of each other on both tasks). Only the targeted push travels toward the runner-up. The two-dimensional geometry of the forced prompts does not separate escaping from non-escaping pushes.

*[Figure fig_plan_geometry: generated in the report phase, not in this draft]*

**No novel algorithm.** Of 10,651 correct programs from push arms, 4 carried the label other. Two blind judges and a tie-breaker, with a manual pass, classed them as a catalogue variant (a two sum hash map), a labeller miss (a Dijkstra), a plain linear scan for first and last position and a random guess-and-count loop for find duplicate; their nearest known solutions had AST Jaccard of at most 0.22. No novel algorithm appeared (rule-of-three bound 3/10,651, 0.03%).

**Table 158.** Committed rules and outcomes. Source: plan subspace experiment `README.md`, `predictions/c2_commit.json`.

| Committed rule | Outcome |
|---|---|
| Subspace pass: Spearman of at least 0.4 with p below 0.01 on at least 4 of 6 tasks, and the masked variant on at least 3 of 6 | no: 0 of 6 and 0 of 6 |
| Subspace disconfirmed: Spearman below 0.2 on most tasks | yes: 6 of 6 (pooled 0.03 [-0.03, 0.09]) |
| Subspace not needed if the median ratio with the subspace part removed is at least 0.7 with a lower bound of at least 0.4 | yes on kth smallest, the shortest path task, two sum, connected components and pooled; grid mixed; find duplicate not evaluable |
| Subspace sufficient when amplified if the in-subspace part alone passes the escape rule at N = 20 | kth smallest (4 of 4); 9 of 20 directions overall |
| Targeted pass: Hill-1 of at least 1.8 within 5 points at the midpoint on at least 4 of 6, at no more than half the median escaping norm | no: 1 of 6 (connected components, 0.15 ban units) |
| Targeted disconfirmed: targeted minus random interval includes 0 on at least 4 of 6 | yes: 5 of 6 |
| Knob projections: pooled Spearman of at least 0.5 and projection beats the norm | no: 0.27; difference interval [-0.02, 0.31] |
| The frozen movable set is movable at alpha 2 or less | 5 of 6: find duplicate is not movable (0 of 80 cells) |
| Departure rule: where the top-1 of the ban differs from the push top-1, the direct route share is at least 0.25 | connected components: pooled direct share 0.74 (pooled, not per task) |
| No correct novel algorithm, claimed only at 0 verdicts | 0 among 10,651 correct push programs |

**Checks.** Every GPU unit test passed: 20-row decoding is independent of row position, batch-1 prefill is bit-identical on repeat, a zero-magnitude edit reproduces the twin token for token, the realised displacement matched its target within 0.07% over 3,480 prefills, and clamping left at most 6.7e-4 of the displacement in the subspace. The engine reproduced the screen exactly on 180 of 180 slots. Every bench labeller recovered all its references. An independent recomputation matched every headline, and each placebo (projection shares shuffled across directions, among others) behaved as a null should.

## 4.7 Movability and compliance in a held-out model family (gemma-3-4b-it)

**Why.** The movability claim was frozen on Qwen3-8B data that had also produced it: in the screen every movable task obeyed its ban and no immovable task did. Only a family never run in this

study could test it. This experiment froze the screen rule, measured movability in gemma-3-4b-it [76] without any ban text, committed every per-task prediction by hash chain before any ban prompt was built (a grep audit of the 460 prompts fed in the ban-free stage found no ban frame), and then sampled 1,500 modal ban programs.

**Design.** For each of 15 tasks (the 12 bench tasks, the grid and shortest path tasks, and cycle detection), the model wrote 50 neutral twin programs, and 8 fixed isotropic directions at alpha 1 and 2 were added to the task-specification states of blocks 11 to 22, the middle third of its 34 blocks, during prefill. A direction escaped by the rule of Section 4.1, confirmed at $N = 30$ by an exact binomial test, and a task predicted to obey its modal ban when its escape fraction was above zero. The ban unit had to change before the commitment. In Gemma the five largest channels carry 99% of the squared residual norm at the specification positions, against 40% in Qwen3-8B, so the committed unit erased the task in a 24-row calibration probe; an outlier-excluded unit was adopted before the first commitment, and after the bans the measured ban displacement turned out to be 1.29 times it (0.92 to 2.1 by block). The rival predictors, the plan noise and temperature arms, the novelty instruction and four landing forecasts ran in the same engine under the same common random numbers.

**The movability claim is inconclusive in the held-out family.** Movability relates to compliance in the predicted direction but does not separate the tasks well enough (Table 159). The confusion matrix is 3, 3, 1, 8 (predicted obey or fail against observed): balanced accuracy 0.74 [0.40, 0.95], Fisher $p = 0.24$ (0.71 under the committed three-claim bound), Spearman 0.54 (one-sided permutation $p = 0.025$) and Kendall 0.47. The rule needed balanced accuracy of at least 0.8, Fisher p below 0.05 and Spearman of at least 0.5. Four tasks broke the prediction: kth smallest obeyed although no direction moved it, and substring search, connected components and longest increasing subsequence were movable but did not obey. A task-permutation placebo has mean balanced accuracy 0.50 (probability 0.13 of reaching the observed value), and leaving one task out gives 0.70 to 0.86.

**Table 159.** Held-out family: escape fraction over 8 directions, prediction, catalogue-only modal ban avoidance ($N = 100$), outcome, and the top labels under the ban. Source: held-out family experiment `results/c3a.json`, `README.md`.

| Task (Gemma default) | Escape fraction | Predicted | Ban avoidance | Observed | Labels under the ban (top) |
|---|---|---|---|---|---|
| grid (DFS; corrected in iteration 5: NO DEFAULT, twin BFS 25 and DFS 25; DFS was frozen from a 10-sample pilot with correct BFS 4 and DFS 5, attempts 5 and 5) | 0.125 | obeys | 0.83 | obeys, but against no default | BFS 83, DFS 17 |
| shortest path (Dijkstra) | 0.25 | obeys | 0.96 | obeys (labeller caveat) | Bellman-Ford/ SPFA 96, mostly array-based Dijkstra |
| kth smallest (full sort) | 0 | fails | 0.63 | obeys (miss) | size-k heap 61, full sort 37 |
| two sum (hash map) | 0.25 | obeys | 0.59 | obeys | brute force 59, hash map 41 |
| substring search (naive) | 0.25 | obeys | 0.19 | fails (miss) | naive 76, built-in find 13, Rabin-Karp 6 |
| connected components (DFS) | 0.125 | obeys | 0.00 | fails (miss) | DFS 100 |

| | | | | | |
|---|---|---|---|---|---|
| longest increasing subsequence (patience sorting with bisect) | 0.125 | obeys | 0.00 | fails (miss) | patience sorting 100 |
| find duplicate (hash set) | 0 | fails | 0.34 | fails | other 38, hash set 28, sort 21 |
| Fibonacci, nonoverlapping intervals, topological order, merge intervals, top-k frequent, cycle detection | 0 each | fails | 0.00 each | fails | default 87 to 100 |
| first and last position (binary search, flagged) | 0 | fails | 0.00 | fails | binary search 99 |

Every sensitivity stays inconclusive (Table 160) *(correction, iteration 5: except the preregistered sensitivity that measures compliance as the drop in the default share, which disconfirms the claim: balanced accuracy 0.57, Fisher p 1.0, Spearman 0.39; see below and Table 169)*. The shortest path task carries one correct prediction on a labelling failure found after the bans. The frozen labeller, gated on heap-based Dijkstra in Qwen3-8B, reads the array-based Dijkstra of Gemma (a linear minimum scan over unvisited nodes with a visited array) as Bellman-Ford/SPFA or unknown, and the inline judge followed it. A second blind judge re-judged all 374 programs labelled Bellman-Ford/SPFA and called 284 of them Dijkstra, including all 219 that match a deterministic structural rule. Under mechanism labels the ban on Dijkstra mostly makes Gemma drop the heap and keep Dijkstra: avoidance falls from 0.96 to 0.11, and the escape fraction from 0.25 to 0.125. The frozen labels stay primary as committed, and the verdict is inconclusive either way. A labeller gated on one family needs a style re-gate on the next.

**Table 160.** Sensitivities of the movability test in the held-out family. Source: held-out family experiment `results/c3a.json`, `results/posthoc_t2_relabel.json`.

| Variant | Tasks | Balanced accuracy | Fisher p | Spearman |
|---|---|---|---|---|
| primary | 15 | 0.74 | 0.24 | 0.54 |
| without cycle detection, or without the flagged first and last position | 14 | 0.73 | 0.24 | 0.52 |
| without the grid and shortest path tasks | 13 | 0.61 | 1.00 | 0.31 |
| correct-only avoidance | 15 | 0.74 | 0.24 | 0.51 |
| ban N = 180 with extra slots | 15 | 0.74 | 0.24 | 0.54 |
| without the tasks flagged by the label audit (shortest path, top-k frequent) | 13 | 0.68 | 0.51 | 0.39 |
| compliance as the drop in the share of the default | 15 | 0.57 (disconfirmed) | 1.00 | 0.39 |
| post hoc: shortest path relabelled by a second blind judge | 15 | 0.67 | 0.53 | 0.44 |
| post hoc: shortest path relabelled by the structural rule | 15 | 0.74 | 0.24 | 0.46 |
| without tasks that have no default (added in iteration 5; the grid task) | 14 | 0.70 | 0.51 | 0.48 |

*(Correction, iteration 5: the row on the drop in the default share is the preregistered sensitivity, and under it the claim is DISCONFIRMED, not inconclusive. The two compliance measures are separated by two tasks: the grid task, whose twin is tied (BFS 25, DFS 25), so that 'obeys' (catalogue-only avoidance 0.83) is measured against a non-default baseline of 0.50 and the default share drops by only 0.33; and find duplicate, where 38 failing programs labelled other count as leaving the default under the drop measure (catalogue-only avoidance 0.34, drop 0.72). Two of the three true positives behind the Spearman of 0.54 are compromised, the grid task by its tied baseline and the shortest path task by the mislabelled array-based Dijkstra; without both, balanced accuracy is 0.61 and Spearman 0.31 (one-sided permutation p*

*0.144). Table 169 gives every variant.)*

**Asking in words predicts compliance better than pushing.** The off-default share under the novelty instruction tracks ban avoidance at Spearman 0.77, against 0.54 for the escape fraction (Table 161); the paired task-bootstrap difference is -0.24 [-0.82, 0.23], and the partial correlation of the escape fraction given the novelty share is 0.52. This matches the prior in the field that prompting is the baseline for an activation-level method to beat [34].

**Table 161.** Predictors of catalogue-only ban avoidance in the held-out family (Spearman rho, oriented so that positive means the predicted direction). Source: held-out family experiment `results/rivals.json`.

| Predictor | Spearman rho |
|---|---|
| novelty instruction off-default share | 0.77 |
| escape fraction (the frozen screen) | 0.54 |
| plan noise alpha 2 off-default share | 0.51 |
| plan noise alpha 1 off-default share | 0.47 |
| best forced-prompt success rate | 0.35 |
| temperature 1.3 off-default share | 0.34 |
| twin modal share | 0.24 |
| opening margin | -0.19 |

**Landings.** On the 4 gated events, the push-landing forecast scored a mean log score of -0.11 with 4 of 4 top-1 hits, against -1.20 for proximity, -1.77 for association and -3.66 for the opening ladder (Table 162). The contest needed at least 3 events on which the push forecast and association disagree, and only 2 existed (kth smallest: size-k heap against quickselect; two sum: brute force against two pointers), so the landing test is inconclusive by rule, with the push forecast ahead by 2.81 nats on those two.

**Table 162.** Landing forecasts on the 4 gated held-out events (grid, shortest path, kth smallest, two sum; the shortest path event carries the labeller caveat). Source: held-out family experiment `results/c3b.json`.

| Forecast | Mean log score | Top-1 hits |
|---|---|---|
| push landing | -0.11 | 4 of 4 (corrected in iteration 5: of the 4, 2 disagree with association, kth smallest and two sum; the other 2 are compromised, the grid task having no default and the shortest path landing being mostly array-based Dijkstra) |
| proximity | -1.20 | 3 of 4 |
| association | -1.77 | 2 of 4 |
| opening ladder | -3.66 | 0 of 4 |

**The price claim, replicated descriptively.** In Gemma too, plan noise raises correct-only diversity within 5 pass points on tasks where temperature does not (Table 163); the novelty instruction diversifies some tasks more, at a large pass cost on others (find duplicate pass 0.08, substring search 0.60). The Hill-1 of the twin is 1.0 on these tasks and 1.31 on substring search.

**Table 163.** 5-point Hill-1 in the held-out family (randomisation hull, 5 pass points). Source: held-out family experiment `results/c1_prices.json`.

| Task | Plan noise | Temperature 1.3/1.6 | Novelty instruction |
|---|---|---|---|

| | | | |
|---|---|---|---|
| shortest path (labeller caveat added in iteration 5: the frozen labeller reads array-based Dijkstra as Bellman-Ford/SPFA; plan noise 1.62 rests on 5 and 12 programs labelled Bellman-Ford/SPFA at alpha 1 and 2, and the novelty instruction 1.96 on 29) | 1.62 [1.37, 1.87] | 1.00 | 1.96 |
| substring search | 1.64 | 1.32 | 1.42 |
| connected components | 1.58 | 1.00 | 1.10 |
| kth smallest | 1.38 | 1.10 | 1.61 |
| longest increasing subsequence | 1.25 | 1.00 | 1.00 |

**Other observations and checks.** The fourth template is again the weakest: its avoidance is 0.15 on the grid task against 1.0 for the other templates, and 0.05 on kth smallest. On first and last position the twin writes a linear scan 86% of the time, yet the ban `Do not use a hand-written binary search` makes it write binary search 99% of the time, an ironic rebound of the kind iteration 3 saw on kth smallest. On the shortest path task, 48 of 50 twin slots flip from the default to a survivor under the ban. The 34 correct programs labelled other were 27 degenerate programs from push arms and 7 catalogue variants, with no novel algorithm (rule-of-three bound 0.088). All 13 bench labellers reproduced their gates, the six escape counts of the screen were reproduced, the GPU tests passed, and an independent rederivation from raw samples matched every escape fraction, avoidance, balanced accuracy, Fisher p, Spearman correlation, rival correlation, landing log score and post hoc number with no mismatch; its placebos rejected at 1.7% (Fisher) and 4.9% (Spearman).

*[Figure fig_movability: generated in the report phase, not in this draft]*

## 4.8 Positioning: the iteration-3 and iteration-4 claims against published work

The positioning audit, dated 2026-10-21, checked 112 verbatim quotes from 74 references (101 against full text, 11 against venue abstracts; none mismatched, and an independent pass re-matched all 112). A saturation search over eight query families found 1,729 unique records, scored 63 on an occupancy rubric and found 44 adjacent, 19 irrelevant and none occupying. Two flags it was asked to raise did not fire: no paper measures test-checked, algorithm-level diversity under activation perturbation, and no paper predicts prompt-ban compliance from perturbation sensitivity. Table 92b sets the iteration-3 claims against their nearest neighbours and Table 92c the iteration-4 claims; each finding is paraphrased here and was verified against a quote. Several contemporaneous submissions under review, a workshop poster without an arXiv record and one workshop paper that the bibliography script could not retrieve are adjacent in the audit but are not cited by number.

**Table 92b.** Iteration-3 claims, their nearest published neighbours and what this run adds. Source: positioning audit `research_report.md`.

| Claim in this run | Nearest neighbour | Finding of the neighbour (paraphrased; verified against a quote) | What this run adds, or replication | Lane |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Snap-back under exposure bans (default 0.05 under the copy twin, 0.72 named, 0.82 unnamed; Llama present minor bans 0.72 to 0.93) | Lu et al. [23] | Under successive denial prompts, models copy previous solutions and constraint following decays | The replacement is the unconditional default of the model itself, measured against a copy twin, in two families | adjacent |
| (same) | Yoon et al. [4] | Under factual negation, heads and neurons suppress the original answer and promote a favoured in-category candidate | Algorithm level, execution-checked labels; the promoted alternative is the unconditional default | closest analogue |
| (same) | Rana [7] | Mentioning a forbidden word primes it in most violations | The unnamed ban snaps back more (0.82) than the named (0.72), so naming is not needed | adjacent |
| Wording dominates compliance (template SD 2.0 logits; likelihood ratio 374; the fourth template fails on 10 of 12 tasks) | Sclar et al. [69] | Formatting changes alone move accuracy by up to 76 points | Replication in prohibition compliance, as a template random effect with a placebo | partly occupied |
| (same) | Heo et al. [58] | An input-embedding dimension predicts instruction following and tracks phrasing | Algorithm prohibitions on code; a variance-components estimate | adjacent |
| The ban is enforced at the opening of the body (forcing the opening of the default undoes it; a neutral `def` line does not) | Qi et al. [63] | Safety alignment acts mostly on the first few output tokens | Replication of shallow enforcement for a user-imposed prohibition, located at the first content token after a neutral signature | adjacent |
| (same) | Zhu et al. [32]; Andriushchenko et al. [79] | Hard code tokens sit at block openings; prefilling defeats refusals | Prefilling used as a measurement rather than an attack | supporting |
| Signature vetoes are respelled around (exact veto, the walk stalled on 6 of 6 tasks) | Biagiola et al. [62]; Park et al. [85] | Incomplete constraints push generation into low-probability regions and cost correctness; constrained decoding distorts the distribution | The algorithm survives a lexical veto by respelling, with a sham-veto control | adjacent |
| No correct novel algorithm (0 of 116 correct out-of-catalogue programs) | Zhang et al. [44]; Jangra and Muresan [3]; Lu et al. [23] | Models produce less diversity than humans; prompting steers toward human strategies without their diversity | A verified-novelty audit against a known-solution catalogue, which these benchmarks do not run | adjacent |

| | | | | |
|---|---|---|---|---|
| Random pushes buy correct diversity on movable tasks | Khalid et al. [70]; Zhu et al. [71]; Kim et al. [84] | Residual-stream noise raises narrative diversity; activation steering raises test coverage; random embedding injection widens pass@N on math | Diversity of algorithms, test-checked and mechanism-labelled, task-dependent and calibrated to the norm of the ban; the method itself is occupied | partly occupied |
| Movability predicts compliance (screen) | Fan et al. [75] | Hidden-state features predict whether a steering intervention succeeds | A black-box prompt prohibition forecast from white-box movability | adjacent |

**Table 92c.** Iteration-4 claims, their nearest published neighbours and what this run adds. Source: positioning audit `research_report.md`.

| **Claim in this run** | **Nearest neighbour** | **Finding of the neighbour (paraphrased; verified against a quote)** | **What this run adds, or what it lacks** | **Lane** |
|---|---|---|---|---|
| Plan noise is priced against temperature and prompt knobs | Ahmed et al. [73]; Wang et al. [74] | Sampling an intent at high temperature and the response at low temperature, or searching over natural-language plans, raises conceptual diversity in code | Activation-level plan noise on test-checked labels with seed-coupled contrasts; the comparison set lacks these text-level plan samplers | partly occupied |
| (same) | Lee et al. [72] | Temperature above 1 raises algorithmic diversity on other problems, at a correctness cost | Qualifies the temperature null: price temperature per task | contradicting in part |
| (same) | Peeperkorn et al. [37]; Zeng et al. [86]; Wu et al. [2] | Temperature correlates weakly with novelty; token-level perturbation gives lexical, not strategic, variation; sampling and prompting changes do not cure monoculture | Consistent with the null on ideas | supporting |
| Site specificity at matched dose | Zhang et al. [83]; Khalid et al. [70]; Zhu et al. [71]; Nagarajan et al. [80] | Diversity injected at different text levels, generated-token noise sites, intra-layer steering sites, and input-layer noise against output temperature | Sites inside the network at matched dose on algorithm labels | partly occupied |
| The same seed design | Ravfogel et al. [77] | Coupling sampling noise across interventions yields counterfactual strings | Token seeds coupled across temperatures and plan noise draws to split idea changes from phrasing changes | adjacent |
| A plan subspace read from forced prompts predicts push effects (disconfirmed here) | Dong et al. [78] | Prompt representations encode global attributes of the coming response | A categorical, test-checked object and a ban-free subspace tested as a predictor of random-push effects; the prediction failed | partly occupied |

| | | | | |
|---|---|---|---|---|
| Geometry and targeted pushes | Wurgaft et al. [27] | Geometry-aware steering with geodesic distances and an MDS isometry | Algorithm-level geometry on test-checked labels; exact crossing on one task | partly occupied |
| Movability forecasts ban compliance in a held-out family | Tang [38]; Fan et al. [75] | Constraints opposing defaults fail at 10% to 100%; steerability is predictable from hidden states | A pre-registered white-box forecast of black-box prompt compliance (inconclusive here) | open |
| Push landings forecast ban landings | Yoon et al. [4]; Zhou et al. [56] | Negation suppresses and promotes, or constructs and suppresses, at the fact level | A ban-free forecast of which alternative a prohibition installs | open |
| No correct novel algorithm up to breakage | Romera-Paredes et al. [81]; Novikov et al. [82] | Search with an evaluator produces new programs | Scope the claim to single-sample generation under knobs, without search or an evaluator | adjacent, scoped |
| Plan noise (row added in iteration 5) | Jerge, Pelczar and Downes, arXiv 2608.21595 | Gaussian noise calibrated to the scale of the last hidden layer, added in the pass that encodes the prompt, diversifies reinforcement learning rollouts of vision-language models | Test-checked mechanism labels on code, a site comparison at matched dose, and seed coupling; theirs is rollouts scored by reward | partly occupied |
| Plan noise against temperature (row added in iteration 5) | Nagarajan et al. [80] | Noise injected at the input layer works as well as, and sometimes better than, temperature sampling at the output layer | The nearest prior answer; this run adds noise in the task-specification states against temperature with the token seed held fixed, on execution-checked algorithm labels | adjacent |
| Novelty boundary (row added in iteration 5) | Bhushan, Zhang and Wang, arXiv 2608.30047 | Agents with iterative search explore novel regions of the solution space but do not convert novelty into better task performance | Scope: single-sample generation without search; the bound on novelty (0 among 14,206 correct programs) | contradicting in part |
| Novelty boundary (row added in iteration 5) | Maveli, Vergari and Cohen, arXiv 2610.02438 | Code generation is better measured as parametric retrieval of a named algorithm than as synthesis of a novel one | Execution-checked labels and the bound under activation pushes | supporting |
| Novelty boundary and knob (row added in iteration 5) | ICLR 2027 submission oag3wgNLRh (OpenReview) | Activation steering redirects generation among trajectories the model already supports, and decoding matched to the departure of steering from greedy decoding reaches comparable accuracy | Algorithm-level labels; decoding at matched departure was not run in iteration 4 and ran in iteration 5 (Section 5.7) | partly occupied |

| | | | | |
|---|---|---|---|---|
| Plan subspace (row added in iteration 5) | ICLR 2027 submission knirOusGcq (OpenReview) | A shared causal subspace for planning the length of a response, tested by decomposing interventions into parallel and orthogonal components | A categorical, test-checked object; the forced-prompt subspace does not predict random-push effects (0.03) and removing it keeps the effect (ratio 1.00) | partly occupied |

## 4.9 Comparisons across the experiments of this iteration

**Movability on fresh directions.** Three engines and three direction sets measured whether a task is movable: the frozen screen (fixed directions from iteration 3), the plan subspace experiment (40 fresh fixed directions per dose) and the plan noise experiment (a fresh direction per sample). They agree on five of the six screen-movable tasks and on every immovable one, and they disagree on find duplicate (Table 164). The single escaping direction of the screen on find duplicate at alpha 2 did not replicate in 80 fresh cells, and per-sample plan noise left the hash set in place at alpha 2 or less. On fresh draws the movable set is kth smallest, two sum, connected components and the two path tasks, and the table of the screen against ban compliance (movable and obeying, movable and failing, immovable and obeying, immovable and failing) moves from 6, 0, 0, 8 to 5, 0, 1, 8: find duplicate obeys its ban (catalogue-only avoidance 0.79) without being movable. Across the 14 tasks the escape fraction still ranks ban avoidance at Spearman 0.86.

**Table 164.** Movability of the screen tasks in three measurements: screen escapes (escaping of run, alpha 1 / alpha 2), the escape fraction of the plan subspace experiment on 40 fresh directions (alpha 1 / alpha 2), and per-sample plan noise (off-default share at alpha 1 / alpha 2, N = 50), with catalogue-only modal ban avoidance. Sources: record audit `screen.json`; plan subspace experiment `results/escape.json`; plan noise experiment `results/alpha_sweep.json`.

| Task | Screen escapes | Fresh directions | Per-sample plan noise | Ban avoidance (catalogue only) |
|---|---|---|---|---|
| kth smallest | 8/18 / 3/3 | 0.30 / 0.63 | 0.28 / 0.50 | 0.97 |
| two sum | 0/6 / 2/3 | 0.00 / 0.13 | 0.00 / 0.02 | 0.86 |
| find duplicate | 0/6 / 1/3 | 0.00 / 0.00 | 0.00 / 0.02 | 0.79 |
| connected components | 1/3 / not run | 0.38 / 0.73 | 0.12 / 0.60 | 1.00 |
| grid | 1/15 / 0/3 | 0.05 / 0.13 | 0.06 / 0.24 | 0.86 |
| shortest path | 3/15 / 0/3 | 0.18 / 0.20 | 0.26 / 0.54 | 0.83 |
| the eight immovable bench tasks | 0 of 3 or 6 each | 0 of 8 each | off-default excess at most 0.04 | 0.00 to 0.47 |

The fixed-direction escape fraction and the per-sample off-default share measure different things: a task with a few strong directions (two sum, where 5 of 40 directions escape at alpha 2) can show almost no movement under per-sample noise, which averages over directions. *(Correction, iteration 5: alpha does not mean the same push in every column. The screen uses the displacement of the ban itself; the bench unit of the plan subspace experiment is 9% to 16% smaller (its deviation 'ban_unit_scale'); the kth smallest unit of the same seed experiment is 0.77 of the screen's; the site experiment uses per-position ban units (blocks 0 to 11 at 4.5% to 7.2% of the residual norm against 12.7% to 18.3% for blocks 12 to 23); cycle detection and Gemma use a ban-free unit, which across tasks is 0.30 to 0.97 of the ban unit. Substring search has by far the largest ban unit (58.5 against 11.6 to 19.0 elsewhere) and does not move. Table 174 gives the units.)*

**Temperature, five times.** Every measurement of temperature in the run says the same thing, now

including the thinking-on arm the review asked for. In iteration 1, 797 of 800 grid programs and 797 of 800 shortest path programs kept the default across temperatures 0.3 to 1.3 with thinking off, and all 1,600 with thinking on. In iteration 2, with seeds coupled, the algorithm changed in 0 of 30 seeds in 10 of 12 cells. In this iteration temperature 1.3 and 1.6, top-k 10 and min-p reach a 5-point Hill-1 of at most 1.17 on the six movable tasks; the same seed design changed the algorithm in 0 of 20 seeds on three tasks with thinking off and on the shortest path task with thinking on; and in gemma-3-4b-it temperature reaches 5-point Hill-1 of at most 1.32. A positioning caveat applies: on other problems temperature above 1 does raise algorithmic diversity at a correctness cost [72].

**Novelty, audited everywhere.** No artifact of this iteration found a correct novel algorithm (Table 165). Together with iterations 1 to 3, every correct program outside the catalogue in the pools audited so far has been read by two blind judges and the executor, and none was new: iterations 1 to 3, the 78 correct programs outside the catalogue among 11,763 unique correct programs (Section 3.2.3, Table 87); iteration 4, the audits of Table 165; iteration 5, its 130 correct programs outside the catalogue, with the 14 over-calls re-adjudicated in iteration 6 (Section 6.2.6); and iteration 6, the pools of Table 278, including the engine control's 3 correct programs labelled other, judged in iteration 7. Two pools were never audited by judges: about 9,000 programs of the forking token experiment, as recorded in Section 3.2.3, and the engine control's nearest-known pass of iteration 6 (up to 237 correct programs per engine and task, compared with known solutions by token overlap only; Table 278). *(Correction, iteration 6: the 14 that judges over-called in iteration 5 were re-adjudicated blind in iteration 6 (judges 14 of 48, manual 0, second adjudication 1 of 14 (Mistral Medium 3.1, three calls at temperature 0, control gate 9 of 10, passed; Kimi (kimi-k2) second view 0 of 14), final count after the executor's full read 0; the contested control, repeated maximum selection for top-k frequent, which the iteration-5 manual pass itself classed as a catalogue variant, was called catalogue variant by Mistral and catalogue variant by Kimi (Mistral gate 10 of 10 under that convention); the final read corrects 1 manual class (CATALOGUE VARIANT to DEGENERATE), none to novel).)* (iteration-7 correction)

**Table 165.** Novelty audits of iteration 4. Sources: the novelty audit files of the six artifacts.

| Artifact | Pool | Correct programs labelled other | Final classes | Novel |
|---|---|---|---|---|
| Record audit | other programs under iteration-3 bans (226 unique, 6 passing) | 6 | 68 catalogue variants, 156 degenerate, 2 labeller misses (all 226) | 0 |
| Plan noise experiment | every correct program labelled other | 12 | 12 labeller misses | 0 |
| Site experiment | 3,555 correct programs | 1 | catalogue variant (quadratic brute force) | 0 |
| Same seed experiment | every correct program labelled other | 1 | labeller miss (naive Dijkstra) | 0 |
| Plan subspace experiment | 10,651 correct push programs | 4 | 1 catalogue variant, 1 labeller miss, 2 degenerate | 0 |
| Held-out family experiment | every correct program labelled other | 34 | 27 degenerate, 7 catalogue variants | 0 |

## 4.10 What iteration 4 established, what it ruled out, and what it abandoned

Everything below is one model per claim except the held-out test, which is one family at one size,

with labels from labellers that passed their gates (and one, the path labeller on Gemma, that needed a style re-gate). Every Qwen3-8B number in this iteration comes from the 8-bit engine of the screen. (Engine class: Hugging Face with W8A8 linear layers in 8-bit floating point, e4m3 weights per output channel and e4m3 activations per row through the GPU's native 8-bit floating-point matrix multiply. Each item below carries its iteration-6 engine tag, and Section 6.3 decides which of them hold under bf16 at the same absolute dose.)

**Established.**

1. *Temperature changes phrasing and not the algorithm, with the seed held fixed, thinking off and on.* Across four temperatures the algorithm changed in 0 of 20 seeds on three tasks with thinking off and on the shortest path task with thinking on, while phrasings per idea grew on the grid task from 3 to 13 clusters. Temperature 1.3 and 1.6, top-k 10 and min-p buy no correct algorithmic diversity on the movable tasks (5-point Hill-1 at most 1.17).
2. *A random displacement of the task-specification states changes the algorithm, and changing the plan noise draw under a fixed seed changes it often.* One direction at alpha 1 changed the algorithm in 17.25 of 20 seeds on the shortest path task, 4.55 on kth smallest and 2.5 on the grid task (pooled contrast against temperature 0.545 [0.41, 0.67], p = 1.0e-5). *(Engine tag, iteration 6: 8-bit floating-point W8A8; the same-seed contrast itself was not rerun in bf16. Per-sample plan noise on the shortest path task did not replicate in iteration 5's bf16 cells at the ban-free unit (2 of 47 correct programs off the default at bf16 alpha 2 against 9 of 43 at 8-bit alpha 1, bracket p 0.017), but at iteration 4's own absolute unit and seeds the bf16 engine reproduces it (22 of 50 valid attempts off the default at alpha 2 under both engines; Section 6.3).)*
3. *The site is the task-specification states at early to middle blocks.* The same displacement at the read position or the condition sentence changed no algorithm at any dose (0 of 526 and 0 of 527 correct programs off the default), the late blocks were almost inert (6 of 521), and at matched downstream displacement the plan site keeps a large advantage (indicator +2.15 [1.79, 2.51]; odds ratio 5.72 [3.94, 8.30]) *(engine tag, iteration 6: 8-bit floating-point W8A8, not rerun in that form. The bf16 rerun at the ban-free unit of Section 6.4 withdrew the committed site rule, because the condition-span arm never reached the plan site's downstream displacement and alpha 8 broke every plan-site program; a post-hoc model restricted to alpha of 4 or less puts the read position below the task-specification site on both tasks.)*. Blocks 12 to 23 are not distinguished from blocks 0 to 11. *(Correction, iteration 5: two wordings here say more than the experiment supports, that the read position and the condition sentence change nothing at any dose, and that the task-specification states are the site of the choice. Restated: among prefill positions, perturbing the task-specification states changes the algorithm; the read position and the condition sentence are inert at per-position doses up to 4 ban units and far weaker at matched energy (energy-matched read position: +0.23 on find duplicate, Floyd cycle detection 6 of 26 correct programs, and +0.18 on connected components, 5 of 28 off the default); noise in generated tokens 1 to 32 also redirects the choice, more than the plan site on find duplicate (12:0 paired flips at alpha 2 over the 30 pairs, exact McNemar p 0.0005; 6:0 among pairs where both programs are correct). Blocks 0 to 11 against 12 to 23: -0.31 [-0.75, 0.13]. This is a prefill site at which noise is effective, not a locus of the decision, and it is reconciled with the PARTIAL site rule under Inconclusive below.)*
4. *Plan noise is the cheapest knob on some tasks and not on others.* It wins kth smallest (5-point Hill-1 2.72) and connected components (2.09) at no pass cost *(engine tag, iteration 6: 8-bit floating-point W8A8 at the ban unit; under bf16 and the ban-free unit of iteration 5 the same prices are 1.98 and 1.09, and on connected components 1 of 50 correct programs left the default*

*at bf16 alpha 2 against 30 of 50 at 8-bit alpha 2 (Fisher p 7.2e-11). The engine control of Section 6.3 traces that gap to the unit and the direction seeds, not to the numerics: at iteration 4's absolute unit and seeds, bf16 plan noise moves connected components in 25 of 50 valid attempts.)* and loses find duplicate to the iterated veto and the shortest path task to the semantic seed; the preregistered price claim is PARTIAL.

5. *Which random directions move the algorithm is a stable property of the direction, and the forced-prompt geometry does not explain it.* Template halves rank 40 directions alike (0.76 to 0.90), the projection share on the plan subspace ranks them at 0.03, and removing the in-subspace component of a push leaves its effect intact (median ratio 1.00). *(Correction, iteration 5: the range 0.76 to 0.90 covers four tasks. On all six: kth smallest 0.80 (34 movers, 27 escaping cells of direction and dose), shortest path 0.81 (37, 14), two sum 0.76 (10, 5), connected components 0.90 (36, 30), find duplicate 0.59 (3, 0) and the grid task 0.39 (22, 6); Table 176.)*
6. *Where pushes escape, they land on one runner-up per task, and that runner-up forecasts where a ban lands.* The landing replicates across 40 fresh directions and the screen on every movable task, and on the gated bench bans, scored descriptively because they were seen before, the push forecast scores -0.76 against -3.46 for association; in Gemma it hits 4 of 4, on too few disagreeing events for the rule. *(Correction, iteration 5: this item moves from Established to the descriptive results. Where pushes escape, the landing replicates across 40 fresh directions and the screen at the plan site (kth smallest quickselect 0.66 and size-k heap 0.31; connected components DFS 0.83 and BFS 0.17; shortest path Bellman-Ford/SPFA 0.99; two sum brute force; grid DFS). As a forecast of the seen bench ban landings it hits 2 of 3 defined Qwen3-8B events (it misses connected components) and scores -0.76 against -3.11 for association on those events, and -1.37 against -3.46 over all 5 with undefined forecasts scored as uniform; in Gemma it hits 4 of 4, of which 2 disagree with association and 2 are compromised. The landing depends on the site on find duplicate.)*
7. *Route landings.* On connected components either route of a ban alone lands mostly on DFS, the generic neighbour, and only both routes together choose BFS. On kth smallest the size-k heap share travels by the prompt states. *(Correction, iteration 5: an observation on a single task, with find duplicate as its counterexample: there the direct route alone lands on index marking, 74 of 78, neither the push landing nor the association answer; on two sum pushes and association disagree.)*
8. *Snap-back replicates in Llama.* On five of six present minor events the modal landing is the default (0.93, 0.73 and 0.72 of attempts not banned on three of them).
9. *No correct novel algorithm appeared under any push, up to breakage.* Correct-only diversity peaks at alpha 2 to 4, and alpha 8 erases the task; 0 novel among 14,206 correct programs of the site and subspace experiments alone.

**Ruled out, or not supported.**

- *A plan subspace read from forced prompts as the channel of push effects.* Disconfirmed by its preregistered rule on 6 of 6 tasks, and the decomposition shows the subspace is not needed.
- *Targeted pushes as a general cheap knob.* Indistinguishable from a matched random push on 5 of 6 tasks; the exception, connected components, crosses exactly at the predicted midpoint.
- *Knob displacements projecting in proportion to their effect.* Pooled 0.27, not better than the displacement norm.
- *The top-five prohibition applied at the plan.* The push never left BFS (avoidance 0.00 at alpha 1 and 2), as the prompt and the token veto had not.

- *Find duplicate as a movable task.* Its screen escape did not replicate in 80 fresh cells.

**Inconclusive.**

- *Movability as a predictor of ban compliance in a held-out family.* Balanced accuracy 0.74 [0.40, 0.95], Fisher p 0.24, Spearman 0.54; the novelty instruction predicts better (0.77), and one correct prediction rests on a labelling failure. *(Correction, iteration 5: INCONCLUSIVE under the primary measure, catalogue-only avoidance, and DISCONFIRMED under the preregistered sensitivity on the drop in the default share (balanced accuracy 0.57, Fisher p 1.0, Spearman 0.39). The grid task (tied twin, BFS 25 and DFS 25; no default under the iteration-5 rule) and find duplicate (38 failing other programs; 0.34 against 0.72) separate the two measures. Without the two compromised true positives, balanced accuracy is 0.61 and Spearman 0.31, one-sided permutation p 0.144.)*
- *Push landings against association in the held-out family.* Ahead by 2.81 nats, but on 2 disagreeing events, below the 3 the rule needs.
- *The site claim.* PARTIAL: the plan site is best on 3 of 6 tasks, with every margin but one a bootstrap close call. *(Added in iteration 5: the site rule is PARTIAL, so no site is established as best; what is established is the prefill specificity at matched displacement, odds ratio 5.72 [3.94, 8.30] for the task-specification states against the other prefill sites. The price of 1.5, the pass cost at which a site's randomisation hull first reaches correct-only Hill-1 of 1.5, is 0.12 for generated-token noise against 0.53 for the plan site on two sum.)*

**Dead ends, with the evidence that closed them.**

- *The plan subspace as the internal account of the choice.* Closed by Tables 153 and 155: the share does not rank directions, and removing it does not remove the effect.
- *Verbalized sampling as a diversity knob.* 46 to 49 of 50 programs dispatch at run time to a randomly chosen algorithm, so their labels measure the trace, not the choice.
- *The frozen counting rule for forecast contests.* It advances some family in 85% of no-information simulations; forecasts are now scored by paired tests and defined-only hit rates.

## 4.11 What the next iteration takes from this one

The iteration answered the first question of the request in its strictest form and narrowed the second. Temperature, at any setting and with the seed fixed, changes how the code is written and not which algorithm it implements, with thinking off and on. The algorithm can be influenced, cheaply on some tasks, by a random displacement of the task-specification states at early to middle blocks, and the read position, the condition sentence and the late blocks are inert. *(Correction, iteration 5: the read position and the condition sentence are inert at per-position doses up to 4 ban units, weaker but not inert at matched energy, and the late blocks almost inert.)* What the iteration did not find is a representation of the choice: the forced-prompt subspace does not explain which displacements work, although which displacements work is reproducible. The open question is therefore what property of a direction makes it move the algorithm. The exploratory signed-progress correlations (up to 0.57) point at direction toward the runner-up rather than share, and the targeted push on connected components shows that one axis can be causally exact. A preregistered test of signed progress on fresh directions, with the reproducible ranking of the directions as the target, is the next step. *(Correction, iteration 5: that step was not taken, and signed progress is closed. Its positive cells concentrate on one task, it is negative on average on two, and on connected components the committed signed component was matched by a within-class control (0.33*

*against 0.37). Per task over 24 block cells: kth smallest 1 cell with p below 0.05 and a mean of -0.02 over the edited blocks (-0.29 at block 15); shortest path 3 and +0.20; two sum 7 and +0.31; connected components 22 and +0.37; find duplicate 0 and -0.12; grid 2 and +0.05 (Table 177; the record's count of 22 of 35 cells was 22 of 24). The next budget goes to more tasks, not to a further predictor at the level of directions on the same five: new tasks with the base model's repertoire committed before any push, so that which tasks move can be predicted from the base model and tested on units never seen.)*

The movability claim needs more than one held-out family at one size, a labeller re-gated on the style of each family, and a comparison with the novelty instruction, which predicted compliance better in Gemma. The landing claim needs events on which the push forecast and association disagree: Gemma gave 2 of 4, and the Qwen3-8B bench gave 3 defined events. The contrast set of the price claim lacks the text-level plan samplers that the positioning audit names [73, 74] and a decoding baseline matched to the departure of plan noise from greedy decoding; until both are run, the claim is that plan noise beats the token-level and prompt-seed knobs tested, on the tasks where it moves the model. The route-landing lead from Section 4.2.2 ties these threads together: where one route alone removes the default, the landing is the generic neighbour that pushes reach, and the ban departs from it only when both routes act. *(Correction, iteration 5: this lead rests on connected components alone; find duplicate is the counterexample and two sum has no agreed neighbour, Section 4.2.2. Movability as a forecast of ban compliance and the contest over where a ban lands are closed beyond descriptive reporting; the next test is the narrowed repertoire account, Section 5.1.)*

# Iteration 5: the base model's repertoire against nudge movability, a transplant of base plan states, text-level plan samplers and a second family

## 5.1 Why this iteration ran what it ran

**The review of the iteration-4 record.** The review was blocking again. It raised five major items, all marked must fix, and three minor ones, and each pointed at a place where the record said more than its own artifacts. The held-out test of the movability claim had been written up as inconclusive under every sensitivity, but one preregistered sensitivity, compliance measured as the drop in the share of the default, disconfirms it, and two of the three true positives behind the positive rank correlation were compromised: the grid task had no default (its Gemma twin wrote BFS 25 and DFS 25), and the shortest path task's compliance rested on array-based Dijkstra read as Bellman-Ford/ SPFA. The site conclusions said the read position and condition sentence changed no algorithm *at any dose*, although the energy-matched read position had moved two tasks, and they called the task-specification states the site of the choice, although noise in generated tokens 1 to 32 redirected it as well and beat the plan site on find duplicate. Several landing conclusions generalised from connected components alone or compared forecasts on unmatched event sets. Executed results bearing on the primary claim were missing: plan noise on the plain prompt, the hull deviation restricted to edges, the Llama block of the knob table, the dose unit of each experiment and one artifact's GPU hours. And the closing summary quoted maxima over directions as if they were typical effects, while the lead it chose for the next iteration, signed progress toward the runner-up, rested mostly on one task. The minor items asked for the positioning rows the iteration-4 audit had verified but not carried over, for exposure arms to be declared controls, and for the literal cell of the request (thinking on, top-k 10, same seed). Table 166 maps every item to its answer; the corrections are applied in place in Sections 3.2.5 and 4.2 to 4.11 and marked as iteration-5

corrections.

**Table 166.** Review items on the iteration-4 record and where this report answers them.

| Review item | What the review found | Where it is answered |
|---|---|---|
| Held-out verdict per compliance measure | The drop sensitivity disconfirms; grid task has no default; shortest path true positive rests on a mislabel | Corrections in Sections 4.7 and 4.10; Tables 169 and 170, Section 5.2; closing section |
| Site conclusions | The phrase *at any dose* contradicted by the energy-matched read position; generated-token noise also redirects the choice; PARTIAL rule listed as established | Corrections in Sections 4.4, 4.10 and 4.11; closing section |
| Landing conclusions | Route-landing lead from one task; Gemma 4 of 4 includes two compromised events; forecasts compared on unmatched events; landing depends on the site | Corrections in Sections 4.2.2, 4.6 and 4.10; Table 171, Section 5.2 |
| Missing executed results | Plan noise on the plain prompt; hull deviation; Llama block; dose units; GPU hours | Corrections in Sections 4.1, 4.2.1, 4.3 and 4.9; Tables 126c, 139, 173 and 174 |
| Best cells and the next lead | Maxima quoted as effects; split-half on four of six tasks; signed progress concentrated on one task; alpha-4 depths | Corrections in Sections 4.2.1, 4.4, 4.10 and 4.11; Tables 175 to 178; the next budget went to new tasks (Sections 5.4 and 5.5) |
| Minor: positioning rows | Jerge et al., Nagarajan et al., Bhushan et al., Maveli et al. and two ICLR 2027 submissions verified but not carried | Rows added to Table 92c in Section 4.8 |
| Minor: exposure arms as knobs | Exposure prompts supply the alternative's text | Class rule corrected in Section 3.2.5; Table 172 |
| Minor: the literal 1a-c cell | Thinking-on arm ran without top-k 10 | Correction in Section 4.5; cell run in Section 5.7 |

**The hypothesis update.** The update kept the plan noise lead of iteration 4 and asked what sets its boundary. Plan noise moved the algorithm on some tasks and on no others, and where it moved it, it landed on one runner-up per task. On the two path tasks those runner-up algorithms were exactly the algorithms the untuned base model writes next to its default: under the chat template Qwen3-8B-Base wrote BFS in 49 of 57 correct grid programs and Dijkstra in 63 of 90 correct shortest path programs, with DFS and Bellman-Ford/SPFA second (Section 1.9). The update therefore proposed what it called the *narrowed repertoire account*: post-training narrowed a choice that the base model makes more evenly, the narrowed choice is set while the model reads the task specification, and a nudge to those states undoes part of the narrowing, reopening the base model's alternatives and only those. A task that no nudge moves would then be one on which the base model itself writes almost one algorithm, and post-training [35, 57, 87] would have narrowed only what pretraining left open.

We use the following quantities throughout. The *base off-default share* of a task, beta, is the share of the base model's correct programs that use an algorithm other than the instruct model's default, with both models given identical token ids under the chat template; the *base runner-up* is the base model's most frequent correct algorithm other than that default. We call a task *diverse in the base* when beta is at least 0.2 and *collapsed in the base* when beta is below 0.05; between the two it is intermediate. The instruct default comes from a twin of at least 50 samples as the modal correct label with a share of at least 0.6; a task below that has *no default* and enters no movability or landing test. One *ban-free unit* is the residual norm of the task-specification states times the median ratio, in Qwen3-8B at the same relative depth, of a ban's displacement to the residual norm; it

replaces the ban's own displacement as the dose unit everywhere, so that a task's dose no longer depends on how far its ban moved it. We call the edit h ← h + lambda (h_base − h_instruct), applied during the instruct prefill at every task-specification position of blocks 12 to 23 with the base states computed on the same prompt, a *transplant of base plan states*, in the spirit of cross-model activation patching [88].

Eight claims were preregistered, each with its rule committed by sha256 before its first decisive sample.

- *The movability half: the base repertoire predicts which tasks move* (primary): across new tasks, beta ranks the escape fraction of 8 fixed random directions at Spearman of at least 0.6 with a permutation p within parent clusters below 0.01, at least 80% of tasks collapsed in the base are immovable, and beta is ahead of each rival. Disconfirmed if Spearman is below 0.3 or most tasks diverse in the base are immovable.
- *The landing half: landings follow the base runner-up* (primary): on movable new tasks the push landing is the base runner-up in at least 70% of events (exact test against per-event chance, p below 0.05), and the base composition beats the association forecast by at least 0.10 nats in log score.
- *The transplant claim* (primary, causal): on at least two thirds of the tasks diverse in the base among the 15 existing ones, the transplant at its best lambda (with pass of at least 0.8 times the twin's) reaches a correct off-default share of at least half of beta, beats a matched-norm random displacement and what we call a *borrowed shift* (another task's base minus instruct direction at the same norm) on paired flips, and lands on the base runner-up; tasks collapsed in the base and a transplant at the read position stay within 0.05 of the twin.
- *The plan sampler comparison* (secondary): plan noise prices at least as well as the best of four text-level plan samplers on at least 3 of the 5 tasks movable on fresh draws: intent factored generation [73], plan search [74], a sampler we call *enumerate and pick* (the model lists algorithms and an external random draw picks one) and what we call *base intent* (the base model writes an intent of one sentence and the instruct model writes the code).
- *The push toward the base*: a push along the base minus instruct direction, which we call a push toward the base, escapes at no more than half the norm a random push needs (secondary).
- *The Gemma replication* (secondary): both halves in gemma-3-4b-pt against gemma-3-4b-it [76], a replication on fresh samples because Gemma's iteration-4 outcomes were seen.
- *The literal cell* of the request (descriptive) and *the novelty count* (secondary).

**The strategy.** Seven artifacts ran (Table 167). One evaluation on CPU only repaired the iteration-4 record from rows on disk and froze eight rival task predictors by hash before reading any outcome. The transplant experiment measured the base repertoire of all 15 existing tasks, committed it, and only then ran the transplant with its controls. Two panels built new coding tasks, committed each task's base repertoire, and only then pushed the instruct model; they split the new tasks between them so that the pooled confirmation would have more units than either could build alone. The knob experiment ran the text-level plan samplers and what we call *decoding at matched departure* (a decoding temperature set so that the output departs from the twin as much as under plan noise) on all 15 existing tasks, and the literal cell. The Gemma experiment repeated both halves and the transplant in a second family, and a research artifact searched the literature for each new claim. The 16 GB card of iteration 4 was replaced by cards of 23 to 24 GB, so every Qwen3-8B experiment of this iteration ran in bf16 rather than with 8-bit linears; this change of engine, together with the

change from the ban unit to the ban-free unit, turns out to matter (Section 5.10).

**Table 167.** The seven artifacts of iteration 5, the question each was asked, and its hardware and spend.

| Artifact | Question | Model and hardware | OpenRouter spend |
|---|---|---|---|
| Record audit | Repair the iteration-4 record (35 corrections); freeze eight ban-free rival task predictors | No generation; CPU only | USD 0.021 |
| Transplant experiment | Base repertoire of the 15 existing tasks; does a transplant of base plan states reopen it (the transplant claim); push toward the base; reverse transplant | Qwen3-8B and Qwen3-8B-Base, bf16, one RTX 4090; 14,280 programs; 1.70 GPU hours | USD 0.29 |
| Panel A | The movability and landing halves on new tasks (majority element to median of two sorted lists); the 15 existing tasks remeasured on the ban-free unit | Same models, bf16, one RTX 4090; 8,390 programs; 0.98 GPU hours of batch time | USD 0.357 |
| Panel B | The movability and landing halves on seven more new tasks; disguised variants of eight bench tasks | Same models, bf16, one RTX 4090; 14,600 programs; 1.74 GPU hours | USD 0.192 |
| Knob experiment | Plan noise against text-level plan samplers and decoding at matched departure (the plan sampler comparison); the literal cell | Qwen3-8B bf16 (HF) and the official 8-bit checkpoint (vLLM, literal cell), one RTX 4090; 7,060 labelled knob rows and 100 traces; 1.88 GPU hours plus 15 minutes | USD 0.12 |
| Gemma experiment | Both halves and the transplant in gemma-3-4b-pt against gemma-3-4b-it | Both models bf16, one 23 GB NVIDIA GPU; 12,290 programs; 2.16 GPU hours | USD 0.39 |
| Positioning audit | Has prior work restored base-model variety, forecast it, or transplanted base states? | Web search and full-text quote checks | none recorded |

## 5.2 The record audit: corrections from rows on disk, and frozen rival predictors

The record audit generated no model text. It answered every review item by recomputing the values behind it from the row files of eight earlier artifacts, and wrote each fix as an exact replacement for the record, with every value tagged by provenance and checked against the record within fixed tolerances (counts exact, rates 0.005, Hill-1 and log scores 0.01, intervals 0.02). Of 187 values that had a record value, 186 matched and 1 was corrected: the record's count of *22 of 35 cells* for signed progress on connected components is 22 of 24, because the staged file holds 24 block cells per quantity. A further 15 values were new and 4 were carried from the iteration-4 record because their source was not a declared dependency (the knob table cells, the route landings, the Llama block and the iteration-3 direction counts). All six quotes behind the rows added to Table 92c were verified again live on 2026-10-21. An independent script using only the standard library recomputed 195 values with no failure, and its four placebos failed as they should (balanced accuracy under permuted escape fractions 0.50; the site odds ratio 1.01 when the plan site and read position are swapped, against 101.7 real; a landing shuffle that shrinks the push advantage from 2.35 to 0.46 nats; rejection of permuted escape fractions at 3.5% to 4.8%). Table 168 lists the 35 corrections and the status of the values behind each.

**Table 168.** The 35 corrections of the iteration-4 record, their location and the values behind them (MATCH: recomputed equal to the record within tolerance; NEW: no record value; CARRIED: from the iteration-4 record). Source: record audit `corrections.json`.

| Item | Report location | Values behind the correction (status against the record) |
|---|---|---|
| 0a | Table 159 grid row | MATCH 7 |
| 0a | Table 159 and Table 163 shortest path rows | MATCH 5 |
| 0a | Table 162, the count of 4 of 4 | MATCH 2 |
| 0a | Table 160 | MATCH 21 |
| 0a | Section 4.10, the entry under Inconclusive | MATCH 7 |
| 0a | Closing paragraph on push movability as a forecast of which bans a model obeys (last sentence) | wording only |
| 0b | Section 4.10 item 3 | MATCH 23, NEW 2 |
| 0b | Section 4.4 summary | MATCH 2 |
| 0b | Closing paragraph 2 | MATCH 2 |
| 0b | Closing paragraph 1 | wording only |
| 0b | Closing paragraph 3 (first sentence) | wording only |
| 0b | Section 4.10, the site claim under Inconclusive (append) | MATCH 3, NEW 2 |
| 0c | Table 157 | MATCH 37, NEW 1 |
| 0c | Section 4.6 | MATCH 6 |
| 0c | Section 4.10 item 6 (moves from Established to the descriptive results) | MATCH 10 |
| 0c | Section 4.10 item 7, Section 4.2.2 route-landing lead, Section 4.11 paragraph 2 last sentence | CARRIED 1 |
| 0c | Closing landing paragraph | wording only |
| 0c | Every statement that the landing is a property of the task (and the statement in closing paragraph 2 that where it can is a property of the task) | wording only |
| 0d | Section 4.3 (plan noise on the plain prompt at alpha 1, executed but not reported) | MATCH 20 |
| 0d | Table 139 (two rows) | MATCH 1, NEW 2 |
| 0d | Tables 123/126 Llama row block and Section 4.2.1 | CARRIED 1 |
| 0d | Beside Table 164 (dose unit) | MATCH 1, NEW 4 |
| 0d | Table 122, row of the experiment of Section 4.5 | MATCH 3 |
| 0e | Seed-coupled plan noise counts (Section 4.5) | NEW 1 |
| 0e | Table 123 two sum row and closing paragraph | MATCH 3, NEW 1 |
| 0e | Tables 123, 126, 164 and closing paragraph 2, numbers of the best cells | CARRIED 1 |
| 0e | Section 4.10 item 5 and closing paragraph 3 | MATCH 6 |
| 0e | Section 4.11 and the open questions of the closing (signed progress) | CARRIED 1 |
| 0e | The open questions of the closing | wording only |
| 0e | Table 147 | MATCH 1 |
| 0e | Section 4.11 paragraph 1 (last two sentences) | wording only |
| 0e | Section 4.11 | wording only |
| 0g | Section 3.2.5 class rule | NEW 1 |
| 0g | Section 4.2.1, the sentence on reaching 1.5 within 20 points | wording only |
| 0h | Section 4.5 | NEW 1 |

**The held-out verdict, per compliance measure.** Table 169 restates the Gemma test of iteration 4 under every variant the audit could compute. Under the primary measure, catalogue-only avoidance, the verdict stays INCONCLUSIVE (balanced accuracy 0.74, Fisher p 0.235, Spearman 0.54). Under the preregistered sensitivity that measures compliance as the drop in the default share, which we call the *drop sensitivity*, it is DISCONFIRMED (0.57, Fisher p 1.0, Spearman 0.39). Two tasks separate the measures, and Table 170 shows why. The grid twin is tied, BFS 25 and DFS 25 of 50 attempts, so the default DFS came from a 10-sample pilot (correct BFS 4 and DFS 5, attempts 5 and 5) and the ban's avoidance of 0.83 is measured against a baseline that already left DFS half the time; the default share drops by only 0.33. On find duplicate, 38 failing programs labelled other count as leaving the default under the drop measure (catalogue-only avoidance 0.34, drop 0.72). Under the iteration-5 default rule the grid task is the only one with no default, and the first and last position task's rule default (linear scan, 0.86 of the twin) differs from the frozen one. Removing the grid task leaves the verdict inconclusive (0.70); removing both compromised true positives gives 0.61 and Spearman 0.31 (one-sided p 0.144).

**Table 169.** The held-out movability test under every variant (15 tasks unless stated). Source: record audit `tables/table160_variants.csv`.

| Variant | Tasks | Confusion table (rows: predicted obey, predicted fail; columns: observed obey, observed fail) | Balanced accuracy [95%] | Fisher p | Spearman | One-sided p | Kendall | Verdict under the frozen rule |
|---|---|---|---|---|---|---|---|---|
| primary: catalogue-only avoidance | 15 | [[3, 3], [1, 8]] | 0.739 [0.42, 0.95] | 0.235 | 0.537 | 0.025 | 0.470 | INCONCLUSIVE |
| sensitivity: drop in default share | 15 | [[2, 4], [2, 7]] | 0.568 [0.25, 0.88] | 1.000 | 0.391 | 0.076 | 0.332 | DISCONFIRMED |
| without the grid and shortest path tasks (avoidance) | 13 | [[1, 3], [1, 8]] | 0.614 [0.27, 0.95] | 1.000 | 0.313 | 0.144 | 0.293 | INCONCLUSIVE |
| descriptive: without the task that has no default (avoidance) | 14 | [[2, 3], [1, 8]] | 0.697 [0.33, 0.96] | 0.505 | 0.484 | 0.048 | 0.434 | INCONCLUSIVE |
| descriptive: without the task that has no default (drop) | 14 | [[2, 3], [2, 7]] | 0.600 [0.29, 0.91] | 0.580 | 0.360 | 0.106 | 0.305 | DISCONFIRMED |
| descriptive: without the task that has no default and the task whose rule default differs from the frozen one (avoidance) | 13 | [[2, 3], [1, 7]] | 0.683 [0.32, 0.95] | 0.510 | 0.468 | 0.062 | 0.417 | INCONCLUSIVE |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| descriptive: same subset (drop) | 13 | [[2, 3], [2, 6]] | 0.583 [0.28, 0.89] | 1.000 | 0.316 | 0.149 | 0.260 | DISCONFIRMED | |

**Table 170.** The Gemma twin, the frozen default and the iteration-5 default rule per task, with both compliance measures under the modal ban. Source: record audit `tables/table159_grid.csv`.

| Task | Frozen default | Twin attempts (N = 50) | Twin modal correct (share) | Iteration-5 default rule | Catalogue-only avoidance | Drop in default share | Failing other programs under the ban | Escape fraction | Ban composition (N = 100) |
|---|---|---|---|---|---|---|---|---|---|
| grid | DFS | DFS 25, BFS 25 | DFS (0.59) | NO DEFAULT | 0.83 | 0.33 | 0 | 0.125 | BFS 83, DFS 17 |
| shortest path | Dijkstra | Dijkstra 50 | Dijkstra (1.00) | DEFAULT | 0.96 | 0.97 | 1 | 0.250 | Bellman-Ford/ SPFA 96, Dijkstra 3, other 1 |
| kth smallest | full sort | full sort 50 | full sort (1.00) | DEFAULT | 0.63 | 0.63 | 0 | 0.000 | size-k heap 61, full sort 37, quickselect 1, heapify and pop 1 |
| substring search | naive | naive 46, built-in find 4 | naive (0.92) | DEFAULT | 0.19 | 0.16 | 5 | 0.250 | naive 76, built-in find 13, Rabin-Karp 6, other 5 |
| find duplicate | hash set | hash set 50 | hash set (1.00) | DEFAULT | 0.34 | 0.72 | 38 | 0.000 | other 38, hash set 28, sort and adjacent check 21, index marking 7, Floyd cycle detection 6 |
| two sum | hash map | hash map 50 | hash map (1.00) | DEFAULT | 0.59 | 0.59 | 0 | 0.250 | brute force 59, hash map 41 |
| connected components | DFS | DFS 50 | DFS (1.00) | DEFAULT | 0.00 | 0.00 | 0 | 0.125 | DFS 100 |
| nonoverlapping intervals | greedy by end | greedy by end 50 | greedy by end (1.00) | DEFAULT | 0.00 | 0.00 | 0 | 0.000 | greedy by end 100 |
| Fibonacci | iteration | iteration 50 | iteration (1.00) | DEFAULT | 0.00 | 0.13 | 12 | 0.000 | iteration 87, other 12, disputed 1 |
| topological order | Kahn | Kahn 50 | Kahn (1.00) | DEFAULT | 0.00 | 0.00 | 0 | 0.000 | Kahn 100 |
| merge intervals | sort and sweep | sort and sweep 50 | sort and sweep (1.00) | DEFAULT | 0.00 | 0.00 | 0 | 0.000 | sort and sweep 100 |
| first and last position | binary search | linear scan 43, binary search 7 | linear scan (0.86) | DEFAULT (differs from the frozen default) | 0.00 | -0.85 | 0 | 0.000 | binary search 99, binary expand 1 |
| top-k frequent | sort by count | sort by count 50 | sort by count (1.00) | DEFAULT | 0.00 | 0.00 | 0 | 0.000 | sort by count 100 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| longest increasing subsequence | patience sorting with bisect | patience sorting with bisect 50 | patience sorting with bisect (1.00) | DEFAULT | 0.00 | 0.00 | 0 | 0.125 | patience sorting with bisect 100 |
| cycle detection | DFS with parent | DFS with parent 50 | DFS with parent (1.00) | DEFAULT | 0.00 | 0.03 | 0 | 0.000 | DFS with parent 97, disputed 3 |

**Landing forecasts on matched events.** The iteration-4 record compared the push forecast's mean over its 3 defined events with the association mean over 5. Table 171 sets every forecast on the same three event sets. On the 3 events where this run's push forecast is defined, it scores -0.76 against -3.11 for association; over all 5 gated events, with an undefined forecast scored as uniform, it scores -1.37 against -3.46. The review's -1.37 is this 5-event value, and it coincides at two decimals with a different cell, the screen push's route replication on its 4 defined events (-1.369).

**Table 171.** Mean log score of each forecast on matched event sets of the five gated iteration-3 bench bans, with top-1 hits. Source: record audit `tables/table157_matched.csv`.

| Forecast | 3 events where this run's push is defined: bench ban (top-1) | same, route full-ban replication | 4 events where the screen push is defined: bench ban (top-1) | same, route replication | all 5 gated events, undefined scored as uniform: bench ban (top-1) | same, route replication |
|---|---|---|---|---|---|---|
| push landing of this run's confirmed escapes | -0.761 (2 of 3) | -0.705 | -0.922 (2 of 3) | -0.888 | -1.366 (2 of 3) | -1.330 |
| push landing of the iteration-3 screen | -1.396 (2 of 3) | -1.280 | -1.480 (2 of 4) | -1.369 | -1.812 (2 of 4) | -1.715 |
| opening ladder | -1.613 (2 of 3) | -1.370 | -2.243 (2 of 4) | -2.071 | -2.391 (2 of 5) | -2.243 |
| association | -3.115 (1 of 3) | -2.857 | -3.363 (1 of 4) | -3.223 | -3.462 (1 of 5) | -3.350 |
| proximity | -2.918 (1 of 3) | -2.903 | -3.270 (1 of 4) | -3.271 | -3.212 (1 of 5) | -3.203 |

**Exposure arms as controls.** Table 172 gives the per-task verdict with exposure arms counted as controls, read from the plan noise experiment's knob rows (one engine) with the iteration-3 vetoes and the record's ban rows carried. Four task verdicts of Table 123 had rested on an exposure arm. Connected components keeps a qualifying knob (plan noise at alpha 2, 2.10 at no cost); longest increasing subsequence, nonoverlapping intervals and first and last position join the tasks on which no knob outside the exposure class reaches 1.5 within 20 pass points.

**Table 172.** Best knob outside the exposure class per task. Source: record audit `tables/table123_exposure.csv` (knob rows of the plan noise experiment).

| Task | Best knob outside the exposure class within 5 points | Best within 20 points | Best within 20 points, same engine only | A knob outside the exposure class reaches 1.5 within 20 points | Record verdict that rested on an exposure arm (now a control) |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| grid | plan noise alpha 2 1.42 at 2 points | novelty instruction 1.84 at 14 points | novelty instruction 1.84 at 14 points | yes | unchanged |
| shortest path | semantic seed 1.95 at 4 points | semantic seed 1.95 at 4 points | semantic seed 1.95 at 4 points | yes | unchanged |
| kth smallest | plan noise alpha 1 2.14 at 0 points | plan noise alpha 2 2.87 at 6 points | plan noise alpha 2 2.87 at 6 points | yes | unchanged |
| substring search | novelty instruction 2.62 at 4 points | novelty instruction 2.62 at 4 points | novelty instruction 2.62 at 4 points | yes | unchanged |
| two sum | novelty instruction 1.26 at 2 points | plan noise alpha 4 1.79 at 20 points | plan noise alpha 4 1.79 at 20 points | yes | unchanged |
| find duplicate | first iterated veto round 2.50 at 2 points | first iterated veto round 2.50 at 2 points | plan noise alpha 0.5 1.00 at 0 points | yes | unchanged |
| Fibonacci | plan noise alpha 0.5 1.00 at 0 points | plan noise alpha 0.5 1.00 at 0 points | plan noise alpha 0.5 1.00 at 0 points | no | unchanged |
| connected components | plan noise alpha 2 2.10 at 0 points | plan noise alpha 2 2.10 at 0 points | plan noise alpha 2 2.10 at 0 points | yes | unnamed exposure ban of BFS 1.85 at 0 |
| longest increasing subsequence | plan noise alpha 0.5 1.00 at 0 points | plan noise alpha 0.5 1.00 at 0 points | plan noise alpha 0.5 1.00 at 0 points | no | named exposure ban of the Fenwick tree 1.78 at 4 |
| merge intervals | plan noise alpha 0.5 1.00 at 0 points | plan noise alpha 0.5 1.00 at 0 points | plan noise alpha 0.5 1.00 at 0 points | no | unchanged |
| top-k frequent | plan noise alpha 1 1.15 at 2 points | plan noise alpha 2 1.17 at 10 points | plan noise alpha 2 1.17 at 10 points | no | unchanged |
| nonoverlapping intervals | plan noise alpha 0.5 1.00 at 0 points | plan noise alpha 0.5 1.00 at 0 points | plan noise alpha 0.5 1.00 at 0 points | no | named exposure ban of the DAG longest path 1.26 at 6 |
| cycle detection | novelty instruction 1.38 at -6 points | novelty instruction 1.38 at -6 points | novelty instruction 1.38 at -6 points | no | unchanged |
| topological order | plan noise alpha 0.5 1.00 at 0 points | plan noise alpha 0.5 1.00 at 0 points | plan noise alpha 0.5 1.00 at 0 points | no | unchanged |
| first and last position | temperature 1.6 1.43 at -4 points | temperature 1.6 1.43 at -4 points | temperature 1.6 1.43 at -4 points | no | exposure veto of exponential search 1.78 at 0 and 1.84 at 8 (round 3) |

**Executed results the record had omitted.** Plan noise on the plain prompt, without the prohibition frame of the twin, ran on the same slots and directions as plan noise under the twin frame (Table 173). The frame is not needed on kth smallest, connected components and the shortest path task, and on connected components the plain prompt moved more programs (0.24

against 0.12). The dose units of the experiments differ (Table 174): the ban-free unit is 0.30 to 0.97 of the ban unit across tasks as a ratio of mean norms (Table 174); the knob experiment's mean of per-position ratios is larger (0.83 to 1.39 on the six decisive tasks: kth smallest 1.394, connected components 0.990, shortest path 0.889, two sum 0.995, grid 0.874, find duplicate 0.829), because a mean of ratios is not a ratio of means. By the ratio of means the shortfall is at most about 25% on grid (0.751), shortest path (0.834), kth smallest (0.974), two sum (0.777), connected components (0.745), but 38% on find duplicate (0.620) and first and last position (0.618), 30% on top-k frequent (0.699) and 70% on substring search (0.296) (iteration-6 correction; record audit, Table 227), and substring search, the task that never moves, has by far the largest ban unit. A check made after the rivals were frozen found that the pooled escape fraction does not track the ratio of ban unit to residual norm (Spearman -0.06, AUC 0.49), so movability is not an artefact of the dose unit on the screen tasks.

**Table 173.** Plan noise at alpha 1 on the plain prompt and under the twin frame, same slots and directions. Source: record audit `tables/pnplain.csv`.

| Task | Arm (alpha 1, same slots and direction) | N | Pass | Correct programs | Off-default share | Hill-1 (depth 25) |
|---|---|---|---|---|---|---|
| cycle detection | plan noise, plain prompt | 50 | 0.76 | DFS with parent 38 | 0.00 | 1.00 |
| cycle detection | plan noise, twin frame | 50 | 0.90 | DFS with parent 45 | 0.00 | 1.00 |
| kth smallest | plan noise, plain prompt | 50 | 1.00 | full sort 42, quickselect 5, size-k heap 3 | 0.16 | 1.70 |
| kth smallest | plan noise, twin frame | 50 | 1.00 | full sort 36, quickselect 9, size-k heap 5 | 0.28 | 2.14 |
| two sum | plan noise, plain prompt | 50 | 1.00 | hash map 50 | 0.00 | 1.00 |
| two sum | plan noise, twin frame | 50 | 1.00 | hash map 50 | 0.00 | 1.00 |
| find duplicate | plan noise, plain prompt | 50 | 1.00 | hash set 50 | 0.00 | 1.00 |
| find duplicate | plan noise, twin frame | 50 | 1.00 | hash set 50 | 0.00 | 1.00 |
| grid | plan noise, plain prompt | 50 | 0.80 | BFS 40 | 0.02 | 1.00 |
| grid | plan noise, twin frame | 50 | 0.70 | BFS 34, DFS 1 | 0.06 | 1.13 |
| connected components | plan noise, plain prompt | 50 | 1.00 | union find 38, DFS 12 | 0.24 | 1.72 |
| connected components | plan noise, twin frame | 50 | 1.00 | union find 44, DFS 6 | 0.12 | 1.43 |
| shortest path | plan noise, plain prompt | 50 | 0.86 | Dijkstra 36, Bellman-Ford/SPFA 7 | 0.26 | 1.55 |
| shortest path | plan noise, twin frame | 50 | 0.86 | Dijkstra 34, Bellman-Ford/SPFA 9 | 0.26 | 1.66 |

**Table 174.** Dose units per task: the ban unit (mean norm of the ban displacement), the ratio of the ban-free unit to it, and both relative to the residual norm. Source: record audit `tables/dose_units.csv`.

| Task | Ban unit, mean norm | Ban-free unit / ban unit | Ban unit / residual norm, blocks 12-23 (site experiment) | Ban unit / residual norm, blocks 0-11 (site experiment) | Ban unit / residual norm, Qwen median (held-out experiment) | Ban-free unit / residual norm |
|---|---|---|---|---|---|---|
| grid | 15.26 | 0.75 | 0.140 | 0.060 | 0.119 | 0.105 |
| shortest path | 18.98 | 0.83 | 0.149 | 0.057 | 0.117 | 0.124 |
| kth smallest | 11.58 | 0.97 | 0.127 | 0.045 | 0.082 | 0.123 |

| | | | | | | |
|---|---|---|---|---|---|---|
| substring search | 58.51 | 0.30 | n/a | n/a | 0.113 | n/a |
| two sum | 14.74 | 0.78 | 0.150 | 0.063 | 0.108 | 0.117 |
| find duplicate | 18.67 | 0.62 | 0.183 | 0.072 | 0.145 | 0.113 |
| Fibonacci | 12.88 | 0.88 | n/a | n/a | 0.086 | n/a |
| connected components | 15.12 | 0.75 | 0.154 | 0.066 | 0.112 | 0.115 |
| longest increasing subsequence | 14.32 | 0.80 | n/a | n/a | 0.100 | n/a |
| merge intervals | 11.93 | 0.95 | n/a | n/a | 0.082 | n/a |
| top-k frequent | 16.19 | 0.70 | n/a | n/a | 0.123 | n/a |
| nonoverlapping intervals | 14.90 | 0.76 | n/a | n/a | 0.094 | n/a |
| cycle detection | n/a | n/a | n/a | n/a | n/a | n/a |
| topological order | 13.19 | 0.86 | n/a | n/a | 0.098 | n/a |
| first and last position | 18.58 | 0.62 | n/a | n/a | 0.133 | n/a |

**Distributions, not best cells.** Table 175 summarises the 624 cells of single directions in the escape screens as quantiles, so that a best cell can be read beside the direction it came from. On kth smallest at alpha 1 the median direction moves 3% of programs off the default and the maximum moves all of them; on two sum 0 of 40 directions escape at alpha 1 and 5 of 40 at alpha 2, with a median of 0. Table 176 gives the template split-half reliability of the direction ranking on all six tasks; the grid task (0.39 with 22 movers) and find duplicate (0.59 with 3 movers) were missing from the record. Table 177 gives signed progress toward the runner-up per task: 22 of the 24 realised-progress cells with uncorrected p below 0.05 on connected components, but 1 on kth smallest, whose mean over the edited blocks is -0.02, and 0 on find duplicate, whose mean is -0.12. On connected components the committed signed component (0.33) was matched by the within-class control (0.37). Signed progress is closed. Table 178 gives the number of correct programs behind each cell of Table 147: the alpha-4 peaks on two sum, find duplicate and connected components rest on 9, 5 and 7 programs.

**Table 175.** Escape screens per direction as distributions: directions run, escaping, movers, and quantiles over directions of the off-default share and of correct-only Hill-1. Source: record audit `tables/direction_quantiles.csv` (rows per direction in `tables/direction_distributions.csv`).

| Task | Alpha | Directions | Escaping | Movers (any off-default attempt) | Off-default share: median / q75 / max | Correct-only Hill-1: median / q75 / max |
|---|---|---|---|---|---|---|
| nonoverlapping intervals | 1 | 8 | 0 | 0 | 0.00 / 0.00 / 0.00 | 1.00 / 1.00 / 1.00 |
| nonoverlapping intervals | 2 | 8 | 0 | 0 | 0.00 / 0.00 / 0.00 | 1.00 / 1.00 / 1.00 |
| cycle detection | 1 | 8 | 0 | 0 | 0.00 / 0.00 / 0.00 | 1.00 / 1.00 / 1.00 |
| cycle detection | 2 | 8 | 0 | 2 | 0.00 / 0.03 / 0.10 | 1.00 / 1.00 / 1.51 |
| topological order | 1 | 8 | 0 | 0 | 0.00 / 0.00 / 0.00 | 1.00 / 1.00 / 1.00 |
| topological order | 2 | 8 | 0 | 0 | 0.00 / 0.00 / 0.00 | 1.00 / 1.00 / 1.00 |
| first and last position | 1 | 8 | 0 | 2 | 0.00 / 0.05 / 0.30 | 1.00 / 1.48 / 1.84 |

| | | | | | | |
|---|---|---|---|---|---|---|
| first and last position | 2 | 8 | 0 | 1 | 0.00 / 0.00 / 0.10 | 1.00 / 1.00 / 1.38 |
| kth smallest | 1 | 40 | 12 | 20 | 0.03 / 0.24 / 1.00 | 1.00 / 1.60 / 2.88 |
| kth smallest | 2 | 40 | 25 | 32 | 0.23 / 0.47 / 1.00 | 1.72 / 2.05 / 2.94 |
| substring search | 1 | 8 | 0 | 0 | 0.00 / 0.00 / 0.00 | 1.00 / 1.00 / 1.00 |
| substring search | 2 | 8 | 0 | 0 | 0.00 / 0.00 / 0.00 | 1.00 / 1.00 / 1.00 |
| two sum | 1 | 40 | 0 | 6 | 0.00 / 0.00 / 0.10 | 1.00 / 1.00 / 1.38 |
| two sum | 2 | 40 | 5 | 9 | 0.00 / 0.00 / 0.97 | 1.00 / 1.04 / 1.98 |
| find duplicate | 1 | 40 | 0 | 0 | 0.00 / 0.00 / 0.00 | 1.00 / 1.00 / 1.00 |
| find duplicate | 2 | 40 | 0 | 2 | 0.00 / 0.00 / 0.40 | 1.00 / 1.00 / 1.89 |
| Fibonacci | 1 | 8 | 0 | 0 | 0.00 / 0.00 / 0.00 | 1.00 / 1.00 / 1.00 |
| Fibonacci | 2 | 8 | 0 | 0 | 0.00 / 0.00 / 0.00 | 1.00 / 1.00 / 1.00 |
| grid | 1 | 40 | 2 | 7 | 0.00 / 0.00 / 0.10 | 1.00 / 1.00 / 1.55 |
| grid | 2 | 40 | 5 | 7 | 0.00 / 0.00 / 0.37 | 1.00 / 1.00 / 1.97 |
| connected components | 1 | 40 | 15 | 22 | 0.10 / 0.33 / 0.93 | 1.38 / 1.74 / 2.60 |
| connected components | 2 | 40 | 29 | 35 | 0.58 / 0.87 / 1.00 | 1.68 / 1.93 / 2.78 |
| longest increasing subsequence | 1 | 8 | 0 | 0 | 0.00 / 0.00 / 0.00 | 1.00 / 1.00 / 1.00 |
| longest increasing subsequence | 2 | 8 | 0 | 0 | 0.00 / 0.00 / 0.00 | 1.00 / 1.00 / 1.00 |
| merge intervals | 1 | 8 | 0 | 0 | 0.00 / 0.00 / 0.00 | 1.00 / 1.00 / 1.00 |
| merge intervals | 2 | 8 | 0 | 0 | 0.00 / 0.00 / 0.00 | 1.00 / 1.00 / 1.00 |
| top-k frequent | 1 | 8 | 0 | 0 | 0.00 / 0.00 / 0.00 | 1.00 / 1.00 / 1.00 |
| top-k frequent | 2 | 8 | 0 | 2 | 0.00 / 0.02 / 0.10 | 1.00 / 1.10 / 1.78 |
| shortest path | 1 | 40 | 7 | 17 | 0.00 / 0.10 / 0.67 | 1.00 / 1.44 / 1.99 |
| shortest path | 2 | 40 | 8 | 29 | 0.13 / 0.33 / 0.63 | 1.51 / 1.86 / 2.29 |

**Table 176.** Template split-half reliability of the ranking of 40 fresh directions by off-default share, on all six tasks. Source: record audit `tables/split_half.csv`.

| Task | Template split-half Spearman over 40 directions | Placebo | Movers (any alpha) | Escaping cells (direction and dose) |
|---|---|---|---|---|
| kth smallest | 0.80 | -0.06 | 34 | 27 |
| shortest path | 0.81 | 0.16 | 37 | 14 |
| two sum | 0.76 | -0.06 | 10 | 5 |
| connected components | 0.90 | 0.13 | 36 | 30 |
| find duplicate | 0.59 | -0.05 | 3 | 0 |
| grid | 0.39 | -0.05 | 22 | 6 |

**Table 177.** Signed progress toward the runner-up per task, over 24 block cells (blocks 12 to 35). Source: record audit `tables/signed_progress.csv`.

| Task | Quantity | Runner-up | Cells with p < 0.05 (uncorrected) | Mean, all blocks | Mean, edited blocks 12-23 | Minimum (block) | Largest correlation |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| kth smallest | signed progress toward the runner-up (realised) | quickselect | 1 of 24 | 0.08 | -0.02 | -0.29 (block 15) | 0.26 |
| kth smallest | signed progress toward the runner-up (applied) | quickselect | 5 of 24 | 0.14 | 0.02 | -0.26 (block 15) | 0.26 |
| shortest path | signed progress toward the runner-up (realised) | Bellman-Ford/SPFA | 3 of 24 | 0.15 | 0.20 | -0.06 (block 35) | 0.30 |
| shortest path | signed progress toward the runner-up (applied) | Bellman-Ford/SPFA | 0 of 24 | 0.05 | 0.03 | -0.13 (block 17) | 0.30 |
| two sum | signed progress toward the runner-up (realised) | brute force | 7 of 24 | 0.20 | 0.31 | -0.27 (block 35) | 0.47 |
| two sum | signed progress toward the runner-up (applied) | brute force | 6 of 24 | 0.18 | 0.25 | -0.20 (block 35) | 0.47 |
| connected components | signed progress toward the runner-up (realised) | DFS | 22 of 24 | 0.39 | 0.37 | 0.16 (block 35) | 0.57 |
| connected components | signed progress toward the runner-up (applied) | DFS | 23 of 24 | 0.49 | 0.41 | 0.21 (block 35) | 0.57 |
| find duplicate | signed progress toward the runner-up (realised) | sort and adjacent check | 0 of 24 | 0.02 | -0.12 | -0.27 (block 16) | 0.22 |
| find duplicate | signed progress toward the runner-up (applied) | sort and adjacent check | 0 of 24 | 0.05 | -0.07 | -0.21 (block 16) | 0.22 |
| grid | signed progress toward the runner-up (realised) | DFS | 2 of 24 | 0.12 | 0.05 | -0.12 (block 12) | 0.48 |
| grid | signed progress toward the runner-up (applied) | DFS | 12 of 24 | 0.22 | 0.12 | -0.02 (block 13) | 0.48 |

**Table 178.** Correct programs behind each cell of Table 147 and the depth at which Hill-1 was rarefied. Source: record audit `tables/table147_depths.csv`.

| Task | Alpha 0 | Alpha 1 | Alpha 2 | Alpha 4 | Alpha 8 |
|---|---|---|---|---|---|
| kth smallest | 50 of 50 (depth 25) | 30 of 30 (depth 25) | 28 of 30 (depth 25) | 20 of 30 (depth 20) | 0 of 50 (depth 0) |
| two sum | 50 of 50 (depth 25) | 30 of 30 (depth 25) | 29 of 30 (depth 25) | 9 of 30 (depth 9) | 0 of 50 (depth 0) |
| find duplicate | 50 of 50 (depth 25) | 30 of 30 (depth 25) | 30 of 30 (depth 25) | 5 of 30 (depth 5) | 0 of 50 (depth 0) |
| connected components | 50 of 50 (depth 25) | 30 of 30 (depth 25) | 29 of 30 (depth 25) | 7 of 30 (depth 7) | 0 of 50 (depth 0) |
| grid | 37 of 50 (depth 25) | 36 of 50 (depth 25) | 34 of 50 (depth 25) | 6 of 50 (depth 6) | 0 of 50 (depth 0) |
| shortest path | 50 of 50 (depth 25) | 30 of 30 (depth 25) | 27 of 30 (depth 25) | 1 of 30 (depth 1) | 0 of 50 (depth 0) |

**Rival task predictors, frozen before any target was read.** For the movability half to mean anything, beta had to beat simpler explanations of why a task moves. The audit computed eight ban-free predictors for the 15 existing tasks and catalogue size as a covariate, wrote them to a canonical file, and froze them by sha256 in a hash chain before opening any outcome file (frozen at 11:59:44 UTC, first scored at 11:59:50; two later freezes left the values identical). The predictors are the twin's opening margin (the margin in log probability, under teacher forcing, of the opening of the default over that of the runner-up), the forced can rate, association clarity, the off-default excess under the novelty instruction, the diversity of human solutions (Mercury solutions labelled by the frozen rules with a judge on unknowns; audit agreement 0.97, kappa 0.97, 121 programs), the runtime merit gap between the reference programs of the runner-up and of the default (76 references timed on their hidden tests), the twin modal share and the ratio of ban unit to residual norm. Five of them (the opening margin, the novelty excess, the merit gap, the twin modal share and catalogue size) can be computed on new tasks and were frozen for the confirmation panels. Scored against the pooled escape fraction of the screen, none is informative after Holm correction (every adjusted p is 1.0), and 10 of the 15 tasks are tied at an escape fraction of 0 (Tables 179 to 181). These scores are descriptive: the escape fractions generated the hypothesis, so the freeze protects only the confirmation on new tasks.

**Table 179.** The frozen rival predictors per task, with the screen's escape fraction and plan noise excess. Source: record audit `tables/rivals_values.csv`.

| Task | Opening margin | Forced can rate | Association clarity | Novelty instruction excess | Diversity of human solutions | Runtime merit gap | Twin modal share | Ban unit ratio | Catalogue size | Escape fraction (pooled) | Plan noise excess | Screen movable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| grid | 1.20 | n/a | 1.00 | 0.30 | n/a | 0.94 | 1.00 | 0.75 | 6 | 0.087 | 0.24 | yes |
| shortest path | 0.00 | n/a | 0.75 | 0.06 | 2.50 | 1.01 | 1.00 | 0.83 | 5 | 0.188 | 0.54 | yes |
| kth smallest | 7.44 | 0.95 | 1.00 | 1.00 | 4.82 | 5.67 | 0.98 | 0.97 | 6 | 0.463 | 0.50 | yes |
| substring search | 3.45 | 1.00 | 1.00 | 0.48 | 1.99 | 0.03 | 1.00 | 0.30 | 6 | 0.000 | 0.00 | no |
| two sum | 14.05 | 1.00 | 1.00 | 0.06 | 1.83 | 2.44 | 1.00 | 0.78 | 4 | 0.062 | 0.02 | yes |
| find duplicate | 12.06 | 0.78 | 1.00 | 0.00 | 3.38 | 3.60 | 1.00 | 0.62 | 5 | 0.000 | 0.02 | no |
| Fibonacci | 14.68 | 1.00 | 1.00 | 0.09 | 1.98 | 1.67 | 1.00 | 0.88 | 4 | 0.000 | 0.02 | no |
| connected components | 21.79 | 1.00 | 1.00 | 0.04 | 2.91 | 1.09 | 1.00 | 0.75 | 4 | 0.550 | 0.60 | yes |
| longest increasing subsequence | 16.70 | 1.00 | 1.00 | 0.09 | 1.99 | 0.01 | 1.00 | 0.80 | 3 | 0.000 | 0.08 | no |
| merge intervals | 25.95 | 0.05 | 0.95 | 0.00 | 1.24 | 60.62 | 1.00 | 0.95 | 3 | 0.000 | 0.00 | no |
| top-k frequent | 34.17 | 0.53 | 1.00 | 0.00 | 3.20 | 1.06 | 1.00 | 0.70 | 4 | 0.000 | 0.08 | no |
| nonoverlapping intervals | 1.89 | 0.10 | 1.00 | 0.00 | 1.00 | 3.31 | 1.00 | 0.76 | 4 | 0.000 | 0.00 | no |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cycle detection | n/a | 1.00 | 1.00 | 0.10 | 2.43 | 0.69 | 1.00 | n/a | 4 | 0.000 | 0.02 | no |
| topological order | 23.38 | 0.00 | 1.00 | 0.00 | 2.39 | 1.45 | 1.00 | 0.86 | 4 | 0.000 | 0.00 | no |
| first and last position | 9.40 | 0.97 | 1.00 | -0.05 | 1.99 | 0.46 | 0.91 | 0.62 | 3 | 0.000 | -0.04 | no |

**Table 180.** The rival predictors scored against the screen's pooled escape fraction (descriptive; the predicted sign is the direction the rival should take if it explains movability). Source: record audit `results/rival_scores.json`, `tables/rivals_scores.csv`.

| Rival | Frozen for new tasks | Tasks | Predicted sign | Spearman with pooled escape fraction [95% bootstrap] | Oriented | Permutation p (two sided) | Oriented Spearman, alpha 1 / alpha 2 | Oriented Spearman with plan noise excess | Partial given catalogue size | AUC for screen movable | Holm p | Flag |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| catalogue size (covariate) | yes | 15 | + | +0.46 [0.00, 0.81] | +0.46 | 0.09 | +0.49 / +0.44 | +0.40 | n/a | 0.77 | n/a | |
| diversity of human solutions | no | 14 | + | +0.39 [-0.28, 0.79] | +0.39 | 0.18 | +0.54 / +0.39 | +0.67 | 0.24 | 0.70 | 1.0 | |
| novelty instruction excess | yes | 15 | + | +0.38 [-0.15, 0.78] | +0.38 | 0.16 | +0.36 / +0.37 | +0.42 | 0.15 | 0.74 | 1.0 | |
| forced can rate | no | 13 | + | +0.30 [-0.17, 0.67] | +0.30 | 0.31 | +0.17 / +0.30 | +0.35 | 0.29 | 0.70 | 1.0 | |
| ban unit ratio (dose check) | no | 14 | - | +0.21 [-0.36, 0.69] | -0.21 | 0.48 | -0.20 / -0.21 | -0.19 | 0.29 | 0.38 | 1.0 | |
| runtime merit gap | yes | 15 | - | +0.14 [-0.36, 0.60] | -0.14 | 0.62 | -0.07 / -0.15 | +0.02 | 0.12 | 0.42 | 1.0 | |
| twin modal share | yes | 15 | - | -0.16 [-0.73, 0.31] | +0.16 | 0.76 | +0.20 / +0.16 | -0.13 | -0.20 | 0.54 | 1.0 | uninformative (< = 3 distinct values) |
| association clarity | no | 15 | + | -0.16 [-0.72, 0.31] | -0.16 | 0.71 | -0.20 / -0.16 | -0.10 | -0.23 | 0.44 | 1.0 | uninformative (< = 3 distinct values) |
| twin opening margin | yes | 14 | - | -0.33 [-0.78, 0.25] | +0.33 | 0.25 | +0.34 / +0.32 | +0.10 | -0.08 | 0.73 | 1.0 | |

**Table 181.** Spearman correlations among the rival predictors over the 15 existing tasks. Source: record audit `tables/rivals_intercorr.csv`.

| Rival | twin opening margin | forced can rate | association clarity | novelty instruction excess | diversity of human solutions | runtime merit gap | twin modal share | ban unit ratio (dose check) | catalogue size (covariate) |
|---|---|---|---|---|---|---|---|---|---|
| twin opening margin | 1.00 | -0.24 | 0.09 | -0.38 | 0.07 | 0.19 | 0.20 | 0.16 | -0.61 |
| forced can rate | -0.24 | 1.00 | 0.41 | 0.63 | 0.03 | -0.53 | 0.06 | -0.22 | 0.08 |
| association clarity | 0.09 | 0.41 | 1.00 | 0.17 | 0.12 | -0.20 | -0.15 | -0.39 | 0.09 |
| novelty instruction excess | -0.38 | 0.63 | 0.17 | 1.00 | 0.19 | -0.21 | 0.04 | 0.25 | 0.59 |
| diversity of human solutions | 0.07 | 0.03 | 0.12 | 0.19 | 1.00 | -0.05 | -0.17 | -0.05 | 0.52 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| runtime merit gap | 0.19 | -0.53 | -0.20 | -0.21 | -0.05 | 1.00 | -0.01 | 0.51 | 0.07 |
| twin modal share | 0.20 | 0.06 | -0.15 | 0.04 | -0.17 | -0.01 | 1.00 | -0.01 | 0.04 |
| ban unit ratio (dose check) | 0.16 | -0.22 | -0.39 | 0.25 | -0.05 | 0.51 | -0.01 | 1.00 | -0.11 |
| catalogue size (covariate) | -0.61 | 0.08 | 0.09 | 0.59 | 0.52 | 0.07 | 0.04 | -0.11 | 1.00 |

Catalogue size, the number of algorithms a task's catalogue admits, tracks the screen's movability as well as any rival ($+0.46$), which is a reminder that a task with more alternatives has more ways to move.

## 5.3 Transplanting the base model's plan states into the instruct model (Qwen3-8B)

**Why.** If post-training narrowed the choice at the plan site, the narrowing should be undone by giving the instruct model the base model's own task-specification states. No earlier iteration had tried this: iteration 1 subtracted an instruct minus base direction estimated at the read position only, which destroyed the output format (Section 1.9). This experiment was the first causal test of the transplant claim, and it also measured the base repertoire that every other Qwen3-8B analysis of the iteration needed.

**Design.** Qwen3-8B and Qwen3-8B-Base (revision 49e3418f, the base revision of iteration 1) ran in bf16 on one RTX 4090 with thinking off, temperature 1.0, top-p 1 and no top-k. The base model sampled 100 programs per task under the instruct chat template on identical token ids (200 on the two path tasks, with top-ups to 200 on connected components, nonoverlapping intervals and cycle detection) and 100 in plain completion. Its labels used the iteration-4 labellers with a blind judge on disagreement, and a 15-program judge audit per task agreed at least 0.86 everywhere. The base repertoire (default, beta with a 95% interval, stratum, base runner-up) was committed by hash before any hooked sample. The transplant set the hooked states to h_instruct + lambda (h_base − h_instruct) at blocks 12 to 23 for lambda in {0.25, 0.5, 1}; the states were set, not added, so nothing compounded, and at lambda 1 the hooked blocks carried the base state exactly. The controls on the same common-random-number slots were a matched-norm random displacement (per block and position), the borrowed shift (another task's mean base minus instruct direction at the same norm), and, where the random control broke the model, a random push matched on pass rate. Site controls transplanted at the read position, at blocks 0 to 11 and at blocks 24 to 35. Lambda was selected on one set of slots and confirmed on fresh slots, both under commits made before the samples.

**The base model is more diverse than the instruct model on most tasks** (Table 182). Under the same prompt the instruct twin wrote its default in 96% to 100% of programs on all 15 tasks, while 5 tasks were diverse in the base (kth smallest, shortest path, connected components, grid and top-k frequent), 7 intermediate and 3 collapsed in the base (merge intervals, nonoverlapping intervals and topological order). The base model under the chat template is a poor programmer: it wrote 14 to 55 correct programs per 100 to 200 samples, and on the grid task only 14 of 200, so the grid beta rests on attempts. On connected components it writes DFS more often than union find (32 against 19 correct programs), so the instruct default is the base model's second choice there.

**Table 182.** The base repertoire of the 15 existing tasks, committed before any transplant sample. Source: transplant experiment `results/base_repertoire.json`.

| Task | Instruct default (twin share) | Base correct / N | Base correct programs (chat template) | beta [95% CI] (basis) | Stratum (stability) | Base runner-up | beta, plain completion | Base Hill-1 (depth 25) |
|---|---|---|---|---|---|---|---|---|
| kth smallest | full sort (1.00) | 47 of 100 | full sort 26, quickselect 13, size-k heap 4, heapify and pop 3, other 1 | 0.43 [0.33, 0.49] (correct) | diverse in the base (1.00) | quickselect | 0.10 | 3.03 |
| shortest path | Dijkstra (1.00) | 44 of 200 | Dijkstra 25, Bellman-Ford/ SPFA 18, other 1 | 0.41 [0.33, 0.55] (correct) | diverse in the base (1.00) | Bellman-Ford/SPFA | 0.32 | 2.12 |
| two sum | hash map (1.00) | 49 of 100 | hash map 41, brute force 7, sort and two pointers 1 | 0.16 [0.12, 0.21] (correct) | intermediate (0.92) | brute force | 0.69 | 1.63 |
| connected components | union find (1.00) | 55 of 200 | DFS 32, union find 19, BFS 4 | 0.65 [0.58, 0.76] (correct) | diverse in the base (1.00) | DFS | 0.81 | 2.34 |
| find duplicate | hash set (1.00) | 51 of 100 | hash set 39, index marking 4, sort and adjacent check 3, Floyd cycle detection 2, other 2, value binary search 1 | 0.20 [0.10, 0.29] (correct) | intermediate (0.55) | index marking | 0.33 | 2.34 |
| grid | BFS (1.00) | 14 of 200 | BFS 10, DFS 4 | 0.38 [0.32, 0.44] (attempt) | diverse in the base (1.00) | DFS | 0.21 | 1.82 |
| substring search | naive (1.00) | 52 of 100 | naive 46, built-in find 6 | 0.12 [0.05, 0.16] (correct) | intermediate (0.99) | built-in find | 0.27 | 1.42 |
| Fibonacci | iteration (1.00) | 37 of 100 | iteration 31, memoised recursion 6 | 0.16 [0.09, 0.22] (correct) | intermediate (0.90) | memoised recursion | 0.08 | 1.55 |
| longest increasing subsequence | quadratic DP (1.00) | 43 of 100 | quadratic DP 39, patience sorting with bisect 4 | 0.09 [0.05, 0.12] (correct) | intermediate (0.99) | patience sorting with bisect | 0.00 | 1.36 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| merge intervals | sort and sweep (1.00) | 40 of 100 | sort and sweep 40 | 0.00 [0.00, 0.00] (correct) | collapsed in the base (1.00) | none | 0.00 | 1.00 |
| top-k frequent | sort by count (1.00) | 31 of 100 | sort by count 20, heap 11 | 0.35 [0.26, 0.41] (correct) | diverse in the base (1.00) | heap | 0.07 | 1.91 |
| nonoverlapping intervals | greedy by end (1.00) | 42 of 200 | greedy by end 42 | 0.00 [0.00, 0.00] (correct) | collapsed in the base (1.00) | none | 0.06 | 1.00 |
| topological order | Kahn (1.00) | 40 of 100 | Kahn 39, DFS postorder 1 | 0.03 [0.00, 0.07] (correct) | collapsed in the base (0.88) | DFS postorder | 0.39 | 1.12 |
| first and last position | binary search (0.96) | 40 of 100 | binary search 29, linear scan 8, bisect module 2, other 1 | 0.05 [0.00, 0.12] (correct) | intermediate (0.53) | bisect module | 0.00 | 2.17 |
| cycle detection | DFS with parent (1.00) | 43 of 200 | DFS with parent 36, union find 6, BFS with parent 1 | 0.16 [0.07, 0.23] (correct) | intermediate (0.80) | union find | 0.15 | 1.63 |

As a screen check, the committed beta ranks the 15 tasks almost exactly as iteration 4's random pushes could move them: Spearman 0.78 against the escape fraction at alpha 1 (one-sided permutation p 0.0002), 0.79 at alpha 2 and 0.79 on their mean, and the base runner-up equals the push's top-1 landing on 5 of 5 movable tasks. This is the observation the hypothesis was built on, measured again; it is not a test, because these tasks generated the hypothesis.

**Selection and confirmation.** Table 183 gives every arm of the selection stage. The transplant moved two tasks at an intact pass rate on the selection slots, top-k frequent at lambda 1 (0.47 of correct programs off the default, pass 0.98) and connected components at lambda 0.5 (0.12), and it broke the shortest path and grid tasks at lambda of 0.5 or more (pass 0.72 and 0.54, 0.50 and 0.30). On kth smallest the matched random displacement at lambda 1 moved more programs (0.34) than the transplant (0.06). Early-block transplants moved kth smallest (0.30) and connected components (0.28) more than the main band, the transplant at the read position broke the output on 13 of 15 tasks without moving an algorithm, and late-block transplants were inert.

**Table 183.** Selection stage on the first slots: pass rate / correct off-default share per arm (N = 50 for the transplant, 30 for the controls; T is the transplant at lambda, MR the matched random displacement, READ, EARLY and LATE the site controls at lambda 1). Source: transplant experiment `results/selection.json`.

| Task | stratum | twin | T(0.25) | T(0.5) | T(1) | MR(0.25) | MR(0.5) | MR(1) | READ(1) | EARLY(1) | LATE(1) | lambda* |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| kth smallest | diverse in the base | 1.00 / 0.00 | 1.00 / 0.02 | 1.00 / 0.00 | 1.00 / 0.06 | 1.00 / 0.10 | 1.00 / 0.07 | 0.97 / 0.34 | 0.97 / 0.00 | 1.00 / 0.30 | 1.00 / 0.04 | 1.00 |
| shortest path | diverse in the base | 0.98 / 0.00 | 1.00 / 0.00 | 0.72 / 0.06 | 0.54 / 0.04 | 0.97 / 0.03 | 0.93 / 0.07 | 0.90 / 0.00 | 0.00 / – | 0.98 / 0.00 | 0.88 / 0.02 | 0.25 |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| two sum | intermediate | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.03 | 0.43 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 0.25 |
| connected components | diverse in the base | 1.00 / 0.00 | 1.00 / 0.04 | 1.00 / 0.12 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.03 | 1.00 / 0.10 | 0.17 / 0.20 | 1.00 / 0.28 | 1.00 / 0.00 | 0.50 |
| find duplicate | intermediate | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.17 | 0.37 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 0.25 |
| grid | diverse in the base | 0.86 / 0.00 | 0.72 / 0.00 | 0.50 / 0.00 | 0.30 / 0.00 | 0.87 / 0.04 | 0.77 / 0.00 | 0.73 / 0.09 | 0.13 / 0.00 | 0.40 / 0.00 | 0.90 / 0.00 | 0.25 |
| substring search | intermediate | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 0.25 |
| Fibonacci | intermediate | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 0.97 / 0.00 | 0.90 / 0.00 | 0.37 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 0.25 |
| longest increasing subsequence | intermediate | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 0.93 / 0.04 | 0.53 / 0.06 | 1.00 / 0.00 | 1.00 / 0.00 | 0.25 |
| merge intervals | collapsed in the base | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | – | – | – | 0.37 / 0.00 | – | – | – |
| top-k frequent | diverse in the base | 0.78 / 0.00 | 0.78 / 0.00 | 0.68 / 0.00 | 0.98 / 0.47 | 0.77 / 0.00 | 0.80 / 0.08 | 0.53 / 0.19 | 0.07 / 0.00 | 0.98 / 0.00 | 0.72 / 0.00 | 1.00 |
| nonoverlapping intervals | collapsed in the base | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 0.40 / 0.00 | – | – | – | 0.30 / 0.00 | – | – | – |
| topological order | collapsed in the base | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | 1.00 / 0.00 | – | – | – | 0.13 / 0.00 | – | – | – |
| first and last position | intermediate | 0.90 / 0.00 | 0.88 / 0.00 | 0.90 / 0.00 | 0.64 / 0.00 | 0.83 / 0.00 | 0.80 / 0.00 | 0.87 / 0.00 | 0.23 / 0.00 | 0.96 / 0.00 | 0.94 / 0.00 | 0.25 |
| cycle detection | intermediate | 0.94 / 0.00 | 0.98 / 0.00 | 0.98 / 0.00 | 0.90 / 0.00 | 0.90 / 0.00 | 0.77 / 0.00 | 0.43 / 0.15 | 0.17 / 0.00 | 1.00 / 0.00 | 0.96 / 0.00 | 0.25 |

On fresh slots (Table 184) only top-k frequent passed: the transplant wrote the base runner-up, a heap, in 21 of 49 correct programs against 0 of 39 for the twin, beat all three controls at p of 7e-4 or less after Holm adjustment, and was ahead on 5 of 5 templates. On kth smallest the matched random displacement beat the transplant (2 flips by the transplant alone against 20 by the random displacement alone; exact test, two sided, p 1.2e-4): the base direction disturbs the plan less than an isotropic one of the same size. On connected components the transplant led both controls 3:0, short of significance, and on the shortest path and grid tasks the lambda that survives (0.25) moved nothing. The frozen rule needs a pass on at least 4 of the 5 tasks diverse in the base, so the transplant claim is DISCONFIRMED; the sensitivities restricted to the stable stratum and robust to noise agree. An extension blind to outcomes, to N = 100, on the decisive cells (Table 185) kept top-k frequent (44:0, 35:3 and 49:0 against the three controls) and added only raw significance against the borrowed shift on kth smallest (6:0, p 0.016, Holm 0.11) and connected components (7:0, p 0.0078, Holm 0.06); neither task beat the matched random control. Its verdict reads PARTIAL for a mechanical reason: at N = 100 the grid transplant's pass rate (0.68) fell below 0.8 times the twin's, which removes the precondition for DISCONFIRMED.

**Table 184.** Confirmation on fresh slots (N = 50 per cell; lambda* is the selected lambda; one-sided exact McNemar on correct off-default programs, transplant only : control only, with p after Holm adjustment; rows marked desc. are descriptive tasks outside the decisive set). Source: transplant

experiment `results/confirmation.json`.

| Task | lambda* | twin pass, fresh slots | T: pass / off-default share / modal | control: pass / off-default share | McNemar b:c, p (Holm) | templates T > control |
|---|---|---|---|---|---|---|
| kth smallest | 1.00 | 1.00 | 1.00 / 0.06 / size-k heap | matched random, lambda 1: 1.00 / 0.42 | 2:20, 1 (1) | 0/5 |
| kth smallest | 1.00 | 1.00 | 1.00 / 0.06 / size-k heap | borrowed shift, lambda 1: 1.00 / 0.00 | 3:0, 0.12 (1) | 2/5 |
| shortest path | 0.25 | 1.00 | 1.00 / 0.00 / none | matched random, 0.25: 0.98 / 0.00 | 0:0, 1 (1) | 0/5 |
| shortest path | 0.25 | 1.00 | 1.00 / 0.00 / none | borrowed shift, 0.25: 0.98 / 0.00 | 0:0, 1 (1) | 0/5 |
| connected components | 0.50 | 1.00 | 1.00 / 0.08 / DFS | matched random, 0.5: 1.00 / 0.02 | 3:0, 0.12 (1) | 3/5 |
| connected components | 0.50 | 1.00 | 1.00 / 0.08 / DFS | borrowed shift, 0.5: 1.00 / 0.02 | 3:0, 0.12 (1) | 3/5 |
| grid | 0.25 | 0.88 | 0.76 / 0.00 / none | matched random, 0.25: 0.94 / 0.00 | 0:0, 1 (1) | 0/5 |
| grid | 0.25 | 0.88 | 0.76 / 0.00 / none | borrowed shift, 0.25: 0.86 / 0.00 | 0:0, 1 (1) | 0/5 |
| top-k frequent | 1.00 | 0.78 | 0.98 / 0.43 / heap | matched random, lambda 1: 0.78 / 0.05 | 19:0, 1.9e-06 (1.9e-05) | 5/5 |
| top-k frequent | 1.00 | 0.78 | 0.98 / 0.43 / heap | borrowed shift, lambda 1: 0.88 / 0.11 | 17:1, 7.2e-05 (6.5e-04) | 5/5 |
| top-k frequent | 1.00 | 0.78 | 0.98 / 0.43 / heap | random matched on pass, 0.5: 0.76 / 0.00 | 21:0, 4.8e-07 (5.2e-06) | 5/5 |
| two sum (desc.) | 0.25 | 1.00 | 1.00 / 0.00 / none | matched random, 0.25: 1.00 / 0.00 | 0:0, 1 (–) | 0/5 |
| two sum (desc.) | 0.25 | 1.00 | 1.00 / 0.00 / none | borrowed shift, 0.25: 1.00 / 0.00 | 0:0, 1 (–) | 0/5 |
| find duplicate (desc.) | 0.25 | 1.00 | 1.00 / 0.00 / none | matched random, 0.25: 1.00 / 0.00 | 0:0, 1 (–) | 0/5 |
| find duplicate (desc.) | 0.25 | 1.00 | 1.00 / 0.00 / none | borrowed shift, 0.25: 1.00 / 0.00 | 0:0, 1 (–) | 0/5 |
| substring search (desc.) | 0.25 | 1.00 | 1.00 / 0.00 / none | matched random, 0.25: 1.00 / 0.00 | 0:0, 1 (–) | 0/5 |
| substring search (desc.) | 0.25 | 1.00 | 1.00 / 0.00 / none | borrowed shift, 0.25: 1.00 / 0.00 | 0:0, 1 (–) | 0/5 |
| Fibonacci (desc.) | 0.25 | 1.00 | 1.00 / 0.00 / none | matched random, 0.25: 1.00 / 0.00 | 0:0, 1 (–) | 0/5 |
| Fibonacci (desc.) | 0.25 | 1.00 | 1.00 / 0.00 / none | borrowed shift, 0.25: 1.00 / 0.00 | 0:0, 1 (–) | 0/5 |

| | | | | | | |
|---|---|---|---|---|---|---|
| longest increasing subsequence (desc.) | 0.25 | 1.00 | 1.00 / 0.00 / none | matched random, 0.25: 1.00 / 0.00 | 0:0, 1 (–) | 0/5 |
| longest increasing subsequence (desc.) | 0.25 | 1.00 | 1.00 / 0.00 / none | borrowed shift, 0.25: 1.00 / 0.00 | 0:0, 1 (–) | 0/5 |
| first and last position (desc.) | 0.25 | 0.76 | 0.78 / 0.00 / none | matched random, 0.25: 0.80 / 0.00 | 0:0, 1 (–) | 0/5 |
| first and last position (desc.) | 0.25 | 0.76 | 0.78 / 0.00 / none | borrowed shift, 0.25: 0.56 / 0.00 | 0:0, 1 (–) | 0/5 |
| cycle detection (desc.) | 0.25 | 0.84 | 0.90 / 0.00 / none | matched random, 0.25: 0.80 / 0.00 | 0:0, 1 (–) | 0/5 |
| cycle detection (desc.) | 0.25 | 0.84 | 0.90 / 0.00 / none | borrowed shift, 0.25: 0.86 / 0.00 | 0:0, 1 (–) | 0/5 |

**Table 185.** Extension blind to outcomes, to N = 100, on the decisive cells. Source: transplant experiment `results/confirmation.json#power_extension_N100`.

| Task | Transplant pass | Correct off-default share | Against matched random (transplant only : random only, one-sided p) | Against borrowed shift | Against random matched on pass |
|---|---|---|---|---|---|
| kth smallest | 1.00 | 0.06 | 4:36, p = 1 | 6:0, p = 0.016 | n/a |
| shortest path | 1.00 | 0.00 | 0:0, p = 1 | 0:0, p = 1 | n/a |
| connected components | 1.00 | 0.08 | 6:2, p = 0.14 | 7:0, p = 0.0078 | n/a |
| grid | 0.68 | 0.00 | 0:0, p = 1 | 0:0, p = 1 | n/a |
| top-k frequent | 0.99 | 0.49 | 44:0, p = 5.7e-14 | 35:3, p = 3.3e-08 | 49:0, p = 1.8e-15 |

Top-k frequent is the one task on which the base state works and random noise does not: it is diverse in the base (beta 0.35) and iteration 4's random pushes never moved it. Its transplant also raised the pass rate from 0.78 to 0.98, so part of what the transplant restores may be a more correct program rather than a reopened choice; the experiment did not separate the two. Tasks collapsed in the base did not move under the full transplant (change 0.00). The transplant at the read position shifted the off-default share by more than 0.05 on connected components and longest increasing subsequence, but only among the few programs that still passed (5 and 16 correct of 30) and never significantly, so the specificity check on point estimates fails while its version robust to noise holds.

**The push toward the base and the reverse transplant.** A push along the unit base minus instruct direction never escaped the default on any of the four tasks at doses of 0.5 to 2 ban-free units, while an isotropic push at the same doses escaped on kth smallest at alpha 2 (Table 186). The claim for the push toward the base is DISCONFIRMED. The reverse interchange wrote the instruct model's task-specification states into the base model (Table 187). It raised the base model's correct program counts on all four tasks (kth smallest 20 to 31, shortest path 15 to 20, grid 5 to 11, connected components 18 to 19) and its default share among attempts, while a matched random displacement lowered the counts; the committed prediction (the default share among correct programs rises by at least half the instruct minus base gap) was met only on the grid task (0.80 to 1.00; one-sided p 0.033). Control is asymmetric: instruct states pull the base model toward the instruct default and toward correct code, but base states do not pull the instruct model back.

**Table 186.** Push toward the base along unit(D) against an isotropic push at the same dose (ban-free units, N = 30; pass, correct off-default share). Source: transplant experiment `results/desharpen.json`.

| Task | twin pass | push along unit(D), alpha 0.5 / 1 / 2 (pass, off-default share) | isotropic push, alpha 0.5 / 1 / 2 (pass, off-default share) | escape dose, unit(D) | escape dose, isotropic |
|---|---|---|---|---|---|
| kth smallest | 1.00 | 1.00, 0.00 ; 1.00, 0.00 ; 1.00, 0.00 | 1.00, 0.00 ; 1.00, 0.13 ; 1.00, 0.27 | – | 2.00 |
| shortest path | 1.00 | 1.00, 0.00 ; 1.00, 0.00 ; 0.97, 0.03 | 1.00, 0.00 ; 1.00, 0.00 ; 0.90, 0.07 | – | >2 (censored) |
| connected components | 1.00 | 1.00, 0.03 ; 1.00, 0.03 ; 0.93, 0.11 | 1.00, 0.03 ; 1.00, 0.03 ; 1.00, 0.10 | – | >2 (censored) |
| grid | 0.87 | 0.87, 0.00 ; 0.77, 0.00 ; 0.60, 0.00 | 0.93, 0.00 ; 0.70, 0.00 ; 0.77, 0.00 | – | >2 (censored) |

**Table 187.** Reverse interchange: instruct task-specification states into the base model (blocks 12 to 23, N = 50). Source: transplant experiment `results/reverse.json`.

| Task | base twin default share (correct / attempts) | instruct states into base, lambda 1 | matched random into base | predicted min rise | met |
|---|---|---|---|---|---|
| connected components | 0.50 / 0.38 (correct programs 18) | 0.53 / 0.55 (correct programs 19) | 0.50 / 0.36 (correct programs 16) | 0.25 | no |
| kth smallest | 0.65 / 0.55 (correct programs 20) | 0.74 / 0.66 (correct programs 31) | 0.56 / 0.39 (correct programs 16) | 0.17 | no |
| shortest path | 0.87 / 0.63 (correct programs 15) | 0.80 / 0.74 (correct programs 20) | 0.50 / 0.41 (correct programs 6) | 0.07 | no |
| grid | 0.80 / 0.39 (correct programs 5) | 1.00 / 0.73 (correct programs 11) | 1.00 / 0.50 (correct programs 2) | 0.10 | yes |

**What the base minus instruct difference is.** D = h_base − h_instruct is mostly shared across tasks and templates: the per-task mean D has a cosine across tasks of 0.76 to 0.88 at blocks 12 to 27 and split-half cosine 0.96 to 1.00 between template halves. It is large: at lambda 1, 71% of the hooked cells exceed 4 ban-free units, the dose at which random noise broke programs in iteration 4, and at the read position the difference is as large as the residual itself (Table 188). The transplanted state does not survive downstream: at block 27 the task-specification state stays 88% to 89% of the way to the base twin, but the state at the read position moves only 18% to 31% of the way (Table 189), so blocks 24 and later pull the computation back toward the instruct model.

**Table 188.** Size of D relative to the residual norm, mean over tasks, per block. Source: transplant experiment `results/d_norms.json`.

| Block | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| norm of D / residual norm, task-specification positions | 0.39 | 0.41 | 0.42 | 0.44 | 0.44 | 0.46 | 0.48 | 0.49 | 0.51 | 0.53 | 0.53 | 0.52 |
| norm of D / residual norm, read position | 1.06 | 1.08 | 1.09 | 1.07 | 1.06 | 1.06 | 0.99 | 1.10 | 1.07 | 1.09 | 1.11 | 1.12 |

**Table 189.** Geometry of the transplanted state: distance of the centroid under the transplant at the selected lambda as a fraction of the gap between the instruct and base twins, with MDS stress-1. Source: transplant experiment `results/geometry.json`.

| Task | lambda | Task spec, block 18 (stress-1) | Task spec, block 23 | Task spec, block 27 | Read position, block 18 | Read position, block 23 |
|---|---|---|---|---|---|---|
| kth smallest | 1.0 | 1.00 (0.25) | 1.00 (0.25) | 0.88 (0.22) | 0.18 (0.06) | 0.30 (0.08) |
| shortest path | 0.25 | 0.25 (0.17) | 0.25 (0.22) | 0.23 (0.21) | 0.05 (0.05) | 0.07 (0.08) |
| connected components | 0.5 | 0.50 (0.27) | 0.50 (0.27) | 0.47 (0.23) | 0.09 (0.06) | 0.15 (0.08) |
| grid | 0.25 | 0.25 (0.27) | 0.25 (0.26) | 0.24 (0.23) | 0.05 (0.06) | 0.07 (0.08) |

| | | | | | | |
|---|---|---|---|---|---|---|
| top-k frequent | 1.0 | 1.00 (0.06) | 1.00 (0.06) | 0.89 (0.08) | 0.18 (0.05) | 0.31 (0.06) |

**Checks.** Every bench labeller recovered all its references; the base extraction agreed with iteration-1 labels on 0.95 of 40 programs; the hash chain had no break, and every hooked row followed the commit of the base repertoire. Two independent recomputations matched 56 of 56 and 34 of 34 checks, and the placebos failed as required (arm shuffle within slots rejects at 0.012). The engine check against the 8-bit rows of iteration 4 (Table 190) agrees slot by slot on the twins and within the resampling null on the pushed cells, except the shortest path twin, which is outside a null of zero width. The novelty audit read every one of 48 correct programs outside the catalogue, base output included: the judges called 14 of them novel, and the manual comparison with the nearest known solution found linear scans, count loops and a threshold binary search, known techniques outside the frozen catalogue; 0 novel (rule-of-three bound 0.062). Ten deviations are logged, among them base sampling and the engine check before the protocol commit (both still before every transplant sample) and the conservative label unknown for the probe of Dijkstra with an array scan.

**Table 190.** Engine check of the bf16 engine against the 8-bit rows of iteration 4. Source: transplant experiment `results/engine_check.json`.

| Task | Cell | Label agreement per slot | JS | Null 95th percentile | Within null |
|---|---|---|---|---|---|
| kth smallest | twin | 1.00 | 0.000 | 0.000 | yes |
| kth smallest | random direction 9002 | 0.74 | 0.030 | 0.075 | yes |
| shortest path | twin | 0.98 | 0.010 | 0.000 | no |
| shortest path | random direction 9004 | 0.88 | 0.005 | 0.005 | yes |

## 5.4 Panel A: does the base repertoire predict which new tasks move? (Qwen3-8B)

*Three instruct twins of new units are mixed: trapped rain water (default share 0.64), minimum spanning tree weight (default share 0.66) and valid anagram (default share 0.76); on two units the instruct model is more diverse than the base: trapped rain water (twin off-default 0.36 against beta 0.23) and minimum spanning tree weight (twin off-default 0.34 against beta 0.11) (iteration-6 note).*

**Why.** The screen correlation of 0.78 was measured on the tasks that generated the hypothesis. The movability and landing halves could only be confirmed on tasks never seen, with every forecast committed from base samples before the first push. Panel A built one half of those tasks.

**Design.** Nine new units derived from LeetCode problems [12] were frozen up front in new wording (a grep audit found no algorithm word or catalogue name), each with hidden tests from the dataset's input and output pairs plus seeded edge, random and stress cases, one reference per algorithm, mutants that must fail, quadratic baselines and runtime and AST labellers. A labeller passed its gate at a kappa of at least 0.8 between runtime and AST labels, an agreement of at least 0.9 between runtime and adjudicated labels and 100% of references, with one revision round on a holdout. Six of the seven planned units passed; number of islands failed after its revision (holdout kappa 0.74), and the two reserves, jump game and median of two sorted lists, ran as a logged extension. Per unit, the instruct twin drew 50 programs and the base model 100 on identical ids plus 50 in plain completion, the rivals were computed, and the forecast (default, beta with its interval, base runner-up, rivals) was committed by hash; a guard in the GPU worker refused push jobs until then. Only then did 8 isotropic directions at alpha 1 and 2 ($N = 10$, confirmed at $N = 30$ by an exact binomial test), plan noise at alpha 1 and 2 ($N = 50$) and temperature 1.3 ($N = 50$) run. The 15 existing tasks were remeasured in the same engine on the ban-free unit; we call this the

remeasurement arm.

**The movability half is not supported in panel A** (Table 191). Over the six preregistered units, Spearman between beta and the escape fraction is -0.03 (exact one-sided p 0.56; Kendall 0). The one unit collapsed in the base, maximum block sum (beta 0), is immovable, as the movability half predicts, but 2 of the 5 units diverse in the base are also immovable, among them the most diverse: count primes (beta 0.81) never escapes. A bootstrap over measurement error gives a median of -0.09 (90% interval -0.37 to 0.21). With the two extension units, both diverse in the base (0.47 and 0.71) and both immovable, Spearman falls to -0.33 (p 0.80). Simple rivals tracked movability better than beta (Tables 192 and 193), and so did beta's own variants: Spearman with the escape fraction is -0.09 for beta from plain completion, -0.03 for beta that also counts correct programs labelled other as off-default (identical to beta on these units) and 0.40 for the base Hill-1 (0.16, -0.33 and 0.22 with the extension units). The three units that moved have an alternative at least twice as fast as the default (minimum runtime merit ratio 0.10 to 0.51), the three that did not have ratios of 0.86 to 4.35, and the runtime merit rival reaches -0.88; the extension unit median of two sorted lists breaks this (ratio 0.11, never escapes), and with it the rival falls to -0.66. The twin's own modal share reaches -0.70, ahead of beta by a task-bootstrap interval that excludes zero (-1.90 to -0.09).

**Table 191.** Panel A units: gate, the forecast committed before any push, and the outcome. Source: panel A `results/summary.md`.

| Unit | Gate (kappa / runtime against adjudicated) | Default (twin share) | Base pass | beta [95% CI] | Base runner-up | Predicted class | Escape fraction (alpha 1 / alpha 2) | Push landing | Plan noise excess |
|---|---|---|---|---|---|---|---|---|---|
| majority element | gated (0.92 / 0.95) | hash count (1.00) | 0.29 (29 correct) | 0.41 [0.26, 0.59] | Boyer–Moore majority vote | movable | 0.125 (0.000 / 0.250) | Boyer–Moore majority vote | 0.04 |
| maximum block sum | gated (1.00 / 1.00) | Kadane (1.00) | 0.43 (43 correct) | 0.00 [0.00, 0.08] | none | immovable | 0.000 (0.000 / 0.000) | none | 0.00 |
| trapped rain water | gated (0.82 / 0.90) | prefix and suffix maxima (0.64) | 0.22 (22 correct) | 0.23 [0.10, 0.43] | two pointers | movable | 0.438 (0.375 / 0.500) | two pointers | -0.05 |
| count primes | gated (1.00 / 0.95) | sieve of Eratosthenes (1.00) | 0.31 (31 correct) | 0.81 [0.64, 0.91] | trial division | movable | 0.000 (0.000 / 0.000) | none | 0.00 |
| longest common prefix | gated (0.84 / 0.90) | horizontal scan (1.00) | 0.20 (20 correct) | 0.30 [0.15, 0.52] | vertical scan | movable | 0.000 (0.000 / 0.000) | none | 0.15 |
| number of islands | gate failed (0.74 / 0.89) | – | – | – | – | – | – | – | – |
| longest mirrored substring | gated (1.00 / 1.00) | expand around centre (1.00) | 0.28 (28 correct) | 0.46 [0.30, 0.64] | brute force | movable | 0.188 (0.000 / 0.375) | brute force | 0.13 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| jump game (ext.) | gated (1.00 / 1.00) | greedy (1.00) | 0.19 (19 correct) | 0.47 [0.32, 0.63] | DP | movable | 0.000 (0.000 / 0.000) | none | 0.01 |
| median of two sorted lists (ext.) | gated (1.00 / 1.00) | merge with two pointers (1.00) | 0.24 (24 correct) | 0.71 [0.51, 0.85] | concatenate and sort | movable | 0.000 (0.000 / 0.000) | none | 0.02 |

**Table 192.** Rivals against beta, primary units (Spearman with the escape fraction, and the difference from beta with a 95% task-bootstrap interval). Source: panel A `results/a1.json`.

| Rival (sign) | rho(rival, E) | Delta = rho(beta,E) - sign*rho(rival,E) | 95% task-bootstrap CI |
|---|---|---|---|
| opening margin (-) | -0.516 | -0.546 | [-1.29, 0.00] |
| minimum runtime merit ratio (-) | -0.880 | -0.911 | [-2.00, 0.39] |
| minimum length merit ratio (-) | -0.273 | -0.304 | [-1.50, 1.00] |
| twin modal share (-) | -0.696 | -0.726 | [-1.90, -0.09] |
| novelty instruction share (+) | -0.417 | 0.387 | [-0.38, 1.66] |

**Table 193.** The same with the two extension units. Source: panel A `results/a1_with_extension.json`.

| Rival (sign) | rho(rival, E) | Delta = rho(beta,E) - sign*rho(rival,E) | 95% task-bootstrap CI |
|---|---|---|---|
| opening margin (-) | -0.518 | -0.846 | [-1.48, -0.18] |
| minimum runtime merit ratio (-) | -0.655 | -0.982 | [-1.75, -0.13] |
| minimum length merit ratio (-) | -0.273 | -0.600 | [-1.41, 0.39] |
| twin modal share (-) | -0.661 | -0.989 | [-1.82, -0.39] |
| novelty instruction share (+) | -0.429 | 0.101 | [-0.51, 0.97] |

**The landing half is supported in panel A.** On all three movable units (majority element, trapped rain water, longest mirrored substring) the base runner-up is the push landing (Poisson-binomial p 0.019 under chance $1/(K-1)$), and the base composition scores best (Table 194). The committed association forecast was uniform because of a mapping bug: every association answer restates the default named in the question, and the committed mapping took the earliest catalogue mention. With the mapping corrected post hoc on the same answers given before commitment (Table 195), association names the landing on 2 of 3 events and is a strong rival, but the base composition still scores higher on all three (mean margin 0.85 nats; 1.10 against the committed uniform version). Exact Wilcoxon p is 0.25, the smallest attainable with three events. No correct program from plan noise or from an escape cell landed on an algorithm with zero correct and zero valid base programs (Table 198).

**Table 194.** Panel A landing events: base runner-up, landing and log scores with the committed association forecast. Source: panel A `results/a2.json`.

| Event | Base runner-up | Push landing | Programs realised | log score base | association | proximity |
|---|---|---|---|---|---|---|
| majority element | Boyer–Moore majority vote | Boyer–Moore majority vote | 35 | -0.063 | -1.812 | -1.636 |
| trapped rain water | two pointers | two pointers | 160 | -0.054 | -1.119 | -1.122 |
| longest mirrored substring | brute force | brute force | 33 | -0.625 | -1.119 | -0.945 |

**Table 195.** The association rival with the corrected mapping (post hoc, logged as a deviation). Source: panel A `results/assoc_corrected.json`.

| Unit | Association top-1 | Base runner-up | Corrected counts |
|---|---|---|---|
| majority element | Boyer–Moore majority vote | Boyer–Moore majority vote | Boyer–Moore majority vote 20, sort and take middle 0, divide and conquer 0, randomised verify 0, bit count 0, library mode 0 |
| maximum block sum | prefix minimum | none | prefix minimum 13, divide and conquer 7, segment tree 0 |
| trapped rain water | two pointers | two pointers | two pointers 20, monotonic stack 0, peak split 0 |
| count primes | sieve of Atkin | trial division | trial division 5, linear sieve 0, segmented sieve 0, sieve of Sundaram 0, sieve of Atkin 14 |
| longest common prefix | vertical scan | vertical scan | vertical scan 16, sort and compare ends 0, binary search on length 4, divide and conquer 0, trie 0, library common prefix 0 |
| longest mirrored substring | Manacher | brute force | DP table 0, Manacher 20, brute force 0 |
| jump game | DP | DP | DP 11, BFS over indices 9 |
| median of two sorted lists | partition binary search | concatenate and sort | partition binary search 20, heap merge 0, concatenate and sort 0 |

**The existing tasks on the ban-free unit.** The remeasurement arm remeasured the 15 existing tasks with 8 fixed directions at alpha 1 and 2 on the ban-free unit in bf16 (Table 196). Only kth smallest (escape fraction 0.25) and connected components (0.125) escaped; two sum, find duplicate, the grid task and the shortest path task, which the frozen screen called movable under the 8-bit engine and the ban unit, did not move. A descriptive view that combines the six new units with the 15 existing tasks (Step 1's committed beta, this arm's escape fractions) gives Spearman 0.54 (p 0.0065), and 0.59 (p 0.0096) for the existing tasks alone, but that correlation comes from 13 of 15 existing tasks being immovable with low beta, while tasks diverse in the base (the grid, shortest path and top-k frequent tasks) are still immovable, and the landing check splits (connected components hit, kth smallest miss: Step 1 runner-up quickselect, landing size-k heap).

**Table 196.** The remeasurement arm: the 15 existing tasks under the ban-free unit (8 directions, alpha 1 and 2), against iteration 4 and the frozen screen. Source: panel A `results/extra_arm_a.json`.

| Task | Default | Twin off-default share / pass | E ban-free (a1 / a2) | Landing | Iteration 4 E (a1 / a2) | Frozen screen E (a1 / a2), movable | Verdict change |
|---|---|---|---|---|---|---|---|
| kth smallest | full sort | 0.02 / 1.00 | 0.250 (0.125 / 0.375) | size-k heap | 0.30 / 0.63 | 0.44 / 1.00, True | False |
| connected components | union find | 0.00 / 1.00 | 0.125 (0.000 / 0.250) | DFS | 0.38 / 0.73 | 0.33 / –, True | False |
| two sum | hash map | 0.00 / 1.00 | 0.000 (0.000 / 0.000) | none | 0.00 / 0.13 | 0.00 / 0.67, True | True |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| find duplicate | hash set | 0.00 / 1.00 | 0.000 (0.000 / 0.000) | none | 0.00 / 0.00 | 0.00 / 0.33, True | True |
| grid | BFS | 0.02 / 0.86 | 0.000 (0.000 / 0.000) | none | 0.05 / 0.13 | 0.07 / 0.00, True | True |
| shortest path | Dijkstra | 0.02 / 0.98 | 0.000 (0.000 / 0.000) | none | 0.18 / 0.20 | 0.20 / 0.00, True | True |
| substring search | naive | 0.00 / 1.00 | 0.000 (0.000 / 0.000) | none | 0.00 / 0.00 | 0.00 / 0.00, False | False |
| Fibonacci | iteration | 0.00 / 1.00 | 0.000 (0.000 / 0.000) | none | 0.00 / 0.00 | 0.00 / –, False | False |
| longest increasing subsequence | quadratic DP | 0.00 / 1.00 | 0.000 (0.000 / 0.000) | none | 0.00 / 0.00 | 0.00 / –, False | False |
| merge intervals | sort and sweep | 0.00 / 1.00 | 0.000 (0.000 / 0.000) | none | 0.00 / 0.00 | 0.00 / –, False | False |
| top-k frequent | sort by count | 0.00 / 0.76 | 0.000 (0.000 / 0.000) | none | 0.00 / 0.00 | 0.00 / –, False | False |
| nonoverlapping intervals | greedy by end | 0.00 / 1.00 | 0.000 (0.000 / 0.000) | none | 0.00 / 0.00 | 0.00 / –, False | False |
| topological order | Kahn | 0.00 / 1.00 | 0.000 (0.000 / 0.000) | none | 0.00 / 0.00 | 0.00 / –, False | False |
| first and last position | binary search | 0.00 / 0.88 | 0.000 (0.000 / 0.000) | none | 0.00 / 0.00 | 0.00 / –, False | False |
| cycle detection | DFS with parent | 0.00 / 0.94 | 0.000 (0.000 / 0.000) | none | 0.00 / 0.00 | – / –, none | none |

**Table 197.** Knob rows of the panel A units: pass rate / off-default share / correct-only Hill-1 at depth 25. Source: panel A `results/knob_rows.json`.

| Unit | twin | plan noise alpha 1 | plan noise alpha 2 | temperature 1.3 | novelty instruction | base model | plain completion |
|---|---|---|---|---|---|---|---|
| majority element | 1.00 / 0.00 / 1.00 | 1.00 / 0.04 / 1.18 | 1.00 / 0.04 / 1.17 | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 1.00 | 0.29 / 0.42 / 1.97 | 0.52 / 0.40 / 2.47 |
| maximum block sum | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 1.00 | 0.43 / 0.02 / 1.00 | 0.22 / 0.02 / 1.00 |
| trapped rain water | 1.00 / 0.36 / 1.91 | 1.00 / 0.32 / 1.86 | 1.00 / 0.30 / 1.83 | 1.00 / 0.36 / 1.91 | 0.97 / 0.00 / 1.00 | 0.22 / 0.22 / 1.71 | 0.12 / 0.25 / 2.75 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| count primes | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 1.00 | 0.31 / 0.72 / 1.63 | 0.14 / 0.49 / 1.98 |
| longest common prefix | 0.94 / 0.00 / 1.00 | 0.96 / 0.08 / 1.32 | 0.98 / 0.22 / 1.69 | 0.88 / 0.00 / 1.00 | 0.97 / 0.27 / 1.98 | 0.20 / 0.39 / 2.11 | 0.26 / 0.62 / 1.54 |
| longest mirrored substring | 1.00 / 0.00 / 1.00 | 1.00 / 0.08 / 1.31 | 1.00 / 0.18 / 1.59 | 1.00 / 0.06 / 1.25 | 1.00 / 0.00 / 1.00 | 0.28 / 0.59 / 2.71 | 0.10 / 0.52 / 1.00 |
| jump game | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 1.00 | 0.98 / 0.02 / 1.00 | 0.98 / 0.00 / 1.00 | 1.00 / 0.00 / 1.00 | 0.19 / 0.47 / 1.93 | 0.14 / 0.16 / 1.82 |
| median of two sorted lists | 1.00 / 0.00 / 1.00 | 1.00 / 0.00 / 1.00 | 1.00 / 0.04 / 1.17 | 1.00 / 0.00 / 1.00 | 0.67 / 0.20 / 1.00 | 0.24 / 0.74 / 2.14 | 0.24 / 0.43 / 1.57 |

**Table 198.** Repertoire check: does any nudge reach beyond the base repertoire? Source: panel A `results/repertoire_check.json`.

| Unit | base Hill-1 (depth) | PN a1 | PN a2 | escape cells | correct off-default push programs outside the base repertoire |
|---|---|---|---|---|---|
| majority element | 1.97 (25) | 1.18 | 1.17 | 1.57 | 0 |
| maximum block sum | 1.00 (25) | 1.00 | 1.00 | 1.00 | 0 |
| trapped rain water | 1.71 (22) | 1.86 | 1.83 | 1.95 | 0 |
| count primes | 1.63 (25) | 1.00 | 1.00 | 1.00 | 0 |
| longest common prefix | 2.11 (20) | 1.32 | 1.69 | 1.00 | 0 |
| longest mirrored substring | 2.70 (25) | 1.31 | 1.59 | 1.62 | 0 |
| jump game | 1.93 (19) | 1.00 | 1.00 | 1.03 | 0 |
| median of two sorted lists | 2.14 (24) | 1.00 | 1.17 | 1.11 | 0 |

**Checks.** The labellers recovered every reference and 35 of 35 handwritten variants; independent TheAlgorithms implementations passed 8 of 9 frozen test suites and received the intended runtime label on 7 of 9 (Table 199), the misses being an unloadable file and a Misra–Gries vote that counts candidates in a Counter. The twins reproduced the 8-bit screen rows, while the pushed cells differed under bf16 (Table 200), so every contrast of the panel is within the bf16 engine. Two correct programs labelled other appeared among 6,854 correct programs (one catalogue variant, one degenerate); 0 novel (rule-of-three bound 0.00044 per correct program). Two recomputations using only the standard library matched with no mismatch, and the placebos failed (beta shuffled across units: mean Spearman 0.00; labels shuffled between twin and escape rows: escape fraction at most 0.007 on average). Eleven deviations are logged, including a re-run of the majority element test builder after its commit (identical cases, changed timing fields) and the association mapping bug.

**Table 199.** External check with independent TheAlgorithms implementations (MIT). Source: panel A `results/external_reference_check.json`.

| Unit | Implementation | Intended | Tests | Runtime | AST |
|---|---|---|---|---|---|
| maximum block sum | dynamic_programming/ max_subarray_sum.py | Kadane | 70/70 | Kadane | Kadane |

| | | | | | |
|---|---|---|---|---|---|
| maximum block sum | divide_and_conquer/ max_subarray.py | divide and conquer | 0/70 (load error) | unknown | divide and conquer |
| longest mirrored substring | strings/manacher.py | Manacher | 73/73 | Manacher | unknown |
| count primes | maths/sieve_of_eratosthenes.py | sieve of Eratosthenes | 48/48 | sieve of Eratosthenes | sieve of Eratosthenes |
| count primes | maths/ prime_sieve_eratosthenes.py | sieve of Eratosthenes | 48/48 | sieve of Eratosthenes | sieve of Eratosthenes |
| count primes | maths/odd_sieve.py | sieve of Eratosthenes | 48/48 | sieve of Eratosthenes | sieve of Eratosthenes |
| count primes | maths/segmented_sieve.py | segmented sieve | 48/48 | segmented sieve | sieve of Eratosthenes |
| count primes | maths/sieve_of_atkin.py | sieve of Atkin | 48/48 | sieve of Atkin | sieve of Atkin |
| majority element | other/ majority_vote_algorithm.py | Boyer–Moore majority vote | 71/71 | hash count | hash count |

**Table 200.** Engine check of panel A's bf16 engine against the 8-bit screen rows. Source: panel A `results/engine_check.json`.

| Cell | Label agreement per slot | JSD | Null p95 | This run | Screen |
|---|---|---|---|---|---|
| kth smallest / twin | 1.00 | 0.0000 | 0.0000 | full sort 50 | full sort 50 |
| kth smallest / random direction 9002, alpha 1 | 0.74 | 0.0302 | 0.0766 | size-k heap 29, full sort 9, quickselect 12 | quickselect 16, full sort 15, size-k heap 19 |
| shortest path / twin | 0.98 | 0.0101 | 0.0000 | Dijkstra 49, Bellman-Ford/SPFA 1 | Dijkstra 50 |
| shortest path / random direction 9004, alpha 1 | 0.86 | 0.0074 | 0.0047 | Dijkstra 31, Bellman-Ford/SPFA 19 | Dijkstra 26, Bellman-Ford/SPFA 24 |

## 5.5 Panel B: seven more new tasks and disguised variants (Qwen3-8B)

*Three instruct twins of new units are mixed: trapped rain water (default share 0.64), minimum spanning tree weight (default share 0.66) and valid anagram (default share 0.76); on two units the instruct model is more diverse than the base: trapped rain water (twin off-default 0.36 against beta 0.23) and minimum spanning tree weight (twin off-default 0.34 against beta 0.11) (iteration-6 note).*

**Why.** Panel B built the other half of the confirmation units under the same protocol, and added what we call *disguised variants*, eight bench tasks rewritten as narratives, each with a paired remeasurement of its parent, to separate a task's wording from its algorithmic affordances.

**Design.** Seven units (missing number, single number, minimum spanning tree weight, integer square root, all subsets, valid anagram and count inversions) were built with wording free of algorithm words, hidden tests checked by two independent oracles, three to six references, mutants and excluded brute force methods, and gated labellers; all seven gated, four after one revision round, and the 15-program blind audits of base labels agreed at 0.93 to 1.0. Because the base model was 46% to 61% off task under the chat template, its repertoire was raised to $N = 200$ per unit before any commit. Each unit's beta, base runner-up, base composition and rivals were hash-chained before its first push; then came 8 directions at alpha 1 and 2, plan noise and temperature 1.3.

**The movability half is not supported in panel B either** (Table 201). Spearman between beta and the escape fraction is 0.21 (exact one-sided p 0.33, bootstrap interval over units -0.75 to 0.95). Four units moved (single number, count inversions, minimum spanning tree weight and integer square root) and three did not: missing number (beta 0.06), all subsets (0.37) and valid anagram (0.81), the clearest disconfirming case. No unit was collapsed in the base, so the part of the movability half that says collapsed tasks are immovable is untested here. Beta ranks the graded plan noise excess better (0.82, p 0.017), and the runtime and length merit rivals track the escape fraction more closely than beta (0.54 and 0.58; Table 202), all with wide intervals. Under the preregistered sensitivities, beta from attempts gives 0.39 (p 0.18), beta from plain completion -0.45, beta from correct programs 0.04, and excluding the uncertain class or the flagged twins gives -0.03 and 0.06; a placebo with beta shuffled 1,000 times has a mean Spearman of 0.008 and rejects at 0.048.

**Table 201.** Panel B units: default, beta under three bases, base runner-up and the escape outcome. Source: panel B `README.md`, `results/a1_panel.json`.

| Unit | Default | Catalogue size | beta | beta (attempts) | beta (plain completion) | Base runner-up | Escape, alpha 1 | Escape, alpha 2 | Escape fraction | Push landing | Plan noise excess | Flags |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| missing number | arithmetic sum | 5 | 0.06 | 0.19 | 0.12 | XOR fold | 0.000 | 0.000 | 0.000 | - | 0.00 | - |
| single number | hash count | 4 | 0.56 | 0.60 | 0.43 | XOR fold | 0.375 | 0.500 | 0.438 | XOR fold | 0.30 | - |
| mst weight | Kruskal | 4 | 0.11 | 0.42 | 0.33 | Prim with heap | 0.125 | 0.125 | 0.125 | Prim with heap | 0.06 | - |
| integer sqrt | binary search | 6 | 0.37 | 0.44 | 0.50 | floating point sqrt | 0.125 | 0.125 | 0.125 | linear scan | 0.10 | - |
| all subsets | backtracking | 4 | 0.37 | 0.38 | 0.58 | iterative doubling | 0.000 | 0.000 | 0.000 | - | 0.08 | - |
| valid anagram | sort and compare | 3 | 0.81 | 0.80 | 0.55 | hash count | 0.000 | 0.000 | 0.000 | - | 0.12 | - |
| count inversions | merge sort count | 4 | 0.75 | 0.75 | 0.00 | sorted insertion with bisect | 0.250 | 0.375 | 0.312 | sorted insertion with bisect | 0.18 | default_from_few_correct(correct programs = 3); twin_pass_below_0.7(0.06); base_fewer_than_20_correct(n = 15): attempt composition used |

**Table 202.** Rivals against beta in panel B. Source: panel B `results/rivals.json`.

| predictor | rho with E | one-sided p | rho(beta) - rho(rival), 95% CI |
|---|---|---|---|
| beta | 0.21 | 0.333 | - |
| opening margin | 0.21 | 0.333 | [-1.32, 1.38] |
| runtime merit gap | 0.54 | 0.114 | [-1.32, 0.67] |
| length merit gap | 0.58 | 0.090 | [-1.40, 0.81] |
| twin modal share | 0.11 | 0.429 | [-0.93, 0.88] |
| novelty instruction share | -0.03 | 0.543 | [-0.68, 0.87] |
| novelty instruction share, correct programs | -0.03 | 0.543 | [-0.69, 0.88] |
| catalogue size | 0.07 | 0.448 | [-1.49, 1.59] |

**The landing half in panel B.** The base runner-up named the push landing in 3 of 4 events (expected by chance 1.2; Poisson-binomial p 0.081), and the base composition beat association by 0.68 nats and proximity by 0.13 nats on average (Table 203). The miss is integer square root, where the base runner-up is a floating point square root and the pushes land on a linear scan. With fewer than five events no Wilcoxon test was run.

**Table 203.** Panel B landing events. Source: panel B `results/a2_panel.json`. Base support (valid attempts / correct programs / runner-up count among correct, of 200 base rows): single number 77

/ 62 / 32; minimum spanning tree weight 89 / 28 / 3; integer square root 108 / 76 / 15; count inversions 87 / 15 / 2 (iteration 6).

| Unit | Base runner-up | Push landing | Hit | Log score, base composition | Association | Proximity |
|---|---|---|---|---|---|---|
| single number | XOR fold | XOR fold | 1.0 | -0.17 | -0.10 | -0.95 |
| minimum spanning tree weight | Prim with heap | Prim with heap | 1.0 | -0.30 | -0.10 | -0.94 |
| integer square root | floating point sqrt | linear scan | 0.0 | -1.10 | -3.29 | -1.57 |
| count inversions | sorted insertion with bisect | sorted insertion with bisect | 1.0 | -2.39 | -3.18 | -1.02 |

**Table 203b (iteration 6).** Knob rows of the panel B units, laid out like Table 197: pass rate / off-default share among correct programs / correct-only Hill-1 rarefied at depth 25 (counts: correct off default of correct). Source: panel B `pooled_inputs.json` (outcomes.twin_cell, pn_cells, t13_cell).

| Unit | twin | plan noise alpha 1 | plan noise alpha 2 | temperature 1.3 |
|---|---|---|---|---|
| missing number | 0.98 / 0.00 / 1.00 (0 of 49) | 0.98 / 0.00 / 1.00 (0 of 49) | 0.94 / 0.00 / 1.00 (0 of 47) | 0.96 / 0.00 / 1.00 (0 of 48) |
| single number | 1.00 / 0.08 / 1.30 (4 of 50) | 1.00 / 0.20 / 1.63 (10 of 50) | 1.00 / 0.38 / 1.92 (19 of 50) | 1.00 / 0.12 / 1.49 (6 of 50) |
| minimum spanning tree weight | 1.00 / 0.34 / 1.88 (17 of 50) | 1.00 / 0.40 / 1.94 (20 of 50) | 0.96 / 0.40 / 1.94 (19 of 48) | 0.98 / 0.37 / 1.92 (18 of 49) |
| integer square root | 1.00 / 0.00 / 1.00 (0 of 50) | 1.00 / 0.04 / 1.17 (2 of 50) | 0.98 / 0.08 / 1.33 (4 of 49) | 1.00 / 0.00 / 1.00 (0 of 50) |
| all subsets | 0.86 / 0.00 / 1.00 (0 of 43) | 0.80 / 0.03 / 1.12 (1 of 40) | 0.74 / 0.11 / 1.41 (4 of 37) | 0.80 / 0.00 / 1.00 (0 of 40) |
| valid anagram | 1.00 / 0.24 / 2.01 (12 of 50) | 1.00 / 0.22 / 1.89 (11 of 50) | 1.00 / 0.36 / 2.37 (18 of 50) | 1.00 / 0.26 / 2.03 (13 of 50) |
| count inversions | 0.06 / 0.00 / 1.00* (0 of 3) | 0.08 / 0.00 / 1.00* (0 of 4) | 0.16 / 0.62 / 2.95* (5 of 8) | 0.08 / 0.00 / 1.00* (0 of 4) |

* depth below 25 (rarefied at the number of correct programs). The pushes stayed inside the base repertoire on 6 of 7 units; 3 push programs fell outside it (count inversions; panel B repertoire check).

**Disguised variants.** Of the eight parent and variant pairs, only kth smallest and connected components moved in either form (Table 204), and connected components changed its default under disguise (union find to BFS); the sign of the change in beta and the change in escape fraction agreed in 0 of 1 pair without a default shift, and their Spearman over pairs was 0.13 (p 0.50). Descriptive views that add the remeasured parents (Spearman 0.43, p 0.055, n = 15) or the variants with a permutation that preserves clusters (0.54, p 0.015, n = 22) are positive, as is the screen view (0.79 on the 15 screen tasks), but none is the preregistered test.

**Table 204.** Comparison of disguised variants within parents (P remeasured parent, V variant; McNemar counts on confirmed escape cells). Source: panel B `results/variants.json`.

| parent | default P / V | beta P | beta V | E P | E V | McNemar (confirmed), only P : only V |
|---|---|---|---|---|---|---|
| kth smallest | full sort / full sort | 0.41 | 0.35 | 0.500 | 0.750 | 1 / 5 |
| connected components | union find / BFS | 0.37 | 0.96 | 0.188 | 0.312 | 2 / 4 |
| two sum | hash map / hash map | 0.14 | 0.05 | 0.000 | 0.000 | 0 / 0 |

| | | | | |
|---|---|---|---|---|
| find duplicate | hash set / hash set | 0.20 | 0.32 | 0.000 0.000 0 / 0 |
| substring search | naive / naive | 0.18 | 0.15 | 0.000 0.000 0 / 0 |
| Fibonacci | iteration / iteration | 0.40 | 0.07 | 0.000 0.000 0 / 0 |
| top-k frequent | sort by count / NO_DEFAULT | 0.26 | - | 0.000 0.000 0 / 0 |
| nonoverlapping intervals | greedy by end / greedy by end | 0.23 | 0.08 | 0.000 0.000 0 / 0 |

**Checks.** The engine check reproduced kth smallest within the null from resampling templates (agreement per slot 0.74) and put the shortest path cell outside it (0.64), a precision sensitivity; the twins agreed at 1.00 and 0.98. Twelve distinct correct programs labelled other appeared among 10,366 correct programs (7 catalogue variants, 5 degenerate); 0 novel. Independent recomputations found no mismatch, the placebo that treats the twin as a push gave an escape fraction of 0 on every unit, and no push row preceded its unit's commit. Count inversions carries three flags: its default comes from 3 correct twin programs, the twin passes only 0.06 (47 of 50 twin programs are the quadratic pair scan, which exceeds the time limit), and its base composition uses attempts (15 correct programs).

## 5.6 The two panels pooled (computed for this report)

Neither panel ran the pooled test: both wrote their raw ingredients per unit for a pooled step in the next round. Because the pooled numbers decide how this iteration is read, we computed them here from the two panels' own result files (`work/pooled_a1a2.py`, output `work/pooled_a1a2.json`), with a Monte Carlo permutation test for Spearman and an exact Poisson-binomial tail and exact Wilcoxon signed-rank tests [65] for the landing half. This is a descriptive preview; the preregistered permutation within parent clusters was not run.

**Table 205.** The movability and landing halves over the new units of both panels (13 preregistered units; 15 with panel A's extension). Source: `work/pooled_a1a2.json`, computed from panel A `results/a1*.json`, `results/a2*.json` and panel B `results/a1_panel.json`, `results/a2_panel.json`.

| Quantity | Preregistered units (n = 13) | With the extension units (n = 15) |
|---|---|---|
| Spearman between beta and the escape fraction | 0.146 (one-sided permutation p 0.315) | 0.006 (p 0.492) |
| Units collapsed in the base that are immovable | 1 of 1 | 1 of 1 |
| Units diverse in the base (beta at least 0.2) that are immovable | 4 of 10 | 6 of 12 |
| Movable units | 7 of 13 | 7 of 15 |
| Landing events (movable units with beta at least 0.05) | 7: majority element, trapped rain water, longest mirrored substring, single number, minimum spanning tree weight, integer square root, count inversions | same |
| Base runner-up equals the push landing | 6 of 7 (expected by chance 2.03; Poisson-binomial p 0.0027) | same |
| Base minus association, corrected mapping (nats per event) | mean +0.75; differences +0.21, +0.09, +2.24, -0.07, -0.20, +2.18, +0.79; exact one-sided Wilcoxon p 0.055 | same |

| | | |
|---|---|---|
| Base minus association, committed (uniform) mapping in panel A | mean +0.86; exact one-sided Wilcoxon p 0.039 | same |
| Base minus proximity | mean +0.50; differences +1.57, +1.07, +0.32, +0.78, +0.64, +0.46, -1.37; exact one-sided Wilcoxon p 0.109 | same |

Read against the preregistered thresholds, the movability half meets its disconfirming outcome on the pooled view (Spearman below 0.3), although not the second clause (4 of 10 units diverse in the base are immovable, not most). The landing half meets its hit-rate criterion (6 of 7, 0.86, against a bar of 0.7 at p below 0.05) and its margin over association (0.75 nats against a bar of 0.10), with the paired Wilcoxon at p 0.055 against the corrected association and 0.039 against the committed one.

*[Figure fig_beta_escape: generated in the report phase, not in this draft]*

## 5.7 Plan noise against text-level plan samplers, decoding at matched departure, and the literal cell (Qwen3-8B)

**Why.** The positioning audit of iteration 4 found that the price comparison lacked its strongest competitors: text-level plan samplers, which diversify code by sampling a plan before the program [73, 74], and a decoding baseline matched to plan noise's own departure from the twin, the baseline under which an ICLR 2027 submission (OpenReview oag3wgNLRh) found activation steering no better than ordinary decoding. The review also asked for the literal cell of the request. This experiment ran both.

**The literal cell never changes the algorithm** (Table 206). On the shortest path task with thinking on, top-k 10 and the same 20 token seeds as the iteration-4 thinking arm, moving the temperature over 0.3, 0.7, 1.0 and 1.3 changed the final program's algorithm in 0 of 20 seeds (one-sided Clopper–Pearson 95% upper bound 0.14), and the first approach named in the thinking changed in 0 of 20. All 100 traces completed, every final program was heap-based Dijkstra, and every trace named Dijkstra first. Temperature changed the tokens: a repeat at temperature 0.7 was byte-identical on only 4 of 20 traces while the labels agreed on 20 of 20 (median common prefix 309.5 tokens), and paired with the top-p 1 traces of iteration 4 at the same seed and temperature the labels agreed on 100 of 100 pairs although the traces shared a median of only 34 tokens. A judge agreed with the first-approach lexicon on 19 of 19 audited traces. With this cell, question 1a-c of the request is answered under every sampler the run used: with thinking off at top-k 10 (Table 76), with thinking off and on at top-p 1 (Section 4.5), and with thinking on at top-k 10 here.

**Table 206.** The literal cell: shortest path task, thinking on, top-k 10, 20 seeds per temperature, 16k cap (vLLM, official 8-bit checkpoint). Source: knob experiment `results/literal_cell.json`.

| T | completed | final labels (completed) | first approach | median think tokens | Dijkstra style | phrasing clusters at theta 0.7 / 0.8 / 0.9 | phrasings per idea (theta 0.8) |
|---|---|---|---|---|---|---|---|
| T 0.3 | 20/20 | Dijkstra 20 | Dijkstra 20 | 3754.50 | heap 20 | 2 / 4 / 8 | dijkstra 4 |
| T 0.7 | 20/20 | Dijkstra 20 | Dijkstra 20 | 3962.00 | heap 20 | 3 / 3 / 10 | dijkstra 3 |
| T 1.0 | 20/20 | Dijkstra 20 | Dijkstra 20 | 3576.00 | heap 20 | 2 / 3 / 9 | dijkstra 3 |
| T 1.3 | 20/20 | Dijkstra 20 | Dijkstra 20 | 4149.00 | heap 20 | 5 / 7 / 12 | dijkstra 7 |
| T 0.7, repeat | 20/20 | Dijkstra 20 | Dijkstra 20 | 3500.00 | heap 20 | 3 / 6 / 10 | dijkstra 6 |

At N = 20 with thinking on a rise shows only at T 1.3, and repeat noise is the same size.
*(Iteration-6 addition: the two phrasing columns, computed by the knob experiment and missing from the*

*record until now.)*

**Design of the knob comparison.** Every knob ran in one HF bf16 engine with exact inverse-CDF common random numbers, so all arms of a slot shared the uniform stream of the code stage, on all 15 existing tasks (N = 50 per cell on the five decisive tasks, 30 on the other ten). Plan noise drew a fresh isotropic direction per sample at the task-specification positions of blocks 12 to 23, prefill only, at alpha 1 and 2 ban-free units, under the twin frame and on the plain prompt. The text-level plan samplers used the twin's frame, with the twin's neutral sentence first: intent factored generation at the direction's (1.3, 0.7) and the paper's tuned (0.73, 0.60) temperatures [73]; a lite plan search (five observations, a random pair of them combined into a plan, then code) [74]; enumerate and pick, in which the model lists up to eight algorithms and an external random number picks one; and base intent, in which Qwen3-8B-Base writes the intent sentence and the instruct model the code. Decoding at matched departure applied a temperature to generated tokens 1 to 32, bisected per task so that its mean per-token KL from the twin equalled plan noise's own at each alpha, measured on prefixes used only for calibration. Prices came from a randomisation hull over the full simplex with 2,000 slot bootstraps that select the best mixture again inside every replicate. The committed rule calls the plan sampler comparison supported if the 5-point Hill-1 of plan noise is at least that of the best text sampler on at least 3 of the 5 decisive tasks.

**The plan sampler comparison is DISCONFIRMED** (Table 207) (ban-free unit and iteration-5 seeds; re-tagged by iteration 7's dose-unit experiment, Section 7.6: at the moving design D* = D1 the plan-sampler rule is still disconfirmed, at 2 of 5) (iteration-8 correction). In Table 207 the margin is the 5-point Hill-1 of plan noise minus that of the best text sampler. Plan noise matched or beat the best text sampler on 1 of the 5 decisive tasks, kth smallest (1.98 against 1.83 for enumerate and pick, P(plan noise at least best) 0.58, unresolved). On connected components, the shortest path task and two sum, base intent won (2.53, 1.45 and 1.61 against 1.09, 1.18 and 1.17), and on the grid task nothing moved much. The mean margin is -0.40 (95% interval -0.70 to -0.24), and only 1.8% of bootstrap replicates reach three wins. Every sensitivity keeps the verdict except one (Table 209): with the multistage samplers dropped (they were padded on the left), intent factored generation is the only text sampler left and plan noise beats it on 4 of 5 tasks, so the disconfirmation rests on base intent and on enumerate and pick. On the ten other tasks (Table 208) plan noise moved only cycle detection and first and last position, while text samplers moved substring search, find duplicate, Fibonacci, longest increasing subsequence, top-k frequent and first and last position. Two prompt knobs outside the text-sampler set also beat plan noise where it was strongest: the novelty instruction on kth smallest (2.03) and the semantic seed on the shortest path task (2.12). (iteration-7 correction)

**Table 207.** The plan sampler comparison on the five decisive tasks: 5-point Hill-1 of plan noise (with its bootstrap interval) against each text-level plan sampler. Source: knob experiment `results/claim_c.json`.

| Task | Plan noise, 5-point Hill-1 [95%] | Intent factored generation | Plan search (lite) | Enumerate and pick | Base intent | Best text sampler | Plan noise wins | P(plan noise at least best) | Flag | Margin |
|---|---|---|---|---|---|---|---|---|---|---|
| kth smallest | 1.98 [1.64, 2.28] | 1.19 | 1.09 | 1.83 | 1.73 | enumerate and pick | yes | 0.58 | unresolved | +0.15 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| connected components | 1.09 [1.00, 1.26] | 1.96 | 1.50 | 1.88 | 2.53 | base intent | no | 0.00 | -1.43 |
| shortest path | 1.18 [1.00, 1.34] | 1.00 | 1.07 | 1.24 | 1.45 | base intent | no | 0.06 | -0.27 |
| two sum | 1.17 [1.00, 1.38] | 1.00 | 1.00 | 1.27 | 1.61 | base intent | no | 0.01 | -0.44 |
| grid | 1.06 [1.00, 1.30] | 1.00 | 1.00 | 1.08 | 1.04 | enumerate and pick | no | 0.38 | unresolved -0.03 |

**Table 208.** The same prices on the ten other tasks (descriptive; the plan search arm was not run on topological order and first and last position, the base intent arm not on first and last position). Source: knob experiment `results/claim_c.json#nd10_descriptive`.

| Task (descriptive, N = 30 per cell) | Plan noise, 5-point Hill-1 | Intent factored generation | Plan search (lite) | Enumerate and pick | Base intent | Best text sampler |
|---|---|---|---|---|---|---|
| substring search | 1.00 | 1.97 | 1.00 | 1.44 | 1.00 | intent factored generation |
| find duplicate | 1.00 | 1.97 | 1.66 | 1.08 | 1.08 | intent factored generation |
| Fibonacci | 1.00 | 1.00 | 1.00 | 1.10 | 1.70 | base intent |
| longest increasing subsequence | 1.00 | 1.00 | 1.57 | 1.57 | 1.00 | enumerate and pick |
| merge intervals | 1.00 | 1.00 | 1.00 | 1.03 | 1.00 | enumerate and pick |
| top-k frequent | 1.00 | 1.00 | 1.00 | 3.02 | 1.17 | enumerate and pick |
| nonoverlapping intervals | 1.00 | 1.00 | 1.00 | 1.01 | 1.00 | enumerate and pick |
| cycle detection | 1.18 | 1.00 | 1.00 | 1.05 | 1.04 | enumerate and pick |
| topological order | 1.00 | 1.00 | n/a | 1.00 | 1.00 | intent factored generation |
| first and last position | 1.29 | 1.18 | n/a | 2.62 | n/a | enumerate and pick |

**Table 207b (iteration 6).** Pass cost (fraction of pass rate given up) at which each knob's randomisation hull first reaches correct-only Hill-1 of 1.5 and then of 2.0 (the two prices in each cell, separated by a slash); a dash means never reached. Source: knob experiment `results/prices.json`; 95% percentile intervals over the 2,000 slot-bootstrap replicates of `results/bootstrap.json` that reach the level (the share that reaches it is in tables/prices_p15_p20.csv).

| Task | plan noise, Hill-1 1.5 / 2.0 | intent factored generation, Hill-1 1.5 / 2.0 | plan search (lite), Hill-1 1.5 / 2.0 | enumerate and pick, Hill-1 1.5 / 2.0 | base intent, Hill-1 1.5 / 2.0 | novelty instruction, Hill-1 1.5 / 2.0 |
|---|---|---|---|---|---|---|

| | | | | | | |
|---|---|---|---|---|---|---|
| kth smallest | 0.000 [0.000, 0.000] / – | – / – | – / – | 0.030 [0.014, 0.061] / 0.062 [0.030, 0.120] | 0.033 [0.000, 0.084] / – | 0.006 [0.000, 0.016] / 0.018 [0.000, 0.045] |
| connected components | – / – | 0.000 [0.000, 0.000] / – | 0.050 [0.040, 0.072] / – | 0.029 [0.020, 0.040] / 0.058 [0.035, 0.090] | 0.005 [0.000, 0.018] / 0.011 [0.000, 0.039] | – / – |
| shortest path | – / – | – / – | – / – | 0.103 [0.058, 0.196] / 0.190 [0.117, 0.336] | 0.057 [0.000, 0.133] / – | – / – |
| two sum | – / – | – / – | – / – | 0.086 [0.048, 0.162] / 0.157 [0.096, 0.273] | 0.042 [0.012, 0.099] / 0.080 [0.024, 0.170] | – / – |
| grid | – / – | – / – | – / – | 0.229 [0.144, 0.420] / 0.341 [0.234, 0.576] | 0.421 [0.230, 0.640] / – | 0.096 [0.000, 0.240] / – |
| substring search | – / – | 0.000 [0.000, 0.000] / – | – / – | 0.055 [0.020, 0.133] / 0.098 [0.040, 0.220] | – / – | – / – |
| find duplicate | – / – | 0.000 [0.000, 0.000] / – | 0.023 [0.000, 0.085] / – | 0.240 [0.120, 0.533] / 0.380 [0.202, 0.600] | – / – | – / – |
| Fibonacci | – / – | – / – | – / – | 0.224 [0.120, 0.397] / – | 0.025 [0.000, 0.093] / – | – / – |
| longest increasing subsequence | – / – | – / – | 0.000 [0.000, 0.000] / – | 0.043 [0.017, 0.090] / 0.100 [0.033, 0.180] | – / – | – / – |
| merge intervals | – / – | – / – | – / – | 0.555 [0.260, 0.630] / – | – / – | – / – |
| top-k frequent | – / – | – / – | – / – | 0.014 [0.000, 0.058] / 0.026 [0.000, 0.117] | – / – | – / – |
| non-overlapping intervals | – / – | – / – | – / – | – / – | – / – | – / – |
| cycle detection | – / – | – / – | – / – | 0.359 [0.170, 0.708] / 0.528 [0.267, 0.693] | 0.470 [0.183, 0.630] / – | – / – |
| topological order | – / – | – / – | – / – | – / – | – / – | – / – |
| first and last position | – / – | – / – | – / – | 0.000 [0.000, 0.033] / 0.000 [0.000, 0.090] | – / – | – / – |

**Table 209.** Sensitivities of the plan sampler rule. Source: knob experiment `results/claim_c.json#sensitivities`.

| Sensitivity | Verdict | Tasks where plan noise wins | Mean margin |
|---|---|---|---|
| intent factored generation at (1.3, 0.7) only | DISCONFIRMED | 1 of 5 | -0.40 |

| | | | |
|---|---|---|---|
| plan noise hull adds the plain prompt arms | DISCONFIRMED | 2 of 5 | -0.40 |
| hull restricted to edges | DISCONFIRMED | 1 of 5 | -0.40 |
| multistage samplers dropped, padded on the left (intent factored generation remains) | SUPPORTED | 4 of 5 | +0.07 |
| style re-gate flagged task excluded | DISCONFIRMED | 1 of 5 | -0.40 |
| relabels from the novelty audit applied | DISCONFIRMED | 1 of 5 | -0.40 |

*[Figure fig_knob_h5: generated in the report phase, not in this draft]*

Table 210 shows what each arm actually wrote on the decisive tasks. Enumerate and pick buys the most raw diversity (Hill-1 of 2.9 to 4.7 per cell) at a cost of 16 to 62 pass points, and 40% of its grid rows are invalid. Base intent keeps the pass rate high on connected components (0.98) and two sum (0.88) while writing three and four algorithms. Intent factored generation at its cooler code temperature collapsed the grid pass rate to 0.12 and 0.08, with every correct program BFS. Plan noise is much weaker here than in iteration 4: on connected components it wrote union find 49 times in 50 at both alphas, where iteration 4's 8-bit engine at the ban unit had written DFS 29 times at alpha 2 (Section 4.3); Section 5.10 returns to this.

**Table 210.** Every arm on the five decisive tasks: pass rate, invalid rate, correct programs and correct-only Hill-1 (rarefied at depth 25 or at the available depth). Source: knob experiment `results/cells.json`.

| Task | Arm | N | Pass | Invalid | Correct programs | Hill-1 (depth) |
|---|---|---|---|---|---|---|
| kth smallest | twin | 50 | 1.00 | 0.00 | full sort 49, quickselect 1 | 1.09 (25) |
| kth smallest | plain prompt | 50 | 1.00 | 0.00 | full sort 50 | 1.00 (25) |
| kth smallest | plan noise alpha 1 | 50 | 0.98 | 0.00 | full sort 45, quickselect 4 | 1.33 (25) |
| kth smallest | plan noise alpha 2 | 50 | 1.00 | 0.00 | full sort 36, quickselect 12, size-k heap 2 | 1.99 (25) |
| kth smallest | plan noise alpha 1, plain prompt | 50 | 1.00 | 0.00 | full sort 49, quickselect 1 | 1.09 (25) |
| kth smallest | intent factored generation (1.3, 0.7) | 50 | 0.98 | 0.00 | full sort 48, heapify and pop 1 | 1.10 (25) |
| kth smallest | intent factored generation (0.73, 0.60) | 50 | 1.00 | 0.00 | full sort 48, heapify and pop 1, quickselect 1 | 1.19 (25) |
| kth smallest | plan search (lite) | 50 | 1.00 | 0.00 | full sort 50 | 1.00 (25) |
| kth smallest | enumerate and pick | 50 | 0.80 | 0.00 | quickselect 24, full sort 6, other 4, counting sort 4, size-k heap 2 | 3.24 (25) |
| kth smallest | base intent | 50 | 0.94 | 0.00 | full sort 39, size-k heap 4, quickselect 2, counting sort 1, other 1 | 1.86 (25) |
| kth smallest | decoding at matched departure, alpha 1 match | 50 | 1.00 | 0.00 | full sort 49, quickselect 1 | 1.09 (25) |
| kth smallest | decoding at matched departure, alpha 2 match | 50 | 0.98 | 0.00 | full sort 48, quickselect 1 | 1.10 (25) |
| kth smallest | temperature 1.3 | 50 | 1.00 | 0.00 | full sort 49, quickselect 1 | 1.09 (25) |
| kth smallest | top-k 10 | 50 | 1.00 | 0.00 | full sort 49, quickselect 1 | 1.09 (25) |
| kth smallest | novelty instruction | 50 | 0.96 | 0.00 | quickselect 47, heapify and pop 1 | 1.10 (25) |
| connected components | twin | 50 | 1.00 | 0.00 | union find 50 | 1.00 (25) |
| connected components | plain prompt | 50 | 1.00 | 0.00 | union find 50 | 1.00 (25) |

| | | | | | | |
|---|---|---|---|---|---|---|
| connected components | plan noise alpha 1 | 50 | 1.00 | 0.00 | union find 49, DFS 1 | 1.09 (25) |
| connected components | plan noise alpha 2 | 50 | 1.00 | 0.00 | union find 49, DFS 1 | 1.09 (25) |
| connected components | plan noise alpha 1, plain prompt | 50 | 1.00 | 0.00 | union find 50 | 1.00 (25) |
| connected components | intent factored generation (1.3, 0.7) | 50 | 1.00 | 0.00 | DFS 43, union find 7 | 1.48 (25) |
| connected components | intent factored generation (0.73, 0.60) | 50 | 1.00 | 0.00 | DFS 47, union find 3 | 1.24 (25) |
| connected components | plan search (lite) | 50 | 0.78 | 0.22 | BFS 31, union find 8 | 1.66 (25) |
| connected components | enumerate and pick | 50 | 0.84 | 0.16 | DFS 17, BFS 15, union find 10 | 2.88 (25) |
| connected components | base intent | 50 | 0.98 | 0.00 | union find 24, DFS 20, BFS 5 | 2.53 (25) |
| connected components | decoding at matched departure, alpha 1 match | 50 | 0.98 | 0.02 | union find 47, DFS 2 | 1.18 (25) |
| connected components | decoding at matched departure, alpha 2 match | 50 | 0.98 | 0.02 | union find 46, DFS 3 | 1.26 (25) |
| connected components | temperature 1.3 | 50 | 1.00 | 0.00 | union find 50 | 1.00 (25) |
| connected components | top-k 10 | 50 | 1.00 | 0.00 | union find 50 | 1.00 (25) |
| connected components | novelty instruction | 50 | 1.00 | 0.00 | union find 43, DFS 7 | 1.48 (25) |
| shortest path | twin | 50 | 0.98 | 0.00 | Dijkstra 49 | 1.00 (25) |
| shortest path | plain prompt | 50 | 0.98 | 0.00 | Dijkstra 49 | 1.00 (25) |
| shortest path | plan noise alpha 1 | 50 | 1.00 | 0.00 | Dijkstra 50 | 1.00 (25) |
| shortest path | plan noise alpha 2 | 50 | 0.94 | 0.00 | Dijkstra 45, Bellman-Ford/SPFA 2 | 1.19 (25) |
| shortest path | plan noise alpha 1, plain prompt | 50 | 1.00 | 0.00 | Dijkstra 50 | 1.00 (25) |
| shortest path | intent factored generation (1.3, 0.7) | 50 | 1.00 | 0.00 | Dijkstra 50 | 1.00 (25) |
| shortest path | intent factored generation (0.73, 0.60) | 50 | 1.00 | 0.00 | Dijkstra 50 | 1.00 (25) |
| shortest path | plan search (lite) | 50 | 0.78 | 0.20 | Dijkstra 36, Bellman-Ford/SPFA 3 | 1.31 (25) |
| shortest path | enumerate and pick | 50 | 0.60 | 0.04 | Dijkstra 11, Bellman-Ford/SPFA 10, Floyd-Warshall 9 | 2.97 (25) |
| shortest path | base intent | 50 | 0.92 | 0.02 | Dijkstra 39, Bellman-Ford/SPFA 7 | 1.52 (25) |
| shortest path | decoding at matched departure, alpha 1 match | 50 | 0.98 | 0.00 | Dijkstra 48, Bellman-Ford/SPFA 1 | 1.10 (25) |
| shortest path | decoding at matched departure, alpha 2 match | 50 | 0.82 | 0.04 | Dijkstra 41 | 1.00 (25) |
| shortest path | temperature 1.3 | 50 | 0.98 | 0.00 | Dijkstra 49 | 1.00 (25) |
| shortest path | top-k 10 | 50 | 0.98 | 0.00 | Dijkstra 49 | 1.00 (25) |
| shortest path | novelty instruction | 50 | 1.00 | 0.00 | Dijkstra 50 | 1.00 (25) |
| shortest path | semantic seed | 50 | 0.98 | 0.00 | Dijkstra 32, Bellman-Ford/SPFA 15, other 2 | 2.12 (25) |

| | | | | | | |
|---|---|---|---|---|---|---|
| two sum | twin | 50 | 1.00 | 0.00 | hash map 50 | 1.00 (25) |
| two sum | plain prompt | 50 | 1.00 | 0.00 | hash map 50 | 1.00 (25) |
| two sum | plan noise alpha 1 | 50 | 1.00 | 0.00 | hash map 50 | 1.00 (25) |
| two sum | plan noise alpha 2 | 50 | 1.00 | 0.00 | hash map 48, brute force 2 | 1.17 (25) |
| two sum | plan noise alpha 1, plain prompt | 50 | 1.00 | 0.00 | hash map 50 | 1.00 (25) |
| two sum | intent factored generation (1.3, 0.7) | 50 | 1.00 | 0.00 | hash map 50 | 1.00 (25) |
| two sum | intent factored generation (0.73, 0.60) | 50 | 1.00 | 0.00 | hash map 50 | 1.00 (25) |
| two sum | plan search (lite) | 50 | 0.98 | 0.00 | hash map 49 | 1.00 (25) |
| two sum | enumerate and pick | 50 | 0.68 | 0.00 | hash map 16, sort and two pointers 7, brute force 7, sort and binary search 4 | 3.46 (25) |
| two sum | base intent | 50 | 0.88 | 0.02 | hash map 29, brute force 9, sort and two pointers 4, sort and binary search 2 | 2.55 (25) |
| two sum | decoding at matched departure, alpha 1 match | 50 | 1.00 | 0.00 | hash map 50 | 1.00 (25) |
| two sum | decoding at matched departure, alpha 2 match | 50 | 1.00 | 0.00 | hash map 50 | 1.00 (25) |
| two sum | temperature 1.3 | 50 | 1.00 | 0.00 | hash map 50 | 1.00 (25) |
| two sum | top-k 10 | 50 | 1.00 | 0.00 | hash map 50 | 1.00 (25) |
| two sum | novelty instruction | 50 | 1.00 | 0.00 | hash map 46, brute force 3, other 1 | 1.35 (25) |
| grid | twin | 50 | 0.86 | 0.00 | BFS 43 | 1.00 (25) |
| grid | plain prompt | 50 | 0.96 | 0.00 | BFS 48 | 1.00 (25) |
| grid | plan noise alpha 1 | 50 | 0.80 | 0.00 | BFS 40 | 1.00 (25) |
| grid | plan noise alpha 2 | 50 | 0.76 | 0.00 | BFS 37, Dijkstra 1 | 1.12 (25) |
| grid | plan noise alpha 1, plain prompt | 50 | 0.86 | 0.00 | BFS 43 | 1.00 (25) |
| grid | intent factored generation (1.3, 0.7) | 50 | 0.12 | 0.00 | BFS 6 | 1.00 (6) |
| grid | intent factored generation (0.73, 0.60) | 50 | 0.08 | 0.00 | BFS 4 | 1.00 (4) |
| grid | plan search (lite) | 50 | 0.50 | 0.40 | BFS 25 | 1.00 (25) |
| grid | enumerate and pick | 50 | 0.24 | 0.40 | A* 3, BFS 3, DFS 3, Dijkstra 2, bidirectional BFS 1 | 4.69 (12) |
| grid | base intent | 50 | 0.32 | 0.00 | BFS 12, DFS 4 | 1.75 (16) |
| grid | decoding at matched departure, alpha 1 match | 50 | 0.80 | 0.00 | BFS 40 | 1.00 (25) |
| grid | decoding at matched departure, alpha 2 match | 50 | 0.78 | 0.00 | BFS 39 | 1.00 (25) |
| grid | temperature 1.3 | 50 | 0.92 | 0.00 | BFS 46 | 1.00 (25) |
| grid | top-k 10 | 50 | 0.84 | 0.00 | BFS 42 | 1.00 (25) |
| grid | novelty instruction | 50 | 0.70 | 0.00 | BFS 26, DFS 9 | 1.77 (25) |
| grid | semantic seed | 50 | 0.60 | 0.00 | BFS 30 | 1.00 (25) |

**Plan noise against decoding at matched departure.** At alpha 2, plan noise moves the algorithm and the matched decoding does not (Table 211): over the decisive tasks 19 slots are correct and off

the default under plan noise only and 3 under the matched decoding only (exact p 0.0009), 13 to 0 on kth smallest alone. At alpha 1 there is no difference (4 against 3). The matched decoding's 5-point Hill-1 is at most 1.25 on every decisive task. So plan noise's effect is not its token-level departure from the twin, which answers the objection of the ICLR 2027 submission for these tasks, but as a practical knob it loses to two text samplers. Plan noise on the plain prompt and under the twin frame showed no significant paired difference on any of the 15 tasks (Table 212).

**Table 211.** Plan noise against decoding at matched departure at matched per-token KL (correct off-default share of N; McNemar on slots correct and off the default under one arm only). Source: knob experiment `results/pn_vs_dmd.json`.

| Task | α | Temperature at matched departure | plan noise KL per token | matched departure KL per token | plan noise | matched departure | TWIN | McNemar, plan noise against matched departure (only plan noise : only matched departure, p) |
|---|---|---|---|---|---|---|---|---|
| kth smallest | 1 | 1.46 | 0.0179 | 0.0200 | 0.08 | 0.02 | 0.02 | 3/0, p = 0.25 |
| kth smallest | 2 | 1.64 | 0.0774 | 0.0707 | 0.28 | 0.02 | 0.02 | 13/0, p = 0.000244 |
| connected components | 1 | 1.84 | 0.0096 | 0.0110 | 0.02 | 0.04 | 0.00 | 1/2, p = 1 |
| connected components | 2 | 2.22 | 0.0333 | 0.0370 | 0.02 | 0.06 | 0.00 | 1/3, p = 0.625 |
| shortest path | 1 | 2.01 | 0.0254 | 0.0244 | 0.00 | 0.02 | 0.00 | 0/1, p = 1 |
| shortest path | 2 | 2.66 | 0.1113 | 0.1230 | 0.04 | 0.00 | 0.00 | 2/0, p = 0.5 |
| two sum | 1 | 1.52 | 0.0036 | 0.0039 | 0.00 | 0.00 | 0.00 | 0/0, p = 1 |
| two sum | 2 | 1.81 | 0.0100 | 0.0104 | 0.04 | 0.00 | 0.00 | 2/0, p = 0.5 |
| grid | 1 | 2.15 | 0.0505 | 0.0442 | 0.00 | 0.00 | 0.00 | 0/0, p = 1 |
| grid | 2 | 2.47 | 0.1554 | 0.1246 | 0.02 | 0.00 | 0.00 | 1/0, p = 1 |

**Table 212.** Plan noise on the plain prompt against plan noise under the twin frame, same direction and stream. Source: knob experiment `results/plain_vs_twin.json`.

| Task | plan noise only on plain : only on twin (p) | pass plain | pass twin | Hill plain | Hill twin |
|---|---|---|---|---|---|
| kth smallest | 0/3 (p = 0.25) | 1.00 | 0.98 | 1.09 | 1.33 |
| connected components | 0/1 (p = 1) | 1.00 | 1.00 | 1.00 | 1.09 |
| shortest path | 0/0 (p = 1) | 1.00 | 1.00 | 1.00 | 1.00 |
| two sum | 0/0 (p = 1) | 1.00 | 1.00 | 1.00 | 1.00 |
| grid | 0/0 (p = 1) | 0.86 | 0.80 | 1.00 | 1.00 |
| substring search | 0/0 (p = 1) | 1.00 | 1.00 | 1.00 | 1.00 |
| find duplicate | 0/0 (p = 1) | 1.00 | 1.00 | 1.00 | 1.00 |
| Fibonacci | 0/0 (p = 1) | 0.97 | 1.00 | 1.00 | 1.00 |
| longest increasing subsequence | 0/0 (p = 1) | 1.00 | 1.00 | 1.00 | 1.00 |
| merge intervals | 0/0 (p = 1) | 1.00 | 1.00 | 1.00 | 1.00 |
| top-k frequent | 0/0 (p = 1) | 0.73 | 0.80 | 1.00 | 1.00 |
| nonoverlapping intervals | 0/0 (p = 1) | 0.90 | 0.87 | 1.00 | 1.00 |
| cycle detection | 0/0 (p = 1) | 0.80 | 0.87 | 1.00 | 1.00 |
| topological order | 0/0 (p = 1) | 1.00 | 1.00 | 1.00 | 1.00 |
| first and last position | 0/1 (p = 1) | 0.90 | 0.87 | 1.00 | 1.17 |

**Checks and deviations.** The engine check against the 8-bit rows of iteration 4 was sensitive to precision (Table 213): the twins agreed per slot at 0.98 to 1.00 and the pushed cells at 0.74 to 0.86, so no iteration-4 row was reused and temperature 1.3, top-k 10, the novelty instruction and the semantic seed were rerun. Twelve correct programs labelled other were 10 labeller misses, 1 degenerate and 1 unresolved; 0 novel. The style re-gate flagged only top-k frequent (agreement 0.80). An independent recomputation reproduced the literal cell, every McNemar count and the verdict, with the 5-point Hill-1 of the hull within 0.08 of the committed value and a win count of 2 rather than 1 (a near tie on the grid task), still DISCONFIRMED; relabelling plan noise rows as the twin gave margins of 0 or below on all five tasks, and shuffles of the arm labels centred at 1.6 wins (3 or more wins in 3 of 20 shuffles), so the count rule over five tasks is only weakly selective on its own. The plan's one allowed wording revision was used after the first format pilot failed (the model wrote code at once), the committed drop order fired after stage 100 and the dropped cells were restored when the queue finished early, and cells with fewer than 30 (decisive) or 15 (other) programs were excluded from hulls, a rule decided after a dry run that favours plan noise on kth smallest. Post hoc, with iteration-1 base label names harmonised, landing mass outside the base support is 0.11 for enumerate and pick on the grid task (bidirectional BFS) and 0 for every shortest path landing.

**Table 213.** Engine check of the bf16 knob engine against iteration 4's 8-bit rows. Source: knob experiment `results/engine_check.json`.

| Cell | byte-identical | label agreement per slot | JSD | null p95 |
|---|---|---|---|---|
| kth smallest / twin | 45/50 | 1.00 | 0.00 | 0.00 |
| kth smallest / random direction 9002, alpha 1 | 32/50 | 0.74 | 0.03 | 0.15 |
| shortest path / twin | 35/50 | 0.98 | 0.01 | 0.00 |
| shortest path / random direction 9004, alpha 1 | 10/50 | 0.86 | 0.01 | 0.01 |
| kth smallest / temperature 1.3 | 0/10 | 1.00 | 0.00 | – |
| kth smallest / top-k 10 | 0/10 | 1.00 | 0.00 | – |
| shortest path / temperature 1.3 | 0/10 | 0.90 | 0.05 | – |
| shortest path / top-k 10 | 0/10 | 0.90 | 0.05 | – |

## 5.8 The second family: gemma-3-4b-pt against gemma-3-4b-it

**Why.** The Gemma replication asked whether the account replicates in a second family, with the base repertoire of gemma-3-4b-pt committed before a fresh, never-run push population was applied to gemma-3-4b-it [76]. Gemma's iteration-4 push outcomes were seen, so this experiment is a replication on fresh directions and samples, not a held-out confirmation.

**Design.** Both models ran in bf16 on one 23 GB NVIDIA GPU, one at a time. A fresh it twin (N = 50 per task) fixed each task's default under the iteration-5 rule; the grid twin now gave DFS on 30 of 40 correct programs, so every one of the 15 tasks had a default. The pt model wrote 100 programs under the it chat template on the same token ids and 50 in plain completion of a `def solve(...)` stub. Under the chat template 91% of its programs had no usable code and only 10 of 1,500 passed, so the preregistered chat validity gate failed, both formats were pooled, and every unit fell under the low-correct rule: beta, the runner-up and the landing forecast come from valid attempts, not correct programs. These were committed by hash before 16 never-run isotropic directions (seeds 52001 to 52016) were applied at alpha 1 and 2 to the task-specification positions of blocks 11 to 22 on the outlier-excluded ban-free unit. A transplant from pt into it at lambda 1 then ran on the seven tasks diverse in the base with a matched random control, a borrowed shift and a transplant at the read position on the same slots, and on three controls collapsed in the base.

**In Gemma the movability half is inconclusive and the landing half passes** (Table 214). Spearman between beta and the escape fraction is 0.42 (one-sided permutation p 0.060, n = 15), against a bar of 0.5 at p below 0.01; 2 of 3 tasks collapsed in the base are immovable against a bar of 80%; and beta is ahead of no rival, with the twin modal share (0.67) and the novelty instruction (0.64) tracking movability better (Table 215). The direction is the predicted one: 6 of 7 tasks diverse in the base are movable against 2 of 8 others (one-sided Fisher p 0.032, not registered). The partial Spearman given catalogue size and the forced can count is 0.41 (0.45 given catalogue size alone), and the fresh population agrees with iteration 4's movability on 13 of 15 tasks (8 movable now, 6 then). No sensitivity meets the preregistered bar (Table 216); beta from the chat template alone reaches 0.62 (p 0.0095), but that arm is 91% unusable output, and beta from correct programs only reaches 0.27. The pt runner-up equals the modal landing of confirmed escapes in 5 of 7 events (hit rate 0.71 against a mean chance of 0.30; exact Poisson-binomial p 0.025), and the pt composition beats iteration 4's association forecast by 1.50 nats per event (exact Wilcoxon p 0.023) and AST proximity by 0.51 (Table 217). The two misses are kth smallest (landing size-k heap, runner-up quickselect) and connected components (landing union find, pooled runner-up BFS; union find is the runner-up under the chat template alone).

**Table 214.** Gemma results per unit: it default (twin share), pt beta and runner-up, escape fraction over 16 fresh directions, landing, and the rivals. Source: Gemma experiment `results/e_a1.json`, `results/e_a2.json`, `README.md`. Base support per unit (valid pt attempts / correct pt programs / runner-up count among valid attempts, of 150 pooled pt rows; the pt labels agree with a blind judge at 0.64): grid 73 / 0 / 4; shortest path 78 / 0 / 1; kth smallest 51 / 14 / 6; substring search 55 / 18 / 26; find duplicate 53 / 14 / 9; two sum 55 / 13 / 14; connected components 47 / 1 / 6; non-overlapping intervals 49 / 0 / 1; Fibonacci 54 / 11 / 3; topological order 42 / 1 / 3; merge intervals 53 / 5 / 2; first and last position 48 / 4 / 12; top-k frequent 56 / 9 / 4; longest increasing subsequence 51 / 5 / 20; cycle detection 46 / 0 / 3 (iteration 6).

| Unit | it default (twin share) | beta (pt) | Class | pt runner-up | Escape fraction (alpha 1, alpha 2) | Maximum over alpha | Landing | Runner-up = landing | Plan noise excess | Novelty instruction off-default | Twin modal share | Opening margin |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| grid | DFS (0.75) | 0.05 | intermediate | BFS | 0.156 (0.125, 0.188) | 0.188 | BFS | yes | 0.02 | 0.77 | 0.62 | 0.0 |
| shortest path | Dijkstra (1.00) | 0.03 | collapsed | Bellman-Ford/SPFA | 0.062 (0.000, 0.125) | 0.125 | Bellman-Ford/ SPFA | yes | 0.04 | 0.20 | 1.00 | 21.7 |
| kth smallest | full sort (1.00) | 0.23 | diverse | quickselect | 0.031 (0.000, 0.062) | 0.062 | size-k heap | no | 0.06 | 0.67 | 1.00 | 18.6 |
| substring search | naive (0.88) | 0.51 | diverse | built-in find | 0.062 (0.000, 0.125) | 0.125 | built-in find | yes | 0.06 | 0.70 | 0.88 | -27.0 |
| find duplicate | hash set (1.00) | 0.23 | diverse | sort and adjacent check | 0.000 (0.000, 0.000) | 0.000 | – | – | 0.00 | 0.00 | 1.00 | 34.2 |
| two sum | hash map (1.00) | 0.53 | diverse | brute force | 0.062 (0.000, 0.125) | 0.125 | brute force | yes | 0.04 | 0.00 | 1.00 | 16.8 |
| connected components | DFS (1.00) | 0.28 | diverse | BFS | 0.062 (0.000, 0.125) | 0.125 | union find | no | 0.14 | 0.00 | 1.00 | -20.2 |
| nonoverlapping intervals | greedy by end (1.00) | 0.02 | collapsed | DAG longest path | 0.000 (0.000, 0.000) | 0.000 | – | – | 0.02 | 0.00 | 1.00 | 0.0 |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Fibonacci | iteration (1.00) | 0.10 | intermediate | fast doubling | 0.000 (0.000, 0.000) | 0.000 | – | – | 0.00 | 0.00 | 1.00 | 16.6 |
| topological order | Kahn (1.00) | 0.12 | intermediate | DFS postorder | 0.000 (0.000, 0.000) | 0.000 | – | – | 0.00 | 0.00 | 1.00 | – |
| merge intervals | sort and sweep (1.00) | 0.04 | collapsed | union find | 0.000 (0.000, 0.000) | 0.000 | – | – | 0.00 | 0.00 | 1.00 | 0.0 |
| first and last position | linear scan (0.92) | 0.40 | diverse | binary search | 0.531 (0.688, 0.375) | 0.688 | binary search | yes | 0.36 | 1.00 | 0.92 | 6.0 |
| top-k frequent | sort by count (1.00) | 0.12 | intermediate | quickselect on counts | 0.000 (0.000, 0.000) | 0.000 | – | – | 0.08 | 0.57 | 1.00 | 18.9 |
| longest increasing subsequence | patience sorting with bisect (1.00) | 0.41 | diverse | quadratic DP | 0.031 (0.000, 0.062) | 0.062 | quadratic DP | yes | 0.02 | 0.00 | 1.00 | 4.4 |
| cycle detection | DFS with parent (1.00) | 0.11 | intermediate | edge count | 0.000 (0.000, 0.000) | 0.000 | – | – | 0.00 | 0.00 | 1.00 | 10.9 |

**Table 215.** Rivals against beta in Gemma (90% task-bootstrap intervals). Source: Gemma experiment `results/e_a1.json`.

| Rival | sign | n | ρ(rival, E) | ρ(β,E) − ρ(rival,E) | 90% bootstrap CI | β ahead |
|---|---|---|---|---|---|---|
| twin modal share | -1 | 15 | 0.668 | -0.247 | [-0.755, 0.259] | no |
| novelty instruction off-default | 1 | 15 | 0.640 | -0.219 | [-0.786, 0.430] | no |
| opening margin | -1 | 14 | 0.272 | 0.147 | [-0.458, 0.725] | no |
| runtime merit gap | -1 | 15 | 0.381 | 0.040 | [-0.453, 0.695] | no |
| token merit gap | -1 | 15 | 0.409 | 0.011 | [-0.634, 0.656] | no |

**Table 216.** Sensitivities of the movability half in Gemma under the same rule. Source: Gemma experiment `results/e_a1.json`.

| Sensitivity | n | ρ | p (one-sided) | verdict under the same rule |
|---|---|---|---|---|
| beta from the base model under the chat template only | 15 | 0.617 | 0.0095 | PARTIAL |
| beta from plain completion only | 15 | 0.362 | 0.0927 | INCONCLUSIVE |
| beta pooled over both formats (primary) | 15 | 0.421 | 0.0602 | INCONCLUSIVE |
| beta with other excluded | 15 | 0.388 | 0.0771 | INCONCLUSIVE |
| beta from correct programs only, any N | 11 | 0.274 | 0.2097 | DISCONFIRMED |
| beta with disputed rows re-judged | 15 | 0.389 | 0.0764 | INCONCLUSIVE |
| escape defined as any off-default attempt (iteration-4 definition) | 15 | 0.421 | 0.0602 | INCONCLUSIVE |
| escape counted on correct programs only | 15 | 0.520 | 0.0258 | PARTIAL |
| escape fraction as the maximum over alpha | 15 | 0.421 | 0.0602 | INCONCLUSIVE |
| without the task flagged by the style re-gate | 14 | 0.343 | 0.1125 | INCONCLUSIVE |
| secondary target: plan noise excess | 15 | 0.454 | 0.0458 | INCONCLUSIVE |
| beta against the iteration-4 escape fraction | 15 | 0.354 | 0.1000 | INCONCLUSIVE |

**Table 217.** Landing events in Gemma: pt runner-up against the modal landing of confirmed escapes. Source: Gemma experiment `results/e_a2.json`. Base support (valid attempts / correct programs / runner-up count among valid attempts, with the count among correct programs in

brackets, of 150 pooled pt rows; beta and the runner-up come from valid attempts because every unit is LOW_CORRECT, and the pt labels agree with a blind judge at 0.64): grid 73 / 0 / 4 (0 correct); kth smallest 51 / 14 / 6 (0 correct); substring search 55 / 18 / 26 (13 correct); two sum 55 / 13 / 14 (7 correct); connected components 47 / 1 / 6 (0 correct); first and last position 48 / 4 / 12 (0 correct); longest increasing subsequence 51 / 5 / 20 (5 correct) (iteration 6).

| Event | pt runner-up | Landing | landing composition | n outcomes |
|---|---|---|---|---|
| grid | BFS | BFS | BFS 91 | 91 |
| kth smallest | quickselect | size-k heap | size-k heap 7 | 7 |
| substring search | built-in find | built-in find | built-in find 44 | 44 |
| two sum | brute force | brute force | brute force 38 | 38 |
| connected components | BFS | union find | union find 20 | 20 |
| first and last position | binary search | binary search | binary search 375 | 375 |
| longest increasing subsequence | quadratic DP | quadratic DP | quadratic DP 19 | 19 |

**The transplant moves two of seven tasks diverse in the base** (Table 218). Every transplanted program stayed valid (pass ratio 0.98 to 1.00). Substring search moved from the naive scan to the built-in find in 40 of 50 programs (twin 6 of 50), and first and last position from the linear scan to binary search in 39 of 50 (twin 4 of 50); both land on the pt runner-up. The other five tasks diverse in the base stayed at their default. The two that moved are exactly the tasks diverse in the base whose it twin already writes the pt runner-up 8% to 12% of the time, so the transplant amplified an existing minority mode and did not install an absent one. The pt-minus-it difference at the task-specification positions is large (median 5.2 ban-free units, 4.7 to 5.8 across blocks), above the dose at which random noise broke programs in iteration 4, and five massive channels carry 86% of its squared norm at block 11, falling to 31% to 36% by blocks 21 and 22. Random controls and borrowed shifts that copy the massive-channel move and match the remaining norm broke most programs (pass ratio 0.16 to 0.50 and 0 to 0.22), so the random comparison is confounded by norm and the borrowed shift is itself broken: what the controls show is that the coherent pt content is benign where an equally large random displacement or borrowed shift is destructive. The transplant at the read position and the three controls collapsed in the base stayed within 0.05 of the twin (Table 219); on nonoverlapping intervals the transplant halved the pass rate without moving the algorithm.

**Table 218.** Transplant from pt into it at lambda 1 on the seven tasks diverse in the base (s = correct off-default share; McNemar counts transplant only : control only). Source: Gemma experiment `results/transplant.json`.

| Unit | β | twin s | T s [95% CI] | RND s | XT s | RP s | T pass ratio | RND pass ratio | McNemar T > RND (b10/b01, p) | T > XT (b10/b01, p) | T lands on r_u | JS(T,pt) < JS(twin,pt) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| kth smallest | 0.23 | 0.00 | 0.00 [0.00, 0.07] | 0.02 | 0.00 | 0.00 | 1.00 | 0.44 | 0/1, 1.000 | 0/0, 1.000 | no | no |
| substring search | 0.51 | 0.12 | 0.80 [0.67, 0.89] | 0.04 | 0.00 | 0.10 | 1.00 | 0.16 | 38/0, 0.000 | 40/0, 0.000 | yes | yes |
| find duplicate | 0.23 | 0.00 | 0.00 [0.00, 0.07] | 0.00 | 0.00 | 0.00 | 0.98 | 0.40 | 0/0, 1.000 | 0/0, 1.000 | no | no |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| two sum | 0.53 | 0.00 | 0.00 [0.00, 0.07] | 0.04 | 0.00 | 0.00 | 1.00 | 0.34 | 0/2, 1.000 | 0/0, 1.000 | no | no |
| connected components | 0.28 | 0.00 | 0.00 [0.00, 0.07] | 0.02 | 0.00 | 0.00 | 1.00 | 0.50 | 0/1, 1.000 | 0/0, 1.000 | no | no |
| first and last position | 0.40 | 0.08 | 0.78 [0.65, 0.87] | 0.06 | 0.00 | 0.10 | 1.00 | 0.20 | 36/0, 0.000 | 39/0, 0.000 | yes | no |
| longest increasing subsequence | 0.41 | 0.00 | 0.02 [0.00, 0.10] | 0.16 | 0.00 | 0.00 | 1.00 | 0.48 | 0/7, 1.000 | 1/0, 0.500 | yes | yes |

**Table 219.** Controls collapsed in the base for the transplant. Source: Gemma experiment `results/transplant.json`.

| Collapsed control | twin s | T s | T pass ratio |
|---|---|---|---|
| shortest path | 0.00 | 0.00 | 1.00 |
| merge intervals | 0.00 | 0.00 | 1.00 |
| nonoverlapping intervals | 0.00 | 0.00 | 0.52 |

Plan noise rarely moved the it model (Table 220); the exceptions were first and last position (correct off-default share 0.44 and 0.38 at alpha 1 and 2), the grid task (DFS to BFS) and connected components at alpha 2. Spearman between beta and the plan noise excess is 0.45 (one-sided p 0.046).

**Table 220.** Plan noise in gemma-3-4b-it (a fresh direction per sample, N = 50 per alpha). Source: Gemma experiment `results/pn.json`.

| Unit | twin pass | twin Hill-1 | PN α1 pass | PN α1 off-default correct | PN α1 Hill-1 | PN α2 pass | PN α2 off-default correct | PN α2 Hill-1 |
|---|---|---|---|---|---|---|---|---|
| grid | 0.80 | 1.74 | 0.72 | 0.22 | 1.84 | 0.58 | 0.18 | 1.85 |
| shortest path | 1.00 | 1.00 | 1.00 | 0.04 | 1.17 | 0.94 | 0.04 | 1.18 |
| kth smallest | 1.00 | 1.00 | 1.00 | 0.06 | 1.25 | 1.00 | 0.06 | 1.25 |
| substring search | 1.00 | 1.43 | 1.00 | 0.16 | 1.54 | 1.00 | 0.18 | 1.59 |
| find duplicate | 1.00 | 1.00 | 1.00 | 0.00 | 1.00 | 0.98 | 0.00 | 1.00 |
| two sum | 1.00 | 1.00 | 1.00 | 0.00 | 1.00 | 1.00 | 0.04 | 1.18 |
| connected components | 1.00 | 1.00 | 1.00 | 0.04 | 1.17 | 1.00 | 0.14 | 1.49 |
| nonoverlapping intervals | 1.00 | 1.00 | 0.98 | 0.00 | 1.00 | 0.74 | 0.02 | 1.12 |
| Fibonacci | 1.00 | 1.00 | 1.00 | 0.00 | 1.00 | 1.00 | 0.00 | 1.00 |
| topological order | 1.00 | 1.00 | 1.00 | 0.00 | 1.00 | 0.98 | 0.00 | 1.00 |
| merge intervals | 1.00 | 1.00 | 1.00 | 0.00 | 1.00 | 1.00 | 0.00 | 1.00 |
| first and last position | 1.00 | 1.32 | 0.98 | 0.44 | 2.26 | 0.92 | 0.38 | 3.08 |
| top-k frequent | 0.96 | 1.00 | 0.86 | 0.00 | 1.00 | 0.92 | 0.08 | 1.34 |
| longest increasing subsequence | 1.00 | 1.00 | 1.00 | 0.00 | 1.00 | 1.00 | 0.02 | 1.19 |
| cycle detection | 1.00 | 1.00 | 1.00 | 0.00 | 1.00 | 1.00 | 0.00 | 1.00 |

**Labels and checks.** The base labels are the weakest link: runtime labels agreed with an adjudication by judges from two families on 30 of 30 it programs but on 5 of 30 pt programs (20 of 30 after the inline judge), because the pt code is mostly broken, so the style re-gate failed and its

preregistered fallback, judging every disagreement between runtime and AST labels, applied; a 225-program audit of pt labels agreed at 0.64 (0.47 to 0.93 by task). A structural audit of 92 default and runner-up programs agreed at 0.99. The re-gated shortest path rule, which reads array-based Dijkstra correctly, relabels 258 iteration-4 shortest path programs: their escape fraction falls from 0.25 to 0.125 and their ban compliance from 0.96 to 0.34, which confirms the iteration-4 mislabel. The engine check reproduced iteration-4 slots exactly; 0 novel algorithms among 56 correct programs labelled other (35 catalogue variants, 21 degenerate); the commits preceded the first push and transplant rows; two independent recomputations found no mismatch, with placebos failing as required. Seventeen deviations are logged.

## 5.9 Positioning: variety in the base model, transplants and base-aligned collaboration in published work

The positioning audit, dated 2026-10-21, extended the iteration-4 audit to the new claims. It checked 137 verbatim quotes from 93 references (111 against full text, 21 against ICLR 2027 abstracts, 5 against blog posts; no mismatch, and a live second pass re-matched all 137) and ran a saturation search under a protocol frozen before the first call: 169 logged calls, 4,508 raw hits, 3,506 unique records, 49 scored on the occupancy rubric, with the stop rule met after the third round. Table 221 gives its verdict per claim.

**Table 221.** Lane verdicts for the iteration-5 claims as of 2026-10-21. Source: positioning audit `research_report.md`, `evidence/flags.json`.

| Claim | Lane verdict | Deciding neighbours |
|---|---|---|
| Premise: tuning narrowed algorithm choice on code (premise flag) | partly occupied; contradicting in part | Lee et al. compare a code base and instruct pair and find instruction tuning narrows algorithmic diversity on easy problems and widens it on others [72]; Le Bronnec et al. [107]; support from the text domain [101] |
| Movability half: movability follows the base repertoire (forecast flag) | partly occupied at training time; inference-time movability open | forecasts per prompt from the frozen base model of what reinforcement learning narrows [98, 99, 109] |
| Landing half: the landing is the base runner-up | adjacent only (stylised and retrospective support) | sharpening of the distribution under typicality bias [46]; aligned top tokens within the base top-5 [100] |
| Transplant claim: base prefill states written into the instruct model (transplant flag) | method occupied, diversity outcome open; transplant flag not fired | cross-patching of base states into a descendant [91]; activation patching from base into tuned models [88, 89, 90]; restoration of diversity in weights [92, 93] |
| Push toward the base | method and control with random directions occupied (style outcome) | ablation of an aligned minus base direction during decoding [97] |
| Plan sampler comparison | partly occupied; base-aligned collaboration is an occupied lane | token-level base-aligned collaboration [94, 95, 96]; intent factored generation, plan search and decoding at matched departure [73, 74] |
| Gemma replication | adjacent only | a base and chat Gemma pair can diverge unusually (Kissane et al., blog); template-controlled contrasts [103] |
| Novelty boundary | unchanged from Table 92c | [81, 82, 105, 106] |

Three flags summarise the search. The transplant flag, a paper that writes base-model hidden states into its tuned descendant and reports a change in output diversity, did not fire, narrowly: cross-patching writes base upstream states into a descendant's late stack but measures a next-token margin [91], and earlier patching work measured backdoor removal [89], factual recall [90] or entity tracking [88]; the goal of restoring diversity is published at the weight level, by restoring layer intervals to pretrained weights [92] or merging aligned lower layers with base upper layers [93], and at the token level [94]. The transplant may not be called first. The forecast flag, a forecast from the base model of where post-training narrows, partly fired: forecasts per prompt at training time exist, and one finds that base pass rate and likelihood explain only 0.30 and 0.24 of the explainable variance in post-training gains [99]; inference-time movability and the identity of the landing remain open. The premise flag partly fired: Lee et al. compared DeepSeek Coder base and instruct models with test-checked correctness and found a mixed direction [72], so the premise row reads contradicting in part. The audit recommends further baselines for any later report: weight interpolation between Qwen3-8B-Base and Qwen3-8B [102], layer restoration and splicing at the transplant's block ranges [92, 93], token routing between base and aligned models [94], arithmetic on the logits of base and tuned models, and template control for every base minus instruct contrast [103]. It also lists 27 corrections to earlier citations. Table 92d condenses the neighbour table of the audit; the verbatim quotes are in its `evidence/quotes_iter5.jsonl`.

**Table 92d.** Iteration-5 claims and their nearest published neighbours (the verbatim quote behind each row is in the positioning audit's evidence file). Source: positioning audit `research_report.md`, Section 3.

| Row | Claim | Neighbour | Relation | What this run adds |
|---|---|---|---|---|
| 92d-A0a | Premise: tuning narrowed an algorithm choice the base model makes more evenly (code, algorithm level) | Lee et al., 2025, Findings of EMNLP 2025 [72] | contradicts in part | Test-checked mechanism labels over correct programs (runtime tracer + AST + blind judge) instead of clusters from an LLM judge; same prompt under the chat template for base and instruct; a base and instruct design in two families (Qwen3-8B, gemma-3-4b); not the first comparison of base and tuned models at the algorithm level (premise flag partly fired) |
| 92d-A0b | (same claim) | Lee et al., 2025, Findings of EMNLP 2025 [72] | contradicts in part | Per-task test of the premise; Lee et al.'s difficulty split predicts the premise can fail on harder tasks |
| 92d-A0c | (same claim) | Lee et al., 2025, Findings of EMNLP 2025 [72] | context | Mechanism labels checked by execution traces rather than an LLM judge |
| 92d-A0d | (same claim) | Le Bronnec et al., 2026, arXiv preprint (under review) [107] | adjacent | Diversity at the algorithm level (not structural near duplicates); instruction-tuned rather than RLVR descendant |
| 92d-A0e | (same claim) | Wu et al., 2024, ICLR 2025 (poster) [2] | context | Open-weight base/instruct code pair (none in Monoculture's code arm) |
| 92d-A0f | (same claim) | Jangra et al., 2026, arXiv preprint [3] | context | Pairing of base and instruct models (DivAlgo has none) |

| | | | | |
|---|---|---|---|---|
| 92d-A0g | (same claim) | Yang et al., 2025, TMLR 2026 (v4 comment) [101] | supports | replication of the narrowing from base to aligned models at the algorithm level on code |
| 92d-A0h | (same claim) | Liu, 2026, arXiv preprint | supports | Code, test-checked; not semantic clusters of QA answers |
| 92d-A0i | (same claim) | Anonymous, 2026, ICLR 2027 submission (under review); contemporaneous by rule | supports | Code algorithms instead of Countdown entrance families; base vs instruction-tuned rather than PPO/GRPO |
| 92d-A1a | Movability follows the base repertoire (the tuned model can be pushed off its default only where the base spreads) | Zhou (T.), 2026, arXiv preprint [98] | partly fires (forecast flag) | Inference-time movability of the tuned model under pushes (not loss of coverage at training time); algorithm choice, not correctness; base repertoire (beta, runner-up) committed by sha256 before any push |
| 92d-A1b | (same claim) | Zhou (T.), 2026, arXiv preprint [98] | partly fires (forecast flag) | Forecast scored on matched events at inference time |
| 92d-A1c | (same claim) | Duan, 2026, arXiv preprint [99] | contradicts in part | Must beat (or report against) simple base signals; Duan et al. find base pass rate / likelihood weak predictors of training gains |
| 92d-A1d | (same claim) | Duan, 2026, arXiv preprint [99] | supports | Base-model behaviour on a task (its algorithm distribution) as the predictor |
| 92d-A1e | (same claim) | Anonymous, 2026, ICLR 2027 submission (under review); contemporaneous by rule | partly fires (forecast flag) | Uses the base model alone (no reward model); outcome = movability under activation/prompt pushes |
| 92d-A1f | (same claim) | Nguyen et al., 2025, NeurIPS 2026 (poster) [109] | adjacent | Per-task forecast tested on the deployed instruct model rather than an account of training dynamics |
| 92d-A1g | (same claim) | Anonymous, 2026, ICLR 2027 submission (under review); contemporaneous by rule | contradicts in part | A control with random tasks for any claim of targeting |
| 92d-A1h | (same claim) | Mohammadi, 2024, arXiv preprint [108] | adjacent | Quantified per task with test-checked labels |
| 92d-A2a | Landing: a push or ban of the tuned model lands on the base model's runner-up algorithm | Zhang et al., 2025, arXiv preprint [46] | partly fires (stylised) | A preregistered forecast per task of the landing identity (runner-up) scored on matched events; none found |

| | | | | |
|---|---|---|---|---|
| 92d-A2b | (same claim) | Lin et al., 2023, arXiv preprint (ICLR 2024 per plan; not confirmed here) [100] | supports | Landings at the algorithm level (whole solutions) rather than token ranks |
| 92d-A2c | (same claim) | Anonymous, 2026, ICLR 2027 submission (under review); contemporaneous by rule | adjacent | Forecast before the push, not retrospective inheritance |
| 92d-A2d | (same claim) | Samanta et al., 2026, arXiv preprint | supports | Code algorithms; prospective |
| 92d-A2e | (same claim) | Fortier et al., 2026, arXiv preprint | supports | Algorithm choice on code; landing under pushes |
| 92d-A2f | (same claim) | Zhao et al., 2025, COLM 2025 | supports | Released base/instruct pair instead of models trained from scratch |
| 92d-A2g | (same claim) | Anonymous, 2026, ICLR 2027 submission (under review); contemporaneous by rule | supports | Activation-level pushes; algorithm identity |
| 92d-B1a | Transplant: base prefill states (task specification positions, blocks 12-23, lambda 0.25/0.5/1) written into the instruct model restore the base choice; controls: read position, blocks 0-11/24-35, matched random, borrowed shift | Zhou (Y.), 2026, arXiv preprint [91] | occupies (method) | Outcome = test-checked algorithm choice / solution diversity (not a next-token margin); site = task-specification prefill positions; dose lambda; matched-norm random displacement and patches borrowed from other tasks as controls |
| 92d-B1b | (same claim) | Zhou (Y.), 2026, arXiv preprint [91] | occupies (method) | Site comparison includes read position and early/late block ranges at matched dose |
| 92d-B1c | (same claim) | Oozeer et al., 2025, ICML 2025 [89] | method overlap | Diversity/algorithm outcome instead of backdoor removal |
| 92d-B1d | (same claim) | Du et al., 2025, COLM 2025 [90] | method overlap | Behavioural (solution) outcome instead of log probabilities of factual recall |
| 92d-B1e | (same claim) | Prakash et al., 2024, ICLR 2024 [88] | method overlap | Direction base->tuned (CMAP patched tuned->base) |
| 92d-B1f | (same claim) | Zhang et al., 2026, arXiv preprint [92] | adjacent | States rather than weights; code; matched random control |
| 92d-B1g | (same claim) | Sumita et al., 2026, arXiv preprint [93] | adjacent | Per-position state transplant in the opposite composition (base states into instruct blocks) |

| | | | | |
|---|---|---|---|---|
| 92d-B1h | (same claim) | Anonymous, 2026, ICLR 2027 submission (under review); contemporaneous by rule | adjacent | Hidden-state transplant (not weights) on code |
| 92d-B1i | (same claim) | Wang et al., 2025, ICML 2026 [94] | occupies (goal, token level) | Activation level instead of token routing |
| 92d-B2a | Push toward the base: push the instruct model along the base minus instruct direction against isotropic pushes at matched norm | Anand et al., 2026, arXiv preprint [97] | occupies (method) | replication of the PASTA ablation of the aligned minus base direction in a new setting: algorithm choice in code, against matched-norm isotropic pushes |
| 92d-B2b | (same claim) | Anand et al., 2026, arXiv preprint [97] | occupies (control) | Matched-norm isotropic directions rather than orthogonal unit directions |
| 92d-B2c | (same claim) | Anonymous, 2026, ICLR 2027 submission (under review); contemporaneous by rule | adjacent | Diversity outcome rather than accuracy |
| 92d-B2d | (same claim) | Minder et al., 2025, ICLR 2026 | adjacent | Opposite sign (toward base) and a diversity outcome |
| 92d-B2e | (same claim) | Zheng et al., 2024, ACL 2025 | adjacent (weight space) | Push in activation space; extrapolation in weight space is a reviewer baseline |
| 92d-B2f | (same claim) | Nakamura, 2026, arXiv preprint [103] | supports (method caution) | Direction estimated with base and instruct under the same chat template |
| 92d-Ca | Knobs compared directly: activation plan noise against text-level plan samplers, base-aligned collaboration (base writes the intent, instruct writes the code) and decoding at matched departure | Wang et al., 2025, ICML 2026 [94] | occupies (collaboration lane) | The base intent arm is an instance of base-aligned collaboration at the level of sentences; the addition is pricing it against activation noise on test-checked algorithms |
| 92d-Cb | (same claim) | Peeperkorn et al., 2025, arXiv preprint [95] | occupies (collaboration lane) | Code, algorithm-level outcome |
| 92d-Cc | (same claim) | Feng et al., 2025, arXiv preprint [96] | occupies (collaboration lane) | Code, algorithm-level outcome |
| 92d-Cd | (same claim) | Zhang et al., 2026, arXiv preprint | adjacent | Code; comparison with activation noise |
| 92d-Ce | (same claim) | Sumita et al., 2026, arXiv preprint [93] | adjacent | Code; text-level plan samplers in the contrast set |
| 92d-Cf | (same claim) | Dang et al., 2025, COLM 2025 [102] | baseline to add | Weight interpolation as a contrast arm |

| | | | | |
|---|---|---|---|---|
| 92d-Cg | (see Table 92c) | Intent Factored Generation: Unleashing the Diversity in Your Language [73] | inherited | text plan sampler, hot for the intent and cold for the response (Table 92c) |
| 92d-Ch | (see Table 92c) | Planning In Natural Language Improves LLM Search For Code Generation [74] | inherited | natural-language plan search (Table 92c) |
| 92d-Ci | (see Table 92c) | Where You Inject Diversity Matters: A Unified Framework for Diverse Ge [83] | inherited | Specifications per output at level 2 (Table 92c) |
| 92d-Cj | (see Table 92c) | Gacha Decoding: Eliciting Diverse Generations Through Instruction Foll | inherited | instruction-following diversity (Table 92c) |
| 92d-Ck | (see Table 92c) | Rethinking Activation Steering for Mathematical Reasoning: What Actual | inherited | decoding matched to steering's departure from greedy (Table 92c) |
| 92d-Cl | (see Table 92c) | Perturb the Thought, Not the Pixels: Latent-Space Rollout Diversificat [104] | inherited | latent rollout perturbation in RL (Table 92c) |
| 92d-Ea | Second family: gemma-3-4b pt vs it | Kissane et al., n.d., grey literature (blog/ research update) | context (caution) | Report base/instruct activation-space similarity for gemma-3-4b before transplanting; Gemma base/chat pairs can diverge unusually |
| 92d-Eb | (same claim) | Zhang et al., 2026, arXiv preprint [92] | adjacent | Gemma in a code setting; states not weights |
| 92d-Ec | (same claim) | Nakamura, 2026, arXiv preprint [103] | supports (method caution) | Template-controlled contrasts on the second family |
| 92d-Ga | (see Table 92c) | Mathematical discoveries from program search with large language model [81] | inherited | novelty boundary unchanged; no new 2026 hit |

## 5.10 Comparisons across this iteration's experiments

**The base repertoire against movability: screen, confirmation and second family.** The same quantity, beta against the escape fraction, was measured five ways in this iteration, and the answer depends on which tasks enter. On the 15 existing tasks, with the beta of Step 1 and the escape fractions of iteration 4, Spearman is 0.78 to 0.79; with this iteration's own bf16 escape fractions it is 0.59; on the new tasks it is -0.03 in panel A, 0.21 in panel B and 0.15 pooled; in Gemma it is 0.42. The screen correlation is carried by many tasks with low beta that never move, while on new tasks those diverse in the base that never move are common: count primes and valid anagram (beta 0.81 each), median of two sorted lists (0.71), jump game (0.47), all subsets (0.37) and longest

common prefix (0.30). The figure, corrected in iteration 6, plots both sets like for like under bf16: the existing tasks at their bf16 escape fraction under the ban-free unit (filled; the 8-bit screen value as an open ghost marker joined by a segment) and the new units, both at iteration 5's ban-free unit. Existing tasks (n 15): Spearman 0.59, one-sided permutation p 0.010, Bonett-Wright 95% [0.06, 0.86]; with the 8-bit screen E instead 0.79. New units (n 13): 0.15, p 0.315, [-0.44, 0.65]; with the two reserve units 0.01. Pooled bf16 (n 28): 0.48, p 0.005. The existing-task correlation is carried by many tasks with low beta that never move; open markers show the 8-bit screen value of each existing task. Per set, the base runner-up named the push landing 5 of 5 on the 8-bit screen, 1 of 2 on the bf16 remeasurement of existing tasks, 3 of 3 in panel A, 3 of 4 in panel B and 5 of 7 in Gemma (17 of 21, Poisson-binomial p 5.8e-07); the bf16 remeasurement's kth smallest and connected components events were missing from the earlier count of 16 of 19 (kth smallest a miss: landing size-k heap against runner-up quickselect; connected components a hit). On the 8 held-out events where the instruct twin wrote no correct program off the default the base runner-up named 5 (p 0.053; Qwen3-8B 3 of 4, p 0.052; Gemma 2 of 4, p 0.422) (iteration-6 correction).

**Movability depends on the engine and the unit.** Table 222 sets the existing tasks' movability side by side across the measurements of iterations 4 and 5. Under iteration 4's 8-bit engine and the ban unit, six tasks were movable; under this iteration's bf16 engine and the ban-free unit, only kth smallest and connected components escape with 8 fixed directions, and per-sample plan noise moves more than 10% of programs off the default only on kth smallest (0.28 at alpha 2). On the shortest path task, per-sample plan noise moved 26% of all samples at alpha 1 in iteration 4 (13 of 50 valid attempts off the default; 9 of 43 correct programs) and 0 of 50 correct programs at alpha 1 here (2 of 47 at alpha 2, 1.67 8-bit ban units by the ratio of mean norms and 1.78 by the mean ratio). A dose-bracketed comparison (iteration 6) sets the bf16 alpha-2 cell, which delivers more than the 1.00 ban unit of the 8-bit alpha-1 cell under either dose definition, against the 8-bit alpha-1 cell: a lower bf16 share there cannot be a dose shortfall. By the rule fixed before computing, connected components and shortest path did not replicate in bf16 at the population level (correct-only bracket one-sided Fisher p kth smallest 0.588, connected components 0.056, shortest path 0.017, two sum 1.000, grid 0.732); kth smallest: not established; two sum: not established (no 8-bit effect of at least 10% at alpha of 2 or less to replicate); grid: not established; find duplicate: not established (no 8-bit effect of at least 10% at alpha of 2 or less to replicate). This rules out the dose unit, not the CRN slots, the labels or the kernels (iteration-6 correction; record audit, Table 228). The engine control of Section 6.3 then tested the numerics directly: at iteration 4's own absolute displacement, direction seeds and slots, the bf16 engine reproduces the 8-bit rows (per-sample plan noise at alpha 2: connected components 25 of 50 valid attempts off the default under bf16 against 30 of 50 under the 8-bit engine; shortest path 22 of 50 under both), so the non-replication tagged here belongs to iteration 5's ban-free unit and direction seeds, not to the bf16 numerics. The engine checks pushed two fixed escaping directions at the ban unit in both engines and agreed within or near the resampling null (shortest path: Dijkstra 31 and Bellman-Ford/SPFA 19 in bf16 against 26 and 24 in the 8-bit engine). Those two directions therefore do not explain the difference, and its cause is not established here. *(Correction, iteration 6: the engine control of Section 6.3 establishes that the cause is not the bf16 numerics. At the absolute unit, direction seeds and slots of iteration 4 itself, bf16 escapes on almost the same directions as the 8-bit engine (connected components 26 of 40 against 29 of 40; shortest path 6 of 40 against 8 of 40). The shrinkage belongs to the ban-free unit of iteration 5 or its direction seeds, and the heading should read that movability depends on the unit and the direction draw.)* Movability is therefore a property of a task under a given engine, unit and direction population, and the confirmation panels measured it under bf16 and the ban-free unit only.

**Table 222.** Movability of the 15 existing tasks across measurements (escape fraction of fixed directions at alpha 1 / alpha 2, or off-default share of per-sample plan noise at alpha 1 / alpha 2). The two plan-noise columns are not on one definition: the 8-bit column counts valid attempts off the default over all samples, the bf16 column correct programs off the default over all samples. On one definition (correct programs only, 8-bit against bf16, alpha 1 / alpha 2): kth smallest 0.28 / 0.49 against 0.08 / 0.28; connected components 0.12 / 0.60 against 0.02 / 0.02; shortest path 0.21 / 0.41 against 0.00 / 0.04; two sum 0.00 / 0.02 against 0.00 / 0.04; grid 0.03 / 0.11 against 0.00 / 0.03; on all samples: kth smallest 0.28 / 0.50 against 0.10 / 0.28; connected components 0.12 / 0.60 against 0.02 / 0.02; shortest path 0.26 / 0.54 against 0.00 / 0.10; two sum 0.00 / 0.02 against 0.00 / 0.04; grid 0.06 / 0.24 against 0.00 / 0.04 (iteration-6 correction; record audit, Table 228). At iteration 4's absolute unit, its direction seeds and its slots, the bf16 engine reproduces the 8-bit plan-noise column (Section 6.3; iteration-6 correction). Sources: record audit `screen.json`; panel A `results/extra_arm_a.json` (iteration-4 fresh directions and the remeasurement arm); panel B `results/variants.json` (remeasured parent, pooled over both alphas); plan noise experiment `results/alpha_sweep.json` (iteration 4); knob experiment `results/cells.json` (iteration 5).

| Task | Frozen screen (8-bit, ban unit) | Iteration 4, 40 fresh directions (8-bit, bench unit) | Iteration 5, 8 directions (bf16, ban-free unit) | Iteration 5, panel B parent (bf16, ban-free, pooled) | Iteration 4 per-sample plan noise (8-bit, ban unit) | Iteration 5 per-sample plan noise (bf16, ban-free unit) |
|---|---|---|---|---|---|---|
| kth smallest | movable | 0.30 / 0.63 | 0.125 / 0.375 | 0.500 | 0.28 / 0.50 | 0.08 / 0.28 |
| connected components | movable | 0.38 / 0.73 | 0.000 / 0.250 | 0.188 | 0.12 / 0.60 | 0.02 / 0.02 |
| shortest path | movable | 0.18 / 0.20 | 0.000 / 0.000 | not run | 0.26 / 0.54 | 0.00 / 0.04 |
| two sum | movable | 0.00 / 0.13 | 0.000 / 0.000 | 0.000 | 0.00 / 0.02 | 0.00 / 0.04 |
| grid | movable | 0.05 / 0.13 | 0.000 / 0.000 | not run | 0.06 / 0.24 | 0.00 / 0.02 |
| find duplicate | movable | 0.00 / 0.00 | 0.000 / 0.000 | 0.000 | 0.00 / 0.02 | 0.00 / 0.00 |
| substring search | immovable | 0 of 8 / 0 of 8 | 0.000 / 0.000 | 0.000 | excess 0.00 at alpha 2 or less | 0.00 / 0.00 |
| Fibonacci | immovable | 0 of 8 / 0 of 8 | 0.000 / 0.000 | 0.000 | excess 0.01 | 0.00 / 0.00 |
| top-k frequent | immovable | 0 of 8 / 0 of 8 | 0.000 / 0.000 | 0.000 | excess 0.04 | 0.00 / 0.00 |
| nonoverlapping intervals | immovable | 0 of 8 / 0 of 8 | 0.000 / 0.000 | 0.000 | excess 0.00 | 0.00 / 0.00 |
| longest increasing subsequence | immovable | 0 of 8 / 0 of 8 | 0.000 / 0.000 | not run | excess 0.03 | 0.00 / 0.00 |
| merge intervals | immovable | 0 of 8 / 0 of 8 | 0.000 / 0.000 | not run | excess 0.00 | 0.00 / 0.00 |
| topological order | immovable | 0 of 8 / 0 of 8 | 0.000 / 0.000 | not run | excess 0.00 | 0.00 / 0.00 |
| first and last position | immovable | 0 of 8 / 0 of 8 | 0.000 / 0.000 | not run | excess -0.08 | 0.03 / 0.07 |
| cycle detection | not screened | 0 of 8 / 0 of 8 | 0.000 / 0.000 | not run | excess 0.01 | 0.00 / 0.07 |

**The transplant in two families.** In Qwen3-8B the transplant passed on 1 of 5 tasks diverse in the base and in Gemma it moved 2 of 7, each time onto the base runner-up. The successes share a feature: in Gemma the instruct twin already wrote the runner-up 8% to 12% of the time, and in Qwen3-8B the transplant raised the pass rate from 0.78 to 0.98 while it moved top-k frequent. Where the instruct model never wrote the base runner-up, the base states did not install it. In both families the base minus instruct difference at the task-specification positions was large (Qwen3-8B

0.39 to 0.53 of the residual norm, 71% of cells above 4 ban-free units at lambda 1; Gemma a median of 5.2 ban-free units), and in both an isotropic displacement or borrowed shift of comparable size did more damage than the coherent base content.

*[Figure fig_transplant: generated in the report phase, not in this draft]*

**Novelty, audited again.** No artifact of this iteration found a correct novel algorithm (Table 223), including the base model's own output and the transplanted output.

**Table 223.** Novelty audits of iteration 5. Sources: the novelty audit files of the five experiments.

| Artifact | Correct programs in the pool | Correct programs outside the catalogue | Final classes | Novel |
|---|---|---|---|---|
| Transplant experiment | all arms, base model included | 48 | 27 known techniques outside the frozen catalogue, 12 degenerate, 9 catalogue variants (judges called 14 novel) | 0 |
| Panel A | 6,854 | 2 | 1 catalogue variant, 1 degenerate | 0 |
| Panel B | 10,366 | 12 | 7 catalogue variants, 5 degenerate | 0 |
| Knob experiment | every correct program labelled other | 12 | 10 labeller misses, 1 degenerate, 1 unresolved | 0 |
| Gemma experiment | every correct program labelled other | 56 | 35 catalogue variants, 21 degenerate | 0 |

## 5.11 What iteration 5 established, what it ruled out, and what it abandoned

Everything below is one or two model families at 4B to 8B parameters, memorised tasks in the style of LeetCode, and labels from gated labellers, except the base-model labels of Gemma, which agree with an adjudication at only 0.64. Every Qwen3-8B number of this iteration comes from a bf16 engine.

**Established.**

1. *Temperature does not change the algorithm, under every sampler the run used.* With thinking on, top-k 10 and the same seed, the algorithm changed in 0 of 20 seeds across four temperatures on the shortest path task, and the first approach named in the thinking never changed.
2. *(Moved to the descriptive results, iteration 6.) Where a random nudge moves the instruct model, it often lands on the base model's runner-up.* Per set the base runner-up named the push landing 5 of 5 on the 8-bit screen, 1 of 2 on the bf16 remeasurement of existing tasks, 3 of 3 in panel A, 3 of 4 in panel B and 5 of 7 in Gemma (17 of 21, Poisson-binomial p 5.8e-07); on new tasks 6 of 7 events (Poisson-binomial p 0.003). 6 of the 11 held-out hits are also the instruct twin's own minority: on the 8 held-out events where the instruct twin wrote no correct program off the default the base runner-up named 5 (p 0.053; Qwen3-8B 3 of 4, p 0.052; Gemma 2 of 4, p 0.422), and the twin's own most frequent alternative, defined on 7 of 21 events, named 6 of them against 6 for the base runner-up (exact McNemar p 1.000). Where the twin writes an alternative at all, the landing half is therefore the twin's own minority, and the base model adds a forecast only on the events where the twin writes none (iteration-6 correction; record audit, Table 230), with the base composition ahead of

association by 0.75 nats (exact Wilcoxon p 0.055 with the corrected association) and of proximity by 0.50. In Gemma it named 5 of 7 (p 0.025; 1.50 nats over association with undefined association forecasts filled uniformly, 1.57 on the 6 defined events, p 0.023), and on the existing Qwen3-8B tasks 5 of 5. In panel A no nudged program landed outside the base repertoire.

3. *The base model is more diverse than the instruct model on most tasks under identical prompts.* The instruct twin wrote its default in 96% to 100% of programs on all 15 existing tasks, while 5 tasks were diverse in the base and 7 intermediate, and every new unit but one had beta of at least 0.05. The base model's own correct output is thin under the chat template (14 to 55 correct programs per 100 to 200 on the existing tasks). Three instruct twins of new units are mixed (trapped rain water default share 0.64, minimum spanning tree weight default share 0.66, valid anagram default share 0.76). On two new units the instruct twin is more diverse than the base: trapped rain water (twin off-default 0.36 against beta 0.23) and minimum spanning tree weight (0.34 against 0.11), and on both the landing is the twin's own minority (iteration-6 correction).
4. *Plan noise's effect is not its token-level departure from the twin, on kth smallest.* At alpha 2 it moved 19 slots against 3 for decoding matched to its per-token KL (exact p 8.6e-04), but the discordant counts are kth smallest 13:0, connected components 1:3, shortest path 2:0, two sum 2:0, grid 1:0: without kth smallest the other four tasks give 6:3 (p 0.508); at alpha 1 the pooled count is 4:3 (p 1.000); bf16-conditional (iteration-6 correction; record audit, Table 233).
5. *The record repairs.* The held-out movability test is inconclusive under catalogue-only avoidance and disconfirmed under the drop sensitivity; the read position and condition sentence are inert only at per-position doses up to 4 ban units; landing comparisons on matched events give push -0.76 against association -3.11 on 3 events and -1.37 against -3.46 on 5.

**Ruled out, or not supported.**

- *The movability half: the base repertoire predicts which tasks a nudge moves.* Not supported as a strong predictor, and not disconfirmed (ban-free unit and iteration-5 seeds; re-tagged by iteration 7's dose-unit experiment, Section 7.6: at D* = D1 kth smallest and connected components are movable, and shortest path, two sum and the grid are not) (iteration-8 correction): beta is not a strong predictor of movability (Spearman 0.15 over 13 new units, Bonett-Wright 95% [-0.44, 0.65], bootstrap BCa [-0.50, 0.70], one-sided permutation p 0.317; power of the registered one-sided permutation test about 0.63 at a true 0.6, 0.50 two-sided, and 20 units for 0.8); a moderate relation is not excluded at this n; conditional on the design of iteration 5 (bf16 with the ban-free unit and its direction seeds; the engine control of Section 6.3 shows that the numerics alone do not shrink the escapes). The pooled value rests on the beta of count inversions, 0.75, which the panel committed on the attempt basis because only a few base programs were correct (67% of its attempts fall in the class other); on the correct-only basis used for the other twelve units its beta is 0.13 and the pooled Spearman is 0.035. Per panel Spearman -0.03 (A) and 0.21 (B); 4 of 10 new units diverse in the base never moved. Under one definition of each rival, pooled over the 13 units with signs fixed so that the predicted direction is positive: opening margin 0.08 (Holm p 1.000, delta +0.07 [-0.66, 0.88]); novelty-instruction excess -0.50 (Holm p 1.000, delta +0.65 [-0.06, 1.34]); runtime merit 0.71 (Holm p 0.021, delta -0.57 [-1.26, 0.20]); twin modal share 0.41 (Holm p 0.347, delta -0.26 [-1.05, 0.56]); catalogue size -0.21 (Holm p 1.000, delta +0.35

[-0.57, 1.24]). Under Holm only runtime merit tracks movability, while beta does not; the paired interval for beta minus that rival still includes 0 (iteration-6 correction; record audit, Table 232). The screen correlation of 0.78 did not transfer to new tasks. (iteration-7 correction)

- *The transplant claim: a transplant of base plan states reopens the base repertoire.* Disconfirmed by the frozen rule (1 of 5 tasks diverse in the base); on kth smallest a matched-norm random displacement moved more programs than the base states (2:20). The narrowing installed by tuning is not carried by the task-specification states of blocks 12 to 23 alone.
- *The push toward the base.* A push along the base minus instruct direction never escaped at 0.5 to 2 ban-free units, while an isotropic push did on kth smallest.
- *The plan sampler comparison: plan noise prices at least as well as text-level plan samplers.* Disconfirmed (1 of 5 decisive tasks) (ban-free unit and iteration-5 seeds; re-tagged by iteration 7's dose-unit experiment, Section 7.6: at the moving design D* = D1 the plan-sampler rule is still disconfirmed, at 2 of 5) (iteration-8 correction); base intent won on connected components, the shortest path task and two sum. (iteration-7 correction)

**Inconclusive.**

- *The movability half in Gemma.* Spearman 0.42 (p 0.060), with the twin modal share and the novelty instruction ahead of beta; descriptively 6 of 7 tasks diverse in the base moved against 2 of 8 others.
- *Top-k frequent.* The one Qwen3-8B task where the transplant reopened the base runner-up also gained 20 pass points, so restored correctness and a reopened choice are not separated.

**Dead ends, with the evidence that closed them.**

- *The base minus instruct direction as a knob.* Closed by Table 186: it never escapes, and the difference it encodes is mostly shared across tasks (cosine 0.76 to 0.88) and larger than the dose at which random noise breaks programs.
- *Transplanting at the read position.* The base and instruct states differ there by about the size of the residual, and the transplant broke the output on 13 of 15 tasks without moving an algorithm.
- *Plan noise as the practical knob for algorithmic diversity.* On the decisive tasks two text-level samplers price better, and under the bf16 engine and ban-free unit plan noise moves more than 10% of programs only on kth smallest (conditional on the design of iteration 5, bf16 with the ban-free unit and its direction seeds: at the absolute unit and seeds of iteration 4 the bf16 engine reproduces the 8-bit effects on connected components and the shortest path task, Section 6.3; iteration-6 correction).
- *Signed progress toward the runner-up as the next predictor.* Closed in the record audit (Table 177): concentrated on one task, negative on two, matched by a within-class control.

## 5.12 What the next iteration takes from this one

The narrowed repertoire account split in two. Its landing half held descriptively in both families and on unseen tasks (5 of 5 on the 8-bit screen, 1 of 2 on the bf16 remeasurement of existing tasks, 3 of 3 in panel A, 3 of 4 in panel B and 5 of 7 in Gemma (17 of 21, Poisson-binomial p 5.8e-07)), but it is not yet separated from the instruct twin's own minority (on the 8 held-out events where the instruct twin wrote no correct program off the default the base runner-up named 5 (p 0.053;

Qwen3-8B 3 of 4, p 0.052; Gemma 2 of 4, p 0.422); iteration-6 correction): when a nudge moves the instruct model, it moves it to the algorithm the base model writes next, and a forecast built from base samples, before any push, names that landing better than the model's stated alternative. Its movability half failed on unseen tasks: a base model that writes many algorithms does not mark a task the instruct model can be nudged off, and its states, written into the instruct prefill, reopen the choice on one task in five. The base repertoire marks where a nudge can lead, not whether it can lead anywhere.

The open question is therefore what makes a task diverse in the base immovable. Count primes and valid anagram (beta 0.81 each) never moved, while trapped rain water (0.23) did. Two leads came out of this iteration's data. In panel A the three units that moved had an alternative at least twice as fast as the default and the three that did not had none, although median of two sorted lists broke the pattern. And in the transplant experiment early blocks 0 to 11 moved kth smallest and connected components more than the main band, which suggests the narrowing sits at least partly upstream of blocks 12 to 23. The positioning audit names the baselines a further test needs: weight interpolation between the base and instruct models, layer restoration at the transplant's block ranges, and token-level base and aligned collaboration [92, 93, 94, 102]. Base intent, which beat plan noise on three of five tasks, is the strongest practical knob the run has priced, and it is itself a form of that collaboration at the level of sentences. Finally, movability has to be measured under the engine and unit at which it will be used: the six screen-movable tasks of iteration 4 shrank to two under bf16 and the ban-free unit, for reasons this iteration did not establish. *(Iteration-6 correction: a dose-bracketed contrast rules out a scalar dose shortfall on connected components and shortest path, and the engine control rules out the bf16 numerics, since at iteration 4's absolute unit and seeds bf16 reproduces the 8-bit escapes (Section 6.3). What remains is the per-position and per-block profile of the ban-free unit, or the direction seeds; every verdict that rests on bf16 plan noise or escape fractions at the ban-free unit is conditional on that design.)*

# Iteration 6: an engine control, where a ban and an instruction are carried, a one-token fork in Gemma, and the plan in words

## 6.1 Why this iteration ran what it ran

**The review of the iteration-5 record.** The review was blocking again. It raised five major items, all marked must fix, and four minor ones. The first went to the run's main internal result. Every plan-noise positive of iteration 4 had come from an 8-bit engine whose W8A8 linear layers compute in 8-bit floating point (e4m3 weights per output channel, e4m3 activations with a per-token scale), while iteration 5's bf16 knob experiment found per-sample plan noise nearly inert: connected components moved in 1 of 50 programs at alpha 2 against 30 of 50 under the 8-bit engine. The record had left the cause open, had not reported the knob experiment's own measurement of the dose it delivered, and still listed iteration-4 items 2 to 4 as established. The review asked for a decisive control: the same directions and slots at the same absolute displacement under the 8-bit engine, under 8-bit floating-point weights with bf16 activations and under bf16. The second item found the landing half of the narrowed repertoire account overstated. Its tally of 16 of 19 omitted the bf16 remeasurement made in that same iteration, and it never scored the minority algorithm of the instruct twin itself as a rival, although that rival explained half the hits. The third item asked for the power and interval behind the claim that the movability half was ruled out, and for the

frozen rivals on the pooled set. The fourth listed executed outputs missing from the record: the phrasing columns of the literal cell, the prices at Hill-1 1.5 and 2, the dose file of the knob experiment, the knob rows of panel B and three Llama rows of Table 126c. The fifth said that after five iterations the record still had no internal account of how the choice manifests, which is the request's central question, and that its one internal positive rested on the 8-bit engine. The minor items asked for the per-task counts behind item 4 of Section 5.11, the fill rule for undefined Gemma forecasts, a sentence on the three new tasks whose instruct twins are already mixed, and a second blind adjudication of the 14 transplant programs that two of three judges had called novel. Table 224 maps every item to its answer.

**Table 224.** Review items on the iteration-5 record and where this report answers them.

| Review item | What the review found | Where it is answered |
|---|---|---|
| Engine | The 8-bit plan-noise effect does not replicate in bf16 at near-matched dose; the dose file unreported; iteration-4 items 2 to 4 still established; plan-sampler and movability verdicts not tagged with their engine | Corrections in Sections 4.2.6, 4.10, 5.2, 5.10, 5.11 and 5.12; Section 6.2.2; the engine control, Section 6.3 |
| Landing half | Selective tally; the twin's own minority never scored; thin base support in Gemma; two units where the instruct twin is more diverse than the base | Corrections in Sections 5.5, 5.8, 5.10, 5.11 and 5.12; Section 6.2.3; the landing contest of Section 6.4.5 |
| Movability half | Ruled out without its interval or power; the frozen rivals not computed on the pooled set | Correction in Section 5.11; Section 6.2.4 |
| Missing outputs | Literal-cell phrasing columns; prices at Hill-1 1.5 and 2; dose file; panel B knob rows; Table 126c rows | In place in Tables 206, 207b, 203b and 126c and in Section 5.2; Section 6.2.5 |
| Coverage | No representation of the choice; the internal positive rests on the 8-bit engine | Sections 6.3 to 6.6; Section 6.10 |
| Minor: matched decoding per task | Item 4 of Section 5.11 generalised from kth smallest | Correction in Section 5.11 |
| Minor: Gemma fill rule | The 1.50 nats included an undefined association forecast filled uniformly | Corrections in Sections 5.8 and 5.11 |
| Minor: mixed instruct twins | Three new units are already mixed; on two the instruct twin is more diverse than the base | Notes under Sections 5.4 and 5.5; correction in Section 5.11; the closing section |
| Minor: novelty adjudication | The zero rested on the executor's override of two judges | Section 6.2.6; corrections in Section 4.9 and the closing section |

**The hypothesis update.** The update turned from the outside of the model to its inside. Five iterations had mapped what moves the algorithm from outside: a ban, a meaningful hint, a plan sentence written by the base model, a list-then-pick step. None had located where the model holds its choice. The update proposed one concrete answer, which it called the plan-variable account. While the model reads the task, it computes a small hidden variable that says which algorithm to write. We call this variable the *plan*. On tasks where a ban acts through the prompt states (shortest path, kth smallest, connected components) the plan sits in the residual stream at the task-specification positions; on tasks where generated tokens read the ban directly (two sum, find duplicate) it sits at the first generated tokens. Exchanging a few of its dimensions between two prompts should switch the algorithm the model writes, with correct code. Random noise should change the algorithm only through the part of it that falls inside the plan, which would explain

why most random directions do nothing; temperature should never touch it.

The update tested the account with the instruments the request names and the run had not yet used. An *interchange intervention* replaces a component's activations under one input with those under another and measures the change in output. *Distributed alignment search* (DAS) fits an orthonormal low-rank subspace in which such an exchange reproduces a counterfactual behaviour [110, 111]. A *natural language autoencoder* (NLA) pairs an activation verbalizer, which writes a text explanation of a residual-stream state, with an activation reconstructor, which maps text back to a state; released checkpoints exist for Qwen2.5-7B-Instruct [124] at layer 20, not for Qwen3-8B. Gemma Scope 2, released in December 2025, provides JumpReLU sparse autoencoders [114] and transcoders trained on gemma-3-4b-it itself, the successor of the Gemma Scope release for Gemma 2 [113], so no transfer test from a base model is needed.

Eleven claims were preregistered, each with a rule committed by sha256 before its first decisive sample. We call them by their content rather than by their codes: the engine claim, the localization claim, the mediation claim, the discreteness claim, the token claim, the resampling claim, the verbal readout claim, the feature claim, the site rerun, the landing contest and the novelty count.

- *The engine claim* (primary): on connected components and shortest path, the same 40 directions and common-random-number slots at the same absolute displacement under four numerics. Activation rounding is the cause if the escape fraction under bf16 weights with 8-bit floating-point activations (W16A8) exceeds bf16 by at least 0.2, the full 8-bit engine (W8A8) matches W16A8 within 0.1 and 8-bit floating-point weights with bf16 activations (W8A16) match bf16 within 0.1, on both tasks. The result is called not reproduced if W8A8 minus bf16 is below 0.1.
- *The localization claim* (primary). Stage A, with no training: exchanging the full residual states of the ban prompt into the neutral twin transfers the ban's substitute at a transfer share of at least 0.5 on at least two of shortest path, kth smallest and connected components, at a pass rate of at least 0.8 times the twin's, and at most 0.2 on two sum and find duplicate. Stage B: DAS at one layer recovers at least 0.8 of the full-state transfer at a rank of 16 or less on held-out templates.
- *The mediation claim* (primary): the energy that a random direction puts into the fitted plan subspace predicts which directions switch the algorithm (Spearman at least 0.5).
- *The discreteness claim*, *the token claim* (temperature leaves the plan coordinate flat while state noise at generated tokens beats temperature at matched per-token KL) and *the resampling claim* (resampling the plan as a diversity knob) (secondary).
- *The verbal readout claim* (secondary): the NLA names the substitute of a ban before any code is written at a rate at least 0.3 above a judge reading the visible text; editing the explanation's algorithm phrase and reconstructing transfers the algorithm by at least 0.2.
- *The feature claim* (secondary): in Gemma's natural BFS against DFS split on the grid task, Gemma Scope 2 features decode the upcoming algorithm at least 0.10 balanced accuracy above a classifier on the visible text, and steer at least as well as a difference-in-means direction at matched pass.
- *The site rerun* (secondary): the iteration-4 noise-site comparison in bf16; the prefill-site claim survives if the task-specification site beats the read position and the condition sentence at matched downstream displacement on at least one task, and is withdrawn otherwise.
- *The landing contest* (secondary): the base runner-up against the instruct twin's own teacher-forced ranking of algorithm openings, on every recorded push-landing event.

- *The novelty count* (secondary).

**The strategy.** Seven artifacts were planned (Table 225). One evaluation on CPU, which we call the record audit, repaired the record from rows on disk and ran the second novelty adjudication. An experiment we call the engine control ran the four numerics on a 16 GB RTX 2000 Ada, an Ada-generation card with the 8-bit floating-point tensor cores that the 8-bit engine needs. A bf16 experiment, which we call the exchange and site experiment, ran the Stage A exchanges, the site rerun, state noise against matched temperature, the top-five push and the landing contest. Gemma and the NLA each received their own experiment, which we call the Gemma feature experiment and the NLA experiment, and a research artifact, which we call the positioning audit, searched the literature for each claim. The seventh artifact, which was to fit the DAS subspaces of Stage B and run the mediation, discreteness, resampling and temperature-coordinate tests together with a manifold of plan coordinates, was lost before it existed: the planning agent returned no plan. Section 6.8 records it as planned and not run. Every Qwen3-8B number of this iteration except those of the engine control comes from a bf16 engine.

**Table 225.** The artifacts of iteration 6, the question each was asked, and its hardware and spend.

| Artifact | Question | Model and hardware | OpenRouter spend |
|---|---|---|---|
| Record audit | Repair the iteration-5 record (36 corrections); re-adjudicate the 14 disputed transplant programs blind | No generation; CPU only | USD 0.29 |
| Engine control | Are iteration 4's random-push escapes a property of the model or of the 8-bit floating-point W8A8 engine? | Qwen3-8B under four numerics built from one bf16 snapshot; one 16 GB RTX 2000 Ada; 6,075 generated programs plus reused iteration-4 rows; 4.1 GPU hours of generation | USD 0.085 |
| Exchange and site experiment | Where is a ban carried; does the noise-site claim hold in bf16; state noise against matched temperature; the top-five push; the landing contest | Qwen3-8B bf16 with the input embedding table on CPU, one 16 GB RTX 2000 Ada; 4,320 programs and 1,220 teacher-forced passes; about 2.7 GPU hours | USD 0.055 |
| Gemma feature experiment | Where does Gemma's BFS-or-DFS choice fork; do Gemma Scope 2 features read and steer it better than difference in means, random features and DAS? | gemma-3-4b-it bf16, one 16 GB RTX 2000 Ada; about 2.1 GPU hours of generation | USD 0.034 |
| NLA experiment | Does the residual stream say in words which algorithm will be written, and can a verbal edit steer it? | Qwen2.5-7B-Instruct bf16 with the released layer-20 NLA pair, one 24 GB NVIDIA Ada GPU; about 2.0 GPU hours | about USD 2.0 |
| Positioning audit | Set the iteration-6 claims against published work | Web search and quote checks | none recorded |
| DAS and plan-coordinate experiment | Stage B of the localization claim; the mediation, discreteness, resampling and temperature-coordinate tests; the causal manifold | Planned for Qwen3-8B bf16; lost at planning | not run |

## 6.2 The record audit: corrections, engine tags and a second novelty adjudication

The record audit generated no model text. It recomputed from the row files of the earlier artifacts every number the review touched, tagged each internal result with the numerics that produced it, and wrote each change as a verbatim replacement keyed to the report. Of the values behind its corrections, 78 matched the record, 18 were new, 2 were carried from earlier evaluations and 1 was corrected: the power of the movability test, recorded as 0.46, is 0.63 for the registered one-sided test and 0.50 two-sided. A standard-library re-derivation passed 277 of 277 checks, its five placebos failed as they should, and a second script reproduced 7 of 7 headline numbers from the rawest files. Judge spend was USD 0.29 over four passes.

### 6.2.1 The corrections

Twenty-nine of the 36 corrections are applied in place in Sections 4 and 5 and marked as iteration-6 corrections. The seven that target the closing section are folded into the closing section, which is rewritten at the end of this iteration. The audit ran in parallel with the engine control, so its engine tags were written before the control's result existed; where the control supersedes them, the in-place text adds a pointer to Section 6.3. Three further notes of the same kind mark the frozen screen of Section 4.2.6, the movability paragraph of Section 5.10 and the caption of Table 222. Table 226 lists every correction.

**Table 226.** The 36 corrections of the iteration-5 record (values behind each: M match, N new, C carried, X corrected). Source: record audit `corrections.json`.

| No. | Location | What changed | Values |
|---|---|---|---|
| 1 | Section 5.2, dose units | The two dose definitions (ratio of mean norms against mean of per-position ratios) reconciled per task | 14 M, 6 N |
| 2 | Table 222 caption | The two plan-noise columns put on one definition | 20 M |
| 3 | Section 5.10, movability paragraph | Shortest path counts restated per definition; dose-bracketed contrast and its tags | 2 M, 5 N |
| 4 | Section 4.10, preamble | Engine class of iteration 4 named | 1 N |
| 5 | Section 4.10, item 2 | Same-seed contrast tagged as 8-bit only | 1 C |
| 6 | Section 4.10, item 3 | Site odds ratio tagged as 8-bit only | 1 C |
| 7 | Section 4.10, item 4 | Plan-noise prices tagged with their engine and unit | 4 M |
| 8 | Closing | Internal claims that survive the engine change (folded into the new closing) | none |
| 9 | Closing | Dose bracket removes a scalar dose shortfall (folded) | none |
| 10 | Closing | Site claim tagged as 8-bit (folded) | none |
| 11 | Section 5.10, figure | The base-variety figure redrawn like for like under bf16 | 2 M |
| 12 | Section 5.11, item 2 | Landing half moved to the descriptive results; tally per set and the twin-minority rival | 7 M, 1 N |
| 13 | Section 5.11, item 2 | Gemma fill rule; defined-only mean 1.57 | 2 M, 1 N |
| 14 | Section 5.12 | Landing half restated as descriptive, with its tally | none |
| 15 | Closing | Landing tally per set (folded) | none |
| 16 | Section 5.10 | Landing tally of 16 of 19 replaced by 17 of 21 per set | 4 M |
| 17 | Table 129 | Row label made explicit: counted on all 13 events | none |
| 18 | Table 129 | Same, for the largest-excess row | none |
| 19 | Section 5.11, item 3 | Three mixed instruct twins; two units where the twin is more diverse than the base | 4 M |

| | | | |
|---|---|---|---|
| 20 | Table 203 | Base support counts per panel B event | none |
| 21 | Table 214 | Base support counts per Gemma unit | none |
| 22 | Table 217 | Base support counts per Gemma landing event | none |
| 23 | Section 5.11, movability half | Interval, power and pooled rivals; verdict restated | 1 M, 1 X, 1 N |
| 24 | Closing | Movability interval and power (folded) | none |
| 25 | Section 5.12 | The shrinkage of the movable set: what is ruled out | none |
| 26 | Section 5.11, dead end | Plan noise as a knob tagged with its design | none |
| 27 | Table 206 | Phrasing clusters per temperature in the literal cell | 4 M, 1 N |
| 28 | After Table 208 | Table 207b: prices at Hill-1 1.5 and 2 with bootstrap intervals | 5 M |
| 29 | Section 5.5 | Table 203b: panel B knob rows | 2 M, 1 N |
| 30 | Table 126c | The three Llama present-minor ban rows completed | 3 M |
| 31 | Section 5.11, item 4 | Plan noise against matched decoding restated per task | 2 M |
| 32 | Section 5.4 heading | Mixed instruct twins noted | none |
| 33 | Section 5.5 heading | Same note | none |
| 34 | Closing | Mixed instruct twins (folded) | none |
| 35 | Closing | Second novelty adjudication (folded) | 2 M, 1 N |
| 36 | Section 4.9 | Second novelty adjudication noted | none |

### 6.2.2 Engine and dose

The two artifacts that the review compared recorded off-default shares on different definitions: the bf16 knob experiment counted correct programs off the default over all samples, the 8-bit plan-noise experiment valid attempts off the default over all samples. The dose definitions differ as well. Table 174 gave the ban-free unit as a ratio of mean norms, while the knob experiment measured a mean of per-position ratios, which is larger because a mean of ratios is not a ratio of means (Table 227). On the six decisive tasks, by either definition, the bf16 alpha-2 cell of iteration 5 delivered at least 1.24 ban units, so a dose-bracketed contrast sets it against the 8-bit alpha-1 cell, which delivered 1.00. A lower bf16 share in that comparison cannot come from a shortfall of the scalar dose.

**Table 227.** The ban-free unit as a fraction of the ban unit under the two definitions, and iteration 5's bf16 doses expressed in 8-bit ban units. Source: record audit `tables/dose_reconciliation.csv`.

| Task | Ratio of mean norms (Table 174) | Mean of per-position ratios (knob experiment) | bf16 alpha 1 in ban units (two definitions) | bf16 alpha 2 in ban units (two definitions) |
|---|---|---|---|---|
| grid | 0.751 | 0.874 | 0.75 / 0.87 | 1.50 / 1.75 |
| shortest path | 0.834 | 0.889 | 0.83 / 0.89 | 1.67 / 1.78 |
| kth smallest | 0.974 | 1.394 | 0.97 / 1.39 | 1.95 / 2.79 |
| two sum | 0.777 | 0.995 | 0.78 / 0.99 | 1.55 / 1.99 |
| find duplicate | 0.620 | 0.829 | 0.62 / 0.83 | 1.24 / 1.66 |
| connected components | 0.745 | 0.990 | 0.75 / 0.99 | 1.49 / 1.98 |
| top-k frequent | 0.699 | 0.924 | 0.70 / 0.92 | 1.40 / 1.85 |
| first and last position | 0.618 | 0.887 | 0.62 / 0.89 | 1.24 / 1.77 |
| substring search | 0.296 | 0.976 | 0.30 / 0.98 | 0.59 / 1.95 |

| | | | | |
|---|---|---|---|---|
| Fibonacci | 0.885 | 1.234 | 0.88 / 1.23 | 1.77 / 2.47 |
| longest increasing subsequence | 0.803 | 1.061 | 0.80 / 1.06 | 1.61 / 2.12 |
| merge intervals | 0.951 | 1.365 | 0.95 / 1.37 | 1.90 / 2.73 |
| nonoverlapping intervals | 0.761 | 1.089 | 0.76 / 1.09 | 1.52 / 2.18 |
| topological order | 0.864 | 1.136 | 0.86 / 1.14 | 1.73 / 2.27 |

Table 228 gives the contrast under both share definitions. The rule, fixed before computing, tags a task as not replicated when the bf16 share falls short at the matched alpha of the 8-bit engine (Fisher p) and in the dose bracket. Connected components and shortest path meet it, kth smallest and the grid task do not, and two sum and find duplicate had no 8-bit effect of at least 10% at alpha 2 or less to replicate. These tags compare the design of iteration 5 (bf16 with the ban-free unit and fresh direction seeds) with the design of iteration 4 (the 8-bit engine with the ban unit); they cannot say which of the three differences (the numerics, the dose unit, the direction seeds) is responsible. Section 6.3 separates them.

**Table 228.** Per-sample plan noise, the bf16 cells of iteration 5 against the 8-bit cells of iteration 4, on two definitions (counts off the default over N). Source: record audit `tables/engine_pn_contrast.csv`.

| Task | Definition | bf16 alpha 1 | bf16 alpha 2 | 8-bit alpha 0.5 | 8-bit alpha 1 | 8-bit alpha 2 | Fisher p, alpha 2 | Bracket p (bf16 alpha 2 against 8-bit alpha 1) | Tag |
|---|---|---|---|---|---|---|---|---|---|
| kth smallest | correct only | 4/49 | 14/50 | 4/50 | 14/50 | 23/47 | 0.039 | 0.588 | not established |
| kth smallest | all samples | 5/50 | 14/50 | 4/50 | 14/50 | 25/50 | 0.040 | 0.588 | not established |
| connected components | correct only | 1/50 | 1/50 | 0/50 | 6/50 | 30/50 | 7.2e-11 | 0.056 | did not replicate in bf16 at the population level |
| connected components | all samples | 1/50 | 1/50 | 0/50 | 6/50 | 30/50 | 7.2e-11 | 0.056 | did not replicate in bf16 at the population level |
| shortest path | correct only | 0/50 | 2/47 | 3/49 | 9/43 | 15/37 | 5.3e-05 | 0.017 | did not replicate in bf16 at the population level |
| shortest path | all samples | 0/50 | 5/50 | 4/50 | 13/50 | 27/50 | 4e-06 | 0.033 | did not replicate in bf16 at the population level |
| two sum | correct only | 0/50 | 2/50 | 0/50 | 0/50 | 1/50 | 1.000 | 1.000 | not established (no 8-bit effect to replicate) |
| two sum | all samples | 0/50 | 2/50 | 0/50 | 0/50 | 1/50 | 1.000 | 1.000 | not established (no 8-bit effect to replicate) |
| grid | correct only | 0/40 | 1/38 | 0/43 | 1/35 | 4/36 | 0.194 | 0.732 | not established |
| grid | all samples | 0/50 | 2/50 | 1/50 | 3/50 | 12/50 | 0.008 | 0.500 | not established |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| find duplicate | correct only | 0/30 | 0/30 | 0/50 | 0/50 | 0/49 | 1.000 | 1.000 | not established (no 8-bit effect to replicate) |
| find duplicate | all samples | 0/30 | 0/30 | 0/50 | 0/50 | 1/50 | 1.000 | 1.000 | not established (no 8-bit effect to replicate) |

The audit tagged 24 internal results of iterations 2 to 5 with the numerics that produced them (Table 229). Its tags were fixed before the engine control finished; the last column gives each tag's status after Section 6.3.

**Table 229.** Internal results by engine class, with the audit's tag and the status after the engine control. Source: record audit `tables/engine_labels.csv`; engine control `results/relabel_table.json`.

| No. | Result | Engine class | bf16 counterpart (audit) | Audit tag | Status after Section 6.3 |
|---|---|---|---|---|---|
| 1 | Plan-noise 5-point Hill-1 2.72 on kth smallest | 8-bit W8A8 | 1.98 at the ban-free unit | not established | not rerun (kth smallest dropped from the engine control) |
| 2 | Plan-noise 5-point Hill-1 2.09 on connected components | 8-bit W8A8 | 1.09; 1 of 50 correct off the default | did not replicate in bf16 at the population level | model property: at the absolute unit bf16 plan noise moves 25 of 50 valid attempts against 30 of 50 |
| 3 | Plan noise moves shortest path 26% at alpha 1 | 8-bit W8A8 | 0 of 50 at alpha 1, 2 of 47 at alpha 2 | did not replicate in bf16 at the population level | model property: 10 of 50 against 12 of 50 at alpha 1, 22 of 50 against 22 of 50 at alpha 2 |
| 4 | Plan noise on the grid task, 0.24 at alpha 2 | 8-bit W8A8 | 1 of 38 correct | not established | not rerun |
| 5 | Site odds ratio 5.72 | 8-bit W8A8 | none | 8-bit only, untested in bf16 | bf16 site rerun (Section 6.4.2): committed rule withdrawn; post-hoc model puts the read position below the task-specification site |
| 6 | Same-seed contrast, 17.25 of 20 seeds | 8-bit W8A8 | none | 8-bit only, untested in bf16 | not rerun; its per-sample plan noise reproduces in bf16 at the absolute unit |
| 7 | 40-direction escape fractions | 8-bit W8A8 | 8 bf16 directions at the ban-free unit: kth smallest 0.25, connected components 0.125, others 0 | did not replicate on shortest path, two sum, grid and find duplicate; replicated, smaller, on kth smallest and connected components | model property on connected components (0.65 against 0.725) and shortest path (0.15 against 0.20) at the same 40 directions and absolute unit |

| | | | | | |
|---|---|---|---|---|---|
| 8 | Frozen screen: six movable tasks | 8-bit W8A8 | bf16 remeasure: two movable | did not replicate (4 of 6 tasks no longer movable) | the escapes behind it reproduce in bf16 at the absolute unit on the two tasks tested; the shrinkage belongs to the ban-free unit or its seeds |
| 9 | Plan-subspace tests | 8-bit W8A8 | none | 8-bit only, untested in bf16 | untested (the DAS experiment was not run) |
| 10 | Connected components screen escape from a bf16 direction | bf16 | itself | bf16 result | unchanged |
| 11 | Pre-selected directions 9002 and 9004 move in both engines | 8-bit and bf16 | shortest path Dijkstra 31, Bellman-Ford/SPFA 19 under bf16 against 26 and 24 | replicated in bf16 | unchanged; direction 9004 also moves shortest path under every numerics (Section 6.3) |
| 12 | Plan noise against matched decoding, 19 against 3 | bf16 | itself | bf16 result | unchanged (13 of the 19 on kth smallest) |
| 13 | Transplant results | bf16 | itself | bf16 result | unchanged |
| 14 | Read-position transplant breaks output without moving the algorithm | bf16 | itself | bf16 result | unchanged |
| 15 | Route of a ban | bf16 | itself | bf16 result | extended by the exchange of Section 6.4 |
| 16 | Band patching on kth smallest and connected components | bf16 | itself | bf16 result | extended by the exchange of Section 6.4 |
| 17 | Literal cell, temperature null with thinking on | vLLM 8-bit floating-point W8A16 (weight only) | none needed | behaviour level, engine-robust | unchanged |
| 18 | Temperature changes phrasing, not the algorithm | vLLM 8-bit floating-point W8A16 | bf16 temperature 1.3 cells show no diversity | behaviour level, engine-robust | unchanged |
| 19 | Llama knob rows | vLLM bf16 | itself | bf16 result | unchanged |
| 20 | Novelty audits | sandboxed CPU scoring | not applicable | behaviour level, engine-robust | unchanged |
| 21 | Plan-sampler comparison disconfirmed | bf16 | itself | bf16-conditional | conditional on iteration 5's ban-free unit and direction seeds, not on the numerics |
| 22 | Movability half | bf16 | itself | bf16-conditional | same |

| | | | | | |
|---|---|---|---|---|---|
| 23 | Plan noise as the practical knob is a dead end | bf16 | itself | bf16-conditional | same |
| 24 | Six screen-movable tasks shrank to two | bf16 | itself | bf16-conditional | same |

The figure of base-model variety against movability is redrawn like for like (correction in Section 5.10): the existing tasks at their bf16 escape fraction under the ban-free unit, with the 8-bit screen values as ghost markers, against the new units under the same design. Over the 15 existing tasks Spearman is 0.59 (one-sided permutation p 0.010, Bonett-Wright 95% interval [126] from 0.06 to 0.86), against 0.79 with the 8-bit screen values; over the 13 new units it is 0.15 (p 0.315, interval from -0.44 to 0.65), 0.01 with the two reserve units; pooled over the 28 bf16 measurements it is 0.48 (p 0.005).

### 6.2.3 Landings

Table 230 restates every recorded push-landing event with its base support and the instruct twin's own composition. Per set, the base runner-up named the landing 5 of 5 times on the 8-bit screen, 1 of 2 on the bf16 remeasurement of existing tasks, 3 of 3 in panel A, 3 of 4 in panel B and 5 of 7 in Gemma, 17 of 21 in all (Poisson-binomial p 5.8e-7). On 6 of the 11 held-out hits the landing is also an algorithm the instruct twin already writes. Restricted to the 8 held-out events where the twin wrote no correct program off the default, the base runner-up named 5 (p 0.053; Qwen3-8B 3 of 4, p 0.052; Gemma 2 of 4, p 0.42). On the 7 events where the twin wrote any alternative, the twin's own most frequent alternative named the landing 6 times, as often as the base runner-up (exact McNemar p 1.0). The landing half therefore moves to the descriptive results: where the twin writes an alternative, the landing is that alternative, and the base model adds a forecast only where the twin writes none.

**Table 230.** Every recorded push-landing event: default, landing, base runner-up, the twin's correct composition, and the base support (valid attempts, correct programs, runner-up count among correct and among attempts). Source: record audit `tables/landing_events.csv`.

| Set | Task | Default | Landing | Base runner-up | Hit | Twin correct composition | Base valid / correct | Runner-up among correct / attempts | Held out, twin wrote no alternative |
|---|---|---|---|---|---|---|---|---|---|
| 8-bit screen | kth smallest | full sort | quickselect | quickselect | yes | full sort 50 | 65 / 47 | 13 / 19 | not held out |
| 8-bit screen | shortest path | Dijkstra | Bellman-Ford/SPFA | Bellman-Ford/SPFA | yes | Dijkstra 50 | 129 / 44 | 18 / 40 | not held out |
| 8-bit screen | two sum | hash map | brute force | brute force | yes | hash map 50 | 59 / 49 | 7 / 8 | not held out |
| 8-bit screen | connected components | union find | DFS | DFS | yes | union find 50 | 92 / 55 | 32 / 42 | not held out |
| 8-bit screen | grid | BFS | DFS | DFS | yes | BFS 37 | 112 / 14 | 4 / 38 | not held out |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| bf16 remeasure | kth smallest | full sort | size-k heap | quickselect | no | full sort 49, quickselect 1 | 65 / 47 | 13 / 19 | not held out (the twin writes quickselect once) |
| bf16 remeasure | connected components | union find | DFS | DFS | yes | union find 50 | 92 / 55 | 32 / 42 | not held out |
| panel A | majority element | hash count | Boyer-Moore vote | Boyer-Moore vote | yes | hash count 50 | 38 / 29 | 12 / 14 | yes |
| panel A | trapped rain water | prefix and suffix maxima | two pointers | two pointers | yes | prefix and suffix maxima 32, two pointers 18 | 37 / 22 | 5 / 6 | no |
| panel A | longest mirrored substring | expand around centre | brute force | brute force | yes | expand around centre 50 | 39 / 28 | 8 / 13 | yes |
| panel B | single number | hash count | XOR fold | XOR fold | yes | hash count 46, XOR fold 4 | 77 / 62 | 32 / 34 | no |
| panel B | minimum spanning tree weight | Kruskal | Prim with heap | Prim with heap | yes | Kruskal 33, Prim 17 | 89 / 28 | 3 / 11 | no |
| panel B | integer square root | binary search | linear scan | floating point sqrt | no | binary search 50 | 108 / 76 | 15 / 17 | yes |
| panel B | count inversions | merge sort count | sorted insertion with bisect | sorted insertion with bisect | yes | merge sort count 3 | 87 / 15 | 2 / 5 | yes |
| Gemma | grid | DFS | BFS | BFS | yes | DFS 30, BFS 10 | 73 / 0 | 0 / 4 | no |
| Gemma | kth smallest | full sort | size-k heap | quickselect | no | full sort 50 | 51 / 14 | 0 / 6 | yes |
| Gemma | substring search | naive | built-in find | built-in find | yes | naive 44, built-in find 6 | 55 / 18 | 13 / 26 | no |
| Gemma | two sum | hash map | brute force | brute force | yes | hash map 50 | 55 / 13 | 7 / 14 | yes |
| Gemma | connected components | DFS | union find | BFS | no | DFS 50 | 47 / 1 | 0 / 6 | yes |
| Gemma | first and last position | linear scan | binary search | binary search | yes | linear scan 46, binary search 4 | 48 / 4 | 0 / 12 | no |
| Gemma | longest increasing subsequence | patience sorting with bisect | quadratic DP | quadratic DP | yes | patience sorting with bisect 50 | 51 / 5 | 5 / 20 | yes |

Two units contradict the narrowing premise directly: on trapped rain water and minimum spanning tree weight the instruct twin is more diverse than the base model (twin off-default share 0.36 against a base off-default share of 0.23, and 0.34 against 0.11), and on both the landing is the twin's own minority.

### 6.2.4 Movability

The movability verdict is restated with its interval and power (correction in Section 5.11) (ban-free unit and iteration-5 seeds; re-tagged by iteration 7's dose-unit experiment, Section 7.6: at D* = D1 kth smallest and connected components are movable, and shortest path, two sum and the grid are not) (iteration-8 correction). Over the 13 new units, Spearman between the base off-default share and the escape fraction is 0.15, with a Bonett-Wright 95% interval from -0.44 to 0.65, a Fieller interval from -0.46 to 0.66, a BCa bootstrap interval from -0.50 to 0.70 and a one-sided permutation p of 0.317. At a true correlation of 0.6, the registered one-sided permutation test has power 0.63 (0.50 two-sided, the figure the record had quoted as 0.46), and 20 units would give 0.8. The data exclude a strong relation, not a moderate one. The pooled value also leans on one unit: count inversions' committed base share of 0.75 came from attempts, because only 15 base programs were correct and 58 of its 87 attempts are labelled other; on the correct-only basis of the other twelve units its share is 0.13 and the pooled Spearman falls to 0.035. Table 231 lists every unit with its share on both bases and each rival under one definition. (iteration-7 correction)

**Table 231.** The 15 new units under one definition of each frozen rival (base share beta; escape fraction E; opening margin; novelty-instruction excess; twin modal share; catalogue size K; runtime merit, the natural log of the runtime of the default over that of the fastest reference). Source: record audit `tables/movability_units.csv`.

| Unit | In the 13 | beta (basis) | beta, correct only | E | Opening margin | Novelty excess | Twin modal share | K | Runtime merit | Twin off-default | Other share of base attempts |
|---|---|---|---|---|---|---|---|---|---|---|---|
| majority element | yes | 0.414 (correct) | 0.414 | 0.125 | -8.72 | 0.00 | 1.00 | 7 | 2.19 | 0.00 | n/a |
| maximum block sum | yes | 0.000 (correct) | 0.000 | 0.000 | 22.52 | 0.00 | 1.00 | 4 | 0.18 | 0.00 | n/a |
| trapped rain water | yes | 0.227 (correct) | 0.227 | 0.4375 | 5.21 | -0.36 | 0.64 | 4 | 2.43 | 0.36 | n/a |
| count primes | yes | 0.806 (correct) | 0.806 | 0.000 | -0.53 | 0.00 | 1.00 | 6 | -1.68 | 0.00 | n/a |
| longest common prefix | yes | 0.300 (correct) | 0.300 | 0.000 | 12.58 | 0.27 | 1.00 | 7 | 0.08 | 0.00 | n/a |
| longest mirrored substring | yes | 0.464 (correct) | 0.464 | 0.1875 | -36.88 | 0.00 | 1.00 | 4 | 2.34 | 0.00 | n/a |
| jump game | no (reserve) | 0.474 (correct) | 0.474 | 0.000 | 3.54 | 0.00 | 1.00 | 3 | -0.98 | 0.00 | n/a |
| median of two sorted lists | no (reserve) | 0.708 (correct) | 0.708 | 0.000 | 17.77 | 0.20 | 1.00 | 4 | 0.95 | 0.00 | n/a |
| missing number | yes | 0.056 (correct) | 0.056 | 0.000 | 3.58 | 0.00 | 1.00 | 5 | -0.25 | 0.00 | 0.033 |
| single number | yes | 0.565 (correct) | 0.565 | 0.4375 | 25.37 | -0.08 | 0.92 | 4 | 0.27 | 0.08 | 0.078 |
| minimum spanning tree weight | yes | 0.107 (correct) | 0.107 | 0.125 | 26.89 | -0.07 | 0.66 | 4 | -0.26 | 0.34 | 0.247 |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| integer square root | yes | 0.368 (correct) | 0.368 | 0.125 | 13.99 | 0.00 | 1.00 | 6 | 0.35 | 0.00 | 0.083 |
| all subsets | yes | 0.370 (correct) | 0.370 | 0.000 | 18.92 | 0.00 | 1.00 | 4 | 0.09 | 0.00 | 0.020 |
| valid anagram | yes | 0.806 (correct) | 0.806 | 0.000 | -7.57 | 0.76 | 0.76 | 3 | 0.20 | 0.24 | 0.072 |
| count inversions | yes | 0.747 (attempt) | 0.133 | 0.3125 | -8.85 | 0.33 | 1.00 | 4 | 0.39 | 0.00 | 0.667 |

The preregistered rule required beta to be ahead of each rival. Computed under one definition on the 13 units, with signs fixed so that the predicted direction is positive, only runtime merit tracks movability after Holm correction (Table 232), and the paired interval for beta minus runtime merit still includes 0.

**Table 232.** The frozen rivals on the 13 new units: Spearman with the escape fraction, one-sided permutation p, the paired difference beta minus rival with its bootstrap 95% interval, and Holm p. Source: record audit `tables/rival_scores.csv`.

| Rival | Predicted sign | Spearman with E | One-sided p | beta minus rival [95% CI] | Holm p |
|---|---|---|---|---|---|
| opening margin | negative | 0.079 | 0.399 | 0.067 [-0.66, 0.88] | 1.0 |
| novelty-instruction excess | positive | -0.503 | 0.960 | 0.649 [-0.06, 1.34] | 1.0 |
| runtime merit | positive | 0.713 | 0.004 | -0.568 [-1.26, 0.20] | 0.021 |
| twin modal share | negative | 0.409 | 0.087 | -0.263 [-1.05, 0.56] | 0.347 |
| catalogue size | positive | -0.206 | 0.753 | 0.352 [-0.57, 1.24] | 1.0 |

### 6.2.5 Executed outputs added to the record

The audit added the outputs the review listed, each in place and marked: the phrasing clusters per temperature in the literal cell (Table 206: 2/4/8, 3/3/10, 2/3/9 and 5/7/12 at theta 0.7/0.8/0.9 for temperatures 0.3, 0.7, 1.0 and 1.3, and 3/6/10 for the repeat at 0.7, so at N = 20 with thinking on a rise shows only at 1.3 and is the size of the repeat noise); Table 207b, the pass cost at which each knob first reaches a correct-only Hill-1 of 1.5 and of 2, with bootstrap intervals; Table 203b, panel B's knob rows, where on minimum spanning tree weight temperature 1.3 already gives Prim 18 of 49 correct programs against 20 of 50 under plan noise, because the twin itself is mixed; and the three Llama present-minor ban rows of Table 126c (Hill-1 2.87, 2.96 and 2.92 at pass costs of -12 to -27 points). Panel B's pushes stayed inside the base repertoire on 6 of 7 units; on count inversions 3 pushed programs implemented a Fenwick tree that the base model never wrote. Per task, plan noise against matched decoding at alpha 2 is 13:0 on kth smallest and 6:3 on the other four tasks together (p 0.508), with connected components at 1:3 (Table 233).

**Table 233.** Plan noise against decoding at matched per-token KL: slots correct and off the default under one arm only. Source: record audit `tables/pn_vs_dmd_discordant.csv`.

| Task | Alpha 1: plan noise only / matched decoding only (exact p) | Alpha 2: plan noise only / matched decoding only (exact p) |
|---|---|---|
| kth smallest | 3 / 0 (0.25) | 13 / 0 (0.00024) |
| connected components | 1 / 2 (1.0) | 1 / 3 (0.625) |
| shortest path | 0 / 1 (1.0) | 2 / 0 (0.5) |
| two sum | 0 / 0 (1.0) | 2 / 0 (0.5) |

| | | |
|---|---|---|
| grid | 0 / 0 (1.0) | 1 / 0 (1.0) |
| pooled | 4 / 3 (1.0) | 19 / 3 (0.00086) |

### 6.2.6 A second blind adjudication of the 14 disputed programs

In iteration 5 two of three judges had called 14 of the 48 correct transplant programs outside the catalogue novel, and the zero count rested on a manual read by the executor. The audit stress-tested each program in a sandbox at the maximum size of the specification and on adversarial cases, retrieved its three nearest known solutions by Zhang-Shasha tree edit distance [64], and had two judges from model families the run had never used classify it blind: Mistral Medium 3.1 (three calls at temperature 0, Fleiss kappa [128] 1.00 across the calls) and Kimi (kimi-k2, Moonshot AI). Ten controls of known class were mixed in, and both judges passed the control gate at 9 of 10. All 14 programs pass their hidden tests; one, a repeated maximum selection on top-k frequent, times out at the maximum size. Mistral called 1 of 14 novel and Kimi 0 of 14 (Table 234). The one program called novel, a first-and-last-position solution, is range guards plus a linear scan to the first match and over its run; it passes 56 of 56 hidden tests and every stress group, and a full read by the executor classes it as a linear search. The final count is 0 novel of 14. Agreement with the manual verdict on the 14 is 0.86 raw (95% interval 0.57 to 0.98), Gwet's AC1 [127] 0.83 and a Cohen kappa of 0.71; with the controls included (24 items) kappa is 0.80 [0.57, 1.00]. The final read corrected one manual class: one first-and-last-position program, a linear scan plus a count of each element, is degenerate rather than a catalogue variant.

**Table 234.** The 14 disputed transplant programs: task, iteration-5 manual class, the majority class of Mistral Medium 3.1 (three calls), the class given by Kimi, and the final class. Source: record audit `tables/novelty_readjudication.csv`, `audit/final_read.json`.

| Program | Task | Iteration-5 manual class | Mistral Medium 3.1 | Kimi | Final class |
|---|---|---|---|---|---|
| 1 | first and last position | known, outside the catalogue | known, outside the catalogue (linear search) | known, outside the catalogue | known, outside the catalogue |
| 2 | first and last position | degenerate | degenerate | degenerate | degenerate |
| 3 | first and last position | known, outside the catalogue | known, outside the catalogue | known, outside the catalogue | known, outside the catalogue |
| 4 | first and last position | known, outside the catalogue | known, outside the catalogue | known, outside the catalogue | known, outside the catalogue |
| 5 | first and last position | catalogue variant | degenerate | degenerate | degenerate (corrected) |
| 6 | first and last position | known, outside the catalogue | novel (3 of 3 calls) | known, outside the catalogue | known, outside the catalogue (linear search) |
| 7 | first and last position | known, outside the catalogue | known, outside the catalogue | known, outside the catalogue | known, outside the catalogue |
| 8 | first and last position | known, outside the catalogue | known, outside the catalogue | known, outside the catalogue | known, outside the catalogue |
| 9 | first and last position | known, outside the catalogue | known, outside the catalogue | known, outside the catalogue | known, outside the catalogue |
| 10 | first and last position | known, outside the catalogue | known, outside the catalogue | known, outside the catalogue | known, outside the catalogue |

| | | | | | |
|---|---|---|---|---|---|
| 11 | first and last position | known, outside the catalogue | known, outside the catalogue | known, outside the catalogue | known, outside the catalogue |
| 12 | connected components | catalogue variant | catalogue variant (union find) | known, outside the catalogue | catalogue variant |
| 13 | top-k frequent | catalogue variant | catalogue variant (sort by count) | catalogue variant | catalogue variant |
| 14 | top-k frequent | known, outside the catalogue | known, outside the catalogue (binary search on a frequency threshold) | known, outside the catalogue | known, outside the catalogue |

Across four judge passes the novel calls of Mistral on the 14 were 0, 0, 1 and 1, and those of Kimi 0 in every pass; the first two passes carried run-dependent timings in their prompts. No pass produced a novel call that survived a full read.

## 6.3 The engine control: are the random-push escapes a property of the model or of 8-bit rounding?

**Why.** The engine claim was the primary test of the iteration, because its answer decides which internal results of iteration 4 belong to Qwen3-8B. The review had pointed at a plausible mechanism without establishing it: a displacement changes the per-token absmax of each linear input, so under dynamic per-token 8-bit floating-point scaling it changes the rounding of every downstream activation, and the engine adds its own perturbation to the push. The 8-bit floating-point format in question is e4m3, with a maximum of 448 [115]. The engine control separated weight rounding from activation rounding by running iteration 4's own design under four numerics, all built from one bf16 snapshot of Qwen3-8B (Table 235).

**Table 235.** The four numerics of the engine control.

| Engine | Weights | Activations | Role |
|---|---|---|---|
| W8A8 | e4m3, per-output-channel absmax / 448 | e4m3, per-token absmax / 448 | iteration 4's engine (native 8-bit floating-point matrix multiply) |
| W8A16 | e4m3, the same codes | bf16 | weight rounding only |
| W16A8 | bf16 | e4m3, fake-quantised per token | activation rounding only |
| bf16 | bf16 | bf16 | full-precision reference |

**Design.** Everything except the numerics was held fixed: the 40 direction files of iteration 4 (seeds 9101 to 9140, checked bitwise), the absolute displacement per block and position at blocks 12 to 23 and the task-specification positions (the bench unit of iteration 4), the twin prompts, the common-random-number slots and uniforms, the stop rule, a 1,024-token cap, one labeller version (that of iteration 4 plus the iteration-5 re-gate that reads array-based Dijkstra correctly) applied to every row, and the two-stage escape rule of iteration 4, with each engine scored against its own twin. The W8A8 engine first regenerated the rows of iteration 4 on the same GPU model, driver and library versions: 400 of 400 programs of the plan-subspace experiment and 40 of 40 per-sample plan-noise programs came out byte-identical, the realised downstream displacement had ratio 1.000, and per-slot label agreement was 1.0. Under the reuse rule committed before sampling, the rows of iteration 4 therefore stand in for the W8A8 engine on every other cell. The bf16 twins reproduce the bf16 twin composition of iteration 5 (connected components union find 50 of 50; shortest path Dijkstra 49, Bellman-Ford/SPFA 1). The direction screen ran at alpha 2 on connected components and shortest path, and per-sample plan noise at alpha 1 and 2; the alpha-1 direction arms of the two single-factor engines, a Gaussian mimic of the 8-bit noise and kth smallest were

dropped by the committed drop order, and the W8A16 engine ran on a slower cast path because the fused kernel could not be compiled.

**bf16 escapes on almost the same directions.** At the same absolute displacement the bf16 engine escapes on 26 of 40 directions on connected components and 6 of 40 on shortest path, against 29 and 8 under W8A8 (Table 236). The direction-level discordance is 3:0 and 4:2 (exact McNemar p 0.25 and 0.69), the per-direction off-default shares rank alike (Spearman 0.96 and 0.87; kappa on escape 0.83 and 0.48), and permuting the pairing of directions drops kappa to about 0. The committed verdict on both tasks is the rule's outcome for a high escape fraction that 8-bit rounding does not explain: W8A8 minus bf16 is below 0.1 and bf16 is within 0.15 of iteration 4's escape fraction, so bf16 escapes too and the account specific to 8-bit rounding is not reproduced. Table 237 pairs every two engines direction by direction. The low escape fraction on shortest path is a pass-rate effect common to both engines: in each, 17 of 40 directions reach the $+0.20$ off-default excess but fail the pass clause, because the pushed Bellman-Ford/SPFA programs often fail the hidden tests. At the screen level the shortest path verdict is MIXED (the artifact's `screen_level_verdict`), beside the literal NOT_REPRODUCED_HIGH: W8A8 passes the screen on 15 of 40 directions against 8 to 9 for the other engines, and 17 (W8A8) and 17 (bf16) of 40 directions reach the $+0.20$ excess but fail the pass clause. (iteration-7 correction)

**Table 236.** Escape fraction at alpha 2 over 40 directions, each engine against its own twin (Wilson 95% intervals). Source: engine control `results/escape_by_engine.json`.

| Task | Engine | Twin off-default (k/n) | Twin pass | Screen passes | Confirmed escapes | Escape fraction [95%] | Screen-level escape fraction |
|---|---|---|---|---|---|---|---|
| connected components | W8A8 | 0/50 | 1.00 | 30/40 | 29 | 0.725 [0.57, 0.84] | 0.750 |
| connected components | W8A16 | 0/50 | 1.00 | 28/40 | 27 | incomplete (confirmation rows short of 30 for all 28 screen passers) | 0.700 |
| connected components | W16A8 | 0/50 | 1.00 | 27/40 | 24 | 0.600 [0.45, 0.74] | 0.675 |
| connected components | bf16 | 0/50 | 1.00 | 27/40 | 26 | 0.650 [0.50, 0.78] | 0.675 |
| shortest path | W8A8 | 0/50 | 1.00 | 15/40 | 8 | 0.200 [0.10, 0.35] | 0.375 |
| shortest path | W8A16 | 1/50 | 0.98 | 8/40 | 5 | 0.125 [0.05, 0.26] | 0.200 |
| shortest path | W16A8 | 1/50 | 1.00 | 8/40 | 6 | 0.150 [0.07, 0.29] | 0.200 |
| shortest path | bf16 | 1/50 | 0.98 | 9/40 | 6 | 0.150 [0.07, 0.29] | 0.225 |

**Table 237.** Direction-paired contrasts of confirmed escapes (exact McNemar, Tango 90% interval, equivalence test at a margin of 0.1, kappa, and Spearman of the first-10 off-default shares). Source: engine control `results/mcnemar.json`.

| Task | Pair | First escape fraction | Second escape fraction | Difference | Directions escaping under first only / second only | Two-sided p (Holm) | 90% CI | Equivalent within 0.1 | Kappa | Spearman of shares |
|---|---|---|---|---|---|---|---|---|---|---|
| connected components | W8A8 vs bf16 | 0.725 | 0.650 | 0.075 | 3 / 0 | 0.250 (1.000) | [0.01, 0.17] | no | 0.83 | 0.96 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| connected components | W16A8 vs bf16 | 0.600 | 0.650 | -0.050 | 1 / 3 | 0.625 (1.000) | [-0.15, 0.04] | no | 0.79 | 0.96 |
| connected components | W8A8 vs W16A8 | 0.725 | 0.600 | 0.125 | 5 / 0 | 0.062 | [0.05, 0.24] | no | 0.73 | 0.95 |
| shortest path | W8A8 vs bf16 | 0.200 | 0.150 | 0.050 | 4 / 2 | 0.688 (1.000) | [-0.06, 0.17] | no | 0.48 | 0.87 |
| shortest path | W16A8 vs bf16 | 0.150 | 0.150 | 0.000 | 2 / 2 | 1.000 (1.000) | [-0.10, 0.10] | yes | 0.61 | 0.90 |
| shortest path | W8A16 vs bf16 | 0.125 | 0.150 | -0.025 | 0 / 1 | 1.000 (1.000) | [-0.10, 0.04] | no | 0.89 | 0.83 |
| shortest path | W8A8 vs W16A8 | 0.200 | 0.150 | 0.050 | 4 / 2 | 0.688 | [-0.06, 0.17] | no | 0.48 | 0.81 |
| shortest path | W8A8 vs W8A16 | 0.200 | 0.125 | 0.075 | 5 / 2 | 0.453 | [-0.04, 0.20] | no | 0.36 | 0.91 |

**Per-sample plan noise reproduces in bf16 at the matched unit.** On the slots, uniforms and per-sample direction seeds that iteration 4 used, at the same absolute unit, bf16 plan noise moves connected components in 25 of 50 valid attempts at alpha 2 and shortest path in 22 of 50, against 30 and 22 under W8A8 (Table 238). The realised displacement at block 12, the first edited block, is 13.75 in the bf16 rows at alpha 2 and 13.76 in the rows of iteration 4. The bf16 plan noise of iteration 5 had moved connected components in 1 of 50 programs at alpha 1 and at alpha 2 on the same prompts and uniforms, but at the ban-free unit and with other direction seeds. The size of the push alone does not explain that null: the alpha-2 displacement of iteration 5 at block 12 (9.07) exceeds the bf16 alpha-1 displacement of this control (6.88), which still moves 7 of 50. The difference lies in the profile of the ban-free unit over positions and blocks, or in the direction seeds; which of the two matters was not tested.

**Table 238.** Per-sample plan noise per engine on the slots of iteration 4 at the absolute unit (valid attempts on a catalogue alternative to the default, over 50; programs labelled other are not counted). One labeller, including the iteration-5 re-gate of the shortest path rule, scores every engine, so the shortest path cells of iteration 4 read 12 and 22 of 50 here against 13 and 27 in the all-samples count of the audit (Table 228), which counts programs labelled other and predates the re-gate. Source: engine control `results/pn.json`, `data/samples.jsonl` (iteration-8 correction: file names restored). The two added columns count only programs that pass every hidden test (from the per-row labels of `data/samples.jsonl` (file name restored, iteration-8 correction), the engine-a rows coming from iteration 4), first over the correct programs of each engine and then over all 50. (iteration-7 correction)

| Task | Arm | Iteration-4 8-bit rows | W8A8 | W8A16 | W16A8 | bf16 | Paired discordance W8A8 : bf16 | Correct off-default / correct programs (W8A8; W8A16; W16A8; bf16) | Correct off-default / 50 (W8A8; W8A16; W16A8; bf16) |
|---|---|---|---|---|---|---|---|---|---|
| connected components | alpha 1 | 6/50 (0.12) | 6/50 (0.12) | 9/50 (0.18) | 4/50 (0.08) | 7/50 (0.14) | 1:2 | 6/50; 9/50; 4/50; 7/50 | 6/50; 9/50; 4/50; 7/50 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| connected components | alpha 2 | 30/50 (0.60) | 30/50 (0.60) | 27/50 (0.54) | 22/50 (0.44) | 25/50 (0.50) | 6:1 (p 0.13) | 30/50; 27/50; 21/49; 24/49 | 30/50; 27/50; 21/50; 24/50 |
| shortest path | alpha 1 | 12/50 (0.24) | 12/50 (0.24) | 11/50 (0.22) | 11/50 (0.22) | 10/50 (0.20) | 4:2 | 9/43; 7/43; 6/40; 5/43 | 9/50; 7/50; 6/50; 5/50 |
| shortest path | alpha 2 | 22/50 (0.44) | 22/50 (0.44) | 19/50 (0.38) | 18/50 (0.36) | 22/50 (0.44) | 8:8 | 14/37; 11/39; 6/32; 9/31 | 14/50; 11/50; 6/50; 9/50 |

*[Figure fig_engine_escape: generated in the report phase, not in this draft]*

**The 8-bit engine adds a small boost on connected components, and activation rounding is not its source.** Pairing every pushed sample with the same slot (direction, template, slot index and uniforms) in the other engine, on the 400 screen slots of the first two samples of all 40 directions, W8A8 is off the default in 58.3% of slots against 53.2% under bf16 on connected components (discordant 30:10, exact McNemar p 0.0022), and 43.2% against 40.7% on shortest path (55:45, p 0.37) (Table 239). The factorial locates the boost. Activation rounding alone (W16A8) sits slightly below bf16 on connected components (49.7% against 53.2%, 9:23, p 0.02); weight rounding alone (W8A16) sits between bf16 and W8A8 (55.6%; 17:7 against bf16, p 0.06). The small boost therefore comes from the e4m3 weights, from the native kernel's reduced-precision accumulation (it matches an exact fp64 reference on 94.2% of outputs, against 99.96% for an exact emulation), or from their interaction. The landing algorithms do not change: correct off-default programs are DFS, then BFS, on connected components and Bellman-Ford/SPFA on shortest path under every engine, with compositions compatible across engines (permutation chi-squared p 0.22 and 0.94; Table 240).

**Table 239.** Common-random-number pairs of screen slots under the alpha-2 pushes: off-default share under each engine, discordant slots and exact McNemar p. Source: engine control `results/slot_flips.json`.

| Task | Pair | Slots | Off-default, first | Off-default, second | Discordant first : second | Exact p | Same label |
|---|---|---|---|---|---|---|---|
| connected components | W8A8 vs bf16 | 400 | 0.583 | 0.532 | 30:10 | 0.0022 | 0.873 |
| connected components | W16A8 vs bf16 | 400 | 0.497 | 0.532 | 9:23 | 0.02 | 0.900 |
| connected components | W8A16 vs bf16 | 399 | 0.556 | 0.531 | 17:7 | 0.064 | 0.925 |
| connected components | W8A8 vs W16A8 | 400 | 0.583 | 0.497 | 45:11 | 5.4e-06 | 0.825 |
| connected components | W8A8 vs W8A16 | 399 | 0.581 | 0.556 | 20:10 | 0.099 | 0.895 |
| shortest path | W8A8 vs bf16 | 400 | 0.432 | 0.407 | 55:45 | 0.37 | 0.740 |
| shortest path | W16A8 vs bf16 | 400 | 0.420 | 0.407 | 35:30 | 0.62 | 0.830 |
| shortest path | W8A16 vs bf16 | 400 | 0.390 | 0.407 | 38:45 | 0.51 | 0.782 |
| shortest path | W8A8 vs W16A8 | 400 | 0.432 | 0.420 | 57:52 | 0.70 | 0.723 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| shortest path | W8A8 vs W8A16 | 400 | 0.432 | 0.390 | 47:30 | 0.068 | 0.805 |

**Table 240.** Composition of correct off-default programs under every alpha-2 push, per engine. Source: engine control `results/landings.json`.

| Task | W8A8 | W8A16 | W16A8 | bf16 |
|---|---|---|---|---|
| connected components | DFS 502, BFS 110 | DFS 410, BFS 86 | DFS 406, BFS 67 | DFS 448, BFS 82 |
| shortest path | Bellman-Ford/SPFA 165, Floyd-Warshall 1, other 1 | Bellman-Ford/SPFA 137, Floyd-Warshall 2 | Bellman-Ford/ SPFA 129, other 1 | Bellman-Ford/SPFA 120, Floyd-Warshall 1, other 1 |

**Mechanism diagnostics.** Per-token e4m3 rounding perturbs every linear input by about 2.5% RMS. Under an alpha-2 push, the push-induced difference in rounding error is 6% to 12% of the push-induced change in activation (median over blocks 13 to 35 per site, peaking at 34% at the block-14 attention input), and 70% to 100% of the rows the push moves receive a new per-token scale. Linear inputs at blocks 0 to 12 are bitwise identical in the pushed and the twin pass in every engine, as the edit site requires. The extra rounding barely changes how the push propagates: relative to bf16, the non-nominal part of the perturbation is 1.01 at the task-specification positions, 1.04 to 1.05 at the read position and 1.03 to 1.12 at generated tokens 1 to 32 under W16A8, and 0.995 to 1.006 under W8A16. Teacher-forced on 40 iteration-4 twin programs, every engine is very close to bf16 (Table 241). One preregistered fidelity check failed uninformatively: teacher forcing runs one long prefill instead of 20-row decode steps, which itself changes the numerics, so log probabilities of iteration 4's programs could not be matched closely enough to resolve engines at that level; the engine was validated instead by byte identity.

**Table 241.** Teacher-forced divergence from bf16 on 40 iteration-4 twin programs, and the failed fidelity check (median absolute difference between the summed log probability of a program and the value recorded in iteration 4). Source: engine control `results/diagnostics_summary.json`.

| Engine | Task | Mean per-token KL to bf16 (nats) | Argmax disagreement with bf16 | Median absolute log-probability difference |
|---|---|---|---|---|
| W8A8 | connected components | 0.0036 | 0.0012 | 0.477 |
| W8A8 | shortest path | 0.0015 | 0.0003 | 0.450 |
| W8A16 | connected components | 0.0019 | 0.0000 | 0.315 |
| W8A16 | shortest path | 0.0005 | 0.0003 | 0.703 |
| W16A8 | connected components | 0.0017 | 0.0019 | 0.448 |
| W16A8 | shortest path | 0.0011 | 0.0021 | 0.912 |
| bf16 | connected components | n/a | n/a | 0.383 |
| bf16 | shortest path | n/a | n/a | 0.679 |

**What it changes.** Table 242 assigns each earlier internal result to the numerics class that reproduces it. The 40-direction escape fractions, the per-sample plan-noise rates and the positive-control direction 9004 of the frozen screen are model properties. Only the per-slot off-default rate on connected components carries a significant 8-bit boost, of about 5 points. Iteration 5's bf16 plan-noise null and its finding that the shortest path task stops being movable in bf16 are not reproduced at the absolute unit; they belong to that design's unit and seeds. The plan-subspace claims of iteration 4 (the projection predictor, the targeted runner-up pushes, the decomposition)

and the site and same-seed claims were not rerun here.

**Table 242.** Earlier internal results and the numerics class that reproduces them. Source: engine control `results/relabel_table.json`.

| Earlier result | Class | Deciding cells |
|---|---|---|
| 40-direction escape fraction, connected components, alpha 2 (iteration 4: 0.725) | model | escape fraction W8A8 0.725, W8A16 incomplete (screen level 0.70), W16A8 0.60, bf16 0.65 |
| Per-sample plan noise, connected components, alpha 1 (iteration 4: 6/50) | model | shares 0.12, 0.18, 0.08, 0.14 |
| Per-sample plan noise, connected components, alpha 2 (iteration 4: 30/50) | model | shares 0.60, 0.54, 0.44, 0.50 |
| 40-direction escape fraction, shortest path, alpha 2 (iteration 4: 0.200) | model | escape fraction 0.200, 0.125, 0.15, 0.15 |
| Per-sample plan noise, shortest path, alpha 1 (iteration 4: 12/50) | model | shares 0.24, 0.22, 0.22, 0.20 |
| Per-sample plan noise, shortest path, alpha 2 (iteration 4: 22/50) | model | shares 0.44, 0.38, 0.36, 0.44 |
| Frozen-screen direction 9004 moves shortest path at alpha 1 | model | first-10 off-default share W8A16 0.3, W16A8 0.4, bf16 0.3 |
| Per-slot off-default rate under alpha-2 pushes, connected components | model plus a small 8-bit boost | W8A8 0.583 against bf16 0.532; discordant 30:10, p 0.0022 |
| Per-slot off-default rate under alpha-2 pushes, shortest path | model | W8A8 0.432 against bf16 0.407; 55:45, p 0.37 |
| Iteration 5's bf16 plan-noise null on connected components (1/50 at the ban-free unit; block-12 dose 0.66 of this unit) | unit and seeds, not reproduced at the absolute unit | bf16 25/50 and W8A8 30/50 at alpha 2 |
| Iteration 5: shortest path no longer movable in bf16 | not reproduced at the absolute unit | bf16 escape fraction 6/40 confirmed escapes |
| Site odds ratio 5.72 | untested here | not rerun in this artifact |
| Same-seed agreement 17.25 of 20 | untested here | not rerun |
| Plan-subspace projection predictor | untested here | not rerun |
| Targeted runner-up pushes | untested here | not rerun |
| Decomposition into subspace and orthogonal parts | untested here | not rerun |

**Novelty and checks.** The two-judge novelty audit was replaced by a nearest-known-solution pass over all ten datasets of the declared source data (18,237 code solutions); Table 243 classes every distinct correct off-default or other program by its largest token-shingle Jaccard to a known solution. CPU gates checked the directions bitwise, the slots and seeds, the escape code (it reproduces iteration 4's escape file exactly) and the logged displacement (within 0.074% on all 1,770 iteration-4 prefills); the relabelling with the re-gated shortest path rule changed 18 iteration-4 labels and no escape fraction. GPU unit tests confirmed bitwise e4m3 codes, a bitwise twin at alpha 0, rerun identity, batch invariance for three rows or more and an exact match of the memory-capped sampler. A stdlib re-derivation and a separate audit from raw rows matched every headline, and a placebo that swaps engine labels was not significant. Judge spend was USD 0.085.

**Table 243.** Nearest known solution of the correct off-default and other programs per engine (distinct programs; median largest Jaccard; classes by Jaccard band). Source: engine control `results/known_solutions.json`.

| Engine and task | Distinct programs | Median largest Jaccard | Variant | Distant | Known, close |
|---|---|---|---|---|---|
| W8A8, connected components | 229 | 0.297 | 162 | 64 | 3 |
| W8A8, shortest path | 237 | 0.209 | 25 | 212 | 0 |
| W8A16, connected components | 186 | 0.291 | 137 | 47 | 2 |
| W8A16, shortest path | 162 | 0.215 | 14 | 148 | 0 |
| W16A8, connected components | 159 | 0.304 | 123 | 32 | 4 |
| W16A8, shortest path | 128 | 0.214 | 16 | 112 | 0 |
| bf16, connected components | 185 | 0.302 | 136 | 46 | 3 |
| bf16, shortest path | 134 | 0.214 | 14 | 120 | 0 |

## 6.4 Where a ban and an instruction are carried, and where noise acts, in bf16 (Qwen3-8B)

**Why.** The localization claim predicted, from the route classes of iterations 2 and 3, where the substitute of a ban is carried: in the task-specification states on the prompt-route tasks, at the condition span on the direct-route tasks. Stage A tests that prediction without any training, by moving the states themselves. The same experiment reran the iteration-4 noise-site comparison in bf16, set state noise at generated tokens against a temperature matched on per-token KL, applied the top-five plan push, and ran the landing contest. Everything ran in one bf16 engine with exact common random numbers on a 16 GB card, with the input embedding table kept in CPU memory; per-slot twin labels agreed with iteration 5's bf16 engine on 0.975 of slots for kth smallest and shortest path and 1.00 for the other tasks. Seven commitments formed an unbroken sha256 chain, and a standard-library script re-derived every headline from raw rows (295 checks, 0 mismatches) while a placebo version of each test failed.

### 6.4.1 Stage A: full-state exchange of the ban

We call a *full-state exchange* the replacement, in the forward pass of the neutral twin, of the residual states at a set of positions and blocks by the states the full ban prompt produces at the same positions; because the twin and the ban have identical token ids from the task text on, every task-specification position aligns. The exchanges covered four block bands at the task-specification positions (0 to 11, 12 to 23, 24 to 35, 0 to 35), a splice of the key-value cache of the condition span (the direct route), and the read position at all 36 blocks. The controls were the twin, a matched-norm random exchange (the state of the twin plus a fresh random direction scaled to the displacement that the ban itself causes per slot, block and position), a placebo that takes a second neutral twin as the source, and an identity exchange. The outcome, as committed before sampling (the first commitment: "share(label $==$ Y $|$ cell) - share(label $==$ Y $|$ TWIN), all samples (invalid rows count as not-Y)"), is the *Y share*: the share of all attempted programs labelled Y, invalid rows counting as not-Y; the *transfer* of a cell is that share minus the share under the twin; a cell counts only if its pass rate is at least 0.8 times that of the twin (the validity gate of the first commitment); and its *recovery* is the transfer divided by that of the full ban. The *correct Y* column added to Tables 244 to 250 counts programs labelled Y that pass every hidden test, because an attempted share counts broken programs as Y: on the grid task the exchange at blocks 0-35 has 14 attempts labelled DFS but 3 correct DFS programs. Each cell has N $=$ 30 on paired slots. Tables 244 to 249 give every cell on the six tasks. (iteration-7 correction)

The prediction held where the route data said it should, and failed at one task where they said it should not. On shortest path, exchanging the full ban's task-specification states at all blocks into the twin turns 24 of 30 slots from Dijkstra to Bellman-Ford/SPFA (24 of 30 attempts, 21 of 30 correct

programs; transfer 0.80, McNemar 24:0, Holm p 7e-7, pass ratio 0.90), while the condition-span splice and the read-position exchange transfer nothing. Blocks 12 to 23 alone carry 0.60 (18 of 30 attempts, 11 of 30 correct programs, at a pass ratio of 0.73, below the 0.8 gate) and blocks 24 to 35 alone 0.37 (11 of 30 attempts, 11 of 30 correct programs); blocks 0 to 11 carry none. On kth smallest the same exchange transfers 0.57 to quickselect (17 of 30 attempts, 16 of 30 correct programs) with every slot leaving the full sort, blocks 0 to 11 alone already transfer 0.40 (12 of 30 attempts, 12 of 30 correct programs), and here the condition span also carries the ban (0.93; 28 of 30 attempts, 25 of 30 correct programs). On connected components every slot leaves union find under the exchange at blocks 12 to 23 or 0 to 35 (off-default transfer 1.00), but only 0.37 to 0.47 land on the ban's BFS (11 and 14 of 30 attempts, 11 and 14 of 30 correct programs) and the rest on DFS. On the grid task, whose route was carried by the prompt states in likelihood and mixed in behaviour, the exchange at all blocks transfers 0.47 toward DFS (14 of 30 attempts, 3 of 30 correct programs, pass ratio 0.59, below the 0.8 gate) and the condition-span splice 0.33 (10 of 30 attempts, 7 of 30 correct programs, pass ratio 0.89), so each carries part of the ban in attempts and only the splice passes the committed validity gate. On the direct-route tasks the condition-span splice alone reproduces the ban (two sum 1.00, 30 of 30 attempts, 30 of 30 correct programs; find duplicate 0.90, 27 of 30 attempts, 26 of 30 correct programs), as predicted, but on two sum the task-specification states also transfer brute force at 0.43 (13 of 30 attempts, 13 of 30 correct programs), above the predicted ceiling of 0.2. The committed verdict is therefore not supported. The secondary predictions hold: the condition span carries at least 0.5 on the direct-route tasks and at most 0.2 on shortest path, the read position carries at most 0.2 everywhere (it carries 0.00 on all six tasks), the placebo moves at most 0.03, and the exchange beats the matched-norm random exchange where it transfers (shortest path 23:0, kth smallest 16:2, connected components 11:0, on attempted Y as preregistered); counted on correct Y instead, a post-hoc contrast added in iteration 7, the same pairs give 21:0 (exact p 9.5e-07), 15:2 (p 2.3e-03) and 11:0 (p 9.8e-04). The ban's information, not the size of the perturbation, is what moves the algorithm. (iteration-7 correction)

*[Figure fig_exchange_transfer: generated in the report phase, not in this draft]*

**Table 244.** Stage A on kth smallest (default full sort; Y = quickselect; route class from iterations 2 and 3: either route suffices). Twin on slots 0 to 29: Y share 0.00, off-default 0.00, pass 1.00, full sort 30. Exchange at blocks 0-35 against the matched-norm random exchange: 16:2, exact p 0.0013 (Holm 0.0020). Source: exchange and site experiment `results/stage_a.json`, `results/summary.md`. The Y share is the outcome of the first commitment (all attempted programs labelled Y; invalid rows count as not-Y); the added column *Correct Y / N* counts programs labelled Y that pass every hidden test, with Clopper-Pearson 95% intervals; a cell meets the validity gate of the first commitment when its pass ratio is at least 0.8. Post hoc, on correct Y, the exchange at blocks 0-35 against the matched-norm random exchange gives 15:2 (exact p 2.3e-03). (iteration-7 correction)

| Cell | n | Y share [CP 95%] | Correct Y / N [CP 95%] | Transfer | Off-default transfer | Pass ratio | Recovery | Slot interchange intervention accuracy | McNemar b:c (p, Holm) | Correct composition |
|---|---|---|---|---|---|---|---|---|---|---|
| full ban | 30 | 0.67 [0.47, 0.83] | 18/30 [0.41, 0.77] | 0.67 | 1.00 | 0.93 | 1.00 | 1.00 | 20:0 (1.9e-06, 1.9e-06) | quickselect 18, size-k heap 10 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| exchange, blocks 0-11 | 30 | 0.40 [0.23, 0.59] | 12/30 [0.23, 0.59] | 0.40 | 0.53 | 1.00 | 0.60 | 0.33 | 12:0 (4.9e-04, 0.003) | full sort 14, quickselect 12, size-k heap 4 |
| exchange, blocks 12-23 | 30 | 0.57 [0.37, 0.75] | 15/30 [0.31, 0.69] | 0.57 | 1.00 | 0.93 | 0.85 | 0.90 | 17:0 (1.5e-05, 7.6e-05) | quickselect 15, size-k heap 13 |
| exchange, blocks 24-35 | 30 | 0.03 [8.4e-04, 0.17] | 1/30 [8.4e-04, 0.17] | 0.03 | 0.03 | 1.00 | 0.05 | 0.03 | 1:0 (1.000, 1.000) | full sort 29, quickselect 1 |
| exchange, blocks 0-35 | 30 | 0.57 [0.37, 0.75] | 16/30 [0.34, 0.72] | 0.57 | 1.00 | 0.97 | 0.85 | 0.90 | 17:0 (1.5e-05, 7.6e-05) | quickselect 16, size-k heap 13 |
| condition-span splice | 30 | 0.93 [0.78, 0.99] | 25/30 [0.65, 0.94] | 0.93 | 1.00 | 0.90 | 1.40 | 0.60 | 28:0 (7.5e-09, 3.7e-08) | quickselect 25, size-k heap 2 |
| read-position exchange | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.00 | 1.00 | 0.00 | 0.00 | 0:0 (1.000, 1.000) | full sort 30 |
| matched-norm random exchange, blocks 0-35 | 30 | 0.10 [0.02, 0.27] | 3/30 [0.02, 0.27] | 0.10 | 0.10 | 1.00 | 0.15 | 0.03 | 3:0 (0.250, 0.750) | full sort 27, quickselect 3 |
| matched-norm random exchange, blocks 12-23 | 30 | 0.13 [0.04, 0.31] | 4/30 [0.04, 0.31] | 0.13 | 0.13 | 1.00 | 0.20 | 0.00 | 4:0 (0.125, 0.375) | full sort 26, quickselect 4 |
| placebo twin | 30 | 0.03 [8.4e-04, 0.17] | 1/30 [8.4e-04, 0.17] | 0.03 | 0.03 | 1.00 | 0.05 | 0.00 | 1:0 (1.000, 1.000) | full sort 29, quickselect 1 |

**Table 245.** Stage A on connected components (default union find; Y = BFS). Twin: Y share 0.00, off-default 0.00, pass 1.00, union find 30. Exchange at blocks 0-35 against the matched-norm random exchange: 11:0, exact p 9.8e-4 (Holm 0.0020). The Y share is the outcome of the first commitment (all attempted programs labelled Y; invalid rows count as not-Y); the added column *Correct Y / N* counts programs labelled Y that pass every hidden test, with Clopper-Pearson 95% intervals; a cell meets the validity gate of the first commitment when its pass ratio is at least 0.8. Post hoc, on correct Y, the exchange at blocks 0-35 against the matched-norm random exchange gives 11:0 (exact p 9.8e-04). (iteration-7 correction)

| Cell | n | Y share [CP 95%] | Correct Y / N [CP 95%] | Transfer | Off-default transfer | Pass ratio | Recovery | Slot interchange intervention accuracy | McNemar b:c (p, Holm) | Correct composition |
|---|---|---|---|---|---|---|---|---|---|---|
| full ban | 30 | 0.90 [0.73, 0.98] | 27/30 [0.73, 0.98] | 0.90 | 1.00 | 1.00 | 1.00 | 1.00 | 27:0 (1.5e-08, 3.0e-08) | BFS 27, DFS 3 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| exchange, blocks 0-11 | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.87 | 0.70 | 0.00 | 0.10 | 0:0 (1.000, 1.000) | DFS 17, union find 4 |
| exchange, blocks 12-23 | 30 | 0.47 [0.28, 0.66] | 14/30 [0.28, 0.66] | 0.47 | 1.00 | 1.00 | 0.52 | 0.43 | 14:0 (1.2e-04, 4.9e-04) | DFS 16, BFS 14 |
| exchange, blocks 24-35 | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.63 | 1.00 | 0.00 | 0.10 | 0:0 (1.000, 1.000) | DFS 19, union find 11 |
| exchange, blocks 0-35 | 30 | 0.37 [0.20, 0.56] | 11/30 [0.20, 0.56] | 0.37 | 1.00 | 1.00 | 0.41 | 0.40 | 11:0 (9.8e-04, 0.002) | DFS 19, BFS 11 |
| condition-span splice | 30 | 0.10 [0.02, 0.27] | 3/30 [0.02, 0.27] | 0.10 | 1.00 | 1.00 | 0.11 | 0.20 | 3:0 (0.250, 0.500) | DFS 27, BFS 3 |
| read-position exchange | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.00 | 1.00 | 0.00 | 0.00 | 0:0 (1.000, 1.000) | union find 30 |
| matched-norm random exchange, blocks 0-35 | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.07 | 1.00 | 0.00 | 0.00 | 0:0 (1.000, 1.000) | union find 28, DFS 2 |
| matched-norm random exchange, blocks 12-23 | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.03 | 1.00 | 0.00 | 0.00 | 0:0 (1.000, 1.000) | union find 29, DFS 1 |
| placebo twin | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.03 | 1.00 | 0.00 | 0.00 | 0:0 (1.000, 1.000) | union find 29, DFS 1 |

**Table 246.** Stage A on shortest path (default Dijkstra; Y = Bellman-Ford/SPFA; prompt route). Twin: Y share 0.00, off-default 0.00, pass 1.00, Dijkstra 30. Exchange at blocks 0-35 against the matched-norm random exchange: 23:0, exact p 2.4e-7 (Holm 7.2e-7). The Y share is the outcome of the first commitment (all attempted programs labelled Y; invalid rows count as not-Y); the added column *Correct Y / N* counts programs labelled Y that pass every hidden test, with Clopper-Pearson 95% intervals; a cell meets the validity gate of the first commitment when its pass ratio is at least 0.8. Post hoc, on correct Y, the exchange at blocks 0-35 against the matched-norm random exchange gives 21:0 (exact p 9.5e-07). (iteration-7 correction)

| Cell | n | Y share [CP 95%] | Correct Y / N [CP 95%] | Transfer | Off-default transfer | Pass ratio | Recovery | Slot interchange intervention accuracy | McNemar b:c (p, Holm) | Correct composition |
|---|---|---|---|---|---|---|---|---|---|---|
| full ban | 30 | 0.97 [0.83, 1.00] | 29/30 [0.83, 1.00] | 0.97 | 0.97 | 1.00 | 1.00 | 1.00 | 29:0 (3.7e-09, 1.5e-08) | Bellman-Ford/SPFA 29, Dijkstra 1 |
| exchange, blocks 0-11 | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.00 | 1.00 | 0.00 | 0.03 | 0:0 (1.000, 1.000) | Dijkstra 30 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| exchange, blocks 12-23 | 30 | 0.60 [0.41, 0.77] | 11/30 [0.20, 0.56] | 0.60 | 0.60 | 0.73 | 0.62 | 0.63 | 18:0 (7.6e-06, 4.6e-05) | Dijkstra 11, Bellman-Ford/SPFA 11 |
| exchange, blocks 24-35 | 30 | 0.37 [0.20, 0.56] | 11/30 [0.20, 0.56] | 0.37 | 0.37 | 1.00 | 0.38 | 0.40 | 11:0 (9.8e-04, 0.006) | Dijkstra 19, Bellman-Ford/SPFA 11 |
| exchange, blocks 0-35 | 30 | 0.80 [0.61, 0.92] | 21/30 [0.51, 0.85] | 0.80 | 0.80 | 0.90 | 0.83 | 0.83 | 24:0 (1.2e-07, 7.2e-07) | Bellman-Ford/SPFA 21, Dijkstra 6 |
| condition-span splice | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.00 | 1.00 | 0.00 | 0.03 | 0:0 (1.000, 1.000) | Dijkstra 30 |
| read-position exchange | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.00 | 1.00 | 0.00 | 0.03 | 0:0 (1.000, 1.000) | Dijkstra 30 |
| matched-norm random exchange, blocks 0-35 | 30 | 0.03 [8.4e-04, 0.17] | 0/30 [0.00, 0.12] | 0.03 | 0.03 | 0.97 | 0.03 | 0.07 | 1:0 (1.000, 1.000) | Dijkstra 29 |
| matched-norm random exchange, blocks 12-23 | 30 | 0.03 [8.4e-04, 0.17] | 0/30 [0.00, 0.12] | 0.03 | 0.03 | 0.93 | 0.03 | 0.07 | 1:0 (1.000, 1.000) | Dijkstra 28 |
| placebo twin | 30 | 0.03 [8.4e-04, 0.17] | 1/30 [8.4e-04, 0.17] | 0.03 | 0.03 | 1.00 | 0.03 | 0.07 | 1:0 (1.000, 1.000) | Dijkstra 29, Bellman-Ford/SPFA 1 |

**Table 247.** Stage A on two sum (default hash map; Y = brute force; direct route). Twin: Y share 0.00, pass 1.00, hash map 30. The Y share is the outcome of the first commitment (all attempted programs labelled Y; invalid rows count as not-Y); the added column *Correct Y / N* counts programs labelled Y that pass every hidden test, with Clopper-Pearson 95% intervals; a cell meets the validity gate of the first commitment when its pass ratio is at least 0.8. (iteration-7 correction)

| Cell | n | Y share [CP 95%] | Correct Y / N [CP 95%] | Transfer | Off-default transfer | Pass ratio | Recovery | Slot interchange intervention accuracy | McNemar b:c (p, Holm) | Correct composition |
|---|---|---|---|---|---|---|---|---|---|---|
| full ban | 30 | 1.00 [0.88, 1.00] | 30/30 [0.88, 1.00] | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 30:0 (1.9e-09, 1.1e-08) | brute force 30 |
| exchange, blocks 0-11 | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.00 | 1.00 | 0.00 | 0.00 | 0:0 (1.000, 1.000) | hash map 30 |
| exchange, blocks 12-23 | 30 | 0.43 [0.25, 0.63] | 13/30 [0.25, 0.63] | 0.43 | 0.43 | 1.00 | 0.43 | 0.43 | 13:0 (2.4e-04, 7.3e-04) | hash map 17, brute force 13 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| exchange, blocks 24-35 | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.00 | 1.00 | 0.00 | 0.00 | 0:0 (1.000, 1.000) | hash map 30 |
| exchange, blocks 0-35 | 30 | 0.43 [0.25, 0.63] | 13/30 [0.25, 0.63] | 0.43 | 0.43 | 1.00 | 0.43 | 0.43 | 13:0 (2.4e-04, 7.3e-04) | hash map 17, brute force 13 |
| condition-span splice | 30 | 1.00 [0.88, 1.00] | 30/30 [0.88, 1.00] | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 30:0 (1.9e-09, 1.1e-08) | brute force 30 |
| read-position exchange | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.00 | 1.00 | 0.00 | 0.00 | 0:0 (1.000, 1.000) | hash map 30 |

**Table 248.** Stage A on find duplicate (default hash set; Y = index marking; direct route). Twin: Y share 0.00, pass 1.00, hash set 30. The Y share is the outcome of the first commitment (all attempted programs labelled Y; invalid rows count as not-Y); the added column *Correct Y / N* counts programs labelled Y that pass every hidden test, with Clopper-Pearson 95% intervals; a cell meets the validity gate of the first commitment when its pass ratio is at least 0.8. (iteration-7 correction)

| Cell | n | Y share [CP 95%] | Correct Y / N [CP 95%] | Transfer | Off-default transfer | Pass ratio | Recovery | Slot interchange intervention accuracy | McNemar b:c (p, Holm) | Correct composition |
|---|---|---|---|---|---|---|---|---|---|---|
| full ban | 30 | 0.97 [0.83, 1.00] | 25/30 [0.65, 0.94] | 0.97 | 1.00 | 0.87 | 1.00 | 1.00 | 29:0 (3.7e-09, 1.5e-08) | index marking 25, Floyd cycle detection 1 |
| exchange, blocks 0-11 | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.00 | 1.00 | 0.00 | 0.00 | 0:0 (1.000, 1.000) | hash set 30 |
| exchange, blocks 12-23 | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.10 | 1.00 | 0.00 | 0.03 | 0:0 (1.000, 1.000) | hash set 27, Floyd cycle detection 3 |
| exchange, blocks 24-35 | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.00 | 1.00 | 0.00 | 0.00 | 0:0 (1.000, 1.000) | hash set 30 |
| exchange, blocks 0-35 | 30 | 0.03 [8.4e-04, 0.17] | 1/30 [8.4e-04, 0.17] | 0.03 | 0.13 | 1.00 | 0.03 | 0.07 | 1:0 (1.000, 1.000) | hash set 26, Floyd cycle detection 2, index marking 1, sort and adjacent check 1 |
| condition-span splice | 30 | 0.90 [0.73, 0.98] | 26/30 [0.69, 0.96] | 0.90 | 0.90 | 0.97 | 0.93 | 0.90 | 27:0 (1.5e-08, 6.0e-08) | index marking 26, hash set 3 |
| read-position exchange | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.00 | 1.00 | 0.00 | 0.00 | 0:0 (1.000, 1.000) | hash set 30 |

**Table 249.** Stage A on the grid task (default BFS; Y = DFS; prompt route in likelihood, mixed in

behaviour). Twin: Y share 0.00, pass 0.90, BFS 27 correct. Pass ratios above 1 occur because the twin's own pass rate is below 1. The Y share is the outcome of the first commitment (all attempted programs labelled Y; invalid rows count as not-Y); the added column *Correct Y / N* counts programs labelled Y that pass every hidden test, with Clopper-Pearson 95% intervals; a cell meets the validity gate of the first commitment when its pass ratio is at least 0.8. (iteration-7 correction)

| Cell | n | Y share [CP 95%] | Correct Y / N [CP 95%] | Transfer | Off-default transfer | Pass ratio | Recovery | Slot interchange intervention accuracy | McNemar b:c (p, Holm) | Correct composition |
|---|---|---|---|---|---|---|---|---|---|---|
| full ban | 30 | 1.00 [0.88, 1.00] | 12/30 [0.23, 0.59] | 1.00 | 1.00 | 0.44 | 1.00 | 1.00 | 30:0 (1.9e-09, 1.1e-08) | DFS 12 |
| exchange, blocks 0-11 | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.00 | 1.11 | 0.00 | 0.00 | 0:0 (1.000, 1.000) | BFS 30 |
| exchange, blocks 12-23 | 30 | 0.37 [0.20, 0.56] | 2/30 [8.2e-03, 0.22] | 0.37 | 0.37 | 0.63 | 0.37 | 0.37 | 11:0 (9.8e-04, 0.002) | BFS 15, DFS 2 |
| exchange, blocks 24-35 | 30 | 0.03 [8.4e-04, 0.17] | 0/30 [0.00, 0.12] | 0.03 | 0.03 | 0.81 | 0.03 | 0.03 | 1:0 (1.000, 1.000) | BFS 22 |
| exchange, blocks 0-35 | 30 | 0.47 [0.28, 0.66] | 3/30 [0.02, 0.27] | 0.47 | 0.47 | 0.59 | 0.47 | 0.47 | 14:0 (1.2e-04, 4.9e-04) | BFS 13, DFS 3 |
| condition-span splice | 30 | 0.33 [0.17, 0.53] | 7/30 [0.10, 0.42] | 0.33 | 0.33 | 0.89 | 0.33 | 0.33 | 10:0 (0.002, 0.006) | BFS 17, DFS 7 |
| read-position exchange | 30 | 0.00 [0.00, 0.12] | 0/30 [0.00, 0.12] | 0.00 | 0.00 | 0.96 | 0.00 | 0.00 | 0:0 (1.000, 1.000) | BFS 26 |

The condition-span splice also matches the behavioural route arms of iteration 3: its recovery is 1.00 on two sum and 0.93 on find duplicate, where the direct-route fraction of iteration 3 was 0.96 and 0.99, and 1.40 on kth smallest (direct-route fraction 0.74). On connected components its recovery toward BFS is 0.11, although every slot leaves union find: either route alone removes the default and lands on DFS, as the route landings of iteration 4 had shown.

**Positive instructions are carried the same way.** As an extension, the forced prompt `Implement Bellman-Ford or SPFA for this task.` replaced the ban as the source (Table 250). Its padding twin, the same prompt with the tokens of the instruction replaced by padding, stays on Dijkstra in 29 of 30 programs; exchanging only the task-specification states of the forced prompt at all blocks into that twin gives Bellman-Ford/SPFA in 30 of 30. On connected components the states of the forced BFS prompt give BFS in 28 of 30 against union find 30 of 30 for its twin. The shortest path semantic seed, which moved that task 66 points in iteration 1, moved nothing under this frame (Dijkstra 30 of 30), and neither did its exchange (2 of 30). The fourth ban template, held out of every other cell, transfers through the exchange as the other templates do (shortest path 0.70, kth smallest 0.60 on quickselect with every slot off the full sort, connected components 0.70, N = 10 each).

**Table 250.** Extensions: forced sources, the semantic seed and the fourth template, each exchanged at blocks 0-35 into its own twin. Source: exchange and site experiment `results/summary.md`. The

added column counts programs labelled Y that pass every hidden test. Under the first commitment's definition the forced Bellman-Ford/SPFA exchange transfers 0.967 (30 of 30 against 1 of 30 for its padding twin). (iteration-7 correction)

| Task | Cell | n | Y share (Y) / default share | Correct Y / n | Off-default share | Pass | Composition |
|---|---|---|---|---|---|---|---|
| kth smallest | fourth-template ban | 10 | 0.30 (quickselect) / 0.00 | 3/10 | 1.00 | 1.00 | size-k heap 7, quickselect 3 |
| kth smallest | fourth-template twin | 10 | 0.00 / 1.00 | 0/10 | 0.00 | 1.00 | full sort 10 |
| kth smallest | fourth-template exchange, blocks 0-35 | 10 | 0.60 / 0.00 | 4/10 | 1.00 | 0.80 | quickselect 6, size-k heap 4 |
| connected components | fourth-template ban | 10 | 0.80 (BFS) / 0.00 | 8/10 | 1.00 | 1.00 | BFS 8, DFS 2 |
| connected components | forced BFS | 30 | 1.00 / 0.00 | 30/30 | 1.00 | 1.00 | BFS 30 |
| connected components | padding twin of forced BFS | 30 | 0.00 / 1.00 | 0/30 | 0.00 | 1.00 | union find 30 |
| connected components | forced BFS exchange, blocks 0-35 | 30 | 0.93 / 0.00 | 24/30 | 1.00 | 0.87 | BFS 28, DFS 2 |
| connected components | fourth-template twin | 10 | 0.00 / 1.00 | 0/10 | 0.00 | 1.00 | union find 10 |
| connected components | fourth-template exchange, blocks 0-35 | 10 | 0.70 / 0.00 | 7/10 | 1.00 | 1.00 | BFS 7, DFS 3 |
| shortest path | fourth-template ban | 10 | 0.70 (Bellman-Ford/SPFA) / 0.30 | 7/10 | 0.70 | 1.00 | Bellman-Ford/SPFA 7, Dijkstra 3 |
| shortest path | forced Bellman-Ford or SPFA | 30 | 1.00 / 0.00 | 27/30 | 1.00 | 0.90 | Bellman-Ford/SPFA 30 |
| shortest path | padding twin of the forced prompt | 30 | 0.03 / 0.97 | 0/30 | 0.03 | 0.97 | Dijkstra 29, Bellman-Ford/SPFA 1 |
| shortest path | forced exchange, blocks 0-35 | 30 | 1.00 / 0.00 | 30/30 | 1.00 | 1.00 | Bellman-Ford/SPFA 30 |
| shortest path | semantic seed | 30 | 0.00 / 1.00 | 0/30 | 0.00 | 0.93 | Dijkstra 30 |
| shortest path | padding twin of the seed | 30 | 0.03 / 0.97 | 0/30 | 0.03 | 0.97 | Dijkstra 29, Bellman-Ford/SPFA 1 |
| shortest path | seed exchange, blocks 0-35 | 30 | 0.07 / 0.93 | 1/30 | 0.07 | 0.87 | Dijkstra 28, Bellman-Ford/SPFA 2 |
| shortest path | fourth-template twin | 10 | 0.00 / 1.00 | 0/10 | 0.00 | 1.00 | Dijkstra 10 |
| shortest path | fourth-template exchange, blocks 0-35 | 10 | 0.70 / 0.30 | 7/10 | 0.70 | 1.00 | Bellman-Ford/SPFA 7, Dijkstra 3 |

The unit tests of Stage A passed except one: splicing the cache of the task-specification span

reproduced the top token of the exchange with a logit gap of 0.5 in bf16 and 10 of 10 equal labels, but not byte for byte. The Stage A arms on trapped rain water and minimum spanning tree weight were dropped for GPU time.

### 6.4.2 Where noise acts, in bf16

The site rerun repeated iteration 4's grid on kth smallest and connected components in bf16, at the ban-free unit, with a fresh direction per sample, N = 30 per cell and alpha 1, 2, 4 and 8 (Tables 251 and 252). Noise at the task-specification positions moves the algorithm in bf16: on kth smallest blocks 12 to 23 reach a correct off-default share of 0.37 at alpha 2, and on connected components blocks 0 to 11 reach 0.33 at alpha 4. The read position, the condition span and the late blocks barely move it below alpha 8. On kth smallest the plan site (the task-specification states at blocks 12 to 23, as in iteration 4) beats the read position in paired slots 24:5 (exact p 5.5e-4) and the condition span 22:0, and a Mantel-Haenszel odds ratio [129] within quintiles of the downstream displacement is 4.6 [1.9, 11.3]. Connected components shows a weaker picture: its plan site moves only 0.07 at alpha 2 and 0.17 at alpha 4, the early blocks beat it (4:18, p 0.004), and the read position starts moving at alpha 8 (0.30).

The committed rule fitted a Firth logistic regression [125] of a correct off-default program on site indicators and the log displacements, so that sites are compared at matched downstream displacement (Tables 253 and 254). It returned the verdict withdrawn, for two reasons. The raw condition-span arm never reaches the plan site's downstream displacement, so it has no common support and its energy-matched arm stood in; and at alpha 8 every plan-site program failed (pass 0.00) while the read position began to move, which erases the read-position contrast in the preregistered model. A post-hoc sensitivity restricted to alpha of 4 or less (before breakage) puts the read position clearly below the plan site on both tasks: beta -3.76 (profile interval [-7.29, -0.47], cluster bootstrap [-8.18, -1.40]) on kth smallest and -4.98 ([-9.77, -0.89]; [-11.48, -1.53]) on connected components. The prefill-site claim is therefore withdrawn as preregistered. Post hoc (alpha of at most 4) the read position is below the task-specification site on both tasks; the condition-span clause fails post hoc as well (beta -1.70 [-3.96, 0.42] on kth smallest; -0.56 [-4.93, 2.91] on connected components, cluster bootstrap; profile intervals [-6.82, 1.16] and [-6.15, 3.41]), so the site of the choice remains open, as the artifact concludes; its rule label withdraws the site claim as specific to 8-bit floating point and states that the site of the choice is open. The half of that label about 8-bit floating point cannot apply: Section 6.3 shows that bf16 reproduces the escapes direction by direction (Spearman 0.96 and 0.87) and that activation rounding adds nothing, so the withdrawal comes from the missing common support and the breakage at alpha 8 in bf16, not from the engine. Noise sites and Stage A agree on the band on kth smallest and disagree on connected components (Table 254b): there noise moves the algorithm most at blocks 0 to 11 (mean 0.19 over alpha 1 to 4 against 0.08 at blocks 12 to 23; 0.33 correct off-default at alpha 4; coefficient +2.51 [1.39, 3.80] against the plan site; paired flips 4:18, p 0.004, between the plan site and blocks 0-11), while the exchange transfers BFS at blocks 12 to 23. The removal of the default there is depth-redundant: the exchange at blocks 0-11 alone takes 0.87 of attempts off union find and at blocks 24-35 alone 0.63. (iteration-7 correction)

**Table 251.** The bf16 site grid on kth smallest (twin correct off-default 0.00; displacements are medians in residual units at block 35). Source: exchange and site experiment `results/site_grid.json`.

| Site | Alpha | n | Correct off-default | Off-default | Pass | Median displacement at the read position, block 35 | Median displacement at generated token 1, block 35 | Correct composition |
|---|---|---|---|---|---|---|---|---|
| spec, blocks 12-23 | 1 | 30 | 0.13 | 0.13 | 1.00 | 1.708 | 1.315 | full sort 26, quickselect 3, size-k heap 1 |
| spec, blocks 12-23 | 2 | 30 | 0.37 | 0.37 | 1.00 | 3.353 | 2.478 | full sort 19, quickselect 8, size-k heap 3 |
| spec, blocks 12-23 | 4 | 30 | 0.33 | 0.40 | 0.87 | 6.293 | 4.628 | full sort 16, size-k heap 5, quickselect 5 |
| spec, blocks 12-23 | 8 | 30 | 0.00 | 0.10 | 0.00 | 12.230 | 7.135 | - |
| read position | 1 | 30 | 0.03 | 0.03 | 1.00 | 1.876 | 0.628 | full sort 29, quickselect 1 |
| read position | 2 | 30 | 0.03 | 0.03 | 1.00 | 3.591 | 1.201 | full sort 29, quickselect 1 |
| read position | 4 | 30 | 0.00 | 0.00 | 1.00 | 6.840 | 2.290 | full sort 30 |
| read position | 8 | 30 | 0.13 | 0.13 | 0.93 | 10.654 | 3.466 | full sort 24, quickselect 3, size-k heap 1 |
| read position, energy-matched | 2 | 30 | 0.07 | 0.07 | 0.77 | 16.426 | 3.870 | full sort 21, quickselect 1, size-k heap 1 |
| read position, energy-matched | 4 | 30 | 0.03 | 0.07 | 0.93 | 25.287 | 3.845 | full sort 27, heapify and pop 1 |
| condition span | 1 | 30 | 0.00 | 0.00 | 1.00 | 0.303 | 0.245 | full sort 30 |
| condition span | 2 | 30 | 0.07 | 0.07 | 1.00 | 0.415 | 0.289 | full sort 28, quickselect 2 |
| condition span | 4 | 30 | 0.03 | 0.03 | 1.00 | 0.674 | 0.424 | full sort 29, quickselect 1 |
| condition span | 8 | 30 | 0.00 | 0.00 | 1.00 | 1.056 | 0.606 | full sort 30 |
| condition span, energy-matched | 2 | 30 | 0.00 | 0.00 | 1.00 | 1.220 | 0.664 | full sort 30 |
| condition span, energy-matched | 4 | 30 | 0.00 | 0.00 | 1.00 | 1.110 | 0.646 | full sort 30 |
| generated tokens 1-32 | 1 | 30 | 0.17 | 0.17 | 1.00 | - | - | full sort 25, quickselect 5 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| generated tokens 1-32 | 2 | 50 | 0.26 | 0.34 | 0.94 | - | - | full sort 33, quickselect 12, size-k heap 1, other 1 |
| generated tokens 1-32 | 4 | 50 | 0.18 | 0.42 | 0.50 | - | - | full sort 15, quickselect 4, heapify and pop 3, size-k heap 2 |
| generated tokens 1-32 | 8 | 30 | 0.00 | 0.00 | 0.13 | - | - | full sort 4 |
| spec, blocks 0-11 | 1 | 30 | 0.03 | 0.03 | 1.00 | 0.956 | 0.728 | full sort 29, quickselect 1 |
| spec, blocks 0-11 | 2 | 30 | 0.17 | 0.17 | 1.00 | 1.832 | 1.320 | full sort 25, quickselect 3, size-k heap 2 |
| spec, blocks 0-11 | 4 | 30 | 0.37 | 0.37 | 1.00 | 3.576 | 2.512 | full sort 19, size-k heap 9, quickselect 2 |
| spec, blocks 0-11 | 8 | 30 | 0.00 | 0.13 | 0.10 | 16.027 | 8.332 | full sort 3 |
| spec, blocks 24-35 | 1 | 30 | 0.00 | 0.00 | 1.00 | 0.909 | 0.654 | full sort 30 |
| spec, blocks 24-35 | 2 | 30 | 0.03 | 0.03 | 1.00 | 1.777 | 1.261 | full sort 29, size-k heap 1 |
| spec, blocks 24-35 | 4 | 30 | 0.03 | 0.03 | 1.00 | 3.270 | 2.307 | full sort 29, quickselect 1 |
| spec, blocks 24-35 | 8 | 30 | 0.10 | 0.10 | 1.00 | 5.119 | 3.323 | full sort 27, quickselect 3 |

**Table 252.** The bf16 site grid on connected components (twin correct off-default 0.00).

| Site | Alpha | n | Correct off-default | Off-default | Pass | Median displacement at the read position, block 35 | Median displacement at generated token 1, block 35 | Correct composition |
|---|---|---|---|---|---|---|---|---|
| spec, blocks 12-23 | 1 | 30 | 0.00 | 0.00 | 1.00 | 1.694 | 1.315 | union find 30 |
| spec, blocks 12-23 | 2 | 30 | 0.07 | 0.07 | 1.00 | 3.281 | 2.537 | union find 28, DFS 2 |
| spec, blocks 12-23 | 4 | 30 | 0.17 | 0.17 | 1.00 | 6.322 | 4.829 | union find 25, DFS 3, BFS 2 |
| spec, blocks 12-23 | 8 | 30 | 0.00 | 0.13 | 0.00 | 12.977 | 7.182 | - |
| read position | 1 | 30 | 0.00 | 0.00 | 1.00 | 1.800 | 0.606 | union find 30 |
| read position | 2 | 30 | 0.00 | 0.00 | 1.00 | 3.700 | 1.197 | union find 30 |
| read position | 4 | 30 | 0.03 | 0.03 | 1.00 | 7.200 | 2.234 | union find 29, DFS 1 |
| read position | 8 | 30 | 0.30 | 0.30 | 0.93 | 10.392 | 3.645 | union find 19, DFS 8, BFS 1 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| read position, energy-matched | 2 | 30 | 0.03 | 0.03 | 0.77 | 15.570 | 4.091 | union find 22, DFS 1 |
| read position, energy-matched | 4 | 30 | 0.07 | 0.07 | 0.83 | 22.781 | 4.019 | union find 23, DFS 2 |
| condition span | 1 | 30 | 0.00 | 0.00 | 1.00 | 0.285 | 0.238 | union find 30 |
| condition span | 2 | 30 | 0.00 | 0.00 | 1.00 | 0.395 | 0.278 | union find 30 |
| condition span | 4 | 30 | 0.00 | 0.00 | 1.00 | 0.620 | 0.371 | union find 30 |
| condition span | 8 | 30 | 0.00 | 0.00 | 1.00 | 0.931 | 0.557 | union find 30 |
| condition span, energy-matched | 2 | 30 | 0.00 | 0.00 | 1.00 | 1.003 | 0.568 | union find 30 |
| condition span, energy-matched | 4 | 30 | 0.00 | 0.00 | 1.00 | 1.163 | 0.618 | union find 30 |
| generated tokens 1-32 | 1 | 30 | 0.00 | 0.00 | 1.00 | - | - | union find 30 |
| generated tokens 1-32 | 2 | 50 | 0.14 | 0.20 | 0.92 | - | - | union find 39, DFS 7 |
| generated tokens 1-32 | 4 | 50 | 0.16 | 0.40 | 0.38 | - | - | union find 11, DFS 6, BFS 2 |
| generated tokens 1-32 | 8 | 30 | 0.00 | 0.00 | 0.17 | - | - | union find 5 |
| spec, blocks 0-11 | 1 | 30 | 0.03 | 0.03 | 1.00 | 0.910 | 0.747 | union find 29, DFS 1 |
| spec, blocks 0-11 | 2 | 30 | 0.20 | 0.20 | 1.00 | 1.717 | 1.308 | union find 24, DFS 6 |
| spec, blocks 0-11 | 4 | 30 | 0.33 | 0.37 | 0.97 | 3.315 | 2.467 | union find 19, DFS 6, BFS 4 |
| spec, blocks 0-11 | 8 | 30 | 0.13 | 0.33 | 0.13 | 16.291 | 9.122 | BFS 3, DFS 1 |
| spec, blocks 24-35 | 1 | 30 | 0.00 | 0.00 | 1.00 | 0.930 | 0.647 | union find 30 |
| spec, blocks 24-35 | 2 | 30 | 0.00 | 0.00 | 1.00 | 1.789 | 1.225 | union find 30 |
| spec, blocks 24-35 | 4 | 30 | 0.00 | 0.00 | 1.00 | 3.267 | 2.278 | union find 30 |
| spec, blocks 24-35 | 8 | 30 | 0.03 | 0.03 | 1.00 | 5.089 | 3.429 | union find 29, DFS 1 |

**Table 253.** Firth logistic regression with site indicators on kth smallest at matched downstream displacement (n 387, 48 events, 387 of 720 rows in common support; the condition span enters through its energy-matched arm). Mantel-Haenszel odds ratio, plan site against read position, within quintiles of the log displacement at the read position: 4.61 [1.87, 11.34]. Paired flips, plan site only : other site only: read position 24:5 (p 5.5e-4), condition span 22:0 (p 4.8e-7), blocks 0-11

17:9 (p 0.169), blocks 24-35 23:3 (p 8.8e-5), generated tokens 21:18 (p 0.749).

| Term | Beta | Profile 95% CI | Cluster bootstrap 95% CI |
|---|---|---|---|
| spec, blocks 0-11 (against spec, blocks 12-23) | 0.24 | [-0.54, 1.01] | - |
| spec, blocks 24-35 | -1.52 | [-2.74, -0.51] | - |
| read position | -0.17 | [-2.64, 2.33] | [-2.58, 2.21] |
| condition span (energy-matched arm) | -0.73 | [-5.71, 1.84] | [-2.12, 1.10] |
| log displacement at the read position | -1.92 | [-4.76, 0.77] | - |
| log displacement at generated token 1 | 1.97 | [-0.99, 5.03] | - |

**Table 254.** Firth logistic regression with site indicators on connected components (n 374, 32 events, 376 of 720 rows in common support). Mantel-Haenszel odds ratio, plan site against read position: 0.82 [0.31, 2.18]. Paired flips, plan site only : other site only: read position 7:10 (p 0.629), condition span 7:0 (p 0.016), blocks 0-11 4:18 (p 0.004), blocks 24-35 7:1 (p 0.070), generated tokens 7:9 (p 0.804).

| Term | Beta | Profile 95% CI | Cluster bootstrap 95% CI |
|---|---|---|---|
| spec, blocks 0-11 (against spec, blocks 12-23) | 2.51 | [1.39, 3.80] | - |
| spec, blocks 24-35 | -0.72 | [-3.03, 0.94] | - |
| read position | 1.14 | [-1.30, 3.64] | [-1.07, 3.24] |
| condition span (energy-matched arm) | 2.38 | [-2.72, 5.40] | [0.69, 5.18] |
| log displacement at the read position | 0.02 | [-2.94, 2.92] | - |
| log displacement at generated token 1 | 1.22 | [-1.87, 4.36] | - |

**Table 254b.** Noise sites against Stage A, per task and block band (NEW in iteration 7): the largest correct off-default share of spec-position noise at alpha 1 to 4 (with its alpha), the full-state exchange's Y share (the first commitment) and off-default share at the same band, and whether the band where noise acts most (by the mean over alpha 1, 2 and 4, the artifact's band rule) is the band where the exchange transfers most. Sources: exchange and site experiment `results/site_grid.json`, `data/labels.jsonl`. (iteration-7 correction)

| Task | Band | Noise, mean correct off-default over alpha 1-4 | Noise, max correct off-default (alpha) | Exchange Y share | Exchange off-default share | Noise band / exchange band | Concordant |
|---|---|---|---|---|---|---|---|
| kth smallest | blocks 0-11 | 0.19 | 0.37 (4) | 0.40 | 0.53 | 12-23 / 12-23 | yes |
| | blocks 12-23 | 0.28 | 0.37 (2) | 0.57 | 1.00 | | |
| | blocks 24-35 | 0.02 | 0.03 (4) | 0.03 | 0.03 | | |
| connected components | blocks 0-11 | 0.19 | 0.33 (4) | 0.00 | 0.87 | 0-11 / 12-23 | no |
| | blocks 12-23 | 0.08 | 0.17 (4) | 0.47 | 1.00 | | |
| | blocks 24-35 | 0.00 | 0.00 (4) | 0.00 | 0.63 | | |

### 6.4.3 State noise at generated tokens against matched temperature

The token claim's second half set noise on the residual states of generated tokens 1 to 32 against a temperature applied to the same tokens and bisected to the same per-token KL, which we call

decoding at matched departure (Table 255). At alpha 2 the state noise leads on discordant correct off-default flips, 12:6 on kth smallest and 7:3 on connected components, but neither is significant (p 0.24 and 0.34). At alpha 4 the lead on connected components reaches 8:1 (p 0.039), at a pass rate of 0.38 against 0.70. The rule (p below 0.05 on both tasks at alpha 2) is not met: state noise against token temperature is not established in bf16.

**Table 255.** State noise at generated tokens 1 to 32 against decoding at matched departure (N = 50 paired slots). Source: exchange and site experiment `results/state_vs_token.json`.

| Task, alpha | State-noise KL per token (nats) | Matched temperature | Share of tokens with the temperature on | State only : temperature only | Exact p | State-noise flip share | Temperature flip share | State-noise pass | Temperature pass |
|---|---|---|---|---|---|---|---|---|---|
| kth smallest, alpha 2 | 0.3883 | 2.06 | 0.3762 | 12:6 | 0.238 | 0.26 | 0.14 | 0.94 | 0.98 |
| kth smallest, alpha 4 | 2.3336 | 3.00 | 1.7734 | 8:3 | 0.227 | 0.18 | 0.08 | 0.50 | 0.82 |
| connected components, alpha 2 | 0.2315 | 2.72 | 0.2310 | 7:3 | 0.344 | 0.14 | 0.06 | 0.92 | 0.92 |
| connected components, alpha 4 | 3.0926 | 3.00 | 0.8178 | 8:1 | 0.039 | 0.16 | 0.02 | 0.38 | 0.70 |

### 6.4.4 The top-five push in bf16

On the grid task, iteration 4's top-five direction (from the mean of the five banned centroids toward the mean of the survivors) was applied at the plan site at alpha 1, 2, 4 and 8 (Table 256). At doses of 1 to 8 ban units, which overshoot the forced-prompt manifold, the push breaks programs before it moves the algorithm; an in-manifold dose was not tried: the pass rate falls from 0.67 at alpha 1 to 0.30, 0.03 and 0.00, with at most one correct program other than BFS per cell, so the landing test could not be computed. Projected onto the plane of the first two principal components of the ten forced-prompt centroids (36% and 25% of their variance), the push leaves the forced-prompt manifold: already at alpha 1 it moves farther along the first component than the whole spread of the ten centroids (the four alpha-1 points move from a twin mean of +0.09 to -0.44 on that component, against centroids spread from -0.098 to +0.144). The prompt version of the top-five ban moves the task-specification states from below the centroid cloud into it, between the banned DFS and greedy best-first centroids and the three surviving centroids (bidirectional BFS and two variants of A*), and its behaviour stays inside the banned set (BFS 25, DFS 23 of 50; avoidance 0.00). (iteration-7 correction)

**Table 256.** The top-five plan push and the prompt top-five ban on the grid task. Source: exchange and site experiment `results/top5_push.json`.

| Cell | n | Pass | BFS share | Correct programs other than BFS | Landing test |
|---|---|---|---|---|---|
| twin | 50 | 0.92 | 1.00 | 0 | n/a |
| single BFS ban | 30 | 0.40 | 0.00 | 12 (DFS 12) | n/a |
| top-five push, alpha 1 | 30 | 0.67 | 0.97 | 1 (DFS, banned) | not computable |
| top-five push, alpha 2 | 30 | 0.30 | 1.00 | 0 | not computable |
| top-five push, alpha 4 | 30 | 0.03 | 0.97 | 0 | not computable |
| top-five push, alpha 8 | 30 | 0.00 | 0.07 | 0 | not computable |
| prompt top-five ban | 50 | 0.62 | 0.50 | 10 (DFS 10, banned) | not computable |

*[Figure fig_qwen_top5_geometry: generated in the report phase, not in this draft]*

*Figure note (iteration-7 correction): Actual activation coordinates: mean over the task-specification positions of each prompt (prefill, before any generated token); mean of the block outputs of blocks 12-23, divided by the median block norm; projection: (a, b) PCA fitted on the 10 forced-prompt centroids (first component 36%, second 25% of centroid variance); (c) classical MDS of geodesic distances on a $k = 6$ nearest-neighbour graph (MDS stress not recorded in `top5_push.json`; file name restored, iteration-8 correction). Points: 28 prompt states, one per template and condition: single-name ban 4, pushes of 1, 2, 4 and 8 ban units 4 each, top-five ban 4, twin 4. Centroids: 10 forced-prompt centroids (mean state of the forced prompts that ask for each algorithm), coloured by the prompt-defined top-five ban set (banned five against survivors). $N = 38$. Caveat: The plotted position precedes the fork and the generated code; none: centroids and colours are defined by prompts, not by the programs written.*

### 6.4.5 Who forecasts the landing

The landing contest scored, on the 21 recorded push-landing events, three forecasts: the base runner-up, the most frequent algorithm other than the default in the instruct twin itself (defined on 7 events), and a teacher-forced ranking of the openings of the catalogue algorithms by the instruct twin, the log-mean-exp likelihood of the first 48 tokens of each algorithm over its reference and forced programs, which we call the opening ranking (Table 257). The base runner-up named 17 of 21 landings and the opening ranking 10. On the 9 events where they disagree the base runner-up wins 8:1 (exact one-sided sign test p 0.020), and its mean log score is higher by 1.62 nats (Wilcoxon p 0.032). Where the twin wrote an alternative, its own minority does as well as the base runner-up (6 of 7; log-score difference 0.02, p 0.69). The comparison is retrospective, because the landings were known when the rule was frozen, and the opening ranking rests on 1 to 6 programs per algorithm. The base model adds information beyond the ranking of openings by the instruct model itself; it does not add information beyond the minority algorithm of the twin where one exists.

**Table 257.** The 21 recorded push-landing events: the landing and each forecast's top-1, with log scores (base runner-up / opening ranking / twin minority). Source: exchange and site experiment `results/landing_forecast.json`.

| Event | Landing | Base runner-up | Twin minority | Opening ranking | Log score base / opening / twin |
|---|---|---|---|---|---|
| Qwen3-8B 8-bit screen, kth smallest | quickselect | quickselect (hit) | n/a | quickselect (hit) | -1.36 / -1.67 / n/a |
| Qwen3-8B 8-bit screen, connected components | DFS | DFS (hit) | n/a | DFS (hit) | -0.84 / -0.78 / n/a |
| Qwen3-8B 8-bit screen, shortest path | Bellman-Ford/ SPFA | Bellman-Ford/ SPFA (hit) | n/a | Floyd-Warshall (miss) | -0.96 / -4.59 / n/a |
| Qwen3-8B 8-bit screen, two sum | brute force | brute force (hit) | n/a | brute force (hit) | -0.73 / -0.05 / n/a |
| Qwen3-8B 8-bit screen, grid | DFS | DFS (hit) | n/a | bidirectional BFS (miss) | -1.14 / -1.14 / n/a |
| Qwen3-8B bf16, kth smallest | size-k heap | quickselect (miss) | quickselect (miss) | quickselect (miss) | -1.50 / -2.64 / -1.50 |
| Qwen3-8B bf16, connected components | DFS | DFS (hit) | n/a | DFS (hit) | -0.73 / -0.05 / n/a |

| | | | | | |
|---|---|---|---|---|---|
| panel A, majority element | Boyer-Moore vote | Boyer-Moore vote (hit) | n/a | sort and take middle (miss) | -0.38 / -4.95 / n/a |
| panel A, trapped rain water | two pointers | two pointers (hit) | two pointers (hit) | monotonic stack (miss) | -0.33 / -4.38 / -0.14 |
| panel A, longest mirrored substring | brute force | brute force (hit) | n/a | brute force (hit) | -0.69 / -0.97 / n/a |
| panel B, single number | XOR fold | XOR fold (hit) | XOR fold (hit) | set-sum identity (miss) | -0.18 / -4.39 / -0.38 |
| panel B, minimum spanning tree weight | Prim with heap | Prim with heap (hit) | Prim with heap (hit) | Prim with heap (hit) | -0.44 / -0.05 / -0.15 |
| panel B, integer square root | linear scan | floating point sqrt (miss) | n/a | linear scan (hit) | -1.13 / -0.56 / n/a |
| panel B, count inversions | sorted insertion with bisect | sorted insertion with bisect (hit) | n/a | segment tree (miss) | -0.59 / -4.37 / n/a |
| Gemma, grid | BFS | BFS (hit) | BFS (hit) | bidirectional BFS (miss) | -0.42 / -4.53 / -0.23 |
| Gemma, kth smallest | size-k heap | quickselect (miss) | n/a | quickselect (miss) | -1.70 / -4.80 / n/a |
| Gemma, substring search | built-in find | built-in find (hit) | built-in find (hit) | built-in find (hit) | -0.13 / -0.05 / -0.50 |
| Gemma, two sum | brute force | brute force (hit) | n/a | brute force (hit) | -0.48 / -0.05 / n/a |
| Gemma, connected components | union find | BFS (miss) | n/a | BFS (miss) | -0.96 / -4.39 / n/a |
| Gemma, first and last position | binary search | binary search (hit) | binary search (hit) | exponential search (miss) | -0.12 / -4.38 / -0.38 |
| Gemma, longest increasing subsequence | quadratic DP | quadratic DP (hit) | n/a | quadratic DP (hit) | -0.06 / -0.04 / n/a |

Per set the base runner-up hit 5 of 5 on the 8-bit screen, 7 of 9 on the bf16 Qwen3-8B events and 5 of 7 in Gemma; the opening ranking hit 3 of 5, 4 of 9 and 3 of 7. Two correct programs labelled other appeared in the whole experiment, both catalogue variants. Judge spend was USD 0.055.

## 6.5 Gemma: a one-token fork, and Gemma Scope 2 features against difference in means

**Why Gemma.** Asked one and the same grid-path prompt, gemma-3-4b-it [76] writes DFS about 70% of the time and BFS about 30%, so it offers what Qwen3-8B never does: the request's BFS-against-DFS contrast inside one prompt, with no ban or instruction to confound it. Gemma Scope 2 provides sparse autoencoders on the residual stream after each block, trained on this instruct model, so the feature claim needs no transfer test. Qwen-3 transcoders for circuit tracing exist as well [123], but they would have to study a choice Qwen3-8B does not make naturally. The experiment asked where the choice forks, whether it can be read from the internals beyond what the visible text already says, and whether the SAE features steer it better than plain baselines: a difference-in-means direction, the raw residual, random features, isotropic directions, DAS and prompting. It used the residual-stream SAEs at layers 9, 17, 22 and 29 (16k latents, at the release's medium sparsity), one 16 GB card and exact common random numbers, under two hash-chained commitments.

**Natural splits.** Over 200 samples per task, the grid prompt gave DFS 139 and BFS 61 (Wilson

interval for DFS 0.63 to 0.75), with DFS on 134 of 172 correct programs (Table 258). The choice depends on the template: BFS takes 6 to 18 of 40 attempts per template. Substring search (naive 173, built-in find 27) and first and last position (linear scan 170, binary search 29) split as well. Labels agreed with iteration 5's Gemma rows on 0.94 of the grid slots and 1.00 of the substring slots.

**Table 258.** Natural splits in gemma-3-4b-it (N = 200 per task; 40 per template). Source: Gemma feature experiment `results/natural_split.json`.

| Task | Attempted | Correct | Pass | Correct-only Hill-1 | Default (share of correct) | Attempted per template (first to fifth) |
|---|---|---|---|---|---|---|
| grid | DFS 139, BFS 61 | DFS 134, BFS 38 | 0.86 | 1.68 | DFS (0.78) | DFS 34 / BFS 6; DFS 22 / BFS 18; DFS 22 / BFS 18; DFS 29 / BFS 11; DFS 32 / BFS 8 |
| substring search | naive 173, built-in find 27 | same | 1.00 | 1.47 | naive (0.87) | naive 36 / built-in 4; naive 40; built-in 21 / naive 19; naive 38 / built-in 2; naive 40 |
| first and last position | linear scan 170, binary search 29, other 1 | same | 1.00 | 1.52 | linear scan (0.85) | linear 33 / binary 7; linear 28 / binary 12; linear 39 / other 1; linear 40; linear 30 / binary 10 |

**The instrument gate.** Each SAE was spliced into the forward pass on three corpora: the model's own programs, generic text and LeetCode prompts. A layer passed if its increase in cross-entropy and its fraction of variance unexplained on the model's programs were at most 1.5 times those on generic text (Table 259). Layers 22 and 29 passed; layer 9 failed the cross-entropy ratio and layer 17 the variance ratio by 0.0001, so their results are descriptive. On generic text the mean number of active latents per token was 57 to 64, slightly above the quoted medium band of 30 to 60.

**Table 259.** Gemma Scope 2 residual-stream SAEs spliced in on three corpora (the model's own programs / generic text / LeetCode prompts): mean active latents per token, fraction of variance explained, increase in cross-entropy (nats per token) and loss recovered. Source: Gemma feature experiment `results/gate.json`.

| Layer | Active latents per token | Fraction of variance explained | Increase in cross-entropy | Loss recovered | Gate |
|---|---|---|---|---|---|
| 9 | 52.2 / 57.5 / 57.3 | 0.963 / 0.974 / 0.917 | 0.065 / 0.033 / -0.010 | 0.9968 / 0.9983 / 1.0005 | fail |
| 17 | 59.9 / 64.1 / 61.4 | 0.919 / 0.946 / 0.814 | 0.097 / 0.184 / 0.119 | 0.9968 / 0.9936 / 0.9957 | fail (variance ratio, by 0.0001) |
| 22 | 60.0 / 61.7 / 62.2 | 0.809 / 0.869 / 0.750 | 0.036 / 0.259 / 0.167 | 0.9968 / 0.9789 / 0.9837 | pass |
| 29 | 64.1 / 64.1 / 69.5 | 0.637 / 0.715 / 0.635 | 0.017 / 0.212 / 0.142 | 0.9994 / 0.9932 / 0.9952 | pass |

**The plan is a single-token fork.** Prefix resampling with change-point detection, the method of forking-path analyses [30, 31], located the fork: 24 base paths, 16 prefix lengths, 8 continuations each and a refinement, 3,712 continuations in all, scored by a partial-code labeller that agreed with the full labeller on 200 of 200 programs. Every one of 12 BFS base paths had a significant change point with a rise of at least 0.4, at prefix lengths 16 to 36. In 177 of 200 programs the first 32 tokens are identical (the code fence, `def solve(grid, start, goal):`, `rows = len(grid)`, `cols = len(grid[0])` and an indent), and the next token decides (Table 260): `visited` leads to DFS (BFS probability 0.011 over 1,409 continuations), while a blank line, `if`, `queue` or `q` lead to BFS

(0.95 to 1.00). The first token that identifies the algorithm in code comes a median of 138 tokens after the fork (10th to 90th percentile 80 to 181). The fork index was chosen by the token class with the most mutual information with the outcome (0.62 bits at index 32), after the first median rule landed on a deterministic indentation token; the change was made before the second commitment.

*Executed but not reported until iteration 7.* The empty-prefix unit test of the fork failed mildly: continuations from the empty prefix give BFS 0.38 (73 of 192) against the natural 0.305 [0.245, 0.372] (outside the interval; exact binomial p 0.028 against 0.305), so the resampled continuations are slightly richer in BFS than the natural samples. The deviation log records what did not run: the clamp read-back and matched-norm realisation tests as separate unit tests (the matched norm is imposed by construction in the clamp code, `src/clamp.py` (file name restored, iteration-8 correction), and the realised norm of each edit is logged per row); the clamp-cell overlay on the generated-position panel; the fork and the cross-prompt readouts of two further bench tasks (dropped in order); and per-sample displacement at generated tokens 1-32 (prefill displacement at blocks 11-33 is logged). (iteration-7 correction)

**Table 260.** The fork token at response index 32 and the BFS probability of the continuations that follow it (all continuations; the natural samples alone in the last column). Source: Gemma feature experiment `results/fork.json`.

| Token at index 32 | Continuations | BFS | DFS | BFS probability | Natural samples (BFS / n) |
|---|---|---|---|---|---|
| `visited` | 1,409 | 15 | 1,394 | 0.011 | 2 / 134 |
| newline | 204 | 193 | 11 | 0.946 | 19 / 21 |
| `if` | 177 | 174 | 2 | 0.989 | 16 / 17 |
| `queue` | 82 | 82 | 0 | 1.000 | 4 / 4 |
| `q` | 3 | 3 | 0 | 1.000 | 1 / 1 |

**Readable beyond the text, at the fork, but not specifically through the SAE.** Within one prompt, decodability before the first identifying token is at ceiling for every method, the visible-text control included (balanced accuracy 1.00 at all four layers), so the SAE has no advantage there; at the fork itself the test is undefined, because 185 of 200 samples share their prefix with a sample of the other class. Across prompts, the experiment built 120 neutral prompt variants by changing nouns and measured each variant's fork log-odds of BFS, which we write y (mean -0.92 nats, SD 0.67; the sampled BFS share tracks it at Spearman 0.65 over 60 variants of 10 samples). A Lasso on layer-29 SAE codes at the fork position predicts held-out y with R-squared 0.66, against 0.18 for a text-only model (difference 0.49, 95% interval 0.26 to 0.65, permutation p 0.003), above the difference-in-means readout (0.34) and a 16-component PCA of the raw residual (0.56) (Table 261). That meets the preregistered bar. Two checks outside the preregistration qualify it: a tuned ridge on the raw layer-29 residual reaches 0.72, and ridge on the same SAE codes reaches at most 0.09. Beside the Lasso, the standardised ridge on the same layer-29 SAE codes reaches R-squared 0.15 against 0.086 for a template-plus-noun-norm control (delta 0.064, below the preregistered 0.10 bar); the unstandardised SAE ridge ranges from -0.76 to 0.06 across penalties while the raw ridge stays at 0.70 to 0.72, which is why the SAE advantage depends on the estimator. Moreover, a linear readout of the fork log-odds from the residual at the fork position is close to a next-token readout, so the raw-residual 0.72 shows that the next-token distribution is linearly encoded there, not that a plan is; no logit-lens control of matched capacity was run. At the task-specification positions, before any response token, the SAE codes do no better than the text (0.18). (iteration-7 correction)

**Table 261.** Cross-prompt readout of the fork log-odds over 120 neutral variants: R-squared with

held-out nouns, Spearman with held-out nouns, and R-squared with held-out templates. Post hoc, the standardised ridge on the same layer-29 SAE codes reaches R-squared 0.15 against 0.086 for a template-plus-noun-norm control (delta 0.064, below the preregistered 0.10 bar). At the fork position this readout is close to a next-token readout (no logit-lens control was run). Source: Gemma feature experiment `results/readout.json`, `results/rederive_headlines.json`. (iteration-7 correction)

| Readout | Site | Layer | R-squared, held-out nouns | Spearman, held-out nouns | R-squared, held-out templates |
|---|---|---|---|---|---|
| text only | n/a | n/a | 0.176 | 0.425 | 0.150 |
| template only | n/a | n/a | 0.148 | 0.272 | -0.122 |
| SAE codes (Lasso) | fork | 22 | 0.237 | 0.583 | 0.139 |
| raw PCA-16 | fork | 22 | 0.412 | 0.666 | 0.491 |
| raw PCA-64 | fork | 22 | 0.092 | 0.533 | 0.239 |
| difference in means | fork | 22 | 0.226 | 0.541 | 0.109 |
| SAE codes (Lasso) | task specification | 22 | 0.180 | 0.464 | 0.123 |
| raw PCA-16 | task specification | 22 | -0.090 | 0.354 | 0.352 |
| raw PCA-64 | task specification | 22 | 0.021 | 0.432 | 0.152 |
| difference in means | task specification | 22 | 0.063 | 0.282 | -0.570 |
| SAE codes (Lasso) | fork | 29 | 0.662 | 0.782 | 0.496 |
| raw PCA-16 | fork | 29 | 0.557 | 0.697 | 0.444 |
| raw PCA-64 | fork | 29 | 0.432 | 0.715 | 0.403 |
| difference in means | fork | 29 | 0.338 | 0.620 | 0.181 |
| SAE codes (Lasso) | task specification | 29 | -0.073 | 0.276 | 0.147 |
| raw PCA-16 | task specification | 29 | -0.602 | 0.194 | -0.062 |
| raw PCA-64 | task specification | 29 | 0.029 | 0.381 | 0.083 |
| difference in means | task specification | 29 | 0.064 | 0.302 | -0.438 |

**Steering: the features move the fork, difference in means moves it more, and edits toward BFS cost correctness.** We call an *SAE clamp* setting the k latents that best separate forced-BFS from forced-DFS prompts (input-selected) to their BFS-class mean at the fork position; output-scored latents are chosen instead by their attribution to the fork's logit difference, and the two sets of 20 share 2 latents. At the fork position of layer 22, edits at matched norm moved y by $+4.04$ nats for the difference-in-means direction, $+1.78$ for the SAE clamp of the top 20 input-selected latents, $+0.51$ for 20 random latents and $+0.12$ for an isotropic direction. Table 261b gives the whole dose-response sweep over the 120 prompt variants (11,280 measurements), at the fork and at the task-specification positions. In generation ($N = 50$ paired slots), the twin wrote BFS in 13 programs; the SAE clamp raised this to 30 (top 20 latents) and 36 (top 50), the difference-in-means

edit at the same norm to 46 and 49, and random latents to 21 (Table 262). Edits toward BFS at the fork lowered the pass rate from 0.84 for the twin to 0.58 and 0.48 for the top-20 and top-50 SAE clamps and 0.32 and 0.16 for difference in means (0.60 to 0.78 for the other BFS-ward fork edits, 0.00 for the DAS interchange), except the SAE top-5 clamp (18 BFS of 50, pass 0.84), so the preregistered outcome, correct BFS programs, shows no significant effect anywhere after Holm correction, and no feature claim is made. The preregistered feature rule (SAE target-correct count at least the difference-in-means count, at a pass rate of at least 0.8 times the twin's) is also met by random latents at k = 50 (9 against 7, pass 0.76), so it cannot discriminate here. The informative comparisons are on attempted BFS, and they are descriptive: the SAE clamp beats the twin (17:0 and 23:0 discordant slots, p 1.5e-5 and 2.4e-7) and random latents (15:0, p 6.1e-5), and difference in means beats the SAE clamp at matched norm (16:0, p 3.1e-5). At matched effect on y rather than matched norm, difference in means and the SAE clamp shift the plan equally (32 against 30 BFS, p 0.50; 34 against 36, p 0.63) and difference in means keeps slightly more programs correct (13 against 9, 14 against 10). Edits at the task-specification positions do nothing (12 to 19 BFS against 13). Steering toward DFS works cleanly for both (SAE 47 of 50, difference in means 50 of 50, against 37 for the twin; pass 0.98 to 1.00). Difference-in-means steering of a code model's choice of language and library has shown the same trade-off, with overly strong interventions reducing output quality [130]. AxBench found simple baselines ahead of SAE steering [34], and a 2026 reanalysis puts SAEs close to AxBench's reference when latents are selected and labelled by a supervised pipeline [131]; here, at comparable norms, neither the input-selected nor the output-scored clamp reached the difference-in-means edit (Table 261b). (iteration-7 correction)

**Table 261b.** Mean change in the fork log-odds of BFS (nats) under each edit at layer 22, by number of latents k and dose lambda; the difference-in-means, random and isotropic edits are scaled to the norm of the input-selected clamp with the same k. Source: Gemma feature experiment `results/logit_sweep_summary.json`.

| Edit | Latents (k) | Fork, lambda 0.5 | Fork, lambda 1 | Fork, lambda 2 | Task specification, lambda 0.5 | Task specification, lambda 1 | Task specification, lambda 2 |
|---|---|---|---|---|---|---|---|
| SAE clamp, input-selected | 5 | 0.52 | 0.98 | 1.97 | 0.09 | 0.11 | 0.11 |
| SAE clamp, input-selected | 20 | 0.87 | 1.78 | 3.76 | 0.12 | 0.12 | 0.22 |
| SAE clamp, input-selected | 50 | 1.09 | 2.26 | 4.47 | 0.08 | 0.10 | 0.20 |
| SAE clamp, output-scored | 5 | 0.66 | 1.33 | 2.69 | 0.16 | 0.20 | 0.28 |
| SAE clamp, output-scored | 20 | 0.74 | 1.35 | 2.65 | 0.17 | 0.22 | 0.21 |
| SAE clamp, output-scored | 50 | 1.38 | 2.94 | 6.07 | 0.23 | 0.32 | 0.37 |
| difference in means (norm of the input-selected clamp) | 5 | 1.05 | 2.04 | 4.14 | 0.09 | 0.09 | 0.08 |
| difference in means (norm of the input-selected clamp) | 20 | 1.98 | 4.04 | 7.93 | 0.41 | 0.70 | 1.29 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| difference in means (norm of the input-selected clamp) | 50 | 3.53 | 7.08 | 12.22 | 0.44 | 0.76 | 1.35 |
| random latents (same norm) | 5 | -0.14 | -0.37 | -0.73 | 0.08 | 0.10 | 0.12 |
| random latents (same norm) | 20 | 0.26 | 0.51 | 1.26 | 0.04 | 0.01 | -0.01 |
| random latents (same norm) | 50 | 0.32 | 0.89 | 2.65 | 0.18 | 0.31 | 0.54 |
| isotropic direction (same norm) | 5 | 0.06 | 0.11 | 0.13 | 0.07 | 0.04 | 0.02 |
| isotropic direction (same norm) | 20 | 0.07 | 0.12 | 0.26 | -0.16 | -0.37 | -0.78 |
| isotropic direction (same norm) | 50 | 0.12 | 0.23 | 0.18 | -0.19 | -0.42 | -0.81 |
| SAE clamp with ban-contrast latents | 20 | n/a | 0.95 | n/a | n/a | 0.09 | n/a |
| SAE clamp, input-selected, layer 29 | 20 | n/a | 0.63 | n/a | n/a | n/a | n/a |

**Table 262.** Generation under edits at the fork (layer 22) and at the task-specification positions (N = 50 paired slots; BFS is the target). Source: Gemma feature experiment `results/clamps.json`.

| Site | Edit | Mean realised norm | Attempted BFS | Correct BFS | Pass |
|---|---|---|---|---|---|
| none | twin | 0 | 13 | 6 | 0.84 |
| fork | SAE clamp, top 5 input-selected latents | 307.9 | 18 | 10 | 0.84 |
| fork | SAE clamp, top 20 input-selected latents | 602.2 | 30 | 9 | 0.58 |
| fork | SAE clamp, top 50 input-selected latents | 1069.8 | 36 | 10 | 0.48 |
| fork | difference in means at the top-20 norm | 602.2 | 46 | 12 | 0.32 |
| fork | difference in means at the top-50 norm | 1069.8 | 49 | 7 | 0.16 |
| fork | 50 random latents at the same norm | 1069.8 | 21 | 9 | 0.76 |
| fork | SAE clamp, top 20 output-scored latents | 673.5 | 25 | 14 | 0.78 |
| fork | difference in means matched on effect, top-20 level | 270.6 | 32 | 13 | 0.62 |
| fork | difference in means matched on effect, top-50 level | 342.9 | 34 | 14 | 0.60 |
| fork | DAS rank-4 interchange | 496.0 | 50 | 0 | 0.00 |

| | | | | | |
|---|---|---|---|---|---|
| fork | ablation of the top 5 attribution nodes (N = 30) | 1004.7 | 29 | 0 | 0.03 |
| fork | steering with the top 5 attribution nodes (N = 30) | 115.5 | 19 | 12 | 0.73 |
| task specification | SAE clamp, top 5 | 4.1 | 14 | 6 | 0.84 |
| task specification | SAE clamp, top 20 | 92.4 | 12 | 4 | 0.82 |
| task specification | SAE clamp, top 50 | 104.9 | 15 | 5 | 0.80 |
| task specification | difference in means at the top-20 norm | 92.4 | 17 | 9 | 0.84 |
| task specification | difference in means at the top-50 norm | 104.9 | 19 | 9 | 0.80 |
| task specification | 50 random features | 104.9 | 15 | 9 | 0.88 |
| fork, toward DFS | SAE clamp, top 20 | 1170.1 | 3 (47 DFS, all correct) | n/a | 0.98 |
| fork, toward DFS | difference in means | 1170.1 | 0 (50 DFS, all correct) | n/a | 1.00 |

*[Figure fig_gemma_steer: generated in the report phase, not in this draft]*

**DAS moves the fork log-odds within the controls it had, and forcing it breaks the program.** DAS at the fork position of layer 22, with forced-BFS prompts as sources, moved held-out y by +8.8, +9.5 and +9.8 nats at ranks 1, 2 and 4, against -0.08 to +0.05 for random subspaces of the same rank (Table 263). The shuffled-source control was uninformative as built, because every source was a forced-BFS prompt. In generation the rank-4 interchange made 50 of 50 programs BFS and 0 of 50 passed. Stated within its controls, a rank-1 DAS subspace fitted on forced-BFS sources moves the held-out fork log-odds by 8.8 nats (random subspaces of the same rank -0.08 to +0.05); because every source was forced-BFS, the shuffled-source control is uninformative by construction (10.1 nats); no random-initialisation or target-shuffled control was run; and the rank-4 interchange gives BFS in 50 of 50 attempts with 0 of 50 programs passing, so this does not show that the choice is one-dimensional. With unrestricted alignment maps interchange accuracy reaches 100% even on randomly initialised language models (Sutter et al., Table 277 row 14), so the fit stays uninterpretable until those controls run; pushing hard costs correctness, as subspace interventions can when they leave the model's natural range [122]. (iteration-7 correction)

**Table 263.** DAS at the fork (layer 22): mean change in the fork log-odds of BFS on held-out and training prompts. Every source was a forced-BFS prompt, so the shuffled-source column is uninformative by construction; no random-initialisation or target-shuffled control was run, and the rank-4 interchange gave 0 of 50 passing programs (Table 262), so the table does not show that the choice is one-dimensional. Source: Gemma feature experiment `results/das.json`. (iteration-7 correction)

| Rank | DAS, held out | DAS, train | Share of held-out prompts moved toward BFS | Random subspace, held out | Shuffled source, held out |
|---|---|---|---|---|---|
| 1 | 8.81 | 9.55 | 1.00 | -0.08 | 10.11 |
| 2 | 9.51 | 10.15 | 1.00 | 0.03 | 9.69 |
| 4 | 9.76 | 10.25 | 1.00 | 0.05 | 11.42 |

The planned transcoder attribution graph was the first item of the drop order and was not built; a

first-order attribution on the residual-stream SAEs replaced it (Table 264). Ablating the two strongest nodes moved y by 4.0 and 6.3 nats, against 0.8 and 0.4 for random active latents at matched contribution; two of the other three beat random and one did not. Jointly ablating the five nodes turned 29 of 30 programs into BFS with none passing, and jointly steering with them gave 19 BFS of 30, 12 correct.

**Table 264.** First-order attribution nodes at the fork, validated by ablation and steering in the real model (mean change in y, against random active latents at matched contribution). Source: Gemma feature experiment `results/attribution.json`.

| Layer | Latent | Attribution score | Ablation | Random ablation | Steering | Random steering |
|---|---|---|---|---|---|---|
| 22 | 1774 | -4.68 | 4.01 | 0.78 | 0.55 | 0.07 |
| 29 | 15050 | -4.35 | 6.28 | 0.40 | 0.45 | 0.11 |
| 29 | 735 | 4.11 | -0.51 | 0.40 | 1.38 | 0.09 |
| 29 | 6026 | 3.95 | -0.95 | 0.51 | 0.14 | 0.09 |
| 22 | 142 | 3.79 | -1.26 | 2.10 | 0.24 | 0.09 |

**The top-five prohibition in a model that writes two algorithms.** Banning the five most common grid algorithms (BFS, DFS, IDDFS, A* and greedy best-first, which share one catalogue label here) left no program outside the banned set: avoidance 0.00, as in Qwen3-8B in iteration 2 (Table 265). The ban pushed Gemma from its default DFS to BFS (10 to 41 of 50; McNemar 32:1, p 7.9e-9) and cut the pass rate from 0.86 to 0.38. In actual activation coordinates (PCA, Isomap with k = 10 and classical MDS of layers 17 and 22, raw and SAE codes), the top-five prompt states sit next to the banned A* and greedy cluster (Table 266); among all 15 ways of splitting the six catalogue centroids into four banned and two survivors, the observed split ranks first in three of the four spaces (rank share 0.067) and third in the fourth (0.20). The prohibition lands inside the banned region, not at its edge or on the opposite side. The single-name bans of BFS and DFS left the banned algorithm in 11 and 8 of 50 programs, and the forced prompts gave BFS 50 of 50 (38 correct) and DFS 50 of 50 (45 correct).

*[Figure fig_gemma_geometry_raw_L17: generated in the report phase, not in this draft]*

*Figure note (iteration-7 correction): Actual activation coordinates: mean over the task-specification positions (from the first token of the task statement to the last token of the user message, `src/cfg.py` line 7; file name restored, iteration-8 correction), so neither the read position nor the fork at response index 32 is plotted; output of block 17 (the residual stream after the block), five massive channels zeroed; projection: PCA (first component 0.43, second 0.12 of variance), Isomap (k = 10) and classical MDS (stress-1 0.371), all fitted on the 230 points. Points: prompt states: forced-algorithm prompts by mode, BFS and DFS bans, the top-five ban and its twin, and 120 neutral variants coloured by y_v, the fork log-odds of BFS measured at response index 32. Centroids: per forced mode, the mean of its forced-prompt points (stars), with marker size growing with the natural implemented count out of 200 (DFS 139, BFS 61). N = 230. Caveat: The plotted position precedes the fork and the generated code; centroid positions none (prompt-defined); centroid marker size from realised labels of the 200 natural samples; variant colour from y_v, a quantity measured at the later fork position (readout target, not a partition).*

**Table 265.** Bans and forced prompts on Gemma's grid task (N = 50). Source: Gemma feature experiment `results/top5.json`, `results/geometry.json`.

| Prompt | Attempted | Correct | Pass | Avoidance of the banned set |
|---|---|---|---|---|
| top-five twin | DFS 40, BFS 10 | DFS 40, BFS 3 | 0.86 | n/a |
| top-five ban | BFS 41, DFS 9 | BFS 10, DFS 9 | 0.38 | 0.00 |
| ban BFS | DFS 39, BFS 11 | DFS 39, BFS 2 | 0.82 | 0.78 |

| | | | | |
|---|---|---|---|---|
| ban DFS | BFS 42, DFS 8 | BFS 15, DFS 7 | 0.44 | 0.84 |
| forced BFS | BFS 50 | BFS 38 | 0.76 | n/a |
| forced DFS | DFS 50 | DFS 45 | 0.90 | n/a |

*[Figure fig_gemma_geometry_raw_L22: generated in the report phase, not in this draft]*

*Figure note (iteration-7 correction): Actual activation coordinates: mean over the task-specification positions (from the first token of the task statement to the last token of the user message, `src/cfg.py` line 7; file name restored, iteration-8 correction), so neither the read position nor the fork at response index 32 is plotted; output of block 22 (the residual stream after the block), five massive channels zeroed; projection: PCA (first component 0.37, second 0.17 of variance), Isomap (k = 10) and classical MDS (stress-1 0.369), all fitted on the 230 points. Points: prompt states: forced-algorithm prompts by mode, BFS and DFS bans, the top-five ban and its twin, and 120 neutral variants coloured by y_v, the fork log-odds of BFS measured at response index 32. Centroids: per forced mode, the mean of its forced-prompt points (stars), with marker size growing with the natural implemented count out of 200 (DFS 139, BFS 61). N = 230. Caveat: The plotted position precedes the fork and the generated code; centroid positions none (prompt-defined); centroid marker size from realised labels of the 200 natural samples; variant colour from y_v, a quantity measured at the later fork position (readout target, not a partition).*

**Table 266.** Where the top-five prompt states sit: geodesic distance from their mean to each catalogue centroid, and the rank of the observed banned-survivor split among the 15 possible splits. Source: Gemma feature experiment `results/geometry.json`.

| Space | MDS stress-1 | Nearest centroid | Geodesic distance to BFS / DFS / bidirectional BFS / IDDFS / A* and greedy / Dijkstra | Rank share of the observed split |
|---|---|---|---|---|
| raw, layer 17 | 0.371 | A* and greedy | 341.9 / 330.6 / 410.3 / 222.1 / 188.8 / 496.5 | 0.067 |
| raw, layer 22 | 0.369 | A* and greedy | 734.1 / 684.9 / 900.1 / 365.2 / 318.6 / 990.5 | 0.067 |
| SAE codes, layer 17 | 0.542 | A* and greedy | 27.5 / 28.4 / 41.1 / 32.9 / 26.1 / 47.1 | 0.067 |
| SAE codes, layer 22 | 0.569 | A* and greedy | 43.2 / 44.1 / 61.5 / 83.6 / 43.1 / 72.5 | 0.20 |

*[Figure fig_gemma_geometry_sae_L17: generated in the report phase, not in this draft]*

*Figure note (iteration-7 correction): Actual activation coordinates: mean over the task-specification positions (from the first token of the task statement to the last token of the user message, `src/cfg.py` line 7; file name restored, iteration-8 correction), so neither the read position nor the fork at response index 32 is plotted; output of block 17 (the residual stream after the block), Gemma Scope 2 SAE codes, log1p; projection: PCA (first component 0.23, second 0.08 of variance), Isomap (k = 10) and classical MDS (stress-1 0.542), all fitted on the 230 points. Points: prompt states: forced-algorithm prompts by mode, BFS and DFS bans, the top-five ban and its twin, and 120 neutral variants coloured by y_v, the fork log-odds of BFS measured at response index 32. Centroids: per forced mode, the mean of its forced-prompt points (stars), with marker size growing with the natural implemented count out of 200 (DFS 139, BFS 61). N = 230. Caveat: The plotted position precedes the fork and the generated code; centroid positions none (prompt-defined); centroid marker size from realised labels of the 200 natural samples; variant colour from y_v, a quantity measured at the later fork position (readout target, not a partition).*

The novelty audit found 1 correct program labelled other among 1,450 correct programs, a first-and-last-position program that two of three judges called a catalogue variant: 0 novel (rule-of-three bound 0.0021). The style re-gate agreed with the labeller on 30 of 30 grid programs. The hash chain verified, a standard-library re-derivation matched 32 of 32 headline quantities, and a

shuffled-class decodability placebo sat at chance (0.43). Judge spend was USD 0.034.

*[Figure fig_gemma_geometry_sae_L22: generated in the report phase, not in this draft]*

*Figure note (iteration-7 correction): Actual activation coordinates: mean over the task-specification positions (from the first token of the task statement to the last token of the user message, `src/cfg.py` line 7; file name restored, iteration-8 correction), so neither the read position nor the fork at response index 32 is plotted; output of block 22 (the residual stream after the block), Gemma Scope 2 SAE codes, log1p; projection: PCA (first component 0.19, second 0.08 of variance), Isomap (k = 10) and classical MDS (stress-1 0.569), all fitted on the 230 points. Points: prompt states: forced-algorithm prompts by mode, BFS and DFS bans, the top-five ban and its twin, and 120 neutral variants coloured by y_v, the fork log-odds of BFS measured at response index 32. Centroids: per forced mode, the mean of its forced-prompt points (stars), with marker size growing with the natural implemented count out of 200 (DFS 139, BFS 61). N = 230. Caveat: The plotted position precedes the fork and the generated code; centroid positions none (prompt-defined); centroid marker size from realised labels of the 200 natural samples; variant colour from y_v, a quantity measured at the later fork position (readout target, not a partition).*

## 6.6 The plan in words: a natural language autoencoder on Qwen2.5-7B-Instruct

**Why.** The request names NLA as an instrument, and the release of NLA checkpoints for Qwen2.5-7B-Instruct [124] made it possible to ask, in words, whether the residual stream states before the first code token say which algorithm will be written. The experiment read layer-20 states with the released verbalizer and reconstructor pair, judged the explanations blind, and tested an edit of the explanation as a steering vector. Published verbalizers of the same kind include LatentQA [117] and SelfIE [118], and a 2026 analysis of a released Qwen-2.5-7B verbalizer found that only a small share of an explanation's specific claims are ones the reconstruction depends on [119]. Pre-registration was a sha256 chain of four commitments, each preceding its samples; the parts added after the committed nulls are marked exploratory.

**Behaviour.** Over eight tasks, Qwen2.5-7B-Instruct defaults to BFS, Dijkstra (0.88), a hash map (0.98), a hash set (0.71), DFS for connected components (0.62, where Qwen3-8B writes union find), the naive scan and binary search; kth smallest has no default (quickselect 0.40, size-k heap 0.34) and failed the style re-gate. The model mostly ignores bans (Table 267). Only two sum gave a substitute (brute force in 32 of 46 correct programs). On connected components a ban of DFS raised DFS from 29 to 43 correct programs, with 15 paired slots flipping from union find to DFS, the rebound that naming a forbidden concept can cause [7, 53]. Every one of the 70 programs written under the fourth template attempted the banned default. The shortest path semantic seed barely moved the model (Dijkstra 28, Bellman-Ford 6, against 38 and 5 for the twin). The labellers recognised 51 of 51 references, and the style re-gate passed on 7 of 8 tasks against two blind code judges.

**Table 267.** Behaviour of Qwen2.5-7B-Instruct (twin and ban N = 50, paired slots; forced prompts N = 20 per algorithm). Source: NLA experiment `results/behaviour.json`.

| Task | Default (share of correct twin programs) | Twin pass | Ban: attempts still using the default | Substitute under the ban | Forced adherence of the algorithm used in the causal arm |
|---|---|---|---|---|---|
| grid | BFS (13/13) | 0.26 | 0.68 | none (DFS 3) | none (best other 0.35) |

| | | | | | |
|---|---|---|---|---|---|
| shortest path | Dijkstra (38/43) | 0.86 | 0.50 | none (Bellman-Ford/SPFA 9, below 10) | none (Bellman-Ford 0.65, below 0.7) |
| kth smallest | none (quickselect 0.40) | 0.94 | n/a | n/a | excluded (re-gate 26/30) |
| two sum | hash map (49/50) | 1.00 | 0.28 | brute force (32/46) | brute force 1.00 |
| find duplicate | hash set (35/49) | 0.98 | 0.46 | none (Floyd cycle detection 7) | Floyd cycle detection 0.95 |
| connected components | DFS (29/47) | 0.94 | 0.86 | none (DFS 43, union find 2) | union find 0.95 |
| substring search | naive (50/50) | 1.00 | 0.94 | none | KMP 0.90 |
| first and last position | binary search (38/38) | 0.76 | 0.96 | none | bisect module 1.00 (not run) |

**The instrument gate.** The extraction convention that matches the training code reads the output of block 20; it reaches a fraction of variance explained of 0.695 on the reference set against 0.580 for the convention in the release's example, and 0.695 is within 0.06 of the released 0.752. Generic web text and chat give 0.78 and 0.79. Code is out of distribution for this NLA: LeetCode prompts 0.45, HumanEval+ prompts 0.39, CLRS traces 0.33. On the task states the fraction of variance explained, with the reference-set variance as denominator, is 0.512, so the gate's verdict is partial (Table 268).

**Table 268.** Fraction of variance explained of the layer-20 reconstructions on the task states, by position (reference-set denominator). Source: NLA experiment `results/gate.json`.

| Position | Read position | Task specification | Token 1 | Token 2 | Token 4 | Token 8 | Token 16 |
|---|---|---|---|---|---|---|---|
| Fraction of variance explained | 0.65 | 0.50 | 0.48 | 0.28 | 0.64 | 0.57 | 0.51 |

**The NLA never names the planned algorithm before the code.** Two blind judges from different families (Gemini 2.5 Flash Lite and Mistral Small 3.2, both scoring 1.00 on calibration; a third model resolved 1,064 disagreements) mapped each of 6,180 explanations to a catalogue algorithm or to none named (Table 269). Every preregistered readout at the end-of-prompt positions is exactly zero (Table 270). On ban prompts, explanations named the substitute in 0 of 16 states (only two sum had a substitute, so this test is underpowered); on twin prompts they named the default in 0 of 112, as did the text-only reader; on ban failures they named the default the model still writes in 0 of 96. The positive control failed too: on forced `Implement Y` prompts the explanations called Y planned in 0 of 7 tasks, while the text-only reader did so on all of them, and they mentioned the algorithm named in the prompt in 0.7% of cases. The explanations describe the local format line, such as a request for no explanation or a Python code block to follow, and confabulate the task (Fibonacci, medians, prime numbers); 4,733 of 5,700 judged explanations name no algorithm at all. At generated tokens the NLA sometimes names the realised algorithm before the visible text identifies it (up to 0.28 at token 8 under the ban), but an outside reader who sees only the prompt and prefix predicts the realised algorithm better, per sample (0.25 against 0.07; exact McNemar 20:88, p 2e-11 when the 364 units are treated as independent). The units are nested in 104 samples and 5 task clusters: the task-cluster bootstrap interval of the difference is [-0.35, 0.02], with one unit per sample (the first in position order, 104 samples) the contrast is 0:25 (exact p 6.0e-08), a median over 2,000 draws of one random unit per sample of 4:29 (p 1.9e-05; 1.00 of draws below

0.05), a sign test on the per-sample differences 12:38 (p 3.1e-04) and a Durkalski-adjusted McNemar with samples as clusters p 6.5e-05. The samples share 5 task clusters, and with 5 clusters no cluster-level test can reach a two-sided p below 0.0625, so the evidence at the task level is inconclusive whatever the sample-level p. (iteration-7 correction)

**Table 269.** Agreement between the two judges by mode (EXPL: mapping an explanation; NAME: naming from the visible text; PREDICT: the outside reader's forecast). Source: NLA experiment `results/readout.json`.

| Mode | Raw agreement | Kappa | Unresolved after the third judge |
|---|---|---|---|
| explanation | 0.94 | 0.82 | 0.6% |
| visible text | 0.67 | 0.56 | 1.6% |
| outside reader | 0.77 | 0.75 | 3.9% |

**Table 270.** Preregistered readout tests. Source: NLA experiment `results/readout.json`.

| Test (units) | NLA | Comparator | b : c | p | Verdict |
|---|---|---|---|---|---|
| ban states name the substitute (two sum only; 16 states) | 0/16 | text 0/16 | 0 : 0 | 1 | fail, underpowered |
| same, against swapped, random and outside-reader controls | 0/16 | swapped 0/16, random 0/8, outside reader 0/16 | 0 : 0 | 1 | n/a |
| twin states name the default (7 tasks; 112 states) | 0/112 | text 0/112 | 0 : 0 | 1 | fail |
| forced states name the forced algorithm (7 tasks) | 0.00 on every task | text 1.00 | n/a | n/a | fail |
| ban-failure states name the default still written (6 tasks; 96) | 0/96 | outside reader 0/96, text 0/96 | 0 : 0 | 1 | null |
| per sample, mixed conditions (364 units nested in 104 samples and 5 task clusters) | 0.066 correct | outside reader 0.253 | 20 : 88 per unit; 0 : 25 at one unit per sample | 2e-11 per unit (units treated as independent); 6.0e-08 at one unit per sample; task-cluster bootstrap of the difference [-0.35, 0.02] | outside reader better per sample; inconclusive at the task level (5 clusters: no cluster-level test below p 0.0625) (iteration-7 correction) |
| generated positions before the identifying token | ban token 8: 0.28; twin token 4: 0.18 | text 0 by construction | n/a | n/a | descriptive |

**The information is local and fades within a sentence (exploratory).** To tell an instrument limit from information that fades along the prompt, the same NLA and judges read states at the token that names the algorithm inside the condition sentence and at the final period of that sentence (220 states, 660 explanations; Table 271). At the name token the explanations mention the algorithm in 80% of forced and 89% of ban states, and they respect the negation: they call it planned in 48% of

forced states and in 6% of ban states. Two to eight tokens later, at the end of the sentence, the mention rates fall to 27% and 6%, and about 170 tokens later, at the end of the prompt, to 0.7%. The null therefore says that the pre-code positions do not carry the algorithm choice in a form this NLA puts into words, not that the instrument is blind.

**Table 271.** Local positive control (majority of three explanations per state; 75 forced and 35 ban states at each of the first two positions). The end-of-prompt row is the committed readout and counts single explanations at the last prompt positions (the algorithm is mentioned in 8 of 1,080 forced and 3 of 420 ban explanations). Source: NLA experiment `results/local_control.json`, `README.md`.

| Position | Forced: names Y as planned | Forced: mentions Y | Ban: names the default as planned | Ban: mentions the default | Reconstruction cosine |
|---|---|---|---|---|---|
| at the algorithm's name | 0.48 | 0.80 | 0.06 | 0.89 | 0.92 |
| end of that sentence (2 to 8 tokens later) | 0.11 | 0.27 | 0.00 | 0.06 | 0.90 |
| end of the prompt (committed readout, about 170 tokens later) | 0.00 | 0.007 | 0.00 | 0.007 | 0.82 to 0.87 |

*[Figure fig_nla_local: generated in the report phase, not in this draft]*

**The preregistered verbal edit fails at the prefill site.** The causal arm replaced the default's name in a twin explanation by an algorithm Y the model writes on request, reconstructed both versions, and patched the difference into the layer-20 states at the last four prompt positions at doses of 1, 2 and 4 (N = 30 paired slots per cell). It failed on all four tasks (Table 272). The edits barely move the reconstruction (cosine 0.9992 between the edited and unedited reconstructions; 0.04 of the state norm per position; Table 273), the unedited round trip at 0.8 of the norm propagates downstream but moves nothing, and all 15 prefill cells per task stay within 0 to 3 slots of the unedited control. The edit direction is nearly orthogonal to the forced-state difference in means (median cosine 0.09).

**Table 272.** The preregistered causal arm: Y share in the unedited control and the edited-minus-round-trip difference at doses 1, 2 and 4 (N = 30 paired slots). Source: NLA experiment `results/causal.json`.

| Task (default to Y) | Control Y share | Dose 1 | Dose 2 | Dose 4 | Pass |
|---|---|---|---|---|---|
| connected components (DFS to union find) | 10/30 | -0.03 (0:1) | -0.03 (0:1) | +0.03 (3:2) | no |
| two sum (hash map to brute force) | 0/30 | 0 | 0 | 0 | no |
| find duplicate (hash set to Floyd cycle detection) | 12/30 | 0 | +0.03 (2:1) | +0.10 (3:0, p 0.25) | no |
| substring search (naive to KMP) | 0/30 | 0 | 0 | 0 | no |

**Table 273.** Realised push sizes relative to the state norm at block 20. Source: NLA experiment `results/causal.json`.

| Cell | Push norm over state norm |
|---|---|
| edit, dose 1 | 0.041 |
| unrelated-word edit, dose 1 | 0.036 |
| round trip, dose 1 | 0.201 |

| | |
|---|---|
| round trip, dose 4 | 0.80 (0.76 at block 27) |

**At the generated-token site and large doses, the verbal edit steers two of four tasks (exploratory).** After the committed nulls, the experiment extended the doses and moved the edit to the read position plus generated tokens 1 to 64 (Tables 274 and 275). On find duplicate the edit at dose 8 (about 0.3 of the state norm at each position) raised Floyd cycle detection from 0.28 to 0.76 (25:1 over 50 paired slots), replicated on fresh slots (15 of 20, 13:0, p 2e-4), and beat a matched random push (28:1), an unrelated-word edit built the same way (12:1, p 0.003) and a norm-matched round trip (13:1, p 0.002); at dose 16 the programs broke (pass 0.37). On two sum brute force rose from 0 to 10 of 50 at dose 8 and to 23 of 30 at dose 16 with a pass rate of 0.90, while the random push never moved it. On connected components only the forced-state difference in means steered (union find 0.80 at dose 16, 15:1); on substring search nothing steered short of breaking the program. The prefill site stayed inert even at dose 16 or at the natural norm of the difference in means. Asking for the algorithm in the prompt remained the ceiling (forced adherence 0.90 to 1.00 on all four tasks), as the AxBench comparison of prompting and steering predicts [34]. The NLA paper itself reports that editing a plan explanation and steering at one token succeeds roughly half the time; this run's success on two of four tasks is of the same kind and is not a reliable control knob.

**Table 274.** Exploratory dose extension (Y share out of 30 paired slots). Source: NLA experiment `results/causal_exploratory.json`.

| Task | Control | Prefill edit, dose 16 | Prefill difference in means at natural norm, dose 2 | Generated-site edit, dose 8 | Generated-site random, dose 8 | Generated-site difference in means, dose 8 | Generated-site difference in means at natural norm |
|---|---|---|---|---|---|---|---|
| connected components (to union find) | 10 | 10 | 9 | 8 | 11 | 19 (10:1, p 0.012) | 13 |
| two sum (to brute force) | 0 | 0 | 0 | 4 (4:0, p 0.125) | 0 | 0 | 0 |
| find duplicate (to Floyd cycle detection) | 12 | 11 | 12 | 23 (12:1, p 0.003) | 8 | 2 | 6 |
| substring search (to KMP) | 0 | 0 | 0 | 0 | 0 | 2 (pass 0.83) | 0 |

**Table 275.** Exploratory follow-up at the generated-token site: dose response, controls built the same way and a replication on fresh slots (Y share; exact McNemar on paired slots). Source: NLA experiment `results/causal_exploratory_followup.json`.

| Task | Edit, dose 4 | Edit, dose 8 (N = 50) | Edit, dose 16 | Dose 8, fresh slots (N = 20) | Dose 8 against random | Dose 8 against unrelated-word edit | Dose 8 against matched round trip | Difference in means, dose 16 |
|---|---|---|---|---|---|---|---|---|
| find duplicate (to Floyd cycle detection) | 0.60 (7:1, p 0.07) | 0.76 against control 0.28 (25:1, p < 1e-6) | 0.30, pass 0.37 | 15/20 (13:0, p 2e-4) | 28:1 | 12:1, p 0.003 | 13:1, p 0.002 | 0.07, pass 0.27 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| two sum (to brute force) | 0.00 | 0.20 against 0.00 (10:0, p 0.002) | 0.77, pass 0.90 (23:0) | 6/20 (6:0, p 0.03) | 10:0, p 0.002 | 4:0, p 0.125 | 4:0, p 0.125 | 0.37, pass 0.40 |
| connected components (to union find) | 0.33 | 0.24 against 0.36 (3:9) | 0.17 | 4/20 | 5:13 | 6:6 | 2:6 | 0.80 (15:1, p 5e-4) |
| substring search (to KMP) | 0.03 | 0.00 | 0.07, pass 0.27 | 0/20 | 0:0 | 0:0 | 0:0 | 0.03, pass 0.10 |

*[Figure fig_nla_mds_centroids: generated in the report phase, not in this draft]*

*Figure note (iteration-7 correction): Actual activation coordinates: the read position (last prompt token), before any generated token; output of block 20 (hidden_states[21]); projection: classical MDS of the READ centroids per task (MDS stress not recorded in `results/geometry.json`; file name restored, iteration-8 correction). Points: one point per centroid: centroids of the forced prompts that ask for each algorithm, the twin centroid and the ban centroid. Centroids: means of the READ states per forced algorithm, twin and ban; filled area grows with the number of twin programs that implemented the algorithm, the dashed ring with the number under the ban. N = 54. Caveat: The plotted position precedes the fork and the generated code; centroid positions none (prompt-defined); marker areas from realised labels of the twin and ban programs.*

In activation space (exploratory), read-position states projected on the plane of the forced-prompt states show every forced algorithm in its own region, with twin and ban prompts apart from all of them. Relative to the algorithm-specific directions of the forced prompts, the ban moves the state toward the banned default's own direction on two sum (+0.13), find duplicate (+0.26) and first and last position (+0.15), as the ban sentence names the default, and toward an alternative on shortest path (Bellman-Ford +0.13), connected components (BFS +0.23), substring search and the grid task (DFS +0.18). The novelty audit found 10 correct programs labelled other in every arm: 8 labeller misses (six kth smallest counting sorts built on a counter, two boolean seen-arrays) and 2 degenerate programs, 0 novel. Spend was about USD 2.0 on OpenRouter (NLA judges with calibration, two interrupted passes, the local control and the novelty audit USD 1.82; code judges for the bench tasks USD 0.11 and for the path tasks USD 0.10) and about 2.0 GPU hours on one 24 GB NVIDIA Ada GPU (68 minutes of generation for behaviour and the committed causal arm, about 27 for the exploratory causal cells, 16 for 6,180 explanations and about 8 for reconstruction passes and the state cache).

*[Figure fig_nla_read_geometry: generated in the report phase, not in this draft]*

*Figure note (iteration-7 correction): Actual activation coordinates: the read position (last prompt token), before any generated token; output of block 20 (the hidden state after block 20); projection: per task, the plane of the first two principal components of that task's forced-prompt READ states (SVD after centring on their mean). Points: single prompt states: forced prompts that ask for each algorithm, by mode and template, the twin and ban prompts of each template (arrow twin to ban), the shortest path seed, twin plus random noise, and NLA edit deltas (x1, x2, x4) and a difference-in-means delta (x4) drawn as arrows. Centroids: per forced mode, the mean of its forced READ states (stars); the legend gives forced adherence. N = 310. Caveat: The plotted position precedes the fork and the generated code; none for positions and centroids (prompt-defined); the legend's adherence is behavioural.*

## 6.7 Positioning against published work

The positioning audit, dated 2026-10-22, extended the audits of iterations 4 and 5 to this iteration's claims. A protocol frozen and stamped before the first call drove 126 logged searches over seven lanes, from 2,191 raw hits to 1,820 unique records, 1,401 in the window, and 75 claim rows scored on the occupancy rubric for 67 papers: 3 occupying, 66 adjacent and 6 irrelevant. Its 123 new quotes from 94 sources matched the cached source text and were re-matched by an independent live pass. Table 276 gives the verdict per claim, and Table 277 condenses the 65 rows of its neighbour table; the verbatim quotes are in the audit's evidence files.

**Table 276.** Lane verdicts for the iteration-6 claims as of 2026-10-22. Source: positioning audit `research_report.md`.

| Claim | Lane verdict | Deciding neighbours |
|---|---|---|
| Localization: a low-rank causal plan variable, found by interchange or DAS, decides which algorithm Qwen3-8B writes, validated by generation | partly occupied: the method is occupied, the object is open; the flag for a published localization of algorithm choice did not fire, narrowly | DAS and Boundless DAS [110, 111]; MIB [52]; CDAS, DAS steering validated by generation on refusal and a backdoor [112]; prompt cues only for algorithm choice in code [1] |
| Mediation: the plan subspace mediates which random pushes switch the algorithm | adjacent only | pre-generation planning subspaces; output-sensitive steering components |
| Discreteness under interpolation | adjacent only | categorical and multidimensional feature geometry; manifold steering [27] |
| Ideas not in the tokens, at matched KL | adjacent only | decoding matched to steering's departure; temperature and novelty [37]; forking paths [30] |
| Resampling the plan as a diversity knob | partly occupied: activation noise and latent sampling for diversity are occupied | STARS [71]; noise steering [70]; random embedding injection [84] |
| The plan in words with NLAs | occupied (method): the NLA paper edits a plan explanation and steers with the reconstruction; this run is a replication in a new setting | the NLA paper (Transformer Circuits, May 2026); LatentQA [117]; SelfIE [118]; cautions [119, 120] |
| The plan in Gemma Scope 2 features | partly occupied: SAE-feature steering of reasoning strategies beats prompt and activation baselines | SAE-Steering [121]; AxBench [34] |
| 8-bit floating-point (W8A8) activation rounding inflates perturbation effects | adjacent only in the published record; partly occupied by contemporaneous ICLR 2027 submissions at abstract level | steering under weight-only quantization [116] |

Three of the audit's flags bear on what this run may claim. No source localizes which algorithm a code model will write by interchange or DAS and validates it by scoring the generated programs, so that combination is open; the method itself is DAS. Editing an NLA explanation of a plan and steering with the reconstruction is published, so the NLA arm is a replication on a new object. Steering under quantization has been compared with full precision only for weight-only 8-bit integer and 4-bit NormalFloat quantization, with activations kept in 16-bit floating point [116]; no source found measures how dynamic per-token activation rounding changes the effect of random perturbations, which is the question the engine control answered, in the negative. The audit also

recorded that community NLAs exist for Qwen3-8B at layer 24, that circuit-tracer ships Qwen-3 transcoders including 8B [123] (so the choice of Gemma for the feature claim needs its reason, which is the natural within-prompt split), and that Gemma Scope 2 dates from 22 December 2025 with instruct SAEs initialised from the pretrained ones.

**Table 277.** The positioning audit's neighbour table, condensed (row; claim; neighbour with venue and version; relation; lane verdict; what this run adds; the baseline a reviewer would ask for). The verbatim quotes behind each row are in the audit's evidence file. Source: positioning audit `research_report.md`, Section 5. The audit wrote the last two columns before the experiments finished; for the claims that were not run (Section 6.8) they state what the planned test would add.

| Row | Claim | Neighbour (venue, version) | Relation | Lane verdict | What this run adds | Reviewer baseline implied |
|---|---|---|---|---|---|---|
| 1 | localization | DAS: Geiger, Wu, Potts, Icard, Goodman, CLeaR 2024 (PMLR 236), arXiv v4, v1 2023-03-05, pub; Boundless DAS, Wu et al., NeurIPS 2023, v3 | method the claim uses | PARTLY OCCUPIED (method) | object = algorithm choice in code; validation by tests on emitted programs; Sutter-style controls | DAS with a declared rotation rank, against Boundless DAS |
| 2 | localization | MIB: Mueller et al., ICML 2025, v2, v1 2025-04-17, pub | control the claim must meet (method ordering) | n/a | n/a | report neuron and SAE-feature arms beside DAS (MIB ordering) |
| 3 | localization | CDAS: Bao et al., ICLR 2026, v2, v1 2026-02-05, pub | occupies the method + criterion pair (rubric 2.5) on a non-plan attribute (refusal, CoT backdoor) | PARTLY OCCUPIED | a plan object (algorithm choice), route-derived predictions, random/ shuffled controls | a CDAS-style distribution-matching DAS arm |
| 4 | localization | Programmable Semantic Interventions, ICLR 2027 submission, c | adjacent (rubric 1.5) | n/a | generation-validated plan object | PSI-style declared-program alignment |
| 5 | localization | Finding Algorithms in Neural Networks (DAS vs transfer), ICLR 2027 submission, c | adjacent (1.5); supports declaring the map class (non-linear DAS needed in vision) | n/a | algorithm identity in a code LLM, checked by generation | linear vs non-linear map classes |

| | | | | | | |
|---|---|---|---|---|---|---|
| 6 | localization | Causal Syntax Circuits in CodeLMs, Kumar et al., NeurIPS 2026 workshop, v1 2026-10-14, c4 | adjacent (1.5): a code attribute that is not the algorithm; next-token readout | n/a | algorithm (not language); low-rank DAS; generation | random-control significance test, as there |
| 7 | localization | Steering LLM Coding Behaviors with Concept Vectors, ICLR 2027 submission, c | adjacent (2.0) | n/a | algorithm-level object with matched-norm controls | a difference-in-means direction for the algorithm, with matched-norm random controls |
| 8 | localization | Before Reasoning Begins (Qwen3-8B, final prompt position, SAE targets), ICLR 2027 submission, c | adjacent (2.0): same model and site; object is reasoning, not algorithm | n/a | which algorithm (not whether reasoning happens); DAS rather than SAE suppression | layer-count-matched random null interventions |
| 9 | localization | Committed Before Reasoning, Jo, v1 2026-07-17, c4 | supports (pre-generation commitment in Qwen3-8B) | n/a | n/a | n/a |
| 10 | localization | Random Choice in LMs: linear choice state, ICLR 2027 submission, c | adjacent (2.0) | n/a | code-algorithm object | switch-vs-distribution decomposition |
| 11 | localization | Detecting and Characterizing Planning in LMs, Nainani et al., v1 2025-08-25, anchor | adjacent (1.5); contradicts in part (the same abstract reports that planning is not universal) | n/a | algorithm-level plan, not a future-token plan | a planning-vs-improvisation criterion |
| 12 | localization | Algorithmic Core Extraction, Schiffman, v2, v1 2026-02-26, pub | adjacent (1.0) | n/a | choice among algorithms, not a task core | n/a |
| 13 | localization | Invisible Lottery (algorithm choice in code moved by cues), ICML 2026: see 92c source 30 | adjacent (2.0): object occupied behaviourally | n/a | internal causal variable | prompt-cue baseline |
| 14 | localization | Sutter, Minder, Hofmann, Pimentel, NeurIPS 2025 Spotlight, v2, v1 2025-07-11, pub | control the claim must meet | n/a | declared linear map class and rank; random-init or random-rotation control | fit the same map on a randomly initialized Qwen3-8B, and with shuffled labels |

| | | | | | | |
|---|---|---|---|---|---|---|
| 15 | localization | Makelov, Lange, Geiger, Nanda, ICLR 2024, v2, pub | control the claim must meet | n/a | nullspace/row-space decomposition of the patch | check that the patched subspace is not causally disconnected |
| 16 | localization | Reply: Wu, Geiger, Huang, Arora, Icard, Potts, Goodman, v1 2024-01-23 | cite with CTRL-2 | n/a | n/a | n/a |
| 17 | localization | Divergent representations: Grant et al., v5 (2026-04-22), v1 2025-11-06 | control | n/a | on-distribution check of patched states | report divergence of the intervened states |
| 18 | localization | Right Answer, Wrong Mechanism, v2, v1 2026-09-30, c4 | control (do not rely on distance checks alone) | n/a | n/a | planted-pathway or clamp test |
| 19 | localization | Curse of Multiple Mediators; Many Circuits, One Mechanism (TMLR 2026); identification-assumptions position | controls | n/a | n/a | report interaction effects, cross-condition transfer and identification assumptions |
| 20 | mediation | Before the First Token (causal response-length subspace; parallel/ orthogonal decomposition of interventions), ICLR 2027 submission: see 92c source 64 | adjacent (nearest) | ADJACENT-ONLY | prediction of random-push effects on algorithm choice from projection onto the plan subspace | parallel/ orthogonal decomposition of each push |
| 21 | mediation | AJR; ASTEER; decodability is not actionability | adjacent | n/a | n/a | Jacobian-readout predictor of push effects |
| 22 | mediation | DecodeShare, v1 2026-05-21, pub | adjacent (1.5) | n/a | random pushes classified by plan-subspace energy | matched-budget random-subspace control |

| | | | | | | |
|---|---|---|---|---|---|---|
| 23 | mediation | What Does Activation Steering Control?, v1 2026-08-24, c4 | supports a projection account (1.0) | n/a | n/a | report the share of each push's norm inside the plan subspace |
| 24 | mediation | Jailbreak success coordinate outside the refusal subspace, ICLR 2027 submission, c | contradicts in part: the outcome-deciding coordinate can lie outside an evaluated subspace | n/a | n/a | test the orthogonal complement as well |
| 25 | mediation | Before Reasoning Begins (targeted vs random-null); Latent Planning Emerges with Scale, ICLR 2026 | adjacent / supports | n/a | n/a | random-null arm |
| 26 | mediation | On the Non-Identifiability of Steering Vectors, v4, v1 2026-02-06, pub (found by a supplementary search) | supports a subspace-level (not direction-level) account (1.0) | n/a | n/a | report plan claims at subspace level; orthogonal-perturbation control |
| 27 | mediation | Steering off Course, v1 2025-04-06, anchor; Tele-Lens (ICML 2026) | caution | n/a | n/a | per-sample effect distributions |
| 28 | discreteness | Park, Choe, Jiang, Veitch, ICLR 2025 oral, v3, anchor | supports (categorical geometry) | ADJACENT-ONLY | an interpolation sweep between plan centroids, with the emitted algorithm labelled at each step | polytope/ simplex geometry of the plan centroids |
| 29 | discreteness | Engels et al., ICLR 2025, v3 (title changed in v3), anchor | adjacent (2.0): interventions along circular features | n/a | discreteness vs continuity of a code plan | a test for irreducible multi-dimensional features |
| 30 | discreteness | Manifold Steering (ICML 2026 MI-workshop spotlight): see 92c source 44 | adjacent | n/a | n/a | geodesic vs linear paths |

| | | | | | | |
|---|---|---|---|---|---|---|
| 31 | discreteness | Prolepsis, Jacopin, v2, v1 2026-04-16; Random Choice switch; manifold vs linear steering (ICLR 2027 submission) | adjacent | n/a | n/a | linear vs manifold interpolation |
| 32 | ideas not in the tokens | departure-matched decoding (an ICLR 2027 submission on activation steering for mathematical reasoning): 92c source 66; Peeperkorn; AdapT; Gumbel counterfactuals; Beneath the Diff | adjacent; departure-matched decoding is the reviewer baseline | ADJACENT-ONLY | plan-level (algorithm) readout of temperature vs state noise at matched KL | decoding matched to the perturbation's departure from greedy |
| 33 | ideas not in the tokens | Forking Paths, Bigelow et al., v1 2024-12-10, anchor | adjacent (2.0) | n/a | n/a | forking-token resampling baseline |
| 34 | ideas not in the tokens | HELIX (4-bit Granite; temperature plus steering), v1 2026-02-06; Predicting Future Behaviors, ICLR 2027 submission | adjacent | n/a | n/a | n/a |
| 35 | resampling the plan | STARS (ICLR 2026), Noise Steering, REI, NC-GRPO, 2606.10302, Diversity Without Discovery: 92c sources 15, 14, 16, 52, 31, 63 | occupies the method family (activation noise for diversity) | PARTLY OCCUPIED | resampling a fitted, causally localized plan variable | matched-dose activation noise; text-level plan samplers (IFG, PlanSearch) |
| 36 | resampling the plan | Random Choice in LMs, ICLR 2027 submission, c | adjacent (2.0); closest to flag D | n/a | per-generation resampling for algorithm diversity | distribution-direction intervention |
| 37 | resampling the plan | LP-BD; PPLR; SVP (ICLR 2027 submissions, c) | adjacent (flag D partly) | n/a | a frozen LM; the variable is fitted, not trained | n/a |

| | | | | | | |
|---|---|---|---|---|---|---|
| 38 | resampling the plan | Latent Plan Transformer, v4, anchor (venue not stated in the arXiv record) | adjacent | n/a | n/a | n/a |
| 39 | the plan in words (NLA) | Fraser-Taliente, Kantamneni, Ong et al., the NLA paper, Transformer Circuits, published May 7, 2026, pub | OCCUPIES the method (rubric 2.5) | OCCUPIED (method) | replication in a new setting: algorithm choice in code, Qwen2.5-7B-Instruct at layer 20, text-only judges, an unrelated-word edit, a matched-norm random displacement | the paper's own newline-only steering protocol and success rate |
| 40 | the plan in words (NLA) | same paper, Reasoning about Rewards | occupies | n/a | n/a | AR-difference steering vector, norm-matched |
| 41 | the plan in words (NLA) | same paper, Method and Limitations | control the claim must meet (confabulation) | n/a | n/a | confabulation audit of plan claims |
| 42 | the plan in words (NLA) | Released checkpoints: kitft README; HF card; nla_meta.yaml; TransformerLens issue #1884 | design facts | n/a | n/a | load nla_meta.yaml injection settings |
| 43 | the plan in words (NLA) | LatentQA, Pan, Chen, Steinhardt, ICLR 2026, v2; SelfIE, ICML 2024, PMLR 235 | occupies the similar-verbaliser clause of flag C (rubric 2.0 each) | n/a | n/a | decoder-loss steering arm |
| 44 | the plan in words (NLA) | Activation Oracles, v2; Dreaming Vectors (LessWrong) | readout kin / steering kin (grey literature) | n/a | n/a | n/a |
| 45 | the plan in words (NLA) | Sood, Substack, 2026-08-18, c4; Auditing the Auditor, ICLR 2027 submission, c | control set the readout must meet | n/a | n/a | shuffled-activation control; visible-text baseline; blinded judge |

| | | | | | | |
|---|---|---|---|---|---|---|
| 46 | the plan in words (NLA) | Train the Model, Not the Reader, Dingeto, v2, v1 2026-07-22, c4 | contradicts in part: an edited claim may not move the AR output | n/a | measure the AR displacement caused by the edit before steering | claim-flip reconstruction test |
| 47 | the plan in words (NLA) | Colombian identity with NLA, v1 2026-07-23, c4 | readout replication on the same checkpoint | n/a | n/a | n/a |
| 48 | the plan in words (NLA) | Community Qwen3-8B NLAs: nanoNLA; EasyNLA | corrects the iteration-5 statement that no NLA exists for Qwen3-8B | n/a | n/a | an optional Qwen3-8B arm (unofficial; faithfulness disclaimer) |
| 49 | the plan in words (NLA) | Planning or Improvisation?, Jacopin, v1 2026-09-16, c4 | contradicts in part (no newline-resident plan in small open models) | n/a | n/a | site sweep (newline vs final prompt token) |
| 50 | the plan in words (NLA) | Measuring Activation Control; Understanding Confabulation | context | n/a | n/a | n/a |
| 51 | Gemma Scope 2 features | Gemma Scope 2: announcement 22 Dec 2025; technical report; HF card | design facts | n/a | n/a | report the IT-vs-PT dictionary used |
| 52 | Gemma Scope 2 features | circuit-tracer README; Qwen-3 transcoders (Hanna et al.) | design fact: Qwen3-8B transcoders exist, so a reviewer may ask why S uses Gemma | n/a | n/a | run S on the Qwen3-8B PLTs, or justify Gemma |
| 53 | Gemma Scope 2 features | AxBench, v3, v1 2025-01-28; rebuttal, v1 2026-05-29 | prior the claim must meet | n/a | n/a | difference-in-means and prompting arms |
| 54 | Gemma Scope 2 features | SAE-Steering, Fang et al., ACL 2026 Main, v2; Galichin et al. | OCCUPIES the method on a non-code strategy (rubric 2.5) | PARTLY OCCUPIED | algorithm choice in code; Gemma Scope 2; transcoder attribution | SAE-Steering-style logit-recall feature selection |
| 55 | Gemma Scope 2 features | Do SAEs Identify Reasoning Features? (Gemma Scope 2), ICML 2026, v7 | contradicts in part | n/a | n/a | token-injection falsification of candidate features |

| | | | | | | |
|---|---|---|---|---|---|---|
| 56 | Gemma Scope 2 features | Prolepsis; Latent Planning Emerges with Scale | adjacent; counter-evidence to the AxBench prior for plans | n/a | n/a | CLT/PLT arm against residual difference-in-means |
| 57 | Gemma Scope 2 features | SAE Interventions are Unreliable; variable roles in code LMs (ICLR 2027 submission); attribution-graph caveat | cautions | n/a | n/a | probe baseline; post-intervention recovery check |
| 58 | engine (8-bit floating-point W8A8) | Steering Under Compression, Bhandari and Wade, v1 2026-09-06, c4 | adjacent (1.5); flag B partly | ADJACENT-ONLY (published) | activation quantization; random perturbations; matched bf16 control | iso-effect comparison at matched behavioural effect |
| 59 | engine (8-bit floating-point W8A8) | Stable Directions, Different Effects, ICLR 2027, desk rejected, c | partly occupies the general claim (contemporaneous) | PARTLY OCCUPIED (contemporaneous) | random pushes; 8-bit floating-point W8A8 named; noise-floor account | a first-order local-sensitivity predictor |
| 60 | engine (8-bit floating-point W8A8) | Activation Steering under Low-Bit Quantization, ICLR 2027 submission, c | adjacent (1.0) | n/a | n/a | hold layer, direction and strength fixed |
| 61 | engine (8-bit floating-point W8A8) | Quantization Modulates Batch-Context Non-Determinism, ICLR 2027 submission, c | mechanism neighbour (noise floor) | n/a | perturbation effect sizes against the 8-bit floating-point noise floor | same-batch-context repeat runs |
| 62 | engine (8-bit floating-point W8A8) | PRISM, v1 2026-10-14, c4 | adjacent (4-bit NormalFloat) | n/a | n/a | n/a |
| 63 | engine (8-bit floating-point W8A8) | Readouts under quantization: Findings ACL 2026;;; AACL 2025; TrustNLP 2026 | flag B partly (readouts, not perturbation effect sizes) | n/a | n/a | n/a |
| 64 | engine (8-bit floating-point W8A8) | Precision-induced divergence: TMLR 2026;;; | supports the mechanism (rounding flips trajectories) | n/a | n/a | a zero-norm (unperturbed) re-run under the same engine |

| | | | | | | |
|---|---|---|---|---|---|---|
| 65 | engine (8-bit floating-point W8A8) | Mechanism sources: LLM.int8(); SmoothQuant; massive activations; 8-bit floating-point formats; the dynamic 8-bit floating-point scheme of llm-compressor | mechanism background | n/a | n/a | per-token-scale ablation |

| | | | | | | |
|---|---|---|---|---|---|---|
| 66 | one-token fork (Gemma) | Forking Paths in Neural Text Generation (Bigelow et al.) (ICLR 2025; arXiv 2412.07961 v1 10 Dec 2024); Forking Fast (no venue on the arXiv record; no NeurIPS 2026 paper page found by the dated search; arXiv 2608.19611 v1 20 Aug 2026); Latent Planning Emerges with Scale (Hanna and Ameisen) (ICLR 2026; arXiv 2604.12493 v1 14 Apr 2026); Detecting and Characterizing Planning in Language Models (Nainani et al.) (no venue on the arXiv record; arXiv 2508.18098 v1 25 Aug 2025); Where's the Plan? (Ma and Rui) (Workshop on Mechanistic Interpretability @ ICML 2026; arXiv 2605.07984 v1 8 May 2026, v2 11 Jun 2026); Planning or Improvisation? (Jacopin) (no venue on the arXiv record; arXiv 2609.18440 v1 16 Sep 2026) | adjacent: forking-token resampling and planning detection are occupied as methods; none of these reports an algorithm-level fork in generated code at a token that does not identify the algorithm | ADJACENT-ONLY (to our knowledge, mini search dated 2026-10-22) | an algorithm-level fork (BFS or DFS) in generated code at one token that does not identify the algorithm, about 138 tokens before the code that does; its fork log-odds varies across 120 prompt variants (SD 0.67 nats) and is linearly decodable from the residual at the fork (R-squared 0.66 Lasso on SAE codes, 0.72 raw ridge), which at that position is close to a next-token readout, not evidence of a plan; steering costs are measured (pass 0.84 to 0.16-0.58) | forking-token resampling baseline; a logit-lens control of matched capacity (iteration-7 correction) |

## 6.8 Planned but not run

The seventh artifact of the strategy never existed. It was to fit DAS subspaces of rank 1 to 16 at one layer of the task-specification positions in bf16 Qwen3-8B on three training templates and confirm them on two held-out templates (Stage B of the localization claim), test whether a fresh random

direction's energy in that subspace predicts its effect (the mediation claim), interpolate and extrapolate the plan coordinate between algorithms (the discreteness claim), measure the plan coordinate's variance across temperatures (the first half of the token claim), resample the plan per generation as a diversity knob (the resampling claim), and draw the request's manifold of algorithms in the fitted plan subspace, with the top-five prohibition placed on it. It was lost at the planning stage: the planning agent returned no plan, so no experiment ran. These claims are untested and open; nothing in this report is a result for them, null or otherwise. The only DAS result of the iteration is Gemma's, at the fork position (Section 6.5). The deviation log of the Gemma experiment adds what it did not run: the clamp read-back and matched-norm realisation tests as separate unit tests (the matched norm is imposed by construction in the clamp code, `src/clamp.py` (file name restored, iteration-8 correction), and the realised norm of each edit is logged per row); the clamp-cell overlay on the generated-position panel; the fork and the cross-prompt readouts of two further bench tasks (dropped in order); and per-sample displacement at generated tokens 1-32 (prefill displacement at blocks 11-33 is logged). (iteration-7 correction)

## 6.9 Comparisons across this iteration's experiments

**Where the choice is held, model by model.** The three internal experiments measured different quantities at different sites, so their results are stated per model and per quantity (Table 279) rather than as one map. In Qwen3-8B what was measured is where an external ban or instruction is carried: exchanging the task-specification states alone reproduces the substitute of the ban on shortest path (Bellman-Ford/SPFA in 24 of 30 attempts, 21 of 30 correct programs) and a forced instruction on shortest path (30 of 30 attempts, 30 of 30 correct) and connected components (BFS in 28 of 30 attempts, 24 of 30 correct), while on the direct-route tasks the condition span carries it; Qwen3-8B's natural choice was not measured, because it writes one algorithm per prompt. In gemma-3-4b-it, which writes two algorithms for one prompt, the natural choice is made at one generated token after a shared opening; task-specification edits and readouts barely reach it, and the read position was never read or edited (Gemma's SPEC site is a mean over the task-specification positions, which exclude the read position). In Qwen2.5-7B-Instruct no verbalisable plan was found at pre-code positions: the NLA names the algorithm at the token where the prompt names it and not 170 tokens later, and edits work only at the generated-token site. The read position is inert as a carrier in Qwen3-8B (its exchange transfers nothing on six tasks) and reads only the format line in the NLA; Gemma's read position was not measured. The finding of iteration 3 that a ban gives way at the opening of the program body and the fork of Gemma at response index 32 point at the same moment from two models: the first statement of the body that is specific to an algorithm, after an opening shared by every algorithm.

**Table 279.** What was measured where, per model and quantity (NEW in iteration 7). Sources: exchange and site experiment `results/stage_a.json`; Gemma feature experiment `results/fork.json`, `results/readout.json`, `src/cfg.py`; NLA experiment `results/readout.json`, `results/local_control.json`. (iteration-7 correction)

| Model | Quantity | Site measured | Instrument | Result | Sites never measured |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| Qwen3-8B | where an external ban or instruction is carried | task-specification states (blocks 0-11, 12-23, 24-35, 0-35), condition span, read position | full-state exchange and KV splice into a token-matched twin | carried by the task-specification states on prompt-route tasks (shortest path 24 of 30 attempts, 21 of 30 correct; kth smallest 17 of 30, 16 of 30) and by the condition span on direct-route tasks; read position 0 on six tasks | generated tokens; any low-rank subspace (Stage B not run) |
| Qwen3-8B | its natural choice | not measured (one algorithm per prompt) | n/a | n/a | every site |
| gemma-3-4b-it | its natural choice (BFS or DFS) | generated token at response index 32; mean over the task-specification positions | prefix resampling; cross-prompt readout; SAE, difference-in-means and DAS edits | one generated token decides; task-spec edits and readouts barely reach it (SAE codes = text, 0.18) | the read position |
| gemma-3-4b-it | the read position | never read or edited | n/a | n/a | the read position |
| Qwen2.5-7B-Instruct | a verbalisable plan | end-of-prompt positions (SPEC-3 to READ), the name token, generated tokens 1 to 16 | NLA explanations judged blind | none at pre-code positions; the algorithm is named at its name token (80% forced, 89% ban); per sample the outside reader is ahead (0:25 at one unit per sample) | non-verbal carriers at any position |

**The top-five prohibition in two families.** Both models answer a five-name ban by writing a banned algorithm (avoidance 0.00 in Qwen3-8B in iteration 2, in its bf16 rerun here and in Gemma). In Gemma the ban flips the model from its default DFS to its minority BFS and halves its pass rate; in activation space the ban's prompt states sit beside a banned centroid. At doses of 1 to 8 ban units, which overshoot the forced-prompt manifold, the activation version of the prohibition breaks programs before it moves the algorithm; an in-manifold dose was not tried. (iteration-7 correction)

**Novelty, audited again.** No artifact of this iteration found a correct novel algorithm (Table 278).

**Table 278.** Novelty audits of iteration 6. Sources: the novelty audit files of the five artifacts.

| Artifact | Pool | Correct programs outside the catalogue | Final classes | Novel |
|---|---|---|---|---|
| Record audit | the 14 transplant programs that two of three judges had called novel in iteration 5, plus 10 controls and 17 related programs | 14 disputed | known outside the catalogue 10, catalogue variant 2, degenerate 2 | 0 |

| | | | | |
|---|---|---|---|---|
| Engine control | every correct off-default or other program, against 18,237 known solutions (nearest-known pass only); its 3 correct programs labelled other judged blind in iteration 7 | up to 237 per engine and task; 3 labelled other | nearest-known pass: variants and distant programs of known algorithms; the 3 programs: catalogue variant 3 (blind judges: gpt-4.1-mini gate 8/10; deepseek-chat-v3.1 gate 10/10; gemini-2.5-flash-lite gate 8/10; gated deepseek-chat-v3.1; kappa n/a on merged classes; manual read) | 0 (iteration-7 correction) |
| Exchange and site experiment | every correct program labelled other | 2 | catalogue variant 2 | 0 |
| Gemma feature experiment | 1,450 correct programs | 1 | catalogue variant | 0 |
| NLA experiment | every correct program labelled other, all arms | 10 | labeller miss 8, degenerate 2 | 0 |

## 6.10 What iteration 6 established, what it ruled out, and what it abandoned

Everything below is one model per claim except where a second family is named, at 4B to 8B parameters, on memorised tasks in the style of LeetCode.

**Established.**

1. *Iteration 4's random-push escapes are a property of Qwen3-8B, not of 8-bit floating-point rounding.* At the same absolute displacement, the same 40 directions and the same slots, bf16 escapes on connected components at 0.65 against 0.725 under W8A8 and on shortest path at 0.15 against 0.20, with per-direction shares ranked alike (Spearman 0.96 and 0.87), and per-sample plan noise reproduces (25 and 22 of 50 against 30 and 22). Activation rounding alone adds nothing; a 5-point per-slot boost on connected components comes from the 8-bit floating-point weights or the native kernel. Iteration 5's bf16 null belongs to its ban-free unit or its direction seeds.
2. *In Qwen3-8B the substitute of a ban and the algorithm of an instruction are carried by the residual states at the task-specification positions on the prompt-route tasks.* Exchanging those states alone transfers the ban on shortest path (24 of 30 attempts, 21 of 30 correct programs) and kth smallest (17 of 30 attempts, 16 of 30 correct), and a forced instruction on shortest path (30 of 30 attempts, 30 of 30 correct) and connected components (28 of 30 attempts, 24 of 30 correct); matched-norm random exchanges and a placebo twin transfer at most 0.13. The condition span carries the ban on the direct-route tasks (two sum 1.00, find duplicate 0.90). The read position carries nothing. (iteration-7 correction)
3. *Gemma's BFS-or-DFS choice is made at one generated token,* `visited` against a blank line, `if` or `queue` at response index 32, about 138 tokens before any algorithm-identifying code, and its fork log-odds is linearly decodable from the residual stream at that position across prompts (Lasso on SAE codes R-squared 0.66 against 0.18 for text, preregistered; ridge on the raw state 0.72, post hoc); at the fork position that is close to a next-token readout, so it shows that the next-token distribution is linearly encoded there, not that a plan is, and the standardised ridge on the same layer-29 SAE codes reaches R-squared 0.15 against 0.086 for a template-plus-noun-norm control (delta 0.064, below the preregistered 0.10 bar).

(iteration-7 correction)

4. *Steering the fork works and costs correctness.* Difference in means moves the fork 2.3 times as far as the SAE clamp at matched norm, both move the plan equally at matched effect, and edits toward the minority algorithm lowered the pass rate from 0.84 to 0.16-0.58, except the SAE top-5 clamp (18 BFS, pass 0.84); edits toward the default are clean. (iteration-7 correction)
5. *The NLA does not verbalise the planned algorithm at pre-code positions in Qwen2.5-7B-Instruct,* although it names an algorithm at the token where the prompt names it; an outside reader of the text predicts the realised algorithm better than the NLA per sample (0:25 at one unit per sample, p 6.0e-08), and at the task level the evidence is inconclusive (5 task clusters; bootstrap interval [-0.35, 0.02]). (iteration-7 correction)
6. *Record repairs.* The landing half is descriptive (17 of 21 per set, 5 of 8 where the twin wrote no alternative, p 0.053); the movability half is not a strong predictor with a moderate relation not excluded (0.15, interval -0.44 to 0.65); the 14 disputed programs are not novel.

**Ruled out, or not supported.**

- *Activation rounding as the cause of the escapes* (the engine claim's preregistered account): W16A8 escapes like bf16 or slightly less, on both tasks.
- *The Stage A rule of the localization claim* as preregistered: the task-specification states of two sum also transfer brute force at 0.43, above the predicted 0.2.
- *A feature claim for Gemma Scope 2:* the SAE clamp never beats difference in means, and the preregistered feature rule is met by random latents.
- *The verbal readout claim:* 0 of 16, 0 of 112 and 0 of 96 at the end of the prompt; the preregistered verbal edit moves nothing at the prefill site.
- *State noise against token temperature at matched KL* in bf16: 12:6 and 7:3 at alpha 2, not significant.

**Withdrawn.**

- *The prefill-site claim in its preregistered form* in bf16: no common support for the condition span, and breakage at alpha 8. Post hoc (alpha 4 or less) the read position is below the task-specification site on both tasks, but the condition-span clause fails (-1.70 [-3.96, 0.42] and -0.56 [-4.93, 2.91]), so the site of the choice is open, as the rule label of the artifact says (withdrawn as specific to 8-bit floating point, with the site of the choice open); its engine half is refuted by Section 6.3. (iteration-7 correction)

**Dead ends, with the evidence that closed them.**

- *8-bit floating-point numerics as an explanation of anything in this run's internal results:* closed by Table 237 and Table 239.
- *The read position as a carrier of the choice:* zero transfer in six Qwen3-8B tasks (Tables 244 to 249) and only the format line in the NLA explanations. Gemma's read position was never read or edited (its task-specification site is a mean over the task-specification positions, from the first token of the task statement to the last token of the user message, `src/cfg.py` line 7; file name restored, iteration-8 correction), so Gemma does not bear on this dead end. (iteration-7 correction)
- *The released layer-20 NLA as a readout of algorithm plans in code:* code is out of distribution for

it (fraction of variance explained 0.39 to 0.45), and its pre-code explanations name no algorithm.

**Planned but not run.** DAS in Qwen3-8B (Stage B), the mediation, discreteness and resampling claims, the temperature half of the token claim, and the causal manifold (Section 6.8).

### 6.11 What the next iteration takes from this one

The plan-variable account was tested in part, and its two halves came out differently. Three different quantities were measured, one per model (Table 279): where a ban or instruction is carried in Qwen3-8B (the task-specification states on the prompt-route tasks), where the natural choice of Gemma is made (one generated token), and whether the plan of Qwen2.5-7B-Instruct is verbalisable before the code (not to this NLA); the natural choice of Qwen3-8B and the read position of Gemma were never measured. Whether the choice is a low-rank variable in Qwen3-8B, and whether random pushes act through it, is untested, because the DAS experiment was lost at planning. The engine control has removed the obstacle that made such a test look premature: random pushes at iteration 4's absolute unit move the algorithm in bf16 as they did in 8-bit floating point, so the escapes are a stable target that a fitted plan subspace can be asked to explain. The next measurements are DAS at the task-specification positions of shortest path and kth smallest, with the Stage A exchanges as the full-state ceiling, and the projection of iteration 4's 40 directions onto the fitted subspace. (iteration-7 correction)

Three leads come with it. The ban-free unit moves fewer programs than the ban unit at a larger scalar dose, so its per-position and per-block profile, or the direction seeds, decide movability; a controlled comparison of the two units on the same seeds would say which. Gemma's one-token fork offers a clean site where a rank-1 DAS subspace moves the fork log-odds by 8.8 nats, a fit that needs the Sutter controls (random initialisation, shuffled targets, forced-DFS sources) before it says anything about dimensionality; whether a smaller push there switches the algorithm without breaking the program is the question for a usable knob. And two sum's task-specification states carry part of a ban that iteration 3 had classed as direct, so the route classes are not all-or-none. (iteration-7 correction)

# Iteration 7: a fitted plan subspace, the fork where the algorithm is decided, temperature through the fork, the dose unit, evolutionary search, and a circuit in Gemma

### 7.1 Why this iteration ran what it ran

**The review of the iteration-6 record.** The review was blocking. It raised five major items and seven minor ones. The first major item went to Stage A, the cleanest internal positive of the run. The record had reported the transfer counts of Stage A as counts of correct programs when they were counts of attempts labelled with the ban's substitute, so a reader could not tell how many of the transferred programs passed. The second said the post hoc site ordering had been stated as if it held on both clauses when its clause on the condition span failed. The third went to Gemma: the record had called a rank-1 DAS subspace one dimensional without the controls that would license the word, had quoted an unstandardised SAE ridge, and had said every edit lowered the pass rate when the SAE top five clamp did not. The fourth said the closing merged quantities measured in three models at three sites into one account. The fifth was coverage: the DAS experiment had been

lost at planning, and the geometry plots that the request asks for in activation coordinates had no figure markers in the record. The minor items asked for the dose of the top five push against the forced prompt manifold, a version of Table 238 restricted to correct programs, a fix for pseudoreplication in the NLA contrast, three executed checks that the record named but did not report, a wrap of the overlong Section 4.9, a positioning row for the fork, and the 2x2 of dose unit against direction seeds before any repricing of plan noise.

**Table 280.** Review items on the iteration-6 record and where this report answers them.

| Review item | What the review found | Where it is answered |
|---|---|---|
| Stage A metric | Attempts reported as correct programs; no correct column | Corrections in Sections 6.4.1 and 6.10; Section 7.2 (Tables 284 and 285) |
| Post hoc site | The ordering overstated; the clause on the condition span fails | Corrections in Sections 6.4.2 and 6.10 and in the closing; Table 254b; Section 7.2 |
| Gemma claims | A single dimension claimed without controls; unstandardised SAE ridge; the claim that every edit lowered the pass rate | Corrections in Sections 6.5, 6.10 and 6.11; the controlled DAS of Section 7.8 |
| Synthesis | Quantities from three models merged | Corrections in Sections 6.9, 6.10 and 6.11; Table 279 in Section 6.9; the closing section |
| Coverage | DAS lost; no geometry figure markers | Section 7.3 (DAS); seven geometry figures placed in Sections 6.4.4, 6.5 and 6.6 |
| Minor: top five push dose | Doses of 1 to 8 ban units overshoot the forced prompt manifold | Corrections in Sections 6.4.4 and 6.9 and in the closing; the push inside the manifold of Section 7.4.7 |
| Minor: Table 238 | Attempts only | Rows restricted to correct programs and the MIXED verdict in Section 6.3; Table 286 |
| Minor: NLA pseudoreplication | Several units per sample counted as independent | Corrections in Sections 6.6 and 6.10; Table 287 |
| Minor: executed checks | The resampling from the empty prefix test at the Gemma fork, the list of what the Gemma experiment did not run, and three programs of the engine control labelled other, all unreported | Corrections in Sections 6.5, 6.8 and 6.9; Table 288 |
| Minor: Section 4.9 | Unwrapped | Rewrapped in place |
| Minor: fork positioning | No row for the fork against published work | Row 66 of Table 277 in Section 6.7 |
| Minor: 2x2 | Dose unit and direction seeds confounded in iteration 5 | Verdict tags in Sections 5.7, 5.11 and 6.2.4 and in the closing; Section 7.6 |

**The hypothesis update.** The update kept the plan and added a fork. We call it the *plan-and-fork account*. While the model reads the task, the residual states at the task specification positions carry a variable of low rank that sets which algorithm the model will write; during generation, the program opens with tokens that every algorithm shares, and the variable is realised at the first token where continuations of different algorithms part. We call that token the *fork*, a forking token in the sense of Bigelow et al. [30]. The *fork margin* is the teacher-forced log odds, at the fork, of the

default class of tokens against the class off the default. If the account holds, three things follow. A subspace of low rank fitted by DAS [110, 111] should carry the transfer of a ban or a forced instruction, under the controls that would expose a spurious fit. A random push should change the algorithm in proportion to how far it moves the fork margin. And temperature should act on the algorithm only through the fork: at temperature T, the share off the default at the fork is the softmax of the fork logits divided by T, so the account predicts each temperature's share from the logits alone, before sampling.

Ten claims were preregistered, each with a rule committed by sha256 before its first decisive sample. We call them by their content.

- *The fork claim* (primary): the teacher-forced fork margin under each of 40 equal-dose random pushes predicts which pushes change the algorithm, on connected components and shortest path.
- *The DAS claim* (primary): a DAS map of rank 16 or less at the site of Stage A recovers at least 0.8 of the full state transfer (the transfer when every residual state at that site is exchanged, which we call the *full state exchange*) in held-out generation, beats random subspaces of the same rank, and survives a fit on permuted labels and a fit on a randomly initialised network [132].
- *The routing claim*: whether an edit in the fitted subspace lands where the forced source points (categorical) or follows a margin toward the ban's usual substitute.
- *The axis claim*: the projection of a random push on an axis from the default to the second choice predicts and mediates its effect; we call *second choice* the algorithm a model ranks second on a task, the runner-up of earlier sections.
- *The temperature claim*: at mixed units, the algorithm given the fork token's class is nearly certain, the share off the default at each temperature follows the fork logits divided by T, and on the collapsed tasks the fork margin is large enough that no temperature in range reaches the alternative.
- *The manifold claim*: in the fitted subspace, bans sit outside a paraphrase null and seeds and instructions sit inside it.
- *The discreteness claim*: scaling the fitted edit from 0 to 1.5 changes the algorithm as a step, not a ramp.
- *The dose unit claim*: the 2x2 of dose unit (the absolute unit of iteration 4 against the ban-free unit of iteration 5; we call the latter the *relative unit*, because it scales the noise to a fixed fraction of the twin's residual norm) by direction seeds (those of iteration 4 against those of iteration 5) says which explains iteration 5's weak plan noise, and plan noise is then repriced at the unit that moves.
- *The Gemma claims*: temperature through the fork in a second family, DAS at Gemma's fork with the full set of controls, and a validated circuit for the fork margin.
- *The search claim*: an evolutionary program search with an executable evaluator writes correct programs outside the catalogue at least 10 times as often per call as the best knob applied to single samples, on at least two tasks.

**The strategy.** Seven artifacts ran (Table 281). The record audit repaired the record from rows on disk and ran the checks the review asked for. A DAS experiment fitted the plan subspace on Qwen3-8B [5] and tested it in generation. A fork experiment located forks by prefix resampling and tested the fork and axis claims. A temperature experiment tested the temperature claim on the

three units where the instruct twin is already mixed. A dose unit experiment ran the 2x2 and repriced plan noise. A search experiment ran evolution in the style of FunSearch [81] over a MAP-Elites archive [133]. A Gemma experiment ran temperature, DAS and a circuit at the fork of Gemma.

**Table 281.** The artifacts of iteration 7, the question each was asked, and its compute and spend.

| Artifact | Question | Model and compute | OpenRouter spend |
|---|---|---|---|
| Record audit | Repair the iteration-6 record (54 corrections); run the checks the review named | No generation; CPU only | USD 0.077 |
| DAS experiment | Does a fitted subspace of low rank at the task specification states carry the choice, under controls with permuted labels, a randomly initialised network and dormancy? | Qwen3-8B bf16, thinking off; 4,500 programs; about 1.6 GPU hours | USD 0.13 |
| Fork experiment | Where is the algorithm decided in generation; does the fork margin, or an axis projection, predict which random pushes escape? | Qwen3-8B bf16; 4,220 programs and 4,560 continuations; 0.63 GPU hours | USD 0.028 |
| Temperature experiment | Does temperature act on the algorithm only through the fork logits? | Qwen3-8B bf16; 2,290 programs and 12,676 continuations; about 0.75 GPU hours | USD 0.022 |
| Dose unit experiment | Is iteration 5's weak plan noise the dose unit or the direction seeds; what does plan noise cost at the unit that moves? | Qwen3-8B bf16, one RTX 4090; about 0.7 GPU hours | USD 0.10 |
| Search experiment | Does evolutionary search with an evaluator and a novelty critic find correct programs outside the catalogue faster than knobs applied to single samples? | Qwen3-8B bf16 on vLLM 0.11.0, one RTX A4500; about 1.0 GPU hour | USD 0.89 |
| Gemma experiment | Temperature, controlled DAS and a validated circuit at Gemma's fork | gemma-3-4b-it [76] bf16, one 16 GB GPU | USD 0.044 |

## 7.2 The record audit: 54 corrections and the checks the review named

The record audit read the iteration-6 report against the rows on disk and wrote 54 corrections, each a replacement, a table replacement or an insertion. All 54 apply exactly once to the report as it stood, and all of them are now in place, each marked as an iteration-7 correction with what changed and why. Of the 724 values the corrections touch, 603 match the rows, 117 are new (the correct columns of Stage A, the NLA tests with one unit per sample, the executed checks), one is carried from an earlier artifact, and 3 are corrected. The three corrected values are venue expectations of the plan that the live records contradict. A lint for residual claims flagged 16 sentences in the uncorrected report and none after correction. An independent recomputation with the standard library ran 520 checks with no mismatch, and seven placebos, each a deliberately wrong input, failed as they must (Table 282).

**Table 282.** The record audit's own checks.

| Check | Result |
|---|---|
| Corrections | 54, all applying exactly once (apply test 54/54), with no overlaps |
| Values ledgered | 603 MATCH, 117 NEW, 1 CARRIED, 3 CORRECTED (the 3 are venue expectations of the plan that the live records contradict) |

| | |
|---|---|
| Lint for residual claims | 16 flags in the uncorrected report, 0 in the corrected report |
| Independent audit (standard library only) | 520 checks, 0 mismatches; 7 of 7 placebos fail as they must |
| Headline recomputation (standard library, its own code path) | 35 checks, 0 failed; 7 of 7 placebos fail |
| Check against the dependencies' own prediction fields | 137 checks, 0 mismatches |
| Inputs | 69 files vendored and hashed with SHA-256, 0 missing |
| Spend and time | OpenRouter USD 0.077; a full offline pass takes about 20 s of wall time, and the judge stage about 2 minutes |

Table 283 lists every correction with the review item behind it, where it sits and the first sentence of its rationale. Corrections 27, 28, 44, 45, 48, 53 and 54 touched the closing section, which this report then rewrites in full.

**Table 283.** The 54 corrections of the iteration-6 record.

| No. | Review item | Location | Kind | Values behind it | What changed (first sentence of the rationale) |
|---|---|---|---|---|---|
| 01 | Stage A metric | 6.4.1, Table 244 | table replace | MATCH 10 | the first commitment's outcome restored as primary; |
| 02 | Stage A metric | 6.4.1, Table 245 | table replace | MATCH 10 | the first commitment's outcome restored as primary; |
| 03 | Stage A metric | 6.4.1, Table 246 | table replace | MATCH 10 | the first commitment's outcome restored as primary; |
| 04 | Stage A metric | 6.4.1, Table 247 | table replace | MATCH 7 | the first commitment's outcome restored as primary; |
| 05 | Stage A metric | 6.4.1, Table 248 | table replace | MATCH 7 | the first commitment's outcome restored as primary; |
| 06 | Stage A metric | 6.4.1, Table 249 | table replace | MATCH 7 | the first commitment's outcome restored as primary; |
| 07 | Stage A metric | 6.4.1, Table 250 | table replace | MATCH 36 | column of correct Y programs for the extension cells |
| 08 | post hoc site | 6.4.2, Table 254b | insert after | wording only | NEW concordance table of noise sites against Stage A |
| 09 | Table 238 | 6.3, Table 238 | table replace | MATCH 48, NEW 16 | columns restricted to correct programs beside the attempted counts |
| 10 | NLA pseudoreplication | 6.6, Table 270 | replace | wording only | the NLA contrast restated with its clustered interval |
| 11 | executed checks | 6.5 | insert after | MATCH 5, NEW 2 | the resampling from the empty prefix test and the Gemma list of what did not run, executed or logged but never reported |
| 12 | executed checks | 6.8 | insert after | wording only | the Gemma list of what did not run, into Section 6.8 |
| 13 | Section 4.9 | 4.9 | replace | CARRIED 1, NEW 1 | iteration-6 insertion wrapped; |
| 14 | geometry figures | 6.4.4 | insert after | NEW 1 | captioned figure markers for the geometry in activation space plots |
| 15 | geometry figures | 6.5 | insert after | NEW 1 | captioned figure markers for the geometry in activation space plots |

| | | | | | |
|---|---|---|---|---|---|
| 16 | geometry figures | 6.5 | insert after | NEW 1 | captioned figure markers for the geometry in activation space plots |
| 17 | geometry figures | 6.5 | insert after | NEW 1 | captioned figure markers for the geometry in activation space plots |
| 18 | geometry figures | 6.5 | insert after | NEW 1 | captioned figure markers for the geometry in activation space plots |
| 19 | geometry figures | 6.6 | insert after | NEW 1 | captioned figure markers for the geometry in activation space plots |
| 20 | geometry figures | 6.6 | insert after | NEW 1 | captioned figure markers for the geometry in activation space plots |
| 21 | executed checks | 6.9, Table 278 | replace | MATCH 1, NEW 1 | checks that ran but were never reported, and the judged programs of the engine control |
| 22 | positioning | 6.7, Table 277 | insert after | CORRECTED 3, MATCH 5, NEW 6 | positioning row, verified by quotation, for the fork of a single token |
| 23 | Stage A metric | 6.4.1 | replace | wording only | Stage A outcome returned to the first commitment's definition (attempted share, invalid as not Y, gate on the pass ratio) with a count of correct Y programs beside every share |
| 24 | Stage A metric | 6.4.1 | replace | wording only | Stage A outcome returned to the first commitment's definition (attempted share, invalid as not Y, gate on the pass ratio) with a count of correct Y programs beside every share |
| 25 | post hoc site | 6.4.2 | replace | wording only | the site of the choice stated as open: the clause on the condition span fails post hoc; |
| 26 | post hoc site | 6.10 | replace | wording only | the site of the choice stated as open: the clause on the condition span fails post hoc; |
| 27 | post hoc site | What we have learned so far | replace | wording only | the site of the choice stated as open: the clause on the condition span fails post hoc; |
| 28 | post hoc site | What we have learned so far | replace | wording only | the site of the choice stated as open: the clause on the condition span fails post hoc; |
| 29 | Table 238 | 6.3 | replace | wording only | columns restricted to correct programs for Table 238 and the MIXED verdict at the screen level |
| 30 | Gemma claims | 6.5 | replace | wording only | Gemma's DAS and readout stated within the controls they had |
| 31 | Gemma claims | 6.5 | replace | wording only | Gemma's DAS and readout stated within the controls they had |
| 32 | Gemma claims | 6.5 | replace | wording only | Gemma's DAS and readout stated within the controls they had |
| 33 | Gemma claims | 6.5 | replace | wording only | Gemma's DAS and readout stated within the controls they had |
| 34 | Gemma claims | 6.5 | replace | wording only | Gemma's DAS and readout stated within the controls they had |
| 35 | Gemma claims | 6.10 | replace | wording only | Gemma's DAS and readout stated within the controls they had |

| | | | | | |
|---|---|---|---|---|---|
| 36 | Gemma claims | 6.10 | replace | wording only | Gemma's DAS and readout stated within the controls they had |
| 37 | Gemma claims | 6.11 | replace | wording only | Gemma's DAS and readout stated within the controls they had |
| 38 | NLA pseudoreplication | 6.6 | replace | wording only | NLA per-sample test quoted with its clustered interval and at one unit per sample |
| 39 | NLA pseudoreplication | 6.10 | replace | wording only | NLA per-sample test quoted with its clustered interval and at one unit per sample |
| 40 | synthesis | 6.9 | replace | wording only | synthesis per model and per quantity in place of one map across models |
| 41 | synthesis | 6.11 | replace | wording only | synthesis per model and per quantity in place of one map across models |
| 42 | synthesis | 6.10 | replace | wording only | synthesis per model and per quantity in place of one map across models |
| 43 | Stage A metric | 6.10 | replace | wording only | Stage A outcome returned to the first commitment's definition (attempted share, invalid as not Y, gate on the pass ratio) with a count of correct Y programs beside every share |
| 44 | synthesis | What we have learned so far | replace | wording only | synthesis per model and per quantity in place of one map across models |
| 45 | synthesis | What we have learned so far | replace | wording only | synthesis per model and per quantity in place of one map across models |
| 46 | top five dose | 6.4.4 | replace | wording only | top five push dose caveat (1 to 8 ban units overshoot the forced prompt manifold) |
| 47 | top five dose | 6.9 | replace | wording only | top five push dose caveat (1 to 8 ban units overshoot the forced prompt manifold) |
| 48 | top five dose | What we have learned so far | replace | wording only | top five push dose caveat (1 to 8 ban units overshoot the forced prompt manifold) |
| 49 | verdict tags | 5.7 | replace | wording only | iteration-5 verdicts tagged as conditional |
| 50 | verdict tags | 5.11 | replace | wording only | iteration-5 verdicts tagged as conditional |
| 51 | verdict tags | 5.11 | replace | wording only | iteration-5 verdicts tagged as conditional |
| 52 | verdict tags | 6.2.4 | replace | wording only | iteration-5 verdicts tagged as conditional |
| 53 | verdict tags | What we have learned so far | replace | wording only | iteration-5 verdicts tagged as conditional |
| 54 | verdict tags | What we have learned so far | replace | wording only | iteration-5 verdicts tagged as conditional |

**Stage A in correct programs.** The review's first item is answered by counting correct programs beside attempts (Table 284). The ban's substitute transfers into the neutral twin on shortest path as 24 attempts and 21 correct programs of 30, and on kth smallest as 17 and 16. The exchange through blocks 12 to 23 alone on shortest path gives 18 attempts but 11 correct programs, at a pass ratio of 0.73 against the twin, below the gate; the grid exchange through all blocks gives 14 attempts and 3 correct programs at a pass ratio of 0.59. Five cells fail the pass gate. A post hoc McNemar test on correct programs against random exchanges of matched norm keeps the

preregistered conclusion on the three tasks (Table 285).

**Table 284.** Stage A exchanges of iteration 6, attempts and correct programs.

| Cell | Attempts labelled Y | Correct programs | Note |
|---|---|---|---|
| shortest path, blocks 0-35 | 24/30 | 21/30 | |
| shortest path, blocks 12-23 | 18/30 | 11/30 | pass ratio 0.73, below the gate |
| kth smallest, blocks 0-35 | 17/30 | 16/30 | |
| grid, blocks 0-35 | 14/30 | 3/30 | pass ratio 0.59, below the gate |
| grid, condition span | 10/30 | 7/30 | |
| forced Bellman-Ford/SPFA exchange | 30/30 | 30/30 | twin 1/30, so the first commitment's transfer is 0.967 |
| forced BFS exchange | 28/30 | 24/30 | pass 0.87 |

**Table 285.** Stage A against random exchanges of matched norm, on correct programs (post hoc) beside the preregistered attempted contrast.

| Task | Correct Y | Exact p | Preregistered attempted contrast |
|---|---|---|---|
| shortest path | 21:0 | 9.5e-7 | 23:0 |
| kth smallest | 15:2 | 0.0023 | 16:2 |
| connected components | 11:0 | 9.8e-4 | 11:0 |

**The site, Gemma and the synthesis.** The site ordering is now stated as open: a post hoc model at alpha 4 or less puts the read position below the task specification site on both tasks but not the condition span, whose coefficient intervals cross zero (beta -1.70 [-3.96, 0.42] on kth smallest and -0.56 [-4.93, 2.91] on connected components). The engine half of the rule label, which withdrew the site claim as specific to 8-bit floating point, is refuted by Section 6.3, and the corrected text says so. A new Table 254b sets, for each task, the band where noise acts against the band where the exchange transfers: they agree on kth smallest (blocks 12 to 23) and disagree on connected components, where noise at blocks 0 to 11 under the iteration-4 seeds raises the log odds of escape by 2.51 [1.39, 3.80] while the exchange transfers at blocks 12 to 23. The Gemma paragraph now quotes the standardised SAE ridge (0.15 against 0.086 for the control built from the template and the norm of the noun, a gain of 0.064 below the preregistered 0.10) beside the unstandardised one, and says that the edits toward BFS lowered the pass rate from 0.84 to between 0.16 and 0.58 except the SAE top five clamp (18 BFS at pass 0.84). Table 279, inserted at the end of iteration 6, keeps each quantity with its own model and site.

**Correct programs in Table 238.** On shortest path, per-sample plan noise under the 8-bit engine (W8A8) gives 14 correct programs off the default among its 37 correct programs at alpha 2, against 9 of 31 in bf16, and 9 of 43 against 5 of 43 at alpha 1 (Table 286). The screen verdict on 8-bit rounding becomes MIXED: under W8A8, 15 of 40 directions pass the screen against 8 to 9 under the other numerics, and 17 of 40 meet the excess but fail the pass clause.

**Table 286.** Table 238 restated on correct programs (shortest path): correct programs off the default over all correct programs, per-sample plan noise at the absolute unit.

| Alpha | W8A8 | bf16 |
|---|---|---|
| 2 | 14/37 | 9/31 |
| 1 | 9/43 | 5/43 |

**The NLA contrast without pseudoreplication.** The contrast per unit of iteration 6 (20:88 over 364 units, p 2.3e-11) counted several units from one sample. With one unit per sample, the outside reader of the visible text still beats the NLA in every version of the test (Table 287); at the level of

the four tasks the contrast is 1:3 with p 0.625, and the smallest p four tasks can give is 0.0625, so the task level is uninformative by construction.

**Table 287.** The NLA against the outside reader with one unit per sample.

| Test | Result |
|---|---|
| First unit per sample | 0:25, p 6.0e-8 |
| 2,000 random draws of one unit per sample | median 4:29, p 1.9e-5; every draw below 0.05 |
| Sign test on per-sample differences | 12:38, p 3.1e-4 |
| McNemar with the Durkalski adjustment | p 6.5e-5 |
| Task level | 1:3, p 0.625 |

**The executed checks.** Three checks the review named were reported. The test from the empty prefix of iteration 6's Gemma fork resampling (a BFS share of 0.38, 73 of 192, from the empty prefix against the natural 0.305 [0.245, 0.372]) gave an exact p of 0.028, and the Gemma experiment's own list of what it did not run (two unit tests replaced by checks built into the edit code, one figure overlay, a fork and two readouts across prompts on further tasks, and per-sample displacement at generated tokens) is now quoted in Sections 6.5 and 6.8. The three correct programs of the engine control labelled other are two Dijkstra programs that scan an array and run in quadratic time (one of them an iteration-4 row of the 8-bit engine) and a Bellman-Ford of n rounds whose comment calls it Floyd-Warshall. All three pass the hidden suite (45 of 45) and the stress inputs. Three judge families were gated on ten controls of known class (Table 288). One passed with classes merged; the declared replacement for the third hit rate limits on 12 of its 39 calls. Agreement between judges was kappa 0.38 on exact classes and 0.61 merged. Every judge and the manual read called all three programs catalogue variants, so the count of novel algorithms stays 0.

**Table 288.** The judge gate for the three programs of the engine control labelled other.

| Judge | Gate (classes merged) | Exact classes | Status |
|---|---|---|---|
| DeepSeek chat (version 3.1) | 10/10 | 7/10 | passed |
| GPT-4.1-mini | 8/10 | | failed |
| Gemini 2.5 Flash Lite | 8/10 | | failed; the declared replacement hit rate limits on 12 of its 39 calls |

**Section 4.9 and positioning.** Section 4.9 is rewrapped without changing a value. The audit checked the venues of the five neighbours the plan named (Table 289) and ran a 14-query search for work on forks in code generation (137 hits, 136 unique). None occupies the run's question; 14 are adjacent and one is irrelevant. Row 66 of Table 277 now places the fork as only adjacent to Forking Paths [30].

**Table 289.** Venues of the neighbours the plan named, against the live records.

| Neighbour | Venue and version found | Against the plan |
|---|---|---|
| Forking Paths | ICLR 2025 (OpenReview and the proceedings; not on the arXiv record) | matches |
| Forking Fast | v1 2026-08-20; no venue; no NeurIPS 2026 paper page found | CORRECTED |
| Hanna and Ameisen | ICLR 2026 | matches |
| Ma and Rui | Mechanistic Interpretability Workshop at ICML 2026, not the main conference | CORRECTED |

| | | |
|---|---|---|
| A Tale of Two Temperatures (seed) | COLM (arXiv v2 comment 'V2: accepted as a full conference paper at COLM', no year) and two ICML 2026 workshop posters (AdaptFM, icml.cc/virtual/2026/75333; 4th Structured Probabilistic Inference & Generative Modeling, icml.cc/virtual/2026/74361); a paper on diffusion language models, not autoregressive code (iteration-8 correction, verified 2026-10-23) | CORRECTED |

The audit also tagged the six verdicts that rest on the relative unit and direction seeds of iteration 5 as pending the 2x2 of dose unit against seeds. Section 7.6 resolves that tag.

## 7.3 A fitted plan subspace in Qwen3-8B

### 7.3.1 Design

The DAS experiment asked whether the choice Stage A transfers is carried by a linear subspace of the task specification states that has low rank. DAS fits one orthonormal map R (4096 by 16, with nested ranks 1, 2, 4, 8 and 16) such that exchanging only the coordinates of R between a source prompt and the neutral twin reproduces the source's behaviour [110, 111]. The map is fitted on teacher-forced pairs: a source prompt (a ban of the default or an instruction forcing an algorithm Y) and the twin, scored on the log likelihood of programs of Y. It is tested in held-out generation scored by hidden tests, on held-out templates, against the full state exchange at the same site, which is the ceiling. Three tasks were used: shortest path (default Dijkstra, ban substitute Bellman-Ford/SPFA), connected components (default union find, substitute BFS or DFS) and kth smallest (default full sort, substitute quickselect or a size k heap).

The controls follow the published objections to DAS. Random subspaces of the same rank test whether any subspace would do. A *fit on permuted labels* trains the same map with the source labels permuted; if it transfers as well, the fitted subspace does not encode the label. We call *random-initialisation control* a fit of the same map on a randomly initialised copy of the network; if the untrained network admits an equally good teacher-forced fit, fit quality says nothing about a learned variable [132]. Two dormancy checks follow Makelov et al. [122]: the share of the twin's natural variance across prompts that lies in the subspace, and a *projected map*, the fitted map restricted to the top 64 principal components of natural variation, whose retention of the teacher-forced effect should be high if the subspace acts through directions the model uses. A *reverse interchange* writes the twin's coordinates into the ban prompt; a variable should restore the default.

### 7.3.2 Site

A full state exchange at a single block peaks at block 18 on shortest path and at block 16 on connected components and kth smallest (Table 290). The two kinds of source behave differently in generation. A forced instruction is carried by one block: exchanging the forced prompt's states at that block alone transfers Bellman-Ford/SPFA in 30 of 30 slots on shortest path and BFS in 28 of 30 on connected components. A ban on the two core tasks is not: the same exchange at a single block moves 0 of 30 slots on both, and only the band of blocks 12 to 23 transfers it (11 of 30 on shortest path, against 1 at base; 17 of 30 on connected components, against 0). The committed fallback to the band therefore fired on both. On kth smallest one block carries the ban (27 of 30, against 28 for the band and 5 at base).

**Table 290.** Site selection: the teacher-forced profile and its generation check.

| Task | Block chosen by teacher forcing | Teacher-forced gain at that single block (nats) | Band 12-23 teacher-forced gain (nats) | Ban source, correct Y of 30: full state at that block / band 12-23 / base | Forced source (family 3), correct Y of 30: at that block / band / base | Site used |
|---|---|---|---|---|---|---|
| shortest path | 18 | 11.34 | 14.20 | 0 / 11 / 1 | 30 / 30 / 0 (Bellman-Ford/SPFA) | band |
| connected components | 16 | 27.26 | 35.33 | 0 / 17 / 0 | 28 / 10 / 0 (BFS) | band |
| kth smallest | 16 | 14.02 | 20.46 | 27 / 28 / 5 | 27 / 29 / 0 (quickselect) | layer |

### 7.3.3 The committed objective diverged, and was amended

The committed objective was contrastive: maximise the log likelihood of Y programs minus that of default programs. It grew without bound on validation, to 1,873 nats on shortest path, 3,868 on connected components and 177 on kth smallest, against full state ceilings of 14 to 35 nats; the fit on permuted labels reached 880 (Table 291). A generation pilot of the contrastive maps broke programs (Table 301). The first amendment replaced the objective with the likelihood of Y programs alone, which stays within the ceiling. Every result below uses the amended map unless the table says otherwise.

**Table 291.** Every DAS fit: steps, best validation gain and orthonormality.

| Task | Objective | Fit | Steps run (budget) | Best step | Best validation gain (nats) | Rank-16 validation gain at the last step | Orthonormality error | Seconds |
|---|---|---|---|---|---|---|---|---|
| shortest path | contrastive (committed) | main | 225 (225) | 225 | 1873.42 | 2492.39 | 4.8e-07 | 208.9 |
| shortest path | contrastive (committed) | permuted labels | 225 (225) | 225 | 879.98 | 1221.58 | 4.8e-07 | 206.2 |
| shortest path | contrastive (committed) | read position | 225 (225) | 200 | 3229.69 | 4099.86 | 4.8e-07 | 207.6 |
| shortest path | likelihood (amended) | main | 225 (225) | 225 | 14.86 | 14.96 | 4.8e-07 | 208.8 |
| shortest path | likelihood (amended) | permuted labels | 225 (225) | 225 | 14.42 | 14.62 | 3.6e-07 | 204.8 |
| shortest path | likelihood (amended) | read position | 225 (225) | 225 | 14.35 | 14.94 | 3.6e-07 | 207.0 |
| connected components | contrastive (committed) | main | 150 (150) | 125 | 3867.95 | 4444.07 | 3.6e-07 | 121.4 |
| connected components | likelihood (amended) | main | 150 (150) | 100 | 24.76 | 23.65 | 4.8e-07 | 120.3 |
| connected components | likelihood (amended) | permuted labels | 150 (150) | 75 | 24.73 | 22.68 | 4.8e-07 | 118.1 |
| connected components | likelihood (amended) | read position | 150 (150) | 75 | 20.67 | 17.16 | 4.8e-07 | 120.2 |
| kth smallest | contrastive (committed) | main | 150 (150) | 125 | 176.69 | 215.55 | 4.8e-07 | 95.9 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| kth smallest | likelihood (amended) | main | 225 (225) | 175 | 6.83 | 6.87 | 4.8e-07 | 144.0 |
| kth smallest | likelihood (amended) | permuted labels | 225 (225) | 175 | 5.57 | 4.92 | 3.6e-07 | 138.6 |

### 7.3.4 Recovery in held-out generation

A rank-1 or rank-2 edit of the ban source moves the algorithm about as far as the full state exchange on all three tasks (forced sources by source in Table 297b; iteration-8 correction), and random subspaces of the same rank move nothing (Tables 292 to 297). On shortest path the rank-2 edit of the fifth template ban gives 12 correct Bellman-Ford/SPFA programs of 30, against 11 for the full state exchange, 1 at base and 1, 0 and 1 for three random rank-16 subspaces (McNemar 11:0 against random, Holm p 9.8e-4). On connected components the rank-16 edit gives 29 correct programs, more than the full state exchange's 17, against 0 for random (61:0 against random, pooled over sources, Holm p 2.6e-18). On kth smallest the rank-1 edit gives 26 against 27 for the full state, 5 at base and 6 for random (20:0, Holm p 1.9e-6). Recovery is the ratio of the edit's transfer to the full state transfer, and its summary is per source and rank: by source it matches or exceeds the full state exchange for the ban sources (except shortest path at rank 16, 6 against 11) and for the forced Floyd-Warshall, quickselect, counting sort and DFS sources, recovers 0.17 to 0.43 of forced Bellman-Ford/SPFA and 0.06 to 0.53 of forced BFS, and rank 2 is untested for forced sources (Table 297b). Pooled over sources as a ratio of sums (secondary), it lies between 0.73 and 1.71 on shortest path and connected components and between 0.99 and 1.01 on kth smallest after the counting-sort relabel, above 1 where a trained direction moves the model further than the full exchange (iteration-8 correction).

The forced sources behave differently by task. On shortest path the forced Floyd-Warshall source transfers at rank 1 in 28 of 30 slots while the full state exchange of the same prompt manages 8 of 20; on kth smallest the forced counting sort source transfers 30 of 30 at every rank, and the full state exchange of the same prompt gives 29 of 30: the 29 programs once labelled other are counting sort in dictionary form (counts in a dictionary walked in sorted key order), which the extended labeller now names counting sort, and the remaining program is a binary search on the value range (Table 300) (iteration-8 correction). On connected components the forced BFS source transfers poorly at low rank (1 of 30 at rank 1, 9 at rank 16, against 17 for the full state), while the forced DFS source transfers 30 of 30 at rank 1.

**Table 292.** Shortest path: correct programs of the target algorithm (of 30) by source and rank, with pass rates.

| Source | Target Y | Base (twin) | Full state exchange | DAS rank 1 | rank 2 | rank 4 | rank 8 | rank 16 | Random subspaces, rank 16 | DAS with permuted labels | Pass rate base / full state |
|---|---|---|---|---|---|---|---|---|---|---|---|
| fifth template ban source | Bellman-Ford/ SPFA | 1 | 11 | 8 (pass 0.77) | 12 (pass 0.83) | 10 (pass 0.57) | 10 (pass 0.83) | 6 (pass 0.80) | 1, 0, 1 | permuted rank 16: 3, permuted rank 2: 15 | 0.97/0.73 |
| forced Bellman-Ford/ SPFA source | Bellman-Ford/ SPFA | 0 | 30 | 13 (pass 0.70) | - | 9 (pass 0.67) | - | 5 (pass 0.80) | 0 | | 0.97/1.00 |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| forced Floyd-Warshall source | Floyd-Warshall | 0 | 8 | 28 (pass 0.93) | - | 28 (pass 0.93) | - | 25 (pass 0.83) | 0 | permuted rank 16: 6 | 0.97/0.67 |

**Table 293.** Shortest path: pooled recovery and its controls by rank.

| Rank | Pooled recovery [slot bootstrap 95%] | Recovery of random subspaces | McNemar DAS against random b:c (p) | Transfer under permuted labels (max) | Smallest pass ratio |
|---|---|---|---|---|---|
| 1 | 1.00 [0.79, 1.24] | - | - | - | - |
| 2 | 1.10 [0.53, 2.43] | 0.00 | 11:0 (9.8e-04) | 0.47 | 0.86 |
| 4 | 0.96 [0.73, 1.25] | - | - | - | - |
| 8 | 0.90 [0.36, 2.00] | - | - | - | - |
| 16 | 0.73 [0.56, 0.93] | -0.01 | 36:1 (5.5e-10) | 0.20 | 0.83 |

**Table 294.** Connected components: correct programs of the target algorithm (of 30) by source and rank.

| Source | Target Y | Base (twin) | Full state exchange | DAS rank 1 | rank 2 | rank 4 | rank 8 | rank 16 | Random subspaces, rank 16 | DAS with permuted labels | Pass rate base / full state |
|---|---|---|---|---|---|---|---|---|---|---|---|
| fifth template ban source | BFS | 0 | 17 | 18 (pass 0.97) | 23 (pass 1.00) | 21 (pass 1.00) | 29 (pass 1.00) | 29 (pass 1.00) | 0, 0, 0 | permuted rank 16: 21, permuted rank 1: 15 | 1.00/1.00 |
| forced BFS source | BFS | 0 | 17 | 1 (pass 1.00) | - | 7 (pass 1.00) | - | 9 (pass 1.00) | 0 | permuted rank 16: 10 | 1.00/1.00 |
| forced DFS source | DFS | 0 | 30 | 30 (pass 1.00) | - | 26 (pass 1.00) | - | 23 (pass 1.00) | 0 | | 1.00/1.00 |

**Table 295.** Connected components: pooled recovery and its controls by rank.

| Rank | Pooled recovery [slot bootstrap 95%] | Recovery of random subspaces | McNemar DAS against random b:c (p) | Transfer under permuted labels (max) | Smallest pass ratio |
|---|---|---|---|---|---|
| 1 | 0.77 [0.65, 0.90] | 0.00 | 18:0 (7.6e-06) | 0.50 | 0.97 |
| 2 | 1.35 [0.94, 2.08] | - | - | - | - |
| 4 | 0.84 [0.71, 1.00] | - | - | - | - |
| 8 | 1.71 [1.29, 2.50] | - | - | - | - |
| 16 | 0.95 [0.82, 1.12] | 0.00 | 61:0 (8.7e-19) | 0.70 | 1.00 |

**Table 296.** Kth smallest: correct programs of the target algorithm (of 30) by source and rank. (iteration-8 correction: forced counting sort source, full state 0 -> 29 after the labeller was extended to counting sort in dictionary form)

| Source | Target Y | Base (twin) | Full state exchange | DAS rank 1 | rank 2 | rank 4 | rank 8 | rank 16 | Random subspaces, rank 16 | DAS with permuted labels | Pass rate base / full state |
|---|---|---|---|---|---|---|---|---|---|---|---|

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| fifth template ban source | quickselect or size k heap | 5 | 27 | 26 (pass 0.90) | - | 27 (pass 0.90) | - | 27 (pass 0.90) | 6 | permuted rank 16: 19, permuted rank 1: 23 | 1.00/1.00 |
| forced quickselect source | quickselect | 0 | 29 | 30 (pass 1.00) | - | 28 (pass 0.93) | - | 27 (pass 0.90) | | | 1.00/0.97 |
| forced counting sort source | counting sort | 0 | 29 | 30 (pass 1.00) | - | 30 (pass 1.00) | - | 30 (pass 1.00) | | | 1.00/1.00 |

**Table 297.** Kth smallest: pooled recovery and its controls by rank. (iteration-8 correction: pooled recovery recomputed after the counting-sort relabel with exp33's own slot bootstrap, 2,000 resamples; the old values 1.59 / 1.57 / 1.55 are reproduced exactly under the old labels; the permuted-label column is an absolute transfer, (count - base)/30, and the random-subspace recovery uses only the ban source, so neither changes; per-source recovery in Table 297b is the summary)

| Rank k | Pooled recovery [slot bootstrap 95%] | Recovery of random subspaces | McNemar DAS against random b:c (p) | Transfer under permuted labels (max) | Smallest pass ratio |
|---|---|---|---|---|---|
| 1 | 1.01 [0.95, 1.08] | 0.05 | 20:0 (1.9e-06) | 0.60 | 0.90 |
| 2 | - [-, -] | - | - | - | - |
| 4 | 1.00 [0.93, 1.08] | - | - | - | - |
| 8 | - [-, -] | - | - | - | - |
| 16 | 0.99 [0.92, 1.06] | 0.05 | 21:0 (9.5e-07) | 0.47 | 0.90 |

**Table 297b.** Recovery by source and rank (summary statistic; pooled recovery is secondary). Each cell is the count of correct programs of the target algorithm of 30 and, in parentheses, the recovery (DAS count - base) / (full state count - base), undefined when the denominator is 0. Forced sources were generated at ranks 1, 4 and 16 only (deviation D-C-k), and the kth smallest ban source at ranks 1, 4 and 16. The kth smallest forced counting sort row uses the relabelled full state (29). Source: `results/p1.json` per_source of the DAS experiment, recomputed from `data/labels.jsonl` (iteration-8 correction: new table)

| Task | Source | Target Y | Base | Full state | Rank 1 | Rank 2 | Rank 4 | Rank 8 | Rank 16 |
|---|---|---|---|---|---|---|---|---|---|
| shortest path | fifth template ban | Bellman-Ford/ SPFA | 1 | 11 | 8 (0.70) | 12 (1.10) | 10 (0.90) | 10 (0.90) | 6 (0.50) |
| shortest path | forced Bellman-Ford/SPFA | Bellman-Ford/ SPFA | 0 | 30 | 13 (0.43) | not generated | 9 (0.30) | not generated | 5 (0.17) |
| shortest path | forced Floyd-Warshall | Floyd-Warshall | 0 | 8 | 28 (3.50) | not generated | 28 (3.50) | not generated | 25 (3.12) |
| connected components | fifth template ban | BFS | 0 | 17 | 18 (1.06) | 23 (1.35) | 21 (1.24) | 29 (1.71) | 29 (1.71) |
| connected components | forced BFS | BFS | 0 | 17 | 1 (0.06) | not generated | 7 (0.41) | not generated | 9 (0.53) |
| connected components | forced DFS | DFS | 0 | 30 | 30 (1.00) | not generated | 26 (0.87) | not generated | 23 (0.77) |
| kth smallest | fifth template ban | quickselect, size k heap | 5 | 27 | 26 (0.95) | not generated | 27 (1.00) | not generated | 27 (1.00) |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| kth smallest | forced quickselect | quickselect | 0 | 29 | 30 (1.03) | not generated | 28 (0.97) | not generated | 27 (0.93) |
| kth smallest | forced counting sort | counting sort | 0 | 29 | 30 (1.03) | not generated | 30 (1.03) | not generated | 30 (1.03) |

### 7.3.5 The controls that make it a handle, not a variable

The DAS claim is not supported on any task. Three controls fail. The fits on permuted labels transfer 0.47 on shortest path, 0.70 on connected components and 0.60 on kth smallest as absolute transfers ((count - base)/30; relative to the full state effect 1.40, 1.24 and 0.82), against a committed limit of 0.1 (iteration-8 correction): on connected components a permuted fit at rank 16 still gives 21 correct BFS programs of 30. The permuted fit can also write its own label: at rank 16 it turns the shortest path ban into Floyd-Warshall in 15 of 30 slots, because one training ban had been relabelled Floyd-Warshall by the permutation. With only two kept algorithms per task the permutation changed 44% of the training labels (56% on kth smallest), so the signal to leave the default survives any such shuffle, which weakens this control as a test of label content. The untrained network admits a teacher-forced fit as good as the trained one or better: the untrained-network control ran on shortest path only; the committed comparison with the diverged contrastive fit (ratio 0.013) proved meaningless and was replaced post hoc; under the matched likelihood fit the untrained network gains 43.6 nats against 8.6 for the trained one (ratio 5.1), while the fit-value ratio is 0.92 (13.31 against 14.45 nats); three different ratios exist (committed 0.013, fit value 0.92, likelihood gain 5.08), each against a limit of 0.5, and only the last is measured in the objective both maps were fitted with (LME_Y) (iteration-8 correction). And the dormancy checks flag two of three tasks: the reverse interchange on shortest path restores the default in 0 of 30 slots, against 21 of 30 for the full state reverse, and on kth smallest the projected map keeps only 0.20 to 0.26 of the teacher-forced effect. The connected components reverse was not testable. The subspace is not dormant by variance: it holds 0.7% to 18% of the twin's natural variance across prompts against 0.03% to 0.4% for random subspaces of the same rank, at the 100th percentile of 1,000 random draws on every task and rank (Table 307).

The reading is that the fitted subspace is a strong steering handle of low rank on the choice, and that this experiment cannot say it is the variable the model computes. A subspace that permuted labels also find, and that an untrained network also admits, is a direction along which the downstream computation is sensitive, which is what a steering vector needs and less than what a causal variable needs. The control arms behave as before: DAS and the full state exchange at the read position, and a rank-1 difference-in-means direction and a PCA-16 subspace at the task specification states, each transfer at most 2 of 30 (Tables 298 and 299). Under the fourth template, whose ban the model mostly ignores, the full state exchange still transfers 4 of 10 on shortest path and 9 of 10 on connected components.

**Table 298.** Shortest path: readout controls, null source controls and fourth template controls.

| Arm | Correct Y (or correct off the default) count of 30 | Base | Transfer | Pass |
|---|---|---|---|---|
| DAS at the read position, fifth template ban source, rank 16 | 2 | 1 | 0.03 | 0.90 |
| full state exchange at the read position, fifth template ban source | 1 | 1 | 0.00 | 0.97 |
| difference in means (rank 1), fifth template ban source | 0 | 1 | -0.03 | 0.97 |
| PCA-16 subspace, fifth template ban source | 1 | 1 | 0.00 | 0.83 |

| | | | | |
|---|---|---|---|---|
| null source (forced default), DAS rank 16 | 1 (off the default) | 0 | 0.03 | 0.83 |
| null source (forced default), full state exchange | 0 (off the default) | 0 | 0.00 | 0.97 |
| neutral twin of template 4 (base) (N = 10) | 0 | - | - | 1.00 |
| full state exchange, fourth template ban source (N = 10) | 4 | - | - | 0.50 |
| DAS (likelihood objective), fourth template ban source, rank 2 (N = 10) | 4 | - | - | 0.90 |

**Table 299.** Connected components: readout controls, null source controls and fourth template controls.

| Arm | Correct Y (or correct off the default) count of 30 | Base | Transfer | Pass |
|---|---|---|---|---|
| DAS at the read position, fifth template ban source, rank 16 | 2 | 0 | 0.07 | 1.00 |
| full state exchange at the read position, fifth template ban source | 0 | 0 | 0.00 | 1.00 |
| difference in means (rank 1), fifth template ban source | 0 | 0 | 0.00 | 1.00 |
| PCA-16 subspace, fifth template ban source | 0 | 0 | 0.00 | 0.93 |
| null source (forced default), DAS rank 16 | 0 (off the default) | 0 | 0.00 | 1.00 |
| null source (forced default), full state exchange | 0 (off the default) | 0 | 0.00 | 1.00 |
| neutral twin of template 4 (base) (N = 10) | 0 | - | - | 1.00 |
| full state exchange, fourth template ban source (N = 10) | 9 | - | - | 1.00 |
| DAS (likelihood objective), fourth template ban source, rank 1 (N = 10) | 0 | - | - | 1.00 |

### 7.3.6 Routing

The routing claim asks whether an edit lands where the forced source points or slides toward the ban's usual substitute. By the committed rule the routing is categorical on shortest path and kth smallest: the Floyd-Warshall source lands on Floyd-Warshall in 28 of 28 correct programs at rank 1, while the full state exchange of the same prompt lands 12 programs on Bellman-Ford/SPFA and 8 on Floyd-Warshall, and the counting sort source lands on counting sort in 30 of 30 at rank 16, where the full state exchange of the same prompt also lands 29 of 30 on counting sort once its dictionary form is labelled (Table 300). Connected components is not categorical: the forced BFS source's rank-16 edit lands 21 DFS and 9 BFS. The subspace thus carries which algorithm, more cleanly than the full state on one task (shortest path: Floyd-Warshall 28/28 against 8/20); on kth smallest the DAS edit changes the implementation form (mostly an array of counts, 28 of 30 at rank 16, against the full state's dictionary with sorted keys, 29 of 30), not the algorithm (iteration-8 correction).

**Table 300.** Routing: where correct programs off the default land under the full state exchange and under DAS. (iteration-8 correction: kth smallest | counting sort row relabelled; the committed C rule still scores kth smallest categorical, because it reads the DAS columns)

| Task | Forced Y | Full state exchange: correct Y / correct off the default [CP 95%] | DAS rank 1 | DAS rank 4 | DAS rank 16 | Landing under the full state exchange | Landing under DAS rank 16 |
|---|---|---|---|---|---|---|---|
| shortest path | Bellman-Ford/SPFA | 1.00 (30/30) [0.88, 1.00] | 1.00 (13/13) [0.75, 1.00] | 1.00 (9/9) [0.66, 1.00] | 1.00 (5/5) [0.48, 1.00] | Bellman-Ford/ SPFA 30 | Dijkstra 19, Bellman-Ford/ SPFA 5 |
| shortest path | Floyd-Warshall | 0.40 (8/20) [0.19, 0.64] | 1.00 (28/28) [0.88, 1.00] | 1.00 (28/28) [0.88, 1.00] | 1.00 (25/25) [0.86, 1.00] | Bellman-Ford/ SPFA 12, Floyd-Warshall 8 | Floyd-Warshall 25 |
| connected components | BFS | 0.57 (17/30) [0.37, 0.75] | 0.03 (1/30) [8.4e-04, 0.17] | 0.23 (7/30) [0.10, 0.42] | 0.30 (9/30) [0.15, 0.49] | BFS 17, DFS 13 | DFS 21, BFS 9 |
| connected components | DFS | 1.00 (30/30) [0.88, 1.00] | 1.00 (30/30) [0.88, 1.00] | 0.87 (26/30) [0.69, 0.96] | 0.77 (23/30) [0.58, 0.90] | DFS 30 | DFS 23, BFS 7 |
| kth smallest | quickselect | 1.00 (29/29) [0.88, 1.00] | 1.00 (30/30) [0.88, 1.00] | 1.00 (28/28) [0.88, 1.00] | 1.00 (27/27) [0.87, 1.00] | quickselect 29 | quickselect 27 |
| kth smallest | counting sort | 0.97 (29/30) [0.83, 1.00] | 1.00 (30/30) [0.88, 1.00] | 1.00 (30/30) [0.88, 1.00] | 1.00 (30/30) [0.88, 1.00] | counting sort (dictionary form) 29, binary search on the value range 1 | counting sort 30 (array of counts 28, dictionary walked over the value range 2) |

**Table 301.** The pilot of the committed contrastive maps in generation (base cells in parentheses). (iteration-8 correction: in the adjudicated column of these 100 programs (rule, two gated judges, manual read; `tables/counting_sort_adjudication.csv`), 56 of the 67 labelled other are degenerate text and 11 are attempted Dijkstra that crash on the edge list, and 19 of the 30 labelled Dijkstra are degenerate text that the frozen path labeller names Dijkstra; no relabelled program passes, so the correct-Y counts stand)

| Task | Arm (committed contrastive map) | Correct Y (base) | Pass (base) | Labels | Mean new tokens | Finish reason |
|---|---|---|---|---|---|---|
| shortest path | DAS (committed contrastive objective), forced Bellman-Ford/SPFA source, rank 1 | 1 (0) | 0.40 (0.90) | other 6, Dijkstra 3, Bellman-Ford/SPFA 1 | 189 | end of text 6, fence end of text 4 |
| shortest path | DAS (committed contrastive objective), forced Bellman-Ford/SPFA source, rank 16 | 0 (0) | 0.00 (0.90) | other 5, Dijkstra 5 | 512 | length cap 10 |
| shortest path | DAS (committed contrastive objective), forced Bellman-Ford/SPFA source, rank 4 | 0 (0) | 0.00 (0.90) | Dijkstra 7, other 2, Bellman-Ford/SPFA 1 | 486 | length cap 9, end of text 1 |

| | | | | | | |
|---|---|---|---|---|---|---|
| shortest path | DAS (committed contrastive objective), forced Floyd-Warshall source, rank 1 | 0 (0) | 0.50 (0.90) | Dijkstra 5, other 4, Bellman-Ford/SPFA 1 | 193 | end of text 9, fence end of text 1 |
| shortest path | DAS (committed contrastive objective), forced Floyd-Warshall source, rank 16 | 0 (0) | 0.00 (0.90) | other 9, Dijkstra 1 | 512 | length cap 10 |
| shortest path | DAS (committed contrastive objective), forced Floyd-Warshall source, rank 4 | 0 (0) | 0.00 (0.90) | other 9, Dijkstra 1 | 512 | length cap 10 |
| shortest path | DAS (committed contrastive objective), fifth template ban source, rank 1 | 0 (1) | 0.10 (0.90) | other 6, Dijkstra 4 | 342 | length cap 5, end of text 4, fence end of text 1 |
| shortest path | DAS (committed contrastive objective), fifth template ban source, rank 16 | 0 (1) | 0.00 (0.90) | other 6, Dijkstra 4 | 467 | length cap 9, fence end of text 1 |
| shortest path | DAS (committed contrastive objective), fifth template ban source, rank 2 | 0 (1) | 0.00 (0.90) | Dijkstra 6, other 4 | 431 | length cap 7, end of text 3 |
| shortest path | DAS (committed contrastive objective), fifth template ban source, rank 4 | 0 (1) | 0.00 (0.90) | other 9, Bellman-Ford/ SPFA 1 | 482 | length cap 7, end of text 2, fence end of text 1 |
| shortest path | DAS (committed contrastive objective), fifth template ban source, rank 8 | 0 (1) | 0.00 (0.90) | other 9, Dijkstra 1 | 512 | length cap 10 |
| connected components | DAS (committed contrastive objective), forced BFS source, rank 1 | 0 (0) | 1.00 (1.00) | union find 10 | 125 | end of text 8, fence end of text 2 |
| connected components | DAS (committed contrastive objective), forced BFS source, rank 16 | 0 (0) | 0.00 (1.00) | invalid 10 | 322 | length cap 8, end of text 1, fence end of text 1 |
| connected components | DAS (committed contrastive objective), forced BFS source, rank 4 | 2 (0) | 0.20 (1.00) | invalid 8, BFS 2 | 304 | length cap 5, fence end of text 3, end of text 2 |
| connected components | DAS (committed contrastive objective), forced DFS source, rank 1 | 0 (0) | 1.00 (1.00) | union find 10 | 125 | end of text 8, fence end of text 2 |
| connected components | DAS (committed contrastive objective), forced DFS source, rank 16 | 0 (0) | 0.20 (1.00) | invalid 8, BFS 2 | 202 | fence end of text 6, length cap 3, end of text 1 |
| connected components | DAS (committed contrastive objective), forced DFS source, rank 4 | 0 (0) | 0.10 (1.00) | invalid 9, BFS 1 | 110 | fence end of text 9, end of text 1 |
| connected components | DAS (committed contrastive objective), fifth template ban source, rank 1 | 0 (0) | 1.00 (1.00) | union find 10 | 152 | fence end of text 7, end of text 3 |

| | | | | | | |
|---|---|---|---|---|---|---|
| connected components | DAS (committed contrastive objective), fifth template ban source, rank 16 | 5 (0) | 0.50 (1.00) | invalid 5, BFS 5 | 267 | end of text 5, length cap 3, fence end of text 2 |
| connected components | DAS (committed contrastive objective), fifth template ban source, rank 2 | 5 (0) | 0.60 (1.00) | BFS 5, invalid 4, DFS 1 | 169 | end of text 6, fence end of text 3, length cap 1 |
| connected components | DAS (committed contrastive objective), fifth template ban source, rank 4 | 3 (0) | 0.40 (1.00) | invalid 6, BFS 3, DFS 1 | 244 | fence end of text 4, end of text 4, length cap 2 |
| connected components | DAS (committed contrastive objective), fifth template ban source, rank 8 | 3 (0) | 0.30 (1.00) | invalid 7, BFS 3 | 256 | fence end of text 4, end of text 3, length cap 3 |

### 7.3.7 Teacher-forced evaluation on held-out pairs

Tables 302 to 306 report every edit's teacher-forced effect on the held-out pairs, under the amended likelihood objective and under the committed contrastive one. *Teacher-forced recovery* is the edit's mean gain divided by the full state gain on the same pairs; the last two columns split the gain into the change in log likelihood of Y programs and of default programs. The contrastive maps reach recoveries far above 1 by lowering the likelihood of default programs, which is the signature of the divergence of Section 7.3.3.

**Table 302.** Shortest path, likelihood objective: held-out teacher-forced evaluation.

| Pairs | Edit | n | Mean gain (nats) | Full state gain | Teacher-forced recovery | Change in log likelihood of Y programs | Change in log likelihood of default programs |
|---|---|---|---|---|---|---|---|
| ban | full state exchange | 12 | 5.14 | 5.14 | 1.00 | 3.89 | -1.24 |
| ban | DAS, likelihood objective, rank 1 | 12 | 4.20 | 5.14 | 0.82 | 4.27 | 0.07 |
| ban | DAS, likelihood objective, rank 2 | 12 | 4.12 | 5.14 | 0.80 | 4.26 | 0.13 |
| ban | DAS, likelihood objective, rank 4 | 12 | 4.06 | 5.14 | 0.79 | 4.29 | 0.23 |
| ban | DAS, likelihood objective, rank 8 | 12 | 3.98 | 5.14 | 0.78 | 4.30 | 0.32 |
| ban | DAS, likelihood objective, rank 16 | 12 | 4.02 | 5.14 | 0.78 | 4.30 | 0.28 |
| ban | DAS with permuted labels, likelihood objective, rank 1 | 12 | 10.57 | 5.14 | 2.06 | 3.09 | -7.48 |
| ban | DAS with permuted labels, likelihood objective, rank 2 | 12 | 6.72 | 5.14 | 1.31 | 3.93 | -2.79 |
| ban | DAS with permuted labels, likelihood objective, rank 4 | 12 | 8.52 | 5.14 | 1.66 | 3.63 | -4.90 |
| ban | DAS with permuted labels, likelihood objective, rank 8 | 12 | 16.40 | 5.14 | 3.19 | 2.10 | -14.30 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| ban | DAS with permuted labels, likelihood objective, rank 16 | 12 | 15.06 | 5.14 | 2.93 | 1.55 | -13.52 |
| ban | DAS (likelihood) projected on top-64 natural PCs, rank 1 | 12 | 3.51 | 5.14 | 0.68 | 2.41 | -1.10 |
| ban | DAS (likelihood) projected on top-64 natural PCs, rank 2 | 12 | 3.47 | 5.14 | 0.68 | 2.71 | -0.76 |
| ban | DAS (likelihood) projected on top-64 natural PCs, rank 4 | 12 | 4.45 | 5.14 | 0.87 | 4.07 | -0.39 |
| ban | DAS (likelihood) projected on top-64 natural PCs, rank 8 | 12 | 4.39 | 5.14 | 0.85 | 4.01 | -0.37 |
| ban | DAS (likelihood) projected on top-64 natural PCs, rank 16 | 12 | 3.48 | 5.14 | 0.68 | 3.23 | -0.25 |
| forced | full state exchange | 4 | 35.50 | 35.50 | 1.00 | 18.20 | -17.30 |
| forced | DAS, likelihood objective, rank 1 | 4 | 25.56 | 35.50 | 0.72 | 21.47 | -4.09 |
| forced | DAS, likelihood objective, rank 2 | 4 | 26.53 | 35.50 | 0.75 | 21.45 | -5.08 |
| forced | DAS, likelihood objective, rank 4 | 4 | 35.08 | 35.50 | 0.99 | 21.44 | -13.64 |
| forced | DAS, likelihood objective, rank 8 | 4 | 41.88 | 35.50 | 1.18 | 21.46 | -20.42 |
| forced | DAS, likelihood objective, rank 16 | 4 | 45.75 | 35.50 | 1.29 | 21.47 | -24.28 |
| forced | DAS with permuted labels, likelihood objective, rank 1 | 4 | 19.68 | 35.50 | 0.55 | 17.23 | -2.45 |
| forced | DAS with permuted labels, likelihood objective, rank 2 | 4 | 22.13 | 35.50 | 0.62 | 17.40 | -4.73 |
| forced | DAS with permuted labels, likelihood objective, rank 4 | 4 | 23.87 | 35.50 | 0.67 | 17.67 | -6.21 |
| forced | DAS with permuted labels, likelihood objective, rank 8 | 4 | 28.72 | 35.50 | 0.81 | 17.76 | -10.96 |
| forced | DAS with permuted labels, likelihood objective, rank 16 | 4 | 25.64 | 35.50 | 0.72 | 16.15 | -9.49 |
| forced | DAS (likelihood) projected on top-64 natural PCs, rank 1 | 4 | 16.32 | 35.50 | 0.46 | 12.16 | -4.16 |
| forced | DAS (likelihood) projected on top-64 natural PCs, rank 2 | 4 | 15.09 | 35.50 | 0.42 | 11.42 | -3.67 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| forced | DAS (likelihood) projected on top-64 natural PCs, rank 4 | 4 | 17.69 | 35.50 | 0.50 | 14.08 | -3.60 |
| forced | DAS (likelihood) projected on top-64 natural PCs, rank 8 | 4 | 21.29 | 35.50 | 0.60 | 14.69 | -6.60 |
| forced | DAS (likelihood) projected on top-64 natural PCs, rank 16 | 4 | 23.52 | 35.50 | 0.66 | 14.94 | -8.57 |
| null | full state exchange | 2 | -2.12 | -2.12 | 1.00 | -2.10 | 0.03 |
| null | DAS, likelihood objective, rank 1 | 2 | 5.57 | -2.12 | -2.62 | 5.52 | -0.05 |
| null | DAS, likelihood objective, rank 2 | 2 | 5.47 | -2.12 | -2.58 | 5.51 | 0.04 |
| null | DAS, likelihood objective, rank 4 | 2 | 5.54 | -2.12 | -2.61 | 5.56 | 0.01 |
| null | DAS, likelihood objective, rank 8 | 2 | 5.47 | -2.12 | -2.58 | 5.56 | 0.09 |
| null | DAS, likelihood objective, rank 16 | 2 | 5.46 | -2.12 | -2.57 | 5.54 | 0.09 |
| null | DAS with permuted labels, likelihood objective, rank 1 | 2 | 6.10 | -2.12 | -2.87 | 4.70 | -1.40 |
| null | DAS with permuted labels, likelihood objective, rank 2 | 2 | 6.43 | -2.12 | -3.03 | 4.97 | -1.46 |
| null | DAS with permuted labels, likelihood objective, rank 4 | 2 | 7.01 | -2.12 | -3.30 | 5.12 | -1.88 |
| null | DAS with permuted labels, likelihood objective, rank 8 | 2 | 9.45 | -2.12 | -4.45 | 5.02 | -4.43 |
| null | DAS with permuted labels, likelihood objective, rank 16 | 2 | 7.73 | -2.12 | -3.64 | 5.30 | -2.43 |
| null | DAS (likelihood) projected on top-64 natural PCs, rank 1 | 2 | 1.73 | -2.12 | -0.81 | 1.72 | -0.00 |
| null | DAS (likelihood) projected on top-64 natural PCs, rank 2 | 2 | 1.69 | -2.12 | -0.79 | 1.70 | 0.01 |
| null | DAS (likelihood) projected on top-64 natural PCs, rank 4 | 2 | 3.93 | -2.12 | -1.85 | 3.87 | -0.06 |
| null | DAS (likelihood) projected on top-64 natural PCs, rank 8 | 2 | 4.21 | -2.12 | -1.98 | 4.09 | -0.11 |
| null | DAS (likelihood) projected on top-64 natural PCs, rank 16 | 2 | 5.41 | -2.12 | -2.55 | 5.14 | -0.28 |

**Table 303.** Shortest path, committed contrastive objective: held-out teacher-forced evaluation.

| Pairs | Edit | n | Mean gain (nats) | Full state gain | Teacher-forced recovery | Change in log likelihood of Y programs | Change in log likelihood of default programs |
|---|---|---|---|---|---|---|---|
| ban | full state exchange | 12 | 5.14 | 5.14 | 1.00 | 3.89 | -1.24 |
| ban | DAS, committed contrastive objective, rank 1 | 12 | 463.81 | 5.14 | 90.30 | -628.61 | -1092.42 |
| ban | DAS, committed contrastive objective, rank 2 | 12 | 996.81 | 5.14 | 194.06 | -973.75 | -1970.56 |
| ban | DAS, committed contrastive objective, rank 4 | 12 | 2532.26 | 5.14 | 492.99 | -250.99 | -2783.25 |
| ban | DAS, committed contrastive objective, rank 8 | 12 | 2313.90 | 5.14 | 450.48 | -589.79 | -2903.70 |
| ban | DAS, committed contrastive objective, rank 16 | 12 | 2049.86 | 5.14 | 399.07 | -957.27 | -3007.13 |
| ban | random subspace, rank 1 | 12 | 0.06 | 5.14 | 0.01 | 0.07 | 0.01 |
| ban | random subspace, rank 2 | 12 | 0.01 | 5.14 | 0.00 | 0.04 | 0.03 |
| ban | random subspace, rank 4 | 12 | 0.11 | 5.14 | 0.02 | 0.15 | 0.04 |
| ban | random subspace, rank 8 | 12 | -0.15 | 5.14 | -0.03 | -0.12 | 0.04 |
| ban | random subspace, rank 16 | 12 | -0.18 | 5.14 | -0.04 | -0.16 | 0.02 |
| ban | DAS with permuted labels, contrastive objective, rank 1 | 12 | 455.36 | 5.14 | 88.65 | -127.12 | -582.48 |
| ban | DAS with permuted labels, contrastive objective, rank 2 | 12 | 732.31 | 5.14 | 142.57 | -148.09 | -880.40 |
| ban | DAS with permuted labels, contrastive objective, rank 4 | 12 | 1160.84 | 5.14 | 226.00 | -123.31 | -1284.15 |
| ban | DAS with permuted labels, contrastive objective, rank 8 | 12 | 1431.45 | 5.14 | 278.68 | -113.25 | -1544.69 |
| ban | DAS with permuted labels, contrastive objective, rank 16 | 12 | 1477.81 | 5.14 | 287.70 | -120.82 | -1598.63 |
| ban | DAS (contrastive) projected on top-64 natural PCs, rank 1 | 12 | 2.84 | 5.14 | 0.55 | 2.66 | -0.18 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| ban | DAS (contrastive) projected on top-64 natural PCs, rank 2 | 12 | 3.06 | 5.14 | 0.60 | 2.77 | -0.29 |
| ban | DAS (contrastive) projected on top-64 natural PCs, rank 4 | 12 | 3.19 | 5.14 | 0.62 | 2.84 | -0.34 |
| ban | DAS (contrastive) projected on top-64 natural PCs, rank 8 | 12 | 2.98 | 5.14 | 0.58 | 2.59 | -0.39 |
| ban | DAS (contrastive) projected on top-64 natural PCs, rank 16 | 12 | 2.35 | 5.14 | 0.46 | 2.05 | -0.30 |
| ban | difference in means (rank 1) | 12 | -1.07 | 5.14 | -0.21 | -1.04 | 0.03 |
| ban | PCA-16 subspace | 12 | 2.81 | 5.14 | 0.55 | 2.72 | -0.08 |
| ban | full state exchange at the read position | 12 | 0.30 | 5.14 | 0.06 | 0.28 | -0.02 |
| ban | DAS at the read position, contrastive objective, rank 1 | 12 | -0.09 | 5.14 | -0.02 | -0.09 | 0.00 |
| ban | DAS at the read position, contrastive objective, rank 2 | 12 | 80.50 | 5.14 | 15.67 | -38.25 | -118.75 |
| ban | DAS at the read position, contrastive objective, rank 4 | 12 | 3936.45 | 5.14 | 766.36 | -115.28 | -4051.73 |
| ban | DAS at the read position, contrastive objective, rank 8 | 12 | 4213.08 | 5.14 | 820.21 | -97.87 | -4310.95 |
| ban | DAS at the read position, contrastive objective, rank 16 | 12 | 4234.74 | 5.14 | 824.43 | -98.39 | -4333.13 |
| ban | DAS at the read position, likelihood objective, rank 1 | 12 | 4.11 | 5.14 | 0.80 | 3.57 | -0.54 |
| ban | DAS at the read position, likelihood objective, rank 2 | 12 | 4.56 | 5.14 | 0.89 | 4.10 | -0.46 |
| ban | DAS at the read position, likelihood objective, rank 4 | 12 | 4.60 | 5.14 | 0.89 | 4.19 | -0.41 |
| ban | DAS at the read position, likelihood objective, rank 8 | 12 | 4.53 | 5.14 | 0.88 | 4.24 | -0.29 |
| ban | DAS at the read position, likelihood objective, rank 16 | 12 | 4.46 | 5.14 | 0.87 | 4.24 | -0.22 |
| forced | full state exchange | 4 | 35.50 | 35.50 | 1.00 | 18.20 | -17.30 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| forced | DAS, committed contrastive objective, rank 1 | 4 | 8.88 | 35.50 | 0.25 | 2.51 | -6.37 |
| forced | DAS, committed contrastive objective, rank 2 | 4 | 31.17 | 35.50 | 0.88 | -20.88 | -52.06 |
| forced | DAS, committed contrastive objective, rank 4 | 4 | 2157.71 | 35.50 | 60.78 | -80.24 | -2237.95 |
| forced | DAS, committed contrastive objective, rank 8 | 4 | 2588.34 | 35.50 | 72.90 | -108.23 | -2696.57 |
| forced | DAS, committed contrastive objective, rank 16 | 4 | 2739.67 | 35.50 | 77.17 | -219.43 | -2959.10 |
| forced | random subspace, rank 1 | 4 | 0.23 | 35.50 | 0.01 | 0.23 | 0.00 |
| forced | random subspace, rank 2 | 4 | 0.19 | 35.50 | 0.01 | 0.19 | -0.01 |
| forced | random subspace, rank 4 | 4 | 0.23 | 35.50 | 0.01 | 0.23 | -0.01 |
| forced | random subspace, rank 8 | 4 | 0.14 | 35.50 | 0.00 | 0.13 | -0.01 |
| forced | random subspace, rank 16 | 4 | 0.36 | 35.50 | 0.01 | 0.37 | 0.01 |
| forced | DAS with permuted labels, contrastive objective, rank 1 | 4 | 397.15 | 35.50 | 11.19 | -44.96 | -442.11 |
| forced | DAS with permuted labels, contrastive objective, rank 2 | 4 | 756.48 | 35.50 | 21.31 | -114.93 | -871.41 |
| forced | DAS with permuted labels, contrastive objective, rank 4 | 4 | 1612.38 | 35.50 | 45.42 | -62.94 | -1675.32 |
| forced | DAS with permuted labels, contrastive objective, rank 8 | 4 | 1681.33 | 35.50 | 47.36 | -44.97 | -1726.30 |
| forced | DAS with permuted labels, contrastive objective, rank 16 | 4 | 1674.07 | 35.50 | 47.15 | -51.28 | -1725.35 |
| forced | DAS (contrastive) projected on top-64 natural PCs, rank 1 | 4 | 3.95 | 35.50 | 0.11 | 3.90 | -0.04 |
| forced | DAS (contrastive) projected on top-64 natural PCs, rank 2 | 4 | 5.14 | 35.50 | 0.14 | 5.01 | -0.13 |
| forced | DAS (contrastive) projected on top-64 natural PCs, rank 4 | 4 | 7.99 | 35.50 | 0.23 | 7.26 | -0.74 |

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|---|---|---|---|---|---|---|---|
| forced | DAS (contrastive) projected on top-64 natural PCs, rank 8 | 4 | 12.16 | 35.50 | 0.34 | 8.86 | -3.30 |
| forced | DAS (contrastive) projected on top-64 natural PCs, rank 16 | 4 | 12.78 | 35.50 | 0.36 | 10.68 | -2.10 |
| forced | difference in means (rank 1) | 4 | 0.65 | 35.50 | 0.02 | 0.63 | -0.02 |
| forced | PCA-16 subspace | 4 | 22.69 | 35.50 | 0.64 | 15.54 | -7.15 |
| forced | full state exchange at the read position | 4 | 0.19 | 35.50 | 0.01 | 0.13 | -0.06 |
| forced | DAS at the read position, contrastive objective, rank 1 | 4 | 0.02 | 35.50 | 6.7e-04 | -0.00 | -0.03 |
| forced | DAS at the read position, contrastive objective, rank 2 | 4 | 2469.59 | 35.50 | 69.56 | -65.64 | -2535.23 |
| forced | DAS at the read position, contrastive objective, rank 4 | 4 | 4358.44 | 35.50 | 122.76 | -75.27 | -4433.71 |
| forced | DAS at the read position, contrastive objective, rank 8 | 4 | 4394.54 | 35.50 | 123.78 | -74.34 | -4468.89 |
| forced | DAS at the read position, contrastive objective, rank 16 | 4 | 4376.68 | 35.50 | 123.28 | -76.97 | -4453.65 |
| forced | DAS at the read position, likelihood objective, rank 1 | 4 | 12.06 | 35.50 | 0.34 | 11.21 | -0.86 |
| forced | DAS at the read position, likelihood objective, rank 2 | 4 | 19.38 | 35.50 | 0.55 | 18.86 | -0.53 |
| forced | DAS at the read position, likelihood objective, rank 4 | 4 | 19.63 | 35.50 | 0.55 | 19.08 | -0.55 |
| forced | DAS at the read position, likelihood objective, rank 8 | 4 | 21.00 | 35.50 | 0.59 | 20.40 | -0.60 |
| forced | DAS at the read position, likelihood objective, rank 16 | 4 | 20.55 | 35.50 | 0.58 | 20.18 | -0.37 |
| null | full state exchange | 2 | -2.12 | -2.12 | 1.00 | -2.10 | 0.03 |
| null | DAS, committed contrastive objective, rank 1 | 2 | 1.69 | -2.12 | -0.80 | 1.65 | -0.04 |
| null | DAS, committed contrastive objective, rank 2 | 2 | 2.23 | -2.12 | -1.05 | -7.64 | -9.87 |

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|---|---|---|---|---|---|---|---|
| null | DAS, committed contrastive objective, rank 4 | 2 | 1049.72 | -2.12 | -494.49 | -282.10 | -1331.82 |
| null | DAS, committed contrastive objective, rank 8 | 2 | 1685.10 | -2.12 | -793.79 | -635.32 | -2320.41 |
| null | DAS, committed contrastive objective, rank 16 | 2 | 1637.36 | -2.12 | -771.30 | -1277.76 | -2915.13 |
| null | random subspace, rank 1 | 2 | 0.16 | -2.12 | -0.08 | 0.17 | 0.00 |
| null | random subspace, rank 2 | 2 | -0.18 | -2.12 | 0.08 | -0.18 | -0.00 |
| null | random subspace, rank 4 | 2 | -0.25 | -2.12 | 0.12 | -0.23 | 0.02 |
| null | random subspace, rank 8 | 2 | -0.40 | -2.12 | 0.19 | -0.44 | -0.04 |
| null | random subspace, rank 16 | 2 | -0.54 | -2.12 | 0.25 | -0.53 | 0.01 |
| null | DAS with permuted labels, contrastive objective, rank 1 | 2 | 14.92 | -2.12 | -7.03 | 3.89 | -11.03 |
| null | DAS with permuted labels, contrastive objective, rank 2 | 2 | 293.52 | -2.12 | -138.26 | -79.50 | -373.01 |
| null | DAS with permuted labels, contrastive objective, rank 4 | 2 | 962.14 | -2.12 | -453.23 | -118.46 | -1080.60 |
| null | DAS with permuted labels, contrastive objective, rank 8 | 2 | 1385.44 | -2.12 | -652.63 | -99.32 | -1484.77 |
| null | DAS with permuted labels, contrastive objective, rank 16 | 2 | 1467.64 | -2.12 | -691.35 | -143.13 | -1610.77 |
| null | DAS (contrastive) projected on top-64 natural PCs, rank 1 | 2 | 1.42 | -2.12 | -0.67 | 1.37 | -0.05 |
| null | DAS (contrastive) projected on top-64 natural PCs, rank 2 | 2 | 1.44 | -2.12 | -0.68 | 1.40 | -0.04 |
| null | DAS (contrastive) projected on top-64 natural PCs, rank 4 | 2 | 1.72 | -2.12 | -0.81 | 1.72 | 7.5e-04 |
| null | DAS (contrastive) projected on top-64 natural PCs, rank 8 | 2 | 3.33 | -2.12 | -1.57 | 3.26 | -0.08 |
| null | DAS (contrastive) projected on top-64 natural PCs, rank 16 | 2 | 3.19 | -2.12 | -1.50 | 3.12 | -0.07 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| null | difference in means (rank 1) | 2 | -0.71 | -2.12 | 0.34 | -0.70 | 0.01 |
| null | PCA-16 subspace | 2 | 3.60 | -2.12 | -1.70 | 3.53 | -0.07 |
| null | full state exchange at the read position | 2 | 0.27 | -2.12 | -0.13 | 0.23 | -0.03 |
| null | DAS at the read position, contrastive objective, rank 1 | 2 | 0.10 | -2.12 | -0.05 | 0.10 | 0.00 |
| null | DAS at the read position, contrastive objective, rank 2 | 2 | 1245.42 | -2.12 | -586.67 | -80.01 | -1325.42 |
| null | DAS at the read position, contrastive objective, rank 4 | 2 | 4341.05 | -2.12 | -2044.91 | -86.97 | -4428.02 |
| null | DAS at the read position, contrastive objective, rank 8 | 2 | 4385.48 | -2.12 | -2065.84 | -85.40 | -4470.87 |
| null | DAS at the read position, contrastive objective, rank 16 | 2 | 4368.99 | -2.12 | -2058.07 | -90.63 | -4459.61 |
| null | DAS at the read position, likelihood objective, rank 1 | 2 | 6.07 | -2.12 | -2.86 | 5.36 | -0.71 |
| null | DAS at the read position, likelihood objective, rank 2 | 2 | 6.08 | -2.12 | -2.86 | 5.48 | -0.60 |
| null | DAS at the read position, likelihood objective, rank 4 | 2 | 6.01 | -2.12 | -2.83 | 5.49 | -0.52 |
| null | DAS at the read position, likelihood objective, rank 8 | 2 | 5.86 | -2.12 | -2.76 | 5.50 | -0.36 |
| null | DAS at the read position, likelihood objective, rank 16 | 2 | 5.70 | -2.12 | -2.69 | 5.45 | -0.25 |

**Table 304.** Connected components, likelihood objective: held-out teacher-forced evaluation.

| Pairs | Edit | n | Mean gain (nats) | Full state gain | Teacher-forced recovery | Change in log likelihood of Y programs | Change in log likelihood of default programs |
|---|---|---|---|---|---|---|---|
| ban | full state exchange | 12 | 32.27 | 32.27 | 1.00 | 19.21 | -13.06 |
| ban | DAS, likelihood objective, rank 1 | 12 | 36.68 | 32.27 | 1.14 | 20.19 | -16.49 |
| ban | DAS, likelihood objective, rank 2 | 12 | 38.97 | 32.27 | 1.21 | 20.24 | -18.73 |
| ban | DAS, likelihood objective, rank 4 | 12 | 40.63 | 32.27 | 1.26 | 20.29 | -20.34 |
| ban | DAS, likelihood objective, rank 8 | 12 | 45.13 | 32.27 | 1.40 | 20.28 | -24.85 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| ban | DAS, likelihood objective, rank 16 | 12 | 47.57 | 32.27 | 1.47 | 20.28 | -27.30 |
| ban | DAS with permuted labels, likelihood objective, rank 1 | 12 | 25.70 | 32.27 | 0.80 | 19.26 | -6.43 |
| ban | DAS with permuted labels, likelihood objective, rank 2 | 12 | 35.00 | 32.27 | 1.08 | 20.23 | -14.78 |
| ban | DAS with permuted labels, likelihood objective, rank 4 | 12 | 35.96 | 32.27 | 1.11 | 20.24 | -15.71 |
| ban | DAS with permuted labels, likelihood objective, rank 8 | 12 | 37.53 | 32.27 | 1.16 | 20.28 | -17.25 |
| ban | DAS with permuted labels, likelihood objective, rank 16 | 12 | 36.39 | 32.27 | 1.13 | 20.26 | -16.13 |
| ban | DAS (likelihood) projected on top-64 natural PCs, rank 1 | 12 | 12.07 | 32.27 | 0.37 | 10.12 | -1.96 |
| ban | DAS (likelihood) projected on top-64 natural PCs, rank 2 | 12 | 17.28 | 32.27 | 0.54 | 12.83 | -4.45 |
| ban | DAS (likelihood) projected on top-64 natural PCs, rank 4 | 12 | 20.93 | 32.27 | 0.65 | 14.53 | -6.40 |
| ban | DAS (likelihood) projected on top-64 natural PCs, rank 8 | 12 | 23.90 | 32.27 | 0.74 | 15.27 | -8.63 |
| ban | DAS (likelihood) projected on top-64 natural PCs, rank 16 | 12 | 22.01 | 32.27 | 0.68 | 14.99 | -7.02 |
| forced | full state exchange | 4 | 33.44 | 33.44 | 1.00 | 19.17 | -14.27 |
| forced | DAS, likelihood objective, rank 1 | 4 | 37.07 | 33.44 | 1.11 | 23.78 | -13.29 |
| forced | DAS, likelihood objective, rank 2 | 4 | 42.16 | 33.44 | 1.26 | 23.87 | -18.29 |
| forced | DAS, likelihood objective, rank 4 | 4 | 47.19 | 33.44 | 1.41 | 23.98 | -23.21 |
| forced | DAS, likelihood objective, rank 8 | 4 | 47.48 | 33.44 | 1.42 | 24.02 | -23.46 |
| forced | DAS, likelihood objective, rank 16 | 4 | 49.79 | 33.44 | 1.49 | 24.03 | -25.76 |
| forced | DAS with permuted labels, likelihood objective, rank 1 | 4 | 31.72 | 33.44 | 0.95 | 23.90 | -7.82 |
| forced | DAS with permuted labels, likelihood objective, rank 2 | 4 | 38.64 | 33.44 | 1.16 | 23.95 | -14.69 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| forced | DAS with permuted labels, likelihood objective, rank 4 | 4 | 40.57 | 33.44 | 1.21 | 23.97 | -16.60 |
| forced | DAS with permuted labels, likelihood objective, rank 8 | 4 | 44.02 | 33.44 | 1.32 | 24.02 | -20.00 |
| forced | DAS with permuted labels, likelihood objective, rank 16 | 4 | 50.70 | 33.44 | 1.52 | 24.03 | -26.67 |
| forced | DAS (likelihood) projected on top-64 natural PCs, rank 1 | 4 | 10.68 | 33.44 | 0.32 | 10.55 | -0.13 |
| forced | DAS (likelihood) projected on top-64 natural PCs, rank 2 | 4 | 14.64 | 33.44 | 0.44 | 14.14 | -0.50 |
| forced | DAS (likelihood) projected on top-64 natural PCs, rank 4 | 4 | 14.03 | 33.44 | 0.42 | 13.69 | -0.34 |
| forced | DAS (likelihood) projected on top-64 natural PCs, rank 8 | 4 | 26.74 | 33.44 | 0.80 | 17.62 | -9.11 |
| forced | DAS (likelihood) projected on top-64 natural PCs, rank 16 | 4 | 30.51 | 33.44 | 0.91 | 18.58 | -11.93 |
| null | full state exchange | 2 | -23.96 | -23.96 | 1.00 | -23.93 | 0.02 |
| null | DAS, likelihood objective, rank 1 | 2 | 18.66 | -23.96 | -0.78 | 18.03 | -0.63 |
| null | DAS, likelihood objective, rank 2 | 2 | 24.92 | -23.96 | -1.04 | 21.58 | -3.34 |
| null | DAS, likelihood objective, rank 4 | 2 | 27.58 | -23.96 | -1.15 | 22.22 | -5.36 |
| null | DAS, likelihood objective, rank 8 | 2 | 23.17 | -23.96 | -0.97 | 19.96 | -3.22 |
| null | DAS, likelihood objective, rank 16 | 2 | 18.49 | -23.96 | -0.77 | 17.75 | -0.74 |
| null | DAS with permuted labels, likelihood objective, rank 1 | 2 | 33.80 | -23.96 | -1.41 | 23.86 | -9.94 |
| null | DAS with permuted labels, likelihood objective, rank 2 | 2 | 28.46 | -23.96 | -1.19 | 23.61 | -4.85 |
| null | DAS with permuted labels, likelihood objective, rank 4 | 2 | 31.31 | -23.96 | -1.31 | 23.84 | -7.47 |
| null | DAS with permuted labels, likelihood objective, rank 8 | 2 | 24.77 | -23.96 | -1.03 | 21.56 | -3.21 |
| null | DAS with permuted labels, likelihood objective, rank 16 | 2 | 17.10 | -23.96 | -0.71 | 16.27 | -0.83 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| null | DAS (likelihood) projected on top-64 natural PCs, rank 1 | 2 | 3.05 | -23.96 | -0.13 | 3.02 | -0.03 |
| null | DAS (likelihood) projected on top-64 natural PCs, rank 2 | 2 | 7.12 | -23.96 | -0.30 | 7.11 | -0.02 |
| null | DAS (likelihood) projected on top-64 natural PCs, rank 4 | 2 | 0.74 | -23.96 | -0.03 | 0.80 | 0.05 |
| null | DAS (likelihood) projected on top-64 natural PCs, rank 8 | 2 | -11.83 | -23.96 | 0.49 | -11.84 | -0.01 |
| null | DAS (likelihood) projected on top-64 natural PCs, rank 16 | 2 | -16.77 | -23.96 | 0.70 | -16.82 | -0.05 |

**Table 305.** Connected components, committed contrastive objective: held-out teacher-forced evaluation.

| Pairs | Edit | n | Mean gain (nats) | Full state gain | Teacher-forced recovery | Change in log likelihood of Y programs | Change in log likelihood of default programs |
|---|---|---|---|---|---|---|---|
| ban | full state exchange | 12 | 32.27 | 32.27 | 1.00 | 19.21 | -13.06 |
| ban | DAS, committed contrastive objective, rank 1 | 12 | -1.93 | 32.27 | -0.06 | -1.80 | 0.13 |
| ban | DAS, committed contrastive objective, rank 2 | 12 | 2648.52 | 32.27 | 82.08 | -11.45 | -2659.97 |
| ban | DAS, committed contrastive objective, rank 4 | 12 | 5084.81 | 32.27 | 157.58 | -48.30 | -5133.11 |
| ban | DAS, committed contrastive objective, rank 8 | 12 | 5246.28 | 32.27 | 162.58 | -64.49 | -5310.77 |
| ban | DAS, committed contrastive objective, rank 16 | 12 | 5207.68 | 32.27 | 161.39 | -87.20 | -5294.88 |
| ban | random subspace, rank 1 | 12 | 0.17 | 32.27 | 0.01 | 0.21 | 0.04 |
| ban | random subspace, rank 2 | 12 | 0.23 | 32.27 | 0.01 | 0.22 | -0.02 |
| ban | random subspace, rank 4 | 12 | 0.31 | 32.27 | 0.01 | 0.31 | -0.01 |
| ban | random subspace, rank 8 | 12 | 0.17 | 32.27 | 0.01 | 0.16 | -0.01 |
| ban | random subspace, rank 16 | 12 | 0.49 | 32.27 | 0.02 | 0.46 | -0.03 |
| ban | DAS (contrastive) projected on top-64 natural PCs, rank 1 | 12 | 10.30 | 32.27 | 0.32 | 6.14 | -4.16 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| ban | DAS (contrastive) projected on top-64 natural PCs, rank 2 | 12 | 11.42 | 32.27 | 0.35 | 6.88 | -4.55 |
| ban | DAS (contrastive) projected on top-64 natural PCs, rank 4 | 12 | 10.87 | 32.27 | 0.34 | 7.20 | -3.67 |
| ban | DAS (contrastive) projected on top-64 natural PCs, rank 8 | 12 | 18.12 | 32.27 | 0.56 | 11.95 | -6.17 |
| ban | DAS (contrastive) projected on top-64 natural PCs, rank 16 | 12 | 18.52 | 32.27 | 0.57 | 12.14 | -6.38 |
| ban | difference in means (rank 1) | 12 | 2.12 | 32.27 | 0.07 | 1.95 | -0.17 |
| ban | PCA-16 subspace | 12 | 13.01 | 32.27 | 0.40 | 9.32 | -3.69 |
| ban | full state exchange at the read position | 12 | 1.43 | 32.27 | 0.04 | 1.42 | -0.01 |
| ban | DAS at the read position, likelihood objective, rank 1 | 12 | 18.05 | 32.27 | 0.56 | 16.63 | -1.42 |
| ban | DAS at the read position, likelihood objective, rank 2 | 12 | 32.20 | 32.27 | 1.00 | 19.86 | -12.34 |
| ban | DAS at the read position, likelihood objective, rank 4 | 12 | 36.22 | 32.27 | 1.12 | 20.18 | -16.05 |
| ban | DAS at the read position, likelihood objective, rank 8 | 12 | 39.08 | 32.27 | 1.21 | 20.18 | -18.90 |
| ban | DAS at the read position, likelihood objective, rank 16 | 12 | 39.75 | 32.27 | 1.23 | 20.18 | -19.56 |
| forced | full state exchange | 4 | 33.44 | 33.44 | 1.00 | 19.17 | -14.27 |
| forced | DAS, committed contrastive objective, rank 1 | 4 | -4.42 | 33.44 | -0.13 | -4.42 | 0.01 |
| forced | DAS, committed contrastive objective, rank 2 | 4 | 4718.94 | 33.44 | 141.12 | -446.34 | -5165.28 |
| forced | DAS, committed contrastive objective, rank 4 | 4 | 4863.38 | 33.44 | 145.44 | -461.14 | -5324.53 |
| forced | DAS, committed contrastive objective, rank 8 | 4 | 4924.37 | 33.44 | 147.27 | -494.32 | -5418.69 |
| forced | DAS, committed contrastive objective, rank 16 | 4 | 4904.10 | 33.44 | 146.66 | -501.67 | -5405.77 |
| forced | random subspace, rank 1 | 4 | 0.14 | 33.44 | 0.00 | 0.12 | -0.02 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| forced | random subspace, rank 2 | 4 | -0.19 | 33.44 | -0.01 | -0.14 | 0.05 |
| forced | random subspace, rank 4 | 4 | 0.06 | 33.44 | 0.00 | 0.02 | -0.05 |
| forced | random subspace, rank 8 | 4 | -0.04 | 33.44 | -0.00 | -0.08 | -0.04 |
| forced | random subspace, rank 16 | 4 | 0.13 | 33.44 | 0.00 | 0.12 | -0.01 |
| forced | DAS (contrastive) projected on top-64 natural PCs, rank 1 | 4 | 20.32 | 33.44 | 0.61 | 15.48 | -4.84 |
| forced | DAS (contrastive) projected on top-64 natural PCs, rank 2 | 4 | 19.94 | 33.44 | 0.60 | 14.79 | -5.15 |
| forced | DAS (contrastive) projected on top-64 natural PCs, rank 4 | 4 | 20.09 | 33.44 | 0.60 | 15.17 | -4.93 |
| forced | DAS (contrastive) projected on top-64 natural PCs, rank 8 | 4 | 22.03 | 33.44 | 0.66 | 14.67 | -7.36 |
| forced | DAS (contrastive) projected on top-64 natural PCs, rank 16 | 4 | 25.82 | 33.44 | 0.77 | 16.10 | -9.72 |
| forced | difference in means (rank 1) | 4 | 0.91 | 33.44 | 0.03 | 0.85 | -0.06 |
| forced | PCA-16 subspace | 4 | 26.00 | 33.44 | 0.78 | 16.35 | -9.64 |
| forced | full state exchange at the read position | 4 | 0.60 | 33.44 | 0.02 | 0.60 | 8.8e-05 |
| forced | DAS at the read position, likelihood objective, rank 1 | 4 | 9.64 | 33.44 | 0.29 | 9.56 | -0.08 |
| forced | DAS at the read position, likelihood objective, rank 2 | 4 | 27.31 | 33.44 | 0.82 | 19.79 | -7.51 |
| forced | DAS at the read position, likelihood objective, rank 4 | 4 | 35.93 | 33.44 | 1.07 | 24.08 | -11.84 |
| forced | DAS at the read position, likelihood objective, rank 8 | 4 | 41.51 | 33.44 | 1.24 | 24.10 | -17.41 |
| forced | DAS at the read position, likelihood objective, rank 16 | 4 | 41.15 | 33.44 | 1.23 | 24.11 | -17.04 |
| null | full state exchange | 2 | -23.96 | -23.96 | 1.00 | -23.93 | 0.02 |
| null | DAS, committed contrastive objective, rank 1 | 2 | -3.31 | -23.96 | 0.14 | -3.29 | 0.02 |
| null | DAS, committed contrastive objective, rank 2 | 2 | 36.56 | -23.96 | -1.53 | 12.82 | -23.74 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| null | DAS, committed contrastive objective, rank 4 | 2 | 5081.85 | -23.96 | -212.12 | -89.75 | -5171.60 |
| null | DAS, committed contrastive objective, rank 8 | 2 | 5252.86 | -23.96 | -219.26 | -102.50 | -5355.36 |
| null | DAS, committed contrastive objective, rank 16 | 2 | 5220.64 | -23.96 | -217.91 | -103.60 | -5324.23 |
| null | random subspace, rank 1 | 2 | -0.10 | -23.96 | 0.00 | 0.00 | 0.10 |
| null | random subspace, rank 2 | 2 | 0.21 | -23.96 | -0.01 | 0.21 | -0.00 |
| null | random subspace, rank 4 | 2 | -0.24 | -23.96 | 0.01 | -0.25 | -0.01 |
| null | random subspace, rank 8 | 2 | -0.76 | -23.96 | 0.03 | -0.71 | 0.05 |
| null | random subspace, rank 16 | 2 | -0.95 | -23.96 | 0.04 | -0.89 | 0.06 |
| null | DAS (contrastive) projected on top-64 natural PCs, rank 1 | 2 | -4.02 | -23.96 | 0.17 | -3.97 | 0.05 |
| null | DAS (contrastive) projected on top-64 natural PCs, rank 2 | 2 | -3.39 | -23.96 | 0.14 | -3.40 | -0.01 |
| null | DAS (contrastive) projected on top-64 natural PCs, rank 4 | 2 | 0.19 | -23.96 | -0.01 | 0.22 | 0.03 |
| null | DAS (contrastive) projected on top-64 natural PCs, rank 8 | 2 | -6.35 | -23.96 | 0.26 | -6.34 | 0.01 |
| null | DAS (contrastive) projected on top-64 natural PCs, rank 16 | 2 | -14.47 | -23.96 | 0.60 | -14.49 | -0.02 |
| null | difference in means (rank 1) | 2 | -1.13 | -23.96 | 0.05 | -1.16 | -0.03 |
| null | PCA-16 subspace | 2 | -25.31 | -23.96 | 1.06 | -25.28 | 0.03 |
| null | full state exchange at the read position | 2 | 1.25 | -23.96 | -0.05 | 1.16 | -0.09 |
| null | DAS at the read position, likelihood objective, rank 1 | 2 | 16.17 | -23.96 | -0.67 | 16.04 | -0.13 |
| null | DAS at the read position, likelihood objective, rank 2 | 2 | 33.53 | -23.96 | -1.40 | 23.45 | -10.08 |
| null | DAS at the read position, likelihood objective, rank 4 | 2 | 38.14 | -23.96 | -1.59 | 23.74 | -14.40 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| null | DAS at the read position, likelihood objective, rank 8 | 2 | 42.28 | -23.96 | -1.76 | 23.74 | -18.54 |
| null | DAS at the read position, likelihood objective, rank 16 | 2 | 42.36 | -23.96 | -1.77 | 23.75 | -18.61 |

**Table 306.** Kth smallest, likelihood objective: held-out teacher-forced evaluation.

| Pairs | Edit | n | Mean gain (nats) | Full state gain | Teacher-forced recovery | Change in log likelihood of Y programs | Change in log likelihood of default programs |
|---|---|---|---|---|---|---|---|
| ban | full state exchange | 12 | 7.00 | 7.00 | 1.00 | 3.85 | -3.15 |
| ban | DAS, likelihood objective, rank 1 | 12 | 13.61 | 7.00 | 1.95 | 4.08 | -9.54 |
| ban | DAS, likelihood objective, rank 2 | 12 | 17.25 | 7.00 | 2.46 | 4.09 | -13.15 |
| ban | DAS, likelihood objective, rank 4 | 12 | 19.60 | 7.00 | 2.80 | 4.13 | -15.47 |
| ban | DAS, likelihood objective, rank 8 | 12 | 20.51 | 7.00 | 2.93 | 4.09 | -16.42 |
| ban | DAS, likelihood objective, rank 16 | 12 | 20.16 | 7.00 | 2.88 | 4.09 | -16.07 |
| ban | DAS with permuted labels, likelihood objective, rank 1 | 12 | 13.24 | 7.00 | 1.89 | 3.14 | -10.10 |
| ban | DAS with permuted labels, likelihood objective, rank 2 | 12 | 14.40 | 7.00 | 2.06 | 2.89 | -11.52 |
| ban | DAS with permuted labels, likelihood objective, rank 4 | 12 | 16.01 | 7.00 | 2.29 | 2.92 | -13.10 |
| ban | DAS with permuted labels, likelihood objective, rank 8 | 12 | 18.07 | 7.00 | 2.58 | 2.86 | -15.21 |
| ban | DAS with permuted labels, likelihood objective, rank 16 | 12 | 21.67 | 7.00 | 3.10 | 3.43 | -18.24 |
| ban | DAS (likelihood) projected on top-64 natural PCs, rank 1 | 12 | 1.25 | 7.00 | 0.18 | 1.01 | -0.24 |
| ban | DAS (likelihood) projected on top-64 natural PCs, rank 2 | 12 | 2.46 | 7.00 | 0.35 | 1.94 | -0.52 |
| ban | DAS (likelihood) projected on top-64 natural PCs, rank 4 | 12 | 3.67 | 7.00 | 0.52 | 2.75 | -0.92 |
| ban | DAS (likelihood) projected on top-64 natural PCs, rank 8 | 12 | 4.61 | 7.00 | 0.66 | 3.11 | -1.50 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| ban | DAS (likelihood) projected on top-64 natural PCs, rank 16 | 12 | 3.60 | 7.00 | 0.51 | 2.54 | -1.06 |
| forced | full state exchange | 4 | 22.31 | 22.31 | 1.00 | 11.76 | -10.55 |
| forced | DAS, likelihood objective, rank 1 | 4 | 24.53 | 22.31 | 1.10 | 15.79 | -8.73 |
| forced | DAS, likelihood objective, rank 2 | 4 | 29.62 | 22.31 | 1.33 | 15.87 | -13.75 |
| forced | DAS, likelihood objective, rank 4 | 4 | 31.91 | 22.31 | 1.43 | 15.84 | -16.07 |
| forced | DAS, likelihood objective, rank 8 | 4 | 44.78 | 22.31 | 2.01 | 15.86 | -28.92 |
| forced | DAS, likelihood objective, rank 16 | 4 | 47.90 | 22.31 | 2.15 | 15.87 | -32.03 |
| forced | DAS with permuted labels, likelihood objective, rank 1 | 4 | 22.74 | 22.31 | 1.02 | 14.08 | -8.66 |
| forced | DAS with permuted labels, likelihood objective, rank 2 | 4 | 20.39 | 22.31 | 0.91 | 14.09 | -6.29 |
| forced | DAS with permuted labels, likelihood objective, rank 4 | 4 | 23.62 | 22.31 | 1.06 | 11.47 | -12.15 |
| forced | DAS with permuted labels, likelihood objective, rank 8 | 4 | 24.98 | 22.31 | 1.12 | 11.80 | -13.18 |
| forced | DAS with permuted labels, likelihood objective, rank 16 | 4 | 27.26 | 22.31 | 1.22 | 7.75 | -19.51 |
| forced | DAS (likelihood) projected on top-64 natural PCs, rank 1 | 4 | 7.54 | 22.31 | 0.34 | 7.52 | -0.02 |
| forced | DAS (likelihood) projected on top-64 natural PCs, rank 2 | 4 | 9.99 | 22.31 | 0.45 | 9.94 | -0.05 |
| forced | DAS (likelihood) projected on top-64 natural PCs, rank 4 | 4 | 10.70 | 22.31 | 0.48 | 10.44 | -0.26 |
| forced | DAS (likelihood) projected on top-64 natural PCs, rank 8 | 4 | 14.24 | 22.31 | 0.64 | 13.09 | -1.15 |
| forced | DAS (likelihood) projected on top-64 natural PCs, rank 16 | 4 | 16.00 | 22.31 | 0.72 | 13.43 | -2.57 |
| null | full state exchange | 2 | -0.36 | -0.36 | 1.00 | -0.33 | 0.04 |
| null | DAS, likelihood objective, rank 1 | 2 | 6.28 | -0.36 | -17.25 | 6.09 | -0.19 |
| null | DAS, likelihood objective, rank 2 | 2 | 9.83 | -0.36 | -26.97 | 7.54 | -2.28 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| null | DAS, likelihood objective, rank 4 | 2 | 11.22 | -0.36 | -30.79 | 7.74 | -3.48 |
| null | DAS, likelihood objective, rank 8 | 2 | 13.31 | -0.36 | -36.53 | 7.73 | -5.58 |
| null | DAS, likelihood objective, rank 16 | 2 | 13.06 | -0.36 | -35.85 | 7.74 | -5.33 |
| null | DAS with permuted labels, likelihood objective, rank 1 | 2 | 5.52 | -0.36 | -15.15 | 5.44 | -0.08 |
| null | DAS with permuted labels, likelihood objective, rank 2 | 2 | 6.69 | -0.36 | -18.36 | 6.41 | -0.28 |
| null | DAS with permuted labels, likelihood objective, rank 4 | 2 | 12.53 | -0.36 | -34.39 | 7.17 | -5.37 |
| null | DAS with permuted labels, likelihood objective, rank 8 | 2 | 15.61 | -0.36 | -42.85 | 5.90 | -9.71 |
| null | DAS with permuted labels, likelihood objective, rank 16 | 2 | 22.65 | -0.36 | -62.16 | 6.81 | -15.85 |
| null | DAS (likelihood) projected on top-64 natural PCs, rank 1 | 2 | 0.61 | -0.36 | -1.67 | 0.65 | 0.04 |
| null | DAS (likelihood) projected on top-64 natural PCs, rank 2 | 2 | 1.50 | -0.36 | -4.11 | 1.53 | 0.03 |
| null | DAS (likelihood) projected on top-64 natural PCs, rank 4 | 2 | 2.04 | -0.36 | -5.58 | 2.06 | 0.03 |
| null | DAS (likelihood) projected on top-64 natural PCs, rank 8 | 2 | 2.21 | -0.36 | -6.07 | 2.25 | 0.03 |
| null | DAS (likelihood) projected on top-64 natural PCs, rank 16 | 2 | 1.26 | -0.36 | -3.46 | 1.29 | 0.03 |

**Table 307.** Dormancy checks (Makelov): natural variance share, held-out displacement share and retention of the projected map by rank.

| Task | Rank k | Natural variance share | Random mean | Percentile against 1,000 random subspaces | Top-k PCA (upper bound) | Share for the twin alone | Held-out displacement share (random k/4096) | Teacher-forced retention of the projected map |
|---|---|---|---|---|---|---|---|---|
| shortest path | 1 | 0.0305 | 2.5e-04 | 100.0 | 0.597 | 0.0185 | 0.0471 (2.4e-04) | 0.74 |
| shortest path | 2 | 0.0331 | 5.0e-04 | 100.0 | 0.723 | 0.0186 | 0.0496 (4.9e-04) | 0.71 |
| shortest path | 4 | 0.0413 | 9.8e-04 | 100.0 | 0.834 | 0.0277 | 0.0608 (9.8e-04) | 0.80 |
| shortest path | 8 | 0.0548 | 0.0019 | 100.0 | 0.918 | 0.0383 | 0.0773 (0.0020) | 0.80 |
| shortest path | 16 | 0.0684 | 0.0039 | 100.0 | 0.967 | 0.0447 | 0.0883 (0.0039) | 0.69 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| connected components | 1 | 0.0280 | 2.5e-04 | 100.0 | 0.407 | 0.0118 | 0.0373 (2.4e-04) | 0.31 |
| connected components | 2 | 0.0381 | 4.9e-04 | 100.0 | 0.542 | 0.0166 | 0.0551 (4.9e-04) | 0.40 |
| connected components | 4 | 0.0553 | 9.6e-04 | 100.0 | 0.709 | 0.0227 | 0.0701 (9.8e-04) | 0.41 |
| connected components | 8 | 0.0625 | 0.0020 | 100.0 | 0.848 | 0.0277 | 0.0774 (0.0020) | 0.55 |
| connected components | 16 | 0.0763 | 0.0039 | 100.0 | 0.931 | 0.0393 | 0.0933 (0.0039) | 0.54 |
| kth smallest | 1 | 0.0069 | 2.4e-04 | 100.0 | 0.986 | 0.0063 | 0.0264 (2.4e-04) | 0.20 |
| kth smallest | 2 | 0.0304 | 5.0e-04 | 100.0 | 0.992 | 0.0274 | 0.0862 (4.9e-04) | 0.24 |
| kth smallest | 4 | 0.0510 | 0.0010 | 100.0 | 0.995 | 0.0489 | 0.1085 (9.8e-04) | 0.26 |
| kth smallest | 8 | 0.0917 | 0.0019 | 100.0 | 0.998 | 0.0893 | 0.1391 (0.0020) | 0.27 |
| kth smallest | 16 | 0.1848 | 0.0039 | 100.0 | 0.999 | 0.1839 | 0.1674 (0.0039) | 0.26 |

### 7.3.8 The manifold in the fitted coordinates

The manifold claim placed prompts of shortest path in the rank-16 coordinates of the fitted map. The *paraphrase null* is the distribution of the distance D from the default centroid over neutral paraphrases of the twin (median 0.52, 95th percentile 1.13). Every ban template lies outside it (D 1.73 to 2.62, at the 100th percentile; the first three, W1-W3, are training templates of the map, so the conclusion that bans leave the null rests on the held-out fifth template, D 2.62, and the fourth, D 1.73; iteration-8 correction), each displaced toward the second choice along the axis from the default centroid to the Bellman-Ford/SPFA centroid ($+1.10$ to $+1.76$). The two random seeds, the short instruction to make a new algorithm and the 2,000-word novelty instruction lie inside it (76th to 88th percentile), as predicted. The semantic seed was predicted outside and lies on the boundary, at the 93.9th percentile (D 1.127 against a null 95th percentile of 1.130; Table 309) (iteration-8 correction). Behaviour agrees with the placement: the neutral prompt, both random seeds and the short instruction write Dijkstra in every correct program, the semantic seed adds one program labelled other, and the long novelty instruction is the only prompt that reaches a named alternative (6 Bellman-Ford/SPFA programs of 28 correct; Table 308), a shift that happens while the model writes its rationale, not in the prompt states. Multidimensional scaling of the centroids has stress 0.079; centroids and implemented counts are in Table 310, and geodesic distances on neighbour graphs in Table 311. Bans that name all but one of the five algorithms (bans that leave one survivor) are not placed near their survivor: the survivor's centroid ranks 3.10 of 5 on average by geodesic distance (random relabelling p 0.63), and every such ban sits nearest the forced Bellman-Ford/SPFA prompts. The geometry does not predict where a ban lands.

*[Figure fig_das_manifold_sp: generated in the report phase, not in this draft]*

**Table 308.** Shortest path: what each seed and instruction prompt writes (30 programs each).

| Prompt | Pass | Correct composition | Correct off the default [CP 95%] | Truncated |
|---|---|---|---|---|
| short instruction to make a new algorithm | 1.00 | Dijkstra 30 | 0 [0.00, 0.12] | 0 |
| 2,000-word novelty instruction | 0.93 | Dijkstra 22, Bellman-Ford/SPFA 6 | 6 [0.08, 0.39] | 0 |
| neutral prompt | 0.97 | Dijkstra 29 | 0 [0.00, 0.12] | 0 |
| random seed 1 (seed = d32qxk4nt43) | 1.00 | Dijkstra 30 | 0 [0.00, 0.12] | 0 |

| | | | | | |
|---|---|---|---|---|---|
| random seed 2 (seed = (horse chair roof love)) | 1.00 | Dijkstra 30 | 0 | [0.00, 0.12] | 0 |
| semantic seed | 0.97 | Dijkstra 28, other 1 | 1 | [8.4e-04, 0.17] | 0 |

**Table 309.** Shortest path: placement of prompts in the rank-16 DAS coordinates against the paraphrase null. (iteration-8 correction: the first three ban templates are training templates of the map; the semantic seed sits on the boundary of the null 95%)

| Prompt (shortest path) | Distance D to the default centroid (rank-16 DAS coordinates) | Null percentile | Inside the null 95% | Coordinate along the axis toward the second choice |
|---|---|---|---|---|
| ban, first template (training template of the map) | 2.25 | 100.0 | no | 1.55 |
| ban, second template (training template of the map) | 1.95 | 100.0 | no | 1.39 |
| ban, third template (training template of the map) | 1.79 | 100.0 | no | 1.15 |
| ban, fourth template | 1.73 | 100.0 | no | 1.10 |
| ban, fifth template | 2.62 | 100.0 | no | 1.76 |
| neutral prompt | 0.45 | 40.9 | yes | 0.11 |
| random seed 1 (seed = d32qxk4nt43) | 0.95 | 87.9 | yes | -0.58 |
| random seed 2 (seed = (horse chair roof love)) | 0.69 | 77.3 | yes | -0.21 |
| short instruction to make a new algorithm | 0.69 | 75.8 | yes | 0.49 |
| 2,000-word novelty instruction | 0.85 | 86.4 | yes | 0.03 |
| semantic seed | 1.13 | 93.9 | yes | 0.44 |

**Table 310.** Shortest path: algorithm centroids along the axis toward the second choice and how often each algorithm has been implemented.

| Algorithm | Centroid coordinate along the axis toward the second choice | Implemented under the twin | Implemented by Qwen3-8B-Base | Implemented under this artifact's knobs | Implemented under the 40 random directions of the engine control | All knobs |
|---|---|---|---|---|---|---|
| Dijkstra | 1.13 | 20 | 25 | 701 | 1608 | 2309 |
| Bellman-Ford/SPFA | 2.30 | 0 | 18 | 343 | 489 | 832 |
| bidirectional Dijkstra | 0.95 | 0 | 0 | 0 | 0 | 0 |
| Floyd-Warshall | 1.53 | 0 | 0 | 139 | 3 | 142 |
| Dial | 0.67 | 0 | 0 | 0 | 0 | 0 |

**Table 311.** Shortest path: geodesic distances between centroids on neighbour graphs of the fitted coordinates.

| k of the neighbour graph | Mean survivor rank of bans that leave one survivor (of 5) | Random relabelling p | Unreachable pairs | Geodesic Dijkstra to Dijkstra (spread) | to Bellman-Ford/ SPFA | to bidirectional Dijkstra | to Floyd-Warshall | to Dial | Bellman-Ford/ SPFA to Floyd-Warshall |
|---|---|---|---|---|---|---|---|---|---|
| 5 | 3.10 | 0.63 | 3080 | 0.24 | 3.20 | 1.77 | 3.87 | 2.55 | 4.67 |
| 10 | 3.10 | 0.63 | 0 | 0.24 | 2.30 | 1.42 | 1.59 | 2.12 | 2.82 |
| 15 | 3.20 | 0.71 | 0 | 0.24 | 1.45 | 1.11 | 1.50 | 1.82 | 1.85 |

### 7.3.9 Seeds, discreteness and the full cell list

A second initialisation seed fits a different subspace (principal angles of 38 to 90 degrees to the first) with similar behaviour: no discordance between seeds reaches significance (Table 312). The subspace is therefore not unique, as in Gemma (Section 7.8). The discreteness claim fails. Scaling the rank-2 edit of the ban by lambda from 0 to 1.5 raises correct Bellman-Ford/SPFA programs from 1 of 20 (lambda 0 to 0.2) through 4 of 20 (0.4 to 0.5) to a plateau of 6 to 9 of 20 from lambda 0.6 to 1.5; the committed logistic width is 3.25 in lambda (midpoint 1.47), against 0.3 or less for a step, and the one candidate hybrid among 114 correct programs was judged a single algorithm (Table 313). Of the 38 correct programs labelled other across the experiment, 36 are catalogue variants and 2 are labeller misses; none is novel. 29 of the 36 come from the full state exchange of the forced counting sort source and 6 more from the forced counting sort prompts: they are counting sort in dictionary form, which the extended labeller now names counting sort (iteration-8 correction). The engine reproduced the iteration-6 twin labels in 20 of 20, 19 of 20 and 20 of 20 slots; two GPU unit tests missed their tolerance because of bf16 noise from kernel shapes (packed against separate forward passes differ by up to 0.273 nats per program, cached against full forward passes by 0.160), so every compared quantity is measured within one batch shape. Table 314 lists every generation cell.

**Table 312.** A second DAS initialisation seed.

| Arm | Seed 1 correct Y of 30 | Seed 2 correct Y of 30 | Discordant seed-1-only : seed-2-only (p) | Pass, seed 2 |
|---|---|---|---|---|
| DAS (likelihood objective), second seed, forced Bellman-Ford/SPFA source, rank 16 | 5 | 1 | 5:1 (0.219) | 0.67 |
| DAS (likelihood objective), second seed, forced Floyd-Warshall source, rank 16 | 25 | 27 | 0:2 (0.500) | 0.90 |
| DAS (likelihood objective), second seed, fifth template ban source, rank 1 | 8 | 3 | 6:1 (0.125) | 0.73 |
| DAS (likelihood objective), second seed, fifth template ban source, rank 16 | 6 | 7 | 3:4 (1.000) | 0.67 |
| DAS (likelihood objective), second seed, fifth template ban source, rank 2 | 12 | 8 | 6:2 (0.289) | 0.60 |

**Table 313.** Discreteness: the rank-2 DAS edit of the ban scaled by lambda (shortest path, 20 slots each).

| Lambda | Correct Bellman-Ford/SPFA (of 20) | Correct Dijkstra | Pass | Labels |
|---|---|---|---|---|
| 0.0 | 1 | 18 | 0.95 | Dijkstra 18, Bellman-Ford/SPFA 2 |
| 0.2 | 1 | 16 | 0.85 | Dijkstra 17, Bellman-Ford/SPFA 3 |
| 0.4 | 4 | 11 | 0.75 | Dijkstra 11, Bellman-Ford/SPFA 9 |
| 0.5 | 4 | 11 | 0.75 | Dijkstra 11, Bellman-Ford/SPFA 9 |
| 0.6 | 6 | 9 | 0.80 | Bellman-Ford/SPFA 10, Dijkstra 9, other 1 |
| 0.7 | 8 | 9 | 0.85 | Bellman-Ford/SPFA 11, Dijkstra 9 |
| 0.8 | 7 | 10 | 0.85 | Dijkstra 10, Bellman-Ford/SPFA 10 |
| 1.0 | 9 | 8 | 0.85 | Bellman-Ford/SPFA 12, Dijkstra 8 |
| 1.25 | 7 | 10 | 0.85 | Dijkstra 10, Bellman-Ford/SPFA 10 |
| 1.5 | 8 | 9 | 0.85 | Bellman-Ford/SPFA 11, Dijkstra 9 |

**Table 314.** Every generation cell of the DAS experiment.

| Task | Arm | n | Pass | Truncated | Correct composition |
|---|---|---|---|---|---|
| shortest path | padding twin (base for forced sources) | 30 | 0.97 | 0 | Dijkstra 29 |
| shortest path | neutral twin of template 4 (base) | 10 | 1.00 | 0 | Dijkstra 10 |
| shortest path | neutral twin of template 5 (base) | 30 | 0.97 | 0 | Dijkstra 28, Bellman-Ford/SPFA 1 |
| shortest path | DAS (likelihood objective), second seed, forced Bellman-Ford/SPFA source, rank 16 | 30 | 0.67 | 0 | Dijkstra 19, Bellman-Ford/SPFA 1 |
| shortest path | DAS (likelihood objective), second seed, forced Floyd-Warshall source, rank 16 | 30 | 0.90 | 0 | Floyd-Warshall 27 |
| shortest path | DAS (likelihood objective), second seed, fifth template ban source, rank 1 | 30 | 0.73 | 0 | Dijkstra 19, Bellman-Ford/SPFA 3 |
| shortest path | DAS (likelihood objective), second seed, fifth template ban source, rank 16 | 30 | 0.67 | 0 | Dijkstra 13, Bellman-Ford/SPFA 7 |
| shortest path | DAS (likelihood objective), second seed, fifth template ban source, rank 2 | 30 | 0.60 | 0 | Dijkstra 9, Bellman-Ford/SPFA 8, other 1 |
| shortest path | DAS (likelihood objective), forced Bellman-Ford/SPFA source, rank 1 | 30 | 0.70 | 0 | Bellman-Ford/SPFA 13, Dijkstra 8 |
| shortest path | DAS (likelihood objective), forced Bellman-Ford/SPFA source, rank 16 | 30 | 0.80 | 0 | Dijkstra 19, Bellman-Ford/SPFA 5 |
| shortest path | DAS (likelihood objective), forced Bellman-Ford/SPFA source, rank 4 | 30 | 0.67 | 0 | Dijkstra 11, Bellman-Ford/SPFA 9 |
| shortest path | DAS (likelihood objective), forced Floyd-Warshall source, rank 1 | 30 | 0.93 | 0 | Floyd-Warshall 28 |
| shortest path | DAS (likelihood objective), forced Floyd-Warshall source, rank 16 | 30 | 0.83 | 0 | Floyd-Warshall 25 |
| shortest path | DAS (likelihood objective), forced Floyd-Warshall source, rank 4 | 30 | 0.93 | 0 | Floyd-Warshall 28 |
| shortest path | DAS (likelihood objective), fourth template ban source, rank 2 | 10 | 0.90 | 0 | Dijkstra 5, Bellman-Ford/SPFA 4 |

| | | | | | |
|---|---|---|---|---|---|
| shortest path | DAS (likelihood objective), fifth template ban source, rank 1 | 30 | 0.77 | 0 | Dijkstra 15, Bellman-Ford/SPFA 8 |
| shortest path | DAS (likelihood objective), fifth template ban source, rank 16 | 30 | 0.80 | 0 | Dijkstra 18, Bellman-Ford/SPFA 6 |
| shortest path | DAS (likelihood objective), fifth template ban source, rank 2 | 30 | 0.83 | 0 | Dijkstra 13, Bellman-Ford/SPFA 12 |
| shortest path | DAS (likelihood objective), fifth template ban source, rank 4 | 30 | 0.57 | 0 | Bellman-Ford/SPFA 10, Dijkstra 7 |
| shortest path | DAS (likelihood objective), fifth template ban source, rank 8 | 30 | 0.83 | 0 | Dijkstra 15, Bellman-Ford/SPFA 10 |
| shortest path | DAS (committed contrastive objective), forced Bellman-Ford/SPFA source, rank 1 | 10 | 0.40 | 0 | Dijkstra 3, Bellman-Ford/SPFA 1 |
| shortest path | DAS (committed contrastive objective), forced Bellman-Ford/SPFA source, rank 16 | 10 | 0.00 | 10 | none |
| shortest path | DAS (committed contrastive objective), forced Bellman-Ford/SPFA source, rank 4 | 10 | 0.00 | 9 | none |
| shortest path | DAS (committed contrastive objective), forced Floyd-Warshall source, rank 1 | 10 | 0.50 | 0 | Dijkstra 4, Bellman-Ford/SPFA 1 |
| shortest path | DAS (committed contrastive objective), forced Floyd-Warshall source, rank 16 | 10 | 0.00 | 10 | none |
| shortest path | DAS (committed contrastive objective), forced Floyd-Warshall source, rank 4 | 10 | 0.00 | 10 | none |
| shortest path | DAS (committed contrastive objective), fifth template ban source, rank 1 | 10 | 0.10 | 5 | Dijkstra 1 |
| shortest path | DAS (committed contrastive objective), fifth template ban source, rank 16 | 10 | 0.00 | 9 | none |
| shortest path | DAS (committed contrastive objective), fifth template ban source, rank 2 | 10 | 0.00 | 7 | none |
| shortest path | DAS (committed contrastive objective), fifth template ban source, rank 4 | 10 | 0.00 | 7 | none |
| shortest path | DAS (committed contrastive objective), fifth template ban source, rank 8 | 10 | 0.00 | 10 | none |
| shortest path | difference in means (rank 1), fifth template ban source | 30 | 0.97 | 0 | Dijkstra 29 |
| shortest path | full state exchange, forced Bellman-Ford/SPFA (family 3), block 18 only | 30 | 1.00 | 0 | Bellman-Ford/SPFA 30 |
| shortest path | full state exchange, forced Bellman-Ford/SPFA (family 3), blocks 12-23 | 30 | 1.00 | 0 | Bellman-Ford/SPFA 30 |

| | | | | | |
|---|---|---|---|---|---|
| shortest path | full state exchange, forced Bellman-Ford/SPFA source | 30 | 1.00 | 0 | Bellman-Ford/SPFA 30 |
| shortest path | full state exchange, forced Floyd-Warshall source | 30 | 0.67 | 0 | Bellman-Ford/SPFA 12, Floyd-Warshall 8 |
| shortest path | full state exchange, fourth template ban source | 10 | 0.50 | 0 | Bellman-Ford/SPFA 4, Dijkstra 1 |
| shortest path | full state exchange, ban source, block 18 only | 30 | 1.00 | 0 | Dijkstra 30 |
| shortest path | full state exchange, ban source, blocks 12-23 | 30 | 0.73 | 0 | Dijkstra 11, Bellman-Ford/SPFA 11 |
| shortest path | rank-2 DAS edit scaled by lambda 0.0, ban source | 20 | 0.95 | 0 | Dijkstra 18, Bellman-Ford/SPFA 1 |
| shortest path | rank-2 DAS edit scaled by lambda 0.2, ban source | 20 | 0.85 | 0 | Dijkstra 16, Bellman-Ford/SPFA 1 |
| shortest path | rank-2 DAS edit scaled by lambda 0.4, ban source | 20 | 0.75 | 0 | Dijkstra 11, Bellman-Ford/SPFA 4 |
| shortest path | rank-2 DAS edit scaled by lambda 0.5, ban source | 20 | 0.75 | 0 | Dijkstra 11, Bellman-Ford/SPFA 4 |
| shortest path | rank-2 DAS edit scaled by lambda 0.6, ban source | 20 | 0.80 | 0 | Dijkstra 9, Bellman-Ford/SPFA 6, other 1 |
| shortest path | rank-2 DAS edit scaled by lambda 0.7, ban source | 20 | 0.85 | 0 | Dijkstra 9, Bellman-Ford/SPFA 8 |
| shortest path | rank-2 DAS edit scaled by lambda 0.8, ban source | 20 | 0.85 | 0 | Dijkstra 10, Bellman-Ford/SPFA 7 |
| shortest path | rank-2 DAS edit scaled by lambda 1.0, ban source | 20 | 0.85 | 0 | Bellman-Ford/SPFA 9, Dijkstra 8 |
| shortest path | rank-2 DAS edit scaled by lambda 1.25, ban source | 20 | 0.85 | 0 | Dijkstra 10, Bellman-Ford/SPFA 7 |
| shortest path | rank-2 DAS edit scaled by lambda 1.5, ban source | 20 | 0.85 | 0 | Dijkstra 9, Bellman-Ford/SPFA 8 |
| shortest path | short instruction to make a new algorithm | 30 | 1.00 | 0 | Dijkstra 30 |
| shortest path | 2,000-word novelty instruction | 30 | 0.93 | 0 | Dijkstra 22, Bellman-Ford/SPFA 6 |
| shortest path | neutral prompt | 30 | 0.97 | 0 | Dijkstra 29 |
| shortest path | random seed 1 (seed = d32qxk4nt43) | 30 | 1.00 | 0 | Dijkstra 30 |
| shortest path | random seed 2 (seed = (horse chair roof love)) | 30 | 1.00 | 0 | Dijkstra 30 |
| shortest path | semantic seed | 30 | 0.97 | 0 | Dijkstra 28, other 1 |
| shortest path | null source (forced default), DAS rank 16 | 30 | 0.83 | 0 | Dijkstra 24, Bellman-Ford/SPFA 1 |
| shortest path | null source (forced default), full state exchange | 30 | 0.97 | 0 | Dijkstra 29 |
| shortest path | padding twin of forced Bellman-Ford/SPFA (family 3) | 30 | 0.97 | 0 | Dijkstra 29 |
| shortest path | PCA-16 subspace, fifth template ban source | 30 | 0.83 | 0 | Dijkstra 24, Bellman-Ford/SPFA 1 |
| shortest path | DAS with permuted labels, forced Floyd-Warshall source, rank 16 | 30 | 0.63 | 0 | Bellman-Ford/SPFA 8, Floyd-Warshall 6, Dijkstra 5 |

| | | | | | |
|---|---|---|---|---|---|
| shortest path | DAS with permuted labels, fifth template ban source, rank 16 | 30 | 0.63 | 0 | Floyd-Warshall 15, Bellman-Ford/SPFA 3, Dijkstra 1 |
| shortest path | DAS with permuted labels, fifth template ban source, rank 2 | 30 | 0.97 | 0 | Bellman-Ford/SPFA 15, Dijkstra 12, Floyd-Warshall 2 |
| shortest path | random subspace, forced Bellman-Ford/SPFA source, rank 16 | 30 | 0.97 | 0 | Dijkstra 29 |
| shortest path | random subspace, forced Floyd-Warshall source, rank 16 | 30 | 0.97 | 0 | Dijkstra 29 |
| shortest path | random subspace 1, fifth template ban source, rank 16 | 30 | 0.97 | 0 | Dijkstra 28, Bellman-Ford/SPFA 1 |
| shortest path | random subspace 2, fifth template ban source, rank 16 | 30 | 0.97 | 0 | Dijkstra 29 |
| shortest path | random subspace 3, fifth template ban source, rank 16 | 30 | 0.97 | 0 | Dijkstra 28, Bellman-Ford/SPFA 1 |
| shortest path | random subspace, fifth template ban source, rank 2 | 30 | 0.97 | 0 | Dijkstra 28, Bellman-Ford/SPFA 1 |
| shortest path | reverse interchange, DAS rank 2 | 30 | 1.00 | 0 | Bellman-Ford/SPFA 30 |
| shortest path | reverse interchange, full state | 30 | 1.00 | 0 | Dijkstra 21, Bellman-Ford/SPFA 9 |
| shortest path | full state exchange at the read position, fifth template ban source | 30 | 0.97 | 0 | Dijkstra 28, Bellman-Ford/SPFA 1 |
| shortest path | DAS at the read position, fifth template ban source, rank 16 | 30 | 0.90 | 0 | Dijkstra 25, Bellman-Ford/SPFA 2 |
| connected components | padding twin (base for forced sources) | 30 | 1.00 | 0 | union find 30 |
| connected components | neutral twin of template 4 (base) | 10 | 1.00 | 0 | union find 10 |
| connected components | neutral twin of template 5 (base) | 30 | 1.00 | 0 | union find 29, DFS 1 |
| connected components | DAS (likelihood objective), forced BFS source, rank 1 | 30 | 1.00 | 0 | DFS 29, BFS 1 |
| connected components | DAS (likelihood objective), forced BFS source, rank 16 | 30 | 1.00 | 0 | DFS 21, BFS 9 |
| connected components | DAS (likelihood objective), forced BFS source, rank 4 | 30 | 1.00 | 0 | DFS 23, BFS 7 |
| connected components | DAS (likelihood objective), forced DFS source, rank 1 | 30 | 1.00 | 0 | DFS 30 |
| connected components | DAS (likelihood objective), forced DFS source, rank 16 | 30 | 1.00 | 0 | DFS 23, BFS 7 |
| connected components | DAS (likelihood objective), forced DFS source, rank 4 | 30 | 1.00 | 0 | DFS 26, BFS 4 |
| connected components | DAS (likelihood objective), fourth template ban source, rank 1 | 10 | 1.00 | 0 | DFS 10 |
| connected components | DAS (likelihood objective), fifth template ban source, rank 1 | 30 | 0.97 | 0 | BFS 18, DFS 11 |
| connected components | DAS (likelihood objective), fifth template ban source, rank 16 | 30 | 1.00 | 0 | BFS 29, DFS 1 |
| connected components | DAS (likelihood objective), fifth template ban source, rank 2 | 30 | 1.00 | 0 | BFS 23, DFS 7 |

| | | | | | |
|---|---|---|---|---|---|
| connected components | DAS (likelihood objective), fifth template ban source, rank 4 | 30 | 1.00 | 0 | BFS 21, DFS 9 |
| connected components | DAS (likelihood objective), fifth template ban source, rank 8 | 30 | 1.00 | 0 | BFS 29, DFS 1 |
| connected components | DAS (committed contrastive objective), forced BFS source, rank 1 | 10 | 1.00 | 0 | union find 10 |
| connected components | DAS (committed contrastive objective), forced BFS source, rank 16 | 10 | 0.00 | 8 | none |
| connected components | DAS (committed contrastive objective), forced BFS source, rank 4 | 10 | 0.20 | 5 | BFS 2 |
| connected components | DAS (committed contrastive objective), forced DFS source, rank 1 | 10 | 1.00 | 0 | union find 10 |
| connected components | DAS (committed contrastive objective), forced DFS source, rank 16 | 10 | 0.20 | 3 | BFS 2 |
| connected components | DAS (committed contrastive objective), forced DFS source, rank 4 | 10 | 0.10 | 0 | BFS 1 |
| connected components | DAS (committed contrastive objective), fifth template ban source, rank 1 | 10 | 1.00 | 0 | union find 10 |
| connected components | DAS (committed contrastive objective), fifth template ban source, rank 16 | 10 | 0.50 | 3 | BFS 5 |
| connected components | DAS (committed contrastive objective), fifth template ban source, rank 2 | 10 | 0.60 | 1 | BFS 5, DFS 1 |
| connected components | DAS (committed contrastive objective), fifth template ban source, rank 4 | 10 | 0.40 | 2 | BFS 3, DFS 1 |
| connected components | DAS (committed contrastive objective), fifth template ban source, rank 8 | 10 | 0.30 | 3 | BFS 3 |
| connected components | difference in means (rank 1), fifth template ban source | 30 | 1.00 | 0 | union find 25, DFS 5 |
| connected components | full state exchange, forced BFS (family 3), block 16 only | 30 | 1.00 | 0 | BFS 28, DFS 2 |
| connected components | full state exchange, forced BFS (family 3), blocks 12-23 | 30 | 1.00 | 0 | DFS 20, BFS 10 |
| connected components | full state exchange, forced BFS source | 30 | 1.00 | 0 | BFS 17, DFS 13 |
| connected components | full state exchange, forced DFS source | 30 | 1.00 | 0 | DFS 30 |
| connected components | full state exchange, fourth template ban source | 10 | 1.00 | 0 | BFS 9, DFS 1 |
| connected components | full state exchange, ban source, block 16 only | 30 | 0.93 | 0 | DFS 24, union find 4 |
| connected components | full state exchange, ban source, blocks 12-23 | 30 | 1.00 | 0 | BFS 17, DFS 13 |

| | | | | | |
|---|---|---|---|---|---|
| connected components | null source (forced default), DAS rank 16 | 30 | 1.00 | 0 | union find 30 |
| connected components | null source (forced default), full state exchange | 30 | 1.00 | 0 | union find 30 |
| connected components | padding twin of forced BFS (family 3) | 30 | 1.00 | 0 | union find 30 |
| connected components | PCA-16 subspace, fifth template ban source | 30 | 0.93 | 0 | DFS 21, union find 7 |
| connected components | DAS with permuted labels, forced BFS source, rank 16 | 30 | 1.00 | 0 | DFS 20, BFS 10 |
| connected components | DAS with permuted labels, fifth template ban source, rank 1 | 30 | 1.00 | 0 | BFS 15, DFS 14, union find 1 |
| connected components | DAS with permuted labels, fifth template ban source, rank 16 | 30 | 1.00 | 0 | BFS 21, DFS 9 |
| connected components | random subspace, forced BFS source, rank 16 | 30 | 1.00 | 0 | union find 30 |
| connected components | random subspace, forced DFS source, rank 16 | 30 | 1.00 | 0 | union find 30 |
| connected components | random subspace, fifth template ban source, rank 1 | 30 | 1.00 | 0 | union find 29, DFS 1 |
| connected components | random subspace 1, fifth template ban source, rank 16 | 30 | 1.00 | 0 | union find 29, DFS 1 |
| connected components | random subspace 2, fifth template ban source, rank 16 | 30 | 1.00 | 0 | union find 29, DFS 1 |
| connected components | random subspace 3, fifth template ban source, rank 16 | 30 | 1.00 | 0 | union find 29, DFS 1 |
| connected components | reverse interchange, DAS rank 1 | 30 | 0.93 | 0 | BFS 18, DFS 10 |
| connected components | reverse interchange, full state | 30 | 0.93 | 0 | BFS 23, DFS 5 |
| connected components | full state exchange at the read position, fifth template ban source | 30 | 1.00 | 0 | union find 29, DFS 1 |
| connected components | DAS at the read position, fifth template ban source, rank 16 | 30 | 1.00 | 0 | DFS 28, BFS 2 |
| kth smallest | padding twin (base for forced sources) | 30 | 1.00 | 0 | full sort 30 |
| kth smallest | neutral twin of template 5 (base) | 30 | 1.00 | 0 | full sort 25, quickselect 5 |
| kth smallest | DAS (likelihood objective), forced counting sort source, rank 1 | 30 | 1.00 | 0 | counting sort 30 |
| kth smallest | DAS (likelihood objective), forced counting sort source, rank 16 | 30 | 1.00 | 0 | counting sort 30 |
| kth smallest | DAS (likelihood objective), forced counting sort source, rank 4 | 30 | 1.00 | 0 | counting sort 30 |
| kth smallest | DAS (likelihood objective), forced quickselect source, rank 1 | 30 | 1.00 | 0 | quickselect 30 |
| kth smallest | DAS (likelihood objective), forced quickselect source, rank 16 | 30 | 0.90 | 0 | quickselect 27 |
| kth smallest | DAS (likelihood objective), forced quickselect source, rank 4 | 30 | 0.93 | 0 | quickselect 28 |

| | | | | |
|---|---|---|---|---|
| kth smallest | DAS (likelihood objective), fifth template ban source, rank 1 | 30 | 0.90 | 0 | quickselect 25, counting sort 1, size k heap 1 |
| kth smallest | DAS (likelihood objective), fifth template ban source, rank 16 | 30 | 0.90 | 0 | quickselect 26, size k heap 1 |
| kth smallest | DAS (likelihood objective), fifth template ban source, rank 4 | 30 | 0.90 | 0 | quickselect 26, size k heap 1 |
| kth smallest | full state exchange, forced quickselect (family 3), block 16 only | 30 | 0.90 | 0 | quickselect 27 |
| kth smallest | full state exchange, forced quickselect (family 3), blocks 12-23 | 30 | 0.97 | 0 | quickselect 29 |
| kth smallest | full state exchange, forced counting sort source | 30 | 1.00 | 0 | counting sort (dictionary form) 29, other 1 (binary search on the value range; iteration-8 correction) |
| kth smallest | full state exchange, forced quickselect source | 30 | 0.97 | 0 | quickselect 29 |
| kth smallest | full state exchange, ban source, block 16 only | 30 | 1.00 | 0 | quickselect 26, full sort 3, size k heap 1 |
| kth smallest | full state exchange, ban source, blocks 12-23 | 30 | 0.93 | 0 | quickselect 26, size k heap 2 |
| kth smallest | padding twin of forced quickselect (family 3) | 30 | 1.00 | 0 | full sort 30 |
| kth smallest | DAS with permuted labels, fifth template ban source, rank 1 | 30 | 0.93 | 0 | quickselect 22, counting sort 5, size k heap 1 |
| kth smallest | DAS with permuted labels, fifth template ban source, rank 16 | 30 | 0.90 | 0 | quickselect 18, counting sort 8, size k heap 1 |
| kth smallest | random subspace, fifth template ban source, rank 1 | 30 | 1.00 | 0 | full sort 24, quickselect 5, size k heap 1 |
| kth smallest | random subspace 1, fifth template ban source, rank 16 | 30 | 1.00 | 0 | full sort 24, quickselect 5, size k heap 1 |

### 7.3.10 Not run and deviations

The DAS experiment's deviation log records what was not run and what departed from the plan; the four that bear on the claims above are quoted verbatim from `results/deviations.json` (iteration-8 correction: new subsection).

- **D-drop.** "Drop order applied: plan resampling (1) and the T1 extension (2: T1 DAS and the literal top-five placement on T1) were not run; M's top-k placement uses T2's leave-one-survivor bans only."
- **D-tfsets.** "Held-out teacher-forced evaluation scores held-out PROMPT pairs with the training target program sets (no separate held-out target sets)."
- **D-targets.** "Distinct 48-token openings are few (the model repeats openings), so the default pool was extended to TRAIN-family padding-twin samples (commit1_<task>_v2 amendment); validation target sets are filled from training programs where < 8 distinct programs exist; T10 dfs has 3 distinct programs, T03 heapify_pop 1 and full_sort 2 (T03 short-program fallback: >= 16 tokens)."
- **D-cap.** "Generation caps are exp30's 384 (bench) / 512 (T2) tokens, not 1,024/1,536: engine-check comparability with exp30 and the time budget; truncations are counted per cell (results/cells.json, n_length)."

Two further deviations are used above: D-randinit (the committed untrained-network comparison against the diverged fit, replaced post hoc by the matched likelihood fit; Section 7.3.5) and D-C-k (forced sources were generated at ranks 1, 4 and 16 only, so the routing claim is scored at rank 16 and rank 2 is untested for forced sources; Table 297b). Plan resampling and the literal top-five placement in the fitted coordinates on the grid task remain open items (Sections 7.10 and 7.11).

## 7.4 Where the algorithm is decided in generation (Qwen3-8B)

### 7.4.1 Design

The fork experiment asked whether one early generated token decides the algorithm, and whether the decision at that token explains which random pushes of iteration 4 change the algorithm. Forks were located by *prefix resampling*: from each base program (one written under the neutral twin, under a push of plan noise or under a ban), the model was restarted at positions along the program's first 128 tokens, and eight continuations were sampled and labelled at each. A *change point* is a position where the share of continuations off the default changes significantly; a fork is a change point on the *shared opening*, the token sequence that most twin programs share before they part. Where no fork exists the committed fallback is the *opening log likelihood ratio* (opening LLR): the teacher-forced log likelihood ratio of the opening of the second choice to the opening of the default over the first L tokens (48 on shortest path, 20 on kth smallest, whose programs are about 20 tokens long). The 40 random directions of iteration 4 were then scored for their fork margin under the push and compared with the correct share off the default (y) that each direction produced in bf16, the share of its programs that pass the hidden tests and implement a named algorithm other than the default; the fork margin under a push is written m.

### 7.4.2 Forks

Connected components has two forks, at generated tokens 10 and 11 of the opening `def solve(n, edges):`, which 94% of twin programs share (Tables 315 and 316). At token 10, a comment token leads to BFS or DFS in all 159 continuations, `parent` leads to union find in all 130, `def` leads off the default in 0.69 of 72, and `from` leads off in all 11. At token 11, after `def`, the routine's name settles it: `dfs` leads to DFS in all 45 continuations and `find` to union find in all 32. The entropy of the label given the token falls from 0.97 bits to 0.00 at token 11. Shortest path has no single fork: pushed programs stay undecided through token 60, and its only change point (token 92) lies off the shared opening. On kth smallest a ban fixes the algorithm before the first generated token: at the first position only 0 to 2 of 8 continuations differ from the base program's class, and no change occurs in 128 tokens.

**Table 315.** Forks found by prefix resampling.

| Task | Status | Forks along the shared opening (token index) | Change points off the shared opening | Base paths | Paths with a significant change point | Base paths by cell | Majority opening length (tokens) | Label entropy at the forks |
|---|---|---|---|---|---|---|---|---|

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| connected components | fork found | 10, 11 | BFS at tokens 12 and 39 | 18 | 4 | neutral twin 6 (0 with a fork); plan noise 4 (3 with a fork); ban 8 (1 with a fork) | 129 | token 10: H(label) 1.55 -> H(label given token) 0.54 bits (n 372); token 11: H(label) 0.97 -> H(label given token) 0.00 bits (n 80) |
| shortest path | no single fork | none | Bellman-Ford/SPFA at token 92 | 12 | 1 | neutral twin 6 (0 with a fork); plan noise 3 (1 with a fork); ban 3 (0 with a fork) | 34 | - |
| kth smallest | no single fork | none | none | 18 | 0 | neutral twin 6 (0 with a fork); ban 12 (0 with a fork) | 22 | - |

**Table 316.** Connected components: the tokens at the two forks and what follows them.

| Generated token index | Token | Continuations | Labels of the continuations | Share off the default (q) | Mapped |
|---|---|---|---|---|---|
| 10 | `#` | 159 | BFS 121, DFS 38 | 1.00 | yes |
| 10 | `parent` | 130 | union find 130 | 0.00 | yes |
| 10 | `def` | 72 | DFS 50, union find 22 | 0.69 | yes |
| 10 | `from` | 11 | BFS 8, DFS 3 | 1.00 | yes |
| 11 | `dfs` | 45 | DFS 45 | 1.00 | yes |
| 11 | `find` | 32 | union find 32 | 0.00 | yes |
| 11 | `count` | 2 | DFS 2 | 1.00 | no |
| 11 | `build` | 1 | DFS 1 | 1.00 | no |

### 7.4.3 The fork claim

On connected components the fork margin under each push predicts which of the 40 directions change the algorithm, at Spearman 0.80 (95% interval 0.62 to 0.91; Table 317). It separates iteration 4's 26 confirmed escapers from the rest with AUC 0.81; its Spearman with the attempted share is 0.82, and the 48-token opening LLR in place of the fork margin reaches 0.87. Both are readouts taken where the outcome is already expressed, teacher-forced at the fork token and at the opening, so their agreement with the sampled outcomes measures the outcome there rather than predicting it from the push; no property of the push direction itself predicted which pushes move the fork (axis projection 0.17 and 0.01, Section 7.4.4; plan-subspace projection 0.03 in iteration 4) (iteration-8 correction). On shortest path the committed fallback fails: the 48-token opening LLR reaches 0.15 (p 0.18). The fork claim therefore fails as a two-task claim (Holm p 0.357) and is recorded as not supported, inconclusive. A binomial model calibrated on all 104 cells of both tasks predicts held-out shares with mean absolute error 0.25, against a committed 0.10. The record of each cell is in Tables 318 and 319. Two checks qualify the connected components result. A placebo that shuffles the map from fork tokens to outcomes still reaches the observed 0.80 in 18% of 200

permutations (median 0.22): with four or five mapped tokens per fork, the margin's power comes from the probability of staying on the default fork tokens, not from the exact outcome rates of the other tokens. And the resampling screen is sparse: in simulation, a jump of 0.4 in the share off the default with only three screen positions before it was detected in 54% of runs, so weak or very early forks can be missed.

On kth smallest a prospective test was committed before generation. The 20-token opening LLR of 24 fresh directions predicts their correct share off the default at Spearman 0.60 (one sided p 0.0014), and 18 of the 24 directions move at least one program (Table 320). The failure on shortest path is a matter of where the decision falls: the Spearman of the opening LLR with the share rises from 0.15 at 48 tokens to 0.42 at 160 (p 0.0032; Table 325), as it should if shortest path decides late.

*[Figure fig_fork_margin: generated in the report phase, not in this draft]*

**Table 317.** The fork claim on connected components (fork margin) and shortest path (48-token opening LLR).

| Task | Directions | Spearman (margin, correct share off the default) | Permutation p | 95% CI | Kendall tau | Tie ceiling | Directions with y > 0 | AUC for confirmed escapers | Confirmed escapers | Spearman with the attempted share | Spearman with the 48-token opening LLR | Twin margin (nats) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| connected components | 40 | 0.80 | 1.0e-04 | [0.62, 0.91] | 0.64 | 0.99 | 31 | 0.81 | 26 | 0.82 | 0.87 | -3.51 |
| shortest path | 40 | 0.15 | 0.1787 | [-0.19, 0.46] | 0.10 | 0.97 | 27 | 0.64 | 6 | 0.37 | 0.15 | -6.02 |

*Note to Table 317.* Pooled Spearman within tasks 0.48 (permutation p 1.0e-04); binomial GLM over 104 cells: intercept -0.74, slope 0.16, held-out MAE 0.25 (10 splits 0.25, 0.27, 0.25, 0.28, 0.24, 0.23, 0.24, 0.24, 0.28, 0.24); MAE without fitted parameters 0.33 (correct) and 0.25 (attempted); Spearman over all cells 0.51.

**Table 318.** Every cell other than the directions scored for the fork claim.

| Task | Cell | Teacher-forced margin m (nats; 48-token LLR on shortest path) | Implied probability off the default | Correct share off the default (y) | Attempted share off the default | n |
|---|---|---|---|---|---|---|
| connected components | ban (fifth template) | 19.37 | 1.000 | 1.00 | 1.00 | 30 |
| connected components | prompt forcing the second choice | 24.38 | 1.000 | 1.00 | 1.00 | 30 |
| connected components | padding twin of the forced prompt | -6.50 | 0.002 | 0.00 | 0.00 | 30 |
| connected components | forced prompt's task specification states exchanged in, blocks 0-35 | 18.75 | 1.000 | 0.87 | 1.00 | 30 |
| connected components | placebo twin | -1.68 | 0.157 | 0.03 | 0.03 | 30 |
| connected components | per-sample plan noise, alpha 1 | -0.78 | 0.315 | 0.14 | 0.14 | 50 |
| connected components | per-sample plan noise, alpha 2 | 0.58 | 0.642 | 0.48 | 0.50 | 50 |

| | | | | | | |
|---|---|---|---|---|---|---|
| connected components | neutral twin | -3.51 | 0.029 | 0.01 | 0.01 | 80 |
| connected components | ban states exchanged in, blocks 0-11 | 1.28 | 0.783 | 0.57 | 0.87 | 30 |
| connected components | ban states exchanged in, blocks 0-35 | 13.16 | 1.000 | 1.00 | 1.00 | 30 |
| connected components | ban states exchanged in, blocks 12-23 | 12.89 | 1.000 | 1.00 | 1.00 | 30 |
| connected components | ban states exchanged in, blocks 24-35 | 1.50 | 0.817 | 0.63 | 0.63 | 30 |
| shortest path | ban (fifth template) | 5.37 | 0.995 | 0.97 | 0.97 | 30 |
| shortest path | prompt forcing the second choice | 13.99 | 1.000 | 0.90 | 1.00 | 30 |
| shortest path | padding twin of the forced prompt | -5.68 | 0.003 | 0.00 | 0.03 | 30 |
| shortest path | forced prompt's task specification states exchanged in, blocks 0-35 | -5.57 | 0.004 | 1.00 | 1.00 | 30 |
| shortest path | placebo twin | -5.97 | 0.003 | 0.03 | 0.03 | 30 |
| shortest path | per-sample plan noise, alpha 1 | -1.59 | 0.170 | 0.10 | 0.20 | 50 |
| shortest path | per-sample plan noise, alpha 2 | 2.06 | 0.887 | 0.18 | 0.44 | 50 |
| shortest path | neutral twin | -6.02 | 0.002 | 0.00 | 0.01 | 105 |
| shortest path | ban states exchanged in, blocks 0-11 | -6.00 | 0.002 | 0.00 | 0.00 | 30 |
| shortest path | ban states exchanged in, blocks 0-35 | -5.82 | 0.003 | 0.70 | 0.80 | 30 |
| shortest path | ban states exchanged in, blocks 12-23 | -5.81 | 0.003 | 0.37 | 0.60 | 30 |
| shortest path | ban states exchanged in, blocks 24-35 | -5.91 | 0.003 | 0.37 | 0.37 | 30 |

**Table 319.** The 40 random directions of iteration 4 at alpha 2: margin and outcome.

| Direction seed | Connected components: margin m (nats) | correct y off the default | attempted n | Shortest path: 48-token LLR (nats) | correct y off the default | attempted n |
|---|---|---|---|---|---|---|
| 9101 | -3.62 | 0.00 | 0.00 10 | 5.98 | 0.10 | 0.40 10 |
| 9102 | -0.24 | 0.67 | 0.67 30 | -5.49 | 0.00 | 0.00 10 |
| 9103 | 0.38 | 0.47 | 0.47 30 | 3.33 | 0.30 | 0.60 10 |
| 9104 | 7.25 | 0.90 | 0.90 30 | -2.80 | 0.63 | 0.87 30 |
| 9105 | 2.77 | 0.43 | 0.53 30 | 3.22 | 0.00 | 0.00 10 |
| 9106 | 0.60 | 0.00 | 0.00 10 | 2.70 | 0.27 | 0.40 30 |
| 9107 | -0.10 | 0.30 | 0.30 30 | 1.49 | 0.33 | 0.40 30 |
| 9108 | 1.36 | 0.57 | 0.57 30 | 5.29 | 0.40 | 0.80 10 |
| 9109 | 1.76 | 0.70 | 0.70 30 | 3.62 | 0.00 | 0.60 10 |
| 9110 | 1.07 | 0.63 | 0.63 30 | 14.37 | 0.63 | 0.90 30 |
| 9111 | 13.62 | 0.87 | 1.00 30 | 16.62 | 0.10 | 0.20 10 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 9112 | 0.68 | 0.20 | 0.70 | 10 11.52 | 0.00 | 0.70 | 10 |
| 9113 | -1.98 | 0.10 | 0.10 | 10 6.13 | 0.13 | 0.27 | 30 |
| 9114 | 6.57 | 0.63 | 0.73 | 30 1.24 | 0.10 | 0.40 | 10 |
| 9115 | -4.13 | 0.00 | 0.00 | 10 5.81 | 0.10 | 0.10 | 10 |
| 9116 | 14.39 | 0.97 | 1.00 | 30 -6.79 | 0.00 | 0.00 | 10 |
| 9117 | 1.22 | 0.83 | 0.83 | 30 8.93 | 0.00 | 0.20 | 10 |
| 9118 | 0.56 | 0.70 | 0.70 | 30 -0.21 | 0.03 | 0.10 | 30 |
| 9119 | 0.39 | 0.23 | 0.30 | 30 0.36 | 0.10 | 0.50 | 10 |
| 9120 | 3.45 | 0.40 | 0.40 | 30 -0.65 | 0.20 | 0.60 | 10 |
| 9121 | -1.27 | 0.00 | 0.00 | 10 -6.25 | 0.00 | 0.00 | 10 |
| 9122 | 8.46 | 0.97 | 1.00 | 30 6.58 | 0.10 | 0.60 | 10 |
| 9123 | 5.90 | 0.13 | 0.13 | 30 1.13 | 0.50 | 0.80 | 10 |
| 9124 | -5.31 | 0.00 | 0.00 | 10 -2.14 | 0.20 | 0.60 | 10 |
| 9125 | 4.13 | 0.83 | 0.83 | 30 3.48 | 0.23 | 0.23 | 30 |
| 9126 | 1.44 | 0.63 | 0.63 | 30 0.90 | 0.10 | 0.20 | 10 |
| 9127 | 5.43 | 0.83 | 1.00 | 30 4.94 | 0.00 | 0.00 | 10 |
| 9128 | 6.85 | 0.87 | 1.00 | 30 3.28 | 0.40 | 0.90 | 10 |
| 9129 | 0.41 | 0.00 | 0.00 | 10 2.43 | 0.00 | 0.50 | 10 |
| 9130 | 0.41 | 0.63 | 0.63 | 30 -0.18 | 0.40 | 0.50 | 30 |
| 9131 | -3.22 | 0.00 | 0.00 | 10 -0.30 | 0.10 | 0.30 | 10 |
| 9132 | -0.84 | 0.37 | 0.37 | 30 -2.09 | 0.00 | 0.00 | 10 |
| 9133 | 8.10 | 0.70 | 1.00 | 10 0.49 | 0.10 | 0.20 | 10 |
| 9134 | 0.75 | 0.10 | 0.10 | 10 13.69 | 0.23 | 0.27 | 30 |
| 9135 | 8.26 | 0.97 | 0.97 | 30 1.12 | 0.00 | 0.20 | 10 |
| 9136 | 10.67 | 0.97 | 1.00 | 30 11.53 | 0.00 | 0.90 | 10 |
| 9137 | -4.33 | 0.00 | 0.00 | 10 -3.81 | 0.10 | 0.10 | 10 |
| 9138 | -6.57 | 0.00 | 0.00 | 10 5.24 | 0.30 | 0.70 | 10 |
| 9139 | 1.47 | 0.27 | 0.27 | 30 -2.18 | 0.10 | 0.40 | 10 |
| 9140 | 0.59 | 0.53 | 0.53 | 30 -10.03 | 0.00 | 0.00 | 10 |

**Table 320.** Kth smallest: the prospective confirmation on 24 fresh directions.

| Fresh direction seed | Axis projection b | Opening margin m (20-token LLR, nats) | Correct y off the default (of 10) | Attempted |
|---|---|---|---|---|
| 97001 | 0.288 | -0.48 | 0.00 | 0.00 |
| 97002 | 0.161 | -5.60 | 0.00 | 0.00 |
| 97003 | 0.228 | 3.88 | 1.00 | 1.00 |
| 97004 | -0.093 | -1.60 | 0.50 | 0.50 |
| 97005 | 0.378 | -6.94 | 0.30 | 0.30 |
| 97006 | 0.051 | -2.55 | 0.00 | 0.00 |
| 97007 | -0.142 | -6.27 | 0.20 | 0.20 |
| 97008 | 0.378 | -3.15 | 0.20 | 0.30 |
| 97009 | 0.544 | 1.25 | 1.00 | 1.00 |
| 97010 | 0.073 | -2.89 | 0.30 | 0.30 |
| 97011 | 0.533 | 4.51 | 1.00 | 1.00 |
| 97012 | -0.087 | 4.38 | 1.00 | 1.00 |
| 97013 | -0.015 | -4.14 | 0.50 | 0.50 |
| 97014 | -0.150 | -1.43 | 0.50 | 0.50 |

| | | | | |
|---|---|---|---|---|
| 97015 | -0.020 | 1.39 | 0.50 | 0.50 |
| 97016 | 0.004 | -2.99 | 0.00 | 0.00 |
| 97017 | -0.028 | -6.12 | 0.00 | 0.00 |
| 97018 | -0.038 | -4.88 | 0.00 | 0.00 |
| 97019 | 0.393 | 0.12 | 0.90 | 1.00 |
| 97020 | 0.847 | 0.80 | 0.20 | 0.20 |
| 97021 | 0.210 | -3.42 | 0.40 | 0.40 |
| 97022 | 0.061 | -2.78 | 0.10 | 0.10 |
| 97023 | 0.120 | -7.75 | 0.50 | 0.50 |
| 97024 | 0.325 | 0.93 | 0.60 | 0.60 |

*Note to Table 320.* Directions 24, with y $>$ 0 on 18; Spearman of m with y 0.60 (one sided p 0.0014), of b with y 0.22 (p 0.150); with the attempted share m 0.59, b 0.23; tie ceiling 0.98.

### 7.4.4 The axis claim

The axis claim proposed that random pushes act through a direction from the default centroid toward the centroid of the second choice in the prompt states. It is disconfirmed. The projection b of each push on the axis predicts the outcome at Spearman 0.17 on connected components and 0.01 on shortest path (Holm p 0.458), and it is ahead of none of its committed competitors, which include the push's energy in the forced prompt subspace of iteration 4 and its raw norm (Table 321). The layer profile shows no block where it would do better (Table 322). The axis is not dormant: neutral prompts vary along it 220 to 400 times more than along random directions. It is simply not the channel the random pushes use.

**Table 321.** The axis claim and its competitors.

| Task | Spearman of the axis projection b with y | Permutation p | 95% CI | Kendall tau | Spearman of b with the fork margin | Forced prompt subspace energy | Raw displacement norm | Energy share of the iteration-4 subspace | b minus forced-subspace energy [95%] | b with the attempted share | AUC for confirmed escapers | Ahead of every competitor |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| connected components | 0.17 | 0.141 | [-0.19, 0.51] | 0.13 | 0.12 | 0.06 | 0.03 | 0.14 | 0.11 [-0.39, 0.62] | 0.20 | 0.65 | no |
| shortest path | 0.01 | 0.458 | [-0.30, 0.34] | 0.01 | -0.17 | 0.12 | -0.05 | -0.03 | -0.11 [-0.28, 0.06] | 0.06 | 0.70 | no |

**Table 322.** The axis projection's Spearman with the outcome, block by block.

| Block | Connected components: Spearman of b with y | Shortest path |
|---|---|---|
| 12 | 0.17 | 0.01 |
| 13 | 0.30 | -0.05 |
| 14 | 0.41 | -0.13 |
| 15 | 0.38 | -0.13 |
| 16 | 0.29 | 0.05 |
| 17 | 0.25 | 0.06 |
| 18 | 0.12 | 0.12 |
| 19 | 0.23 | 0.10 |
| 20 | 0.11 | 0.09 |
| 21 | 0.08 | 0.10 |
| 22 | 0.10 | 0.05 |
| 23 | -0.03 | 0.06 |
| 24 | -0.17 | 0.01 |

| | | |
|---|---|---|
| 25 | -0.18 | -0.03 |
| 26 | -0.18 | -0.08 |
| 27 | -0.15 | -0.18 |
| 28 | -0.17 | -0.17 |
| 29 | -0.11 | -0.14 |
| 30 | -0.13 | -0.15 |
| 31 | -0.12 | -0.17 |
| 32 | -0.07 | -0.18 |
| 33 | -0.07 | -0.18 |
| 34 | -0.08 | -0.14 |
| 35 | -0.04 | -0.14 |

### 7.4.5 Necessity and sufficiency of the axis

Clamping the axis component of each escaping push at blocks from the axis layer to 35 removes nothing: 105 escapes with the clamp against 104 without it and 106 with a clamp along a random direction (Table 323). Pushing along the axis itself is a different matter (Table 324). Doses s are counted in units of the distance between the centroids of the prompts forcing the default and the second choice along the axis. At the committed doses (s up to 1) nothing escapes on any task. At four times that distance, beyond the committed range, connected components switches to DFS in 29 of 30 slots and BFS in the 30th, with all 30 programs passing, while the matched random direction moves none (30:0, p 1.9e-9). The axis edit at that dose has 0.55 of the norm of the median escaping random direction. On shortest path the axis does nothing even at eight times the distance (1 of 30), and on kth smallest the axis at eight times the distance moves 5 of 30, exactly as many as the random direction. On connected components the axis is a sufficient control direction without being the path random pushes take.

**Table 323.** The mediator clamp, per direction.

| Task | Direction seed | Push without clamp: y | Axis component clamped: y | Random direction clamped: y | n |
|---|---|---|---|---|---|
| connected components | 9116 | 0.90 | 1.00 | 0.95 | 20 |
| connected components | 9122 | 0.95 | 0.95 | 0.95 | 20 |
| connected components | 9135 | 1.00 | 0.85 | 1.00 | 20 |
| connected components | 9136 | 1.00 | 0.90 | 1.00 | 20 |
| shortest path | 9130 | 0.35 | 0.40 | 0.45 | 20 |
| shortest path | 9107 | 0.20 | 0.25 | 0.20 | 20 |
| shortest path | 9106 | 0.40 | 0.55 | 0.35 | 20 |
| shortest path | 9125 | 0.40 | 0.35 | 0.40 | 20 |

*Note to Table 323.* Pooled escapes: no clamp 104, axis clamp 105, random clamp 106; removed share -0.010 (axis) and -0.019 (random); McNemar axis clamp 9:10 (p 1.00), random clamp 2:4 (p 0.69); verdict NOT SUPPORTED.

**Table 324.** A targeted push along the axis against a matched random direction, by dose.

| Task | Dose s (forced prompt distance) | Axis: correct share of the second choice | Axis: correct off the default | Axis: pass | Axis: labels | Matched random: correct second choice | Random: pass | Random: labels | Axis only : random only (exact p) | Edit norm over the median escaping direction |
|---|---|---|---|---|---|---|---|---|---|---|
| connected components | 0.1 | 0.00 | 0.00 | 1.00 | union find 30 | 0.00 | 1.00 | union find 30 | 0:0 (p 1.000) | 0.01 |
| connected components | 0.25 | 0.00 | 0.00 | 1.00 | union find 30 | 0.00 | 1.00 | union find 30 | 0:0 (p 1.000) | 0.03 |
| connected components | 0.5 | 0.00 | 0.00 | 1.00 | union find 30 | 0.00 | 1.00 | union find 30 | 0:0 (p 1.000) | 0.07 |
| connected components | 1 | 0.03 | 0.03 | 1.00 | union find 29, DFS 1 | 0.00 | 1.00 | union find 30 | 1:0 (p 1.000) | 0.14 |
| connected components | 2 | 0.17 | 0.17 | 1.00 | union find 25, DFS 5 | 0.00 | 1.00 | union find 30 | 5:0 (p 0.062) | 0.28 |
| connected components | 4 | 0.97 | 1.00 | 1.00 | DFS 29, BFS 1 | 0.00 | 1.00 | union find 30 | 30:0 (p 1.9e-09) | 0.55 |
| connected components | 8 | 0.90 | 1.00 | 1.00 | DFS 27, BFS 3 | 0.00 | 1.00 | union find 30 | 30:0 (p 1.9e-09) | 1.11 |
| shortest path | 0.1 | 0.00 | 0.00 | 1.00 | Dijkstra 30 | 0.00 | 1.00 | Dijkstra 30 | 0:0 (p 1.000) | 0.02 |
| shortest path | 0.25 | 0.00 | 0.00 | 1.00 | Dijkstra 30 | 0.00 | 1.00 | Dijkstra 30 | 0:0 (p 1.000) | 0.04 |
| shortest path | 0.5 | 0.00 | 0.00 | 1.00 | Dijkstra 30 | 0.00 | 0.97 | Dijkstra 29, Bellman-Ford/SPFA 1 | 0:0 (p 1.000) | 0.09 |
| shortest path | 1 | 0.00 | 0.00 | 0.97 | Dijkstra 29, Bellman-Ford/SPFA 1 | 0.00 | 0.97 | Dijkstra 29, Bellman-Ford/SPFA 1 | 0:0 (p 1.000) | 0.18 |
| shortest path | 2 | 0.00 | 0.00 | 1.00 | Dijkstra 30 | 0.00 | 0.97 | Dijkstra 29, Bellman-Ford/SPFA 1 | 0:0 (p 1.000) | 0.36 |
| shortest path | 4 | 0.00 | 0.00 | 0.97 | Dijkstra 29, Bellman-Ford/SPFA 1 | 0.00 | 1.00 | Dijkstra 30 | 0:0 (p 1.000) | 0.71 |
| shortest path | 8 | 0.03 | 0.03 | 0.97 | Dijkstra 28, Bellman-Ford/SPFA 2 | 0.00 | 0.97 | Dijkstra 29, Bellman-Ford/SPFA 1 | 1:0 (p 1.000) | 1.42 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| kth smallest 0.1 | 0.03 | 0.03 | 1.00 | full sort 29, quickselect 1 | 0.03 | 1.00 | full sort 29, quickselect 1 | 0:0 (p 1.000) | 0.01 |
| kth smallest 0.25 | 0.03 | 0.03 | 1.00 | full sort 29, quickselect 1 | 0.00 | 1.00 | full sort 30 | 1:0 (p 1.000) | 0.03 |
| kth smallest 0.5 | 0.03 | 0.03 | 1.00 | full sort 29, quickselect 1 | 0.00 | 1.00 | full sort 30 | 1:0 (p 1.000) | 0.06 |
| kth smallest 1 | 0.03 | 0.03 | 1.00 | full sort 29, quickselect 1 | 0.03 | 1.00 | full sort 29, quickselect 1 | 0:0 (p 1.000) | 0.13 |
| kth smallest 2 | 0.07 | 0.07 | 1.00 | full sort 28, quickselect 2 | 0.00 | 1.00 | full sort 30 | 2:0 (p 0.500) | 0.25 |
| kth smallest 4 | 0.10 | 0.10 | 1.00 | full sort 27, quickselect 3 | 0.03 | 1.00 | full sort 29, quickselect 1 | 2:0 (p 0.500) | 0.50 |
| kth smallest 8 | 0.17 | 0.17 | 1.00 | full sort 25, quickselect 5 | 0.17 | 1.00 | full sort 25, quickselect 5 | 2:2 (p 1.000) | 1.01 |

*Note to Table 324.* Reference norms. Connected components: axis at block 12, median RMS of escaping random directions 21.26, axis distance RMS 2.94; shortest path: axis at block 24, median RMS of escaping random directions 213.41, axis distance RMS 37.94; kth smallest: axis at block 22, median RMS of escaping random directions 77.14, axis distance RMS 9.74.

### 7.4.6 Forcing the fork token

Forcing tokens under the unchanged neutral twin separates the routine's name from the fork position (Table 326). On connected components, forcing `def dfs` at tokens 10 and 11 gives DFS in 30 of 30 slots, and `def find` gives union find in 30. Forcing a comment token at position 10, which led off the default in every resampled continuation in which it appeared, gives union find in all 30 twin slots; forcing `from` moves 10 of 30. The token that names the routine decides; a token that only correlates with pushed states in resampling does not decide on its own. On kth smallest a forced comment as the first body token moves 5 of 30 (p 0.0625), and on shortest path no forced token moves anything.

**Table 325.** Shortest path: Spearman of the opening LLR with the outcome, by opening length.

| Opening length L (tokens) | Spearman of the opening LLR over L tokens with y over 40 directions | Permutation p | Spearman over all cells |
|---|---|---|---|
| 16 | 0.27 | 0.0462 | 0.30 |
| 32 | 0.16 | 0.1662 | 0.22 |
| 48 | 0.15 | 0.1758 | 0.22 |

| | | | |
|---|---|---|---|
| 96 | 0.25 | 0.0650 | 0.36 |
| 128 | 0.28 | 0.0434 | 0.37 |
| 160 | 0.42 | 0.0032 | 0.53 |

**Table 326.** Forced opening tokens under the unchanged twin.

| Task | Forced token under the unchanged neutral twin | n | Labels | Pass | Correct share off the default | Flips against the twin, to off : to default (exact p) |
|---|---|---|---|---|---|---|
| kth smallest | first body token forced to `#` (comment) | 30 | full sort 25, quickselect 4, heapify and pop 1 | 1.00 | 0.17 | 5:0 (p 0.062) |
| kth smallest | first body token forced to the twin's own token | 30 | full sort 29, quickselect 1 | 1.00 | 0.03 | 1:0 (p 1.000) |
| connected components | token 10 forced to `#` (comment) | 30 | union find 30 | 1.00 | 0.00 | 0:0 (p 1.000) |
| connected components | token 10 forced to `def` | 30 | union find 30 | 1.00 | 0.00 | 0:0 (p 1.000) |
| connected components | token 10 forced to `from` | 30 | union find 20, DFS 10 | 1.00 | 0.33 | 10:0 (p 0.002) |
| connected components | token 10 forced to `parent` | 30 | union find 30 | 1.00 | 0.00 | 0:0 (p 1.000) |
| connected components | tokens 10-11 forced to `def dfs` | 30 | DFS 30 | 1.00 | 1.00 | 30:0 (p 1.9e-09) |
| connected components | tokens 10-11 forced to `def find` | 30 | union find 30 | 1.00 | 0.00 | 0:0 (p 1.000) |
| shortest path | first body token forced to `#` (comment) | 30 | Dijkstra 30 | 1.00 | 0.00 | 0:0 (p 1.000) |
| shortest path | first body token forced to the twin's own token | 30 | Dijkstra 30 | 1.00 | 0.00 | 0:0 (p 1.000) |

### 7.4.7 The top five prohibition inside the manifold

The review asked for the top five push at a dose that stays on the forced prompt manifold of the grid task. The fork experiment pushed the twin's states toward the centroid of the survivor region, toward a random survivor, away from the banned centroid and toward IDA*, at a quarter, a half and one times the distance between centroids (Table 327). All 300 programs implement BFS, at pass rates between 0.73 and 0.93; none implements an algorithm outside the banned five. The prompt top five ban of iteration 6 had at least moved the model between BFS and DFS (25 and 23 of 50). On the manifold at block 18, three survivors (bidirectional BFS, Dijkstra with unit weights and IDA*) sit at its edge, and the nearest centroids to the banned region are greedy best first, BFS and A* (Table 328; Isomap stress 0.24, metric MDS stress 0.38). Moving the states toward where those survivors' forced prompts sit does not make the model write them. The engine reproduced the engine control's outcomes per direction (Spearman 0.95 on connected components and 0.93 on shortest path; Table 329), and the two programs sent to novelty audit were labeller misses. One GPU unit test missed its tolerance: teacher-forced log probabilities and those from a cached prefill differ by up to 0.25 on the top 50 tokens, with the same argmax, so every margin was computed on the teacher-forced path.

**Table 327.** The grid task: pushes inside the forced prompt manifold toward the survivors of the top five ban (30 slots each).

| Arm (grid task) | n | Pass | Labels (catalogue of ten algorithms) | Correct programs outside the top five | Attempted programs outside the top five |
|---|---|---|---|---|---|
| neutral twin | 30 | 0.87 | BFS 29, DFS 1 | 0 | 0 |
| away from the banned centroid, s 0.25 | 30 | 0.90 | BFS 30 | 0 | 0 |
| away from the banned centroid, s 0.5 | 30 | 0.93 | BFS 30 | 0 | 0 |
| away from the banned centroid, s 1 | 30 | 0.90 | BFS 30 | 0 | 0 |
| toward IDA*, s 0.5 | 30 | 0.87 | BFS 30 | 0 | 0 |
| toward a random survivor, s 0.25 | 30 | 0.90 | BFS 30 | 0 | 0 |
| toward a random survivor, s 0.5 | 30 | 0.83 | BFS 30 | 0 | 0 |
| toward a random survivor, s 1 | 30 | 0.73 | BFS 30 | 0 | 0 |
| toward the centroid of the survivor region, s 0.25 | 30 | 0.83 | BFS 30 | 0 | 0 |
| toward the centroid of the survivor region, s 0.5 | 30 | 0.90 | BFS 30 | 0 | 0 |
| toward the centroid of the survivor region, s 1 | 30 | 0.90 | BFS 30 | 0 | 0 |
| prompt top five ban (iteration 6, N = 50) | 50 | 0.62 | BFS 25, DFS 23, unknown 1, unit weight Dijkstra 1 | 0 | 1 |

**Table 328.** The grid task: geometry of the forced prompt manifold at blocks 18 and 22.

| Block | Points | Isomap (k 15) stress-1 | MDS stress-1 | PCA variance (first and second components) | Geodesic distance from the banned region, per survivor | Edge survivors | Distance from the centroid of the banned region, every centroid |
|---|---|---|---|---|---|---|---|
| 18 | 100 | 0.24 | 0.38 | 0.30, 0.19 | bidirectional BFS 4.61, unit weight Dijkstra 4.51, IDA* 4.13, jump point search 6.44, bidirectional A* 4.74 | bidirectional BFS, unit weight Dijkstra, IDA* | greedy best first 1.84, BFS 1.93, A* 2.22, DFS 2.43, IDA* 2.89, unit weight Dijkstra 3.04, IDDFS 3.18, bidirectional BFS 3.43, bidirectional A* 3.63, jump point search 4.59 |
| 22 | 100 | 0.26 | 0.40 | 0.31, 0.16 | bidirectional BFS 6.54, unit weight Dijkstra 6.54, IDA* 6.58, jump point search 9.78, bidirectional A* 7.63 | bidirectional BFS, unit weight Dijkstra, IDA* | greedy best first 2.65, BFS 2.82, A* 3.07, DFS 3.27, IDDFS 4.26, IDA* 4.55, unit weight Dijkstra 4.67, bidirectional BFS 4.82, bidirectional A* 5.33, jump point search 6.60 |

**Table 329.** The fork and axis statistics remeasured on this engine's own outcomes.

| Task | Directions | Spearman of the fork margin on the outcomes of this engine | p | Spearman of the axis projection | p | Spearman of y with the bf16 y of the engine control | Mean y here | Mean y, engine control |
|---|---|---|---|---|---|---|---|---|
| connected components | 40 | 0.77 | 1.0e-04 | 0.25 | 0.062 | 0.95 | 0.492 | 0.460 |
| shortest path | 40 | 0.12 | 0.2181 | -0.04 | 0.603 | 0.93 | 0.185 | 0.158 |

## 7.5 Temperature acts through the fork (Qwen3-8B)

### 7.5.1 Design

The temperature experiment tested the temperature claim where it can be tested: on the three units where the instruct twin already writes two algorithms. Minimum spanning tree weight mixes Kruskal and Prim, trapped rain water mixes prefix and suffix maxima with two pointers, and valid anagram mixes sorting with counting. Iteration 5 had recorded minority shares of 0.34, 0.36 and 0.24; this experiment's twins give 0.19, 0.31 and 0.13 on prompts identical byte for byte (Table 330). The claim has three committed tests. Test 1: the entropy of the algorithm given the class of the token at the fork is at most 0.3 bits on at least two units. Test 2: predictions made from the teacher-forced fork logits divided by T, committed before any temperature sample, match the observed minority share (drop from T 1.0 to 0.3 of at least 0.15, mean absolute error at most 0.10, and the change from 1.0 to 1.3 within 0.10 of prediction) on at least two units. Test 3: on the four collapsed tasks the fork margin is at least 4 nats, large enough that no temperature in range reaches the alternative. Every temperature cell shares its random numbers across temperatures (same seed), at temperatures 0.3, 0.7, 1.0 and 1.3, under top-p 1 and top-k 10.

**Table 330.** The twins of the three mixed units.

| Unit | Labels of 100 twin programs | Minority algorithm | Minority share | Pass rate | Iteration-5 record of the minority share |
|---|---|---|---|---|---|
| minimum spanning tree weight | Kruskal 80, Prim with heap 19, disputed 1 | Prim with heap | 0.19 | 0.97 | 0.34 |
| trapped rain water | prefix and suffix maxima 68, two pointers 31, disputed 1 | two pointers | 0.31 | 1.00 | 0.36 |
| valid anagram | sort and compare 77, fixed count array 13, hash count 10 | fixed count array | 0.13 | 1.00 | 0.24 |

### 7.5.2 One sampled token decides, at a predictable place

Prefix resampling finds a fork of a single token on every unit, at the first token after the shared opening (Table 331). On minimum spanning tree weight it is the word after `#` once the adjacency list is built: `Kr` has probability 0.72 and leads to Kruskal in every continuation, `Prim` has 0.27 and leads to Prim in 97%. On trapped rain water, after `if not height: return 0, n` leads to prefix and suffix arrays and `left` to two pointers. On valid anagram, after the length check, `return` leads to sorting and `count` to counting. Test 1 passes on all three units: the entropy of the algorithm given the fork class is 0.050, 0.153 and 0.189 bits (Table 332).

**Table 331.** Fork location and the calibration of the sampler at the fork.

| Unit | Mode | Shared opening length (tokens) | Coverage of the opening clusters | Confirmed forks / base paths | Forks on the opening confirmed one token after the opening | Sampler calibration (minority share at k = 0 against the twin, Fisher p) |
|---|---|---|---|---|---|---|
| minimum spanning tree weight | one token | 61 | c0 61 tok 0.51, c1 61 tok 0.20 | 5/12 | 3/3 | 0.16 vs 0.19 (p 0.58, pass) |
| trapped rain water | one token | 18 | c0 18 tok 0.48, c1 19 tok 0.31 | 2/12 | 1/1 | 0.22 vs 0.31 (p 0.20, pass) |
| valid anagram | one token | 23 | c0 23 tok 0.78 | 1/8 | 1/1 | 0.22 vs 0.13 (p 0.19, pass) |

**Table 332.** Test 1: the algorithm given the fork token's class.

| Unit | H(algorithm given fork class) (bits) | Bootstrap 95% CI | Class mass at the fork | Mass on undecided tokens | Pass (at most 0.3) |
|---|---|---|---|---|---|
| minimum spanning tree weight | 0.050 | [0.014, 0.096] | Kruskal 0.741, Prim with heap 0.259 | 0.003 | yes |
| trapped rain water | 0.153 | [0.091, 0.203] | prefix and suffix maxima 0.571, two pointers 0.429, other 0.0 | 0.116 | yes |
| valid anagram | 0.189 | [0.123, 0.239] | sort and compare 0.738, fixed count array 0.261, hash count 0.001, other 0.0 | 0.000 | yes |

### 7.5.3 The fork logits predict the temperature response

Test 2 passes on minimum spanning tree weight and fails on the other two units, so the temperature claim is recorded as PARTIAL (Tables 333 and 334). On minimum spanning tree weight, lowering T from 1.0 to 0.3 removes Prim from 17 slots and adds it to none (one sided p below 1e-4); the observed drop of 0.34 is twice the predicted 0.17, and the mean absolute error across temperatures is 0.083. On trapped rain water the mean absolute error is 0.105 and the drop 0.08 (7:3), and on valid anagram the minority share does not drop (2:5), with an error of 0.043. Pooled over the first two units the drop is 24:3 (p 2.5e-5), which survives a Bonferroni correction over the seven committed tests. Three post hoc readings qualify the misses. The replay per slot, which walks the opening with each slot's own random numbers and reads the class of the fork token it picks, names the algorithm of an individual program in 0.90 to 0.98 of slots on the opening in every cell (Table 333, last column). On trapped rain water the shares implied by the replay of the actual 50 streams (0.08, 0.18, 0.18 and 0.16) reproduce the observed sweep (0.10, 0.18, 0.18 and 0.20), so the miss there is the gap between a cell of 50 streams and its expectation, not a failure of the mechanism. On minimum spanning tree weight the sweep's 50 streams lean toward Prim at T 1.0 (0.40, against 0.20 for the twin's own slots, Fisher p 0.011), which is why the observed drop is twice the predicted one. On valid anagram the prediction itself had said the drop would be small: one template prefers `count` even greedily, and about 19% of samples leave the opening at every temperature. Above T 1.0 nothing is added on any unit: the observed change from 1.0 to 1.3 is between -0.02 and +0.06, and the predicted change is at most 0.013, because what the minority gains at the fork is offset by samples that leave the shared opening as T rises (on minimum spanning tree weight the predicted share on the opening falls from 0.72 at T 1.0 to 0.55

at 1.3). Across all temperatures the ideas stay the two or three known algorithms of each unit; only phrasings multiply (Table 340). Top-k 10 gave the same program as top-p 1 in all 600 paired slots, because every token at the fork lies in the top 10.

*[Figure fig_temperature_fork: generated in the report phase, not in this draft]*

**Table 333.** Test 2: predicted against observed minority share by temperature and sampler.

| Unit | Cell | Predicted minority share | Observed [Wilson 95%] | Correct minority observed / predicted | Share on the opening observed / predicted | JS (bits) | Replay per slot accuracy |
|---|---|---|---|---|---|---|---|
| minimum spanning tree weight | T 0.3, top-p 1 | 0.068 | 0.060 [0.02, 0.16] | 0.060 / 0.068 | 1.00 / 0.99 | 0.000 | 0.98 (n 50) |
| minimum spanning tree weight | T 0.3, top-k 10 | 0.068 | 0.060 [0.02, 0.16] | 0.060 / 0.068 | 1.00 / 0.99 | 0.000 | 0.98 (n 50) |
| minimum spanning tree weight | T 0.7, top-p 1 | 0.193 | 0.220 [0.13, 0.35] | 0.220 / 0.193 | 0.92 / 0.87 | 0.003 | 0.96 (n 47) |
| minimum spanning tree weight | T 0.7, top-k 10 | 0.193 | 0.220 [0.13, 0.35] | 0.220 / 0.193 | 0.92 / 0.87 | 0.003 | 0.96 (n 47) |
| minimum spanning tree weight | T 1.0, top-p 1 | 0.237 | 0.400 [0.28, 0.54] | 0.400 / 0.237 | 0.76 / 0.72 | 0.026 | 0.95 (n 37) |
| minimum spanning tree weight | T 1.0, top-k 10 | 0.237 | 0.400 [0.28, 0.54] | 0.400 / 0.237 | 0.76 / 0.72 | 0.026 | 0.95 (n 37) |
| minimum spanning tree weight | T 1.3, top-p 1 | 0.247 | 0.380 [0.26, 0.52] | 0.360 / 0.247 | 0.60 / 0.55 | 0.022 | 0.94 (n 32) |
| minimum spanning tree weight | T 1.3, top-k 10 | 0.247 | 0.380 [0.26, 0.52] | 0.360 / 0.247 | 0.60 / 0.55 | 0.022 | 0.91 (n 32) |
| trapped rain water | T 0.3, top-p 1 | 0.166 | 0.100 [0.04, 0.21] | 0.100 / 0.166 | 0.88 / 0.86 | 0.007 | 0.98 (n 45) |
| trapped rain water | T 0.3, top-k 10 | 0.166 | 0.100 [0.04, 0.21] | 0.100 / 0.166 | 0.88 / 0.86 | 0.007 | 0.98 (n 45) |
| trapped rain water | T 0.7, top-p 1 | 0.285 | 0.180 [0.10, 0.31] | 0.180 / 0.285 | 0.84 / 0.82 | 0.011 | 0.98 (n 41) |
| trapped rain water | T 0.7, top-k 10 | 0.285 | 0.180 [0.10, 0.31] | 0.180 / 0.285 | 0.84 / 0.82 | 0.011 | 0.98 (n 41) |
| trapped rain water | T 1.0, top-p 1 | 0.311 | 0.180 [0.10, 0.31] | 0.180 / 0.311 | 0.74 / 0.76 | 0.026 | 0.92 (n 38) |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| trapped rain water | T 1.0, top-k 10 | 0.311 | 0.180 [0.10, 0.31] | 0.180 / 0.311 | 0.74 / 0.76 | 0.026 | 0.92 (n 38) |
| trapped rain water | T 1.3, top-p 1 | 0.319 | 0.200 [0.11, 0.33] | 0.200 / 0.319 | 0.70 / 0.70 | 0.022 | 0.97 (n 35) |
| trapped rain water | T 1.3, top-k 10 | 0.319 | 0.200 [0.11, 0.33] | 0.200 / 0.319 | 0.70 / 0.70 | 0.022 | 0.97 (n 35) |
| valid anagram | T 0.3, top-p 1 | 0.161 | 0.180 [0.10, 0.31] | 0.180 / 0.161 | 0.80 / 0.81 | 0.000 | 0.93 (n 40) |
| valid anagram | T 0.3, top-k 10 | 0.161 | 0.180 [0.10, 0.31] | 0.180 / 0.161 | 0.80 / 0.81 | 0.000 | 0.93 (n 40) |
| valid anagram | T 0.7, top-p 1 | 0.182 | 0.140 [0.07, 0.26] | 0.140 / 0.182 | 0.76 / 0.82 | 0.003 | 0.95 (n 39) |
| valid anagram | T 0.7, top-k 10 | 0.182 | 0.140 [0.07, 0.26] | 0.140 / 0.182 | 0.76 / 0.82 | 0.003 | 0.95 (n 39) |
| valid anagram | T 1.0, top-p 1 | 0.200 | 0.120 [0.06, 0.24] | 0.120 / 0.200 | 0.78 / 0.81 | 0.010 | 0.90 (n 39) |
| valid anagram | T 1.0, top-k 10 | 0.200 | 0.120 [0.06, 0.24] | 0.120 / 0.200 | 0.78 / 0.81 | 0.010 | 0.90 (n 39) |
| valid anagram | T 1.3, top-p 1 | 0.212 | 0.180 [0.10, 0.31] | 0.180 / 0.212 | 0.80 / 0.79 | 0.002 | 0.95 (n 39) |
| valid anagram | T 1.3, top-k 10 | 0.212 | 0.180 [0.10, 0.31] | 0.180 / 0.212 | 0.80 / 0.79 | 0.002 | 0.95 (n 39) |

**Table 334.** Test 2 by unit.

| Unit | MAE | Observed drop T 1.0 to 0.3 [slot bootstrap] | Predicted drop | McNemar b:c, one sided p | Observed change T 1.0 to 1.3 | Predicted change | Slots changing the idea across T (top-p 1) | Pass |
|---|---|---|---|---|---|---|---|---|
| minimum spanning tree weight | 0.083 | 0.340 [0.22, 0.48] | 0.169 | 17:0, 0.0000 | -0.020 | 0.010 | 20/50 | yes |
| trapped rain water | 0.105 | 0.080 [-0.04, 0.22] | 0.144 | 7:3, 0.1719 | 0.020 | 0.009 | 12/50 | no |
| valid anagram | 0.043 | -0.060 [-0.16, 0.04] | 0.039 | 2:5, 0.9375 | 0.060 | 0.013 | 11/50 | no |

### 7.5.4 The collapsed tasks

Test 3 passes: on the grid, shortest path, connected components and kth smallest the fork margin of the twin is 7.46, 4.85, 17.04 and 6.10 nats (Table 335). At T 1.3 the predicted rate off the default is

at most 0.02 on every task, and 100 samples at that temperature produced no program off the default on any of them; a further check of 20 samples found one program off the default on the grid and one on shortest path; the shortest path one fits its predicted rate, and the grid one is unlikely under its margin (binomial p 0.04). The record of iterations 1 to 4, 797 of 800 default programs and no flips under the same seed, implies margins of about 6 nats. The temperature nulls of iterations 1 to 4 are what the fork logits predict: on the collapsed tasks the alternative lies 5 to 17 nats behind at the fork, and dividing by 1.3 does not bring it within reach.

**Table 335.** Test 3: fork margins of the collapsed tasks and the bound they imply.

| Task | Fork margin (nats) | At least 4 | Predicted rate off the default at T 0.3 / 1.0 / 1.3 | Predicted change 0.3 to 1.3 | Bound check at T 1.3 (off of 100 against the logit prediction) | T 1.3 check (off of 20) | 48-token opening LLR |
|---|---|---|---|---|---|---|---|
| grid | 7.46 | yes | 0.0000 / 0.0006 / 0.0021 | 0.0021 | 0/100 vs 0.0021 | 1/20 | 8.79 |
| shortest path | 4.85 | yes | 0.0000 / 0.0078 / 0.0204 | 0.0204 | 0/100 vs 0.0204 | 1/20 | 5.15 |
| connected components | 17.04 | yes | 0.0000 / 0.0000 / 0.0000 | 0.0000 | 0/100 vs 0.0000 | 0/20 | 24.58 |
| kth smallest | 6.10 | yes | 0.0000 / 0.0022 / 0.0085 | 0.0085 | 0/100 vs 0.0085 | 0/20 | 7.55 |

### 7.5.5 What does move the fork

Exchanging the task specification states of a prompt that names the minority algorithm into the twin gives that algorithm in 30 of 30 slots on all three units, against 0 to 7 under the padding twin, the forced prompt with its instruction replaced by padding tokens so that positions align (exact one sided p of 1.2e-7 or less; Tables 336 and 337). The argmax token at the fork flips with it ( `Kr` to `Prim`), and the fork margin moves by 4.9 to 22 nats toward the minority, against 2 nats or less for a matched random exchange. Eight random pushes of the task specification states at alpha 2 move the minority share in the direction their change in fork margin predicts (Spearman -0.69, -0.44 and -0.94 between margin change and share change; Tables 338 and 339). The committed permutation test used the upper tail, the side opposite to the predicted negative relation; the lower tail is reported beside it as a correction made after the commitment.

**Table 336.** Exchanging the forced minority prompt's task specification states into the twin.

| Unit | Arm | n | Attempted minority share | Correct minority share | Pass | Labels |
|---|---|---|---|---|---|---|
| minimum spanning tree weight | padding twin | 30 | 0.23 | 0.23 | 1.00 | Kruskal 23, Prim with heap 7 |
| minimum spanning tree weight | forced minority prompt | 30 | 1.00 | 0.97 | 0.97 | Prim with heap 30 |
| minimum spanning tree weight | forced prompt's task specification states exchanged in | 30 | 1.00 | 1.00 | 1.00 | Prim with heap 30 |

| | | | | | | |
|---|---|---|---|---|---|---|
| minimum spanning tree weight | random exchange of matched norm | 30 | 0.40 | 0.40 | 1.00 | Kruskal 18, Prim with heap 12 |
| trapped rain water | padding twin | 30 | 0.13 | 0.13 | 1.00 | prefix and suffix maxima 25, two pointers 4, peak split 1 |
| trapped rain water | forced minority prompt | 30 | 1.00 | 1.00 | 1.00 | two pointers 30 |
| trapped rain water | forced prompt's task specification states exchanged in | 30 | 1.00 | 1.00 | 1.00 | two pointers 30 |
| trapped rain water | random exchange of matched norm | 30 | 0.13 | 0.13 | 1.00 | prefix and suffix maxima 25, two pointers 4, peak split 1 |
| valid anagram | padding twin | 30 | 0.00 | 0.00 | 1.00 | sort and compare 30 |
| valid anagram | forced minority prompt | 30 | 1.00 | 1.00 | 1.00 | fixed count array 30 |
| valid anagram | forced prompt's task specification states exchanged in | 30 | 1.00 | 1.00 | 1.00 | fixed count array 30 |
| valid anagram | random exchange of matched norm | 30 | 0.00 | 0.00 | 1.00 | sort and compare 30 |

**Table 337.** The exchange against the padding twin and against a matched random exchange.

| Unit | Exchange against padding twin, b:c (one sided p) | Exchange against random exchange | Fork margin change, exchange (nats) | Fork margin change, random exchange | Predicted minority share under the exchange | Pass ratio against the padding twin |
|---|---|---|---|---|---|---|
| minimum spanning tree weight | 23:0 (p 1.2e-07) | 18:0 (p 3.8e-06) | -4.87 | -1.25 | 0.17 | 1.00 |
| trapped rain water | 26:0 (p 1.5e-08) | 26:0 (p 1.5e-08) | -5.50 | -2.00 | 0.40 | 1.00 |
| valid anagram | 30:0 (p 9.3e-10) | 30:0 (p 9.3e-10) | -22.12 | 0.26 | 0.78 | 1.00 |

**Table 338.** Random pushes of the task specification states: margin change and minority share per direction.

| Unit | Direction seed | Fork margin change (nats) | Observed minority share (of 10) | Predicted share | Observed change against the twin | Pass | Labels |
|---|---|---|---|---|---|---|---|
| minimum spanning tree weight | 9101 | 0.45 | 0.00 | 0.17 | -0.20 | 0.90 | Kruskal 10 |
| minimum spanning tree weight | 9102 | -0.29 | 0.30 | 0.17 | 0.10 | 1.00 | Kruskal 7, Prim with heap 3 |
| minimum spanning tree weight | 9103 | 1.89 | 0.00 | 0.17 | -0.20 | 0.80 | Kruskal 10 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| minimum spanning tree weight | 9104 | 3.04 | 0.00 | 0.12 | -0.20 | 1.00 | Kruskal 10 |
| minimum spanning tree weight | 9105 | 2.20 | 0.00 | 0.16 | -0.20 | 0.90 | Kruskal 10 |
| minimum spanning tree weight | 9106 | -0.19 | 0.30 | 0.20 | 0.10 | 1.00 | Kruskal 7, Prim with heap 3 |
| minimum spanning tree weight | 9107 | -1.21 | 0.40 | 0.30 | 0.20 | 1.00 | Kruskal 6, Prim with heap 4 |
| minimum spanning tree weight | 9108 | -0.59 | 0.00 | 0.20 | -0.20 | 0.70 | Kruskal 10 |
| trapped rain water | 9101 | -0.20 | 0.00 | 0.21 | -0.40 | 1.00 | prefix and suffix maxima 6, peak split 4 |
| trapped rain water | 9102 | 2.80 | 0.00 | 0.21 | -0.40 | 1.00 | prefix and suffix maxima 10 |
| trapped rain water | 9103 | -2.45 | 0.20 | 0.28 | -0.20 | 1.00 | prefix and suffix maxima 8, two pointers 2 |
| trapped rain water | 9104 | 0.80 | 0.30 | 0.16 | -0.10 | 1.00 | prefix and suffix maxima 7, two pointers 3 |
| trapped rain water | 9105 | 7.05 | 0.00 | 0.22 | -0.40 | 0.90 | prefix and suffix maxima 10 |
| trapped rain water | 9106 | 2.00 | 0.50 | 0.23 | 0.10 | 1.00 | prefix and suffix maxima 5, two pointers 5 |
| trapped rain water | 9107 | 1.10 | 0.20 | 0.13 | -0.20 | 1.00 | prefix and suffix maxima 8, two pointers 2 |
| trapped rain water | 9108 | -6.25 | 0.90 | 0.59 | 0.50 | 1.00 | two pointers 9, prefix and suffix maxima 1 |
| valid anagram | 9101 | -7.65 | 1.00 | 0.19 | 0.90 | 1.00 | fixed count array 10 |
| valid anagram | 9102 | -6.70 | 0.20 | 0.19 | 0.10 | 1.00 | hash count 8, fixed count array 2 |
| valid anagram | 9103 | 2.11 | 0.00 | 0.05 | -0.10 | 1.00 | sort and compare 9, hash count 1 |
| valid anagram | 9104 | -7.75 | 0.80 | 0.60 | 0.70 | 1.00 | fixed count array 8, hash count 2 |
| valid anagram | 9105 | 3.21 | 0.00 | 0.01 | -0.10 | 1.00 | sort and compare 10 |
| valid anagram | 9106 | 0.80 | 0.00 | 0.12 | -0.10 | 1.00 | sort and compare 9, hash count 1 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| valid anagram | 9107 | -1.31 | 0.20 | 0.40 | 0.10 | 1.00 | hash count 4, sort and compare 4, fixed count array 2 |
| valid anagram | 9108 | 2.05 | 0.00 | 0.03 | -0.10 | 1.00 | sort and compare 9, hash count 1 |

**Table 339.** Random pushes by unit.

| Unit | Spearman (margin change, share change) | Lower tail permutation p | Committed upper tail p | MAE of predicted shares | Log-score gain per sample over no push effect | Pass ratio against the twin | Twin minority share on these slots |
|---|---|---|---|---|---|---|---|
| minimum spanning tree weight | -0.69 | 0.0271 | 0.979 | 0.146 | 0.053 | 0.912 | 0.20 (pred 0.24) |
| trapped rain water | -0.44 | 0.1161 | 0.888 | 0.188 | 0.109 | 0.988 | 0.40 (pred 0.19) |
| valid anagram | -0.94 | 2.6e-04 | 1.000 | 0.178 | 0.162 | 1.000 | 0.10 (pred 0.20) |

### 7.5.6 Question 1a-c of the request

The request's question 1a-c asks whether, with the seed fixed and top-k 10, a tight temperature gives one idea in many phrasings or many ideas, and whether loosening it multiplies ideas or phrasings. On the mixed units a tight temperature collapses onto the default idea, and loosening it restores the second idea at its natural rate, never more, while the phrasings within each idea multiply (Table 340): Kruskal programs on minimum spanning tree weight form 3 phrasing clusters at T 0.3 and 7 at T 1.3. No third idea appears at any temperature. In Gemma, where few samples leave the opening, T 1.3 does raise the minority above its natural rate, as its fork logits predicted (Section 7.8.2). Hill-1 of the correct ideas on minimum spanning tree weight rises from 1.25 at T 0.3 to 1.95 at T 1.0 and stays at 1.92 at 1.3; ten literal trajectories at T 0.3 give 9 Kruskal and 1 Prim. The only correct programs off the catalogue are 40 continuations from one fork token of low probability on valid anagram; they form 2 distinct programs, both the catalogue's extra counting mode, and are catalogue variants.

**Table 340.** Ideas and phrasings across temperature (same seed; Hill-1 at depth 25).

| Unit | Cell | Ideas among correct programs | Hill-1 at depth 25 | Phrasing clusters per idea (Jaccard 0.8) | Ten literal trajectories: ideas | Their phrasing clusters |
|---|---|---|---|---|---|---|
| minimum spanning tree weight | T 0.3, top-p 1 | Kruskal, Prim with heap | 1.25 | Kruskal 3, Prim with heap 1 | Kruskal 9, Prim with heap 1 | Kruskal 3, Prim with heap 1 |
| minimum spanning tree weight | T 0.7, top-p 1 | Kruskal, Prim with heap | 1.69 | Kruskal 4, Prim with heap 1 | Kruskal 6, Prim with heap 4 | Kruskal 2, Prim with heap 1 |
| minimum spanning tree weight | T 1.0, top-p 1 | Kruskal, Prim with heap | 1.95 | Prim with heap 5, Kruskal 4 | Prim with heap 6, Kruskal 4 | Prim with heap 3, Kruskal 2 |

| | | | | | | |
|---|---|---|---|---|---|---|
| minimum spanning tree weight | T 1.3, top-p 1 | Kruskal, Prim with heap | 1.92 | Kruskal 7, Prim with heap 4 | Prim with heap 6, Kruskal 4 | Kruskal 3, Prim with heap 2 |
| minimum spanning tree weight | T 0.3, top-k 10 | Kruskal, Prim with heap | 1.25 | Kruskal 3, Prim with heap 1 | Kruskal 9, Prim with heap 1 | Kruskal 3, Prim with heap 1 |
| minimum spanning tree weight | T 0.7, top-k 10 | Kruskal, Prim with heap | 1.69 | Kruskal 4, Prim with heap 1 | Kruskal 6, Prim with heap 4 | Kruskal 2, Prim with heap 1 |
| minimum spanning tree weight | T 1.0, top-k 10 | Kruskal, Prim with heap | 1.95 | Prim with heap 5, Kruskal 4 | Prim with heap 6, Kruskal 4 | Prim with heap 3, Kruskal 2 |
| minimum spanning tree weight | T 1.3, top-k 10 | Kruskal, Prim with heap | 1.92 | Kruskal 7, Prim with heap 4 | Prim with heap 6, Kruskal 4 | Kruskal 3, Prim with heap 2 |
| trapped rain water | T 0.3, top-p 1 | prefix and suffix maxima, two pointers | 1.37 | prefix and suffix maxima 1, two pointers 1 | prefix and suffix maxima 9, two pointers 1 | prefix and suffix maxima 1, two pointers 1 |
| trapped rain water | T 0.7, top-p 1 | prefix and suffix maxima, two pointers | 1.58 | prefix and suffix maxima 2, two pointers 1 | prefix and suffix maxima 9, two pointers 1 | prefix and suffix maxima 1, two pointers 1 |
| trapped rain water | T 1.0, top-p 1 | peak split, prefix and suffix maxima, two pointers | 1.74 | prefix and suffix maxima 2, two pointers 2, peak split 1 | prefix and suffix maxima 9, two pointers 1 | prefix and suffix maxima 1, two pointers 1 |
| trapped rain water | T 1.3, top-p 1 | peak split, prefix and suffix maxima, two pointers | 1.79 | prefix and suffix maxima 3, two pointers 2, peak split 1 | prefix and suffix maxima 8, two pointers 2 | prefix and suffix maxima 3, two pointers 1 |
| trapped rain water | T 0.3, top-k 10 | prefix and suffix maxima, two pointers | 1.37 | prefix and suffix maxima 1, two pointers 1 | prefix and suffix maxima 9, two pointers 1 | prefix and suffix maxima 1, two pointers 1 |
| trapped rain water | T 0.7, top-k 10 | prefix and suffix maxima, two pointers | 1.58 | prefix and suffix maxima 2, two pointers 1 | prefix and suffix maxima 9, two pointers 1 | prefix and suffix maxima 1, two pointers 1 |
| trapped rain water | T 1.0, top-k 10 | peak split, prefix and suffix maxima, two pointers | 1.74 | prefix and suffix maxima 2, two pointers 2, peak split 1 | prefix and suffix maxima 9, two pointers 1 | prefix and suffix maxima 1, two pointers 1 |
| trapped rain water | T 1.3, top-k 10 | peak split, prefix and suffix maxima, two pointers | 1.79 | prefix and suffix maxima 3, two pointers 2, peak split 1 | prefix and suffix maxima 8, two pointers 2 | prefix and suffix maxima 3, two pointers 1 |
| valid anagram | T 0.3, top-p 1 | fixed count array, hash count, sort and compare | 1.83 | fixed count array 2, sort and compare 2, hash count 1 | sort and compare 8, fixed count array 2 | sort and compare 2, fixed count array 1 |
| valid anagram | T 0.7, top-p 1 | fixed count array, hash count, sort and compare | 1.72 | fixed count array 3, sort and compare 2, hash count 1 | sort and compare 8, fixed count array 2 | sort and compare 2, fixed count array 1 |

| | | | | | | |
|---|---|---|---|---|---|---|
| valid anagram | T 1.0, top-p 1 | fixed count array, hash count, sort and compare | 1.84 | fixed count array 2, sort and compare 2, hash count 1 | sort and compare 8, fixed count array 2 | sort and compare 2, fixed count array 1 |
| valid anagram | T 1.3, top-p 1 | fixed count array, hash count, sort and compare | 1.85 | fixed count array 5, sort and compare 2, hash count 1 | sort and compare 8, fixed count array 2 | sort and compare 2, fixed count array 1 |
| valid anagram | T 0.3, top-k 10 | fixed count array, hash count, sort and compare | 1.83 | fixed count array 2, sort and compare 2, hash count 1 | sort and compare 8, fixed count array 2 | sort and compare 2, fixed count array 1 |
| valid anagram | T 0.7, top-k 10 | fixed count array, hash count, sort and compare | 1.72 | fixed count array 3, sort and compare 2, hash count 1 | sort and compare 8, fixed count array 2 | sort and compare 2, fixed count array 1 |
| valid anagram | T 1.0, top-k 10 | fixed count array, hash count, sort and compare | 1.84 | fixed count array 2, sort and compare 2, hash count 1 | sort and compare 8, fixed count array 2 | sort and compare 2, fixed count array 1 |
| valid anagram | T 1.3, top-k 10 | fixed count array, hash count, sort and compare | 1.85 | fixed count array 5, sort and compare 2, hash count 1 | sort and compare 8, fixed count array 2 | sort and compare 2, fixed count array 1 |

## 7.6 The dose unit, not the seeds, made iteration 5's plan noise weak (Qwen3-8B)

### 7.6.1 Design

Iteration 4 found plan noise strong under its *absolute unit*, in which each cell (block and position) receives a random direction scaled to the displacement a ban itself causes at that cell. Iteration 5 found it nearly inert under its *relative unit*, which scales the direction to a fixed fraction of the twin's own residual norm at each cell, and with different direction seeds. Section 6.3 showed that the 8-bit engine was not the cause. The dose unit experiment crossed the two units with the two seed sets in bf16, on the same 50 slots per design, and committed a decision rule before the first row: the answer is the unit if the main effect of the unit is positive with its interval above zero and the effect of the seeds lies within plus or minus 0.10. Two anchors reproduce the earlier artifacts on the same slots: the absolute unit with iteration-4 seeds gives 26 of 50 attempts against 25 in the engine control (agreement per slot 0.98), and the relative unit with iteration-5 seeds gives 1 of 50 against 1. Iteration 5's null is thus reproduced without its left padding, so the decoding path is not the cause either.

### 7.6.2 The 2x2

The answer is the unit (Table 341). On connected components at alpha 2, the absolute unit gives a correct share off the default of 0.50 and 0.52 under the two seed sets, and the relative unit 0.10 and 0.02: a unit effect of +0.45 [0.35, 0.55], a seed effect of +0.03 [-0.07, 0.14] and an interaction of -0.10 [-0.28, 0.08]. A conditional logistic regression with slot strata agrees (unit coefficient 4.34, p 1e-4; seeds 1.68, p 0.13). On shortest path, at 100 slots, the unit effect is smaller (+0.115 [0.06, 0.17]) and most attempts under the absolute unit fail the hidden tests (13 correct of 41 attempted at pass 0.66; Table 342). On 50 fresh connected components slots the unit effect replicates (+0.42 [0.32, 0.52]), and on kth smallest it is +0.27 [0.18, 0.37] with a seed effect of +0.03 (Tables 346 and 347). On kth smallest, unlike connected components, the relative unit does move the algorithm, at half the absolute unit's rate.

*[Figure fig_unit_seed: generated in the report phase, not in this draft]*

**Table 341.** The 2x2 on connected components at alpha 2.

| connected components, α 2, N = 50 per design | share Y | 95% CI |
|---|---|---|
| absolute unit, iteration-4 seeds | 0.50 | [0.37, 0.63] |
| absolute unit, iteration-5 seeds | 0.52 | [0.39, 0.65] |
| relative unit, iteration-4 seeds | 0.10 | [0.04, 0.21] |
| relative unit, iteration-5 seeds | 0.02 | [0.00, 0.10] |
| unit main effect (risk difference) | +**0.45** | [0.35, 0.55] |
| seed main effect (risk difference) | +0.03 | [−0.07, 0.14] |
| interaction | −0.10 | [−0.28, 0.08] |

**Table 342.** Every 2x2 cell (Y, correct programs off the default with Wilson 95% interval; A, attempted off the default).

| task | arm | N | Y (Wilson 95%) | Y, any named algorithm | A | pass |
|---|---|---|---|---|---|---|
| connected components | neutral twin | 50 | 0 [0.00, 0.07] | 0 | 0 | 1.00 |
| connected components | the four designs at α 1 (absolute/ iteration-4, absolute/iteration-5, relative/iteration-4, relative/iteration-5) | 50 each | 6 / 7 / 0 / 1 | same | 6 / 7 / 0 / 1 | 1.00 |
| connected components | absolute unit, iteration-4 seeds, α 2 | 50 | 25 [0.37, 0.63] | 25 | 26 | 0.98 |
| connected components | absolute unit, iteration-5 seeds, α 2 | 50 | 26 [0.39, 0.65] | 26 | 26 | 1.00 |
| connected components | relative unit, iteration-4 seeds, α 2 | 50 | 5 [0.04, 0.21] | 5 | 5 | 1.00 |
| connected components | relative unit, iteration-5 seeds, α 2 | 50 | 1 [0.00, 0.10] | 1 | 1 | 1.00 |
| connected components | profile of the relative unit at absolute energy per block, α 2 | 50 | 14 [0.17, 0.42] | 14 | 17 | 0.94 |
| shortest path | neutral twin | 100 | 0 [0.00, 0.04] | 0 | 1 | 0.99 |
| shortest path | the four designs at α 1 (absolute/ iteration-4, absolute/iteration-5, relative/iteration-4, relative/iteration-5) | 50 each | 7 / 7 / 0 / 0 | same | 12 / 12 / 2 / 0 | 0.84 / 0.88 / 0.96 / 1.00 |
| shortest path | absolute unit, iteration-4 seeds, α 2 | 100 | 13 [0.08, 0.21] | 13 | 41 | 0.66 |
| shortest path | absolute unit, iteration-5 seeds, α 2 | 100 | 20 [0.13, 0.29] | 20 | 38 | 0.74 |
| shortest path | relative unit, iteration-4 seeds, α 2 | 100 | 5 [0.02, 0.11] | 5 | 9 | 0.96 |
| shortest path | relative unit, iteration-5 seeds, α 2 | 100 | 5 [0.02, 0.11] | 5 | 10 | 0.95 |
| shortest path | profile of the relative unit at absolute energy per block, α 2 | 100 | 8 [0.04, 0.15] | 8 | 16 | 0.90 |

### 7.6.3 Why the units differ

The two units differ in where the energy sits. The absolute unit is peaky: on connected components its top quartile of cells holds 81% of its energy, on bands of a few task tokens, and its largest cell

receives about 2.1 times that cell's residual norm. The relative unit is flat by construction: its top quartile holds 42%, and its largest cell receives 0.25 times the residual norm. A fifth design applies the per-position profile of the relative unit at the energy per block of the absolute unit; it moves 0.28 on connected components, between the two, so the profile (+0.22 [0.08, 0.36], McNemar 13:2) and the energy (+0.18 [0.06, 0.30], 10:1) each carry part of the effect.

A split of the absolute unit by cell confirms that the peaky cells matter (Tables 343 and 344). At the same total energy, the cells of the top quartile alone move connected components in 29 of 50 slots, the other three quartiles in 15 (18:4, Holm p 0.01), and the same energy at shuffled positions in 4 (21:0 against the full unit). On shortest path the other three quartiles include the cell with a massive activation at position 87 (the token `0`, norm about 10,290, 82% of the unit's energy there); noise on them breaks every program (pass 0.00), and the top quartile alone moves 5 of 50. Sweeping the dose of the relative unit shows that it comes close to the effect of the absolute unit on connected components only near alpha 4 (20 of 50 at pass 0.98, against 25 for the absolute unit at alpha 2), and that at alpha 6 the programs break (pass 0.46; Table 345). At alpha 4 it pays 1.8 times the downstream displacement of the absolute unit at the read position (423 against 240), and at a matched dose at block 12 (alpha 3, 13.6 against 13.8) it moves only 9 of 50 (McNemar 1:17 against the absolute unit). On shortest path it never moves more than 5 of 50 before the programs break.

The size of the downstream displacement does not explain the difference. At alpha 2 the realised displacement at block 35 is about the same under the two units, at the read position and at the first generated token, yet the absolute unit escapes 5 to 25 times as often. Adding the log block-35 norms to the per-sample logit leaves the unit coefficients unchanged on connected components (-2.35 to -2.54 for the relative unit with iteration-4 seeds, -3.31 to -3.49 with iteration-5 seeds). And the arm of the top quartile has the smallest downstream displacement of all pushed arms while moving the most. An exploratory analysis of individual samples found that which samples escape is close to a coin flip for a given direction and token stream: two directions on the same common random numbers agree weakly (phi coefficient 0.16 to 0.20 on connected components, about 0 on shortest path), the same direction is not monotone in dose (escape at alpha 2 given escape at alpha 1 in 2 of 6 samples, against a marginal 0.52), and neither the size nor the direction of the block-35 displacement predicts escape within the absolute unit (AUC under cross-validation 0.40 to 0.53). The unit sets the rate of escape; single outcomes behave like per-sample random variables.

**Table 343.** The absolute unit split by cell, at the full unit's total energy.

| Task | Arm (all at the absolute unit's total energy, alpha 2) | N | Correct Y off the default (share) | Attempted off the default | Pass | Correct composition | Realised displacement at block 12 (spec mean) | Realised displacement at block 35 (read position) |
|---|---|---|---|---|---|---|---|---|
| connected components | absolute unit, full | 50 | 25 (0.50) | 26 | 0.98 | union find 24, DFS 24, BFS 1 | 13.76 | 240.50 |
| connected components | absolute unit, cells of the top quartile only | 50 | 29 (0.58) | 29 | 1.00 | DFS 29, union find 21 | 4.64 | 159.51 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| connected components | absolute unit, other three quartiles only | 50 15 (0.30) | 16 | 0.92 | union find 31, DFS 11, BFS 4 | 21.86 | 367.51 |
| connected components | absolute unit, positions shuffled | 50 4 (0.08) | 5 | 0.98 | union find 45, DFS 4 | 13.75 | 248.24 |
| shortest path | absolute unit, full | 50 8 (0.16) | 24 | 0.62 | Dijkstra 23, Bellman-Ford/SPFA 8 | 13.85 | 247.88 |
| shortest path | absolute unit, cells of the top quartile only | 50 5 (0.10) | 11 | 0.84 | Dijkstra 37, Bellman-Ford/SPFA 5 | 2.38 | 135.06 |
| shortest path | absolute unit, other three quartiles only | 50 0 (0.00) | 36 | 0.00 | none | 60.18 | 865.72 |
| shortest path | absolute unit, positions shuffled | 50 0 (0.00) | 5 | 0.86 | Dijkstra 43 | 13.85 | 240.91 |

**Table 344.** Tests of the cell split.

| Task | Contrast (correct off the default) | Discordant first : second | Exact p | Holm p |
|---|---|---|---|---|
| connected components | top quartile against rest | 18:4 | 0.0043 | 0.01 |
| connected components | full unit against shuffled | 21:0 | 9.5e-07 | 2.9e-06 |
| connected components | full unit against top quartile | 4:8 | 0.3877 | 0.39 |
| shortest path | top quartile against rest | 5:0 | 0.0625 | 0.12 |
| shortest path | full unit against shuffled | 8:0 | 0.0078 | 0.02 |
| shortest path | full unit against top quartile | 6:3 | 0.5078 | 0.51 |

**Table 345.** The relative unit swept in dose, beside the absolute unit.

| Task | Arm | N | Correct off the default | Attempted off the default | Pass | Correct composition | Displacement at block 12 | Displacement at block 35, read position | Against the absolute unit at alpha 2: arm only : absolute only (p) |
|---|---|---|---|---|---|---|---|---|---|
| connected components | neutral twin | 50 | 0 | 0 | 1.00 | union find 50 | - | - | 0:25 (p 6.0e-08) |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| connected components | absolute unit, iteration-4 seeds, alpha 1 | 50 6 | 6 | 1.00 | union find 44, DFS 6 | 6.88 | 129.33 | 4:23 (p 3.1e-04) |
| connected components | absolute unit, iteration-4 seeds, alpha 2 | 50 25 | 26 | 0.98 | union find 24, DFS 24, BFS 1 | 13.76 | 240.50 | - |
| connected components | relative unit, iteration-4 seeds, alpha 1 | 50 0 | 0 | 1.00 | union find 50 | 4.53 | 117.14 | 0:25 (p 6.0e-08) |
| connected components | relative unit, alpha 2 | 50 5 | 5 | 1.00 | union find 45, DFS 5 | 9.07 | 222.26 | 1:21 (p 1.1e-05) |
| connected components | relative unit, alpha 3 | 50 9 | 10 | 0.98 | union find 40, DFS 7, BFS 2 | 13.60 | 320.76 | 1:17 (p 1.4e-04) |
| connected components | relative unit, alpha 4 | 50 20 | 21 | 0.98 | union find 29, DFS 14, BFS 6 | 18.13 | 423.31 | 3:8 (p 0.2266) |
| connected components | relative unit, alpha 6 | 50 14 | 22 | 0.46 | union find 9, DFS 9, BFS 5 | 27.20 | 692.77 | 4:15 (p 0.0192) |
| connected components | profile of the relative unit at absolute energy per block, alpha 2 | 50 14 | 17 | 0.94 | union find 33, DFS 9, BFS 5 | 21.17 | 381.96 | 2:13 (p 0.0074) |
| shortest path | neutral twin | 50 0 | 1 | 0.98 | Dijkstra 49 | - | - | 0:8 (p 0.0078) |
| shortest path | absolute unit, iteration-4 seeds, alpha 1 | 50 7 | 12 | 0.84 | Dijkstra 35, Bellman-Ford/SPFA 7 | 6.93 | 134.52 | 5:6 (p 1.0000) |
| shortest path | absolute unit, iteration-4 seeds, alpha 2 | 50 8 | 24 | 0.62 | Dijkstra 23, Bellman-Ford/SPFA 8 | 13.85 | 247.88 | - |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| shortest path | relative unit, iteration-4 seeds, alpha 1 | 50 0 | 2 | 0.96 | Dijkstra 48 | 4.52 | 119.49 | 0:8 (p 0.0078) |
| shortest path | relative unit, alpha 2 | 50 1 | 3 | 0.96 | Dijkstra 47, Bellman-Ford/SPFA 1 | 9.04 | 228.62 | 1:8 (p 0.0391) |
| shortest path | relative unit, alpha 3 | 50 3 | 10 | 0.82 | Dijkstra 38, Bellman-Ford/SPFA 3 | 13.56 | 329.85 | 3:8 (p 0.2266) |
| shortest path | relative unit, alpha 4 | 50 5 | 19 | 0.66 | Dijkstra 28, Bellman-Ford/SPFA 5 | 18.08 | 434.64 | 5:8 (p 0.5811) |
| shortest path | relative unit, alpha 6 | 50 0 | 27 | 0.14 | Dijkstra 7 | 27.12 | 672.06 | 0:8 (p 0.0078) |
| shortest path | profile of the relative unit at absolute energy per block, alpha 2 | 50 2 | 4 | 0.92 | Dijkstra 44, Bellman-Ford/SPFA 2 | 17.55 | 238.77 | 2:8 (p 0.1094) |

**Table 346.** Connected components on 50 fresh slots (exploratory replication). (iteration-8 correction: profile rows and the fresh-slot seed effect added from the dose-unit experiment's `results/e3_replicate.json`, block T10_fresh_k10_19)

| design | share Y |
|---|---|
| absolute unit, iteration-4 seeds | 0.42 |
| absolute unit, iteration-5 seeds | 0.60 |
| relative unit, iteration-4 seeds | 0.08 |
| relative unit, iteration-5 seeds | 0.10 |
| top-quartile cells only, absolute unit, iteration-4 seeds | 0.48 (24/50; Wilson [0.35, 0.61]) |
| the other three quartiles at the same energy | 0.18 (9/50); McNemar top quartile against rest 15:0 |
| same energy at shuffled positions | 0.08 (4/50); McNemar full unit against shuffled 17:0 |
| seed effect on the fresh slots (RD_seed) | -0.10 [-0.22, 0.02], at the edge of the committed +-0.10 band |

**Table 347.** Kth smallest at alpha 2 (exploratory third task).

| design | share Y |
|---|---|
| absolute unit, iteration-4 seeds | 0.56 |
| absolute unit, iteration-5 seeds | 0.52 |
| relative unit, iteration-4 seeds | 0.28 |
| relative unit, iteration-5 seeds | 0.26 |

### 7.6.4 Plan noise repriced at the unit that moves

At the absolute unit with iteration-4 seeds, plan noise was priced against the five knobs of iteration

5 that act on the text on five tasks (Tables 348 to 351). We call the price measure the 5-point Hill-1: the Hill-1 number of distinct correct algorithms reachable within a pass cost of 5 points along the dose range of each knob. Beside it we report the pass cost of reaching Hill-1 of 1.5 and 2. Plan noise beats an intent sentence written by the base model (base intent) on kth smallest (3.22 against 1.73) and the grid (1.39 against 1.05), and loses on connected components (2.13 against 2.53), shortest path (1.21 against 1.45) and two sum (1.10 against 1.61). The committed prediction was 3 of 5 and the result is 2 of 5. Iteration 5's comparison, which plan noise had lost at the relative unit, is reversed in its mean (the margin of plan noise over the best text sampler moves from -0.40 to +0.12 [-0.18, 0.27]) but not in its rule: the iteration-5 rule replayed at the absolute unit is still disconfirmed, 2 of 5. Temperature 1.3 and top-k 10 reach Hill-1 of at most 1.09 on every task. The six verdicts that the record audit tagged as resting on the relative unit are now retagged: at the design that moves (the absolute unit with iteration-4 seeds at alpha 2), kth smallest (+0.58) and connected components (+0.50) are movable, and shortest path (+0.16, at pass 0.62 against the twin's 0.98), two sum (+0.02) and the grid (+0.08) are not. Five correct kth smallest programs labelled other count frequencies in a hash map and walk the sorted distinct keys by cumulative count; they are counting sort with key compression, a catalogue variant. Of the 25 pairs of task and text knob, 17 reproduced the iteration-5 labels in 10 of 10 check slots and reuse its rows, and 8 were regenerated in this engine; the 250 base-model intents were identical in text and token ids to iteration 5's. Against the exchange experiment's engine, labels and pass agreed in 15 of 16 slots on shortest path and kth smallest, 40 of 40 on connected components and two sum, and 34 of 40 on the grid (all BFS, with 6 pass flips), which the artifact logs as kernel sensitivity across GPUs.

**Table 348.** The 5-point Hill-1 of plan noise against the knobs that act on the text.

| task | plan noise (absolute unit) | base intent | enumerate and pick | intent factored generation (1.3, 0.7) | T 1.3 | top-k 10 | plan noise (absolute unit) vs base intent |
|---|---|---|---|---|---|---|---|
| kth smallest | **3.22** | 1.73 | 1.77 | 1.09 | 1.09 | 1.09 | WIN (P = 1.00) |
| connected components | 2.13 | 2.53 | 1.79 | 1.96 | 1.00 | 1.00 | LOSS (P = 0.05) |
| shortest path | 1.21 | 1.45 | 1.28 | 1.00 | 1.00 | 1.00 | LOSS (P = 0.15) |
| two sum | 1.10 | 1.61 | 1.16 | 1.00 | 1.00 | 1.00 | LOSS (P = 0.00) |
| grid | 1.39 | 1.05 | 1.11 | 1.00 | 1.00 | 1.00 | WIN (P = 0.99) |

**Table 349.** The 5-point Hill-1 with bootstrap intervals and the pass cost of reaching Hill-1 of 1.5 and 2 (as a fraction of the pass rate).

| Task | Knob | 5-point Hill-1 | Bootstrap 95% CI | Price of 1.5 (pass cost) | Price of 2 (pass cost) |
|---|---|---|---|---|---|
| kth smallest | plan noise (absolute unit) | 3.216 | [2.73, 3.53] | 0.0032 | 0.0072 |
| kth smallest | base intent | 1.729 | [1.25, 2.25] | 0.0336 | not reached |
| kth smallest | enumerate and pick | 1.765 | [1.40, 2.56] | 0.0320 | 0.0680 |
| kth smallest | intent factored generation (1.3, 0.7) | 1.095 | [1.00, 1.30] | not reached | not reached |
| kth smallest | temperature 1.3 | 1.093 | [1.00, 1.30] | not reached | not reached |
| kth smallest | top-k 10 | 1.092 | [1.00, 1.26] | not reached | not reached |
| connected components | plan noise (absolute unit) | 2.129 | [1.91, 2.37] | 0.0014 | 0.0132 |

| | | | | | |
|---|---|---|---|---|---|
| connected components | base intent | 2.527 | [2.11, 2.77] | 0.0050 | 0.0108 |
| connected components | enumerate and pick | 1.791 | [1.51, 2.26] | 0.0324 | 0.0648 |
| connected components | intent factored generation (1.3, 0.7) | 1.962 | [1.96, 1.97] | 0.0000 | not reached |
| connected components | temperature 1.3 | 1.000 | [1.00, 1.00] | not reached | not reached |
| connected components | top-k 10 | 1.000 | [1.00, 1.00] | not reached | not reached |
| shortest path | plan noise (absolute unit) | 1.214 | [1.10, 1.55] | 0.1246 | not reached |
| shortest path | base intent | 1.450 | [1.13, 1.72] | 0.0570 | not reached |
| shortest path | enumerate and pick | 1.276 | [1.13, 1.56] | 0.0884 | 0.1666 |
| shortest path | intent factored generation (1.3, 0.7) | 1.000 | [1.00, 1.00] | not reached | not reached |
| shortest path | temperature 1.3 | 1.000 | [1.00, 1.00] | not reached | not reached |
| shortest path | top-k 10 | 1.000 | [1.00, 1.00] | not reached | not reached |
| two sum | plan noise (absolute unit) | 1.099 | [1.00, 1.25] | not reached | not reached |
| two sum | base intent | 1.605 | [1.25, 2.57] | 0.0420 | 0.0804 |
| two sum | enumerate and pick | 1.165 | [1.08, 1.34] | 0.1332 | 0.2304 |
| two sum | intent factored generation (1.3, 0.7) | 1.000 | [1.00, 1.00] | not reached | not reached |
| two sum | temperature 1.3 | 1.000 | [1.00, 1.00] | not reached | not reached |
| two sum | top-k 10 | 1.000 | [1.00, 1.00] | not reached | not reached |
| grid | plan noise (absolute unit) | 1.390 | [1.08, 1.62] | not reached | not reached |
| grid | base intent | 1.053 | [1.01, 1.14] | 0.3504 | not reached |
| grid | enumerate and pick | 1.109 | [1.04, 1.22] | 0.1776 | 0.2736 |
| grid | intent factored generation (1.3, 0.7) | 1.000 | [1.00, 1.00] | not reached | not reached |
| grid | temperature 1.3 | 1.000 | [1.00, 1.00] | not reached | not reached |
| grid | top-k 10 | 1.000 | [1.00, 1.00] | not reached | not reached |

**Table 350.** Plan noise against base intent and against the best text sampler.

| Task | Plan noise against base intent | Bootstrap P(plan noise ahead) | Best text sampler | Margin over the best text sampler | Plan noise at least the best (iteration-5 rule) | Bootstrap P |
|---|---|---|---|---|---|---|
| kth smallest | WIN | 0.9995 | enumerate and pick | 1.45 | yes | 0.998 |
| connected components | LOSS | 0.0525 | base intent | -0.40 | no | 0.049 |
| shortest path | LOSS | 0.1540 | base intent | -0.24 | no | 0.086 |
| two sum | LOSS | 0.0025 | base intent | -0.51 | no | 0.003 |
| grid | WIN | 0.9905 | enumerate and pick | 0.28 | yes | 0.945 |

**Table 351.** Every pricing cell.

| Task | Arm | N | Pass | Correct composition | Correct off the default |
|---|---|---|---|---|---|
| kth smallest | neutral twin | 50 | 1.00 | full sort 49, quickselect 1 | 1 |

| | | | | | |
|---|---|---|---|---|---|
| kth smallest | plan noise, alpha 1 | 50 | 0.98 | full sort 30, size k heap 12, quickselect 7 | 19 |
| kth smallest | plan noise, alpha 2 | 50 | 0.98 | full sort 19, quickselect 17, size k heap 11, other 2 | 30 |
| kth smallest | reused iteration-5 base intent | 50 | 0.94 | full sort 39, size k heap 4, quickselect 2, counting sort 1, other 1 | 8 |
| kth smallest | enumerate and pick | 50 | 0.80 | quickselect 25, full sort 7, counting sort 4, other 2, median of medians 1, heapify and pop 1 | 33 |
| kth smallest | intent factored generation | 50 | 1.00 | full sort 49, heapify and pop 1 | 1 |
| kth smallest | temperature 1.3 | 50 | 1.00 | full sort 49, quickselect 1 | 1 |
| kth smallest | reused iteration-5 top-k 10 | 50 | 1.00 | full sort 49, quickselect 1 | 1 |
| connected components | neutral twin | 50 | 1.00 | union find 50 | 0 |
| connected components | plan noise, alpha 1 | 50 | 1.00 | union find 44, DFS 6 | 6 |
| connected components | plan noise, alpha 2 | 50 | 0.98 | union find 24, DFS 24, BFS 1 | 25 |
| connected components | reused iteration-5 base intent | 50 | 0.98 | union find 24, DFS 20, BFS 5 | 25 |
| connected components | enumerate and pick | 50 | 0.82 | DFS 18, BFS 14, union find 9 | 32 |
| connected components | reused iteration-5 intent factored generation | 50 | 1.00 | DFS 43, union find 7 | 43 |
| connected components | reused iteration-5 temperature 1.3 | 50 | 1.00 | union find 50 | 0 |
| connected components | reused iteration-5 top-k 10 | 50 | 1.00 | union find 50 | 0 |
| shortest path | neutral twin | 50 | 0.98 | Dijkstra 49 | 0 |
| shortest path | plan noise, alpha 1 | 50 | 0.84 | Dijkstra 35, Bellman-Ford/SPFA 7 | 7 |
| shortest path | plan noise, alpha 2 | 50 | 0.62 | Dijkstra 23, Bellman-Ford/SPFA 8 | 8 |
| shortest path | reused iteration-5 base intent | 50 | 0.92 | Dijkstra 39, Bellman-Ford/SPFA 7 | 7 |
| shortest path | enumerate and pick | 50 | 0.64 | Dijkstra 12, Bellman-Ford/SPFA 10, Floyd-Warshall 10 | 20 |
| shortest path | reused iteration-5 intent factored generation | 50 | 1.00 | Dijkstra 50 | 0 |
| shortest path | reused iteration-5 temperature 1.3 | 50 | 0.98 | Dijkstra 49 | 0 |
| shortest path | reused iteration-5 top-k 10 | 50 | 0.98 | Dijkstra 49 | 0 |
| two sum | neutral twin | 50 | 1.00 | hash map 50 | 0 |
| two sum | plan noise, alpha 1 | 50 | 1.00 | hash map 50 | 0 |
| two sum | plan noise, alpha 2 | 50 | 1.00 | hash map 49, brute force 1 | 1 |
| two sum | reused iteration-5 base intent | 50 | 0.88 | hash map 29, brute force 9, sort and two pointers 4, sort and binary search 2 | 15 |
| two sum | enumerate and pick | 50 | 0.64 | hash map 19, brute force 7, sort and two pointers 4, sort and binary search 2 | 13 |

| | | | | | |
|---|---|---|---|---|---|
| two sum | reused iteration-5 intent factored generation | 50 | 1.00 | hash map 50 | 0 |
| two sum | reused iteration-5 temperature 1.3 | 50 | 1.00 | hash map 50 | 0 |
| two sum | reused iteration-5 top-k 10 | 50 | 1.00 | hash map 50 | 0 |
| grid | neutral twin | 50 | 0.76 | BFS 38 | 0 |
| grid | plan noise, alpha 1 | 50 | 0.66 | BFS 33 | 0 |
| grid | plan noise, alpha 2 | 50 | 0.78 | BFS 35, DFS 4 | 4 |
| grid | base intent | 50 | 0.28 | BFS 10, DFS 4 | 4 |
| grid | enumerate and pick | 50 | 0.28 | BFS 6, A* or greedy best first 3, DFS 2, Dijkstra 2, bidirectional BFS 1 | 8 |
| grid | reused iteration-5 intent factored generation | 50 | 0.12 | BFS 6 | 0 |
| grid | reused iteration-5 temperature 1.3 | 50 | 0.92 | BFS 46 | 0 |
| grid | reused iteration-5 top-k 10 | 50 | 0.84 | BFS 42 | 0 |

**Table 352.** The dose unit experiment's committed predictions.

| Prediction | What was predicted | Observed | Verdict |
|---|---|---|---|
| Unit effect | unit risk difference on connected components α 2 > 0, CI excludes 0 (point +0.35, range 0.20–0.50) | +0.45 [0.35, 0.55] | confirmed |
| Seed effect | seed risk difference within ±0.10, CI includes 0 | +0.03 [−0.07, 0.14] | confirmed |
| Interaction | interaction CI includes 0 | −0.10 [−0.28, 0.08] | confirmed |
| Shortest path unit effect | shortest path α 2 unit risk difference on correct programs ≈ +0.10 | +0.115 [0.06, 0.17] | confirmed |
| Profile of the relative unit at absolute energy | profile of the relative unit at absolute energy intermediate, connected components α 2 share 0.05–0.35 | 0.28 | confirmed |
| Top quartile against rest | top quartile > rest by ≥ 0.10 | connected components 0.58 vs 0.30 | confirmed |
| Full unit against shuffled positions | full absolute unit > shuffled positions by ≥ 0.10 | connected components 0.50 vs 0.08 | confirmed |
| Pricing against base intent | plan noise (absolute unit) matches or beats base intent on 3 of 5 | 2 of 5 (kth smallest, grid WIN; connected components, shortest path, two sum LOSS) | missed by one |
| Pricing, 5-point Hill-1 | 5-point Hill-1 of plan noise (absolute unit) ≥ 2 on kth smallest and connected components | 3.22, 2.13 | confirmed |

## 7.7 Evolutionary search with an evaluator (Qwen3-8B)

### 7.7.1 Design

Six iterations of generation one sample at a time had produced no correct novel algorithm. The search experiment tested the documented alternative [81, 82]: an evolutionary program search in which the model mutates parent programs and an executable evaluator keeps the correct ones. The

search uses four islands, each with a MAP-Elites archive [133] whose cells are the runtime label and a signature of the program's data structures; each round draws two parents per prompt by a Boltzmann choice over cells, asks for four children per prompt, and resets the worst island every 500 calls. The *evaluator* runs each program in an isolated subprocess with memory and time limits through 8 quick cases, the full hidden suite and 10 stress inputs of maximum size; a program that embeds hidden answers as constants fails the stress stage. A *novelty critic* compares each program, as hashed 4-gram shingles of its tokens after normalisation of identifiers, with 18,237 known solutions from 10 datasets, and a program passes the *catalogue screen* if it is correct and either labelled other or less similar to every known solution than the 5th percentile of the forced catalogue programs (Table 353). Two levels are counted: level a, distinct programs passing the screen, and level b, programs that adjudication places outside the catalogue (a known technique outside it, or novel).

The comparison arms use the same evaluator, labeller and critic: no evolution (the same prompt with frozen seed parents, the AlphaEvolve ablation [82]), the novelty instruction with no parents, enumerate and pick, base intent, the neutral twin, and a larger mutator (qwen3-235b through OpenRouter, 500 calls). The committed prediction was that search reaches at least 10 times the level a rate of the better of enumerate and pick and base intent, on at least two of shortest path, kth smallest and trapped rain water.

**Table 353.** The similarity threshold of the screen, per task.

| Task | s* |
|---|---|
| shortest path | 0.359 |
| kth smallest | 0.303 |
| trapped rain water | 0.608 |

### 7.7.2 Results

The search claim is not met at either level (Tables 354 to 356). At level a, search beats enumerate and pick by 3.83 times on shortest path, 6.65 times on kth smallest (lower bound 4.86) and 0.91 times on trapped rain water. At level b only kth smallest produces any program: 78 distinct programs from search against 0 from enumerate and pick (lower bound 12.8), 0 from no evolution and 2 from the instruction alone. The level b gain on kth smallest comes from evolution: against no evolution with the same prompt it is 78 to 0 (lower bound 5.1), and 66 of the 93 level b rows sit at depth 2 or more with parents other than the seeds (Table 362). At level a on kth smallest most of the gain is the instruction's: the instruction alone beats enumerate and pick by 4.6 times, and search beats the instruction alone by 1.45 times.

The programs outside the catalogue are known techniques (Tables 357 and 358). On kth smallest, 103 distinct programs implement order statistics on a Fenwick tree, binary search on the value range, bucket select, partial selection sort and a treap with order statistics; 540 are catalogue variants. On shortest path the 15-fold lead of search over no evolution at level a is restyled SPFA: every adjudicated program is a catalogue variant or degenerate, and two of them pass every test while being wrong in general, because they deduplicate on distance modulo a constant. On trapped rain water search loses to both no evolution and the instruction alone. No program is novel: the rule-of-three bound [67] is 7.5% over the 40 merged adjudicated clusters and 0.16% per correct search program on kth smallest.

**Table 354.** Counts per task and arm.

| Task | Arm | N | Correct | distinct programs passing the screen (rate) | distinct adjudicated programs outside the catalogue | idea clusters passing the screen |
|---|---|---|---|---|---|---|
| kth smallest | **search** | 2500 | 1917 | **412 (0.165)** | **78** | 63 |
| kth smallest | no evolution | 500 | 444 | 52 (0.104) | 0 | 13 |
| kth smallest | instruction only | 500 | 471 | 57 (0.114) | 2 | 11 |
| kth smallest | enumerate and pick | 1250 | 787 | 31 (0.025) | 0 | 10 |
| kth smallest | base intent | 1250 | 947 | 23 (0.018) | 0 | 8 |
| kth smallest | neutral twin | 200 | 198 | 1 (0.005) | 0 | 1 |
| kth smallest | larger mutator (qwen3-235b) | 500 | 441 | 67 (0.134) | 23 | 45 |
| shortest path | **search** | 1500 | 1079 | **46 (0.031)** | 0 | 11 |
| shortest path | no evolution | 500 | 414 | 1 (0.002) | 0 | 1 |
| shortest path | instruction only | 500 | 270 | 7 (0.014) | 0 | 3 |
| shortest path | enumerate and pick | 750 | 499 | 6 (0.008) | 0 | 2 |
| shortest path | base intent | 750 | 715 | 5 (0.007) | 0 | 4 |
| shortest path | neutral twin | 200 | 200 | 0 | 0 | 0 |
| trapped rain water | search | 1000 | 817 | 31 (0.031) | 0 | 7 |
| trapped rain water | no evolution | 300 | 158 | 33 (0.110) | 0 | 4 |
| trapped rain water | instruction only | 300 | 94 | 17 (0.057) | 0 | 7 |
| trapped rain water | enumerate and pick | 500 | 305 | 17 (0.034) | 0 | 6 |
| trapped rain water | base intent | 500 | 486 | 9 (0.018) | 0 | 3 |
| trapped rain water | neutral twin | 100 | 100 | 1 (0.010) | 0 | 1 |

**Table 355.** The primary test: search against the better text knob (enumerate and pick on every task).

| Task | Level | Rate ratio | One sided lower 95% bound | Verdict |
|---|---|---|---|---|
| shortest path | a | 3.83 | 1.82 | NOT MET |
| kth smallest | a | 6.65 | 4.86 (CI 4.61–9.91) | NOT MET |
| trapped rain water | a | 0.91 | 0.54 | NOT MET |
| kth smallest | b | ∞ (78 vs 0) | 12.8 | STRONG |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| shortest path, trapped rain water b | | - | - | | | | | NOT ESTIMABLE (0 vs 0) |

**Table 356.** Every level a and level b rate test.

| Level | Task and contrast | First arm X/N | Second arm X/N | Rate ratio | One sided lower 95% bound | One sided p | Holm p | Verdict under the primary rule |
|---|---|---|---|---|---|---|---|---|
| a | shortest path | 46/1500 | 6/750 | 3.83 | 1.82 | 2.9e-04 | 5.8e-04 | NOT MET |
| a | shortest path, secondary: search against no evolution | 46/1500 | 1/500 | 15.33 | 3.10 | 2.2e-05 | - | MET |
| a | shortest path, secondary: search against instruction only | 46/1500 | 7/500 | 2.19 | 1.09 | 0.0283 | - | NOT MET |
| a | shortest path, secondary: no evolution against instruction only | 1/500 | 7/500 | 0.14 | 0.01 | 0.9961 | - | NOT MET |
| a | shortest path, secondary: instruction only against enumerate and pick | 7/500 | 6/750 | 1.75 | 0.60 | 0.2288 | - | NOT MET |
| a | shortest path, secondary: enumerate and pick against neutral twin | 6/750 | 0/200 | not reached | 0.41 | 0.2421 | - | UNRESOLVED |
| a | kth smallest | 412/2500 | 31/1250 | 6.65 | 4.86 | 2.4e-40 | 7.3e-40 | NOT MET |
| a | kth smallest, secondary: search against no evolution | 412/2500 | 52/500 | 1.58 | 1.24 | 6.1e-04 | - | NOT MET |
| a | kth smallest, secondary: search against instruction only | 412/2500 | 57/500 | 1.45 | 1.14 | 0.0040 | - | NOT MET |
| a | kth smallest, secondary: no evolution against instruction only | 52/500 | 57/500 | 0.91 | 0.65 | 0.7171 | - | NOT MET |
| a | kth smallest, secondary: instruction only against enumerate and pick | 57/500 | 31/1250 | 4.60 | 3.12 | 2.0e-12 | - | NOT MET |
| a | kth smallest, secondary: enumerate and pick against neutral twin | 31/1250 | 1/200 | 4.96 | 0.98 | 0.0530 | - | NOT MET |
| a | trapped rain water | 31/1000 | 17/500 | 0.91 | 0.54 | 0.6816 | 0.6816 | NOT MET |
| a | trapped rain water, secondary: search against no evolution | 31/1000 | 33/300 | 0.28 | 0.18 | 1.0000 | - | NOT MET |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| a | trapped rain water, secondary: search against instruction only | 31/1000 | 17/300 | 0.55 | 0.32 | 0.9826 | - | NOT MET |
| a | trapped rain water, secondary: no evolution against instruction only | 33/300 | 17/300 | 1.94 | 1.15 | 0.0164 | - | NOT MET |
| a | trapped rain water, secondary: instruction only against enumerate and pick | 17/300 | 17/500 | 1.67 | 0.89 | 0.0934 | - | NOT MET |
| a | trapped rain water, secondary: enumerate and pick against neutral twin | 17/500 | 1/100 | 3.40 | 0.64 | 0.1728 | - | NOT MET |
| b | shortest path | 0/1500 | 0/750 | - | - | 1.0000 | 1.0000 | NOT ESTIMABLE |
| b | shortest path, secondary: search against no evolution | 0/1500 | 0/500 | - | - | 1.0000 | - | NOT ESTIMABLE |
| b | shortest path, secondary: search against instruction only | 0/1500 | 0/500 | - | - | 1.0000 | - | NOT ESTIMABLE |
| b | shortest path, secondary: no evolution against instruction only | 0/500 | 0/500 | - | - | 1.0000 | - | NOT ESTIMABLE |
| b | shortest path, secondary: instruction only against enumerate and pick | 0/500 | 0/750 | - | - | 1.0000 | - | NOT ESTIMABLE |
| b | shortest path, secondary: enumerate and pick against neutral twin | 0/750 | 0/200 | - | - | 1.0000 | - | NOT ESTIMABLE |
| b | kth smallest | 78/2500 | 0/1250 | not reached | 12.77 | 1.8e-14 | 5.5e-14 | STRONG |
| b | kth smallest, secondary: search against no evolution | 78/2500 | 0/500 | not reached | 5.11 | 6.7e-07 | - | MET |
| b | kth smallest, secondary: search against instruction only | 78/2500 | 2/500 | 7.80 | 2.41 | 6.6e-05 | - | NOT MET |
| b | kth smallest, secondary: no evolution against instruction only | 0/500 | 2/500 | 0.00 | 0.00 | 1.0000 | - | NOT MET |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| b | kth smallest, secondary: instruction only against enumerate and pick | 2/500 | 0/1250 | not reached | 0.72 | 0.0816 | - | UNRESOLVED |
| b | kth smallest, secondary: enumerate and pick against neutral twin | 0/1250 | 0/200 | - | - | 1.0000 | - | NOT ESTIMABLE |
| b | trapped rain water | 0/1000 | 0/500 | - | - | 1.0000 | 1.0000 | NOT ESTIMABLE |
| b | trapped rain water, secondary: search against no evolution | 0/1000 | 0/300 | - | - | 1.0000 | - | NOT ESTIMABLE |
| b | trapped rain water, secondary: search against instruction only | 0/1000 | 0/300 | - | - | 1.0000 | - | NOT ESTIMABLE |
| b | trapped rain water, secondary: no evolution against instruction only | 0/300 | 0/300 | - | - | 1.0000 | - | NOT ESTIMABLE |
| b | trapped rain water, secondary: instruction only against enumerate and pick | 0/300 | 0/500 | - | - | 1.0000 | - | NOT ESTIMABLE |
| b | trapped rain water, secondary: enumerate and pick against neutral twin | 0/500 | 0/100 | - | - | 1.0000 | - | NOT ESTIMABLE |

**Table 357.** Final classes of the distinct programs passing the screen.

| Task | Distinct programs passing the catalogue screen with a final class (all arms) | Catalogue variant | Known technique outside the catalogue | Degenerate | Labeller miss | Novel algorithm |
|---|---|---|---|---|---|---|
| shortest path | 65 | 59 | 0 | 6 | 0 | 0 |
| kth smallest | 643 | 540 | 103 | 0 | 0 | 0 |
| trapped rain water | 108 | 108 | 0 | 0 | 0 | 0 |

**Table 358.** The known techniques outside the catalogue (kth smallest).

| Known technique outside the catalogue (kth smallest; as named in the manual read) | Distinct programs |
|---|---|
| Fenwick tree (BIT) order statistics with coordinate compression | 25 |
| binary search on the value range (binary search on the answer) with frequency counts indexed by value | 22 |
| binary search on the value range (binary search on the answer) | 16 |
| partial selection sort (k rounds of minimum extraction) | 7 |
| Fenwick tree (BIT) order statistics | 7 |
| Fenwick tree (BIT) order statistics (binary lifting to find the kth) | 5 |

| | |
|---|---|
| Fenwick tree (BIT) order statistics (descent to find the kth) | 4 |
| bucket select (histogram localisation, then select inside one bucket) | 3 |
| Fenwick tree (BIT) order statistics over the raw value range | 3 |
| partial selection sort (repeated extreme extraction) | 2 |
| recursive bucket select (sqrt(n) value buckets, recurse into the bucket holding rank k) | 1 |
| bucket select (sqrt(n) value buckets, then sort one bucket) | 1 |
| iterative bucket select (value buckets of fixed width, narrow to the bucket holding rank k) | 1 |
| treap with order statistics (randomised balanced BST with subtree sizes) | 1 |
| bucket counts to localise the rank, then binary search on the value range (bucket select) | 1 |
| binary search on the value range (binary search on the answer) (recursive) | 1 |
| binary search on the value range (recursive) | 1 |
| bucket select (histogram localisation, then select inside one bucket) (histogram in two levels) | 1 |
| binary search on the value range (counts in a frequency table) | 1 |

### 7.7.3 Adjudication, the larger mutator and robustness

The judge gate failed: no judge family reached 9 of 10 on the controls of known class (Table 359). Every judge called the D'Esopo–Pape control a variant of SPFA, and three of five called novel a control that processes heights in decreasing order. Under the committed fallback the final class of every candidate is the executor's manual read (121 full reads and 123 structured reads). The reads were not blind to arm, so a rule-based reclassification blind to arm was run; it agrees with the manual classes on 94.4% of 816 distinct programs and reproduces the level b contrast on kth smallest (87 of 2,500 against 2 of 1,250, lower bound 6.7). The pair of judges fixed in advance was still run as advisory votes, with kappa 0.42; all 11 of their novel calls were overturned on reading (dead-code tournaments, partial selection sort and SPFA on a lifted graph).

The larger mutator, over its 500 calls, reaches level a at 0.84 times the rate of the first 500 calls of search and level b at 2.6 times (23 against 9, lower bound 1.28; Table 360). It writes mostly quickselect (402 of 441 correct programs), while search writes counting sort, quickselect and programs labelled other in nearly equal numbers. The search rate on kth smallest is stable across islands (bootstrap 0.173 to 0.227), and at matched effort no other arm approaches search's 63 idea clusters (Table 361). A placebo that shuffles the arms is significant in 1% to 6% of shuffles with median rate ratio near 1.

**Table 359.** The judge gate on ten controls of known class.

| Judge | Control accuracy | Gate (at least 9 of 10) | Control 1 | Control 2 | Control 3 | Control 4 | Control 5 | Control 6 | Control 7 | Control 8 | Control 9 | Control 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Mistral Medium 3.1 | 6/10 | failed | variant | variant | known outside | degenerate | variant | variant | degenerate | variant | variant | novel |
| Kimi k2 from Moonshot AI (release 0905) | 8/10 | failed | variant | known outside | known outside | degenerate | variant | variant | degenerate | known outside | variant | variant |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| google/ gemini-2.5-flash | 6/10 | failed | variant | variant | known outside | degenerate | variant | variant | degenerate | variant | variant | novel |
| DeepSeek chat (version 3.1) | 6/10 | failed | variant | variant | known outside | degenerate | variant | variant | degenerate | variant | variant | variant |
| openai/ gpt-4.1-mini | 5/10 | failed | variant | variant | variant | degenerate | variant | variant | degenerate | variant | variant | novel |

**Table 360.** The larger mutator against the first 500 calls of search (kth smallest).

| Arm (kth smallest) | N | Valid | Correct | Distinct programs passing the screen | Rate (level a) | 95% CI | Distinct programs outside the catalogue | Rate (level b) | Clusters a | Clusters b | Hill-1 of correct labels | Label mix |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| larger mutator (qwen3-235b), 500 calls | 500 | 492 | 441 | 67 | 0.134 | [0.104, 0.170] | 23 | 0.046 | 45 | 7 | 1.44 | quickselect 402, other 30, counting sort 6, heapify and pop 1, full sort 1, median of medians 1 |
| search, first 500 calls | 500 | 489 | 409 | 80 | 0.160 | [0.127, 0.199] | 9 | 0.018 | 24 | 5 | 3.27 | counting sort 139, quickselect 136, other 126, full sort 6, size k heap 2 |

*Note to Table 360.* Level a: 67/500 against 80/500, rate ratio 0.84, one sided lower 95% bound 0.63, p 0.876; level b: 23/500 against 9/500, rate ratio 2.56, one sided lower 95% bound 1.28, p 0.0100.

**Table 361.** Matched effort: idea clusters per arm, extrapolated to search's number of calls.

| Task | Search calls | Search clusters observed | Search programs passing the screen per island | Search rate, island bootstrap median [95%] | No evolution: clusters (Chao-1 extrapolated to the search's effort) | Instruction only | Enumerate and pick | Base intent | Neutral twin |
|---|---|---|---|---|---|---|---|---|---|
| shortest path | 1500 | 11 | island 0: 17/384, island 1: 5/372, island 2: 6/372, island 3: 21/372 | 0.033 [0.015, 0.050] | 1 (1.0) | 3 (4.0) | 2 (2.0) | 4 (5.3) | 0 (0.0) |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| kth smallest | 2500 | 63 | island 0: 150/636, island 1: 102/624, island 2: 126/624, island 3: 123/616 | 0.200 [0.173, 0.227] | 13 (32.0) | 11 (13.2) | 10 (16.1) | 8 (11.9) | 1 (1.0) |
| trapped rain water | 1000 | 7 | island 0: 10/252, island 1: 6/252, island 2: 11/252, island 3: 15/244 | 0.042 [0.029, 0.056] | 4 (5.0) | 7 (12.5) | 6 (7.9) | 3 (3.9) | 1 (1.0) |

**Table 362.** Lineage of the search programs that pass the screen.

| Task | Search programs passing the screen, by depth (generations) | Level b search programs by depth | Level b search programs | Level b at depth 2 or more with parents other than the seeds | Passing the screen at depth 2 or more with parents other than the seeds |
|---|---|---|---|---|---|
| shortest path | 1: 6, 2: 20, 3: 12, 4: 14, 5: 11, 6: 11, 7: 13, 8: 7, 9: 2 | none | 0 | 0 | 15 |
| kth smallest | 1: 114, 2: 61, 3: 60, 4: 53, 5: 79, 6: 72, 7: 103, 8: 60, 9: 65, 10: 42, 11: 52, 12: 36, 13: 13, 14: 10 | 2: 5, 3: 2, 4: 14, 5: 12, 6: 18, 7: 11, 8: 12, 9: 11, 10: 1, 11: 6, 13: 1 | 93 | 66 | 442 |
| trapped rain water | 1: 54, 2: 6, 3: 1, 4: 1 | none | 0 | 0 | 1 |

*[Figure fig_search_rates: generated in the report phase, not in this draft]*

## 7.8 Gemma: temperature, a controlled DAS and a validated circuit at the fork

### 7.8.1 Engine and calibration

The Gemma experiment returned to the fork of gemma-3-4b-it on the grid task, where a shared 32-token opening ends at a token that decides between DFS and BFS (Section 6.5). The engine check on 50 twin slots gave 19 BFS programs (0.38), just outside iteration 6's interval for 61 of 200 (0.245 to 0.372), so the check fails as preregistered; iteration 6's own share on the same 50 slots is 0.36, so the slots, not the engine, lean toward BFS. The discrepancy from the empty prefix of iteration 6 did not reproduce: 200 continuations from the empty prefix give 65 BFS on the path over several prompts and 72 on the path for a single prompt, both consistent with the natural 61 (Fisher p 0.75 and 0.29). The two paths disagree on 9 of 200 slots (1 against 8, p 0.039), so the batch shape changes a few coupled programs; by the rule committed beforehand, the probability of BFS given each fork token comes from a calibration run on the main engine (Table 363). The fork token fixes both the algorithm and its implementation: `visited` leads to DFS (BFS 1 of 125), and `queue` leads to a BFS that carries whole paths and pops from the front of a list, which is BFS but too

slow for the 30 by 30 hidden tests (0 of 125 pass).

**Table 363.** Calibration at the fork: the shared opening prefilled and the fork token forced (125 programs each).

| Fork token | BFS | Pass |
|---|---|---|
| `visited` | 1/125 | 124/125 |
| blank line | 111/125 | 85/125 |
| `if` | 122/125 | 99/125 |
| `queue` | 125/125 | **0/125** |

### 7.8.2 Temperature through the fork in a second family

The temperature claim holds in Gemma (Table 364). The predicted attempted BFS share, computed from the teacher-forced fork logits divided by T and committed before any temperature sample, matches the observed share in every top-p 1 cell: 0.015 against 0.02 at T 0.3, 0.166 against 0.24 at 0.7, 0.323 against 0.34 at 1.0 and 0.450 against 0.46 at 1.3 (mean absolute error 0.026; every prediction inside its Wilson interval). The entropy of the algorithm given the fork class is 0.153 bits. Lowering T from 1.0 to 0.3 switches 16 slots from BFS to DFS and none back (p 1.5e-5), and half of all slots change their algorithm somewhere across temperature. Top-k 10 cells are identical to top-p 1 program for program. A placebo that shuffles the temperatures of the sweep gives an error of 0.123 and fails the 0.10 bar.

**Table 364.** Gemma: predicted against observed BFS share across temperature (same seed, N = 50 per cell).

| Cell | Predicted attempted BFS | Observed (N = 50) | Wilson 95% | P(shared opening) predicted / observed | Correct BFS | Pass |
|---|---|---|---|---|---|---|
| T 0.3, top-p 1 | 0.015 | 1/50 = 0.02 | 0.00-0.10 | 1.00 / 1.00 | 1 | 1.00 |
| T 0.7, top-p 1 | 0.166 | 12/50 = 0.24 | 0.14-0.37 | 0.96 / 0.94 | 8 | 0.92 |
| T 1.0, top-p 1 | 0.323 | 17/50 = 0.34 | 0.22-0.48 | 0.88 / 0.86 | 9 | 0.80 |
| T 1.3, top-p 1 | 0.450 | 23/50 = 0.46 | 0.33-0.60 | 0.78 / 0.72 | 15 | 0.80 |

### 7.8.3 DAS at the fork with the controls iteration 6 lacked

DAS at the fork was fitted on held-out templates and held-out forced sources (768 pairs). The layer was chosen by the committed rule, the argmax of rank-1 recovery under cross-validation, at block 17. Swapping the whole fork state moves the held-out margin by 3.6 nats at block 17, rising to 8.2 to 9.1 nats at blocks 25 to 29; DAS at block 17 recovers 1.98, 2.36, 2.58 and 2.81 of that at ranks 1, 2, 4 and 8 (Tables 364b and 364c), because a trained direction moves the margin further than the weak full state swap at an early block. These controls hold (iteration-8 correction: within their limits below). Fifty random subspaces recover nothing (0.000). Permuting the source labels gives recovery between -0.27 and -0.30 against the true targets (one permutation of three reaches 0.17 at rank 1). A randomly initialised Gemma moves its margin at most 0.25 times as far (interchange intervention accuracy 0.50 to 0.72 against 1.00). The DAS claim's Gemma version is therefore supported at every rank by the random-subspace, permuted-label and untrained-network controls; the dormancy checks (reverse interchange, projected-map retention) were not run, a recovery of 1.98 to 2.81 times the full-state swap is the off-manifold signature Makelov et al. [122] warn about,

and the Qwen3-8B and Gemma controls used different objectives, readouts and permutation schemes, so their outcomes are not directly comparable (Table 364d) (iteration-8 correction).

Two limits remain. The committed wording rule does not license calling the subspace one dimensional: rank-1 recovery passes, but the subspace coordinate tracks the natural fork log odds of 120 twin variants at Spearman -0.50, just short of the 0.5 the rule required, and the subspace holds less of the twin's natural variance across prompts than the random 95th percentile at ranks 1, 2 and 4 (0.00093 against 0.00098, 0.00116 against 0.00185, 0.00187 against 0.00275) and more only at rank 8 (0.00565 against 0.00486), so the dial's rank-4 subspace lies below it; at block 22 the rank-1 share lies above it (0.00294 against 0.00078). The committed dormant flag (exp38 `src/g2.py` line 618: rank-1 share at or below the random 95th percentile AND $|\text{rho}| < 0.2$) was false only because $|\text{rho}| = 0.497$, the coordinate tracking the natural fork log odds, not because of variance (iteration-8 correction). And the subspace is not unique: across three seeds only one direction is shared (cosine 0.50 to 0.52), and a difference-in-means direction sits at cosine 0.39 to it.

**Table 364b.** Gemma: DAS at the fork by rank, with its controls (held-out templates and held-out forced sources, 768 pairs; block 17, and block 22 for comparison). (iteration-8 correction: the natural variance share lies below the random 95th percentile at ranks 1, 2 and 4 and above it at rank 8; at block 22, rank 1, it lies above (0.00294 against 0.00078); dormancy checks were not run; the controls are not comparable with those of Qwen3-8B, Table 364d) Recovery is the edit's change in the fork margin divided by that of the full state swap at the same block; the directional effect is the change in the fork margin toward the source's class.

| Rank k | Recovery, median of 3 seeds (each seed) | Directional effect (nats) | Interchange intervention accuracy | 50 random subspaces, recovery | Permuted labels, recovery (median of 3) | Randomly initialised Gemma, directional effect (nats) | Natural variance share (random 95th percentile) | Block 22: recovery / directional effect (nats) | Block 22: random recovery |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1.98 (2.04, 1.98, 1.98) | 5.57 | 1.00 | 0.000 | -0.27 | 0.00 | 0.00093 (0.00098) | 1.08 / 6.03 | 0.000 |
| 2 | 2.36 (2.36, 2.33, 2.36) | 6.68 | 1.00 | 0.000 | -0.29 | 1.69 | 0.00116 (0.00185) | 1.32 / 7.39 | 0.000 |
| 4 | 2.58 (2.58, 2.54, 2.58) | 7.57 | 1.00 | 0.000 | -0.30 | 1.69 | 0.00187 (0.00275) | 1.51 / 8.58 | 0.000 |
| 8 | 2.81 (2.81, 2.80, 2.88) | 8.55 | 1.00 | 0.0001 | -0.28 | 1.74 | 0.00565 (0.00486) | 1.71 / 9.76 | 0.0028 |

The subspace coordinate at rank 1 tracks the natural fork log odds of the 120 twin variants at Spearman -0.50 (permutation p 0.0005) at block 17 and -0.67 at block 22; the difference-in-means coordinate tracks it at 0.43. A rank-1 difference-in-means interchange recovers 1.01 at block 17 (directional effect 2.84 nats) and 1.06 at block 22.

**Table 364c.** Gemma: the full state ceiling and the rank-1 layer scan by block (held-out pairs). The committed rule chose the block with the largest rank-1 recovery under cross-validation; ties go to the earlier block.

| Block | Full state swap, forced BFS sources (nats) | Full state swap, forced DFS sources (nats) | Share of pairs moved by at least 1 nat | Rank-1 recovery under cross-validation, median (three folds) |
|---|---|---|---|---|
| 17 | +3.62 | -1.94 | 0.97 | 2.15 (2.15, 2.26, 2.05) |
| 18 | +3.80 | -1.99 | 0.98 | not scanned |
| 19 | +4.34 | -1.89 | 0.98 | 2.07 (2.07, 2.10, 1.94) |

| | | | | |
|---|---|---|---|---|
| 20 | +5.67 | -2.28 | 1.00 | not scanned |
| 21 | +5.36 | -2.31 | 1.00 | 1.67 (1.67, 1.80, 1.60) |
| 22 | +7.36 | -3.77 | 1.00 | 1.17 (1.17, 1.22, 1.17) |
| 23 | +6.87 | -4.16 | 1.00 | 1.20 (1.20, 1.22, 1.17) |
| 24 | +7.62 | -5.18 | 1.00 | not scanned |
| 25 | +8.19 | -5.23 | 1.00 | 1.07 (1.07, 1.07, 1.07) |
| 26 | +9.11 | -5.19 | 1.00 | not scanned |
| 27 | +8.99 | -5.65 | 1.00 | 1.09 (1.09, 1.07, 1.09) |
| 28 | +8.90 | -5.72 | 1.00 | not scanned |
| 29 | +8.81 | -5.80 | 1.00 | 1.10 (1.15, 1.10, 1.09) |

**Table 364d.** The controls of the two families were run under different protocols, so their outcomes are not directly comparable (iteration-8 correction: new table; `tables/control_protocols.csv`).

| Family | Objective | Site and readout | Permutation scheme | Untrained network | Pairs |
|---|---|---|---|---|---|
| Qwen3-8B (Section 7.3) | whole-program likelihood LME_Y of the source's programs (amended; the committed contrastive objective diverged) | task-specification states at blocks 12-23 (one map over the band); held-out generation of whole programs and teacher-forced LME_Y | labels permuted over the two kept algorithms per task (44% of training labels changed; 56% on kth smallest) | random-init Qwen3-8B, shortest path only; committed comparison against the diverged fit (0.013), replaced post hoc by the matched LME_Y fit (5.08) | all source x base pairs (bans W1-W3, forced families 0-2 -> neutral and padding twins); held-out W5 ban and forced families 3-4, N = 30 slots |
| gemma-3-4b-it (Section 7.8) | single-token fork margin (logsumexp over the BFS tokens minus the ' visited' logit) | block 17 at the fork position (response index 32, opening prefilled); teacher-forced margin, then generation | BFS/DFS source labels swapped (3 permutations) | randomly initialised Gemma on the same fork-margin readout | 768 held-out pairs (held-out templates and forced sources) |

### 7.8.4 A dose dial that keeps the programs

In generation the DAS edit at block 17, applied once at the fork position with the opening prefilled, acts as a dose dial (Table 365). From the twin's 11 correct BFS programs of 50, the edit raises correct BFS to 20, 27, 31, 37, 38 and 42 as lambda rises from 0.1 to 1, at pass rates of 0.68 to 0.88; the explicit instruction to use BFS gives 35, but it was sampled from index 0 without the prefilled opening and is not comparable: the DAS, difference-in-means and twin cells run with the canonical 32-token opening prefilled (exp38 `src/g2.py` line 877, `run.run_cell(ids, cs, rows_w, raw, ...)`), while the forced-BFS ceiling is sampled from index 0 with no prefix (lines 878-885, `run.run_cell(pfz["ids"], [], rows, raw, "G2:forced")`) on the same CRN slots (iteration-8 correction). A difference-in-means direction scaled to the same teacher-forced effect flips the algorithm just as well (up to 50 of 50 attempts) and destroys the programs (pass 0.70 down to 0.00, never significantly more correct BFS than the twin). A version of the DAS edit restricted to the logits, which changes the fork token's distribution but not the state that persists in the cache, gives 35 correct programs: the edited fork state helps the later code. The teacher-forced prediction of the attempted share (the fork distribution times the probability of BFS given each token) matches the

observed share within 0.10 in all 26 cells (mean gap 0.029). Toward DFS, the edit gives 41 to 50 correct DFS programs of 50 against the twin's 32 (Table 365b). The strong difference-in-means edit toward DFS, left unmatched by a bisection bug, keeps most programs (45 correct of 49 attempted, pass 0.90), so the difference-in-means edit breaks the programs when it pushes toward BFS, not toward DFS. In iteration 6, a rank-4 interchange at block 22 had produced 50 BFS attempts and 0 valid programs; a dosed edit to the coordinate of the class mean at block 17 keeps them.

*[Figure fig_gemma_dose: generated in the report phase, not in this draft]*

**Table 365.** Gemma: DAS against difference in means as a dose dial toward BFS (N = 50 per cell, same seed). (iteration-8 correction: every cell except the forced BFS prompt has the shared 32-token opening prefilled; the forced BFS prompt is sampled from index 0 without it, on the same slots, so its row is not comparable with the edits)

| Cell | Attempted BFS | Correct BFS | Pass | p (correct > twin, one sided McNemar, Holm over 6 doses) | Teacher-forced predicted attempted share |
|---|---|---|---|---|---|
| twin (no edit) | 18 | 11 | 0.86 | - | 0.26 |
| DAS lambda 0.1 | 27 | 20 | 0.86 | 0.006 | 0.47 |
| DAS lambda 0.2 | 38 | 27 | 0.78 | 4e-4 | 0.75 |
| DAS lambda 0.35 | 46 | 31 | 0.68 | 5e-5 | 0.92 |
| DAS lambda 0.5 | 47 | 37 | 0.78 | 2e-6 | 0.96 |
| DAS lambda 0.75 | 47 | 38 | 0.80 | 1e-6 | 0.96 |
| DAS lambda 1 | 47 | **42** | **0.88** | 2e-8 | 0.96 |
| difference in means 0.1 (matched teacher-forced effect) | 30 | 15 | 0.70 | 0.87 | 0.52 |
| difference in means 0.2 | 36 | 15 | 0.58 | 0.87 | 0.78 |
| difference in means 0.35 | 46 | 12 | 0.32 | 1.00 | 0.95 |
| difference in means 0.5 | 48 | 11 | 0.26 | 1.00 | 0.99 |
| difference in means 0.75 | 50 | 2 | 0.04 | 1.00 | 1.00 |
| difference in means 1 | 50 | 0 | 0.00 | 1.00 | 1.00 |
| random rank-4 subspace, lambda 1 | 17 | 12 | 0.90 | 0.50 (unadjusted) | 0.25 |
| forced BFS prompt, fifth forced source (sampled from index 0 without the prefilled opening; not comparable) | 50 | 35 | 0.70 | 4e-6 (unadjusted) | - |
| DAS on the logits only, lambda 1 | 44 | 35 | 0.76 | 2e-6 (unadjusted) | - |
| DAS on the logits only, lambda 0.5 | 46 | 35 | 0.76 | 2e-6 (unadjusted) | - |

**Table 365b.** Gemma: the same edits toward DFS (N = 50 per cell, same seed; the twin writes DFS in 32 of 50). The six difference-in-means cells toward DFS were left unmatched by a bisection bug and are identical; they are reported once, as a strong reference edit.

| Cell | Attempted DFS | Correct DFS | Pass | McNemar on correct DFS against the twin (cell only : twin only) |
|---|---|---|---|---|
| twin (no edit) | 32 | 32 | 0.86 | - |
| DAS lambda 0.1 | 41 | 41 | 0.94 | 9:0 |
| DAS lambda 0.2 | 46 | 46 | 0.98 | 14:0 |
| DAS lambda 0.35 | 50 | 50 | 1.00 | 18:0 |
| DAS lambda 0.5 | 50 | 50 | 1.00 | 18:0 |
| DAS lambda 0.75 | 50 | 50 | 1.00 | 18:0 |

| | | | | |
|---|---|---|---|---|
| DAS lambda 1 | 50 | 50 | 1.00 | 18:0 |
| difference in means toward DFS (one strong edit) | 49 | 45 | 0.90 | 14:1 |

### 7.8.5 A circuit for the fork margin

*Path patching* replaces a component's contribution along one path with its value under a contrasting input and measures the effect on the output [134, 135]; *attribution patching* (AtP) approximates the same effect for every node with one gradient pass [136]. The experiment screened 1,224 nodes (272 attention heads and 34 MLPs at four position classes: the fork position, the shared opening, the task specification and the condition sentence) by AtP on 15 contrasting pairs, patched the top 60 exactly by *noising*, which writes a node's value from the run of a pair that leans toward BFS into the run that leans toward DFS (AtP against exact noising: Pearson 0.65, Spearman 0.73; the reverse, denoising, has the opposite sign for every node), and edges measured by path patching among the top 15. Half of the exact noising effect enters at the fork position (53%) and a third at the condition sentence (36%). The top ten nodes are MLPs at the condition sentence (layers 0, 2, 6 and 14) and a stack at the fork position (MLPs at layers 17, 21, 22, 25 and 26, and head 5 of layer 22; Table 366). Kept edges, those whose effect under path patching is at least 5% of the total effect, form a chain at the fork position (MLP 17 to MLP 19, MLP 21 to MLP 22, MLP 25 to MLP 26, and head 5 of layer 22 to MLP 25), with direct paths to the margin from head 2 of layer 33 and MLPs 22, 25 and 26.

Each top node was validated in the real model against 50 random components of its position class matched on write norm (Table 367). Seven of ten nodes pass when ablated to the opposite class's mean, and eight pass when steered; the joint sets of the top five and the top ten beat all 50 matched random sets. In generation, steering the top five toward BFS doubles correct BFS programs (16 of 30 against 8 for the twin, p 0.019) at pass 0.83, while a matched random five does nothing (Table 368). Ablating the same nodes to the natural BFS mean does not move generation, because the natural classes differ by only about 2 nats at the fork. The reading is a circuit hypothesis for this one fork: MLPs at the condition sentence carry the prompt's lexical cue, and the stack at the fork position turns it into the token at index 32. An *attribution graph*, the graph of direct linear effects between transcoder features from the prompt to the fork margin, built on the Gemma Scope 2 transcoders [113], would test the circuit on features; it was attempted three times and dropped: the container lacked a C compiler for Triton, and with one installed, attribution exceeded the 16 GB GPU at batch 64 and again at batch 16 (16.4 GB). No correct program among 2,325 was outside the catalogue (rule-of-three bound 0.0013).

The deviation log of the Gemma experiment records five departures that bear on these results. One unit test missed its tolerance: the cache path for a single position differs from a full recompute by up to 0.27 nats on the fork margin, an effect of bf16 kernels and batch shapes, so every DAS effect is measured against the base's own margin on the same path and in the same batch shape. The DAS optimiser ran at a learning rate of 2e-2 instead of the planned 2e-3, because a toy check showed that 400 steps at 2e-3 leave the subspace near random. A bisection bug left the six difference-in-means cells toward DFS unmatched, so they are reported as one strong reference edit (45 of 50 correct DFS). The mediation test of the fork subspace on random pushes was not run (dropped in order), so the DAS claim covers forced prompt transfer only. And the transcoder graph was dropped after the three attempts described above.

**Table 366.** Gemma: the top ten nodes by exact noising effect (mean over 15 pairs).

| Node | Noising effect (nats) |
|---|---|

| | |
|---|---|
| MLP of layer 0 at the condition sentence | +4.43 |
| MLP of layer 6 at the condition sentence | +2.01 |
| MLP of layer 25 at the fork position | +1.96 |
| MLP of layer 21 at the fork position | +1.54 |
| MLP of layer 2 at the condition sentence | +1.00 |
| attention head 5 of layer 22 at the fork position | +1.00 |
| MLP of layer 26 at the fork position | +0.97 |
| MLP of layer 14 at the condition sentence | +0.93 |
| MLP of layer 22 at the fork position | +0.86 |
| MLP of layer 17 at the fork position | +0.84 |

**Table 367.** Gemma: validation of each node against 50 matched random components. The effect d of a node is its oriented change in the fork margin minus the mean change under the 50 matched random components.

| Node | ablate median d (Holm p) | steer median d (Holm p) | Claim (ablate / steer) |
|---|---|---|---|
| MLP of layer 0 at the condition sentence | +0.28 (< 1e-4) | +0.49 (4e-4) | yes / yes |
| MLP of layer 6 at the condition sentence | +0.16 (0.09) | +0.25 (0.25) | no / no |
| MLP of layer 25 at the fork position | +0.28 (3e-4) | +0.62 (< 1e-4) | yes / yes |
| MLP of layer 21 at the fork position | +0.23 (5e-4) | +0.48 (< 1e-4) | yes / yes |
| MLP of layer 2 at the condition sentence | +0.07 (0.15) | +0.10 (0.13) | no / no |
| attention head 5 of layer 22 at the fork position | +0.19 (0.002) | +0.39 (< 1e-4) | yes / yes |
| MLP of layer 26 at the fork position | +0.15 (5e-4) | +0.61 (< 1e-4) | yes / yes |
| MLP of layer 14 at the condition sentence | +0.17 (5e-4) | +0.39 (0.001) | yes / yes |
| MLP of layer 22 at the fork position | +0.34 (1e-4) | +0.68 (< 1e-4) | yes / yes |
| MLP of layer 17 at the fork position | +0.05 (0.10) | +0.17 (< 1e-4) | no / yes |

**Table 368.** Gemma: generation arms with the validated nodes (N = 30, opening prefilled).

| Arm | Attempted target | Correct target | Pass | Test |
|---|---|---|---|---|
| twin | 9 BFS | 8 | 0.97 | - |
| top-5 steer toward BFS | 20 BFS | **16** | 0.83 | vs twin: correct 10 vs 2, p = 0.019; attempted 12 vs 1, p = 0.002. vs random-5 steer: p = 0.006 (correct), 0.002 (attempted) |
| random-5 steer (matched) | 9 BFS | 7 | 0.93 | null |
| top-5 ablate to the BFS mean | 12 BFS | 7 | 0.83 | attempted 3 vs 0, p = 0.125 |
| random-5 ablate | 9 BFS | 6 | 0.90 | null |
| top-10 ablate to the BFS mean | 12 BFS | 10 | 0.93 | p = 0.125 (attempted) |
| top-5 ablate to the DFS mean | 22 DFS | 22 | 0.90 | vs twin 21 DFS: n.s. |

## 7.9 Comparisons across this iteration's experiments

**Temperature through the fork, in two families.** The temperature claim was tested on four mixed

units in two models. The fork logits predicted the temperature response within the committed error on two of them (the grid in Gemma, mean absolute error 0.026, and minimum spanning tree weight in Qwen3-8B, 0.083, where the observed drop was twice the predicted one), just outside it on trapped rain water (0.105), and not in direction on valid anagram, where the minority share did not fall at low temperature. Table 369 sets them side by side. On every unit the algorithm given the fork class was nearly certain (0.05 to 0.19 bits) and top-k 10 changed nothing. Temperature above 1 raised the minority only where the fork logits said it would: in Gemma the BFS share rose from 0.34 at T 1.0 to 0.46 at T 1.3 against a predicted rise from 0.323 to 0.450, while on the three Qwen3-8B units the predicted rise was at most 0.013 and the observed change lay between -0.02 and +0.06. On the collapsed tasks the fork margins of 4.85 to 17.04 nats explain why no temperature in iterations 1 to 4 changed the algorithm. Temperature is a dial on the fork's existing odds; it cannot create an alternative that the fork does not already weigh.

**Table 369.** The temperature claim in two model families (top-p 1, same seed).

| Model and unit | Fork token pair | Entropy of the algorithm given the fork class (bits) | Minority share at T 0.3 / 1.0 / 1.3 (observed) | Predicted | Mean absolute error | Drop T 1.0 to 0.3 (McNemar) |
|---|---|---|---|---|---|---|
| Qwen3-8B, minimum spanning tree weight | `Kr` (Kruskal) or `Prim` | 0.050 | 0.06 / 0.40 / 0.38 | 0.068 / 0.237 / 0.247 | 0.083 | 17:0 |
| Qwen3-8B, trapped rain water | `n` (prefix and suffix) or `left` (two pointers) | 0.153 | 0.10 / 0.18 / 0.20 | 0.166 / 0.311 / 0.319 | 0.105 | 7:3 |
| Qwen3-8B, valid anagram | `return` (sort) or `count` | 0.189 | 0.18 / 0.12 / 0.18 | 0.161 / 0.200 / 0.212 | 0.043 | 2:5 |
| gemma-3-4b-it, grid | `visited` (DFS) or a blank line, `if`, `queue` (BFS) | 0.153 | 0.02 / 0.34 / 0.46 | 0.015 / 0.323 / 0.450 | 0.026 | 16:0 |

**Where the choice is decided.** Three different answers came out of three tasks in one model. On connected components the decision falls at generated tokens 10 and 11, where the routine is named; forcing the name decides the algorithm in every slot. On kth smallest a ban decides it before the first generated token. On shortest path no single token decides it within 60 tokens, and the opening's likelihood predicts the outcome better the longer the opening measured. The three mixed units of the temperature experiment and Gemma's grid task each have a fork of a single token right after a shared opening of 18 to 61 tokens. The fork is thus a common place for the decision to surface, not a universal one, and where it exists, the effect of a push is already expressed at the fork token on connected components (a teacher-forced readout of the margin matches the sampled outcomes of random pushes at Spearman 0.80, and the 48-token opening log-likelihood ratio at 0.87; no direction-level property predicted which pushes move the fork), and its logits divided by T predicted the temperature response within the committed error on two of the four mixed units (iteration-8 correction).

**What moves the algorithm, and at what cost.** Table 370 collects the edits of this iteration that move the algorithm in generation, with their correct target programs and pass rates. In Qwen3-8B a

rank-1 or rank-2 DAS edit matches the full state exchange at the same site for most sources (it recovers 0.06 to 0.53 of forced BFS and 0.17 to 0.43 of forced Bellman-Ford/SPFA; Table 297b; iteration-8 correction), plan noise at the absolute unit moves half of connected components programs at no pass cost, and the axis from the default to the second choice moves connected components only at a dose beyond the committed range. In Gemma the DAS edit at the fork, with the shared opening prefilled, raises correct BFS programs from 11 to 42 of 50 (the explicit instruction's 35 was sampled without the prefill and is not comparable), and steering five validated circuit nodes doubles them (iteration-8 correction). In Gemma the difference-in-means edit toward BFS moves the algorithm and breaks the programs (toward the default DFS a strong edit kept them), and in Qwen3-8B a rank-1 difference-in-means interchange at the DAS site transfers nothing (0 of 30 on both core tasks). In this iteration the edits fitted or validated on the model's own contrasts kept the programs; the difference-in-means edit toward the minority algorithm did not.

**Table 370.** Edits that moved the algorithm in generation in iteration 7. (iteration-8 correction: prefill condition of the two Gemma rows stated; kth smallest counting sort row annotated)

| Model and task | Edit | Correct target programs | Twin | Pass |
|---|---|---|---|---|
| Qwen3-8B, shortest path | DAS rank 2, fifth template ban source, blocks 12-23 | 12/30 Bellman-Ford/SPFA | 1/30 | 0.83 |
| Qwen3-8B, shortest path | DAS rank 1, forced Floyd-Warshall source | 28/30 Floyd-Warshall | 0/30 | 0.93 |
| Qwen3-8B, connected components | DAS rank 16, fifth template ban source | 29/30 BFS or DFS | 0/30 | 1.00 |
| Qwen3-8B, connected components | Plan noise, absolute unit, alpha 2 | 25/50 | 0/50 | 0.98 |
| Qwen3-8B, connected components | Axis push at four times the forced prompt distance | 30/30 (29 DFS, 1 BFS) | 0/30 | 1.00 |
| Qwen3-8B, connected components | Tokens 10-11 forced to `def dfs` | 30/30 DFS | 0/30 | 1.00 |
| Qwen3-8B, kth smallest | DAS rank 1, forced counting sort source | 30/30 counting sort (28 an array of counts; the full state exchange gives 29/30 in dictionary form, iteration-8 correction) | 0/30 | 1.00 |
| Qwen3-8B, kth smallest | Plan noise, absolute unit, alpha 2 (pricing cell) | 30/50 | 1/50 | 0.98 |
| Qwen3-8B, three mixed units | Exchange of the forced minority prompt's task specification states | 30/30 on each | 0 to 7/30 | 1.00 |
| gemma-3-4b-it, grid | DAS rank 4 at block 17, lambda 1 (opening prefilled) | 42/50 BFS | 11/50 | 0.88 |
| gemma-3-4b-it, grid | Explicit BFS instruction (no prefill, not comparable) | 35/50 BFS | 11/50 | 0.70 |
| gemma-3-4b-it, grid | Difference in means, lambda 1 | 0/50 BFS | 11/50 | 0.00 |
| gemma-3-4b-it, grid | Steering the five validated nodes toward BFS | 16/30 BFS | 8/30 | 0.83 |

**Novelty.** Table 371 collects the novelty audits of this iteration. Across the seven artifacts, the only

correct programs outside the catalogue are the known techniques reached on kth smallest by evolutionary search, the larger mutator and the instruction alone, and none is novel.

**Table 371.** Novelty audits of iteration 7. (iteration-8 correction: the DAS experiment row names the counting-sort cells)

| Artifact | Programs audited | Final classes | Novel |
|---|---|---|---|
| Record audit | 3 programs of the engine control labelled other | catalogue variants (Dijkstra by array scan, Bellman-Ford of n rounds) | 0 |
| DAS experiment | 38 correct programs labelled other | 36 catalogue variants (35 are counting sort in dictionary form, 29 of them from the full state exchange of the forced counting sort source; relabelled counting sort in iteration 8), 2 labeller misses | 0 |
| Fork experiment | 2 | 2 labeller misses | 0 |
| Temperature experiment | 2 distinct programs (40 continuations) | catalogue variant (the extra counting mode of valid anagram) | 0 |
| Dose unit experiment | 5 kth smallest programs | catalogue variant (counting sort with key compression) | 0 |
| Search experiment | 816 distinct programs passing the screen | 707 catalogue variants, 103 known techniques outside the catalogue, 6 degenerate | 0 |
| Gemma experiment | 2,325 correct programs | none outside the catalogue | 0 |

**Table 371b.** Positioning of the iteration-7 claims (to our knowledge, mini search dated 2026-10-22, re-verified 2026-10-23 with fetch_grep of each abstract and venue page; raw outputs in `audit/web/`). Verdicts come from the record audit's positioning search (art_LzP2qPwUrANz `positioning.json`). (iteration-8 correction: new table)

| Claim | Nearest neighbours (venue) | What they do | What the run adds | Verdict | Evidence (URL, version, quote, status, date) |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| Temperature acts on the algorithm through the fork logits divided by T | Hot or Cold? Adaptive Temperature Sampling for Code Generation (AAAI 2024); Control the Temperature (COLM 2025); Entropy-informed Decoding (ICML 2026 (arXiv 2605.09745 comment)); Beyond the 80/20 Rule (NeurIPS 2025); A Tale of Two Temperatures (COLM (arXiv v2 comment, no year); ICML 2026 workshop posters) | Hot or Cold? Adaptive Temperature Sampling for Code Generation: temperature set by token difficulty in code generation, no algorithm-level prediction; Control the Temperature: switches temperature per token to protect precision; Entropy-informed Decoding: branches decoding at high-entropy tokens, no algorithm-level outcome; Beyond the 80/20 Rule: forking (high-entropy) tokens drive RL for reasoning, not temperature at a code fork; A Tale of Two Temperatures: tempered remasking for sample diversity in diffusion language models, not autoregressive code | a per-algorithm prediction of the temperature response from teacher-forced fork logits, committed before sampling, and fork margins that explain the nulls on the collapsed tasks | adjacent: no neighbour predicts an algorithm-level temperature response | https://arxiv.org/abs/2309.02772 (v1 6 Sep 2023, v3 28 Dec 2023; 'Hot or Cold? Adaptive Temperature Sampling for Code Generation'; 'accepted by AAAI 2024'; VERIFIED, 2026-10-23); https://arxiv.org/abs/2510.01218 (v1 20 Sep 2025; 'Control the Temperature'; 'Second Conference on Language Modeling, 2025'; VERIFIED, 2026-10-23); https://arxiv.org/abs/2605.09745 (v1 10 May 2026; 'Entropy-informed Decoding'; 'Accepted at ICML 2026'; VERIFIED, 2026-10-23); https://arxiv.org/abs/2506.01939 (v1 2 Jun 2025, v2 13 Nov 2025; 'Beyond the 80/20 Rule'; 'Accepted to NeurIPS 2025'; VERIFIED, 2026-10-23); https://arxiv.org/abs/2604.09921 (v1 10 Apr 2026, v2 10 Aug 2026; 'A Tale of Two Temperatures'; 'V2: accepted as a full conference paper at COLM'; VERIFIED, 2026-10-23); https://icml.cc/virtual/2026/75333 (AdaptFM: Resource-Adaptive Foundation Model Inference; VERIFIED, 2026-10-23); https://icml.cc/virtual/2026/74361 (4th Structured Probabilistic Inference & Generative Modeling; VERIFIED, 2026-10-23) |

| | | | | | |
|---|---|---|---|---|---|
| A DAS subspace at Gemma's fork is a dose dial on the algorithm in code | CDAS (ICLR 2026); MIB (ICML 2025) | CDAS: DAS-style interchange steering validated by generation on refusal and a backdoor; MIB: benchmark that ranks DAS first for causal variable localisation | an algorithm choice in code at a generated fork, with a dose curve that preserves the pass rate (with the shared opening prefilled; the instruction cell is not comparable) | adjacent: DAS steering is published; the algorithm-in-code object and the dose curve are not | https://arxiv.org/abs/2602.05234 (v1 5 Feb 2026, v2 15 Mar 2026; 'Faithful Bi-Directional Model Steering'; 'accepted for ICLR 2026'; VERIFIED, 2026-10-23); https://arxiv.org/abs/2504.13151 (v1 17 Apr 2025, v2 9 Jun 2025; 'MIB: A Mechanistic Interpretability Benchmark'; 'Accepted to ICML 2025'; VERIFIED, 2026-10-23) |
| Evaluator-guided search reaches known techniques outside the catalogue at a rate single samples do not | FunSearch (Nature 2023); AlphaEvolve (2025 (arXiv)); DivAlgo (arXiv 2610.19157); Beneath the Diff (EMNLP 2026); Algorithm Discovery With LLMs (arXiv 2504.05108) | FunSearch: evaluator-guided program search finds new mathematical constructions; AlphaEvolve: evaluator-guided evolutionary coding agent (occupies the method family); DivAlgo: benchmarks algorithmic diversity of code models; Beneath the Diff: algorithmic mode collapse in autonomous research loops; Algorithm Discovery With LLMs: evolutionary search plus RL for algorithm discovery | a rate-matched comparison against single-sample knobs on a frozen LeetCode-style catalogue (78 of 2,500 calls against 2 of 500; 0 novel) | adjacent: the method is occupied (FunSearch, AlphaEvolve); the rate-matched comparison is the addition | https://www.nature.com/articles/s41586-023-06924-6 (version n/a; 'Mathematical discoveries from program search with large language models'; 'Published: 14 December 2023'; VERIFIED, 2026-10-23); https://arxiv.org/abs/2506.13131 (v1 16 Jun 2025; 'AlphaEvolve: A coding agent'; VERIFIED, 2026-10-23); https://arxiv.org/abs/2610.19157 (v1 13 Oct 2026; 'DivAlgo'; VERIFIED, 2026-10-23); https://arxiv.org/abs/2609.00077 (v1 31 Aug 2026; 'Beneath the Diff'; 'Accepted by EMNLP 2026'; VERIFIED, 2026-10-23); https://arxiv.org/abs/2504.05108 (v1 7 Apr 2025, v4 4 Aug 2025; 'Algorithm Discovery With LLMs'; VERIFIED, 2026-10-23) |

## 7.10 What iteration 7 established, what it ruled out, and what it abandoned

**Table 372.** Verdicts of the ten claims of iteration 7.

| Claim | Committed verdict | Deciding evidence |
|---|---|---|

| | | |
|---|---|---|
| Fork claim | Not supported (inconclusive) | Spearman 0.80 on connected components; 0.15 on shortest path, which has no single fork; prospective 0.60 on kth smallest |
| DAS claim (Qwen3-8B) | Not supported on any task | By source (Table 297b): matches or exceeds the full-state exchange for ban sources and for the forced Floyd-Warshall, quickselect, counting sort and DFS sources; recovers 0.17 to 0.43 of forced Bellman-Ford/ SPFA and 0.06 to 0.53 of forced BFS; rank 2 untested for forced sources; random subspaces null; permuted labels transfer 0.47 to 0.70 of the 30 slots; the untrained network (shortest path only) gains 43.6 nats against 8.6 under the matched likelihood fit (ratio 5.1; the committed ratio 0.013 was against the diverged fit) (iteration-8 correction) |
| Routing claim | Categorical by the committed rule on two tasks (shortest path, kth smallest), not on connected components (iteration-8 correction) | Floyd-Warshall 28/28 under DAS against 8/20 for the full state; on kth smallest the full state also lands 29 of 30 on counting sort (dictionary form) against 30/30 under DAS (28 of them an array of counts); forced BFS lands on DFS |
| Axis claim | Disconfirmed | Spearman 0.17 and 0.01; clamp removes nothing |
| Temperature claim | Partial | Test 1 passes on 3 of 3 units, test 2 on 1 of 3, test 3 on 4 of 4 tasks; holds in Gemma |
| Manifold claim | Bans outside the null (resting on the held-out templates, fourth D 1.73 and fifth D 2.62; the first three are training templates), seeds and instructions inside; semantic seed on the boundary (93.9th percentile) (iteration-8 correction) | D 1.73 to 2.62 against a null 95th percentile of 1.13 |
| Discreteness claim | Not discrete | Logistic width 3.25 in lambda; 0 hybrids of 114 |
| Dose unit claim | The unit, not the seeds | Unit +0.45 [0.35, 0.55]; seeds +0.03 |
| Repriced plan noise | Ahead of base intent on 2 of 5 tasks (predicted 3) | Kth smallest 3.22 and grid 1.39; behind on connected components, shortest path and two sum |
| Gemma claims | Temperature law holds; DAS passes the random-subspace, permuted-label and untrained-network controls (dormancy checks not run; natural variance below the random 95th percentile at ranks 1, 2 and 4; controls not comparable with Qwen3-8B's), calling it one dimensional not licensed; 7 to 8 of 10 circuit nodes validated (iteration-8 correction) | Error 0.026; random initialisation at most 0.25 times; with the opening prefilled the dial raises correct BFS from 11 to 42 of 50 (the instruction's 35 was sampled without the prefill, not comparable) |

| | | |
|---|---|---|
| Search claim | Not met at levels a and b | 6.65 times on kth smallest at level a; at level b on kth smallest only, search 78 of 2,500 calls (0.0312 [0.0247, 0.0388], 12 merged clusters) against 2 of 500 calls (0.0040 [0.0005, 0.0144], 1 merged cluster) for the instruction alone (rate ratio 7.8, lower 95% bound 2.41, short of the committed 10x), 0 of 1,250 for enumerate and pick and 0 of 500 without evolution; the larger mutator reached 23 of 500 calls (0.0460 [0.0294, 0.0682], 7 merged clusters); 0 novel (iteration-8 correction) |

**Established.** In Qwen3-8B a rank-1 or rank-2 subspace of the task specification states at blocks 12 to 23 is a steering handle on the algorithm that matches or exceeds the full-state exchange for ban sources and for the forced Floyd-Warshall, quickselect, counting sort and DFS sources; recovers 0.17 to 0.43 of forced Bellman-Ford/SPFA and 0.06 to 0.53 of forced BFS; rank 2 untested for forced sources. It routes more cleanly than the full state on one task (shortest path: Floyd-Warshall 28/28 against 8/20); on kth smallest the DAS edit changes the implementation form (mostly an array of counts against the full state's dictionary with sorted keys), not the algorithm. The handle is not a certified variable, because permuted labels and an untrained network admit it (the untrained-network control ran on shortest path only; the committed comparison with the diverged contrastive fit (ratio 0.013) proved meaningless and was replaced post hoc; under the matched likelihood fit the untrained network gains 43.6 nats against 8.6 for the trained one (ratio 5.1), while the fit-value ratio is 0.92 (13.31 against 14.45 nats)) (iteration-8 correction). In Gemma the analogous subspace at the fork passes the random-subspace, permuted-label and untrained-network controls (dormancy checks not run; its natural variance lies below the random 95th percentile at ranks 1, 2 and 4, and it recovers 1.98 to 2.81 times the full-state swap) and, with the shared opening prefilled, acts as a dose dial that keeps the programs; the Qwen3-8B and Gemma controls used different objectives, readouts and permutation schemes, so their outcomes are not directly comparable (iteration-8 correction). Where a fork of a single token exists, temperature acts on the algorithm through the fork logits divided by T, as predicted before sampling on one Qwen3-8B unit and on Gemma; on the collapsed tasks the fork margins put the alternative out of temperature's reach. On connected components the decision falls at the token that names the routine, and the effect of a random push is already expressed at that token: a teacher-forced readout of the fork margin matches the sampled outcomes at Spearman 0.80 (the 48-token opening log-likelihood ratio at 0.87), through the probability that stays on the default fork tokens rather than through which algorithm each other token leads to (a placebo that shuffles the token map reaches 0.80 in 18% of permutations); no direction-level property predicted which pushes move the fork (axis 0.17 and 0.01; plan-subspace projection 0.03 in iteration 4) (iteration-8 correction). The weak plan noise of iteration 5 came from its dose unit; profile and energy each carry part of the effect on connected components (profile $+0.22$ [0.08, 0.36], energy $+0.18$ [0.06, 0.30]); the shortest-path profile contrast is confounded by a massive-activation cell holding 82% of the unit's energy (iteration-8 correction). Evolutionary search with an evaluator reaches known techniques outside the catalogue on kth smallest: search 78 of 2,500 calls (0.0312 [0.0247, 0.0388], 12 merged clusters) against 2 of 500 calls (0.0040 [0.0005, 0.0144], 1 merged cluster) for the instruction alone (rate ratio 7.8, lower 95% bound 2.41, short of the committed 10x), 0 of 1,250 for enumerate and pick and 0 of 500 without evolution; the larger mutator reached 23 of 500 calls (0.0460 [0.0294, 0.0682], 7 merged clusters); about five techniques in all (value-range binary search, Fenwick tree order statistics, a treap, bucket select, partial selection sort) (iteration-8 correction).

**Ruled out.** The axis from the default to the second choice as the channel of random pushes

(disconfirmed and not mediating). Discreteness of the effect of the fitted edit. The seeds as the cause of the null of iteration 5. Temperature above 1 as a source of a new idea: on every unit it only reweighted the algorithms the fork already weighs. Search at ten times the best text knob's level a rate on two tasks.

**Dead ends.** The committed contrastive DAS objective, which diverged and broke programs, was replaced by the likelihood objective. The transcoder attribution graph in Gemma was dropped after three attempts on a 16 GB GPU. The judge gate for novelty failed for every judge family in the search experiment, so its classes rest on manual and rule-based reads. A top five push inside the forced prompt manifold of the grid task produced no algorithm outside the five in 300 programs, which closes the push route for that question at the doses tried.

**Not run.** Plan resampling and the DAS-coordinate top-five placement on the grid task (the T1 extension) were dropped by the DAS experiment's committed drop order (deviation D-drop, Section 7.3.10); the untrained-network control ran on shortest path only (iteration-8 correction).

### 7.11 What the next iteration takes from this one

The next iteration takes four leads. First, the DAS handle in Qwen3-8B needs the control it failed: a fit whose versions with permuted labels and with an untrained network do not transfer, which may require fitting at the fork in generation, where Gemma's controls passed, rather than at the task specification states; because the Qwen3-8B and Gemma controls used different objectives, readouts and permutation schemes, their outcomes are not directly comparable, and a matched protocol is part of that test (iteration-8 correction). Second, shortest path decides late, so the fork claim needs a measure that follows the decision to where it happens (the opening LLR improved from 48 to 160 tokens). Third, the temperature claim held on two of four mixed units; the misses on trapped rain water, which has a second opening with its own split, and valid anagram, where a second decision (fixed array against hash map) follows a few tokens after the fork, call for a prediction that composes more than one fork. Fourth, evolutionary search found known techniques outside the catalogue on kth smallest at 0.031 per call against 0.004 for the instruction alone (ratio 7.8, lower bound 2.41, short of the committed 10x; iteration-8 correction), so a search on a task whose catalogue already covers the known techniques, where a correct program outside it would be new, is the remaining route to a novel algorithm. Two items not run in iteration 7 carry forward: plan resampling and the DAS-coordinate top-five placement on the grid task (Section 7.3.10) (iteration-8 correction).

# Iteration 8: temperature at the fork, one DAS protocol for two families, a circuit in Qwen3-8B, a fair test of Gemma's dial, a temperature law on held-out families, and a top-five prohibition across ten configurations

### 8.1 Why this iteration ran what it ran

**The review of the iteration-7 record.** The review was blocking. It raised eleven items, six of them marked as needing a fix before any new claim. The first said the kth smallest results rested on a labeller gap: the frozen runtime rule sent "count the values, then sort the distinct keys" to the class other, so 35 programs that implement counting sort in dictionary form were read as programs outside the catalogue, and the pooled DAS recovery on that task was inflated. The second asked for

the DAS recovery of iteration 7 by source and rank, because a pooled number hid that forced BFS and forced Bellman-Ford/SPFA sources recovered far less than the ban sources, and it asked what the three untrained-network ratios in the record measured. The third found that the Gemma dial had been compared with a forced-BFS prompt sampled without the 32-token shared opening that every dial cell had prefilled, so "42 against 35" compared two conditionings. The fourth asked that the fork margin be stated as a readout of a push's effect rather than as a predictor of it. The fifth asked for the natural-variance numbers and dormancy checks of the Gemma subspace, which the record had passed by a flag that fired on a correlation threshold alone. The sixth asked that the positioning of iteration 7 be verified live, venue by venue. The five minor items asked for a list of what iteration 7 did not run, fresh-slot rows for the dose unit table, search results stated as rates per call with merged clusters, a per-unit statement of the temperature claim's scope, and the repair of four stale tags and ten lost file names.

**Table 373.** Review items on the iteration-7 record and where this report answers them. Every correction is now in place in Sections 7.1 to 7.11 and in the closing, each marked as an iteration-8 correction.

| Review item | What the review found | Corrections | Where it is answered |
|---|---|---|---|
| Counting sort | 35 dictionary-form counting sort programs labelled other; pooled kth smallest recovery inflated | 17 | Sections 7.3 (Tables 296 to 301), 7.3.9, 7.10, the closing; Section 8.2 (Table 376) |
| Recovery by source | Pooled recovery hid weak forced sources; three untrained-network ratios undefined | 10 | New Table 297b in Section 7.3.4; Section 7.10; Table 377 |
| Gemma prefill | The dial cells prefilled the shared opening, the forced prompt did not | 7 | Sections 7.8 and 7.10 and the closing; the matched test of Section 8.6 |
| Fork margin as readout | Stated as a predictor of which pushes move | 4 | Sections 7.4, 7.10 and the closing |
| Gemma natural variance | Dormancy checks not run; flag passed on a correlation threshold | 10 | Sections 7.8 and 7.10; new Table 364d; Section 8.6 runs the checks |
| Positioning | Venues not verified live | 2 | New Table 371b in Section 7.9 |
| Not run | Dropped items not listed | 3 | New Section 7.3.10 |
| Dose unit rows | No fresh-slot rows in Table 346 | 4 | Table 346 |
| Search as rates | Counts without calls or clusters | 4 | Sections 7.7 and 7.10 and the closing; Table 379 |
| Scope per unit | Temperature claim stated across units; manifold claim on training templates | 8 | Sections 7.5, 7.9, 7.10 and the closing |
| Tags and file names | Four undefined tags, ten lost file names | 15 | In place |

**The hypothesis update.** Iteration 7 left one finding that every later test could build on: on the tasks where it can be located, the algorithm is fixed by one or two generated tokens after a shared opening, the fork of Section 7.1. The update narrowed the plan-and-fork account to that token. We call the narrowed account the *one-token account*: whatever changes the algorithm a model writes (temperature, a prompt, an internal edit) changes the odds at the fork, the odds can be read before sampling from teacher-forced logits, and the components that compute them can be found and steered. The artifacts of this iteration call the fork the *decision token*; this section keeps the word fork. Seven claims were preregistered, each with a rule committed by sha256 before its first decisive sample. We call them by their content.

- *The fork-dosing claim* (primary): temperature applied only at the fork, at a dose read off the

fork's teacher-forced logits and a map from fork tokens to algorithms, gives the alternatives in the shares predicted before sampling and at no correctness cost; on connected components, where the twin writes union-find in every sample, it reaches a correct-only Hill-1 of at least 1.8 at a pass cost of at most 5 points while every global knob stays at or below 1.15. The claim also predicted that the fork's entropy does not stand out among the program's tokens, so that samplers triggered by entropy would miss it.

- *The reachability claim*: forcing each of the fork's 64 most probable tokens reaches at least one correct known idea that no sampling knob reaches, on at least three units, and none of them is novel.
- *The matched DAS claim*: under one protocol for both families, the fork holds a certified low-rank variable for the algorithm in Qwen3-8B and in Gemma, and the task band of Qwen3-8B (blocks 12 to 23 at the task specification positions) works in generation but fails the reverse, retention and clamp tests.
- *The circuit claim*: in Qwen3-8B, a path-patching circuit computes the fork margin on two contrasts, its nodes beat matched random components in the real model, steering its top five nodes changes the written algorithm more than a matched random five, and its shape matches the circuit of Gemma's fork (Section 7.8).
- *The Gemma dial claim*: at Gemma's grid fork, a forced-BFS prompt with the shared opening prefilled reaches at least 40 of 50 correct BFS programs, so that the dial and prompting tie under matched conditioning; the dial also works in free generation; and the subspace passes the dormancy checks that Qwen3-8B's handle failed.
- *The temperature-law claim*: a law composed over one or two forks found by a blind locator predicts the temperature response of the algorithm mix within a mean total variation distance of 0.10 on at least 70% of units, including at least 3 of at least 4 held-out units in two families on which the law was never developed.
- *The top-five prohibition claim*: among ten open configurations, at least one obeys a ban of its own five most used path algorithms with correct programs in at least 20% of samples, and on that model's manifold of ten forced algorithms the survivors land at the edge of the banned five rather than on the opposite side.

**The strategy.** Seven artifacts ran (Table 374). The record audit repaired the iteration-7 record from rows on disk. A fork-dosing experiment tested temperature at the fork against every common knob and ran the forced branches of the reachability claim. A DAS experiment ran one protocol with seven committed controls at the fork of both families and at Qwen3-8B's task band. A circuit experiment ported the Gemma circuit pipeline of iteration 7 to Qwen3-8B. A Gemma experiment ran the prefill-matched prompting cells, a free-generation dial and the dormancy checks. A temperature-law experiment tested the composed law on three families, with Gemma and Llama-3.1-8B-Instruct [22] held out, and compared Qwen3-8B with its base model at the fork. A prohibition experiment screened ten configurations under a ban of five names and placed the survivors on the selected model's manifold.

**Table 374.** The artifacts of iteration 8, the question each was asked, and its compute and spend.

| Artifact | Question | Model and compute | OpenRouter spend |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Record audit | Repair the iteration-7 record (84 corrections); extend the kth smallest labeller to counting sort in dictionary form | No generation; CPU only | USD 0.095 |
| Fork-dosing experiment | Does temperature at the fork give the alternatives in predicted shares at no correctness cost, and better than every global knob? Which correct ideas are reachable only by forcing the fork's branches? | Qwen3-8B [5] and gemma-3-4b-it [76] in bf16; one NVIDIA L4; about 2.8 GPU hours; 5,543 correct programs | about USD 0.16 |
| DAS experiment | Under one protocol and seven controls, is there a certified low-rank variable for the algorithm at the fork of either family, or only a steering handle? | Qwen3-8B and gemma-3-4b-it in bf16; one NVIDIA L4; about 3.5 GPU hours | USD 0.00 |
| Circuit experiment | Which components of Qwen3-8B compute the fork margin, and does steering them change the written algorithm more than matched random components? | Qwen3-8B bf16; one NVIDIA L4; 400 generated programs | USD 0.00 |
| Gemma experiment | Does the Gemma dial beat prompting under matched conditioning, does it work in free generation, and is its subspace dormant? | gemma-3-4b-it bf16; one RTX 4090; about 14 GPU minutes; 1,320 programs | USD 0.0042 |
| Temperature-law experiment | Does a law composed over blindly located forks predict the temperature response on held-out units and families? Is the instruct model's fork the base model's mean? | Qwen3-8B, Qwen3-8B-Base, gemma-3-4b-it and Llama-3.1-8B-Instruct in bf16; one RTX A4500; 4,800 sweep programs | USD 0.046 |
| Prohibition experiment | Which open configurations obey a ban of their five most used path algorithms, and where do the survivors land on the manifold of the model that obeys most? | Eight new configurations through OpenRouter and local vLLM [6], two carried; states of Qwen3-32B with 4-bit NormalFloat (NF4) weights on one RTX 2000 Ada (16 GB); about 1,600 programs | USD 1.53 |

The circuit experiment's agent reached its six-hour limit after its README, results, figures and stdlib re-derivation were written; only its final structured output was not emitted. Every number quoted from it below comes from its result files.

## 8.2 The record audit: 84 corrections, counting sort in dictionary form, and the checks the review named

The record audit read the iteration-7 report against the rows on disk and wrote 84 corrections, each marked as an iteration-8 correction. The report this section is appended to was byte-identical to the audit's input, so all 84 corrections applied exactly once, in place, and the result equals the audit's corrected report. Of the 141 values the corrections touch, 68 match the rows, 63 are new, 7 are corrected and 3 are carried from an earlier artifact (Table 375).

**Table 375.** The record audit's own checks.

| Check | Result |
|---|---|

| | |
|---|---|
| Corrections | 84, each marked as an iteration-8 correction directly or through its table's caption |
| Lint for residual claims | 35 hits before (28 phrases and 4 undefined tags), 0 after |
| Apply test | Passes; a one-character mutation of one replaced passage makes it fail, as it must |
| Value ledger | 141 values: 68 match, 7 corrected, 63 new, 3 carried |
| Independent recomputation with the standard library | 97 checks, 0 mismatches; 6 of 6 placebos fail as required; no new or corrected value unchecked |
| Headline check through separate code paths | 11 of 11 headline numbers re-derived; 5 of 5 placebos fail as required |

**Counting sort in dictionary form.** The audit added one recogniser to the kth smallest labeller, mapped to the existing class counting sort, and validated it before relabelling any row. One revision was needed: the first version required a plain dictionary in the runtime tracer's record of dictionary sizes, which the tracer does not keep for `collections.Counter`. Table 376 gives the validation and the relabel. The relabel changed 35 of the 960 kth smallest rows, all from other to counting sort, and the old rules reproduce every stored label. With it, the full-state exchange of the forced counting sort source gives 29 of 30 counting sort programs (the 30th is a binary search on the value range that the rule calls quickselect), and the pooled kth smallest recovery of the DAS map falls from 1.59, 1.57 and 1.55 to 1.01 [0.95, 1.08], 1.00 [0.93, 1.08] and 0.99 [0.92, 1.06] at ranks 1, 4 and 16. The recovery of random subspaces (0.05) and the transfers of the fits with permuted labels do not change; those transfers are absolute shares of the 30 slots (0.47 to 0.70), and relative to the full-state effect they are 1.40, 1.24 and 0.82. The 36 catalogue variants that Section 7.3.9 listed among the correct programs outside the catalogue are mostly this one cell: 29 of them come from it and 6 more from the forced counting sort prompts.

**Table 376.** The counting sort extension of the kth smallest labeller: validation before the relabel, the relabel, and the adjudication of the cells with many programs labelled other.

| Step | Result |
|---|---|
| Bench references | 6 of 6 agree |
| Gold set | Unchanged: kappa runtime against AST 0.951, runtime against adjudicated 0.964 |
| Constructed controls | 5 of 5: two positives (a cumulative count over a Counter, a dictionary of counts expanded by item) and three negatives (a memo dictionary then `sorted(nums)`, a select over value buckets, a heap over Counter items) |
| Relabel of 960 kth smallest rows | 35 transitions, all other to counting sort; the old rules reproduce every stored label (1.000) |
| Labels after the relabel | quickselect 454, full sort 275, counting sort 153 (118 unchanged and 35 relabelled from other), size k heap 40, heapify and pop 21, median of medians 15, invalid 2; no row remains labelled other |
| Adjudication of 162 programs in the 19 cells with more than 20% other | Judge deepseek-chat-v3.1 passed 10 of 10 gate controls, mistral-medium-3.1 9 of 10; kappa between them 0.96; the rule agrees with each at 0.64; 48 distinct programs read by hand (38 disagreements and a sample of 10 agreements) |
| Full-state exchange of the forced counting sort source | 29 of 30 counting sort in dictionary form |
| Pooled kth smallest recovery at ranks 1, 4 and 16 | 1.01 [0.95, 1.08], 1.00 [0.93, 1.08], 0.99 [0.92, 1.06] (was 1.59, 1.57, 1.55); the port first reproduces every iteration-7 pooled recovery and interval exactly on all three tasks |

The same item settled the routing claim of Section 7.3: DAS routes more cleanly than the full state on one task, shortest path (Floyd-Warshall 28 of 28 against 8 of 20). On kth smallest the DAS edit

changes the implementation form, not the algorithm: at rank 16, 28 of 30 DAS programs are an array of counts, while the full state's programs are a dictionary with sorted keys. In the contrastive pilot on shortest path (Table 301), 56 of the 67 programs labelled other are degenerate text and 11 are Dijkstra attempts that crash, and 19 of the 30 programs labelled Dijkstra are degenerate text that the frozen path labeller calls Dijkstra; no relabelled program passes, so the counts of correct programs stand.

**Recovery by source, the untrained network and the Gemma prefill.** New Table 297b gives the iteration-7 recovery by source and rank: 0.17 to 0.43 of the full-state effect for forced Bellman-Ford/SPFA and 0.06 to 0.53 for forced BFS, with rank 2 never generated for forced sources. The three untrained-network ratios in the record measure different things, all on shortest path (Table 377).

**Table 377.** The three untrained-network ratios of iteration 7 (shortest path only), defined from their keys.

| Ratio | Value | What it compares | Reading |
|---|---|---|---|
| Committed comparison | 0.013 | The untrained network against the diverged contrastive fit | Not meaningful |
| Ratio of fit values | 0.92 | Fit value on the untrained network against the trained one (13.31 against 14.45 nats) | The untrained network fits almost as well |
| Ratio of gains | 5.08 | Gain in the fitted likelihood objective, untrained against trained (43.6 against 8.6 nats) | The untrained network gains more |

The Gemma dial cells of iteration 7 ran with the 32-token opening prefilled, as the raw rows record (prefix length 32), while the forced-BFS cell was sampled from index 0 on the same slots, so the prefill condition is now stated at every mention of "42 against 35"; Section 8.6 runs the comparison under matched conditioning.

**The fork margin and Gemma's natural variance.** The fork margin is now stated as a readout of an effect already expressed at the fork: its Spearman correlation with the escapes of the 40 pushes recomputes to 0.804 [0.62, 0.91], and the 48-token opening log likelihood ratio to 0.869, while no property of the push direction predicted which pushes move the fork (axis projection 0.17 and 0.01; plan-subspace projection of iteration 4, 0.03). For Gemma, the natural-variance share of the iteration-7 subspace lies below the random 95th percentile at ranks 1, 2 and 4 and above it at rank 8 (Table 378). The committed dormancy flag was false only because one correlation reached |rho| 0.497, and the recovery of 1.98 to 2.81 times the full-state swap is the off-manifold signature that Makelov et al. describe [122]. New Table 364d sets the Qwen3-8B and Gemma control protocols side by side and declares them not comparable.

**Table 378.** Natural-variance share of the iteration-7 Gemma subspace against the 95th percentile of random subspaces of the same rank.

| Block and rank | Share of natural variance | Random 95th percentile | Below the random 95th percentile |
|---|---|---|---|
| Block 17, rank 1 | 0.000932 | 0.000976 | yes |
| Block 17, rank 2 | 0.001157 | 0.001846 | yes |
| Block 17, rank 4 | 0.001873 | 0.002751 | yes |
| Block 17, rank 8 | 0.005651 | 0.004858 | no |
| Block 22, rank 1 | 0.002936 | 0.000784 | no |

**Positioning, the items not run, fresh slots and scope.** New Table 371b verifies all twelve neighbour rows of the iteration-7 positioning live on 2026-10-23, including both ICML 2026 workshop posters of the temperature paper on diffusion language models, whose venue row now

reads COLM from the arXiv comment, which gives no year. New Section 7.3.10 quotes the four dropped items of the DAS experiment verbatim; plan resampling and the placement of the top five survivors in DAS coordinates on the grid task carry forward. Table 346 gains its fresh-slot rows: on 50 fresh slots, noise on the top quartile of cells escapes 24 times against 9 for the rest (McNemar 15:0), and the full unit 21 times against 4 for the same energy at shuffled positions (17:0), with a seed effect of -0.10 [-0.22, 0.02]. The 81% and 82% of the dose unit are different quantities: the share of the noise energy in the top quartile of cells on connected components, and the share held by one massive-activation cell on shortest path. Search is now reported as rates per call with merged clusters (Table 379). The scope of the temperature claim is stated per unit: minimum spanning tree weight passes, but its T 1.0 prediction (0.237) lies outside the observed Wilson interval [0.28, 0.54], and the explanation by a lean of the sampling streams (Fisher p 0.011) is post hoc. The manifold claim rests on the two held-out templates (D 1.73 and 2.62), because the first three are training templates of the map, and the semantic seed sits on the boundary of the null (93.9th percentile, D 1.127 against 1.130).

**Table 379.** Evolutionary search of iteration 7 on kth smallest, as rates per call (Clopper-Pearson 95% intervals). A cluster merges programs that implement the same known technique.

| Arm | Correct programs outside the catalogue per call | Rate [95% CI] | Merged clusters; programs by technique |
|---|---|---|---|
| Search | 78 of 2,500 | 0.0312 [0.0247, 0.0388] | 12; order statistics on a Fenwick tree 44, binary search on the value range 30, bucket select 3, treap 1 |
| Larger mutator (qwen3-235b) | 23 of 500 | 0.046 [0.0294, 0.0682] | 7; binary search on the value range 10, partial selection sort 9, bucket select 4 |
| Instruction only | 2 of 500 | 0.004 [0.0005, 0.0144] | 1; binary search on the value range 2 |
| Enumerate and pick | 0 of 1,250 | 0 [0, 0.0029] | 0 |
| Same prompt without evolution | 0 of 500 | 0 [0, 0.0074] | 0 |
| Base model intent sentence | 0 of 1,250 | 0 [0, 0.0029] | 0 |
| Neutral twin | 0 of 200 | 0 [0, 0.0183] | 0 |

The ratio of the search rate to the instruction alone is 7.8, with a lower 95% bound of 2.41, short of the committed tenfold. None of the 103 distinct programs is novel. The four stale tags were replaced with the sentences that the dose unit experiment wrote for them, and ten file names lost in an earlier edit were restored.

**What the audit did not run.** The dormancy checks of the Gemma subspace, the untrained-network control of Qwen3-8B beyond shortest path, and a prompting cell matched to the dial's prefill were left to the experiments of this iteration (Sections 8.4 and 8.6).

## 8.3 Temperature at the fork (Qwen3-8B and Gemma)

The fork-dosing experiment asked whether temperature, applied only where the model commits to an algorithm, is the creativity knob that global temperature failed to be. We call the knob *fork-dosed temperature*: temperature d over the 20 most probable candidates at the fork, temperature 1 at every other position, with the dose read off the fork's teacher-forced logits and a token map (the

algorithm each fork token leads to, from forced continuations) and committed before any sample. Two doses were committed per unit, which we call the low dose and the high dose; the low dose targets a minority share set in advance. Each cell has 50 slots of common random numbers (five prompt templates by ten random streams, the same in every arm), so changes of the fork token and of the algorithm are paired per slot. The comparators were global temperature 1.3, 1.6 and 2.0 with top-p 0.95, and min-p sampling [138] at threshold 0.1 and T 3.0. Two adaptive samplers were added: entropy-triggered temperature in the style of EDT [137] (T 2 wherever the token's entropy exceeds the twin's 90th percentile) and block-start temperature in the style of AdapT [32] (T 2 at the start of each code block). Two placebos put the high dose at the highest-entropy position or at a random position, and class-uniform forcing draws the fork token's algorithm class uniformly. Five commits were hash-chained: the protocol, the dose table with every per-arm prediction, the locator verdict, the reachability predictions, and one amendment.

**Table 380.** The located forks and the committed doses. A token in quotes begins with a space where shown.

| Task (model) | Role | Fork position and tokens | Low dose | High dose | Doses after the amendment |
|---|---|---|---|---|---|
| Connected components (Qwen3-8B) | Collapsed: union-find in 50 of 50 | Generated tokens 10 and 11: ' parent' or ' def', then ' find' or ' dfs' | 3.5 | 7 | 5 and 10 |
| Minimum spanning tree weight (Qwen3-8B) | Mixed: Kruskal 0.62, Prim 0.38 | Token 61: ' Kr' or ' Prim' | 2.25 | 6 | n/a |
| Trapped rain water (Qwen3-8B) | Mixed: prefix and suffix arrays 0.76, two pointers 0.24 | Token 18: ' n' or ' left' | 1.25 | 1.5 | n/a |
| Valid anagram (Qwen3-8B) | Mixed: sort and compare 0.80 | Token 24: ' return' or ' count' | 4 | 13 | n/a |
| Grid path (Gemma) | Mixed: DFS 0.64, BFS 0.36 | Token 32: ' visited' against a blank line, 'if' or ' queue' | 1.75 | 3 | 2.25 and 3 |
| Grid path, shortest path, kth smallest (Qwen3-8B) | Boundary: no single fork; a window of 16 tokens at T 7 | First 16 body tokens | n/a | n/a | n/a |

**Table 381.** The committed tests of the fork-dosing claim.

| Committed test | Result | Verdict |
|---|---|---|
| Composed prediction: mean absolute error at most 0.10 on at least 80% of located units | Connected components 0.017, minimum spanning tree 0.017, trapped rain water 0.039, valid anagram 0.047, Gemma 0.010; per-slot replay 0.013, 0.009, 0.008, 0.031, 0.006 | Pass (5 of 5) |
| Collapsed unit: best dose reaches correct-only Hill-1 of at least 1.8 at a pass cost of at most 5 points; every comparator at most 1.15; paired bootstrap lower bound above 0 against each | Low dose 1.24 at cost 0.02; high dose 1.60 at cost 0.14; best comparator 1.17 (global T 2.0 and entropy-triggered); best fork arm minus best comparator +0.06 [97.5% lower bound -0.24], p 0.30 | Fail |

| | | |
|---|---|---|
| Mixed units: low dose within 0.10 of its target; Hill-1 at least 1.8 on at least 4 located units | Minority share at the low dose against target: minimum spanning tree 0.44 against 0.38, trapped rain water 0.26 against 0.35, valid anagram 0.40 against 0.50, Gemma 0.54 against 0.50; Hill-1 1.98, 1.86, 2.78, 1.97 | Pass (4 of 4) |
| The fork's entropy in the twin is below the 90th percentile of its per-token entropies in at least 90% of samples | 0 of 50 on connected components, 0 of 44 in Gemma, 0% on every unit | Fail (the opposite holds) |
| A window of 16 tokens at the boundary units moves the algorithm by at most 0.10 | Pass rate under the window 0.20, 0.42 and 0.24 (grid path, shortest path, kth smallest) | Untestable by breakage |
| Reversion (a dosed fork token followed by a program of the default algorithm) at most 10%, pooled | 4 of 102 (0.04) | Pass |
| Gemma: pass cost of the high dose within 0.10 of the committed sum over fork tokens of their dosed probability times their failure rate; class-uniform forcing never draws ' queue' and costs no more than the high dose | 0.14 against 0.156; 0 slots draw ' queue'; -0.04, below 0.14 | Pass |
| Disconfirming branch: pass cost above 10 points | Connected components at the high dose (0.14) | Fires |
| Primary p (intersection-union test over validated collapsed units) | 0.30; connected components is the only validated collapsed unit, because grid path failed validation | Not significant |
| Secondary: exact McNemar on correct off-default flips, high dose against global T 2.0, connected components | 7:1, one-sided p 0.035 | Significant unadjusted |
| Fork-dosing claim | The primary test fails and the pass-cost branch fires; the composition, the mixed-unit targets, reversion and the Gemma breakage prediction hold | Not supported as committed |

The five unit tables follow (Tables 382 to 386). Shares are attempted shares among the 50 programs; correct-only Hill-1 is the exponentiated expected Shannon entropy of the algorithm labels of correct programs in a random subset of 25 (1 means one idea); pass cost is the drop in pass rate against the twin with a Newcombe interval [42]; "fork changed" counts slots whose fork token differs from the twin's, and "flips" counts slots whose algorithm differs.

**Table 382.** Connected components, Qwen3-8B (collapsed). Shares of union-find, BFS and DFS, then invalid programs.

| Arm | Union-find | BFS | DFS | Invalid | Pass | Correct-only Hill-1 | Pass cost [95% CI] | Fork changed | Flips |
|---|---|---|---|---|---|---|---|---|---|
| Twin, T 1 | 1.00 | 0.00 | 0.00 | 0.00 | 1.00 | 1.00 | 0.00 [-0.07, 0.07] | 0 | 0 |
| Global T 1.3, top-p 0.95 | 1.00 | 0.00 | 0.00 | 0.00 | 1.00 | 1.00 | 0.00 [-0.07, 0.07] | 3 | 0 |
| Global T 1.6, top-p 0.95 | 1.00 | 0.00 | 0.00 | 0.00 | 1.00 | 1.00 | 0.00 [-0.07, 0.07] | 5 | 0 |
| Global T 2.0, top-p 0.95 | 0.94 | 0.00 | 0.04 | 0.02 | 0.98 | 1.17 | 0.02 [-0.05, 0.10] | 11 | 3 |
| Min-p 0.1, T 3.0 | 0.98 | 0.00 | 0.02 | 0.00 | 1.00 | 1.09 | 0.00 [-0.07, 0.07] | 2 | 1 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Entropy-triggered, T 2 | 0.94 | 0.00 | 0.04 | 0.02 | 0.98 | 1.17 | 0.02 [-0.05, 0.10] | 11 | 3 |
| Block-start, T 2 | 1.00 | 0.00 | 0.00 | 0.00 | 1.00 | 1.00 | 0.00 [-0.07, 0.07] | 11 | 0 |
| High dose at the highest-entropy position | 1.00 | 0.00 | 0.00 | 0.00 | 0.96 | 1.00 | 0.04 [-0.04, 0.13] | 1 | 0 |
| High dose at a random position | 0.86 | 0.00 | 0.00 | 0.14 | 0.80 | 1.00 | 0.20 [0.09, 0.33] | 1 | 7 |
| Fork dose, low (3.5) | 0.94 | 0.00 | 0.06 | 0.00 | 0.98 | 1.24 | 0.02 [-0.05, 0.10] | 18 | 3 |
| Fork dose, high (7) | 0.70 | 0.00 | 0.18 | 0.12 | 0.86 | 1.60 | 0.14 [0.04, 0.26] | 35 | 15 |
| Class-uniform forcing | 0.38 | 0.48 | 0.14 | 0.00 | 0.52 | 1.99 | 0.48 [0.33, 0.61] | 28 | 31 |
| Filtered fork dose, low (5; amendment) | 0.82 | 0.00 | 0.18 | 0.00 | 0.98 | 1.54 | 0.02 [-0.05, 0.10] | 29 | 9 |
| Filtered fork dose, high (10; amendment) | 0.88 | 0.00 | 0.12 | 0.00 | 0.98 | 1.37 | 0.02 [-0.05, 0.10] | 39 | 6 |

On connected components the composition predicted the low dose at union-find 0.91 and DFS 0.07 (per-slot replay 0.93 and 0.06; observed 0.94 and 0.06) and the high dose at union-find 0.72, DFS 0.13 and invalid 0.14 (replay 0.74, 0.12, 0.12; observed 0.70, 0.18, 0.12), with a predicted pass cost of 0.17 against 0.14 observed. The dial reaches DFS where global temperature cannot: 18% at the high dose against at most 4% for every global knob, and 7 correct off-default flips against 1 for global T 2.0. It pays for them, because seven of the fork's 20 candidates are tokens that break the program ('_parent', '-parent' and '父' fail 10 of 10 continuations). Forcing the fork class uniformly writes BFS 24 times, but only one of those BFS programs passes: the model's own BFS for this task is buggy.

**The amendment.** After the primary cells were seen, one amendment was committed before its samples. It applies the same dose rules to the 20 candidates without the tokens whose committed continuations fail at least half the time (8 tokens at index 10 and 4 at index 11 on connected components; 13 tokens in Gemma, including ' queue'). No removed token was ever drawn (0 of 100 slots on connected components). The filter removes the breakage (pass cost 0.02) but not the ceiling: DFS rises to 0.18 at the low dose and 0.12 at the high dose, against predictions of 0.12 and 0.17 (mean absolute error 0.023 and 0.024), and correct-only Hill-1 reaches 1.54, which beats the best comparator family by +0.38 (lower bound +0.01, p 0.022) but stays short of 1.8. The alternative does not hold more than about 19% of the mass among the fork's 20 candidates, so no dose at that token reaches the target.

**Table 383.** Minimum spanning tree weight, Qwen3-8B (mixed). Shares of Kruskal and Prim with a heap, then invalid programs.

| Arm | Kruskal | Prim | Invalid | Pass | Correct-only Hill-1 | Pass cost [95% CI] | Fork changed | Flips |
|---|---|---|---|---|---|---|---|---|
| Twin, T 1 | 0.62 | 0.38 | 0.00 | 1.00 | 1.92 | 0.00 [-0.07, 0.07] | 0 | 0 |
| Global T 1.3, top-p 0.95 | 0.64 | 0.36 | 0.00 | 1.00 | 1.90 | 0.00 [-0.07, 0.07] | 3 | 5 |
| Global T 1.6, top-p 0.95 | 0.62 | 0.38 | 0.00 | 0.96 | 1.92 | 0.04 [-0.04, 0.13] | 1 | 12 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Global T 2.0, top-p 0.95 | 0.68 | 0.32 | 0.00 | 0.94 | 1.82 | 0.06 [-0.02, 0.16] | 1 | 13 |
| Min-p 0.1, T 3.0 | 0.54 | 0.46 | 0.00 | 1.00 | 1.97 | 0.00 [-0.07, 0.07] | 6 | 8 |
| Entropy-triggered, T 2 | 0.60 | 0.38 | 0.02 | 0.88 | 1.91 | 0.12 [0.02, 0.24] | 4 | 15 |
| Block-start, T 2 | 0.72 | 0.28 | 0.00 | 1.00 | 1.79 | 0.00 [-0.07, 0.07] | 0 | 5 |
| High dose at the highest-entropy position | 0.50 | 0.46 | 0.04 | 0.90 | 1.98 | 0.10 [0.01, 0.21] | 20 | 19 |
| High dose at a random position | 0.62 | 0.36 | 0.02 | 0.98 | 1.91 | 0.02 [-0.05, 0.10] | 0 | 2 |
| Fork dose, low (2.25) | 0.56 | 0.44 | 0.00 | 1.00 | 1.97 | 0.00 [-0.07, 0.07] | 9 | 5 |
| Fork dose, high (6) | 0.54 | 0.46 | 0.00 | 0.96 | 1.98 | 0.04 [-0.04, 0.13] | 23 | 12 |
| Class-uniform forcing | 0.56 | 0.44 | 0.00 | 1.00 | 1.97 | 0.00 [-0.07, 0.07] | 24 | 23 |

The composition predicted Prim at 0.38 for the low dose and 0.46 for the high dose (replay 0.44 and 0.51; observed 0.44 and 0.46). The twin is already near two ideas here, and min-p at T 3.0 matches the dial.

**Table 384.** Trapped rain water, Qwen3-8B (mixed). Min-p and block-start temperature were dropped on this unit by the committed drop order.

| Arm | Two pointers | Prefix and suffix arrays | Peak split | Pass | Correct-only Hill-1 | Pass cost [95% CI] | Fork changed | Flips |
|---|---|---|---|---|---|---|---|---|
| Twin, T 1 | 0.24 | 0.76 | 0.00 | 1.00 | 1.72 | 0.00 [-0.07, 0.07] | 0 | 0 |
| Global T 1.3, top-p 0.95 | 0.26 | 0.72 | 0.02 | 1.00 | 1.90 | 0.00 [-0.07, 0.07] | 1 | 3 |
| Global T 1.6, top-p 0.95 | 0.24 | 0.76 | 0.00 | 1.00 | 1.72 | 0.00 [-0.07, 0.07] | 4 | 4 |
| Global T 2.0, top-p 0.95 | 0.28 | 0.70 | 0.02 | 1.00 | 1.93 | 0.00 [-0.07, 0.07] | 5 | 8 |
| Entropy-triggered, T 2 | 0.28 | 0.70 | 0.02 | 1.00 | 1.93 | 0.00 [-0.07, 0.07] | 5 | 8 |
| High dose at the highest-entropy position | 0.24 | 0.74 | 0.02 | 1.00 | 1.86 | 0.00 [-0.07, 0.07] | 0 | 2 |
| High dose at a random position | 0.22 | 0.76 | 0.02 | 1.00 | 1.81 | 0.00 [-0.07, 0.07] | 0 | 3 |
| Fork dose, low (1.25) | 0.24 | 0.74 | 0.02 | 1.00 | 1.86 | 0.00 [-0.07, 0.07] | 1 | 4 |
| Fork dose, high (1.5) | 0.24 | 0.74 | 0.02 | 1.00 | 1.86 | 0.00 [-0.07, 0.07] | 4 | 6 |
| Class-uniform forcing | 0.42 | 0.58 | 0.00 | 1.00 | 1.95 | 0.00 [-0.07, 0.07] | 19 | 17 |

On trapped rain water the composition predicted two pointers at 0.33 and 0.34, the replay agreed with the observed cells to 0.008, and the observed share stayed at 0.24: the doses were too small to

move many forks (1 and 4 slots changed), and the second decision inside the ' left' branch (Section 8.7) absorbs part of what moves. The fork arms do not beat the best comparator here (paired bootstrap p 0.96).

**Table 385.** Valid anagram, Qwen3-8B (mixed). Min-p and block-start temperature were dropped on this unit.

| Arm | Sort and compare | Hash count | Fixed count array | Alphabet scan | Invalid | Pass | Correct-only Hill-1 | Pass cost [95% CI] | Fork changed | Flips |
|---|---|---|---|---|---|---|---|---|---|---|
| Twin, T 1 | 0.80 | 0.04 | 0.16 | 0.00 | 0.00 | 1.00 | 1.78 | 0.00 [-0.07, 0.07] | 0 | 0 |
| Global T 1.3, top-p 0.95 | 0.80 | 0.06 | 0.14 | 0.00 | 0.00 | 1.00 | 1.82 | 0.00 [-0.07, 0.07] | 2 | 3 |
| Global T 1.6, top-p 0.95 | 0.78 | 0.04 | 0.18 | 0.00 | 0.00 | 1.00 | 1.83 | 0.00 [-0.07, 0.07] | 1 | 3 |
| Global T 2.0, top-p 0.95 | 0.76 | 0.04 | 0.20 | 0.00 | 0.00 | 1.00 | 1.89 | 0.00 [-0.07, 0.07] | 2 | 5 |
| Entropy-triggered, T 2 | 0.76 | 0.04 | 0.20 | 0.00 | 0.00 | 1.00 | 1.89 | 0.00 [-0.07, 0.07] | 2 | 5 |
| High dose at the highest-entropy position | 0.38 | 0.22 | 0.34 | 0.00 | 0.06 | 0.94 | 2.87 | 0.06 [-0.02, 0.16] | 18 | 26 |
| High dose at a random position | 0.70 | 0.06 | 0.18 | 0.00 | 0.06 | 0.92 | 1.95 | 0.08 [-0.01, 0.19] | 2 | 10 |
| Fork dose, low (4) | 0.60 | 0.06 | 0.34 | 0.00 | 0.00 | 1.00 | 2.27 | 0.00 [-0.07, 0.07] | 16 | 11 |
| Fork dose, high (13) | 0.42 | 0.18 | 0.38 | 0.00 | 0.02 | 0.98 | 2.78 | 0.02 [-0.05, 0.10] | 32 | 23 |
| Class-uniform forcing | 0.42 | 0.12 | 0.18 | 0.28 | 0.00 | 1.00 | 3.50 | 0.00 [-0.07, 0.07] | 33 | 30 |

Valid anagram is the unit where the dial earns its place: correct-only Hill-1 rises from 1.78 to 2.78 at a pass cost of 2 points, global temperature and the entropy-triggered sampler stay at or below 1.89, and the paired bootstrap beats each comparator family run there (lower bound of the best-comparator difference +0.19, p 0.008). The high dose at the highest-entropy position does as well (2.87), because that position is the fork in 21 of 39 slots. Class-uniform forcing reaches four ideas with every program passing (Hill-1 3.50).

**Table 386.** Grid path, gemma-3-4b-it (mixed). Attempted counts with correct counts in parentheses, out of 50; ' queue' counts slots whose fork token is ' queue', whose programs never pass.

| Arm | BFS attempted (correct) | DFS attempted (correct) | Pass | Correct-only Hill-1 | Pass cost [95% CI] | Fork changed | Flips | ' queue' |
|---|---|---|---|---|---|---|---|---|
| Twin, T 1 | 18 (8) | 32 (32) | 0.80 | 1.64 | 0.00 [-0.06, 0.06] | 0 | 0 | 5 |
| Global T 1.3, top-p 0.95 | 24 (16) | 26 (24) | 0.80 | 1.94 | 0.00 [-0.12, 0.12] | 5 | 8 | 2 |
| Global T 2.0, top-p 0.95 | 36 (25) | 14 (11) | 0.72 | 1.84 | 0.08 [-0.09, 0.25] | 7 | 20 | n/a |
| Entropy-triggered, T 2 (A* 1) | 35 (22) | 14 (9) | 0.62 | 1.82 | 0.18 [-0.01, 0.36] | 7 | 22 | n/a |
| High dose at the highest-entropy position | 29 (19) | 21 (14) | 0.66 | 1.97 | 0.14 [-0.01, 0.29] | 23 | 15 | n/a |
| High dose at a random position | 18 (12) | 32 (31) | 0.86 | 1.79 | -0.06 [-0.17, 0.04] | 4 | 0 | n/a |
| Fork dose, low (1.75) | 27 (15) | 23 (20) | 0.70 | 1.97 | 0.10 [-0.05, 0.24] | 17 | 13 | 7 |
| Fork dose, high (3) | 29 (19) | 21 (14) | 0.66 | 1.97 | 0.14 [-0.01, 0.29] | 23 | 15 | 4 |
| Class-uniform forcing | 26 (21) | 24 (21) | 0.84 | 1.98 | -0.04 [-0.16, 0.08] | 24 | 20 | 0 |
| Filtered fork dose, low (2.25; amendment) | 27 | 23 | 0.84 | 1.98 | -0.04 [-0.14, 0.05] | 15 | 11 | 0 |
| Filtered fork dose, high (3; amendment) | 28 | 22 | 0.82 | 1.98 | -0.02 [-0.12, 0.08] | 18 | 12 | 0 |

In Gemma the composition predicted BFS at 0.48 and DFS at 0.46 for the low dose (replay 0.58 and 0.42; observed 0.54 and 0.46) and BFS 0.58, DFS 0.36 for the high dose (replay 0.62 and 0.38; observed 0.58 and 0.42), with predicted pass costs of 0.08 and 0.16 against 0.10 and 0.14 observed. The pass cost comes from ' queue', the BFS opening without a wall check (Section 8.6); filtering it out restores the pass rate (0.84 and 0.82) at the same BFS share. The filtered dial beats global T by $+0.08$ (p 0.23) and min-p, entropy-triggered and block-start temperature with p 0.016 to 0.023. The twin is already diverse, and global T 1.3 does about as well as the dial.

*[Figure fig_forkdose_hill: generated in the report phase, not in this draft]*

**Where the fork sits in the entropy profile.** The claim predicted that the fork would not stand out by entropy, so that entropy-triggered samplers would miss it. The opposite holds. Code tokens are nearly deterministic, so the 90th percentile of the twin's per-token entropies is only 0.0003 to 0.05 nats, and the fork's entropy at T 1 lies above it in every sample of every unit (median 0.13 nats on connected components and 0.53 to 0.81 nats on the mixed units). The fork is also the single

highest-entropy position among the first 64 tokens on 44 of 44 Gemma slots that reach it, 32 of 39 minimum spanning tree slots and 21 of 39 valid anagram slots, but on only 2 of 50 connected components slots and 0 of 37 trapped rain water slots. A rule that doses the highest-entropy position therefore finds the fork on the mixed units (in Gemma it reproduces the fork-dosed arm exactly) and misses it on the collapsed one. This places the fork among the high-entropy minority tokens that Wang et al. find drive reinforcement learning on reasoning traces [139]. The threshold rule of the entropy-triggered sampler fires on about 10% of all tokens and spends most of its temperature away from the fork (DFS 4% on connected components; a 12-point pass cost and no gain on minimum spanning tree).

**The boundary units.** The grid path task on Qwen3-8B failed validation (no token at index 3 gives at least 5 of 10 off-default programs), so it joined shortest path and kth smallest as a boundary unit, where a window of 16 body tokens at T 7 over the top 20 replaced the fork dose (Table 387). All three windows fall below the committed pass-rate floor of 0.5, so the test is untestable by breakage. On kth smallest the surviving programs scatter across the catalogue, an effect on the program's form rather than a located choice.

**Table 387.** The boundary units under a window of 16 tokens at T 7.

| Task (Qwen3-8B) | Twin | Window | Pass, twin to window | Flips |
|---|---|---|---|---|
| Grid path | BFS 0.98, DFS 0.02 | BFS 0.94, DFS 0.02, other 0.04 | 0.88 to 0.20 | 2 |
| Shortest path | Dijkstra 0.98, Bellman-Ford/SPFA 0.02 | Dijkstra 0.98, Bellman-Ford/SPFA 0.02 | 1.00 to 0.42 | 0 |
| Kth smallest | Full sort 1.00 | Full sort 0.30, quickselect 0.14, size k heap 0.06, heapify and pop 0.02, counting sort 0.06, other 0.06, invalid 0.36 | 1.00 to 0.24 | 35 |

**The blind locator.** Prefix resampling at T 1 used 6 base paths per unit, prefixes 2 to 64 and 8 continuations each (3,072 continuations in 16 GPU minutes), with the Bayesian change-point rule of iteration 7. It located minimum spanning tree's fork exactly at token 61 on the three paths that show a change point. The other three show none, one short of the preregistered 4 of 6, because a Kruskal path starts near a 0.65 share of its own class and the share-jump rule cannot fire on it. On connected components no path shows a change point: the twin writes union-find in 50 of 50 programs and resampling at T 1 never leaves it, so the fork at tokens 10 and 11 was known only from iteration 7's pushed and banned prompts. On a collapsed task the knob needs knowledge that a locator at T 1 cannot supply. The opening classifier that stops continuations agrees with the full labels on 0.989 and 0.985 of completed programs.

**The reachability claim.** At every fork node, each of the 64 most probable tokens was forced and continued five times at T 1; an idea counts when at least one forced continuation is a correct program of it (Table 388). Forcing adds one known idea on three units, as predicted: BFS on connected components, Prim with an array scan on minimum spanning tree weight, and a scan over the alphabet on valid anagram. Each appears at T 1 about once in 2 million to 50 million samples, and the fork dose lifts its mass to at most 0.010, far below the 0.05 the plan asked for, because these ideas sit at deep ranks (ranks 34 and 52 on minimum spanning tree weight, rank 18 on valid anagram) or behind a token whose programs mostly fail (' def bfs' on connected components, whose mapped continuations pass 2 of 10).

**Table 388.** Ideas reached by forcing the fork's top 64 tokens against ideas reached by any sampling knob. Counts are correct forced continuations.

| Task (model) | Ideas reached by forcing (correct programs) | Ideas reached by any knob | Idea reached only by forcing | Its reachability at T 1 | Largest fork-dose mass |
|---|---|---|---|---|---|
| Connected components (Qwen3-8B) | union-find 294, DFS 153, BFS 2 | DFS, union-find | BFS | 5.4e-07 (about 2 million samples) | 0.005 |
| Minimum spanning tree weight (Qwen3-8B) | Kruskal 290, Prim with a heap 285, Prim with an array 2 | Kruskal, Prim with a heap | Prim with an array | 2.0e-08 (about 50 million samples) | 0.000 |
| Valid anagram (Qwen3-8B) | fixed count array 163, hash count 99, sort and compare 20, alphabet scan 5 | the first three | alphabet scan | 2.9e-08 (about 40 million samples) | 0.010 |
| Trapped rain water (Qwen3-8B) | prefix and suffix arrays 288, two pointers 143, peak split 4 | all three | none | n/a | n/a |
| Grid path (Gemma) | DFS 77, BFS 57 | BFS, DFS | none | n/a | n/a |
| Grid path (Qwen3-8B) | BFS 29 | BFS, DFS | none | n/a | n/a |

**Geometry at the fork and novelty.** The engine stored the final-norm hidden states of the 16 positions from the fork on, and the artifact plots their first two principal components (61% to 81% of the variance) in actual activation coordinates. The states separate by the algorithm the program implements, and a nearest-centroid check with centroids from the class-forced slots assigns the fork-dosed slots to their own program's class at 0.90 on connected components (94 slots), 0.97 on minimum spanning tree weight (73), 0.975 on trapped rain water (80), 0.83 on valid anagram (77) and 0.69 in Gemma (86), where BFS and DFS programs share most of the opening. The dial reaches the model's own states for its alternative ideas, not new regions. After adjudication only 2 correct programs from any stage carry the label other, both from the kth smallest window and both counting sort in dictionary form; two blind judges (gemini-2.5-flash, 10 of 10 gate controls; deepseek-chat-v3.1, 8 of 10, calling both known controls outside the catalogue novel) and a manual read called both catalogue variants, with nearest known solutions at tree edit distance 35 to 55 among 17,442 solutions. No correct program of 5,543 is novel (rule-of-three upper bound 0.0005).

**Checks and departures.** Every prediction was committed before its samples and the chain verifies. Labels agree with earlier engines on 30 of 30 connected components slots, 30 of 30 grid path, 49 of 50 minimum spanning tree, 29 of 30 shortest path, 30 of 30 kth smallest, 46 of 50 trapped rain water, 47 of 50 valid anagram and 47 of 50 Gemma slots. The run used a different GPU from the one planned (Table 374), built token maps from plug-in estimates, did not run two planned arms, dropped min-p and block-start temperature on trapped rain water and valid anagram, did not run the kth smallest labeller extension inside this artifact, and committed the locator verdict after the dose table (as its own commit). Of 574 programs sent to adjudication, 572 were resolved.

## 8.4 One DAS protocol at the fork of both families (Qwen3-8B and Gemma)

The DAS experiment answered the first lead of Section 7.11: the Qwen3-8B and Gemma handles had been tested under different objectives, readouts and permutation schemes, so a matched protocol was part of any test of a variable. One protocol was declared and hash-committed before any GPU use. The map class is a linear orthonormal map of rank 1, 2, 4 or 8 (nested); the interchange adds the map's projection of the source state minus that of the base state; the objective is bounded in the style of CDAS [112], the KL divergence from the source's own class distribution to the edited base's; and the readout is the class margin, the log probability of the source's class

minus that of the base's class at the fork. It ran at three sites: the fork of connected components in Qwen3-8B (token 11, after ' def'; classes BFS, DFS and union-find), the fork of Gemma's grid task (token 32; BFS and DFS, because forced A* and IDDFS prompts failed adherence), and Qwen3-8B's task band (blocks 12 to 23 at all 145 task specification positions, one shared map). Seven controls were committed with thresholds before any fit (Table 389). Teacher-forced results are scored on held-out pairs, which need both a held-out source family and a held-out noun; generation is scored by hidden tests on slots of common random numbers.

**Table 389.** The seven committed controls and their thresholds.

| Control | What it tests | Pass threshold |
|---|---|---|
| Random subspaces | 50 random subspaces per rank | Median recovery at most 0.05 |
| Label control | Refits on deranged labels (in Gemma, balanced shuffles of two classes) | At most 0.10 of the true fit's effect |
| Random-initialisation control | The same map class and budget on a randomly initialised network | At most 0.25 of the trained effect |
| Reverse interchange | Writing the default class's coordinates into a forced base | At least 0.5 of the full-state reverse |
| Retention | The map projected on the top 64 principal components of natural twin states | At least 0.5 of the effect kept |
| Natural-variance share | Share of natural twin variance in the subspace | Above the 95th percentile of random subspaces |
| Seed stability | Principal angles between seeds | Median first cosine at least 0.5 and above the random 99th percentile |

**Where the decision is written.** Exchanging the full residual state at the fork does almost nothing up to block 14 in either model (under 0.5 nats), so the choice is not yet written into the fork position there. The committed scan rule takes the block that maximises recovery normalised by the full-state effect, and it chose block 22 in Qwen3-8B (full-state effect 1.5 nats) and block 16 in Gemma (2.7 nats), the first blocks where the full state starts to carry the class. A rank-1 subspace there moves the margin 7.9 times (Qwen3-8B) and 2.5 times (Gemma) as much as the full state. A DAS edit that moves the margin far more than exchanging the whole state ever does raises the committed off-manifold flag, the signature of an edit that writes states the model does not produce, as Makelov et al. describe [122]. From block 24 in Qwen3-8B and block 22 in Gemma on, the full-state exchange carries 15 to 25 and 7 to 9 nats, and a rank-1 subspace recovers 0.6 to 0.9 and 1.2 to 1.4 of it. Iteration 7's Gemma block 17 sat on the same weak-ceiling edge.

**Table 390.** Teacher-forced recovery on held-out pairs, by rank and by source class, at each site (recovery is the DAS effect over the full-state effect; seed 0).

| Site | Chosen block and rank | Recovery at the chosen rank (95% CI) | Rank 1 | Rank 2 | Rank 4 | Rank 8 | By class at the chosen rank | Mean effect, DAS against full state (nats) | Off-manifold flag |
|---|---|---|---|---|---|---|---|---|---|
| Qwen3-8B, connected components fork | 22, rank 4 | 8.43 (5.94 to 11.30) | 7.67 | 8.12 | 8.43 | 9.26 | BFS 15.38, DFS 5.47, union-find -0.07 | 17.37 against 2.06 | Raised |
| Qwen3-8B, task band | Blocks 12 to 23, rank 2 | 0.82 (0.48 to 1.16) | 0.76 | 0.82 | 0.86 | 0.88 | BFS 1.57, DFS 0.68, union-find -0.04 | 14.89 against 18.19 | Not raised |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Gemma, grid fork | 16, rank 4 | 3.07 (2.85 to 3.41) | 2.21 | 2.72 | 3.07 | 3.25 | BFS 3.07 | 9.25 against 3.02 | Raised |
| Qwen3-8B, connected components fork, block 24 (post hoc) | 24, rank 2 | 1.22 (0.98 to 1.43) | 1.10 | 1.22 | 1.24 | 1.29 | BFS 1.72, DFS 0.77, union-find 0.41 | 19.82 against 16.24 | Not raised |

At the same sites, a difference-in-means direction of rank 1 recovers 0.44 to 0.61 on connected components and 0.92 in Gemma, and the top 8 principal components recover 0.97 and 1.01: the directions of natural variance reproduce the full-state effect, and only DAS overshoots it. The fitted first direction is nearly orthogonal to the unembedding difference of the class tokens (cosine 0.07 in Qwen3-8B, 0.03 in Gemma).

**Table 391.** The seven controls at each site. Ratios are effects relative to the true or trained fit.

| Control (threshold) | Qwen3-8B fork, block 22 | Qwen3-8B task band | Gemma fork, block 16 | Qwen3-8B fork, block 24 (post hoc) |
|---|---|---|---|---|
| Random subspaces (median at most 0.05) | Median -0.005, 95th percentile 0.015: pass | Median 0.003, 95th percentile 0.008: pass | Median 0.013, 95th percentile 0.033: pass | Median 0.000, 95th percentile 0.003: pass |
| Label control (at most 0.10) | 0.87, 0.63 and 0.77: fail | 0.71, 0.79 and 0.72: fail | 0.11, 0.11 and 0.20: fail | 0.61, 0.57 and 0.50: fail |
| Random-initialisation control (at most 0.25) | 0.69 (12.08 against 17.53 nats): fail | 0.18 (2.71 against 14.89 nats): pass | 0.41 (3.70 against 9.10 nats): fail | 0.54 (10.66 against 19.82 nats): fail |
| Reverse interchange (at least 0.5) | -2.72 (16 pairs): fail | 0.77 (16 pairs): pass | 2.83 (8 pairs): pass, as an overshoot | 1.17 (16 pairs): pass |
| Retention on 64 twin principal components (at least 0.5) | 0.18: fail | 0.43: fail | 0.12: fail | 0.52: pass |
| Retention with forced prompts and bans in the basis (at least 0.5) | 0.19: fail | 0.90: pass | 0.44: fail | 0.74: pass |
| Natural-variance share (above the random 95th percentile) | 0.0076 against 0.0016: pass | 0.0030 against 0.0006: pass | 0.0014 against 0.0030: fail | 0.0075 against 0.0010: pass |
| Seed stability (first cosine at least 0.5) | 0.87 (random 99th percentile 0.07): pass | 0.49 (0.05): fail | 0.53 (0.09): pass | 0.62 (0.05): pass |
| Verdict | No handle by the pooled rule (toward DFS alone the generation test passes) | Sufficient handle | Sufficient handle | Post hoc, teacher forcing only; fails the label and random-initialisation controls |

The label controls on connected components used two cyclic derangements and a derangement at the prompt level, changing every retained target; they still move held-out pairs toward the true source class, and the first of them also fits its permuted targets (training KL 0.0002) and writes DFS for BFS sources in generation (20 of 20). Gemma's two-class site could only use balanced shuffles that change 33% to 58% of labels, a weaker control, so the matched claim carries the qualifier "three-class label control tested in Qwen3-8B only". The randomly initialised networks have next-token entropies of 11.1 to 12.0 nats, and the full-state exchange on them does little (1.2 and 0.45 nats), yet a linear map at the fork recovers 0.41 to 0.69 of the trained effect: a linear map at one

token can manufacture most of the effect with no trained computation behind it, the concern Sutter et al. raise for non-linear maps [132].

*[Figure fig_das_controls: generated in the report phase, not in this draft]*

**Verdict.** The committed rule confirms the matched DAS claim if both forks are certified and disconfirms it if a fork fails the label control or the random-initialisation control in either family. Both forks fail both, so the claim is disconfirmed; the largest one-sided p over sites of the pair-level tests that the true and trained fits beat their controls is 0.037, so the fits do beat their controls, but by margins far below the committed thresholds. The committed prediction (both forks certified; the band passes generation but fails the reverse, retention and clamp tests) was wrong on both forks and on the band's reverse test. What the protocol finds in both families is a strong steering handle of low rank at the fork, not a certified variable: a certified variable would pass every control, while a steering handle moves the output without passing them.

**Generation.** Table 392 gives every generation cell (held-out pairs; the canonical opening prefilled for the fork sites; 30 slots of common random numbers; correct means the target class's label and the hidden tests both pass).

**Table 392.** Generation under the fitted maps (correct target programs). Base is the unedited twin, full the full-state exchange, random an equal-rank random subspace and permuted the first label-control map. The two reverse columns write the twin's coordinates, or its full state, into a forced base and count programs that still implement the forced class; no program in any reverse cell returns to the twin's class. Source is the forced prompt itself. McNemar p-values are two-sided.

| Site and target | Base | Full | DAS | Random | Permuted | Reverse DAS | Reverse full | Source | Other ranks | Pass, DAS against base | McNemar, DAS against random |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Qwen3-8B fork, toward DFS | 0 of 30 | 2 of 30 | 29 of 30 | 0 of 30 | 20 of 20 | 20 of 20 | 20 of 20 | 20 of 20 | Rank 1: 25 of 30; rank 2: 29 of 30 | 1.00 against 0.83 | 29:0, p 3.7e-9 |
| Qwen3-8B fork, toward BFS | 0 of 30 | 0 of 30 | 5 of 30 (29 write BFS) | 0 of 30 | 0 of 20 | 12 of 20 | 13 of 20 | 14 of 20 | n/a | 0.20 against 0.83 | 5:0, p 0.063 |
| Qwen3-8B band, toward BFS | 0 of 30 | 23 of 30 | 28 of 30 (30 write BFS) | 0 of 30 | 3 of 20 | 20 of 20 | 20 of 20 | 20 of 20 | n/a | 0.93 against 1.00 | 28:0, p 7.5e-9 |
| Qwen3-8B band, toward DFS | 0 of 30 | 20 of 30 | 3 of 30 (24 write BFS) | 0 of 30 | 17 of 20 | 19 of 20 | 20 of 20 | 13 of 20 | Rank 1: 5 of 30; rank 4: 4 of 30 | 0.70 against 1.00 | 3:0, p 0.25 |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Gemma fork, toward BFS | 6 of 30 | 10 of 30 | 22 of 30 (27 write BFS) | 5 of 30 | 10 of 20 | 0 of 20 | 0 of 20 | 1 of 20 | Rank 1: 22 of 30; rank 2: 21 of 30 | 0.73 against 0.90 | 17:0, p 1.5e-5 |

Three readings follow. First, a rank-4 edit at one token moves the algorithm: 29 of 30 correct DFS programs on connected components where the twin writes union-find, and 22 of 30 correct BFS programs in Gemma against 6 unedited and 5 for a random subspace. Second, the edit toward BFS in Qwen3-8B writes the fork token and BFS in 29 of 30 programs, but 24 of them fail; a typical failure returns `len(visited)` from the helper while the caller does `visited.update(bfs(i))`. The model's program around the written token is incoherent, one more sign of an edit off the natural states. Third, no edit restores the default: neither the reverse DAS edit nor the reverse full-state exchange returns a forced prompt to its twin's class in generation (0 of 20 in each cell), because with the opening prefilled through ' def' a forced prompt keeps its class whatever is written at the fork. The band's map toward DFS writes BFS in 24 of 30 programs, and its label-control map writes DFS for both source classes: the rank-2 band handle encodes "leave union-find, toward the algorithm this map was fitted to favour" rather than the source's class, which is the generation face of the label-control failure. A full-state exchange over the band flips the teacher-forced class on only 58% of held-out pairs, below the planned 90%, with positions verified by tests, and the first principal cosines between this band handle and iteration 7's handle are 0.12 to 0.17.

**Do iteration 4's random pushes act through the fitted subspaces? No.** The 40 random directions of iteration 4 (alpha 2, absolute unit, blocks 12 to 23 at the task specification positions) were projected on the band subspace, and the projection, scaled by the forced-class displacement, was committed as the predictor of each direction's correct off-default share before any outcome was read. The retrospective association is Spearman 0.39 (p 0.006, n 40; 0.36 and 0.39 for the two engines; partial correlation controlling the push norm 0.42 over the band and 0.39 at block 35; a permuted predictor gives 0.14), below the committed 0.5. On 24 prospective directions committed in advance it falls to 0.09 (p 0.33; 208 of 240 programs ran before the time cap). Clamping the subspaces removes nothing (Table 393), so the committed clamp test fails: random pushes move the algorithm through other directions than the fitted handles.

**Table 393.** Clamping the fitted subspaces under the ten most-escaping random pushes (270 paired slots, 27 to 30 per direction).

| Clamp | Escapes under the push | Escapes with the clamp | Removal | McNemar (push only : clamp only) | Pass rate with the clamp |
|---|---|---|---|---|---|
| Band subspace clamped to the twin at every band block | 218 of 270 | 221 of 270 | -0.01 | 8:11, p 0.65 | 0.87 |
| Random clamp of equal rank | 218 of 270 | 213 of 270 | 0.02 | 8:3, p 0.23 | 0.82 |
| Fork coordinates clamped at block 22 (53 slots that reproduce the opening; reach rate 0.53) | 44 of 53 | 46 of 53 | -0.05 | 1:3, p 0.63 | n/a |

**Geometry in actual activation coordinates.** The artifact plots every prompt's coordinates in the fitted rank-8 subspace at the chosen block and the fork, centred on the twin mean and embedded by metric MDS (Kruskal stress-1 0.015 for connected components and 0.027 for Gemma) and Isomap [25] (k 8), beside a PCA of the raw states. Class centroids are sized by implemented, test-passing programs (connected components: union-find 87, DFS 35, BFS 34; Gemma: DFS 21, BFS 7).

On connected components the forced BFS, DFS and union-find prompts form three separated clusters; the neutral twins, the two seed strings, the short instruction to make a new algorithm and the long novelty instruction sit together, away from every forced cluster; the five bans of union-find sit just outside the twin cluster, toward the forced clusters. In Gemma, forced A* sits beside forced BFS (its ' open' continues as BFS under a neutral prompt), forced IDDFS sits near the twin cluster on the BFS side (its programs were labelled DFS 7 of 20 and disputed 13 of 20), and the top-five prohibition prompt and the novelty instruction stay inside the twin cluster. The figure titles say "handle coordinates (not certified)".

**The post hoc fit at block 24.** Block 24 is the first block where the full-state exchange carries the class in Qwen3-8B (16.2 nats, flipping every held-out pair). The same protocol there, reported as a sensitivity and never substituted for the committed block, recovers 1.22 (0.98 to 1.43), below the off-manifold line, and passes the reverse interchange (1.17), retention (0.52; 0.74 with forced prompts in the basis), natural variance (0.0075 against 0.0010) and seed stability (0.61 to 0.63). The label controls still reach 0.61, 0.57 and 0.50 of the true fit (p 0.0005 each for true above permuted), the randomly initialised network reaches 0.54, and a difference-in-means direction recovers 0.88 to 0.99 and the top four principal components 0.99, the same as DAS. Once the site has a real full-state ceiling, the dormancy and off-manifold failures go away and the lack of specificity stays: the class information at the fork is carried by the natural difference-in-means direction, which a linear map of almost any labelling partly recovers.

**Novelty, inputs and departures.** None of 1,857 correct programs (1,708 on connected components, 149 in Gemma) is labelled other, so no judge was called (rule-of-three bound 0.0016); 19 Gemma programs are disputed (18 with runtime IDDFS against AST BFS, all failing) and count as neither class. The 20 new single-token twin nouns came from the concreteness norms of the run's dataset dependency, and 76 connected components programs from four of its code datasets served as the nearest known solutions. The run used one 23 GB GPU (Table 374) for about 5.5 hours in all. Minimum spanning tree weight and Gemma's task band were dropped at the first commit, and band fits ran 109 steps with 2 seeds instead of 3. Blocks whose full-state exchange moved fewer than 10% of pairs by 1 nat (blocks 0 to 14) were recorded as undefined. Three stages were rerun, after a NaN-ordering bug, a crash in the two-class label control and an out-of-memory stop. The engine check against iterations 6 and 7 and the band geometry figure were not run, labeller disagreements were not sent to the blind judge, and mediation ran 270 of 300 clamp slots and 208 of 240 prospective programs before the time caps. Eight commits form a chain with no break, and a standard-library re-derivation reproduces all 38 checked headline numbers from the raw rows, with placebos that fail as they should.

**Positioning.** DAS [110] and Boundless DAS [111] use the same map class; CDAS supplies the bounded objective and the bidirectional test [112]; MIB ranks DAS above difference in means and PCA under teacher forcing but scores no specificity control [52]. Makelov et al. predict the off-manifold overshoot, the failed retention and the failed reverse at the forks [122], Grant et al. describe the divergent representations that interventions produce [141], and Sutter et al. show that maps of enough capacity fit untrained networks [132]. This run adds the full set of specificity controls under one protocol in two families, and the result that even a linear map at the fork fails them.

## 8.5 The circuit behind Qwen3-8B's fork

The circuit experiment asked which components of Qwen3-8B compute the odds at the fork, and whether steering them rewrites the algorithm in generated, hidden-test-checked programs. It ported

iteration 7's Gemma pipeline so that both families are measured the same way. A screen by attribution patching [136] covers 4,752 nodes: 32 heads and one MLP in each of 36 layers, at four position classes (the condition span, the task specification, the shared opening and the fork position). The top nodes are then patched exactly, by noising and denoising, and path patching [135] runs among the top 15, followed by faithfulness curves. Each top-10 node is validated in the real model against 50 matched random components of the same type and position class with matched write norm, under ablation and under steering (exact sign-flip tests, Holm over all 20 node and test p-values). The top five nodes are then steered in generation (30 slots of common random numbers, hidden tests, exact McNemar against a matched random five and against the twin), and a committed motif test compares the circuit with Gemma's. Two contrasts were studied. On minimum spanning tree weight the metric is the logit of ' Kr' minus the logit of ' Prim' after the 61-token opening; on connected components it is ' find' minus ' dfs' after ' def'.

**The primary contrast changed before any screen row.** The natural contrast on minimum spanning tree weight paired 24 neutral-noun variants per template (nouns from the concreteness norms of the dataset dependency), the variant with the highest metric against the lowest; its mean total effect was 0.89 nats, below 1 nat, which triggered the preregistered fallback: forced-Prim paraphrases against their padding twins became the primary contrast, committed as an amendment before any screen row. On connected components the corrupt run exchanges the forced-DFS prompt's residual states into the twin at every task specification position of all 36 blocks, which flips the teacher-forced argmax to ' dfs' in every pair; because that corruption is an intervention, condition-span nodes have zero effect there by construction, and the contrast maps routing from the task specification to the fork. Table 394 lists the contrasts.

**Table 394.** Contrasts and their circuits. The correlation of attribution patching with exact patching is given on 160 random nodes and on the top set.

| Contrast | Pairs | Mean total effect (nats) | Attribution against exact patching, random nodes / top set | Edges kept | Completeness / sufficiency of the top 10 |
|---|---|---|---|---|---|
| Minimum spanning tree, forced Prim against padding twin (primary) | 10 | 9.91 | 0.29 / 0.89 | 12 | 0.90 / 1.01 |
| Connected components, forced-DFS states exchanged into the twin | 15 | 19.33 | 0.25 / 0.91 | 17 | 0.94 / 0.96 |
| Minimum spanning tree, natural noun variants (descriptive) | 15 | 0.89 | 0.11 / 0.66 | 20 | 1.05 / 1.02 |
| Minimum spanning tree, mixed set | 15 | 4.00 | n/a / 0.79 | 0 | n/a |
| Minimum spanning tree, disjoint noun set for stability | 12 | 0.46 | n/a / 0.51 | 33 | 0.82 / 0.99 |

Attribution patching ranks the strong nodes well but not the near-zero random ones, whose exact effects sit at the bf16 noise floor of about 0.06 nats; attribution patching with integrated gradients over the input did not improve that correlation, so an exhaustive exact sweep over every fork-position node, all 144 MLP nodes and the screen's top 120 replaced the screen's ranking. The sweep found the largest node of the primary contrast, the layer-0 MLP at the condition span, with 7.7 of the 9.9 nats, which attribution patching had estimated at 0.1. Completeness (the top k noised together) and sufficiency (the top k restored together, with the rest of the network live) recover

most of the effect; freezing every node outside the circuit to its corrupt value leaves almost nothing at any k up to 60, because the fork nodes need upstream processing of the instruction spread over many small nodes.

**Table 395.** Validation of the top 10 nodes against 50 matched random components (minimum spanning tree, forced Prim; 30 prompts, 10 held out). Effects are mean metric changes in nats; p-values are Holm-adjusted over 20 tests. Node names give type, layer, head and position class.

| Node | Ablation effect | Ablation p | Steering effect | Steering p | Both pass |
|---|---|---|---|---|---|
| MLP, layer 0, condition span | 5.429 | 3.8e-06 | 4.938 | 3.8e-06 | yes |
| Head 14 of layer 33, fork | 2.050 | 1.9e-08 | 4.251 | 1.9e-08 | yes |
| MLP, layer 35, fork | -2.638 | 1 | -5.068 | 1 | no |
| Head 28 of layer 34, fork | 1.964 | 1.9e-08 | 3.824 | 1.9e-08 | yes |
| MLP, layer 27, fork | 0.777 | 1.9e-08 | 1.896 | 1.9e-08 | yes |
| Head 6 of layer 29, fork | 1.403 | 1.9e-08 | 2.827 | 1.9e-08 | yes |
| MLP, layer 30, fork | 0.716 | 1.9e-08 | 1.529 | 1.9e-08 | yes |
| Head 10 of layer 23, opening | 0.783 | 1.9e-08 | 1.531 | 1.9e-08 | yes |
| MLP, layer 28, fork | 0.414 | 3.4e-07 | 0.904 | 1.9e-08 | yes |
| MLP, layer 11, task specification | 1.162 | 1.9e-08 | 1.288 | 1.9e-08 | yes |

**Table 396.** Validation of the top 10 nodes on connected components (40 prompts, 10 held out).

| Node | Ablation effect | Ablation p | Steering effect | Steering p | Both pass |
|---|---|---|---|---|---|
| MLP, layer 27, fork | 9.596 | 1.8e-11 | 16.058 | 1.8e-11 | yes |
| MLP, layer 26, fork | 5.088 | 1.8e-11 | 10.606 | 1.8e-11 | yes |
| MLP, layer 32, fork | 1.924 | 3.7e-10 | 7.967 | 1.8e-11 | yes |
| MLP, layer 30, fork | 0.738 | 0.0015 | 5.851 | 1.8e-11 | yes |
| Head 10 of layer 23, fork | 3.627 | 1.8e-11 | 7.964 | 1.8e-11 | yes |
| Head 31 of layer 27, fork | 3.047 | 1.8e-11 | 6.869 | 1.8e-11 | yes |
| MLP, layer 28, fork | 0.485 | 0.0077 | 1.685 | 3.4e-10 | yes |
| MLP, layer 31, fork | -1.527 | 1 | 0.684 | 0.0079 | no |
| MLP, layer 13, task specification | 1.712 | 2.4e-06 | 2.847 | 1.3e-07 | yes |
| Head 13 of layer 11, task specification | -1.802 | 1 | -3.351 | 1 | no |

**Table 397.** Validation of the top 10 nodes on the natural noun-variant contrast (descriptive; 15 clean and 5 held-out prompts under the committed time rule).

| Node | Ablation effect | Ablation p | Steering effect | Steering p | Both pass |
|---|---|---|---|---|---|
| MLP, layer 0, condition span | 0.478 | 1.9e-05 | 0.523 | 1.9e-05 | yes |
| MLP, layer 27, fork | 0.194 | 1.9e-05 | 0.456 | 1.9e-05 | yes |
| MLP, layer 20, task specification | 0.092 | 0.014 | 0.176 | 1.9e-05 | yes |
| MLP, layer 12, task specification | 0.058 | 0.21 | 0.158 | 2.3e-05 | no |
| MLP, layer 30, fork | 0.110 | 8.4e-05 | 0.240 | 1.9e-05 | yes |
| MLP, layer 8, task specification | 0.088 | 0.05 | 0.168 | 0.00012 | no |
| MLP, layer 19, task specification | 0.064 | 0.21 | 0.138 | 0.00039 | no |
| MLP, layer 5, condition span | -0.015 | 0.73 | 0.037 | 0.48 | no |
| Head 28 of layer 34, fork | 0.178 | 1.9e-05 | 0.443 | 1.9e-05 | yes |
| MLP, layer 16, task specification | 0.024 | 0.51 | 0.081 | 0.087 | no |

On the primary contrast 9 of 10 nodes pass both tests, and on connected components 8 of 10; the failures are components whose effect runs against the decision, brakes rather than drivers. On the natural contrast 5 of 10 pass. In every contrast the top five and top ten as sets beat all 50 matched

random sets under ablation and steering (0.96 for the top ten under ablation on the natural contrast). The top fork heads are outliers in write norm, so their matched pools fell back to the 50 nearest-norm heads; a sensitivity committed before its rows rescales each draw to the target's write norm and leaves every head significant (uncorrected one-sided p 9.3e-10 for heads 6 of layer 29, 14 of layer 33 and 28 of layer 34; 9.1e-13 for head 31 of layer 27). Recomputed from its p-values, Gemma's circuit of iteration 7 has 7 nodes passing both tests under either multiplicity convention.

**Steering the top five in generation.** The top five non-condition nodes (all at the fork, by the alignment rule committed in the amendment, because the condition-span MLP of the forced contrast has no counterpart in the twin) were steered toward the minority class: each node's own activation plus twice the difference between the minority-class and default-class means. The matched random five apply the same operation to one matched component per node. Prefill fixes the prompt and opening for every arm; free generation applies the opening and fork edits only while the sampled prefix equals the opening (Table 398).

**Table 398.** Steering the top five at the committed dose 2. Correct target programs are Prim (heap or array) on minimum spanning tree weight and DFS on connected components.

| Task and condition | Arm | Correct target | Attempted target | Pass rate | Reached the fork | Labels |
|---|---|---|---|---|---|---|
| Minimum spanning tree, prefilled | Twin | 8 of 30 | 8 | 1.00 | 30 | Prim 8, Kruskal 21, disputed 1 |
| Minimum spanning tree, prefilled | Top five | 29 of 30 | 29 | 1.00 | 30 | Prim 29, disputed 1 |
| Minimum spanning tree, prefilled | Random five | 23 of 30 | 24 | 0.97 | 30 | Prim 24, Kruskal 5, disputed 1 |
| Minimum spanning tree, free | Twin | 7 of 30 | 7 | 0.97 | 14 | Prim 7, Kruskal 22, disputed 1 |
| Minimum spanning tree, free | Top five | 16 of 30 | 16 | 0.97 | 14 | Prim 16, Kruskal 14 |
| Minimum spanning tree, free | Random five | 12 of 30 | 12 | 0.97 | 14 | Prim 12, Kruskal 17, disputed 1 |
| Connected components, prefilled | Twin | 1 of 30 | 1 | 1.00 | 30 | union-find 29, DFS 1 |
| Connected components, prefilled | Top five | 29 of 30 | 29 | 1.00 | 30 | DFS 29, union-find 1 |
| Connected components, prefilled | Random five | 30 of 30 | 30 | 1.00 | 30 | DFS 30 |
| Connected components, free | Twin | 0 of 30 | 0 | 1.00 | 3 | union-find 30 |
| Connected components, free | Top five | 3 of 30 | 3 | 1.00 | 3 | union-find 27, DFS 3 |
| Connected components, free | Random five | 3 of 30 | 3 | 1.00 | 3 | union-find 27, DFS 3 |

With the opening prefilled, the top five add 21 correct Prim programs over the twin on minimum spanning tree weight and 28 correct DFS programs on connected components, with no loss of pass rate. But at the committed dose the matched random five also flip most programs (23 and 30 of 30), so the comparison of the top five with the random five has little room: 7:1 discordant slots on minimum spanning tree weight (two-sided exact McNemar p 0.070, one-sided 0.035) and 0:1 on connected components (p 1). In free generation few slots reproduce the opening (14 and 3 of 30),

so the edits rarely apply. The protocol commit did not state the sidedness of the McNemar test; the analysis code fixed by hash at that commit used the one-sided test, while the plan's power calibration (8:0 gives 0.0078, 7:1 gives 0.07) is two-sided. The artifact takes the two-sided test as primary because it is the more conservative, which makes both verdicts of the circuit claim partial; with the one-sided test the minimum spanning tree verdict would pass.

**A post hoc dose sensitivity.** Committed before its rows and labelled post hoc, lower doses on the same 30 slots separate the circuit from the random five on connected components but only modestly on minimum spanning tree weight (Table 399 and the figure below).

**Table 399.** Prefilled generation at lower doses (correct target programs of 30).

| Task | Dose | Twin | Top five | Random five | Discordant, top five : random five | McNemar, two-sided | Pass, top five / random five |
|---|---|---|---|---|---|---|---|
| Minimum spanning tree | 0.5 | 8 | 25 | 20 | 6:1 | 0.125 | 30 / 30 |
| Minimum spanning tree | 1.0 | 8 | 27 | 28 | 1:2 | 1 | 30 / 30 |
| Connected components | 0.5 | 1 | 30 | 6 | 24:0 | 1.19e-07 | 30 / 30 |
| Connected components | 1.0 | 1 | 30 | 18 | 12:0 | 0.000488 | 30 / 30 |

At the logit level, the mean probability of the minority fork token over the five twins rises under the top five from 0.263 to 0.542, 0.785, 0.962 and 0.997 at doses 0.25, 0.5, 1 and 2 on minimum spanning tree weight, against 0.439, 0.638, 0.883 and 0.983 under the random five; on connected components it rises from 0.029 to 0.978 already at dose 0.25 under the top five, against 0.089, 0.194, 0.695 and 1.000 under the random five.

*[Figure fig_circuit_dose: generated in the report phase, not in this draft]*

**Table 400.** The committed predictions of the circuit experiment.

| Prediction | Observed | Outcome |
|---|---|---|
| At least 5 of 10 nodes pass both tests on the primary contrast | 9 of 10 (natural contrast, descriptive: 5 of 10) | Met |
| Top five prefilled: twin plus at least 6 correct Prim programs, beats the random five (McNemar p below 0.05), pass rate at least 0.8 of the twin's | +21; two-sided p 0.070 (one-sided 0.035); pass ratio 1.00 | Not met two-sided; met one-sided |
| The same on connected components toward DFS | +28; p 1; pass ratio 1.00 | Not met: the random five also saturate |
| Motif: fork share in [0.38, 0.68], condition share in [0.21, 0.51], effect-weighted depth of the fork stack in [0.45, 0.80] | Mixed set: fork 0.63, condition 0.01, depth 0.86 | Not met (condition share and depth) |
| Top-10 Jaccard between two disjoint pair sets at least 0.4 | 0.25 | Not met |

**The motif against Gemma.** The committed motif function (the share of absolute mean noising effect by position class over the top 40 nodes by attribution plus 20 random calibration nodes) first reproduced Gemma's printed shares (fork 0.531, condition 0.357). Table 401 gives the shares. The committed comparison, Qwen3-8B's mixed set against Gemma, reads "different motif" on the condition share and the depth of the fork stack. The committed composition depends on the attribution screen, which missed Qwen3-8B's largest node; with the exhaustive sweep, the condition share of the primary contrast rises to 0.17 and the picture matches Gemma qualitatively: a layer-0

MLP at the condition span injects the instruction, and a late stack of fork-position MLPs and heads turns it into the fork logit. In Qwen3-8B that stack sits deeper (effect-weighted depth about 0.86 against 0.69). The two decisions share components: the MLPs of layers 27, 28 and 30 at the fork position appear in both top-10 lists, and head 10 of layer 23 appears in both (at the opening for minimum spanning tree weight, at the fork for connected components).

**Table 401.** Share of the absolute noising effect by position class, and the effect-weighted depth of the fork stack. Rows marked sweep rank the top 60 nodes by exact effect; rows marked without injection drop layer 0 and 1 nodes at the corrupted span.

| Model and contrast | Condition span | Task specification | Opening | Fork | Fork-stack depth |
|---|---|---|---|---|---|
| Gemma (15 pairs) | 0.36 | 0.06 | 0.05 | 0.53 | 0.69 |
| Qwen3-8B, natural variants | 0.08 | 0.46 | 0.13 | 0.34 | 0.87 |
| Qwen3-8B, forced Prim | 0.01 | 0.20 | 0.18 | 0.62 | 0.86 |
| Qwen3-8B, mixed set | 0.01 | 0.22 | 0.14 | 0.63 | 0.86 |
| Qwen3-8B, disjoint noun set | 0.14 | 0.48 | 0.10 | 0.28 | 0.93 |
| Qwen3-8B, connected components | 0.00 | 0.32 | 0.01 | 0.67 | 0.79 |
| Qwen3-8B, natural variants, sweep | 0.27 | 0.39 | 0.10 | 0.24 | 0.87 |
| Qwen3-8B, forced Prim, sweep | 0.17 | 0.18 | 0.17 | 0.48 | 0.86 |
| Qwen3-8B, disjoint noun set, sweep | 0.23 | 0.28 | 0.06 | 0.43 | 0.96 |
| Qwen3-8B, connected components, sweep | 0.00 | 0.37 | 0.02 | 0.60 | 0.79 |
| Gemma, without injection | 0.23 | 0.07 | 0.06 | 0.64 | n/a |
| Qwen3-8B, natural variants, without injection | 0.04 | 0.47 | 0.13 | 0.35 | n/a |
| Qwen3-8B, forced Prim, without injection | 0.01 | 0.20 | 0.18 | 0.62 | n/a |
| Qwen3-8B, mixed set, without injection | 0.01 | 0.22 | 0.14 | 0.63 | n/a |
| Qwen3-8B, disjoint noun set, without injection | 0.14 | 0.48 | 0.10 | 0.28 | n/a |

**Stability.** Between the natural pair set and a disjoint set, the top-10 Jaccard is 0.25 and the top-15 Jaccard 0.36, the Spearman correlation of exact effects over the 36 nodes measured in both top-60 sets is 0.55, and 4 edges are kept in both; the disjoint set has a smaller total effect (0.46 against 0.89 nats).

**Checks and departures.** CPU tests (positions, node keys, the motif on Gemma, sign-flip against brute force, Holm and McNemar against standard libraries, the random-pool ladder, the labeller) all pass; the labeller re-check passes 4 of 4 references per task, 22 of 22 gold rows and 30 of 30 iteration-7 programs. The probability of ' Kr' on the first twin template is 0.724, against 0.72 in iteration 7, and labels agree with the iteration-7 engine on 20 of 20 slots per task (11 of 20 token-identical on minimum spanning tree weight). The metric was read in fp32 from the final-norm state after the first GPU tests showed that bf16 logits are quantised to 0.125 to 0.25 nats at these magnitudes; suffix forwards with a reused prefix cache differ from full recomputation by bf16 path noise (mean 0.076 nats, against 0.061 nats between full forwards in two batch shapes), so every effect is measured against an unedited row of its own batch. The transcoder attribution graph was dropped at the precheck, because Qwen3-8B's per-layer transcoders (about 96.5 GB) exceed both the 23 GB GPU and the 57 GB of RAM, so no feature-level account exists. A second contrast of a ban against the twin on connected components was not run, three natural pairs with total effects below 0.2 nats were dropped from the disjoint set, two out-of-memory stops were resumed from files without dropping rows, and one GPU test (path patching against direct logit attribution) was

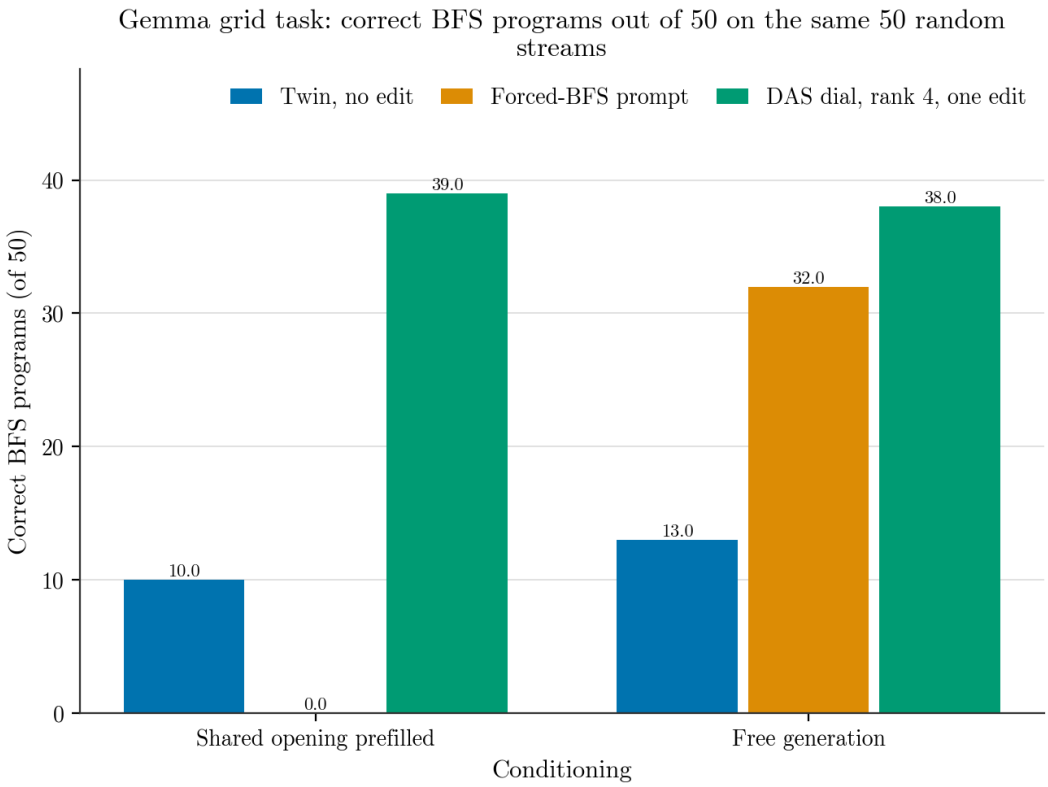
not run. Of 395 correct generated programs, none is labelled other (rule-of-three bound 0.0076). The nine-link hash chain orders every validation row after its commit and every steered program after its commit.

## 8.6 A fair test of Gemma's dial: prompting under matched conditioning, free generation and dormancy

Iteration 7 found a rank-4 DAS subspace at block 17 of gemma-3-4b-it that, edited once at the fork after the shared 32-token opening, raised correct BFS programs on the grid task from 11 of 50 to 42 of 50, and it compared this with a forced-BFS prompt sampled without that opening (35 of 50). The Gemma experiment ran the missing cells on the same 50 slots (five templates by ten streams, iteration 7's streams; T 1.0, top-p 1, at most 1,024 new tokens), giving the 2x2 of {prompt, dial} by {opening prefilled, free generation}. It added a dial applied in free generation at the fork's position in each free sample, placed by a rule committed in advance, and ran the dormancy checks at ranks 1, 2, 4 and 8 under the rule already applied to Qwen3-8B. One model, one task and one fork: every claim below holds at Gemma's grid fork.

**Table 402.** The 2x2 toward BFS on the same 50 slots (Wilson 95% intervals [8]).

| Cell | What | Attempted BFS | Correct BFS (95% CI) | Pass ' queue' at the fork |
|---|---|---|---|---|
| Twin, prefilled | Twin prompt with the opening prefilled | 15 | 10 (0.11 to 0.33) | 0.88 0.04 |
| Forced prompt, prefilled | Forced-BFS prompt with the opening prefilled | 50 | 0 (0.00 to 0.07) | 0.00 1.00 |
| Best wording, prefilled | The forced wording best by teacher forcing, with the opening prefilled | 50 | 0 (0.00 to 0.07) | 0.00 1.00 |
| Dial, prefilled | Twin with the opening prefilled, DAS rank 4, lambda 1 at the fork | 47 | 39 (0.65 to 0.87) | 0.82 0.00 |
| Twin, free | Twin, free from index 0 | 18 | 13 (0.16 to 0.40) | 0.88 0.00 |
| Forced prompt, free | Forced-BFS prompt, free from index 0 | 50 | 32 (0.50 to 0.76) | 0.64 0.00 |
| Dial, free | Twin, free, DAS rank 4, lambda 1 at the position the committed rule picks | 48 | 38 (0.63 to 0.86) | 0.78 0.00 |




**Gemma's dial against prompting under matched conditioning.** Correct BFS programs out of 50 on Gemma's grid task, on the same 50 slots of common random numbers, with the 32-token shared opening prefilled or with free generation. With the opening prefilled the forced-BFS prompt writes BFS in every slot but every program fails, because the prefill drives the fork to ' queue', an opening without a wall check; the DAS dial gives 39. In free generation the dial (38) and the committed forced-BFS wording (32) cannot be told apart (11 to 5 discordant slots, p 0.21), and the best of eight wordings reaches 49.

**Table 403.** Paired contrasts on correct BFS (exact McNemar; discordant slots, first cell only against second cell only).

| Contrast | Discordant | p | Paired difference (Newcombe 95%) |
|---|---|---|---|
| Dial against prompt, prefilled | 39 / 0 | 3.6e-12 (two-sided) | +0.78 (0.63 to 0.87) |
| Dial against prompt, free | 11 / 5 | 0.21 (two-sided) | +0.12 (-0.04 to +0.27) |
| Dial, prefilled against free | 2 / 1 | 1.0 | +0.02 (-0.06 to +0.11) |
| Prompt, prefilled against free | 0 / 32 | 4.7e-10 | -0.64 (-0.76 to -0.48) |
| Twin, prefilled against free | 1 / 4 | 0.38 | -0.06 (-0.16 to +0.04) |
| Interaction (slot bootstrap, 10,000) | n/a | n/a | 0.66 (0.50 to 0.82) |

**Why prompting fails under the prefill.** The committed prediction that the forced prompt with the opening prefilled would reach at least 40 of 50 correct BFS programs failed completely: it writes BFS 50 of 50 times and every program fails. The prefilled opening is unlikely under the forced prompt: its log probability is -11.3 to -14.1 nats under the five forced-BFS wordings (1e-6 to 1e-5), against -0.11 to -0.17 nats (0.85 to 0.90) under the twins and the forced-DFS prompts. Forcing it drives the fork token to ' queue' (probability 0.96 to 0.998 for all five wordings, 50 of 50 programs), and that opening writes a list-based BFS that carries the path, `queue = [(start, [start])]`, without checking whether start or goal is a wall. Every such program fails the hidden case of a one-by-one wall with a wrong answer, not a timeout, which corrects iteration 7's explanation that the 0 of 125 ' queue' programs failed by slowness. The teacher-forced account of the fork predicted this before any sample: 1.0 correct and 49.97 attempted BFS programs of 50. In free generation the forced prompt never writes the shared opening (index 32 is '[' inside `from collections import deque` in 46 of 50 programs), and the forced-DFS prompt with the opening

prefilled is natural and gives 50 of 50 correct DFS. So the prefill is a confound, but it hurts the prompt, not the dial: under matched prefill the dial beats prompting 39 to 0.

**The dial in free generation.** The rule that places the edit was committed before any edited sample. If the free twin reproduces the shared opening, the edit goes at index 32. Otherwise it goes at the first index from 8 to 96 after the body indent where BFS and DFS together hold at least half the probability, ' visited' holds at least 0.01, and no string naming an algorithm has appeared. 47 of 50 slots reproduce the opening and 3 never reach a qualifying position (73 and 7 over 80 slots with the held-out set); the literal rule of the plan would have fired at index 2 in every slot, as anticipated. No edited row differs from the free twin before the edit (0 of 480), and the edit's decode-time margin at index 32 is 0.42 nats above the prefilled path (12.09 against 11.67 on the first template), above the 0.3 tolerance; the prefilled and decoded paths agree on labels in 46 of 47 dial slots and 42 of 47 twin slots, one slot short of the committed 0.90 for the twin.

**Table 404.** Free-generation cells against the free twin (edit positions placed by the committed rule; one-sided McNemar against the free twin, Holm across the four DAS cells).

| Free cell | Correct target (of 50) | Pass | Against the free twin | Committed prediction |
|---|---|---|---|---|
| Twin | 13 BFS | 0.88 | n/a | 10.0 |
| DAS rank 4, lambda 1, toward BFS | 38 | 0.78 | 26/1, Holm p 8.3e-7 | 37.0 |
| DAS rank 4, lambda 0.5, toward BFS | 34 | 0.72 | 22/1, Holm p 5.7e-6 | 30.6 |
| DAS rank 1, lambda 1, toward BFS | 35 | 0.72 | 23/1, Holm p 4.5e-6 | 31.1 |
| DAS rank 1, lambda 0.5, toward BFS | 31 | 0.68 | 19/1, Holm p 2.0e-5 | 27.7 |
| DAS of seed 2, rank 4, lambda 1, toward BFS (optional) | 41 | 0.82 | 28/0, p 3.7e-9 | 33.5 |
| Random subspace, rank 4, lambda 1 (optional) | 11 | 0.84 | 1/3, p 0.94 | 6.9 |
| Difference in means at matched effect, toward BFS | 0 (attempted 49) | 0.00 | 0/13 | 1.4 |
| DAS rank 4, lambda 1, toward DFS | 47 DFS | 0.94 | 16/0, p 1.5e-5 | 47.0 |
| Difference in means at matched effect, toward DFS | 47 DFS | 0.94 | 16/0, p 1.5e-5 | 46.5 |

The free dial reaches 38 of 50 correct BFS against 13 for the free twin, and at this N it cannot be told apart from the prefilled dial (2 against 1 discordant slots). It holds on held-out templates. On the two templates outside the map's training set plus 30 fresh slots of them, it gives 33 of 50 against 11 for the twin (24 against 2 discordant, p 5.2e-6); on the 30 fresh slots alone 20 against 6; on the three training templates 25 of 30 against 8. Its pass cost against the free twin is 0.10 (Newcombe -0.22 to +0.01); of its 20 failing programs over the 80 slots, 15 are timeouts of slow BFS variants and 5 are wrong answers on a wall case. Toward DFS the dial and the forced-DFS prompt with the prefill both give 50 of 50 correct DFS, and the free dial and free difference-in-means edit 47 of 50; each beats its twin by 16 to 0 discordant slots.

**In free generation the dial does not beat prompting.** Against the committed free prompt (32 of 50) the dial's advantage is 11 to 5 discordant slots (p 0.21). Across eight forced-BFS wordings, an extension committed before its samples and selected post hoc, free prompting gives 14 to 49 correct

BFS (mean 39.5), and the best wording beats the dial (Table 405). This matches the prior from AxBench that prompting beats steering [34]. The slot-bootstrap interaction, (dial minus prompt, prefilled) minus (dial minus prompt, free), is 0.66 (0.50 to 0.82).

**Table 405.** Free prompting across eight forced-BFS wordings against the free dial (38 of 50). The third column gives dial-only and prompt-only slots.

| Wording, free from index 0 | Correct BFS (of 50) | Pass | Against the free dial |
|---|---|---|---|
| "Use breadth-first search (BFS) for this task." | 40 | 0.80 | 6 / 8, p 0.79 |
| "Your solution must rely on breadth-first search (BFS)." | 45 | 0.90 | 2 / 9, p 0.065 |
| "Use breadth-first search (BFS) throughout." | 49 | 0.98 | 1 / 12, p 0.0034 (Holm 0.024) |
| "Breadth-first search (BFS) is required here." | 14 | 0.28 | 25 / 1, p below 1e-5 |
| "... uses breadth-first search (BFS)." (the committed free prompt) | 32 | 0.64 | 11 / 5, p 0.21 |
| "... uses BFS." | 44 | 0.88 | 3 / 9, p 0.15 |
| "... uses breadth-first search." | 45 | 0.90 | 2 / 9, p 0.065 |
| "... uses a breadth-first search." | 47 | 0.94 | 2 / 11, p 0.022 |

**Dormancy.** By the committed rule shared with Qwen3-8B, the block-17 subspace is a dormant or off-manifold handle at every rank (Table 406); dormant means that the subspace lies in directions the model's natural states barely use, so that an edit there writes a state the model would not produce on its own. The flag comes through the retention arm alone: projected on the top 64 principal components of the twins' natural fork states, the subspace keeps 0.28, 0.23, 0.21 and 0.18 of its effect at ranks 1, 2, 4 and 8, while the residual orthogonal to them keeps 0.51 to 0.55. The reverse arm does not flag: with forced-BFS bases and twin sources, reverse recovery is 1.53 to 2.39, Haar random subspaces give about 0, and DAS sits at the 100th percentile. The DAS reverse moves the margin -5.58 nats at rank 4, more than the same-site full-state reverse (-2.62); a late-band reverse replacing blocks 17 to 33 moves it -10.4, and DAS is 0.54 of that. Natural variance is low at ranks 1, 2 and 4, as iteration 7 found (reproduced on its cached states to 1.1e-9), and above the random 95th percentile at rank 8. The flag is specific to a natural set of twins only: with the 40 forced sources added, retention rises to 0.62, 0.53, 0.46 and 0.39, the orthogonal residual carries nothing, and the variance share is about 50 times the random 95th percentile. The subspace is almost unused by the twins' own noun-driven variation, but it lies inside the variation that prompts induce. The account's prediction (reverse at least 0.5 and retention at least 0.5 at rank 4) fails on retention. Reverse generation (N 30) is undefined: replacing even the whole block-17 fork state restores 0 DFS programs, and all 30 programs in both reverse cells still take ' queue'.

**Table 406.** Dormancy at block 17 (seed 0; teacher forcing at one position; 768 held-out pairs for the forward recovery).

| Rank | Forward recovery (median ratio) | Reverse recovery, held out (95% CI) | Retention on 64 principal components of 120 twins (95% CI) | Residual orthogonal to them | Retention on 64 principal components of twins and 40 forced prompts | Natural share against random 95th percentile (percentile) | Largest seed cosine | Verdict |
|---|---|---|---|---|---|---|---|---|
| 1 | 2.04 | 1.53 (1.42 to 1.70) | 0.28 (0.27 to 0.29) | 0.51 | 0.62 | 0.00088 against 0.00096 (94th) | 0.36 | Dormant or off-manifold |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 2 | 2.36 | 1.74 (1.63 to 1.92) | 0.23 (0.23 to 0.24) | 0.52 | 0.53 | 0.00110 against 0.00176 (81st) | 0.44 | Dormant or off-manifold |
| 4 | 2.54 | 2.13 (1.98 to 2.36) | 0.21 (0.20 to 0.21) | 0.54 | 0.46 | 0.00182 against 0.00267 (73rd) | 0.52 | Dormant or off-manifold |
| 8 | 2.79 | 2.39 (2.21 to 2.66) | 0.18 (0.17 to 0.18) | 0.55 | 0.39 | 0.00530 against 0.00479 (98th) | 0.60 | Dormant or off-manifold |

The forward recovery reproduces iteration 7 (seed medians 1.98, 2.36, 2.58 and 2.81 there). Retention with 32 and 100 principal components is 0.10 and 0.34 at rank 4, and the dial's own version of retention (a class-mean edit over 72 training bases and 5 twins) is 0.15; planted checks pass (a direction inside the span keeps 1.000, an orthogonal one 0.000). The spread of each rank-4 direction's projection over that of random directions is 1.6, 0.81, 0.84 and 1.18 on the twins and 12.3, 5.6, 4.1 and 4.2 with the forced sources. Between seeds one direction is shared (cosines 0.50 to 0.60 at ranks 4 and 8) and the rest lie below 0.15; the fraction of the subspace inside the 64 twin components is 0.10, 0.07, 0.05 and 0.04.

**What the subspace carries.** An exploratory layer sweep, committed before its rows, separates the algorithm from the forced prompt's implementation. Transplanting the forced-BFS fork state into the twin at a single block sets the probability of ' queue' to 0.06 to 0.09 up to block 14, 0.43 at block 15, 0.55 at block 17 and 0.94 to 0.98 from block 26. In the other direction, the twin's state written into the forced prompt leaves ' queue' at 0.96 to 0.99 up to block 21, and it falls only from block 22 (0.88; 0.79 at block 24, 0.34 at 26, 0.06 at 33). The block-17 dial reaches the same BFS margin (about 12 nats) with ' queue' at only 0.006 to 0.07, its BFS mass sitting on a blank line and 'if', the openings that check walls first; a difference-in-means edit at the same margin puts 0.94 to 0.98 on ' queue' and destroys the programs (0 of 50 correct, pass 0.00). The subspace moves the algorithm, BFS against DFS, but not the forced prompt's choice of implementation.

**Table 407.** The committed scorecard of the Gemma experiment (the last prediction comes from teacher-forced numbers committed before any sample of its cells).

| Prediction | Observed | Holds |
|---|---|---|
| Forced prompt with the opening prefilled: at least 40 of 50 correct BFS | 0 of 50 (attempted 50, pass 0.00) | No |
| Prefilled dial and prefilled prompt not distinguishable (two-sided McNemar p at least 0.05) | 39 to 0, p 3.6e-12: the dial beats matched prompting | No |
| Free dial at least 30 of 50, pass at least 0.8 of the free twin's, above the free twin | 38 of 50; pass 0.78 against a floor of 0.70; p 2.1e-7 | Yes |
| Share of slots where the rule places an edit at least 0.80 | 0.94, all at index 32 | Yes |
| Labels agree between decoded and prefilled paths on at least 0.90 of slots, twin and dial | Twin 0.894 (42 of 47), dial 0.979 (46 of 47) | No (twin, by 0.006) |
| Account: reverse and retention at least 0.5 at rank 4; iteration 7: low natural variance at ranks 1, 2 and 4 | Reverse 2.13, retention 0.21; low variance at ranks 1, 2 and 4 | Account no; iteration-7 numbers yes |
| Free difference in means toward BFS: pass below 0.8 of the free twin's | Pass 0.00 | Yes |

| | | |
|---|---|---|
| Teacher-forced prediction of attempted BFS within 0.10 in every prefilled cell | Largest gap 0.068 (twin 0.23 against 0.30) | Yes |

**Engine, labels, audit and departures.** On an RTX 4090 against iteration 7's RTX 2000 Ada, the prefilled twin gives 10 of 50 (iteration 7: 11) with label agreement 0.94, below 0.95 (kernel sensitivity; every contrast stays inside this engine), the prefilled dial 39 (42) with agreement 1.00, and the free forced prompt 32 (35) with agreement 1.00; recomputed fork states differ from iteration 7's by up to 1.3% in norm (0.73 nats of margin). The frozen labeller labelled 1,320 programs, runtime and AST agree on 1,254, and a blind judge (gpt-4.1-mini, mostly from iteration 7's cache) settled the other 66; a blind re-gate agrees 30 of 30 with the labeller, and no program was truncated or labelled other. None of 913 correct programs is novel (rule-of-three bound 0.0033). A standard-library audit re-derives 139 quantities from the labels with 0 mismatches, an independent re-scoring of the 8 headline cells from raw text with its own test runner and classifier agrees on 400 of 400 programs and reproduces every headline test and the interaction, and within-slot permutation placebos give p below 0.05 in 0% to 3.1% of 1,000 permutations. Generation used DAS seed 0, class targets at ranks up to 4 came from iteration 7's commit, the decision rule was refined as described, the stage that pools known solutions was rebuilt in the novelty stage (2,158 graph, grid and path programs from 7 code datasets), and DAS refits and the label and random-initialisation controls were not re-run, so iteration 7's passed controls stand. Spend was USD 0.0042 and about 14 GPU minutes.

**Positioning.** CDAS already validates DAS-based steering by generation and compares it with prompting [112], and prefix steering shows that an intervention over a short span, even one token, retains much of full steering's control [140]. What this experiment adds is a causally localised subspace edited at a decision position read off the model's own trajectory, prompting cells matched to its prefill on the same slots, hidden-test correctness, and the dormancy and implementation-channel analysis. The permitted wording is: at Gemma's grid fork, a one-shot DAS edit at the fork works in free generation as it does after a prefill; prompting under the prefill fails because the prefill is off-policy for the prompt; in free generation the dial lies within the range of prompt wordings.

## 8.7 A temperature law composed over forks, tested on held-out families

Section 7.5 found on Qwen3-8B that dividing the fork logits by T predicted the temperature response on one unit of three. The temperature-law experiment turned this into a preregistered test across three families, with held-out units on which the law was never developed. For each unit (a task on one model) the pipeline found the forks blind: prefix resampling along stratified base programs, a preregistered Bayesian change-point rule, single-token refinement, and forced-token confirmation, with a second fork searched inside any branch whose algorithm stays impure. It then committed, by hash and before any temperature sample, the law's prediction of the attempted-label distribution at T 0.3, 0.7, 1.0 and 1.3 under top-p 1 and top-k 10, computed only from teacher-forced logits along the openings and at the forks and from the T 1 conditionals of the algorithm given each fork token. We call this the *composed law*: the share of each algorithm is the probability of each opening at temperature T, times the tempered distribution of the fork token given the opening, times the probability of the algorithm given that token, where a token followed by a second fork contributes the same mixture over the second fork's tokens. Four rivals were committed with it: a flat law (the fit twin's shares at every T), algorithm-level tempering (the twin's shares raised to the power 1/T and renormalised), a single-fork law without the second level, and the same law at the highest-entropy position. The same 100 fresh random streams were sampled at every temperature, and each predictor is scored by its mean absolute error, the mean over

temperatures of the total variation distance between predicted and observed label distributions. A unit passes at a mean absolute error of at most 0.10 with the entropy of the algorithm given the fork classes at most 0.3 bits; the claim passes on at least 70% of scored units including at least 3 held-out units, with at least 4 scored.

The units were three Qwen3-8B tasks used to develop the law (trapped rain water, valid anagram, minimum spanning tree weight), two held-out Gemma tasks (substring search and first and last position in a sorted array) and one held-out Llama-3.1-8B-Instruct task (the grid path). The two Llama bench units selected from its pilot (kth smallest and find duplicate) were dropped under the plan's time box.

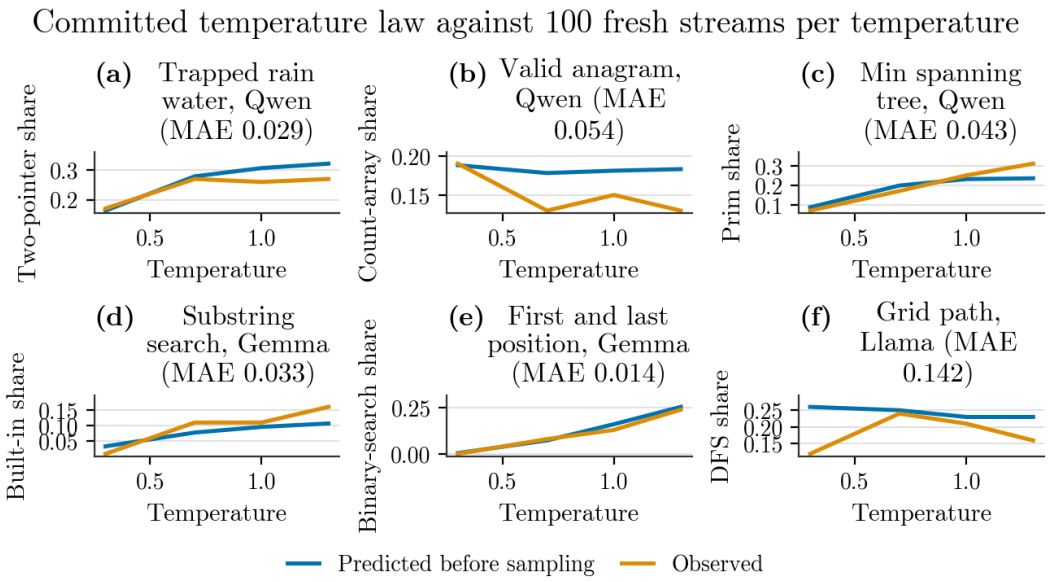
**Table 408.** The verdict of the temperature-law claim. Mean absolute error over the four temperatures; the lowest per row is the best predictor.

| Task | Family | Held out | Composed law | Flat | Algorithm-level tempering | Single fork | Highest-entropy position | Entropy of the algorithm given the fork classes (bits) | Pass |
|---|---|---|---|---|---|---|---|---|---|
| Trapped rain water | Qwen3-8B | no | 0.029 | 0.055 | 0.070 | 0.038 | 0.037 | below 0.001 | yes |
| Valid anagram | Qwen3-8B | no | 0.054 | 0.030 | 0.091 | 0.054 | 0.071 | below 0.001 | yes |
| Minimum spanning tree weight | Qwen3-8B | no | 0.043 | 0.093 | 0.058 | 0.043 | 0.045 | 0.004 | yes |
| Substring search | gemma-3-4b-it | yes | 0.033 | 0.043 | 0.023 | 0.033 | 0.036 | 0.001 | yes |
| First and last position | gemma-3-4b-it | yes | 0.014 | 0.073 | 0.018 | 0.051 | 0.017 | 0.001 | yes |
| Grid path | Llama-3.1-8B-Instruct | yes | 0.142 | 0.138 | 0.143 | 0.142 | 0.134 | 0.63 | no |

**Table 409.** The composed law's predicted share of the minority algorithm against the observed share, at T 0.3, 0.7, 1.0 and 1.3 (100 fresh streams per cell).

| Task (family) | Minority algorithm | Forks found blind | Predicted | Observed | Drop from T 1.0 to 0.3 (McNemar) |
|---|---|---|---|---|---|
| Trapped rain water (Qwen3-8B) | Two pointers | Token 18 (' n' to arrays, ' left' mostly to two pointers) and, inside ' left', token 19 (' =' to two pointers, '_max' to arrays) | 0.163 / 0.279 / 0.307 / 0.322 | 0.17 / 0.27 / 0.26 / 0.27 | 14:5, p 0.03 (observed 0.09, predicted 0.14) |
| Valid anagram (Qwen3-8B) | Fixed count array | Token 24 inside the counting branch (' =' to the 26-slot array, '_s' mostly to a hash map) and token 26 ('{}' or 'dict' to a hash map, '[' to the array) | 0.188 / 0.178 / 0.181 / 0.183 | 0.19 / 0.13 / 0.15 / 0.13 | n/a |

| | | | | | |
|---|---|---|---|---|---|
| Minimum spanning tree weight (Qwen3-8B) | Prim | Token 61 (' Kr' or ' Prim') | 0.087 / 0.198 / 0.231 / 0.235 | 0.07 / 0.17 / 0.25 / 0.31 | 18:0, p 3.8e-6 (observed 0.18, predicted 0.14) |
| Substring search (Gemma) | Built-in `haystack.index` | Token 31, after the length check ('for' 0.73 and a blank line 0.17 to the window scan; 'try' 0.10 to the built-in) | 0.034 / 0.078 / 0.096 / 0.107 | 0.01 / 0.11 / 0.11 / 0.16 | 10:0, p 0.001 |
| First and last position (Gemma) | Binary search | Token 12 ('first' 0.95, impure at 9 linear to 1 binary) and, inside it, token 24 ('for' to a linear scan, 'left', 'l' or 'low' to binary search) | 0.004 / 0.073 / 0.161 / 0.255 | 0.00 / 0.08 / 0.13 / 0.24 | 13:0, p 1.2e-4 |
| Grid path (Llama) | DFS | After a 52-token opening: ' visited' 0.45 and ' stack' 0.15 to DFS, ' queue' 0.25 to BFS; the two confirmed openings cover about 2% of the fit twin | 0.26 / 0.25 / 0.23 / 0.23 | 0.12 / 0.24 / 0.21 / 0.16 (adjudicated at T 1.3) | 18:9, p 0.06 |




**The composed temperature law against held-out sweeps.** Predicted (committed before sampling) and observed share of each task's minority algorithm at T 0.3, 0.7, 1.0 and 1.3, 100 fresh streams per temperature. Panels a to c are the Qwen3-8B units on which the law was developed; d and e are held-out Gemma units, where the law passes; f is the held-out Llama grid task, where the fork is reached through many openings and programs degenerate at T 1.3 (adjudicated labels), and every predictor fails. MAE is the mean total variation distance over temperatures.

**The held-out Gemma units pass, and the second level matters.** On substring search the blind locator puts the fork right after `if len(needle) > len(haystack): return -1`, and the law's error is 0.033. On first and last position the first fork, 'first', is impure, so the second-fork search ran inside it and found token 24; with it the law's error is 0.014, against 0.051 for the same law without the second level (bootstrap 95% CI of the difference -0.051 to -0.011) and 0.073 for the flat law. On both Gemma units the drop from T 1.0 to 0.3 is one-directional (10:0 and 13:0).

**The development units pass, and iteration 7's misses do not recur.** On trapped rain water the

locator rebuilt iteration 7's structure blind: two 18-token openings, ' n' (0.80) to the arrays and ' left' (0.20, or 0.62 in the second cluster) mostly to two pointers, with the second fork at token 19 separating ' left =' from ' left_max'. Iteration 7's error of 0.105 on 50 streams falls to 0.029 on 100 fresh streams. On minimum spanning tree weight the fresh-stream Prim share at T 1.0 is 25 of 100 against a prediction of 0.231 (binomial p 0.64), against iteration 7's 20 of 50 (Fisher p 0.09) and its twin's 19 of 100 (p 0.39): the lean that iteration 7 attributed to its streams was a property of those streams. Valid anagram passes, but the flat law beats the composed law (0.030 against 0.054); as in iteration 7, temperature barely moves this unit.

**The held-out Llama unit fails, for a reason outside the law's scope.** Llama's opening classifier failed its gate (the frozen labeller calls some path-storing BFS programs IDDFS), so every locator continuation was completed and labelled (3,276 continuations, on a grid thinned by the plan's cost rule). The locator found forks of the same kind as Gemma's, but Llama reaches them through many openings: the two confirmed openings cover about 2% of the fit twin, so the committed law is almost the twin's flat shares, and the entropy of the algorithm given the fork classes is 0.63 bits, above the 0.3 criterion. Observed BFS is 0.76, 0.66 and 0.61 and DFS 0.12, 0.24 and 0.21 at T 0.3, 0.7 and 1.0. At T 1.3 the programs degenerate: 72 of 100 run to the 1,536-token cap, the pass rate falls to 0.04, and the frozen labeller calls 87 of them other; that cell exceeded the preregistered 20% bar for other and was adjudicated by two judge families with a tiebreak (43 BFS, 15 DFS, 29 other), and the adjudicated column is primary (BFS 0.53, DFS 0.16, other 0.29). Every predictor fails on this unit (0.134 to 0.143; 0.28 to 0.29 with the frozen labels): no algorithm-level law predicts the collapse into unusable programs. Llama's top-k cells, built by keeping each top-p program up to its first token outside the top 10 and regenerating from there (shadow sampling), were reported but not used, because the construction failed its validation (1 of 20 directly sampled programs at T 1.3 token-identical).

**Verdict and rivals.** Five of six scored units pass (0.83, above 0.70), including both held-out Gemma units, but only 3 held-out units were scored where the claim needs at least 4, so the verdict is partial: the held-out criterion is underpowered, not met, and no held-out unit exceeds 0.2, so the disconfirming branch does not fire. The composed law has the lowest error on 3 of 6 units and beats the flat law on 4 of 6, tempering on 5 of 6, the highest-entropy rival on 5 of 6 and the single-fork law on 3 of 6, equalling it where no second fork exists; the exact one-sided sign test of the law against the flat law over units gives p 0.34. Slot-bootstrap 95% intervals of the error difference against the flat law exclude 0 on minimum spanning tree weight [-0.050, -0.025], first and last position [-0.081, -0.018] and substring search [-0.041, -0.002]. Algorithm-level tempering is a strong rival: it beats the law on substring search (0.023 against 0.033) and comes close on first and last position and minimum spanning tree weight, so much of the temperature response is sharpening or flattening the known mix. The law adds most where an impure first fork needs the second level (first and last position, trapped rain water) and where the default is not the share being sharpened (minimum spanning tree weight). Three placebos behave as they should. Permuting the predictions across temperatures puts the true error below the 5th percentile on trapped rain water, minimum spanning tree weight and first and last position (not on the two nearly flat units nor on Llama). Shuffling the conditional of the algorithm given the token across tokens raises the error (p at most 0.02 on four units, 0.10 on substring search). Shuffling temperature labels within slots rejects the drop test at the nominal rate (0.02 to 0.04). Replaying each slot's own uniforms through the committed token classes names the algorithm of 0.87 to 0.99 of Qwen3-8B slots on the opening (0.94 to 1.00 in Gemma), and the replay-implied shares lie within 0.00 to 0.04 of the observed cells on every passing unit, so the remaining error is the law's expectation against 100 realised streams.

**Is the instruct model's fork the base model's mean answer?** Qwen3-8B and Qwen3-8B-Base were teacher-forced on identical chat-templated ids at the forks and at 64 phrasing positions of 20 programs per task (Table 410). On the three collapsed tasks, post-training widened the fork margin by 2.11 to 13.41 nats while the entropy at phrasing positions moved by less than 0.5 nats (0.23 to 0.31; the base model is uniformly a little less certain), as committed; without chat tokens the gains are 2.69 to 11.05, so the result is not an artefact of the template. The Jensen-Shannon divergence between the two models is 6 to 10 times larger at the fork than at phrasing positions, and 1.5 to 1.9 times larger than at phrasing positions of the same base-entropy decile on the collapsed tasks (equal on minimum spanning tree weight, lower on trapped rain water); random positions show no excess. At phrasing positions the instruct token is the base model's top token at 89% to 95% of positions (the URIAL token categories [100]), but at the fork it is mostly the base model's second or third choice. The instruct default is therefore not the base model's mean: on minimum spanning tree weight the base model's fork distribution, pushed through the instruct model's conditional of the algorithm given each token, would give Prim 0.68 where the instruct model writes Kruskal 0.78; on trapped rain water, two pointers 0.37 against 0.29. Post-training moved and sharpened one token; the repertoire of ideas stayed the same.

**Table 410.** The base model against the instruct model at the fork (Qwen3-8B; five templates).

| Task | Role | Fork margin, instruct (nats) | Base | Gain | Gain without chat tokens | Phrasing entropy, base minus instruct [95% CI] | Jensen-Shannon divergence: fork / phrasing / entropy-matched / random |
|---|---|---|---|---|---|---|---|
| Grid path (BFS) | Collapsed | 4.9 to 7.4 | 2.9 to 3.3 | 3.27 | 5.31 | 0.31 [0.29, 0.33] | 0.51 / 0.05 / 0.27 / 0.04 |
| Connected components (union-find) | Collapsed | 10.5 to 18.7 | 2.4 to 2.7 | 13.41 | 11.05 | 0.28 [0.27, 0.28] | 0.31 / 0.04 / 0.20 / 0.09 |
| Shortest path (Dijkstra) | Collapsed | 4.9 to 6.2 | 3.4 to 3.6 | 2.11 | 2.69 | 0.23 [0.22, 0.24] | 0.44 / 0.04 / 0.29 / 0.05 |
| Minimum spanning tree weight | Mixed, descriptive | 0.5 to 1.5 | -0.8 | 1.84 | 1.69 | 0.34 [0.33, 0.35] | 0.34 / 0.05 / 0.34 / 0.02 |
| Trapped rain water | Mixed, descriptive | 1.0 to 2.5 | -0.2 to 0.1 | 1.53 | 1.91 | 0.21 [0.20, 0.22] | 0.11 / 0.03 / 0.20 / 0.04 |

**Ideas against phrasings (question 1a-c of the request).** Across 4,800 sweep programs no unit produced an algorithm outside its catalogue at any temperature. At T 0.3 each sweep is essentially one idea with one to four phrasing clusters (identifier-normalised AST 4-gram Jaccard 0.8): first and last position 100 linear scans in 2 clusters; minimum spanning tree weight 93 Kruskal in 4 clusters and 7 Prim in 2; trapped rain water 83 array programs in 1 cluster and 17 two-pointer programs in 1. Loosening T does not add ideas; it moves mass to the second known idea at the predicted rate and multiplies its phrasings: at T 1.3 minimum spanning tree weight has 66 Kruskal programs in 22 clusters and 31 Prim in 7, and first and last position 76 linear scans in 2 clusters and 24 binary searches in 8. Correct-only Hill-1 rises accordingly (1.29 to 2.06 and 1.00 to 1.68). Top-k 10 is token-identical to top-p 1 on every Qwen3-8B and Gemma cell except 6 of minimum spanning tree weight's 400 programs, which were regenerated after a divergence. The novelty audit read every correct program labelled outside the catalogue (10 distinct among 4,703 correct programs, checked against an 18,305-solution corpus built from the dataset dependency plus references, with two judge families passing 3 of 3 known-class controls and a manual read): 8 labeller misses of catalogue algorithms, 1 catalogue variant and 1 known brute force outside the

catalogue; none is novel (rule-of-three bound 0.0006).

**Validity checks and departures.** The hash chain verifies end to end, with each family's forks and predictions committed before any sweep sample of that family and four amendments (the adjudication rule, a locator fix for nested openings, content-addressed continuation ids, and a predictor fix for second forks not found) committed before the samples they govern; the held-out families' predictions were committed and swept before any Qwen3-8B sample. CPU tests pass, including a two-level tree exact to 1e-9 and top-k shadow sampling equal to direct top-k on 1,000 synthetic sequences; the change-point rule has a 0.00 false-positive rate on 200 null series and 0.865 power on planted jumps; the scoring code reproduces iteration 7's errors exactly (0.083, 0.105 and 0.043). Gemma prompt ids reproduce earlier hashes on all 10 templates, sweep streams are disjoint from every earlier key (0 overlaps among 1,300 seeds), and per-slot labels agree with earlier engines on 0.92 (trapped rain water, below the 0.95 expectation), 0.95 and 0.98 of Qwen3-8B slots and 0.98 and 0.97 of Gemma slots. Llama's bf16 twin on the grid (BFS 60, DFS 23, IDDFS 14, other 3) differs from iteration 2's twin, served by vLLM with 8-bit floating-point weights (BFS 77, DFS 20, other 3; Fisher p 0.015 on BFS), so every Llama contrast stays within this engine. The Llama label re-gate was weak (agreement 0.10 and 0.71 on the two bench units), so primary scoring stays on the frozen labeller with the judge-adjudicated labels as a sensitivity. The base sub-arm used the 20 stored iteration-7 programs for its three collapsed tasks, and frozen-labeller disputed labels count as other. Spend was USD 0.046.

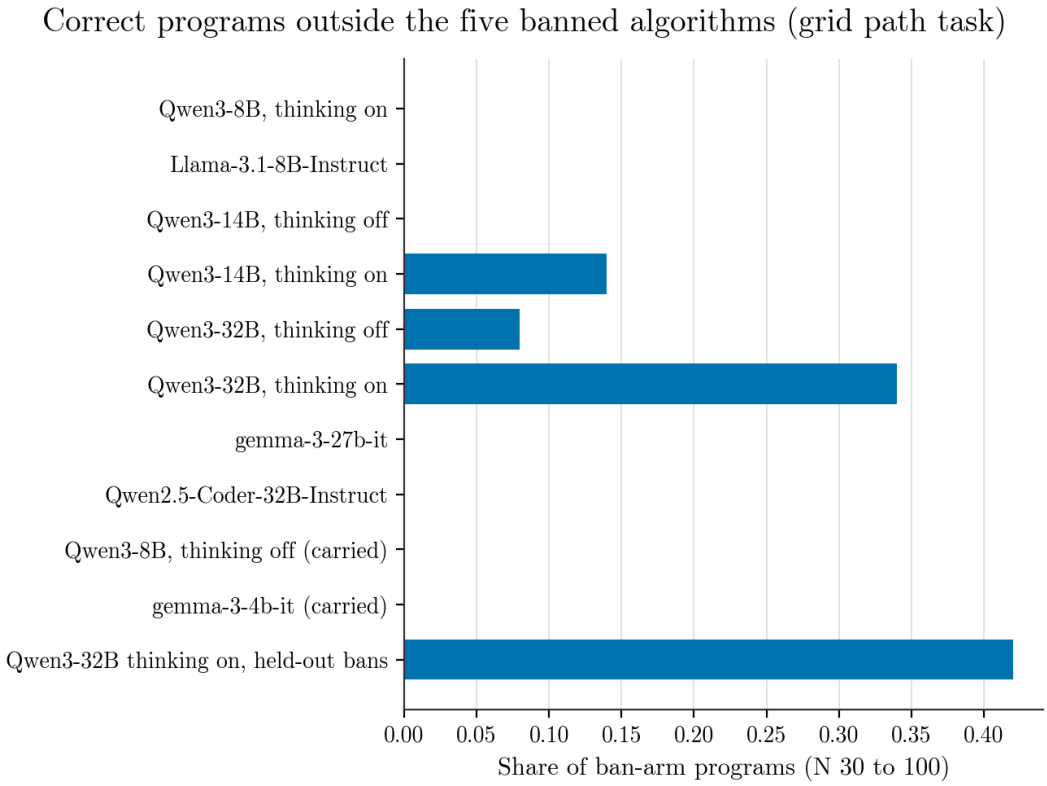
## 8.8 The top-five prohibition across ten configurations, and where the survivors land

The request asks what happens when a model's five most used algorithms are prohibited: do the survivors come from the edge of the banned set or from its opposite side? Earlier iterations found that Qwen3-8B (iteration 2) and gemma-3-4b-it (iteration 6) answer a ban of five names by reimplementing a banned algorithm under another name, so there were no survivors to place. The prohibition experiment first screened eight new open configurations plus two carried rows for one that obeys, then placed that model's survivors on its own manifold of ten forced algorithms, in actual activation coordinates. Each configuration's banned set is its own top five by implemented count, filled from BFS, DFS, greedy best-first, IDDFS and A*; because every baseline wrote only BFS and DFS, the banned set is the same five for every configuration. A correct survivor is a program under the ban that passes the 48-case hidden suite and whose adjudicated mechanism is outside the banned set: a non-banned catalogue algorithm or a named known algorithm outside the catalogue. Four commits fixed the configurations, prompts, selection rule, labelling and adjudication, geometry, landing nulls, robustness rule and the prediction (edge) before the samples they govern.

**Table 411.** The screen. N is the number of programs under the ban; shares carry Wilson 95% intervals.

| Configuration (engine) | Baseline (N) | Ban N | Correct survivors | Share (95% CI) | Share using a banned mechanism | Pass rate under the ban |
|---|---|---|---|---|---|---|
| Qwen3-8B, thinking on (local vLLM, FP8 weights) | BFS 30 of 30 | 30 | 0 | 0.00 (0.00 to 0.11) | 1.00 | 0.13 |
| Llama-3.1-8B-Instruct (CoreWeave, bf16) | BFS 77, DFS 20, other 3 | 50 | 0 | 0.00 (0.00 to 0.07) | 0.90 | 0.02 |
| Qwen3-14B, thinking off (DeepInfra, FP8) | BFS 94, DFS 6 | 50 | 0 | 0.00 (0.00 to 0.07) | 1.00 | 0.46 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Qwen3-14B, thinking on (DeepInfra, FP8) | BFS 96, DFS 4 | 50 | 7 | 0.14 (0.07 to 0.26) | 0.76 | 0.32 |
| Qwen3-32B, thinking off (DeepInfra, FP8) | BFS 81, DFS 19 | 50 | 4 | 0.08 (0.03 to 0.19) | 0.82 | 0.28 |
| Qwen3-32B, thinking on (DeepInfra, FP8) | BFS 91, invalid 9 | 50 | 17 | 0.34 (0.22 to 0.48) | 0.22 | 0.42 |
| gemma-3-27b-it (DeepInfra, FP8) | BFS 99 of 99 | 50 | 0 | 0.00 (0.00 to 0.07) | 1.00 | 0.86 |
| Qwen2.5-Coder-32B-Instruct (Cloudflare) | DFS 52, BFS 48 | 50 | 0 | 0.00 (0.00 to 0.07) | 0.94 | 0.40 |
| Qwen3-8B, thinking off (carried from iteration 2) | BFS 100 | 100 | 0 | 0.00 (0.00 to 0.04) | 1.00 | 0.68 |
| gemma-3-4b-it (carried from iteration 6) | n/a | 50 | 0 | 0.00 (0.00 to 0.07) | 1.00 | 0.38 |




**Who obeys a ban of five path algorithms.** Share of programs under a ban of each configuration's five most used path algorithms (BFS, DFS, IDDFS, A* and greedy best-first for every configuration) that pass the hidden tests with a mechanism outside the banned set. Only Qwen3-32B with thinking on clears the committed bar of 0.20 (17 of 50, Wilson 0.22 to 0.48), and it reaches 21 of 50 on four held-out ban templates (last bar). Bars at zero are configurations with no correct survivor.

**Who obeys.** Almost no open configuration obeys a ban of five names. The committed rule (share at least 0.20) selects one: Qwen3-32B with thinking on, 17 of 50 correct survivors, namely unit-weight Dijkstra 6, Bellman-Ford/SPFA run on the grid 7, a frontier search with random priorities 2, a random walk with restarts 1 and wavefront propagation 1. Thinking is the switch, and size matters: the same 32B weights with thinking off reach 0.08, Qwen3-14B reaches 0.14 with thinking on and 0.00 without, and Qwen3-8B with thinking on reaches 0.00 (it writes BFS 11 and DFS 19 under the ban). Disobedience mostly reimplements a banned mechanism without naming it: among programs that use a banned mechanism, 55% to 100% contain no alias of any banned algorithm in the code, the lesson of iteration 2 now across families. Obeying is expensive: the thinking 32B model's mean reasoning grows from 6,864 to 9,090 tokens under the ban, and 23 of 50 ban

answers hit the 9,728-token cap (7 of 100 at baseline), counted as non-survivors. The ban costs correctness elsewhere too: Llama's pass rate falls to 0.02 and Qwen3-8B with thinking on passes 0.13.

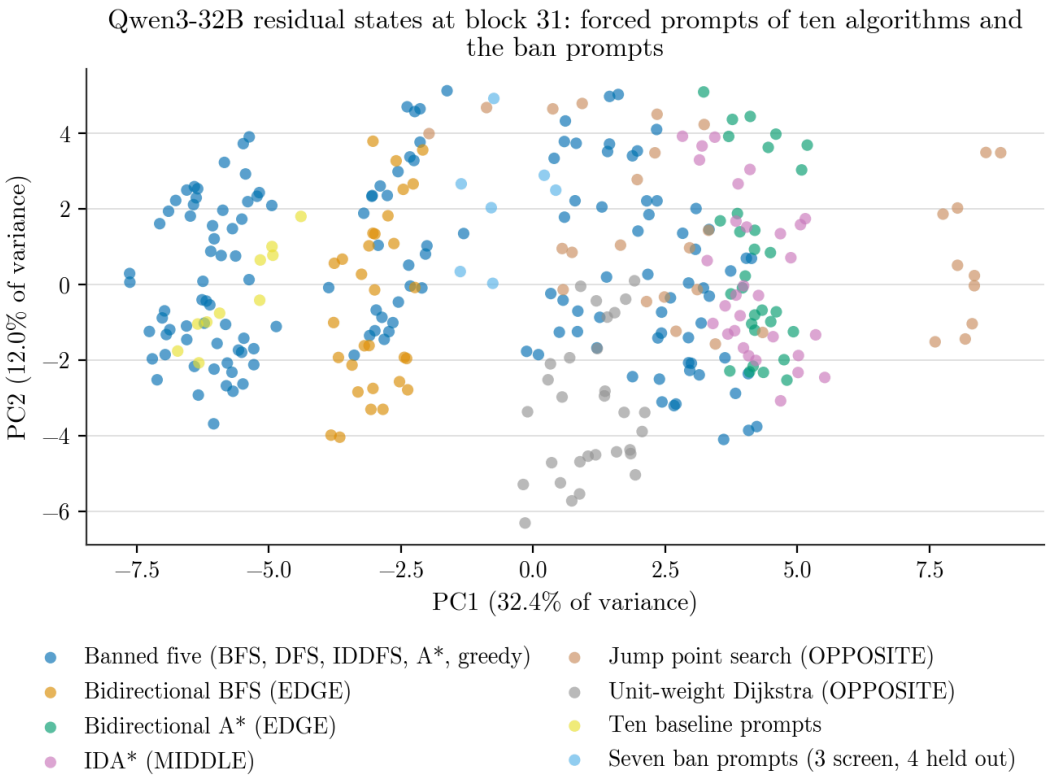
**How robust "obeys" is.** The committed definition counts a correct program as a survivor if its adjudicated mechanism is a non-banned catalogue algorithm or a named known algorithm outside the catalogue. The strictest reading drops every survivor outside the catalogue whose adjudicated name, or any novelty judge's name for it, mentions BFS, DFS or SPFA. It leaves 11 of 50 (0.22) on the screen and 14 of 50 (0.28) on the confirmation, still above the bar. Counting catalogue algorithms only gives 6 of 50 (0.12) and 9 of 50 (0.18), below it. "Obeys" therefore rests on counting genuinely different known algorithms, mostly Bellman-Ford on the grid, as obedience; those programs sit outside the ten-algorithm manifold and are not part of the landing test.

**The manifold of the model that obeys.** States are residual-stream block outputs of Qwen3-32B, read from a checkpoint whose weights are quantised to 4-bit NormalFloat and averaged over the task specification tokens, for 300 forced prompts (ten algorithms by ten templates by three aliases); the decoder is truncated to 48 of 64 blocks, exact for every block used. Every block from 19 to 45 classifies the forced algorithm perfectly (grouped 10-fold balanced accuracy 1.00, holding out template and alias groups), and the committed ceiling rule chose block 31. There the first two principal components explain 32.4% and 12.0% of the variance; classical MDS has stress-1 0.42 on the 300 points and 0.28 on the ten centroids; and the Isomap neighbour graph has 8, 3 and 1 components at k 5, 10 and 15, so the committed primary k is 15. We write g for an algorithm's geodesic distance to the banned five: the smallest, over the five banned algorithms, of the mean geodesic distance between its points and theirs. A non-banned algorithm is called edge when its g lies below the median g of the five non-banned algorithms, opposite when it lies above that median, and middle at the median; the landing statistic is the mean g over the correct catalogue survivors (Table 412).

**Table 412.** Geodesic classes of the non-banned algorithms at block 31, k 15, and their stability under 1,000 bootstrap resamples of paraphrase groups.

| Non-banned algorithm | Geodesic distance to the banned five | Class | Share of resamples in the same class | Share of resamples called edge |
|---|---|---|---|---|
| Bidirectional BFS | 10.9 | Edge | n/a | 0.975 |
| Bidirectional A* | 11.6 | Edge | n/a | 0.936 |
| IDA* | 13.3 | Middle | n/a | n/a |
| Jump point search | 15.9 | Opposite | n/a | n/a |
| Unit-weight Dijkstra | 21.6 | Opposite (the farthest) | 1.00 | 0.00 |

The split-half Spearman correlation of the 10 by 10 geodesic matrices is 0.58 at k 5, 0.95 at k 10 and 0.98 at k 15. The ten centroids carry no loop (the persistence of their one-dimensional loops, measured by persistent homology, is 0.00 against a 95th percentile of 1.19 for covariance-matched Gaussian clouds), and the 300 points carry little (1.59 against 1.29), so edge and opposite are classes of geodesic distance on a manifold that is not a ring.




**Forced prompts and ban prompts in Qwen3-32B's activation space.** Residual-stream states of Qwen3-32B (weights quantised to 4-bit NormalFloat) at block 31, averaged over the task specification tokens, projected on their first two principal components (32.4 and 12.0 per cent of variance). Each forced-algorithm cloud has 30 prompts (ten templates by three aliases); the banned five share one colour. The ten baseline prompts sit with the BFS and DFS clouds; the seven ban prompts move toward the middle; unit-weight Dijkstra, the only catalogue algorithm the model writes under the ban and the farthest non-banned algorithm from the banned five by geodesic distance (21.6 at k 15), forms its own cloud at the bottom. Coordinates are actual activation coordinates; the plot is descriptive, not causal.

The figure plots the 300 forced states in the first two principal components at block 31 (the banned five pooled into one colour, the five non-banned algorithms each in their own), with the ten baseline prompts and the seven ban prompts projected on the same axes. The baseline prompts sit with the BFS and DFS clouds at the left; the ban prompts move right, toward the middle of the plot; unit-weight Dijkstra, the only catalogue algorithm the model writes under the ban, forms its own cloud at the bottom, away from the ban prompts.

**Landing.** The screen's correct catalogue survivors are 6 of 6 unit-weight Dijkstra, so the landing statistic is 21.59, and the committed verdict is opposite: the preregistered edge prediction is not supported. Under the uniform null and the baseline-frequency null, p for opposite is 0.0001; under a null weighted by how often each algorithm is written correctly when forced (the correctness-filter confound), 0.057; under the exact permutation of algorithm identities, 0.20, which is that null's floor, because one landing algorithm cannot reach significance at the level of algorithms. The attempted-survivor, Euclidean-centroid and standardised-coordinate variants also give opposite. Robustness fails: opposite holds in 10 of 15 cells (every block from 29 to 33 at k 10 and k 15), but at k 5 unit Dijkstra is middle in all five blocks, against a committed bar of 12 of 15.

**Held-out confirmation.** The fourth commit froze the block, k, the point set, the geodesic matrices, the classes, the null code and the verdict rule, and the thinking 32B model was then sampled 50 times with four unseen ban templates. Survivors reach 21 of 50 (0.42, Wilson 0.29 to 0.56; 0.34 and 0.28 under the two stricter readings), with pass rate 0.58, banned-mechanism share 0.34 and truncation 0.20; by template, the one that asks to solve the task without using the named algorithms gives 8 of 12 survivors, and the one that states only that they are not allowed here gives

3 of 13. The catalogue survivors are 9 of 9 unit-weight Dijkstra, so the statistic is again 21.59: p for opposite is below 1e-4 under the uniform and frequency nulls (0 of 10,000 draws), 0.017 under the correctness-weighted null and 0.20 under the permutation null. The verdict repeats, so the result is confirmed under the committed rule; it replicates on new templates but adds no power at the level of algorithms.

**Table 413.** Is the geometry from 4-bit NormalFloat weights sensitive to precision? Qwen3-8B forced-prompt states at three precisions (its own chosen block is 17).

| Precision against bf16 | Spearman of the 10 by 10 geodesic matrices (k 10) | Class agreement | State cosine at every block |
|---|---|---|---|
| int8 | 0.954 | 5 of 5 | at least 0.999 |
| 4-bit NormalFloat (NF4) | 0.916 | 5 of 5 | at least 0.997 |

Both lie above the 0.8 flag, and in Qwen3-8B's bf16 geometry unit-weight Dijkstra is also opposite the banned five.

**Other descriptive geometry.** In the full-dimensional states at block 31, the ten baseline prompts sit nearest the DFS and BFS centroids, while the three screen ban prompts and the four held-out ban prompts move toward the IDDFS and greedy best-first centroids, two of the five named and banned algorithms, which is priming as in iteration 2; unit-weight Dijkstra is the farthest of the ten centroids from both sets of ban prompts (rank 10 of 10). Per-program states of 59 programs of the thinking 32B model (all 29 valid ban programs and 30 sampled baseline programs, with reasoning capped at 4,096 tokens) make the adjudicated label decodable at the first token that signals the class (balanced accuracy 0.97) and the token before it (1.00), positions that depend on the label, but only at 0.53 at the label-independent 16th code token, against a chance level of 0.25. The unit-Dijkstra survivors sit there nearest the greedy best-first cloud (geodesic 38 to 79), with which they share the heap signature token, and the off-catalogue survivors (Bellman-Ford, random walk, random-priority search) sit far from every class cloud (231 to 394). Iteration 7 found that pushes inside the manifold moved 0 of 300 programs, so none of this geometry is causal.

**Novelty, audits and departures.** Of 661 correct programs across all arms, 27 were adjudicated other or flagged for a manual read; each was shown to two blind judges with its three nearest known solutions (tree edit distance on normalised ASTs over 1,987 grid and graph search programs). No judge voted novel; 16 are known by both judges, and for the other 11 deepseek-chat-v3.1 says known while kimi-k2 gave no verdict (rule-of-three bound 0.45%). A standard-library re-derivation reproduces every share, interval, geodesic (maximum difference 9e-13), landing statistic, p-value and verdict with 0 mismatches; a second audit through different code paths reproduces them too, and gives the exact uniform p for opposite as 6.4e-5, (1/5) to the sixth. Its placebos explain why the per-program nulls overstate the evidence: random survivor labels on the real geometry reject at 5.0%, nominal, but shuffling the algorithm labels of the 300 points still rejects in 36% of shuffles, because all six survivors are one algorithm and the test mostly asks whether that algorithm happens to have the largest geodesic distance. The algorithm-level p of 0.20 is the honest figure for the landing claim. The run used an RTX 2000 Ada with 16 GB and sampled Llama through OpenRouter, appending a no-think tag where the provider ignored the thinking switch. Before the first commit it replaced gemma-3-12b-it, whose endpoints timed out, with gemma-3-27b-it. A judge that failed its gate (gemini-2.5-flash-lite, 7 of 10) gave way to deepseek-chat-v3.1 (9 of 10) and kimi-k2 (10 of 10). Qwen3-8B's thinking ban arm was cut to its committed floor of 30 because each row took 57 s. Spend was USD 1.53.

## 8.9 Comparisons across this iteration's experiments

**Temperature acts through one or two tokens and only reweights known ideas.** Three experiments measured temperature at the fork from different sides. The fork-dosing experiment predicted the algorithm mix of every dosed cell within a mean absolute error of 0.010 to 0.047 on five units, the temperature-law experiment predicted global temperature sweeps within 0.014 to 0.054 on five of six units, two of them in a held-out family, and both found nothing outside the catalogue. Where the twin is already mixed, any knob that reaches the fork does about as well as the dial (min-p on minimum spanning tree weight, global T 1.3 in Gemma); the dial clearly earns its place on valid anagram, where it raises correct-only Hill-1 from 1.78 to 2.78. On the collapsed connected components task it beats every comparator family only after the amendment removed the tokens that break programs (Hill-1 1.54 against 1.17 for the best comparator, lower bound of the difference $+0.01$), still short of two ideas. Where the twin has collapsed, the fork's top 20 candidates hold at most about 19% for the alternative, so no dose at that token reaches two ideas without breakage. The forced branches show what lies beyond: one known idea per unit on three units, reachable at T 1 once in 2 million to 50 million samples. The base sub-arm explains why: post-training widened the fork margin by 2 to 13 nats on the collapsed tasks while leaving the phrasing positions almost untouched.

**Internal handles act at the fork, and none is a certified variable.** Table 414 sets the internal results side by side. On connected components three different edits at late fork-position sites all write DFS: the rank-4 DAS map at block 22 (29 of 30 correct), the top five circuit components at dose 2 (29 of 30) and the matched random five at dose 2 (30 of 30), while a random rank-4 subspace does nothing (0 of 30). The fork margin there is easy to move with a large enough push at the right place, which is why the controls that ask whether the handle is the model's own variable (permuted labels, an untrained network, matched random components) fail or saturate. The circuit separates from random components only at a dose small enough to leave room (24:0 at dose 0.5 on connected components; 6:1 on minimum spanning tree weight). In Gemma the dial is a dormant handle by retention but lies inside the variation prompts induce, works in free generation and ties prompting there.

**Table 414.** The internal results of iteration 8 side by side.

| Question | Qwen3-8B | Gemma | Source |
|---|---|---|---|
| Certified low-rank variable at the fork under one protocol | No: label controls 0.63 to 0.87, untrained network 0.69, reverse -2.72 | No: label controls 0.11 to 0.20, untrained network 0.41, retention 0.12 | Section 8.4 |
| Steering handle at the fork in generation | Toward DFS 29 of 30 correct; toward BFS 29 of 30 written, 5 correct | Toward BFS 22 of 30 correct (twin 6) | Section 8.4 |
| Task band | Sufficient handle; untrained network passes (0.18); routes DFS sources to BFS | Not run | Section 8.4 |
| Do random pushes act through the fitted subspace? | No: clamp removal -0.01; prospective Spearman 0.09 | n/a | Section 8.4 |
| Validated circuit nodes (both tests, top 10) | 9 (minimum spanning tree), 8 (connected components) | 7 (iteration 7) | Section 8.5 |
| Top five against matched random five in generation | 7:1 at dose 2 (p 0.070), 0:1 on connected components; 24:0 at dose 0.5 (post hoc) | Doubled correct BFS (iteration 7) | Section 8.5 |

| | | | |
|---|---|---|---|
| Shape of the circuit | Layer-0 condition MLP plus a late fork stack, depth about 0.86 | The same shape, depth 0.69 | Section 8.5 |
| Dial against prompting, prefill matched | n/a | 39 to 0 for the dial | Section 8.6 |
| Dial against prompting, free | n/a | 38 against 32 (p 0.21); best wording 49 | Section 8.6 |
| Dormancy | Off-manifold at block 22; passes at block 24 (post hoc) | Dormant by retention at every rank; inside the prompt-induced variation | Sections 8.4 and 8.6 |

**The prohibition answers the request's question on one model.** Only Qwen3-32B with thinking on obeys a five-name ban often enough to place survivors, and its catalogue survivors are all unit-weight Dijkstra, the farthest non-banned algorithm from the banned five, on the screen and on held-out templates. The ban prompts themselves move toward two of the banned algorithms, not toward where the survivors land. With one landing algorithm the evidence at the level of algorithms is p 0.20, so the result reads as a description of one model, not a law.

**No correct novel algorithm anywhere.** Table 415 collects the novelty audits of iteration 8.

**Table 415.** Novelty audits of iteration 8.

| Artifact | Correct programs | Audited as possibly outside the catalogue | Final classes | Novel |
|---|---|---|---|---|
| Record audit | 162 adjudicated kth smallest programs | 35 relabelled | Counting sort in dictionary form | 0 |
| Fork-dosing experiment | 5,543 | 2 | Counting sort in dictionary form (catalogue variant) | 0 |
| DAS experiment | 1,857 | 0 | n/a | 0 |
| Circuit experiment | 395 | 0 | n/a | 0 |
| Gemma experiment | 913 | 0 | n/a | 0 |
| Temperature-law experiment | 4,703 | 10 distinct | 8 labeller misses, 1 catalogue variant, 1 known brute force | 0 |
| Prohibition experiment | 661 | 27 | Known algorithms (Bellman-Ford/SPFA on the grid, random walk, wavefront, random- or coordinate-ordered frontier search) | 0 |

## 8.10 What iteration 8 established, what it ruled out, and what it abandoned

**Table 416.** Verdicts of the claims of iteration 8.

| Claim | Committed verdict | Deciding evidence |
|---|---|---|
| Fork-dosing claim | Not supported as committed | Composition within 0.010 to 0.047 on 5 of 5 units and mixed-unit targets met; on connected components Hill-1 1.24 and 1.60 against the 1.8 target, pass cost 0.14 at the high dose; best fork arm minus best comparator +0.06, p 0.30 |

| | | |
|---|---|---|
| Fork entropy prediction | Opposite holds | The fork lies above the 90th percentile of per-token entropy in every sample; it is the highest-entropy position on 44 of 44 Gemma slots but 2 of 50 on connected components |
| Reachability claim | As predicted | One known idea reachable only by forcing on three units, at 1 in 2 million to 50 million samples at T 1; 0 novel |
| Matched DAS claim | Disconfirmed | Both forks fail the label control (0.11 to 0.87) and the untrained-network control (0.41 and 0.69); off-manifold overshoot at both forks; the band is a sufficient handle that fails the label control |
| Circuit claim | Partial on both contrasts (two-sided McNemar primary) | 9 of 10 and 8 of 10 nodes validated; top five +21 and +28 correct programs, but the matched random five also flip at dose 2 (7:1, p 0.070; 0:1); motif differs on condition share and depth |
| Gemma dial claim | Prefill-matched prompting fails (0 of 50); the dial works in free generation (38 of 50); dormant by retention | Dial against prompt 39:0 prefilled, 11:5 free (p 0.21); best free wording 49 of 50; retention 0.18 to 0.28 |
| Temperature-law claim | Partial | 5 of 6 units pass including both held-out Gemma units; only 3 held-out units scored (4 required); Llama fails (0.142) |
| Top-five prohibition claim | One configuration obeys; survivors land opposite, not at the edge; not robust; confirmed on held-out templates | 17 of 50 and 21 of 50 survivors; unit-weight Dijkstra 6 of 6 and 9 of 9; p 0.20 at the level of algorithms; 10 of 15 robustness cells |

**Established.** Temperature applied at the fork moves the algorithm mix exactly as the fork's logits and token map predict (mean absolute error 0.010 to 0.047 on five units in two families), and a law composed over one or two forks found blind predicts global temperature sweeps on held-out Gemma units (0.033 and 0.014) as well as on Qwen3-8B (0.029 to 0.054). The fork is a high-entropy position relative to the near-deterministic code around it, though its entropy can be small in absolute terms (0.13 nats on connected components). Post-training widens the fork margin by 2 to 13 nats on the collapsed tasks and leaves phrasing positions almost unchanged, so the instruct default is a sharpened second or third choice of the base model, not its mean. A rank-4 linear edit at the fork moves the written algorithm in both families, and under one protocol neither family's handle is a certified variable: permuted labels, an untrained network or matched random components reproduce most of each effect. Late fork-position components of Qwen3-8B (validated MLPs at layers 26, 27, 28, 30 and 32, and heads such as head 14 of layer 33 and head 28 of layer 34) carry the fork logit, with a layer-0 MLP at the condition span injecting the instruction, the same shape as Gemma's circuit. In Gemma, prompting with the dial's prefill fails because the prefill is off-policy for the forced prompt and selects a BFS implementation that skips the wall check; in free generation the dial and prompting are within each other's range. Among ten open configurations, only a 32B model with thinking on obeys a five-name ban often, and its catalogue survivors are all the algorithm farthest from the banned five on its manifold.

**Ruled out.** A certified low-rank variable for the algorithm at the fork under the committed controls, in either family. Temperature at the fork as a route to two ideas on a collapsed task without breakage. The edge prediction for the survivors of the top-five prohibition on the model that obeys. The explanation of iteration 7 that ' queue' programs in Gemma fail by slowness (they fail a wall case). Iteration 7's lean of the minimum spanning tree sweep (a property of its streams). Fitted subspaces as the channel of iteration 4's random pushes (clamping removes nothing; the

prospective association is 0.09).

**Dead ends.** The blind locator at T 1 cannot find the fork of a collapsed task, because resampling never leaves the default. The window of 16 tokens at the boundary units breaks the programs before it can test anything. Steering at dose 2 saturates both the circuit and its random control. The transcoder attribution graph was dropped again, now because Qwen3-8B's transcoders need about 96.5 GB. The two Llama bench units of the temperature law and the band and minimum spanning tree sites of the DAS experiment were dropped under time boxes.

**Not run.** The task band of Gemma and DAS at minimum spanning tree weight; a ban-against-twin contrast for the Qwen3-8B circuit; the engine check of the DAS experiment against iterations 6 and 7; a feature-level account at either fork; plan resampling and the placement of top-five survivors in DAS coordinates on the grid task, carried from iteration 7.

### 8.11 What the next iteration takes from this one

Four leads remain. First, the controls show that a large enough push at a late fork-position site flips a collapsed fork whatever its direction, so a variable, if there is one, has to be sought where the full state carries the class and with doses below saturation: the block-24 fit of Qwen3-8B passes every off-manifold control and fails only specificity, and the circuit separates from random components at dose 0.5. Second, the composed temperature law passed on held-out Gemma units but scored only three held-out units, and it fails where a family reaches its fork through many openings (Llama); more held-out units, and a law that sums over openings, are what would settle it. Third, the top-five prohibition placed survivors on one model with one landing algorithm, which cannot reach significance at the level of algorithms; tasks where an obeying model writes several catalogue survivors are needed for the edge-against-opposite question. Fourth, no route has produced a correct novel algorithm. The forced branches show that the known ideas a model does not write sit at deep ranks of the fork. Evolutionary search on a task whose catalogue already covers the known techniques remains the only route left untried.

## What we have learned so far

After eight iterations the run has a firm picture of the collapse, a measured place where the algorithm is decided in generation, a law that predicts what temperature does there, internal handles in two families that move the algorithm without being the model's own variable, and no correct novel algorithm. Qwen3-8B [5] writes BFS and Dijkstra almost without exception on the two path search tasks and one algorithm on each of 13 further coding tasks (modal share 1.00 on 12, 0.98 on one). The collapse is not universal: three of 15 new tasks are already mixed in the instruct twin (trapped rain water, minimum spanning tree weight and valid anagram, with minority shares of 0.13 to 0.36), and on two of them the instruct model is more diverse than its base model. Llama-3.1-8B-Instruct [22] attempts more algorithms but its correct grid programs are BFS 20 of 20; gemma-3-4b-it [76] writes DFS in about 70% and BFS in about 30% of programs for one grid prompt; Qwen2.5-7B-Instruct [124] shares most of Qwen3-8B's defaults. On the grid task, eight more open configurations (Qwen3-14B and Qwen3-32B with thinking on and off, Qwen3-8B with thinking on, gemma-3-27b-it, Qwen2.5-Coder-32B-Instruct and Llama-3.1-8B-Instruct served in bf16) write BFS or DFS at baseline, apart from a few programs labelled other or invalid.

The algorithm is decided at one or two generated tokens after a shared opening, the fork, where one can be found. On the mixed units programs share an opening of 18 to 61 tokens and the next token settles the algorithm: ' Kr' or ' Prim' on minimum spanning tree weight, ' n' or ' left' on

trapped rain water, ' visited' against a blank line or ' queue' in Gemma. The entropy of the algorithm given the fork token's class is at most 0.19 bits as iteration 7 measured it, and at most 0.004 bits under iteration 8's blind locator, which adds a second fork inside impure branches (' left =' against ' left_max' on trapped rain water; 'for' against 'left' on first and last position in Gemma). A blind locator by prefix resampling finds these forks on mixed units in Qwen3-8B and Gemma, but not on a collapsed task, where resampling at T 1 never leaves the default and the fork of connected components (' def find' against ' def dfs') was found only under pushed or banned prompts; in Llama the grid fork exists but is reached through many openings. The fork is the highest-entropy position of the near-deterministic code around it in every sample, though its entropy can be small in absolute terms (median 0.13 nats on connected components). On kth smallest a ban decides before the first generated token, and on shortest path no single token decides within 60 tokens. The effect of a random push is already expressed at the fork: a teacher-forced readout of the fork margin matches which of 40 pushes change the algorithm on connected components (Spearman 0.80; the 48-token opening log likelihood ratio 0.87), resting on how much probability stays on the default fork tokens, while no property of the push direction predicts which pushes move it (axis 0.17 and 0.01; plan-subspace projection 0.03).

Temperature acts on the algorithm only through those one or two tokens, and only reweights ideas the fork already weighs. A law composed over the forks, committed before sampling from teacher-forced logits, predicted the minority share across T 0.3 to 1.3 within a mean total variation distance of 0.029 to 0.054 on three Qwen3-8B units and 0.033 and 0.014 on two held-out Gemma units. The second fork was needed on first and last position (0.014 against 0.051 without it). The law failed on Llama's grid task (0.142), where the fork is reached through many openings and programs degenerate at T 1.3, and with three held-out units scored instead of four the claim stays partial. Applying temperature only at the fork moves the mix exactly as predicted (0.010 to 0.047 on five units), raises correct-only Hill-1 on valid anagram from 1.78 to 2.78 at a 2-point pass cost, and on the collapsed connected components task reaches DFS in at most 18% of programs, because the alternative holds at most about 19% of the mass among the fork's 20 candidates and several candidates break the program. Forcing each of the fork's top 64 tokens reaches one more known idea on three units (BFS on connected components, Prim with an array, an alphabet scan), each about once in 2 million to 50 million samples at T 1. Lowering T from 1.0 to 0.3 moves the output toward the default (10:0 to 18:0 discordant slots on minimum spanning tree weight and the two Gemma units, 14:5 on trapped rain water), and top-k 10 gives the same programs as top-p 1 because every fork token lies in the top 10. On the collapsed tasks the fork margins of 4.85 to 17.04 nats put the alternative out of reach, which is why iterations 1 to 4 changed the algorithm in 0 of 20 seeds under a shared seed while phrasings multiplied. Post-training made those margins: on the three collapsed tasks the instruct model's fork margin exceeds the base model's by 2.1 to 13.4 nats while phrasing entropy moves by under 0.5 nats, and the instruct default is the base model's second or third choice at the fork, sharpened, not its mean answer. The answer to question 1a-c of the request is that a model has few ideas at tight temperature and the same few at loose temperature, in proportions its fork logits fix, with more phrasings of each as T rises.

A random displacement of the task specification states at early to middle blocks changes the algorithm in Qwen3-8B, and its size has to be counted in the right unit. At the absolute unit of iteration 4 bf16 escapes on 26 of 40 directions on connected components and 6 of 40 on shortest path, close to the 8-bit engine's 29 and 8. Iteration 5's weak plan noise came from its relative unit, not its seeds (unit effect $+0.45$ [0.35, 0.55], seed effect $+0.03$), with profile and energy each carrying part of the effect and the same energy at shuffled positions moving almost nothing (21 of 50 against 4 of 50 on fresh slots). These pushes do not act through any fitted subspace: their

projection on the task band's DAS subspace associates with their escapes at Spearman 0.39 in retrospect and 0.09 on 24 new directions, and clamping the subspace back to the twin removes none of 218 escapes. The axis from the default to the second choice is not their channel either. Priced as a knob, plan noise at the absolute unit beats a base-model intent sentence on kth smallest and the grid and loses on connected components, shortest path and two sum. Where the noise must enter is not settled: in bf16 noise at the task specification positions moves the algorithm while the read position and the condition span barely do, but the preregistered site rule was withdrawn and the condition-span clause of the post hoc ordering fails.

Internal handles exist in both families, and under one protocol none is a certified variable. In Qwen3-8B, exchanging the task specification states of a ban or a forced instruction into a neutral twin transfers the algorithm, and a DAS subspace of rank 1 or 2 at blocks 12 to 23 matches or exceeds that exchange for ban sources and several forced sources, while random subspaces do nothing. But permuted labels transfer 0.47 to 0.70 of the slots, an untrained network admits the fit, the reverse interchange does not restore the default, and the effect of scaling it is a ramp. At the fork itself, one DAS protocol with seven committed controls found a rank-4 handle in both families that writes the algorithm in generation (29 of 30 correct DFS on connected components; 22 of 30 correct BFS in Gemma against 6 unedited). The handle fails specificity in both: fits on permuted labels reach 0.11 to 0.87 of the true fit's effect and fits on a randomly initialised network 0.41 to 0.69, and both forks show the off-manifold overshoot Makelov et al. warn of [122]. At block 24, where the full state carries the class, the off-manifold failures disappear but the specificity failures remain, because the natural difference-in-means direction carries the class and almost any linear fit partly recovers it [132]. In Gemma, the iteration-7 dial is dormant by the shared rule (it keeps 0.18 to 0.28 of its effect inside the twins' natural variation) but lies inside the variation that prompts induce; it works in free generation (38 of 50 correct BFS against 13), and there it is within the range of prompt wordings (14 to 49; the committed wording 32, the best 49). Prompting with the dial's prefill fails (0 of 50), because the prefilled opening is off-policy for the forced prompt and selects a BFS implementation without a wall check. Path-patching circuits for the fork margin, validated against matched random components in the real model, have the same shape in both families: a layer-0 MLP at the condition span injects the instruction, and a late stack of fork-position MLPs and heads turns it into the fork logit, deeper in Qwen3-8B (effect-weighted depth about 0.86) than in Gemma (0.69); 9 of 10 and 8 of 10 Qwen3-8B nodes pass both tests. Steering the top five Qwen3-8B nodes rewrites the algorithm (29 of 30 Prim; 29 of 30 DFS) with every program passing, but at the committed dose matched random components do nearly as much, and the circuit separates from them only at a lower dose (24:0 at dose 0.5 on connected components; 6:1 on minimum spanning tree weight). A natural language autoencoder on Qwen2.5-7B-Instruct names the algorithm only where the prompt names it, and an outside reader of the visible text predicts it better. The read position is inert as a carrier in Qwen3-8B and in that reading.

A single prompt ban, where the model obeys it, removes the default and the output collapses on one replacement: Bellman-Ford/SPFA on shortest path, DFS on the grid, and quickselect and size k heap, brute force, index marking and BFS on the bench tasks where Qwen3-8B obeys. Shown a correct program of another algorithm and told not to use it, the model goes home to its default. How often a ban is obeyed depends on its wording more than on anything else measured, and the template that states only that the algorithm is not allowed here is the weakest in every artifact and model tested. A ban of five names fails by reimplementation without the name in every non-thinking configuration tested (55% to 100% of programs that use a banned mechanism carry no banned name), and among ten open configurations only Qwen3-32B with thinking on obeys it often (17 of 50 correct survivors on the screen, 21 of 50 on held-out templates), at a cost of about 2,200

more reasoning tokens and a 46% truncation rate. Its catalogue survivors are all unit-weight Dijkstra, the farthest non-banned algorithm from the banned five on its manifold of forced prompts at block 31, so they land on the opposite side, not at the edge as preregistered, while the ban prompts themselves move toward two of the banned algorithms; with one landing algorithm the evidence at the level of algorithms is p 0.20, and the verdict holds in 10 of 15 robustness cells. In the fitted coordinates of shortest path, bans sit outside a paraphrase null on the held-out templates while seeds and long novelty instructions sit inside it, and a push inside the forced-prompt manifold toward the survivors of a top-five ban produced BFS in all 300 programs.

Where a ban acts depends on the task. On the path tasks it is carried mostly by the prompt states; on the bench, pooled over five gated tasks, the direct route carries 0.744 [0.38, 1.11] of the behavioural effect and the prompt states 0.611 [0.06, 1.17], with overlapping intervals. The ban is enforced at the opening of the program body, to a single token on kth smallest, the same place where forks surface. The read position predicts how strongly a wording will be obeyed (-0.41 across 110 wordings) but carries none of the effect when exchanged or spliced. What decides the replacement has measured leads and no confirmed rule: the base model's second choice names the push landing in 17 of 21 recorded events and 5 of 8 held-out events (p 0.053), and writing the base model's task specification states into the instruct prefill reopened its second choice on 1 of 5 Qwen3-8B tasks and 2 of 7 Gemma tasks.

No instruction, knob, push, transplant, feature clamp, verbal edit, fitted edit, steered circuit, fork dose, forced branch or prohibition produced a correct novel algorithm, and neither did evolutionary search. Iterations 1 to 3 produced 116 correct programs outside the catalogue and iteration 5 another 130, and none was new; iteration 4 found none among 14,206 correct programs; in iteration 7 an evolutionary search with an executable evaluator reached known techniques outside the catalogue on kth smallest at 0.031 per call against 0.004 for the instruction alone (ratio 7.8, lower bound 2.41, short of the committed tenfold), 103 distinct programs in all and none novel. In iteration 8, none of 5,543, 1,857, 395, 913, 4,703 and 661 correct programs in six experiments was novel; the programs outside the catalogue were counting sort in dictionary form, labeller misses, and known algorithms such as Bellman-Ford on a grid, a random walk with restarts and wavefront propagation.

What remains open follows from this. First, whether a variable for the algorithm exists anywhere: the off-manifold failures vanish where the full state carries the class, the specificity failures do not, and the circuit separates from random components only below saturating doses. Second, the composed temperature law needs more held-out units and a form that sums over openings, for families that reach their fork through many openings. Third, the edge-against-opposite question of the top-five prohibition needs tasks where an obeying model writes several catalogue survivors. Fourth, a feature-level account at either fork, which the transcoder graph has not delivered on the hardware available. Fifth, evolutionary search on a task whose catalogue already covers the known techniques, so that a correct program outside it would be new. Of the items of the request, these have been run. Geometry plots in actual activation coordinates exist at a position that depends on the label (iteration 2), at the plan site (iteration 4), for the top five prohibition and the fork (iteration 6), in the fitted DAS coordinates (iterations 7 and 8), at the fork for every dosed arm (iteration 8) and for the forced prompts of ten algorithms with the ban prompts of the model that obeys (iteration 8), with geodesic distances, Isomap and MDS. The top five prohibition was applied by prompt, token veto, plan push and push inside the manifold, and screened across ten configurations. Natural BFS against DFS internals were read within a ban in Qwen3-8B and within one prompt in Gemma, with circuits for the fork in both. Thought anchors and forks, the logit and

tuned lenses, the test under the same seed with thinking on and off and with top-k 10 (explained by fork margins), the mean-answer question (answered at the fork by the base model comparison), natural language autoencoders, sparse autoencoders (Qwen-Scope on Qwen3-8B, Gemma Scope 2 on Gemma), DAS with the published controls under one protocol, circuits with path patching and evolutionary search with an evaluator were all run.

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